Spring 2024

Algorithm Dynamics in Modern Statistical Learning: Asymptotics, Universality, and Implicit Regularization

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Abstract

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2024

Understanding the dynamics of algorithms is crucial for characterizing the behavior of trained models in modern statistical learning. This thesis presents a few recent results on theoretical analyses of the dynamics of two classes of algorithms: Approximate Message Passing (AMP) algorithms and Stochastic Gradient Descent (SGD). For AMP algorithms, the focus is to derive the precise asymptotic distributional characterization of the iterates, known as the "state evolution" which summarizes the dynamics of AMP iterates, and to understand the universality of such characterization with respect to the underlying data distribution. For SGD, the goal is to perform trajectory analysis to understand its implicit regularization, a key property believed to be essential for the generalization of modern deep learning models. The results presented here provide unified frameworks for analyzing and understanding the dynamics of these algorithms, and can be potentially extended to other algorithms.

Algorithm Dynamics in Modern Statistical Learning: Asymptotics, Universality, and Implicit Regularization

A Dissertation
Presented to the Faculty of the Graduate School
of
Yale University
in Candidacy for the Degree of
Doctor of Philosophy

by
Tianhao Wang

Dissertation Director: Zhou Fan

May 2024

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Acknowledgments

My advisor, Zhou Fan, has been a wonderful mentor throughout my PhD journey, a role model for my development as a researcher, and a guiding light for my future research career. I cannot thank him enough for his unwavering support, encouragement, and guidance. From Zhou, I have learned how to approach research problems by making small cracks that shed light on the path to eventually finding the solution, as well as many beautiful ideas and mathematical insights. His deep understanding of the field, his ability to communicate complex ideas in a simple and clear way, his dedication and perseverance in work are what I aspire to achieve. I am also grateful for the freedom and flexibility he has given me to explore my own research interests. Beyond research, his kindness and genuine warmth have been a great source of comfort and have shaped my own approach to collaboration and personal interactions. Zhou's exemplary mentorship and personality have profoundly influenced both my professional growth and personal development, for which I am deeply grateful.

During my PhD years, I have had the privilege of receiving guidance and support from other research mentors. Quanquan Gu was the first to introduce me to the world of machine learning research, and his continuous support and our collaboration since my undergraduate years have been invaluable. Zhiyuan Li led me into intriguing research problems about implicit regularization of optimization algorithms, and it has been a unique experience to learn something new during every single project we have worked on. I am also grateful to Sanjeev Arora for his guidance during my collaboration with Zhiyuan and a short visit to Princeton. His vision and leadership in the field of deep learning have always been inspiring.

I thank my other collaborators, without whose help and support I would not have been able to complete the works presented in this thesis, as well as those not included. Thanks to Max Lovig and Xinyi Zhong for inspiring conversations about AMP algorithms; to Jason D. Lee, Kaifeng Lyu, Sadhika Malladi, Runzhe Wang, and Dingli Yu for exciting explorations in deep learning; to Jiafan He, Yifei Min, Zhaoran Wang, Ruitu Xu, Zhuoran Yang, and Dongruo Zhou for fruitful discussions on reinforcement learning; to Roy R. Lederman, Yi Sun, Yihong Wu, and Sheng Xu for interesting discussions on orbit recovery problems. Towards the end of my PhD, it has been a great pleasure to work with Siyu Chen and Heejune Sheen on new research projects about transformers.

I am grateful for the support I received from Sanjeev Arora, Joseph Chang, Quanquan Gu, Shuangning Li, Yanlin Qu, Yandi Shen, Yihong Wu, Yuchen Wu, Ilias Zadik, and Harry Zhou during my academic job search.

I cannot imagine a PhD life without the companionship of the friends I met at Yale: Lin Chen, Siwei Chen, Siyu Chen, Siyuan Dong, Han Gu, Jinyi He, Tong Liu, Yifei Min, Yutong Nie, Heejune Sheen, Shanquan Tian, Yitan Wang, Timothy Wee, Ruitu Xu, Yingxi Yang, and Xinyi Zhong. The wonderful time we spent together, playing games, hiking, playing tennis, and enjoying great food, will be cherished forever. I also want to thank my old friends who have been in touch even though we have been far apart: Jinqi Jiang, Haoxiang Huang, Xuening Xu, and Ziji Zhang. Long may our friendship last.

I am extremely fortunate to have met my girlfriend, Jiaqi Cui, at Yale. Her love and support have been my shield against the toughest challenges. I am lucky to have her as a harbor during the dark times and a beacon of joy in the brightest moments. She has illuminated the second half of my PhD journey, as well as our future to come. Many thanks to our dog, Lambo, for being an angel in our life. I also thank my parents and other family members, who have supported me in every possible way.

Chapter 1

Introduction

1.1 Background and motivation

From the classic linear model, to kernel methods, to deep neural networks, and further to the latest large language models, the fields of statistics and machine learning have witnessed a rapid development in the past few decades. The successes of these models are largely attributed to the availability of large-scale and high-quality data and the development of advanced architectures and efficient algorithms to learn from them. However, due to the increasing data dimensionality and ever more complex models, the trained models are often treated as black boxes, where the lack of interpretability leads to concerns about their reliability and beyond. This is particularly true for modern deep learning models that are often so overparameterized that the resulting model search space is much larger than what is needed to fit the data, in which case, classical statistical learning theory fails to explain the generalization. The reason behind such difficulty, to some extent, is the local search nature of the algorithms used in many modern statistical learning problems. It is thus of paramount importance to analyze the algorithm dynamics and characterize the properties of the generated trajectories.

High-dimensionality, overparametrization, and non-convexity altogether complicate the analyses of algorithms used in modern statistical learning. The classical notions of tractability and generalization have to be revisited and possibly redefined in such a context. This requires new mathematical tools and insights to track the evolution of the parameters and to characterize the learned models. In this thesis, we present recent results in pursuit of understanding the algorithms dynamics in modern statistical learning. In particular, we focus on two versatile types of algorithms: Approximate Message Passing (AMP) algorithms and Stochastic Gradient Descent (SGD).

For AMP algorithms, we focus on the asymptotics of the iterates, which admit an exact distributional characterization known as the "state evolution". This provides simple but precise descriptions of the behavior of AMP algorithms, and closely matches finite-sample performance in numerical simulations. AMP has been successfully applied to a wide range of applications in high-dimensional statistics and learning, and has served as a broadly useful theoretical tool for analyzing the asymptotic behavior of statistical methods. The results presented here provide a unified framework for analyzing and understanding the universality of AMP state evolution with respect to the underlying data distribution, thus further strengthening the applicability of AMP algorithms. We also study AMP algorithm with spectral initialization, a popular choice in practice, and provide a rigorous recipe for incorporating informative initialization into AMP algorithms.

For SGD, we focus on its special property known as "implicit regularization". This refers to certain hidden progress made along the SGD trajectory, which is not captured by the decrease in the loss value. Oftentimes, implicit regularization appears as a form of minimization of some regularizer, but the regularizer is not explicitly introduced as a penalty term in the loss function. We will provide a novel mathematical framework to understand the implicit regularization of SGD, which reveals a rich interplay between the data distribution, the model architecture, and the optimization algorithm.

1.2 Contributions

The rest of the thesis is organized as follows.

1. In Chapter 2, we study the universality of AMP algorithm. We extend the state evolution characterization of AMP algorithms to broad classes of random matrices, including generalized Wigner matrices and generalized invariant ensembles. This chapter is drawn from the paper "Universality of Approximate Message Passing algorithms and tensor networks" [WZF22], jointly authored with Xinyi Zhong and Zhou Fan.

2. In Chapter 3, we study AMP algorithms with spectral initialization. We design AMP algorithms with spectral initialization for orthogonally invariant ensembles, derive the corresponding state evolution, and apply the results specifically to the structured PCA problem. This chapter is drawn from the paper "Approximate message passing for orthogonally invariant ensembles: Multivariate non-linearities and spectral initialization" [ZWF21], jointly authored with Xinyi Zhong and Zhou Fan.

3. In Chapter 4, we study the implicit regularization of stochastic gradient descent. We provide a mathematical framework for characterizing SGD's implicit regularization by deriving the corresponding limiting diffusion process describing the long-term behavior of SGD, and in particular, showcase the induced generalization benefit in a concrete example of overparameterized linear model. This chapter is drawn from the paper "What happens after SGD reaches zero loss? — A mathematical framework" [LWA22], jointly authored with Zhiyuan Li and Sanjeev Arora.

4. Chapter 5 presents the results on the implicit regularization of gradient descent for training reparametrized models. Gradient descent is the deterministic coun-

terpart of SGD, and we characterize its implicit regularization via the lens of algorithmic equivalence. This chapter is drawn from the paper "Implicit bias of gradient descent on reparametrized models: On equivalence to mirror descent" [LWLA22], jointly authored with Zhiyuan Li, Jason D. Lee and Sanjeev Arora.

The presentation of the results in this thesis is subject to minor modifications compared to the original manuscripts.

Chapter 2

Universality of Approximate Message Passing Algorithms

Approximate Message Passing (AMP) algorithms provide a valuable tool for studying mean-field approximations and dynamics in a variety of applications. Although these algorithms are often first derived for matrices having independent Gaussian entries or satisfying rotational invariance in law, their state evolution characterizations are expected to hold over larger universality classes of random matrix ensembles.

We develop several new results on AMP universality. For AMP algorithms tailored to independent Gaussian entries, we show that their state evolutions hold over broadly defined generalized Wigner and white noise ensembles, including matrices with heavy-tailed entries and heterogeneous entrywise variances that may arise in data applications. For AMP algorithms tailored to rotational invariance in law, we show that their state evolutions hold over delocalized sign-and-permutation-invariant matrix ensembles that have a limit distribution over the diagonal, including sensing matrices composed of subsampled Hadamard or Fourier transforms and diagonal operators.

We establish these results via a simplified moment-method proof, reducing AMP universality to the study of products of random matrices and diagonal tensors along a tensor network. As a by-product of our analyses, we show that the aforementioned matrix ensembles satisfy a notion of asymptotic freeness with respect to such tensor networks, which parallels usual definitions of freeness for traces of matrix products.

2.1 Introduction on AMP algorithms and the universality

Approximate Message Passing (AMP) algorithms are a general family of iterative algorithms, driven by a random matrix $\mathbf{W}$, whose iterates admit a simple distributional characterization in the asymptotic limit of increasing dimensions. Their origins may be traced separately in the engineering, statistics, and probability literatures [Kab03, DMM09, Bol14], where these algorithms have since provided an important tool for studying mean-field phenomena in many probabilistic models. Without seeking to be exhaustive, we mention here their applications to analyses of spin glass and perceptron models [Bol18, DS19, FW21, BNSX21, FLS22], recovery thresholds and asymptotic phenomena in high-dimensional statistical models [DMM09, Ran11, RF12, DJM13, DM16, BDM+16, BKRS19, SCC19, MV21b, MV21a, ZWF21, LW21], and mean-field dynamics of other first-order optimization algorithms including discrete-time and continuous-time gradient descent [CMW20, CCM21]. We refer readers to [FVRS22] for a recent review.

Asymptotic distributional characterizations of the AMP iterates, known as their *state evolutions*, are often first proved for orthogonally invariant matrices $\mathbf{W}$ using an inductive conditioning technique. For $\mathbf{W}$ with i.i.d. Gaussian entries, this method was developed in [Bol14, BM11] and has been extended to analyze AMP algorithms of increasing generality in [Ran11, JM13, BMN20, MV21b, GB21]. For $\mathbf{W}$ satisfying rotational invariance in law, a similar technique has been applied to analyze various AMP algorithms in [RSF19, SRF16, MP17, Tak17, Tak20, Tak21, LHK21, Fan22], with a parallel line of work [CWF14, OCW16, CO19, CO20] deriving related algorithms using non-rigorous methods of dynamic functional theory.

It is expected—and in some settings known—that the state evolution characterizations of AMP algorithms should extend beyond orthogonally invariant matrices, to describe also the limit distributions of iterates when applied to broader universality classes of random matrix ensembles. For example, it was shown in [BLM15] that AMP algorithms designed for i.i.d. Gaussian matrices and having polynomial non-linearities admit state evolutions that are universal across matrices with sub-Gaussian entries of common variance. In [CL21], universality over a similar matrix class for AMP with Lipschitz non-linearities was proven using a different Gaussian interpolation method, and extended to spectrally initialized algorithms for spiked matrix models. Moving beyond matrices with independent entries, in [DB20] it was shown that the state evolution of a linear AMP algorithm for phase retrieval holds universally for sub-sampled Hadamard matrices. Recently, results of [DLS22, DSL22]—fruit of parallel research efforts—showed universality for AMP algorithms having divergence-free non-linearities over a broad model of semi-random matrices with randomly signed rows/columns and delocalized entries. The latter work [DSL22] also applied these results to establish universality classes of matrices for more general first-order iterative algorithms, including proximal gradient methods and general versions of AMP.

2.1.1 Conventions of notation in this chapter

We denote entries of $\mathbf{x} \in \mathbb{R}^n$ and $\mathbf{W} \in \mathbb{R}^{n\times n}$ as $x[i]$ and $W[i,j]$. For vectors $\mathbf{x}_1, \ldots, \mathbf{x}_k \in \mathbb{R}^n$ and a random vector $(X_1, \ldots, X_k)$, we write

$$(\mathbf{x}_1, \ldots, \mathbf{x}_k) \overset{W_p}{\to} (X_1, \ldots, X_k) \text{ as } n \to \infty$$

for the Wasserstein-p convergence of the empirical distribution of rows of $(\mathbf{x}_1, \ldots, \mathbf{x}_k) \in \mathbb{R}^{n \times k}$ to the joint law of $(X_1, \ldots, X_k)$. This means, for any continuous function $f : \mathbb{R}^k \to \mathbb{R}$ satisfying

$$|f(x_1, \ldots, x_k)| \leq C(1 + \|(x_1, \ldots, x_k)\|_2^p) \text{ for a constant } C > 0, \tag{2.1.1}$$

we have as $n \to \infty$

$$\frac{1}{n} \sum_{i=1}^{n} f(x_1[i], \ldots, x_k[i]) \to \mathbb{E}[f(X_1, \ldots, X_k)]. \tag{2.1.2}$$

We write

$$(\mathbf{x}_1, \ldots, \mathbf{x}_k) \xrightarrow{W} (X_1, \ldots, X_k)$$

to mean that the above Wasserstein-p convergence holds for every order $p \geq 1$.

For a function $f : \mathbb{R}^k \to \mathbb{R}$ and vectors $\mathbf{x}_1, \ldots, \mathbf{x}_k \in \mathbb{R}^n$, we denote by $f(\mathbf{x}_1, \ldots, \mathbf{x}_k) \in \mathbb{R}^n$ the evaluation of $f(\cdot)$ on each row of $(\mathbf{x}_1, \ldots, \mathbf{x}_k) \in \mathbb{R}^{n \times k}$. We write $\langle \cdot \rangle$ for the empirical average of the coordinates of a vector, and introduce the shorthand $\mathbf{x}_{1:k} = (\mathbf{x}_1, \ldots, \mathbf{x}_k)$ and $X_{1:k} = (X_1, \ldots, X_k)$. Thus (2.1.2) may be expressed as $\langle f(\mathbf{x}_{1:k}) \rangle \to \mathbb{E}[f(X_{1:k})]$.

For vectors $\mathbf{x}, \mathbf{y} \in \mathbb{R}^n$, $\mathbf{x} \odot \mathbf{y} \in \mathbb{R}^n$ is their entrywise product. $\mathrm{diag}(\mathbf{x}) \in \mathbb{R}^{n \times n}$ or $\mathrm{diag}(\mathbf{x}) \in \mathbb{R}^{n \times \cdots \times n}$ denotes the diagonal matrix or tensor with $\mathbf{x}$ along the main diagonal, i.e. $\mathrm{diag}(\mathbf{x})[i, \ldots, i] = \mathbf{x}[i]$ and $\mathrm{diag}(\mathbf{x})$ has all other entries equal to 0. For $\mathbf{x} \in \mathbb{R}^{\min(m,n)}$, we write also $\mathrm{diag}(\mathbf{x}) \in \mathbb{R}^{m \times n}$ for the rectangular diagonal matrix where each (i, i) entry is $x[i]$; we will indicate the dimensions if needed to disambiguate these notations. $\|\mathbf{W}\|_{\mathrm{op}}$ is the $\ell_2 \to \ell_2$ operator norm of the matrix $\mathbf{W}$. We denote $[n] = \{1, \ldots, n\}$, and reserve Roman letters $i, j, \ldots$ for indices in $[n]$ and Greek letters $\alpha, \beta, \ldots$ for indices in $[m]$.

2.2 Main results

2.2.1 Universality of AMP algorithms for symmetric matrices

Let $\mathbf{W} \in \mathbb{R}^{n\times n}$ be a symmetric random matrix. Consider an initialization $\mathbf{u}_1 \in \mathbb{R}^n$ and auxiliary "side information" vectors $\mathbf{f}_1, \ldots, \mathbf{f}_k \in \mathbb{R}^n$, independent of $\mathbf{W}$. In applications, such side information vectors may play the role of the external field in spin glass models, the true signal vector in spiked matrix models, or the signal and residual error vectors in regression models. We refer to [BM11, Ran11, RF12] for several examples. Let $u_2, u_3, u_4, \ldots$ be a sequence of non-linear functions, where $u_{t+1} : \mathbb{R}^{t+k} \to \mathbb{R}$. We study a general form for an AMP algorithm with separable non-linearities that computes, for $t = 1, 2, 3, \ldots$

$$\mathbf{z}_t = \mathbf{W}\mathbf{u}_t - \sum_{s=1}^{t} b_{ts}\mathbf{u}_s \tag{2.2.1a}$$

$$\mathbf{u}_{t+1} = u_{t+1}(\mathbf{z}_1, \ldots, \mathbf{z}_t, \mathbf{f}_1, \ldots, \mathbf{f}_k) \tag{2.2.1b}$$

where $\{b_{ts}\}_{s\le t}$ are deterministic scalar "Onsager correction" coefficients. We will characterize the iterates of this algorithm in the large system limit as $n \to \infty$, for fixed $k \ge 0$.

We assume throughout the following conditions for $(\mathbf{u}_1, \mathbf{f}_1, \ldots, \mathbf{f}_k)$.

Assumption 2.2.1. Almost surely as $n \to \infty$,

$$(\mathbf{u}_1, \mathbf{f}_1, \ldots, \mathbf{f}_k) \xrightarrow{W} (U_1, F_1, \ldots, F_k) \tag{2.2.2}$$

for a joint limit law $(U_1, F_1, \ldots, F_k)$ having finite moments of all orders, where $\mathbb{E}[U_1^2] > 0$. Furthermore, multivariate polynomials are dense in the real L^2-space of functions

$f : \mathbb{R}^{k+1} \to \mathbb{R}$ with inner-product

$$(f, g) \mapsto \mathbb{E}[f(U_1, F_1, \ldots, F_k) g(U_1, F_1, \ldots, F_k)].$$

Remark 2.2.2. The convergence (2.2.2) holds, for example, if rows of $(\mathbf{u}_1, \mathbf{f}_1, \ldots, \mathbf{f}_k) \in \mathbb{R}^{n\times(k+1)}$ are i.i.d. and equal in law to $(U_1, F_1, \ldots, F_k)$. The density of polynomials holds if $\|(U_1, F_1, \ldots, F_k)\|_2$ has finite moment generating function in a neighborhood of 0; see [Sch17, Section 14.1 and Corollary 14.24].

In an AMP algorithm, the coefficients $\{b_{ts}\}$ of (2.2.1) are defined so that the iterates $\{\mathbf{z}_t\}$ are described by a simple *state evolution* in the asymptotic limit as $n \to \infty$. For $\mathbf{W} \sim \mathrm{GOE}(n)$ (c.f. Definition 2.2.3), this may be done as follows: Set $\mathbf{\Sigma}_1 = \mathbb{E}[U_1^2] \in \mathbb{R}^{1\times 1}$. Inductively, having defined $\mathbf{\Sigma}_t \in \mathbb{R}^{t\times t}$, let $Z_{1:t} \sim \mathcal{N}(0, \mathbf{\Sigma}_t)$ be independent of $(U_1, F_{1:k})$, set $U_{s+1} = u_{s+1}(Z_{1:s}, F_{1:k})$ for each $s = 1, \ldots, t$, and define

$$\mathbf{\Sigma}_{t+1} = (\mathbb{E}[U_r U_s])_{r,s=1}^{t+1} \in \mathbb{R}^{(t+1)\times(t+1)}. \tag{2.2.3}$$

Let $b_{tt} = 0$, and for each $s < t$, define the coefficient b_{ts} as

$$b_{ts} = \mathbb{E}[\partial_s u_t(Z_{1:t-1}, F_{1:k})] \tag{2.2.4}$$

where $\partial_s u_t$ is the partial derivative of $u_t(\cdot)$ in its s^{th} argument. We will call (2.2.3) and (2.2.4) the *GOE prescriptions* for $\mathbf{\Sigma}_t$ and b_{ts}. Results of [BM11, JM13] (see also [Mon21, Proposition 2.1] for this form) then imply that, for any Lipschitz functions $u_t(\cdot)$, the iterates of (2.2.1) satisfy the state evolution, almost surely as $n \to \infty$ for any fixed $t \geq 1$,

$$(\mathbf{u}_1, \mathbf{f}_1, \ldots, \mathbf{f}_k, \mathbf{z}_1, \ldots, \mathbf{z}_t) \xrightarrow{W} (U_1, F_1, \ldots, F_k, Z_1, \ldots, Z_t).$$

We note that a variant of this algorithm may instead use the empirical average $b_{ts} = \langle \partial_s u_t(\mathbf{z}_{1:t-1}, \mathbf{f}_{1:k}) \rangle$, for which the same state evolution continues to hold (cf. Remark 2.2.9).

In [Fan22], building upon work of [OCW16], an extension of this result was proven for a larger class of orthogonally invariant matrices and non-linear functions: We say that $\mathbf{W}$ is *orthogonally invariant* if it has spectral decomposition $\mathbf{W} = \mathbf{ODO}^\top$ where $\mathbf{O} \sim \mathrm{Haar}(\mathbb{O}(n))$ is Haar-distributed on the orthogonal group and independent of $\mathbf{D} = \mathrm{diag}(\mathbf{d})$. Suppose that $\mathbf{d} \xrightarrow{W} D$ as $n \to \infty$, where D represents the limit spectral law of $\mathbf{W}$. Set $\mathbf{\Sigma}_1 = \mathrm{Var}[D] \cdot \mathbb{E}[U_1^2] \in \mathbb{R}^{1\times 1}$. Having defined $\mathbf{\Sigma}_t \in \mathbb{R}^{t\times t}$, let $Z_{1:t} \sim \mathcal{N}(0, \mathbf{\Sigma}_t)$ be independent of $(U_1, F_{1:k})$, let $U_{s+1} = u_{s+1}(Z_{1:s}, F_{1:k})$ for each $s = 1, \ldots, t$, and define

$$\mathbf{\Sigma}_{t+1} = \mathbf{\Sigma}_{t+1}\Big(\{\mathbb{E}[U_r U_s]\}_{1\le r,s\le t+1}, \{\mathbb{E}[\partial_r u_{s+1}(Z_{1:s}, F_{1:k})]\}_{1\le r\le s\le t} \Big) \in \mathbb{R}^{(t+1)\times(t+1)} \tag{2.2.5}$$

for a continuous function $\mathbf{\Sigma}_{t+1}(\cdot)$ whose form depends only on the law of D. For each $s \le t$ and a continuous function $b_{ts}(\cdot)$ whose form also depends only on the law of D, define

$$b_{ts} = b_{ts}\Big(\{\mathbb{E}[U_q U_r]\}_{1\le q,r\le t}, \{\mathbb{E}[\partial_q u_{r+1}(Z_{1:r}, F_{1:k})]\}_{1\le q\le r<t} \Big). \tag{2.2.6}$$

We will call (2.2.5) and (2.2.6) the *orthogonally invariant prescriptions* for $\mathbf{\Sigma}_t$ and b_{ts}. We refer to [Fan22, Section 4] for their precise functional forms, which will not be important for our current work. When $\mathbf{W} \sim \mathrm{GOE}(n)$ and D has Wigner's semicircle law on $[-2, 2]$, these reduce to the previous GOE prescriptions of (2.2.3) and (2.2.4). It was shown in [Fan22] that for weakly differentiable functions $u_t(\cdot)$ whose derivatives have at most polynomial growth, the iterates of (2.2.1) again satisfy the state evolution, almost surely as $n \to \infty$ for any fixed $t \ge 1$,

$$(\mathbf{u}_1, \mathbf{f}_1, \ldots, \mathbf{f}_k, \mathbf{z}_1, \ldots, \mathbf{z}_t) \xrightarrow{W} (U_1, F_1, \ldots, F_k, Z_1, \ldots, Z_t).$$

Our main results are universality statements that extend the state evolution characterizations of these AMP algorithms to more general random matrix ensembles. Corresponding to $\mathbf{W} \sim \text{GOE}(n)$, we study the following universality class of generalized Wigner matrices, having possibly heteroskedastic entrywise variances and heavy-tailed entries.

Definition 2.2.3. $\mathbf{W} \in \mathbb{R}^{n\times n}$ is a *generalized Wigner matrix* with (deterministic) variance profile $\mathbf{S} \in \mathbb{R}^{n\times n}$ if

(a) $\mathbf{W}$ is symmetric, and entries on and above the diagonal $(W[i,j] : 1 \leq i \leq j \leq n)$ are independent.

(b) Each $W[i,j]$ has mean 0, variance $n^{-1}S[i,j]$, and higher moments satisfying, for each integer $p \geq 3$,

$$\lim_{n\to\infty} n \cdot \max_{i,j=1}^{n} \mathbb{E}[|W[i,j]|^p] = 0.$$

(c) For a constant $C > 0$ independent of n,

$$\max_{i,j=1}^{n} S[i,j] \leq C \quad \text{and} \quad \lim_{n\to\infty} \max_{i=1}^{n} \left| \frac{1}{n} \sum_{j=1}^{n} S[i,j] - 1 \right| = 0.$$

We write $\mathbf{W} \sim \text{GOE}(n)$ for the special case where $W[i,j] \sim \mathcal{N}(0,1/n)$ and $S[i,j] = 1$ for all $i < j$, and $W[i,i] \sim \mathcal{N}(0,2/n)$ and $S[i,i] = 2$ for all i.

The moment assumption in condition (b) weakens a uniform sub-Gaussianity condition for $\sqrt{n}\,W[i,j]$ that is assumed in the previous AMP universality results of [BLM15, CL21] and that would require instead $\mathbb{E}[|W[i,j]|^p] \lesssim n^{-p/2}$ for all $p \geq 3$. This condition (b) is weak enough to encompass centered and normalized adjacency matrices of sparse random graphs with slowly growing average vertex degree. Condition

(c) allows general patterns of entrywise variances whose rows and columns have approximately the same sum, where we also require in (2.2.7) of Theorem 2.2.4 below that these rows and columns are "asymptotically unaligned" with the initialization and side information vectors $\mathbf{u}_1, \mathbf{f}_1, \ldots, \mathbf{f}_k$. We discuss two applications in Examples 2.2.24 and 2.2.25 of Section 2.2.4.

The following theorem shows that the state evolution of AMP algorithms for GOE random matrices remains valid for matrices $\mathbf{W}$ in this generalized Wigner universality class.

Theorem 2.2.4. *Let $\mathbf{W} \in \mathbb{R}^{n\times n}$ be a generalized Wigner matrix with variance profile $\mathbf{S}$, and let $\mathbf{u}_1, \mathbf{f}_1, \ldots, \mathbf{f}_k$ be independent of $\mathbf{W}$ and satisfy Assumption 2.2.1. Suppose that*

1. *Each function $u_{t+1} : \mathbb{R}^{t+k} \to \mathbb{R}$ is continuous, satisfies the polynomial growth condition (2.1.1) for some order $p \geq 1$, and is Lipschitz in its first t arguments.*
2. *$\|\mathbf{W}\|_{\mathrm{op}} < C$ for a constant $C > 0$ almost surely for all large n.*
3. *Let $\mathbf{s}_i$ be the i^{th} row of $\mathbf{S}$. For any fixed polynomial function $q : \mathbb{R}^{k+1} \to \mathbb{R}$, almost surely as $n \to \infty$,*

$$\max_{i=1}^{n} \Big| \langle q(\mathbf{u}_1, \mathbf{f}_1, \ldots, \mathbf{f}_k) \odot \mathbf{s}_i \rangle - \langle q(\mathbf{u}_1, \mathbf{f}_1, \ldots, \mathbf{f}_k) \rangle \cdot \langle \mathbf{s}_i \rangle \Big| \to 0. \tag{2.2.7}$$

Let $\{b_{ts}\}$ and $\{\mathbf{\Sigma}_t\}$ be defined by the GOE prescriptions (2.2.3) and (2.2.4), where each matrix $\mathbf{\Sigma}_t$ is non-singular. Then for any fixed $t \geq 1$, almost surely as $n \to \infty$, the iterates of (2.2.1) satisfy

$$(\mathbf{u}_1, \mathbf{f}_1, \ldots, \mathbf{f}_k, \mathbf{z}_1, \ldots, \mathbf{z}_t) \xrightarrow{W_2} (U_1, F_1, \ldots, F_k, Z_1, \ldots, Z_t)$$

where $(Z_1, \ldots, Z_t) \sim \mathcal{N}(0, \mathbf{\Sigma}_t)$ is independent of $(U_1, F_1, \ldots, F_k)$, i.e. this limit has the same joint law as described by the AMP state evolution for $\mathbf{W} \sim \mathrm{GOE}(n)$.

Next, corresponding to orthogonal invariance, we study universality classes of matrices that are permutation-and-sign-invariant in law and that have limit distributions over the diagonal, in the following sense inspired by [ACD+21]: Let $\Delta : \mathbb{R}^{n\times n} \to \mathbb{R}^{n\times n}$ be the diagonal map that preserves only the entries on the diagonal, i.e.

$$\Delta(\mathbf{M}) = \operatorname{diag}(M[1,1], \ldots, M[n,n]) \in \mathbb{R}^{n\times n}.$$

Let $\Delta\langle \mathbf{x} \rangle$ denote the set of all words in $\mathbf{x}$ and $\Delta(\cdot)$, for example

$$\mathbf{x}\mathbf{x}, \quad \mathbf{x}\Delta(\mathbf{x}\mathbf{x})\mathbf{x}, \quad \Delta(\mathbf{x}\mathbf{x}\Delta(\mathbf{x}))\mathbf{x}, \quad \mathbf{x}\mathbf{x}\mathbf{x}\Delta(\Delta(\mathbf{x}))\Delta(\mathbf{x}\mathbf{x}).$$

We refer to $\Delta\langle \mathbf{x} \rangle$ as the set of *diagonal monomials* in $\mathbf{x}$. For $p(\mathbf{x}) \in \Delta\langle \mathbf{x} \rangle$ and $\mathbf{M} \in \mathbb{R}^{n\times n}$, we write $p(\mathbf{M}) \in \mathbb{R}^{n\times n}$ for its evaluation at $\mathbf{x} = \mathbf{M}$.

Definition 2.2.5. The *distribution over the diagonal* of $\mathbf{M}$ is the mapping[1]

$$p(\mathbf{x}) \in \Delta\langle \mathbf{x} \rangle \mapsto \tfrac{1}{n} \operatorname{Tr} p(\mathbf{M}).$$

Matrices $\mathbf{M} \in \mathbb{R}^{n\times n}$ *converge in diagonal distribution a.s.* if $\lim_{n\to\infty} \frac{1}{n} \operatorname{Tr} p(\mathbf{M})$ exists almost surely (and is finite) for every fixed $p(\mathbf{x}) \in \Delta\langle \mathbf{x} \rangle$. The *limit diagonal distribution* of $\mathbf{M}$, which we will refer to as $\mathcal{D}_{\text{diag}}$, is then the mapping

$$p(\mathbf{x}) \in \Delta\langle \mathbf{x} \rangle \mapsto \lim_{n\to\infty} \tfrac{1}{n} \operatorname{Tr} p(\mathbf{M}).$$

We remark that $\mathcal{D}_{\text{diag}}$ specifies the limit of $\frac{1}{n} \operatorname{Tr} \mathbf{M}^\nu$ for each fixed integer $\nu \geq 1$, and hence also the limit spectral distribution of $\mathbf{M}$ when this distribution has compact support.

We call $\mathbf{\Pi} = \mathbf{P}\mathbf{\Xi} \in \mathbb{R}^{n\times n}$ a *uniformly random signed permutation matrix* if

1. We define the distribution over the diagonal by the values of $\frac{1}{n} \operatorname{Tr} p(\mathbf{M}) \in \mathbb{R}$ rather than $\Delta(p(\mathbf{M})) \in \mathbb{R}^{n\times n}$ as might be more standard in operator-valued free probability.

$\mathbf{\Xi} = \text{diag}(\xi[1], \ldots, \xi[n]) \in \mathbb{R}^{n \times n}$ where each diagonal entry $\xi[i]$ is independently chosen from $\{+1, -1\}$ with equal probability, and $\mathbf{P} \in \mathbb{R}^{n \times n}$ is a uniformly random permutation matrix independent of $\mathbf{\Xi}$. Note that for any symmetric matrix $\mathbf{M}$ and signed permutation matrix $\mathbf{\Pi}$, we have $\Delta(\mathbf{\Pi M \Pi}^\top) = \mathbf{\Pi}\Delta(\mathbf{M})\mathbf{\Pi}^\top$, so also $p(\mathbf{\Pi M \Pi}^\top) = \mathbf{\Pi}\, p(\mathbf{M})\, \mathbf{\Pi}^\top$ for every diagonal monomial $p(\mathbf{x}) \in \Delta\langle \mathbf{x} \rangle$. In particular, $\mathbf{M}$ and $\mathbf{\Pi M \Pi}^\top$ have the same distributions over the diagonal. The following then defines our universality class.

Definition 2.2.6. $\mathbf{W} = \mathbf{\Pi M \Pi}^\top \in \mathbb{R}^{n \times n}$ is a *symmetric generalized invariant matrix*[2] with limit diagonal distribution $\mathcal{D}_{\text{diag}}$ if, as $n \to \infty$,

(a) $\mathbf{M} \in \mathbb{R}^{n \times n}$ converges in diagonal distribution a.s. to a limit $\mathcal{D}_{\text{diag}}$.

(b) For any $\varepsilon > 0$ and any fixed $p(\mathbf{x}) \in \Delta\langle \mathbf{x} \rangle$, almost surely for all large n,

$$\max_{i \neq j} |p(\mathbf{M})[i,j]| < n^{-1/2+\varepsilon}.$$

(c) $\mathbf{\Pi} \in \mathbb{R}^{n \times n}$ is a uniformly random signed permutation, independent of $\mathbf{M}$.

Our result on AMP universality will pertain specifically to such matrices $\mathbf{W}$ whose limit diagonal distribution $\mathcal{D}_{\text{diag}}$ coincides with that of an orthogonally invariant matrix. In this setting, the next proposition clarifies that $\mathcal{D}_{\text{diag}}$ is determined uniquely by the limit spectral law of $\mathbf{W}$, and it also provides simpler conditions inspired by the Spectral Universality Class in [DSL22] that imply Definition 2.2.6. We have stated Definition 2.2.6 for more general limits $\mathcal{D}_{\text{diag}}$ because, as discussed in Remark 2.2.15 to follow, we will in fact prove a general lemma showing the existence and universality of the limit empirical distribution of iterates for first-order iterative algorithms applied

2. More formally, these definitions of generalized Wigner and generalized invariant matrices are describing sequences of matrices $\mathbf{W} \in \mathbb{R}^{n \times n}$ of increasing dimensions $n \to \infty$, rather than a single matrix. We will choose not make this terminological distinction in our work.

to any such matrix $\mathbf{W}$, even if $\mathcal{D}_{\text{diag}}$ does not correspond to an orthogonally-invariant model.

Proposition 2.2.7. *Let $\mathbf{W} \in \mathbb{R}^{n\times n}$ be a symmetric matrix with eigenvalues $\mathbf{d} \in \mathbb{R}^n$ satisfying $\mathbf{d} \xrightarrow{W} D$ almost surely as $n \to \infty$, where D has finite moments of all orders.*

(a) If $\mathbf{W}$ is orthogonally invariant, then $\mathbf{W}$ is a symmetric generalized invariant matrix in the sense of Definition 2.2.6, and its limit diagonal distribution $\mathcal{D}_{\text{diag}}$ is determined uniquely by the law of D.

(b) Suppose that either

1. *$\mathbf{W} = \mathbf{O}\mathbf{D}\mathbf{O}^\top$ where $\mathbf{D} = \operatorname{diag}(\mathbf{d})$ and $\mathbf{O} = \mathbf{\Pi}_V \mathbf{H} \mathbf{\Pi}_E$, such that $\mathbf{\Pi}_V, \mathbf{\Pi}_E \in \mathbb{R}^{n\times n}$ are uniformly random signed permutations independent of each other and of $(\mathbf{D}, \mathbf{H})$, and $\mathbf{H}$ is an orthogonal matrix with entries satisfying*

$$\max_{i,j\in[n]} |H[i,j]| < n^{-1/2+\varepsilon} \tag{2.2.8}$$

for any fixed $\varepsilon > 0$, almost surely for all large n.

2. *$\mathbf{W} = \mathbf{\Pi}\mathbf{M}\mathbf{\Pi}^\top$ such that $\mathbf{\Pi}$ is a uniformly random signed permutation independent of $\mathbf{M}$ (which has eigenvalues $\mathbf{d}$), and for each fixed integer $\nu \geq 1$, the matrix $\mathbf{M}^\nu$ satisfies*

$$\max_{i=1}^{n} \left| M^\nu[i,i] - \tfrac{1}{n} \operatorname{Tr} \mathbf{M}^\nu \right| < n^{-1/2+\varepsilon}, \quad \max_{i\neq j} \left| M^\nu[i,j] \right| < n^{-1/2+\varepsilon} \tag{2.2.9}$$

for any fixed $\varepsilon > 0$, almost surely for all large n.

Then $\mathbf{W}$ is a generalized invariant matrix in the sense of Definition 2.2.6, and its limit diagonal distribution $\mathcal{D}_{\text{diag}}$ coincides with that of the orthogonally invariant matrix in part (a).

We prove Proposition 2.2.7 in Section 2.4. Important examples for applications are when $\mathbf{W}$ is a composition of permutations, deterministic Hadamard/Fourier matrices, and diagonal operators. We discuss one such application to compressed sensing in Example 2.2.26 of Section 2.2.4.

The following is our main theorem on AMP universality in this context, showing that the state evolution of AMP algorithms for orthogonally invariant matrices holds universally over the class of generalized invariant matrices with matching limit diagonal distribution.

Theorem 2.2.8. *Let $\mathbf{W} \in \mathbb{R}^{n\times n}$ be a symmetric generalized invariant matrix whose limit diagonal distribution $\mathcal{D}_{\text{diag}}$ coincides with that of an orthogonally invariant matrix $\mathbf{G}$. Let $\mathbf{u}_1, \mathbf{f}_1, \ldots, \mathbf{f}_k$ be independent of $\mathbf{W}$ and satisfy Assumption 2.2.1. Suppose that*

1. *Each function $u_{t+1} : \mathbb{R}^{t+k} \to \mathbb{R}$ is continuous, satisfies the polynomial growth condition (2.1.1) for some order $p \geq 1$, and is Lipschitz in its first t arguments.*

2. *$\|\mathbf{W}\|_{\text{op}} < C$ for a constant $C > 0$ almost surely for all large n.*

Let $\{b_{ts}\}$ and $\{\mathbf{\Sigma}_t\}$ be defined by the orthogonally invariant prescriptions (2.2.5) and (2.2.6) for the limit spectral distribution D specified by $\mathcal{D}_{\text{diag}}$. Suppose that $\text{Var}[D] > 0$ and each matrix $\mathbf{\Sigma}_t$ is non-singular. Then for any fixed $t \geq 1$, almost surely as $n \to \infty$, the iterates of (2.2.1) satisfy

$$(\mathbf{u}_1, \mathbf{f}_1, \ldots, \mathbf{f}_k, \mathbf{z}_1, \ldots, \mathbf{z}_t) \xrightarrow{W_2} (U_1, F_1, \ldots, F_k, Z_1, \ldots, Z_t)$$

where $(Z_1, \ldots, Z_t) \sim \mathcal{N}(0, \mathbf{\Sigma}_t)$ is independent of $(U_1, F_1, \ldots, F_k)$, i.e. this limit has the same joint law as described by the AMP state evolution for $\mathbf{G}$.

Remark 2.2.9. Theorems 2.2.4 and 2.2.8 hold equally for AMP algorithms where, in the prescriptions (2.2.4) and (2.2.6) for b_{ts}, the quantities $\mathbb{E}[\partial_r u_{s+1}(Z_{1:s}, F_{1:k})]$ and

$\mathbb{E}[U_r U_s]$ are replaced by the empirical averages

$$\langle \partial_r u_{s+1}(\mathbf{z}_{1:s}, \mathbf{f}_{1:k}) \rangle = \frac{1}{n} \sum_{i=1}^{n} \partial_r u_{s+1}(z_{1:s}[i], f_{1:k}[i]), \qquad \langle \mathbf{u}_r \odot \mathbf{u}_s \rangle = \frac{1}{n} \sum_{i=1}^{n} u_r[i] u_s[i].$$

For example, such an AMP algorithm for GOE matrices $\mathbf{W}$ and non-linearities $u_{t+1}(z_{1:t}, f_{1:k}) = u_{t+1}(z_t)$ consists of the iterations

$$\mathbf{z}_t = \mathbf{W}\mathbf{u}_t - \langle u_t'(\mathbf{z}_{t-1}) \rangle \mathbf{u}_{t-1}, \qquad \mathbf{u}_{t+1} = u_{t+1}(\mathbf{z}_t).$$

To see this, note that b_{11} depends only on $\mathbb{E}[U_1^2]$, so these prescriptions for b_{11} asymptotically coincide by Assumption 2.2.1. Then the state evolution holds for $\mathbf{z}_1$. Inductively, validity of the state evolution for $\mathbf{z}_{1:t}$ ensures that, almost surely as $n \to \infty$,

$$\langle \partial_r u_{s+1}(\mathbf{z}_{1:s}, \mathbf{f}_{1:k}) \rangle \to \mathbb{E}[\partial_r u_{s+1}(Z_{1:s}, F_{1:k})] \text{ for all } r \leq s \leq t,$$
$$\langle \mathbf{u}_r \odot \mathbf{u}_s \rangle \to \mathbb{E}[U_r U_s] \text{ for all } r, s \leq t+1$$

where the first statement follows from Wasserstein-2 convergence of $(\mathbf{z}_{1:s}, \mathbf{f}_{1:k})$ and Stein's lemma (c.f. [Fan22, Proposition E.5]). Then the presciptions of (2.2.4) and (2.2.6) for $\{b_{t+1,s}\}_{s \leq t+1}$ asymptotically coincide with their empirical versions defined by $\langle \partial_r u_{s+1}(\mathbf{z}_{1:s}, \mathbf{f}_{1:k}) \rangle$ and $\langle \mathbf{u}_r \odot \mathbf{u}_s \rangle$, which in turn implies validity of the state evolution for $\mathbf{z}_{1:(t+1)}$.

Remark 2.2.10. Theorems 2.2.4 and 2.2.8 show universality of AMP algorithms with an initialization $\mathbf{u}_1$ that is independent of $\mathbf{W}$. For spiked matrix models with a low-rank signal component, alternative AMP algorithms with spectral initializations have been studied for example in [MV21b, MV21a, ZWF21]. Universality for such algorithms may be shown using the preceding results, by approximating the spectral initialization with a large number of linear AMP iterations starting from an initialization $\mathbf{u}_1$ that is independent of $\mathbf{W}$ but correlated with the true signal; we refer to [CL21, Section 8] and [ZWF21, Section A.2] for examples of this type of argument.

Since we allow the non-linearities $u_{t+1}(\cdot)$ to be functions of all preceding iterates $\mathbf{z}_1, \ldots, \mathbf{z}_t$, universality of AMP with matrix-valued iterates in $\mathbb{R}^{n \times J}$ for a fixed dimension $J \geq 1$ may also be deduced from the preceding results, by simulating each iteration of any such algorithm using J iterations of an algorithm with iterates in $\mathbb{R}^n$. We leave the further study of these extensions to future work, as the need arises in applications.

2.2.2 Tensor networks and strategy of proof

We describe here our high-level strategy of proof for Theorems 2.2.4 and 2.2.8. The full proofs of these results are contained in Section 2.3.

Definition 2.2.11. A *diagonal tensor network* $T = (\mathcal{V}, \mathcal{E}, \{q_v\}_{v \in \mathcal{V}})$ in k variables is an undirected tree graph with vertices $\mathcal{V}$ and edges $\mathcal{E} \subset \mathcal{V} \times \mathcal{V}$, each of whose vertices $v \in \mathcal{V}$ is labeled by a polynomial function $q_v : \mathbb{R}^k \to \mathbb{R}$. The *value* of T on a symmetric matrix $\mathbf{W} \in \mathbb{R}^{n \times n}$ and vectors $\mathbf{x}_1, \ldots, \mathbf{x}_k \in \mathbb{R}^n$ is

$$\mathrm{val}_T(\mathbf{W}; \mathbf{x}_1, \ldots, \mathbf{x}_k) = \frac{1}{n} \sum_{\mathbf{i} \in [n]^{\mathcal{V}}} q_{\mathbf{i}|T} \cdot W_{\mathbf{i}|T}$$

where, for each index tuple $\mathbf{i} = (i_v : v \in \mathcal{V}) \in [n]^{\mathcal{V}}$, we set

$$q_{\mathbf{i}|T} = \prod_{v \in \mathcal{V}} q_v(x_1[i_v], \ldots, x_k[i_v]), \qquad W_{\mathbf{i}|T} = \prod_{(u,v) \in \mathcal{E}} W[i_u, i_v].$$

This value may be understood as:

1. Associating to each vertex $v \in \mathcal{V}$ a diagonal tensor $\mathbf{T}_v = \mathrm{diag}(q_v(\mathbf{x}_1, \ldots, \mathbf{x}_k)) \in \mathbb{R}^{n \times \cdots \times n}$, where the order of this tensor equals the degree of v in the tree.[3]

3. We remind readers our notation that $q_v(\mathbf{x}_1, \ldots, \mathbf{x}_k) \in \mathbb{R}^n$ indicates the application of $q_v : \mathbb{R}^k \to \mathbb{R}$

2. Associating to each edge the symmetric matrix $\mathbf{W}$.

3. Iteratively contracting all tensor-matrix-tensor products represented by the edges of the tree.

For example, if $\mathcal{V} = [w+1]$ and T is the line graph $1 - 2 - \cdots - w - (w+1)$, then $\mathbf{T}_1, \mathbf{T}_{w+1} \in \mathbb{R}^n$ are vectors, $\mathbf{T}_v \in \mathbb{R}^{n \times n}$ is a diagonal matrix for each vertex $v \in \{2, \ldots, w\}$, and the value (in usual matrix-vector product notation) is

$$\mathrm{val}_T(\mathbf{W}; \mathbf{x}_1, \ldots, \mathbf{x}_k) = \frac{1}{n}\, \mathbf{T}_1^\top \mathbf{W} \mathbf{T}_2 \mathbf{W} \cdots \mathbf{W} \mathbf{T}_w \mathbf{W} \mathbf{T}_{w+1}.$$

When each tensor $\mathbf{T}_v$ has all 1's along the main diagonal, this definition is an example of the graph sum used to show asymptotic freeness of Wigner and diagonal matrices in [MS12, MS17], and it is also a specific case of a "graph monomial" in the notion of traffic freeness in [Mal20].

We will show in Lemma 2.3.11 that for any AMP algorithm (or more generally, any first-order iterative algorithm of the form (2.2.1)) with polynomial non-linearities $u_2, u_3, u_4, \ldots$, and for any polynomial test function $p(\cdot)$, the coordinate average

$$\langle p(\mathbf{u}_{1:t}, \mathbf{z}_{1:t}, \mathbf{f}_{1:k}) \rangle = \frac{1}{n} \sum_{i=1}^{n} p(u_{1:t}[i], z_{1:t}[i], f_{1:k}[i])$$

of $p(\cdot)$ evaluated on the AMP iterates and side information vectors is a linear combination of values of different tensor networks on $\mathbf{W}$ and $\mathbf{u}_1, \mathbf{f}_1, \ldots, \mathbf{f}_k$. Then, leveraging state evolution results of [Fan22] to perform an inductive polynomial approximation argument, the proof reduces the universality of AMP for Lipschitz non-linearities to the universality of these tensor network values. This reduction is encapsulated in the following lemma.

---

row-wise to $(\mathbf{x}_1, \ldots, \mathbf{x}_k) \in \mathbb{R}^{n \times k}$, and $\mathbf{T}_v$ is then a diagonal tensor with $q_v(\mathbf{x}_1, \ldots, \mathbf{x}_k) \in \mathbb{R}^n$ along its main diagonal.

Lemma 2.2.12. *Let $\mathbf{u}_1, \mathbf{f}_1, \ldots, \mathbf{f}_k \in \mathbb{R}^n$ satisfy Assumption 2.2.1. Let $\mathbf{W}, \mathbf{G} \in \mathbb{R}^{n\times n}$ be symmetric random matrices independent of $\mathbf{u}_1, \mathbf{f}_1, \ldots, \mathbf{f}_k$ such that*

1. *$\mathbf{G} = \mathbf{O}\mathbf{D}\mathbf{O}^\top$ is an orthogonally invariant matrix, where $\mathbf{D} = \operatorname{diag}(\mathbf{d})$ and $\mathbf{d} \stackrel{W}{\to} D$ for a limit law D with compact support and $\operatorname{Var}[D] > 0$.*

2. *$\|\mathbf{W}\|_{\mathrm{op}} < C$ for a constant $C > 0$, almost surely for all large n.*

3. *For every diagonal tensor network T in $k+1$ variables, almost surely as $n \to \infty$,*

$$\operatorname{val}_T(\mathbf{W}; \mathbf{u}_1, \mathbf{f}_1, \ldots, \mathbf{f}_k) - \operatorname{val}_T(\mathbf{G}; \mathbf{u}_1, \mathbf{f}_1, \ldots, \mathbf{f}_k) \to 0.$$

Let $u_{t+1} : \mathbb{R}^{t+k} \to \mathbb{R}$ be continuous functions which satisfy the polynomial growth condition (2.1.1) for some order $p \geq 1$, and are Lipschitz in their first t arguments. Let $\{b_{ts}\}$ and $\{\mathbf{\Sigma}_t\}$ be defined by the orthogonally invariant prescriptions (2.2.5) and (2.2.6) for the limit law D, where each $\mathbf{\Sigma}_t$ is non-singular. Then the iterates (2.2.1) applied to $\mathbf{W}$ satisfy, almost surely as $n \to \infty$ for any fixed $t \geq 1$,

$$(\mathbf{u}_1, \mathbf{f}_1, \ldots, \mathbf{f}_k, \mathbf{z}_1, \ldots, \mathbf{z}_t) \xrightarrow{W_2} (U_1, F_1, \ldots, F_k, Z_1, \ldots, Z_t)$$

where this limit has the same joint law as described by the AMP state evolution for $\mathbf{G}$.

This lemma applies also in the special case of $\mathbf{G} \sim \mathrm{GOE}(n)$, where the definitions of $\{b_{ts}\}$ and $\{\mathbf{\Sigma}_t\}$ reduce to the GOE prescriptions of (2.2.3) and (2.2.4). The lemma does not assume any particular matrix model for $\mathbf{W}$, and thus may be used as a tool to establish AMP universality for matrix models beyond the ones we consider in this work.

Theorems 2.2.4 and 2.2.8 then follow from the next two lemmas, which verify the universality of tensor network values for the classes of generalized Wigner matrices and symmetric generalized invariant matrices.

Lemma 2.2.13. *Let* $\mathbf{x}_1, \ldots, \mathbf{x}_k \in \mathbb{R}^n$ *be (random or deterministic) vectors and let* $(X_1, \ldots, X_k)$ *have finite moments of all orders, such that almost surely as* $n \to \infty$*,*

$$(\mathbf{x}_1, \ldots, \mathbf{x}_k) \xrightarrow{W} (X_1, \ldots, X_k). \tag{2.2.10}$$

Let $\mathbf{W} \in \mathbb{R}^{n \times n}$ *be a generalized Wigner matrix, independent of* $\mathbf{x}_1, \ldots, \mathbf{x}_k$*, with variance profile matrix* $\mathbf{S}$*. Let* $\mathbf{s}_i$ *be the* i^{th} *row of* $\mathbf{S}$*, and suppose for each fixed polynomial function* $q : \mathbb{R}^k \to \mathbb{R}$ *that*

$$\max_{i=1}^{n} \left| \langle q(\mathbf{x}_1, \ldots, \mathbf{x}_k) \odot \mathbf{s}_i \rangle - \langle q(\mathbf{x}_1, \ldots, \mathbf{x}_k) \rangle \cdot \langle \mathbf{s}_i \rangle \right| \to 0. \tag{2.2.11}$$

Then for any diagonal tensor network T *in* k *variables, there is a deterministic value* $\text{lim-val}_T(X_1, \ldots, X_k)$ *depending only on* T *and the joint law of* $(X_1, \ldots, X_k)$ *such that almost surely,*

$$\lim_{n \to \infty} \text{val}_T(\mathbf{W}; \mathbf{x}_1, \ldots, \mathbf{x}_k) = \text{lim-val}_T(X_1, \ldots, X_k).$$

In particular, this limit value is the same for $\mathbf{W}$ *as for* $\mathbf{G} \sim \text{GOE}(n)$*.*

Lemma 2.2.14. *Let* $\mathbf{x}_1, \ldots, \mathbf{x}_k \in \mathbb{R}^n$ *be (random or deterministic) vectors and let* $(X_1, \ldots, X_k)$ *have finite moments of all orders, such that almost surely as* $n \to \infty$*,*

$$(\mathbf{x}_1, \ldots, \mathbf{x}_k) \xrightarrow{W} (X_1, \ldots, X_k). \tag{2.2.12}$$

Let $\mathbf{W} \in \mathbb{R}^{n \times n}$ *be a symmetric generalized invariant matrix, independent of* $\mathbf{x}_1, \ldots, \mathbf{x}_k$*, with limit diagonal distribution* $\mathcal{D}_{\text{diag}}$*. Then for any diagonal tensor network* T *in* k *variables, there is a deterministic limit value* $\text{lim-val}_T(X_1, \ldots, X_k, \mathcal{D}_{\text{diag}})$ *depending* *only on T, the joint law of $(X_1, \ldots, X_k)$, and $\mathcal{D}_{\text{diag}}$ such that almost surely,*

$$\lim_{n\to\infty} \mathrm{val}_T(\mathbf{W}; \mathbf{x}_1, \ldots, \mathbf{x}_k) = \text{lim-val}_T(X_1, \ldots, X_k, \mathcal{D}_{\text{diag}}).$$

In particular, if there exists an orthogonally invariant matrix $\mathbf{G}$ having the same limit diagonal distribution $\mathcal{D}_{\text{diag}}$, then this limit value is the same for $\mathbf{W}$ as for $\mathbf{G}$.

Remark 2.2.15. Lemma 2.2.14 applies to any class of symmetric generalized invariant matrices satisfying Definition 2.2.6, where the limit diagonal distribution $\mathcal{D}_{\text{diag}}$ does not necessarily coincide with that of an orthogonally invariant model.

This has the following implication: Consider any first-order iterative algorithm having the structure (2.2.1), where b_{ts} are arbitrary fixed constants and $u_{t+1} : \mathbb{R}^{t+k} \to \mathbb{R}$ are polynomial functions applied entrywise. Then for any polynomial test function $p(\cdot)$, the value

$$\frac{1}{n}\sum_{i=1}^{n} p(u_{1:t}[i], z_{1:t}[i], f_{1:k}[i])$$

is a linear combination of tensor network values (c.f. Lemma 2.3.11) and hence has a universal limit as $n \to \infty$. Under mild moment assumptions, this implies that there exists a limit law for the empirical distribution of each iterate $\mathbf{u}_t$ and $\mathbf{z}_t$, and this law is universal across such matrices having the same limit diagonal distribution $\mathcal{D}_{\text{diag}}$.

When $\mathcal{D}_{\text{diag}}$ is not described by an orthogonally invariant model, we believe it may be an interesting open question to develop such an algorithm that has a more succinct state-evolution characterization of its iterates in terms of this limit diagonal law.

The proofs of Lemmas 2.2.13 and 2.2.14 result in forms for the limit tensor network values that are, in general, combinatorially complex. However, a by-product of the proofs is that these forms reduce to 0 when all diagonal tensors of the tensor network have vanishing normalized trace. This may be viewed as a version of asymptotic freeness for tensor networks, and we state the result here for independent interest.

Proposition 2.2.16. *(a) In the setting of Lemma 2.2.13, let T be a diagonal tensor network such that, for every vertex of v of T, almost surely*

$$\lim_{n\to\infty} \langle q_v(\mathbf{x}_1, \ldots, \mathbf{x}_k) \rangle = \lim_{n\to\infty} \frac{1}{n} \sum_{i=1}^{n} q_v(x_1[i], \ldots, x_k[i]) = 0. \tag{2.2.13}$$

Then $\text{lim-val}_T(X_1, \ldots, X_k) = 0$.

(b) In the setting of Lemma 2.2.14, suppose T is a diagonal tensor network for which (2.2.13) holds almost surely for every vertex v. Suppose also that there exists an orthogonally invariant matrix having the same limit diagonal distribution $\mathcal{D}_{\text{diag}}$ as $\mathbf{W}$, and $\lim_{n\to\infty} \frac{1}{n} \operatorname{Tr} \mathbf{W} = 0$. Then $\text{lim-val}_T(X_1, \ldots, X_k, \mathcal{D}_{\text{diag}}) = 0$.

2.2.3 Universality of AMP algorithms for rectangular matrices

Let $\mathbf{W} \in \mathbb{R}^{m\times n}$ be a rectangular random matrix. Consider an initialization $\mathbf{u}_1 \in \mathbb{R}^m$ and vectors of side information $\mathbf{f}_1, \ldots, \mathbf{f}_k \in \mathbb{R}^m$ and $\mathbf{g}_1, \ldots, \mathbf{g}_\ell \in \mathbb{R}^n$, all independent of $\mathbf{W}$. Let $v_1, v_2, v_3, \ldots$ and $u_2, u_3, u_4, \ldots$ be two sequences of non-linear functions where $v_t : \mathbb{R}^{t+\ell} \to \mathbb{R}$ and $u_{t+1} : \mathbb{R}^{t+k} \to \mathbb{R}$. We study an AMP algorithm that computes, for $t = 1, 2, 3, \ldots$

$$\mathbf{z}_t = \mathbf{W}^\top \mathbf{u}_t - \sum_{s=1}^{t-1} b_{ts} \mathbf{v}_s \tag{2.2.14a}$$

$$\mathbf{v}_t = v_t(\mathbf{z}_1, \ldots, \mathbf{z}_t, \mathbf{g}_1, \ldots, \mathbf{g}_\ell) \tag{2.2.14b}$$

$$\mathbf{y}_t = \mathbf{W}\mathbf{v}_t - \sum_{s=1}^{t} a_{ts} \mathbf{u}_s \tag{2.2.14c}$$

$$\mathbf{u}_{t+1} = u_{t+1}(\mathbf{y}_1, \ldots, \mathbf{y}_t, \mathbf{f}_1, \ldots, \mathbf{f}_k) \tag{2.2.14d}$$

where $\{b_{ts}\}_{s<t}$ and $\{a_{ts}\}_{s\le t}$ are deterministic "Onsager correction" coefficients. We will characterize the iterates of this algorithm in the limit as $m, n \to \infty$ proportionally

with $m/n \to \gamma \in (0, \infty)$, for fixed $k, \ell \geq 0$. For Gaussian and bi-orthogonally invariant matrices $\mathbf{W}$ (see the definition after Definition 2.2.20), we review the forms for these correction coefficients and the corresponding state evolutions in (2.6.1-2.6.2) and (2.6.4–2.6.5) of Section 2.6.

We assume the following condition for $(\mathbf{u}_1, \mathbf{f}_1, \ldots, \mathbf{f}_k)$ and $(\mathbf{g}_1, \ldots, \mathbf{g}_\ell)$, which is analogous to Assumption 2.2.1.

Assumption 2.2.17. Almost surely as $m, n \to \infty$,

$$(\mathbf{u}_1, \mathbf{f}_1, \ldots, \mathbf{f}_k) \xrightarrow{W} (U_1, F_1, \ldots, F_k) \quad \text{and} \quad (\mathbf{g}_1, \ldots, \mathbf{g}_\ell) \xrightarrow{W} (G_1, \ldots, G_\ell)$$

for joint limit laws $(U_1, F_1, \ldots, F_k)$ and $(G_1, \ldots, G_\ell)$ having finite moments of all orders, where $\mathbb{E}[U_1^2] > 0$. Multivariate polynomials are dense in the real L^2-spaces of functions $f : \mathbb{R}^{k+1} \to \mathbb{R}$ and $g : \mathbb{R}^\ell \to \mathbb{R}$ with the inner-products

$$\begin{aligned}
(f, \tilde{f}) &\mapsto \mathbb{E}[f(U_1, F_1, \ldots, F_k)\tilde{f}(U_1, F_1, \ldots, F_k)] \\
(g, \tilde{g}) &\mapsto \mathbb{E}[g(G_1, \ldots, G_\ell)\tilde{g}(G_1, \ldots, G_\ell)].
\end{aligned}$$

Our main results show that the state evolution characterizations of AMP algorithms for Gaussian and orthogonally invariant matrices are universal across the following matrix ensembles, analogous to Definitions 2.2.3 and 2.2.6 in the symmetric setting.

Definition 2.2.18. $\mathbf{W} \in \mathbb{R}^{m \times n}$ is a *generalized white noise matrix* with (deterministic) variance profile $\mathbf{S} \in \mathbb{R}^{m \times n}$ if

(a) All entries $W[\alpha, i]$ are independent.

(b) Each entry $W[\alpha, i]$ has mean 0, variance $n^{-1}S[\alpha, i]$, and higher moments satisfying, for each integer $p \geq 3$,

$$\lim_{m,n\to\infty} n \cdot \max_{\alpha=1}^{m} \max_{i=1}^{n} \mathbb{E}[|W[\alpha, i]|^p] = 0.$$

(c) For a constant $C > 0$ independent of m, n,

$$\max_{\alpha=1}^{m} \max_{i=1}^{n} S[\alpha, i] \leq C, \qquad \lim_{m,n\to\infty} \max_{\alpha=1}^{m} \left| \frac{1}{n} \sum_{i=1}^{n} S[\alpha, i] - 1 \right| = 0,$$

$$\lim_{m,n\to\infty} \max_{i=1}^{n} \left| \frac{1}{m} \sum_{\alpha=1}^{m} S[\alpha, i] - 1 \right| = 0.$$

We call $\mathbf{W}$ a *Gaussian white noise matrix* in the special case where $W[\alpha, i] \sim \mathcal{N}(0, 1/n)$ and $S[\alpha, i] = 1$ for all $(\alpha, i) \in [m] \times [n]$.

Next, we introduce a notion of diagonal distribution for rectangular matrices, analogous to Definition 2.2.5. Recall the diagonal map $\Delta(\cdot)$, and let $\Delta\langle \mathbf{x}, \mathbf{I}_m, \mathbf{I}_n \rangle$ be the set of all words in $\mathbf{x}, \mathbf{I}_m, \mathbf{I}_n$ and $\Delta(\cdot)$, for example

$$\mathbf{x}\mathbf{I}_m, \quad \mathbf{x}\Delta(\mathbf{I}_n\mathbf{x})\mathbf{I}_m, \quad \Delta(\mathbf{x}\mathbf{x}\Delta(\mathbf{I}_n))\mathbf{x}, \quad \mathbf{I}_m\mathbf{x}\mathbf{I}_n\Delta(\Delta(\mathbf{x}))\Delta(\mathbf{x}\mathbf{I}_m).$$

For $p(\mathbf{x}) \in \Delta\langle \mathbf{x}, \mathbf{I}_m, \mathbf{I}_n \rangle$ and $\mathbf{M} \in \mathbb{R}^{m\times n}$, we write $p(\widetilde{\mathbf{M}}) \in \mathbb{R}^{(m+n)\times(m+n)}$ for its evaluation at $\mathbf{x} = \widetilde{\mathbf{M}}$,

$$\mathbf{I}_m = \begin{pmatrix} \mathrm{Id}_m & 0 \\ 0 & 0 \end{pmatrix} \in \mathbb{R}^{(m+n)\times(m+n)}, \quad \mathbf{I}_n = \begin{pmatrix} 0 & 0 \\ 0 & \mathrm{Id}_n \end{pmatrix} \in \mathbb{R}^{(m+n)\times(m+n)},$$

where we define the symmetric embedding

$$\widetilde{\mathbf{M}} = \begin{pmatrix} 0 & \mathbf{M} \\ \mathbf{M}^\top & 0 \end{pmatrix} \in \mathbb{R}^{(m+n)\times(m+n)} \tag{2.2.15}$$

and the identity matrices $\mathrm{Id}_m \in \mathbb{R}^{m\times m}$ and $\mathrm{Id}_n \in \mathbb{R}^{n\times n}$.

Definition 2.2.19. The distribution over the diagonal of a rectangular matrix $\mathbf{M} \in$

$\mathbb{R}^{m\times n}$ is the mapping

$$p(\mathbf{x}) \in \Delta\langle \mathbf{x}, \mathbf{I}_m, \mathbf{I}_n\rangle \mapsto \tfrac{1}{m+n} \operatorname{Tr} p(\widetilde{\mathbf{M}}).$$

Matrices $\mathbf{M} \in \mathbb{R}^{m\times n}$ converge in diagonal distribution a.s. if $\lim_{m,n\to\infty} \frac{1}{m+n} \operatorname{Tr} p(\widetilde{\mathbf{M}})$ exists almost surely (and is finite) for every fixed $p(\mathbf{x}) \in \Delta\langle \mathbf{x}, \mathbf{I}_m, \mathbf{I}_n\rangle$, as $m, n \to \infty$ with $m/n \to \gamma \in (0, \infty)$. The limit diagonal distribution of $\mathbf{M}$, which we will refer to as $\mathcal{D}_{\text{diag}}$, is then the mapping

$$p(\mathbf{x}) \in \Delta\langle \mathbf{x}, \mathbf{I}_m, \mathbf{I}_n\rangle \mapsto \lim_{m,n\to\infty} \tfrac{1}{m+n} \operatorname{Tr} p(\widetilde{\mathbf{M}}).$$

Note that $\mathcal{D}_{\text{diag}}$ and γ specify the limit of $\frac{1}{m} \operatorname{Tr}(\mathbf{M}\mathbf{M}^\top)^\nu$ for each fixed integer $\nu \geq 1$, and hence also the limit singular value distribution of $\mathbf{M}$ when this distribution has compact support. Note also that, similarly to the symmetric setting, $\mathbf{M}$ and $\mathbf{\Pi}_U \mathbf{M} \mathbf{\Pi}_V^\top$ must have the same limit diagonal distribution $\mathcal{D}_{\text{diag}}$ for any signed permutation matrices $\mathbf{\Pi}_U \in \mathbb{R}^{m\times m}$ and $\mathbf{\Pi}_V \in \mathbb{R}^{n\times n}$.

Definition 2.2.20. $\mathbf{W} = \mathbf{\Pi}_U \mathbf{M} \mathbf{\Pi}_V^\top \in \mathbb{R}^{m\times n}$ is a *rectangular generalized invariant matrix* with limit diagonal distribution $\mathcal{D}_{\text{diag}}$ if, as $m, n \to \infty$ with $m/n \to \gamma \in (0, \infty)$,

(a) $\mathbf{M}$ converges in diagonal distribution a.s. to a limit $\mathcal{D}_{\text{diag}}$.

(b) For any $\varepsilon > 0$ and any fixed $p(\mathbf{x}) \in \Delta\langle \mathbf{x}, \mathbf{I}_m, \mathbf{I}_n\rangle$, almost surely for all large m, n,

$$\max_{i\neq j} |p(\widetilde{\mathbf{M}})[i, j]| < n^{-1/2+\varepsilon}$$

where $\widetilde{\mathbf{M}}$ is the symmetric embedding (2.2.15).

(c) $\mathbf{\Pi}_U \in \mathbb{R}^{m\times m}$ and $\mathbf{\Pi}_V \in \mathbb{R}^{n\times n}$ are uniformly random signed permutations independent of each other and of $\mathbf{M}$.

We call $\mathbf{W} \in \mathbb{R}^{m \times n}$ *bi-orthogonally invariant* if it has singular value decomposition $\mathbf{W} = \mathbf{O}\mathbf{D}\mathbf{Q}^\top$ where $\mathbf{O} \sim \text{Haar}(\mathbb{O}(m))$ and $\mathbf{Q} \sim \text{Haar}(\mathbb{O}(n))$ are Haar-distributed on the orthogonal groups independently of each other and of $\mathbf{D} = \text{diag}(\mathbf{d}) \in \mathbb{R}^{m \times n}$. We verify in Proposition 2.6.1 of Section 2.6 that such bi-orthogonally invariant matrices satisfy Definition 2.2.20, where $\mathcal{D}_{\text{diag}}$ is determined uniquely by $\gamma = \lim_{m,n\to\infty} m/n$ and the limit singular value distribution of $\mathbf{D}$.

The following theorems show that the state evolution of AMP algorithms for Gaussian white noise matrices holds universally for generalized white noise matrices as in Definition 2.2.18, and the state evolution for bi-orthogonally invariant matrices holds universally for rectangular generalized invariant matrices as in Definition 2.2.20.

Theorem 2.2.21. *Let $\mathbf{W} \in \mathbb{R}^{m \times n}$ be a generalized white noise matrix with variance profile matrix $\mathbf{S}$, and let $\mathbf{u}_1, \mathbf{f}_1, \ldots, \mathbf{f}_k, \mathbf{g}_1, \ldots, \mathbf{g}_\ell$ be independent of $\mathbf{W}$ and satisfy Assumption 2.2.17. Suppose that*

1. *Each function $v_t : \mathbb{R}^{t+\ell} \to \mathbb{R}$ and $u_{t+1} : \mathbb{R}^{t+k} \to \mathbb{R}$ is continuous, satisfies the polynomial growth condition (2.1.1) for some order $p \geq 1$, and is Lipschitz in its first t arguments.*

2. *$\|\mathbf{W}\|_{\text{op}} < C$ for a constant $C > 0$ almost surely for all large m, n.*

3. *Let $\mathbf{s}_\alpha$ be the α^{th} row of $\mathbf{S}$ and $\mathbf{s}^i$ be the i^{th} column of $\mathbf{S}$. For any fixed polynomial functions $p : \mathbb{R}^{k+1} \to \mathbb{R}$ and $q : \mathbb{R}^\ell \to \mathbb{R}$, almost surely as $m, n \to \infty$,*

$$\begin{aligned} \max_{\alpha=1}^{m} \Big| \langle p(\mathbf{u}_1, \mathbf{f}_1, \ldots, \mathbf{f}_k) \odot \mathbf{s}_\alpha \rangle - \langle p(\mathbf{u}_1, \mathbf{f}_1, \ldots, \mathbf{f}_k) \rangle \cdot \langle \mathbf{s}_\alpha \rangle \Big| &\to 0, \\ \max_{i=1}^{n} \Big| \langle q(\mathbf{g}_1, \ldots, \mathbf{g}_\ell) \odot \mathbf{s}^i \rangle - \langle q(\mathbf{g}_1, \ldots, \mathbf{g}_\ell) \rangle \cdot \langle \mathbf{s}^i \rangle \Big| &\to 0. \end{aligned} \tag{2.2.16}$$

Let $\{a_{ts}\}, \{b_{ts}\}, \{\mathbf{\Omega}_t\}, \{\mathbf{\Sigma}_t\}$ be defined by the white noise prescriptions (2.6.1) and (2.6.2), where each matrix $\mathbf{\Omega}_t$ and $\mathbf{\Sigma}_t$ is non-singular. Then for any fixed $t \geq 1$, almost *surely as $m, n \to \infty$ with $m/n \to \gamma \in (0, \infty)$, the iterates of (2.2.14) satisfy*

$$(\mathbf{u}_1, \mathbf{f}_1, \ldots, \mathbf{f}_k, \mathbf{y}_1, \ldots, \mathbf{y}_t) \xrightarrow{W_2} (U_1, F_1, \ldots, F_k, Y_1, \ldots, Y_t),$$
$$(\mathbf{g}_1, \ldots, \mathbf{g}_\ell, \mathbf{z}_1, \ldots, \mathbf{z}_t) \xrightarrow{W_2} (G_1, \ldots, G_\ell, Z_1, \ldots, Z_t)$$

where $(Z_1, \ldots, Z_t) \sim \mathcal{N}(0, \mathbf{\Omega}_t)$ and $(Y_1, \ldots, Y_t) \sim \mathcal{N}(0, \mathbf{\Sigma}_t)$ are independent of $(U_1, F_1, \ldots, F_k)$ and $(G_1, \ldots, G_\ell)$, i.e. these limits have the same joint laws as described by the AMP state evolution for a Gaussian white noise matrix $\mathbf{W}$.

Theorem 2.2.22. *Let $\mathbf{W} \in \mathbb{R}^{m \times n}$ be a rectangular generalized invariant matrix whose limit diagonal distribution $\mathcal{D}_{\text{diag}}$ coincides with that of a bi-orthogonally invariant matrix $\mathbf{G}$. Let $\mathbf{u}_1, \mathbf{f}_1, \ldots, \mathbf{f}_k, \mathbf{g}_1, \ldots, \mathbf{g}_\ell$ be independent of $\mathbf{W}$ and satisfy Assumption 2.2.17. Suppose that*

1. *Each function $v_t : \mathbb{R}^{t+\ell} \to \mathbb{R}$ and $u_{t+1} : \mathbb{R}^{t+k} \to \mathbb{R}$ is continuous, satisfies the polynomial growth condition (2.1.1) for some order $p \geq 1$, and is Lipschitz in its first t arguments.*

2. *$\|\mathbf{W}\|_{\text{op}} < C$ for a constant $C > 0$ almost surely for all large m, n.*

Let $\{a_{ts}\}, \{b_{ts}\}, \{\mathbf{\Omega}_t\}, \{\mathbf{\Sigma}_t\}$ be defined by the bi-orthogonally invariant prescriptions (2.6.4) and (2.6.5) for the limit singular value distribution D specified by $\mathcal{D}_{\text{diag}}$ and γ. Suppose that $\mathbb{E}[D^2] > 0$ and each $\mathbf{\Omega}_t$ and $\mathbf{\Sigma}_t$ is non-singular. Then for any fixed $t \geq 1$, almost surely as $m, n \to \infty$ with $m/n \to \gamma \in (0, \infty)$, the iterates of (2.2.14) satisfy

$$(\mathbf{u}_1, \mathbf{f}_1, \ldots, \mathbf{f}_k, \mathbf{y}_1, \ldots, \mathbf{y}_t) \xrightarrow{W_2} (U_1, F_1, \ldots, F_k, Y_1, \ldots, Y_t),$$
$$(\mathbf{g}_1, \ldots, \mathbf{g}_\ell, \mathbf{z}_1, \ldots, \mathbf{z}_t) \xrightarrow{W_2} (G_1, \ldots, G_\ell, Z_1, \ldots, Z_t)$$

where $(Z_1, \ldots, Z_t) \sim \mathcal{N}(0, \mathbf{\Omega}_t)$ and $(Y_1, \ldots, Y_t) \sim \mathcal{N}(0, \mathbf{\Sigma}_t)$ are independent of $(U_1, F_1, \ldots, F_k)$ and $(G_1, \ldots, G_\ell)$, i.e. these limits have the same joint laws as described by the AMP state evolution for $\mathbf{G}$.

Remark 2.2.23. As in Remark 2.2.9, Theorems 2.2.21 and 2.2.22 hold equally for AMP algorithms where, in the prescriptions (2.6.2) and (2.6.5) for a_{ts} and b_{ts}, the quantities $\mathbb{E}[\partial_r v_s(Z_{1:s}, G_{1:\ell})]$, $\mathbb{E}[\partial_r u_{s+1}(Y_{1:s}, F_{1:k})]$, $\mathbb{E}[U_r U_s]$, and $\mathbb{E}[V_r V_s]$ are replaced by the empirical averages

$$\langle \partial_r v_s(\mathbf{z}_{1:s}, \mathbf{g}_{1:\ell})\rangle, \qquad \langle \partial_r u_{s+1}(\mathbf{y}_{1:s}, \mathbf{f}_{1:k})\rangle, \qquad \langle \mathbf{u}_r \odot \mathbf{u}_s\rangle, \qquad \langle \mathbf{v}_r \odot \mathbf{v}_s\rangle.$$

For example, such an AMP algorithm for Gaussian white noise matrices $\mathbf{W}$ and non-linearities $v_t(z_{1:t}, g_{1:\ell}) = v(z_t)$ and $u_{t+1}(y_{1:t}, f_{1:k}) = u(y_t)$ consists of the iterations

$$\mathbf{z}_t = \mathbf{W}^\top \mathbf{u}_t - \gamma\langle u'(\mathbf{y}_{t-1})\rangle \mathbf{v}_{t-1}, \qquad \mathbf{v}_t = v(\mathbf{z}_t),$$

$$\mathbf{y}_t = \mathbf{W}\mathbf{v}_t - \langle v'(\mathbf{z}_t)\rangle \mathbf{u}_t, \qquad \mathbf{u}_{t+1} = u(\mathbf{y}_t).$$

The proofs of Theorems 2.2.21 and 2.2.22 are similar to those of Theorems 2.2.4 and 2.2.8 for symmetric matrices, and we defer them to Section 2.6.

2.2.4 Applications

Example 2.2.24. *AMP algorithms for the Gaussian universality class may be heuristically derived by approximating belief propagation on dense graphical models [Kab03, DMM10a]. Our assumptions in Theorems 2.2.4 and 2.2.21 are sufficiently weak to show that their state evolutions remain valid in sparse random graphs down to sparsity levels of* $(\log n)/n$.

As a concrete example, consider the symmetric stochastic block model where G *is an undirected graph over* n *vertices, divided into two communities* $\mathcal{V}_+$ *and* $\mathcal{V}_-$ *of equal sizes* $n/2$. *For two* n*-dependent probabilities* $p_n > q_n$, *each pair of vertices* (i, j) *in* G *(including self-loops, for simplicity of discussion) is independently connected with* *probability*

$$\mathbb{P}[i \textit{ is connected to } j] = \begin{cases} p_n & \textit{if } i,j \in \mathcal{V}_+ \textit{ or } i,j \in \mathcal{V}_-, \\ q_n & \textit{if } i \in \mathcal{V}_+ \textit{ and } j \in \mathcal{V}_- \textit{ or if } i \in \mathcal{V}_- \textit{ and } j \in \mathcal{V}_+. \end{cases}$$

Let $\mathbf{A} \in \{0,1\}^{n \times n}$ *be the adjacency matrix of* G*, and let* $\bar{p}_n = (p_n + q_n)/2$ *be the mean connectivity. Then the centered and normalized adjacency matrix takes the form*

$$\frac{\mathbf{A} - \bar{p}_n}{\sqrt{n\bar{p}_n(1-\bar{p}_n)}} = \frac{\sqrt{\lambda_n}}{n}\mathbf{f}\mathbf{f}^\top + \mathbf{W} \tag{2.2.17}$$

where $\lambda_n = n(p_n - q_n)^2/[4\bar{p}_n(1-\bar{p}_n)]$ *is a parameter representing the signal-to-noise ratio of the model,* $\mathbf{f} \in \{+1,-1\}^n$ *is the binary indicator vector representing the membership of the vertices, and* $\mathbf{W}$ *is a symmetric noise matrix with independent entries. It may checked for each* $(i,j) \in [n] \times [n]$ *that*

$$\mathbb{E}[W[i,j]] = 0, \quad \mathbb{E}[W[i,j]^2] \in \left\{ \frac{p_n(1-p_n)}{n\bar{p}_n(1-\bar{p}_n)}, \frac{q_n(1-q_n)}{n\bar{p}_n(1-\bar{p}_n)} \right\}, \quad |W[i,j]| \leq \frac{1}{\sqrt{n\bar{p}_n(1-\bar{p}_n)}}.$$

In the asymptotic regime where $n\bar{p}_n(1-\bar{p}_n) \to \infty$ *and* $\lambda_n \to \lambda$ *a positive constant, we have that* $S[i,j] := n \cdot \mathbb{E}[W[i,j]^2] \to 1$ *uniformly over* $(i,j) \in [n] \times [n]$*, so that* $\mathbf{W}$ *is a generalized Wigner matrix in the sense of Definition 2.2.3 (with variance profile* $\mathbf{S}$ *approximately constant in every entry). Furthermore, under a slightly stronger assumption*

$$n\bar{p}_n(1-\bar{p}_n) \geq c \log n \tag{2.2.18}$$

for any constant $c > 0$*, [BGBK20, Theorem 2.7 and Eq. (2.4)] implies that* $\|\mathbf{W}\|_{\mathrm{op}} < C$ *almost surely for all large* n*. This encompasses the stochastic block model in regimes with sparsity* $\bar{p}_n \gtrsim (\log n)/n$*.*[4]

4. Our universality result for AMP with polynomial non-linearities does not require the operator norm bound $\|\mathbf{W}\|_{\mathrm{op}} < C$ and hence holds for any sparsity $\bar{p}_n \gg 1/n$, c.f. Remark 2.3.13. We believe that the operator norm requirement in condition 2 of Theorem 2.2.4 may be an artifact of our

It was shown in [DAM17] that the mutual information between G and $\mathbf{f}$ has an asymptotic limit depending only on the limit signal-to-noise ratio λ, which is non-trivial when $\lambda > 1$. This was proven by interpolating between the model (2.2.17) and a "$\mathbb{Z}_2$-synchronization" model where $\mathbf{W} \sim \mathrm{GOE}(n)$, and applying an AMP analysis in the latter model. Our result of Theorem 2.2.4 implies that, under the additional condition (2.2.18), this AMP analysis may instead be directly applied to the model (2.2.17), bypassing interpolation to the GOE.

Example 2.2.25. *Let $\mathbf{Y} \in \mathbb{R}^{m \times n}$ be a signal-plus-noise data matrix modeled as*

$$\mathbf{Y} = \mathbf{X} + \mathbf{E}$$

where $\mathbf{X} = \mathbb{E}[\mathbf{Y}] = \sum_{j=1}^{k} \mathbf{f}_j \mathbf{g}_j^\top \in \mathbb{R}^{m \times n}$ is a low-rank signal matrix, and $\mathbf{E} = \mathbf{Y} - \mathbf{X}$ is a mean-zero matrix of residual noise. We assume that $\mathbf{E}$ has independent entries, although in many applications involving count observations or missing data, these entries may have a heteroskedastic variance profile $\mathbf{V}$ where

$$V[\alpha, i] := \mathrm{Var}[E[\alpha, i]].$$

Such models where the variance $V[\alpha, i]$ is a quadratic function $a + bX[\alpha, i] + cX[\alpha, i]^2$ of the mean were discussed recently in [LZK21], including Poisson and negative-binomial models for $\mathbf{Y}$ in the context of single-cell RNA sequencing applications [THAI19, HS19, SS21]. Such models encompass also simple models of missing data,

polynomial approximation proof.

In contrast, we do not expect AMP universality to hold for random graph models with sparsity $\bar{p}_n \asymp 1/n$, where the belief propagation recursions on such graphs may not admit asymptotic Gaussian approximations.

where $\mathbf{Y}$ *is a partial observation of an underlying low-rank signal matrix* $\widetilde{\mathbf{X}}$ *so that*

$$Y[\alpha, i] = \begin{cases} \widetilde{X}[\alpha, i] & \text{with probability } p, \\ 0 & \text{with probability } 1-p, \end{cases}$$

independently for each entry. Then $X[\alpha, i] = p \cdot \widetilde{X}[\alpha, i]$ *and* $V[\alpha, i] = p(1-p) \cdot \widetilde{X}[\alpha, i]^2$ *are the corresponding means and variances.*

When the entries $V[\alpha, i]$ *are heteroskedastic, the singular value spectrum of* $\mathbf{E}$ *does not generally conform to the Marcenko-Pastur law. However, row and column normalization is typically applied in practice prior to data analysis, with [LZK21] suggesting the following normalization scheme: Determine via Sinkhorn iteration two diagonal matrices* $\mathbf{D}_1 \in \mathbb{R}^{m \times m}$ *and* $\mathbf{D}_2 \in \mathbb{R}^{n \times n}$ *for which* $\mathbf{S} = \mathbf{D}_1 \mathbf{V} \mathbf{D}_2$ *has all rows summing to* n *and all columns summing to* m*, and use these to standardize* $\mathbf{Y}$ *into the biwhitened matrix*

$$\widetilde{\mathbf{Y}} = \frac{1}{\sqrt{n}} \cdot \mathbf{D}_1^{1/2} \mathbf{Y} \mathbf{D}_2^{1/2} = \frac{1}{\sqrt{n}} \cdot \mathbf{D}_1^{1/2} \mathbf{X} \mathbf{D}_2^{1/2} + \mathbf{W}, \qquad \mathbf{W} = \frac{1}{\sqrt{n}} \cdot \mathbf{D}_1^{1/2} \mathbf{E} \mathbf{D}_2^{1/2}.$$

[LZK21] proved that such biwhitened count matrices have singular value spectra asymptotically described by the Marcenko-Pastur law, and showed a remarkable empirical agreement with the Marcenko-Pastur law for matrices arising in several domains of application, from single-cell biology to topic modeling of text.

In this standardized model $\widetilde{\mathbf{Y}}$*, the error matrix* $\mathbf{W}$ *now has variance profile* $\mathbf{S} = \mathbf{D}_1 \mathbf{V} \mathbf{D}_2$ *which satisfies by construction* $\frac{1}{n} \sum_{i=1}^n S[\alpha, i] = \frac{1}{m} \sum_{\alpha=1}^m S[\alpha, i] = 1$*, and Theorem 2.2.21 describes conditions under which state evolution holds for Gaussian AMP algorithms applied to this matrix* $\mathbf{W}$*. We note that to analyze AMP applied instead to* $\widetilde{\mathbf{Y}}$*, the condition (2.2.16) represents a potentially strong restriction on the relation between the variance profile matrix* $\mathbf{S}$ *and the low-rank mean signal. A modified analysis of AMP may be needed in settings where this restriction does not* *hold, and we leave this as a direction to explore in future work.*

Example 2.2.26. *Much of the early development of AMP algorithms was motivated by compressed sensing applications of reconstructing sparse signals from linear measurements. Consider a model of m measurements*

$$\mathbf{y} = \mathbf{W}\mathbf{x} + \boldsymbol{\varepsilon} \in \mathbb{R}^m$$

where $\mathbf{x} \in \mathbb{R}^n$ is the underlying signal, $\mathbf{W} \in \mathbb{R}^{m\times n}$ is a random sensing matrix, and $\boldsymbol{\varepsilon}$ is measurement noise. For i.i.d. Gaussian sensing matrices $\mathbf{W}$, pioneering work of [DMM09, DMM10a, DMM10b] proposed an AMP algorithm for reconstructing $\mathbf{x}$, where the non-linearities are soft-thresholding functions tailored to the sparsity of $\mathbf{x}$. Analysis of the dynamics of this algorithm leads to a derivation of a sparsity-undersampling phase transition curve that matches a phase transition for ℓ_1-based reconstruction in this model [DT05, DMM09, BM11, BLM15].

Extensive numerical experiments performed in [DT09, MJG+13] suggested that this phase transition curve is universal across broad classes of non-Gaussian sensing matrices. Theorem 2.2.8 provides an extension of the AMP universality shown in [BLM15] for this application, broadening the universality class to matrices composed of subsampled Fourier or Hadamard transforms and diagonal operators. Importantly, matrix-vector multiplication operations for such matrices may be computed in $O(n \log n)$ time without explicitly storing the matrices in memory, allowing applications of AMP at much larger scales than would be possible with i.i.d. sensing designs.

As an example, consider

$$\mathbf{W} = (\mathbf{\Pi}_U \mathbf{H} \mathbf{\Pi}_E)\mathbf{D}(\mathbf{\Pi}_V \mathbf{K} \mathbf{\Pi}_F)^\top \in \mathbb{R}^{m\times n} \tag{2.2.19}$$

where $\mathbf{D} \in \mathbb{R}^{m\times n}$ is diagonal with its diagonal entries sampled i.i.d. from a Marcenko-Pastur law; $\mathbf{H}, \mathbf{K} \in \mathbb{R}^{n\times n}$ are orthogonal matrices representing deterministic Hadamard *or discrete Fourier transforms; and* $\mathbf{\Pi}_U, \mathbf{\Pi}_E, \mathbf{\Pi}_V, \mathbf{\Pi}_F$ *are independent random signed permutations. We verify in Proposition 2.6.1(b1) that this class of matrices satisfies Definition 2.2.20. If the signal vector* $\mathbf{x}$*, residual error* $\boldsymbol{\varepsilon}$*, and initialization* $\mathbf{x}_1$ *are each comprised of i.i.d. entries, then the random permutations in* $\mathbf{\Pi}_U, \mathbf{\Pi}_V$ *may be further absorbed into* $\mathbf{x}_1, \mathbf{x}, \boldsymbol{\varepsilon}$*. Thus the AMP iterates are equal in law to those of AMP applied with a simpler sensing matrix*

$$\widetilde{\mathbf{W}} = \mathbf{\Xi}_U \mathbf{H} \widetilde{\mathbf{D}} \widetilde{\mathbf{K}}^\top \mathbf{\Xi}_V$$

where $\mathbf{\Xi}_U, \mathbf{\Xi}_V$ *are diagonal matrices of i.i.d.* $\{+1, -1\}$ *signs,* $\widetilde{\mathbf{K}}^\top \in \mathbb{R}^{m \times n}$ *is a random subsampling of* m *rows of* $\mathbf{K}^\top$*, and* $\widetilde{\mathbf{D}} \in \mathbb{R}^{m \times m}$ *is a diagonal matrix whose diagonal entries are given by those of* $\mathbf{D}$ *also multiplied by i.i.d.* $\{+1, -1\}$ *signs.*

Theorem 2.2.22 implies that AMP applied to the above matrix $\mathbf{W}$ *admits the same state evolution as when applied to an i.i.d. Gaussian sensing matrix* $\mathbf{G}$*. This universality extends beyond the Gaussian setting, to sensing matrices (2.2.19) where the diagonal entries of* $\mathbf{D}$ *are sampled from an arbitrary compactly supported singular value distribution. Theorem 2.2.22 then shows that the state evolution characterizations for the more general AMP algorithms of [Fan22]—derived originally for bi-orthogonally invariant ensembles—are valid in such settings. For this compressed sensing application, we note that the resulting AMP algorithms are similar to the convolutional AMP algorithms developed and studied recently in [Tak20, Tak21].*

2.3 Proofs for symmetric matrices

2.3.1 Universality for generalized Wigner matrices

In this section, we prove Lemma 2.2.13 on the universality of the tensor network value for generalized Wigner matrices.

Fix a tensor network $T = (\mathcal{V}, \mathcal{E}, \{q_v\}_{v\in\mathcal{V}})$. Let $\mathcal{P}$ be the set of all partitions of $\mathcal{V}$. For each index tuple $\mathbf{i} \in [n]^{\mathcal{V}}$, define its induced partition $\pi(\mathbf{i}) \in \mathcal{P}$ such that vertices $u, v \in \mathcal{V}$ belong to the same block of $\pi(\mathbf{i})$ if and only if $i_u = i_v$. Then we can decompose the value of T as

$$\mathrm{val}_T(\mathbf{W}; \mathbf{x}_1, \ldots, \mathbf{x}_k) = \frac{1}{n} \sum_{\pi\in\mathcal{P}} \sum_{\mathbf{i}\in[n]^{\mathcal{V}}:\pi(\mathbf{i})=\pi} q_{\mathbf{i}|T} \cdot W_{\mathbf{i}|T}. \tag{2.3.1}$$

Definition 2.3.1. Let $(\mathcal{V}, \mathcal{E})$ be an undirected graph. For any partition π of $\mathcal{V}$, the *image of* $(\mathcal{V}, \mathcal{E})$ *under* π is the undirected multi-graph $G_\pi = (\mathcal{K}_\pi, \mathcal{F}_\pi)$ that is the image of $(\mathcal{V}, \mathcal{E})$ under the graph homomorphism sending each vertex $u \in \mathcal{V}$ to the block of π containing u.

I.e., the vertices $\mathcal{K}_\pi \equiv \pi$ of G_π are the blocks of π, and G_π has the same number of edges $|\mathcal{F}_\pi|$ (counting multiplicity and self-loops) as $|\mathcal{E}|$. For each edge $(u, v) \in \mathcal{E}$, there is a corresponding edge $(U, V) \in \mathcal{F}_\pi$ where $U, V \in \pi$ are the blocks for which $u \in U$ and $v \in V$.

For each $\pi \in \mathcal{P}$, let $G_\pi = (\mathcal{K}_\pi, \mathcal{F}_\pi)$ be the image of $(\mathcal{V}, \mathcal{E})$ under π. For each block $U \in \mathcal{K}_\pi$, define the polynomial $Q_U = \prod_{u\in U} q_u$, and for each unique (undirected) edge (U, V) of G_π, let $e(U, V)$ be the number of times it appears in $\mathcal{F}_\pi$. Then, identifying the sum over $\{\mathbf{i} : \pi(\mathbf{i}) = \pi\}$ as a sum over one distinct index in $[n]$ for each block $U \in \mathcal{K}_\pi$, we have

$$\sum_{\mathbf{i}\in[n]^{\mathcal{V}}:\pi(\mathbf{i})=\pi} q_{\mathbf{i}|T} \cdot W_{\mathbf{i}|T} = \sum_{\mathbf{i}\in[n]^{\mathcal{K}_\pi}}^{*} Q_{\mathbf{i}|G_\pi} \cdot W^e_{\mathbf{i}|G_\pi}$$

where $\sum^*$ denotes the restriction of the summation to index tuples $\mathbf{i} = (i_U : U \in \mathcal{K}_\pi) \in [n]^{\mathcal{K}_\pi}$ having all indices distinct, and

$$Q_{\mathbf{i}|G_\pi} = \prod_{U\in\mathcal{K}_\pi} Q_U(x_1[i_U], \ldots, x_k[i_U]), \qquad W^e_{\mathbf{i}|G_\pi} = \prod_{\text{unique edges } (U,V) \text{ of } G_\pi} W[i_U, i_V]^{e(U,V)}.$$

Applying this to (2.3.1), we obtain

$$\mathrm{val}_T(\mathbf{W}; \mathbf{x}_1, \ldots, \mathbf{x}_k) = \frac{1}{n} \sum_{\pi \in \mathcal{P}} \sideset{}{^*}\sum_{\mathbf{i} \in [n]^{\mathcal{K}_\pi}} Q_{\mathbf{i}|G_\pi} \cdot W^e_{\mathbf{i}|G_\pi}. \tag{2.3.2}$$

We will compute the expectation of (2.3.2), and see that the only non-vanishing contributions in the limit $n \to \infty$ arise from partitions π where G_π is itself a tree and $e(U, V) = 2$ for each unique edge (U, V) of G_π. These non-vanishing terms may be related to the values of a reduced tensor network associated to G_π, evaluated on the matrix $\mathbf{S}/n$ in place of $\mathbf{W}$.

In anticipation of this computation, we first show the following lemma which establishes universality of the value of any tensor network evaluated on $\mathbf{S}/n$.

Lemma 2.3.2. *Under the assumptions of Lemma 2.2.13, for any diagonal tensor network $T = (\mathcal{V}, \mathcal{E}, \{q_v\}_{v \in \mathcal{V}})$ in k variables,*

$$\lim_{n \to \infty} \mathrm{val}_T(\mathbf{S}/n; \mathbf{x}_1, \ldots, \mathbf{x}_k) = \prod_{v \in \mathcal{V}} \mathbb{E}[q_v(X_1, \ldots, X_k)].$$

Proof. Observe first that for any diagonal tensor network $T = (\mathcal{V}, \mathcal{E}, \{q_v\}_{v \in \mathcal{V}})$, we have

$$\frac{1}{n} \sum_{\mathbf{i} \in [n]^{\mathcal{V}}} \prod_{v \in \mathcal{V}} |q_v(x_{1:k})[i_v]| \cdot \prod_{(u,v) \in \mathcal{E}} \frac{1}{n} \le C \tag{2.3.3}$$

for a constant $C := C(T) > 0$, almost surely for all large n. Indeed, since T is a tree, we have $\frac{1}{n} \prod_{(u,v) \in \mathcal{E}} \frac{1}{n} = n^{-|\mathcal{V}|}$ in the above. Each function $|q_v|$ is continuous and satisfies the polynomial growth condition (2.1.1), so (2.3.3) follows from the assumption (2.2.10).

Now note that since T is a tree, we can order its vertices as $1, 2, \ldots, |\mathcal{V}|$ such that removing one vertex at a time in this order, the remaining graph is always still a tree. Denote the remaining tensor network after removing vertices $1, \ldots, h-1$ by $T_h = (\mathcal{V}_h, \mathcal{E}_h, \{q_v\}_{v \ge h})$. The vertex h has only one neighbor in T_h, which we denote by

$u_h \in \{h+1, \ldots, |\mathcal{V}|\}$. Then

$$
\begin{aligned}
&\mathrm{val}_T(\mathbf{S}/n; \mathbf{x}_{1:k}) \\
&= \frac{1}{n} \sum_{\mathbf{i} \in [n]^{\mathcal{V}}} \prod_{v \in \mathcal{V}} q_v(x_{1:k}[i_v]) \prod_{(u,v) \in \mathcal{E}} \frac{S[i_u, i_v]}{n} \\
&= \frac{1}{n} \sum_{i_2, \ldots, i_{|\mathcal{V}|}=1}^{n} \left(\prod_{v \in \mathcal{V}_2} q_v(x_{1:k}[i_v]) \prod_{(u,v) \in \mathcal{E}_2} \frac{S[i_u, i_v]}{n} \right) \cdot \left(\sum_{i_1=1}^{n} q_1(x_{1:k}[i_1]) \frac{S[i_1, i_{u_1}]}{n} \right) \\
&= \frac{1}{n} \sum_{i_2, \ldots, i_{|\mathcal{V}|}=1}^{n} \left(\prod_{v \in \mathcal{V}_2} q_v(x_{1:k}[i_v]) \prod_{(u,v) \in \mathcal{E}_2} \frac{S[i_u, i_v]}{n} \right) \cdot \Big(\langle q_1(\mathbf{x}_{1:k}) \rangle \cdot \langle \mathbf{s}_{i_{u_1}} \rangle + \delta(i_{u_1}) \Big) \\
&= \frac{1}{n} \sum_{i_2, \ldots, i_{|\mathcal{V}|}=1}^{n} \left(\prod_{v \in \mathcal{V}_2} q_v(x_{1:k}[i_v]) \prod_{(u,v) \in \mathcal{E}_2} \frac{S[i_u, i_v]}{n} \right) \cdot \Big(\mathbb{E}[q_1(X_{1:k})] + \delta'(i_{u_1}) \Big).
\end{aligned}
$$

Here, $\delta(i), \delta'(i)$ denote errors that satisfy $\lim_{n\to\infty} \max_{i \in [n]} |\delta(i)|, |\delta'(i)| = 0$, as follows from (2.2.11), the conditions of Definition 2.2.3(c), and (2.2.10). Note that

$$
\left| \frac{1}{n} \sum_{i_2, \ldots, i_{|\mathcal{V}|}=1}^{n} \left(\prod_{v \in \mathcal{V}_2} q_v(x_{1:k}[i_v]) \prod_{(u,v) \in \mathcal{E}_2} \frac{S[i_u, i_v]}{n} \right) \cdot \delta'(i_{u_1}) \right| \to 0
$$

as $n \to \infty$ by the condition $|S[i_u, i_v]| \leq C$ of Definition 2.2.3(c), the bound (2.3.3) applied to the network T_2 with vertex 1 removed, and the convergence $\max_{i \in [n]} |\delta'(i)| \to 0$. Thus

$$
\lim_{n\to\infty} \mathrm{val}_T(\mathbf{S}/n; \mathbf{x}_{1:k}) = \lim_{n\to\infty} \frac{1}{n} \sum_{i_2, \ldots, i_{|\mathcal{V}|}=1}^{n} \left(\prod_{v \in \mathcal{V}_2} q_v(x_{1:k}[i_v]) \prod_{(u,v) \in \mathcal{E}_2} \frac{S[i_u, i_v]}{n} \right) \cdot \mathbb{E}[q_1(X_{1:k})].
$$

Repeating the above procedure by removing vertices $1, 2, \ldots, |\mathcal{V}|-1$ sequentially, we are left with the single vertex $|\mathcal{V}|$ and no edges, and

$$
\lim_{n\to\infty} \mathrm{val}_T(\mathbf{S}/n; \mathbf{x}_{1:k}) = \lim_{n\to\infty} \frac{1}{n} \sum_{i_{|\mathcal{V}|}=1}^{n} q_{|\mathcal{V}|}(x_{1:k}[i_{|\mathcal{V}|}]) \cdot \prod_{v=1}^{|\mathcal{V}|-1} \mathbb{E}[q_v(X_{1:k})] = \prod_{v=1}^{|\mathcal{V}|} \mathbb{E}[q_v(X_{1:k})].
$$

■

The next lemma relates summations over distinct indices to ones without the distinctness requirement.

Lemma 2.3.3. *Let* $\mathbf{x}_1, \dots, \mathbf{x}_k \in \mathbb{R}^n$ *and* $(X_1, \dots, X_k)$ *be such that*

$$(\mathbf{x}_1, \dots, \mathbf{x}_k) \xrightarrow{W} (X_1, \dots, X_k).$$

For a finite index set $\mathcal{S}$*, let* $(q_s : s \in \mathcal{S})$ *be* $|\mathcal{S}|$ *continuous functions satisfying the polynomial growth condition (2.1.1) for some order* $p \geq 1$*. Then*

$$\begin{aligned}\lim_{n\to\infty} \frac{1}{n^{|\mathcal{S}|}} \sum_{\mathbf{i}\in[n]^{\mathcal{S}}}^{*} \prod_{s\in\mathcal{S}} q_s(x_1[i_s], \dots, x_k[i_s]) &= \lim_{n\to\infty} \frac{1}{n^{|\mathcal{S}|}} \sum_{\mathbf{i}\in[n]^{\mathcal{S}}} \prod_{s\in\mathcal{S}} q_s(x_1[i_s], \dots, x_k[i_s]) \\ &= \prod_{s\in\mathcal{S}} \mathbb{E}[q_s(X_1, \dots, X_k)].\end{aligned}$$

Proof. Let $\mathcal{P}$ be the set of partitions of $\mathcal{S}$, and let $\pi(\mathbf{i}) \in \mathcal{P}$ be the partition induced by $\mathbf{i} \in [n]^{\mathcal{S}}$. Let $0_{\mathcal{P}}$ the partition having $|\mathcal{S}|$ singleton blocks, corresponding to $\mathbf{i}$ having all indices distinct. Then for the first equality, it suffices to show that

$$\Delta := \frac{1}{n^{|\mathcal{S}|}} \sum_{\pi\in\mathcal{P}:\pi\neq 0_{\mathcal{P}}} \sum_{\mathbf{i}\in[n]^{\mathcal{S}}:\pi(\mathbf{i})=\pi} \prod_{s\in\mathcal{S}} |q_s(x_{1:k}[i_s])| \tag{2.3.4}$$

vanishes as $n \to \infty$.

For any $\pi \in \mathcal{P}$ and block $R \in \pi$, define $Q_R = \prod_{u\in R} q_u$, and let $|\pi|$ be the number of blocks of π. Then, identifying the sum over $\{\mathbf{i} : \pi(\mathbf{i}) = \pi\}$ with the sum over one distinct index in $[n]$ for each block of π,

$$\frac{1}{n^{|\pi|}} \sum_{\mathbf{i}\in[n]^{\mathcal{S}}:\pi(\mathbf{i})=\pi} \prod_{s\in\mathcal{S}} |q_s(x_{1:k}[i_s])| = \frac{1}{n^{|\pi|}} \sum_{\mathbf{i}\in[n]^{\pi}}^{*} \prod_{R\in\pi} |Q_R(x_{1:k}[i_R])|.$$

As an upper bound, adding back the excluded index tuples $\mathbf{i} \in [n]^{\pi}$ where some indices coincide,

$$\frac{1}{n^{|\pi|}} \sum_{\mathbf{i}\in[n]^{\mathcal{S}}:\pi(\mathbf{i})=\pi} \prod_{s\in\mathcal{S}} |q_s(x_{1:k}[i_s])| \leq \prod_{R\in\pi} \left(\frac{1}{n} \sum_{i=1}^{n} |Q_R(x_{1:k}[i])| \right).$$

Since $\mathbf{x}_{1:k} \xrightarrow{W} X_{1:k}$ and $|Q_R|$ is a continuous function satisfying the polynomial growth condition (2.1.1), this upper bound is at most a constant $C(\pi)$ for all large n. For any $\pi \neq 0_{\mathcal{P}}$, we have $|\pi| \leq |\mathcal{S}| - 1$. As the number of partitions $\pi \in \mathcal{P}$ is independent of n, applying these observations to (2.3.4) shows $\Delta \leq C/n$ for a constant $C > 0$ and all large n, and hence $\Delta \to 0$ as desired.

The second equality of the lemma follows from the given condition $\mathbf{x}_{1:k} \xrightarrow{W} X_{1:k}$, hence

$$\frac{1}{n^{|\mathcal{S}|}} \sum_{\mathbf{i} \in [n]^{\mathcal{S}}} \prod_{s \in \mathcal{S}} q_s(x_{1:k}[i_s]) = \prod_{s \in \mathcal{S}} \frac{1}{n} \sum_{i=1}^{n} q_s(x_{1:k}[i]) \to \prod_{s \in \mathcal{S}} \mathbb{E}[q_s(X_{1:k})].$$

■

We now show that the limit tensor network value is universal in expectation over $\mathbf{W}$.

Lemma 2.3.4. *Let $\mathbb{E}$ denote the expectation over $\mathbf{W}$, conditional on $\mathbf{x}_1, \ldots, \mathbf{x}_k$. Then Lemma 2.2.13 holds for $\mathbb{E}[\mathrm{val}_T(\mathbf{W}; \mathbf{x}_1, \ldots, \mathbf{x}_k)]$ in place of $\mathrm{val}_T(\mathbf{W}; \mathbf{x}_1, \ldots, \mathbf{x}_k)$.*

Proof. Recall the decomposition of $\mathrm{val}_T(\mathbf{W}; \mathbf{x}_1, \ldots, \mathbf{x}_k)$ in (2.3.2), where $\mathcal{P}$ is the set of partitions of the vertices $\mathcal{V}$ of T. Taking expectation on both sides yields

$$\mathbb{E}[\mathrm{val}_T(\mathbf{W}; \mathbf{x}_{1:k})] = \frac{1}{n} \sum_{\pi \in \mathcal{P}} \sum_{\mathbf{i} \in [n]^{\mathcal{K}_\pi}}^{*} Q_{\mathbf{i}|G_\pi} \cdot \mathbb{E}[W^e_{\mathbf{i}|G_\pi}]. \tag{2.3.5}$$

First note that, since the indices of $\mathbf{i}$ are distinct and all entries of $\mathbf{W}$ have mean 0, $\mathbb{E}[W^e_{\mathbf{i}|G_\pi}]$ is nonzero only if each unique edge of $G_\pi = (\mathcal{K}_\pi, \mathcal{F}_\pi)$ appears at least twice. Let $|\mathcal{K}_\pi|$ and $|\mathcal{F}_\pi|_*$ be the number of vertices and number of *unique* (undirected) edges of G_π. The graph G_π must be connected since the original tree T was connected, so $|\mathcal{K}_\pi| \leq |\mathcal{F}_\pi|_* + 1$. Then any G_π where each unique edge appears at least twice has $|\mathcal{K}_\pi| \leq |\mathcal{F}_\pi|_* + 1 \leq |\mathcal{E}|/2 + 1$, where $|\mathcal{E}|$ is the number of edges of the original tree T.

Furthermore, we claim that the contribution from partitions π where $|\mathcal{K}_\pi| \leq |\mathcal{E}|/2$

is negligible. To see this, we apply Definition 2.2.3(b) to get

$$\begin{aligned}\left|\mathbb{E}[W^e_{\mathbf{i}|G_\pi}]\right| &= \prod_{(U,V):e(U,V)=2} \mathbb{E}[W[i_U,i_V]^2] \prod_{(U,V):e(U,V)>2} \left|\mathbb{E}[W[i_U,i_V]^{e(U,V)}]\right| \\ &\leq \prod_{(U,V):e(U,V)=2} \frac{C}{n} \prod_{(U,V):e(U,V)>2} \frac{o(1)}{n}.\end{aligned}$$

If there is an edge (U,V) of G_π with $e(U,V) > 2$, then this shows $|\mathbb{E}[W^e_{\mathbf{i}|G_\pi}]| \leq o(1)/n^{|\mathcal{F}_\pi|_*} \leq o(1)/n^{|\mathcal{K}_\pi|-1}$. If, conversely, every edge in G_π appears exactly twice, then by assumption $|\mathcal{F}_\pi|_* = |\mathcal{E}|/2 \geq |\mathcal{K}_\pi|$, so this shows $|\mathbb{E}[W^e_{\mathbf{i}|G_\pi}]| \leq (C/n)^{|\mathcal{F}_\pi|_*} \leq o(1)/n^{|\mathcal{K}_\pi|-1}$ also. Therefore,

$$\left|\frac{1}{n}\sum^*_{\mathbf{i}\in[n]^{\mathcal{K}_\pi}} Q_{\mathbf{i}|G_\pi}\cdot\mathbb{E}[W^e_{\mathbf{i}|G_\pi}]\right| \leq \frac{o(1)}{n^{|\mathcal{K}_\pi|}}\sum^*_{\mathbf{i}\in[n]^{\mathcal{K}_\pi}}\left|Q_{\mathbf{i}|G_\pi}\right|.$$

As an upper bound, adding back the excluded tuples $\mathbf{i}\in[n]^{\mathcal{K}_\pi}$ where not all indices are distinct, we have

$$\frac{1}{n^{|\mathcal{K}_\pi|}}\sum^*_{\mathbf{i}\in[n]^{\mathcal{K}_\pi}} |Q_{\mathbf{i}|G_\pi}| \leq \prod_{U\in\mathcal{K}_\pi}\frac{1}{n}\sum_{i=1}^n |Q_U(x_{1:k}[i])| \tag{2.3.6}$$

By (2.2.10), this upper bound is at most a constant $C(\pi)$ for all large n, so

$$\left|\frac{1}{n}\sum^*_{\mathbf{i}\in[n]^{\mathcal{K}_\pi}} Q_{\mathbf{i}|G_\pi}\cdot\mathbb{E}[W^e_{\mathbf{i}|G_\pi}]\right| \to 0$$

as claimed.

Thus the only non-vanishing contributions to (2.3.5) come from partitions π where $|\mathcal{K}_\pi| = |\mathcal{F}_\pi|_* + 1 = |\mathcal{E}|/2 + 1$. Then each unique edge of G_π appears exactly twice, and these edges form a tree. In this case, we have

$$\frac{1}{n}\sum^*_{\mathbf{i}\in[n]^{\mathcal{K}_\pi}} Q_{\mathbf{i}|G_\pi}\cdot\mathbb{E}[W^e_{\mathbf{i}|G_\pi}] = \frac{1}{n}\sum^*_{\mathbf{i}\in[n]^{\mathcal{K}_\pi}} Q_{\mathbf{i}|G_\pi}\cdot\prod_{\text{unique edges }(U,V)\text{ of }G_\pi}\frac{S[i_U,i_V]}{n}. \tag{2.3.7}$$

Let $\mathcal{I}_*$ be the set of tuples $\mathbf{i} \in [n]^{\mathcal{K}_\pi}$ where all indices are distinct. Then, applying $|S[i_U, i_V]| \leq C$ from Definition 2.2.3(c), for a constant $C' = C(\pi) > 0$,

$$\left| \frac{1}{n} \sum_{\mathbf{i} \in [n]^{\mathcal{K}_\pi} \setminus \mathcal{I}_*} Q_{\mathbf{i}|G_\pi} \prod_{\text{unique edges } (U,V) \text{ of } G_\pi} \frac{S[i_U, i_V]}{n} \right| \leq \frac{C'}{n^{1+|\mathcal{F}_\pi|_*}} \sum_{\mathbf{i} \in [n]^{\mathcal{K}_\pi} \setminus \mathcal{I}_*} |Q_{\mathbf{i}|G_\pi}|.$$

Since $|\mathcal{K}_\pi| = |\mathcal{F}_\pi|_* + 1$, the first equality of Lemma 2.3.3 shows that this vanishes as $n \to \infty$. Then the right side of (2.3.7) has the same limit as

$$\frac{1}{n} \sum_{\mathbf{i} \in [n]^{\mathcal{K}_\pi}} Q_{\mathbf{i}|G_\pi} \cdot \prod_{\text{unique edges } (U,V) \text{ of } G_\pi} \frac{S[i_U, i_V]}{n}.$$

This is the limit value of the tensor network $T_\pi = (\mathcal{K}_\pi, \text{unique edges of } \mathcal{F}_\pi, \{Q_U\}_{U \in \mathcal{K}_\pi})$ applied to $\mathbf{S}/n$, which by Lemma 2.3.2 equals $\prod_{U \in \mathcal{K}_\pi} \mathbb{E}[Q_U(X_{1:k})]$. Applying this back to (2.3.7) and (2.3.5),

$$\begin{aligned} \lim_{n \to \infty} \mathbb{E}[\text{val}_T(\mathbf{W}; \mathbf{x}_{1:k})] &= \sum_{\pi \in \mathcal{P}:\, |\mathcal{K}_\pi| = |\mathcal{F}_\pi|_* + 1 = |\mathcal{E}|/2 + 1} \; \prod_{U \in \mathcal{K}_\pi} \mathbb{E}[Q_U(X_{1:k})] \qquad (2.3.8) \\ &=: \text{lim-val}_T(X_1, \ldots, X_k). \end{aligned}$$

This limit depends only on T and the joint law of $X_1, \ldots, X_k$, concluding the proof. ■

We make a brief interlude to show here the asymptotic freeness result of Proposition 2.2.16(a).

Proof of Proposition 2.2.16(a). From the preceding proof, only partitions π where $|\mathcal{K}_\pi| = |\mathcal{E}|/2 + 1$ contribute to $\text{lim-val}_T(X_1, \ldots, X_k)$. For every such partition, since T has $|\mathcal{E}| + 1$ vertices, this implies that some vertex of $\mathcal{K}_\pi$, i.e. some block U of π, contains only a single vertex v of T. For this block U, we have

$$\mathbb{E}[Q_U(X_{1:k})] = \mathbb{E}[q_v(X_{1:k})] = 0$$

by the condition (2.2.13). Therefore, every summand in (2.3.8) vanishes, implying as desired $\text{lim-val}_T(X_1, \ldots, X_k) = 0$. ∎

To complete the proof of Lemma 2.2.13, it remains to establish the concentration of the value of any tensor network around its mean as $n \to \infty$.

Lemma 2.3.5. *Let $\mathbb{E}$ denote the expectation over $\mathbf{W}$, conditional on $\mathbf{x}_1, \ldots, \mathbf{x}_k$. Under the setting of Lemma 2.2.13, almost surely as $n \to \infty$,*

$$\text{val}_T(\mathbf{W}; \mathbf{x}_1, \ldots, \mathbf{x}_k) - \mathbb{E}[\text{val}_T(\mathbf{W}; \mathbf{x}_1, \ldots, \mathbf{x}_k)] \to 0.$$

Proof. We write $\text{val}(\mathbf{W}) = \text{val}_T(\mathbf{W}; \mathbf{x}_1, \ldots, \mathbf{x}_k)$. We will bound the fourth moment of $\text{val}(\mathbf{W}) - \mathbb{E}[\text{val}(\mathbf{W})]$ and apply the Borel-Cantelli lemma. (Note that $\text{val}(\mathbf{W}) - \mathbb{E}[\text{val}(\mathbf{W})]$ typically fluctuates on the order of $1/\sqrt{n}$, so that bounding the variance would not suffice to show almost-sure convergence.)

First, we expand

$$\begin{aligned} &\mathbb{E}[(\text{val}(\mathbf{W}) - \mathbb{E}[\text{val}(\mathbf{W})])^4] \\ &= \mathbb{E}[\text{val}(\mathbf{W})^4] - 4\mathbb{E}[\text{val}(\mathbf{W})^3]\mathbb{E}[\text{val}(\mathbf{W})] + 6\mathbb{E}[\text{val}(\mathbf{W})^2]\mathbb{E}[\text{val}(\mathbf{W})]^2 - 3\mathbb{E}[\text{val}(\mathbf{W})]^4. \end{aligned} \tag{2.3.9}$$

We introduce four independent copies of the matrix $\mathbf{W}$ as $\mathbf{W}^{(1)}, \mathbf{W}^{(2)}, \mathbf{W}^{(3)}, \mathbf{W}^{(4)}$, define four index tuples $\mathbf{i}_1, \mathbf{i}_2, \mathbf{i}_3, \mathbf{i}_4 \in [n]^{\mathcal{V}}$, and write as shorthand

$$q_{\mathbf{i}_{1:4}} = q_{\mathbf{i}_1|T} \cdot q_{\mathbf{i}_2|T} \cdot q_{\mathbf{i}_3|T} \cdot q_{\mathbf{i}_4|T}, \qquad W^{(a_1,a_2,a_3,a_4)}_{\mathbf{i}_{1:4}} = W^{(a_1)}_{\mathbf{i}_1|T} \cdot W^{(a_2)}_{\mathbf{i}_2|T} \cdot W^{(a_3)}_{\mathbf{i}_3|T} \cdot W^{(a_4)}_{\mathbf{i}_4|T}$$

where each $W_{\mathbf{i}|T}^{(a)}$ is defined by the copy $\mathbf{W}^{(a)}$. Then

$$\begin{aligned}
\mathbb{E}[\mathrm{val}(\mathbf{W})^4] &= \frac{1}{n^4} \sum_{\mathbf{i}_1,\ldots,\mathbf{i}_4 \in [n]^{\mathcal{V}}} q_{\mathbf{i}_{1:4}} \cdot \mathbb{E}[W_{\mathbf{i}_{1:4}}^{(1,1,1,1)}], \\
\mathbb{E}[\mathrm{val}(\mathbf{W})^3]\mathbb{E}[\mathrm{val}(\mathbf{W})] &= \frac{1}{n^4} \sum_{\mathbf{i}_1,\ldots,\mathbf{i}_4 \in [n]^{\mathcal{V}}} q_{\mathbf{i}_{1:4}} \cdot \mathbb{E}[W_{\mathbf{i}_{1:4}}^{(1,1,1,2)}], \\
\mathbb{E}[\mathrm{val}(\mathbf{W})^2]\mathbb{E}[\mathrm{val}(\mathbf{W})]^2 &= \frac{1}{n^4} \sum_{\mathbf{i}_1,\ldots,\mathbf{i}_4 \in [n]^{\mathcal{V}}} q_{\mathbf{i}_{1:4}} \cdot \mathbb{E}[W_{\mathbf{i}_{1:4}}^{(1,1,2,3)}], \\
\mathbb{E}[\mathrm{val}(\mathbf{W})]^4 &= \frac{1}{n^4} \sum_{\mathbf{i}_1,\ldots,\mathbf{i}_4 \in [n]^{\mathcal{V}}} q_{\mathbf{i}_{1:4}} \cdot \mathbb{E}[W_{\mathbf{i}_{1:4}}^{(1,2,3,4)}].
\end{aligned} \tag{2.3.10}$$

Corresponding to each index tuple $\mathbf{i}_{1:4}$, consider a multi-graph $G(\mathbf{i}_{1:4})$ whose vertices are the unique index values in $\mathbf{i}_{1:4}$, with one edge $(i_{a,u}, i_{a,v})$ for every combination of $a = 1,2,3,4$ and edge $(u,v) \in \mathcal{E}$, counting multiplicity. (One may visualize $G(\mathbf{i}_{1:4})$ as a multi-graph whose vertices are a subset of $[n]$, and having edges of 4 colors corresponding to $a = 1,2,3,4$.) Then the edges of $G(\mathbf{i}_{1:4})$ corresponding to each single index $a = 1,2,3,4$ must belong to a single connected component, so the number of connected components in $G(\mathbf{i}_{1:4})$ can be either 1, 2, 3 or 4. Let us partition the index tuples $\mathbf{i}_1, \mathbf{i}_2, \mathbf{i}_3, \mathbf{i}_4 \in [n]^{\mathcal{V}}$ into the three sets

$$\begin{aligned}
\mathcal{I}_2 &= \{\mathbf{i}_{1:4} : G(\mathbf{i}_{1:4}) \text{ has 1 or 2 connected components}\}, \\
\mathcal{I}_3 &= \{\mathbf{i}_{1:4} : G(\mathbf{i}_{1:4}) \text{ has 3 connected components}\}, \\
\mathcal{I}_4 &= \{\mathbf{i}_{1:4} : G(\mathbf{i}_{1:4}) \text{ has 4 connected components}\},
\end{aligned}$$

and define correspondingly for $j = 2,3,4$,

$$A_j = \frac{1}{n^4} \sum_{\mathbf{i}_{1:4} \in \mathcal{I}_j} q_{\mathbf{i}_{1:4}} \Big(\mathbb{E}[W_{\mathbf{i}_{1:4}}^{(1,1,1,1)}] - 4 \cdot \mathbb{E}[W_{\mathbf{i}_{1:4}}^{(1,1,1,2)}] + 6 \cdot \mathbb{E}[W_{\mathbf{i}_{1:4}}^{(1,1,2,3)}] - 3 \cdot \mathbb{E}[W_{\mathbf{i}_{1:4}}^{(1,2,3,4)}] \Big). \tag{2.3.11}$$

Then by (2.3.9) and (2.3.10), we have $\mathbb{E}[(\mathrm{val}(\mathbf{W}) - \mathbb{E}[\mathrm{val}(\mathbf{W})])^4] = A_2 + A_3 + A_4$.

Below, we will show that $A_3 = A_4 = 0$ and $A_2 = O(1/n^2)$ as $n \to \infty$.

For A_4, observe that since $G(\mathbf{i}_{1:4})$ has 4 connected components, the tuples $\mathbf{i}_1, \mathbf{i}_2, \mathbf{i}_3, \mathbf{i}_4$ have no common indices. Then due to the independence between entries of $\mathbf{W}$, we have

$$\begin{aligned}\mathbb{E}[W_{\mathbf{i}_{1:4}}^{(a_1,a_2,a_3,a_4)}] &= \mathbb{E}[W_{\mathbf{i}_1}^{(a_1)}] \cdot \mathbb{E}[W_{\mathbf{i}_2}^{(a_2)}] \cdot \mathbb{E}[W_{\mathbf{i}_3}^{(a_3)}] \cdot \mathbb{E}[W_{\mathbf{i}_4}^{(a_4)}] \\ &= \mathbb{E}[W_{\mathbf{i}_1}^{(1)}] \cdot \mathbb{E}[W_{\mathbf{i}_2}^{(1)}] \cdot \mathbb{E}[W_{\mathbf{i}_3}^{(1)}] \cdot \mathbb{E}[W_{\mathbf{i}_4}^{(1)}] = \mathbb{E}[W_{\mathbf{i}_{1:4}}^{(1,1,1,1)}]\end{aligned}$$

for any a_1, a_2, a_3, a_4. Applying this to A_4 defined in (2.3.11), we get $A_4 = 0$.

Next, for A_3, we write $\mathbf{i}_j \parallel \mathbf{i}_{j'}$ if $\mathbf{i}_j$ and $\mathbf{i}_{j'}$ share at least one index. Note that for any $\mathbf{i}_{1:4} \in \mathcal{I}_3$, there is a unique pair $\mathbf{i}_j, \mathbf{i}_{j'}$ such that $\mathbf{i}_j \parallel \mathbf{i}_{j'}$, so

$$\mathcal{I}_3 = \bigsqcup_{1 \le j < j' \le 4} \{\mathbf{i}_{1:4} \in \mathcal{I}_3 : \mathbf{i}_j \parallel \mathbf{i}_{j'}\}.$$

We will repeatedly apply this six-fold decomposition of $\mathcal{I}_3$ and the independence between the different copies of $\mathbf{W}$. If $\mathbf{i}_3 \parallel \mathbf{i}_4$, we have

$$\mathbb{E}[W_{\mathbf{i}_{1:4}}^{(1,1,1,1)}] = \mathbb{E}[W_{\mathbf{i}_1}^{(1)}] \cdot \mathbb{E}[W_{\mathbf{i}_2}^{(1)}] \cdot \mathbb{E}[W_{\mathbf{i}_3}^{(1)} W_{\mathbf{i}_4}^{(1)}] = \mathbb{E}[W_{\mathbf{i}_{1:4}}^{(1,2,3,3)}]$$

which together with permutation symmetry between the labels 1, 2, 3, and 4 further implies that

$$\sum_{\mathbf{i}_{1:4} \in \mathcal{I}_3} q_{\mathbf{i}_{1:4}} \cdot \mathbb{E}[W_{\mathbf{i}_{1:4}}^{(1,1,1,1)}] = 6 \sum_{\mathbf{i}_{1:4} \in \mathcal{I}_3, \mathbf{i}_3 \parallel \mathbf{i}_4} q_{\mathbf{i}_{1:4}} \cdot \mathbb{E}[W_{\mathbf{i}_{1:4}}^{(1,1,1,1)}] = 6 \sum_{\mathbf{i}_{1:4} \in \mathcal{I}_3, \mathbf{i}_3 \parallel \mathbf{i}_4} q_{\mathbf{i}_{1:4}} \cdot \mathbb{E}[W_{\mathbf{i}_{1:4}}^{(1,2,3,3)}]. \tag{2.3.12}$$

Similarly, considering the two cases $\mathbf{i}_3 \parallel \mathbf{i}_4$ and $\mathbf{i}_2 \parallel \mathbf{i}_3$ and their symmetric equivalents,

$$\sum_{\mathbf{i}_{1:4} \in \mathcal{I}_3} q_{\mathbf{i}_{1:4}} \cdot \mathbb{E}[W_{\mathbf{i}_{1:4}}^{(1,1,1,2)}] = 3 \sum_{\mathbf{i}_{1:4} \in \mathcal{I}_3, \mathbf{i}_3 \parallel \mathbf{i}_4} q_{\mathbf{i}_{1:4}} \cdot \mathbb{E}[W_{\mathbf{i}_{1:4}}^{(1,2,3,4)}] + 3 \sum_{\mathbf{i}_{1:4} \in \mathcal{I}_3, \mathbf{i}_2 \parallel \mathbf{i}_3} q_{\mathbf{i}_{1:4}} \cdot \mathbb{E}[W_{\mathbf{i}_{1:4}}^{(1,2,2,3)}]$$

$$= 3 \sum_{\mathbf{i}_{1:4} \in \mathcal{I}_3, \mathbf{i}_3 \| \mathbf{i}_4} q_{\mathbf{i}_{1:4}} \cdot \mathbb{E}[W_{\mathbf{i}_{1:4}}^{(1,2,3,4)}] + 3 \sum_{\mathbf{i}_{1:4} \in \mathcal{I}_3, \mathbf{i}_3 \| \mathbf{i}_4} q_{\mathbf{i}_{1:4}} \cdot \mathbb{E}[W_{\mathbf{i}_{1:4}}^{(1,2,3,3)}]. \tag{2.3.13}$$

Considering the three cases $\mathbf{i}_3 \parallel \mathbf{i}_4$, $\mathbf{i}_2 \parallel \mathbf{i}_4$, $\mathbf{i}_1 \parallel \mathbf{i}_2$ and their symmetric equivalents,

$$\begin{aligned}
&\sum_{\mathbf{i}_{1:4} \in \mathcal{I}_3} q_{\mathbf{i}_{1:4}} \cdot \mathbb{E}[W_{\mathbf{i}_{1:4}}^{(1,1,2,3)}] \\
&= \sum_{\mathbf{i}_{1:4} \in \mathcal{I}_3, \mathbf{i}_3 \| \mathbf{i}_4} q_{\mathbf{i}_{1:4}} \cdot \mathbb{E}[W_{\mathbf{i}_{1:4}}^{(1,2,3,4)}] + 4 \sum_{\mathbf{i}_{1:4} \in \mathcal{I}_3, \mathbf{i}_2 \| \mathbf{i}_3} q_{\mathbf{i}_{1:4}} \cdot \mathbb{E}[W_{\mathbf{i}_{1:4}}^{(1,2,3,4)}] + \sum_{\mathbf{i}_{1:4} \in \mathcal{I}_3, \mathbf{i}_1 \| \mathbf{i}_2} q_{\mathbf{i}_{1:4}} \cdot \mathbb{E}[W_{\mathbf{i}_{1:4}}^{(1,1,2,3)}] \\
&= 5 \sum_{\mathbf{i}_{1:4} \in \mathcal{I}_3, \mathbf{i}_3 \| \mathbf{i}_4} q_{\mathbf{i}_{1:4}} \cdot \mathbb{E}[W_{\mathbf{i}_{1:4}}^{(1,2,3,4)}] + \sum_{\mathbf{i}_{1:4} \in \mathcal{I}_3, \mathbf{i}_3 \| \mathbf{i}_4} q_{\mathbf{i}_{1:4}} \cdot \mathbb{E}[W_{\mathbf{i}_{1:4}}^{(1,2,3,3)}]. \qquad (2.3.14)
\end{aligned}$$

Finally, by symmetry,

$$\sum_{\mathbf{i}_{1:4} \in \mathcal{I}_3} q_{\mathbf{i}_{1:4}} \cdot \mathbb{E}[W_{\mathbf{i}_{1:4}}^{(1,2,3,4)}] = 6 \sum_{\mathbf{i}_{1:4} \in \mathcal{I}_3, \mathbf{i}_3 \| \mathbf{i}_4} q_{\mathbf{i}_{1:4}} \cdot \mathbb{E}[W_{\mathbf{i}_{1:4}}^{(1,2,3,4)}]. \tag{2.3.15}$$

Collecting (2.3.12), (2.3.13), (2.3.14), and (2.3.15) and applying them to A_3 defined in (2.3.11), we get $A_3 = 0$.

Finally, we bound A_2. Let $|\mathcal{V}_{G(\mathbf{i}_{1:4})}|$ and $|\mathcal{E}_{G(\mathbf{i}_{1:4})}|_*$ be the number of vertices and number of unique (undirected) edges of $G(\mathbf{i}_{1:4})$. Since $G(\mathbf{i}_{1:4})$ has at most 2 connected components, we have $|\mathcal{V}_{G(\mathbf{i}_{1:4})}| \leq |\mathcal{E}_{G(\mathbf{i}_{1:4})}|_* + 2$. By Definition 2.2.3(b), for a constant $C > 0$ and any a_1, a_2, a_3, a_4, we have $|\mathbb{E}[W_{\mathbf{i}_{1:4}}^{(a_1,a_2,a_3,a_4)}]| \leq C/n^{|\mathcal{E}_{G(\mathbf{i}_{1:4})}|_*} \leq C/n^{|\mathcal{V}_{G(\mathbf{i}_{1:4})}|-2}$. Therefore,

$$\frac{1}{n^4} \left| \sum_{\mathbf{i}_{1:4} \in \mathcal{I}_2} q_{\mathbf{i}_{1:4}} \mathbb{E}[\mathbf{W}_{\mathbf{i}_{1:4}}^{(a_1,a_2,a_3,a_4)}] \right| \leq \frac{1}{n^4} \sum_{v=1}^{4|\mathcal{V}|} \frac{C}{n^{v-2}} \sum_{\mathbf{i}_{1:4} \in \mathcal{I}_2 : |\mathcal{V}_{G(\mathbf{i}_{1:4})}| = v} |q_{\mathbf{i}_{1:4}}|.$$

Stratifying the inner summation over $\{\mathbf{i}_{1:4} \in \mathcal{I}_2 : |\mathcal{V}_{G(\mathbf{i}_{1:4})}| = v\}$ by its induced partition $\pi(\mathbf{i}_{1:4})$ of the $4|\mathcal{V}|$ total indices (having exactly v blocks), and applying the same argument as in (2.3.6), this inner summation may be bounded as $\sum_{\mathbf{i}_{1:4} \in \mathcal{I}_2 : |\mathcal{V}_{G(\mathbf{i}_{1:4})}| = v} |q_{\mathbf{i}_{1:4}}| \leq C n^v$ for a constant $C > 0$. Applying this bound for each term of A_2, we obtain $|A_2| \le C/n^2$.

Combining the analyses of A_2, A_3, and A_4, we get $\mathbb{E}[(\mathrm{val}(\mathbf{W})-\mathbb{E}[\mathrm{val}(\mathbf{W})])^4] \le C/n^2$. Then by Markov's inequality, for any $\varepsilon > 0$, $\mathbb{P}[|\,\mathrm{val}(\mathbf{W}) - \mathbb{E}[\mathrm{val}(\mathbf{W})]| > \varepsilon] \le C/(\varepsilon^4 n^2)$. This bound is summable over all $n \ge 1$, so almost-sure convergence follows by the Borel-Cantelli lemma. ■

Combining Lemmas 2.3.4 and 2.3.5 concludes the proof of Lemma 2.2.13.

2.3.2 Universality for symmetric generalized invariant matrices

In this section, we prove Lemma 2.2.14 on the universality of tensor network values for generalized invariant matrices.

Fix the tensor network $T = (\mathcal{V}, \mathcal{E}, \{q_v\}_{v\in\mathcal{V}})$. Expanding the product $\mathbf{W} = \mathbf{\Pi}\mathbf{M}\mathbf{\Pi}^\top$, the tensor network value is given by

$$\mathrm{val}_T(\mathbf{W}; \mathbf{x}_{1:k}) = \frac{1}{n} \sum_{\mathbf{i}\in[n]^{\mathcal{V}}} \sum_{\mathbf{j},\mathbf{l}\in[n]^{\mathcal{E}}} \prod_{v\in\mathcal{V}} q_v(x_{1:k}[i_v]) \prod_{e=(u,v)\in\mathcal{E}} \Pi[i_u, j_e] M[j_e, l_e] \Pi[i_v, l_e].$$

The matrix $\mathbf{\Pi}$ may be written as $\mathbf{\Pi} = \mathbf{\Xi}\mathbf{P}$ where $\mathbf{\Xi}$ is a random sign matrix and $\mathbf{P}$ is a random permutation matrix independent of $\mathbf{\Xi}$. Let σ denote the permutation of $[n]$ for which $P[i, \sigma(i)] = 1$ for all $i \in [n]$. Then $\Pi[i_u, j_e]$ is non-zero if and only if $j_e = \sigma(i_u)$. Therefore, the tensor network value is equivalently expressed as

$$\mathrm{val}_T(\mathbf{W}; \mathbf{x}_{1:k}) = \frac{1}{n} \sum_{\mathbf{i}\in[n]^{\mathcal{V}}} \prod_{v\in\mathcal{V}} q_v(x_{1:k}[i_v]) \prod_{e=(u,v)\in\mathcal{E}} \Xi[i_u] \cdot \Xi[i_v] \cdot M[\sigma(i_u), \sigma(i_v)]. \tag{2.3.16}$$

Let $\mathcal{P}$ be the set of partitions of $\mathcal{V}$. For each $\pi \in \mathcal{P}$, let $G_\pi = (\mathcal{K}_\pi, \mathcal{F}_\pi)$ be the image of $(\mathcal{V}, \mathcal{E})$ under π, in the sense of Definition 2.3.1. For each $\mathbf{i} \in [n]^{\mathcal{V}}$, let $\pi(\mathbf{i}) \in \mathcal{P}$ be the partition induced by $\mathbf{i}$. Stratifying the summation over $\mathbf{i} \in [n]^{\mathcal{V}}$ by its induced partition $\pi(\mathbf{i})$,

$$\mathrm{val}_T(\mathbf{W};\mathbf{x}_{1:k}) = \sum_{\pi\in\mathcal{P}} \frac{1}{n} \sum_{\mathbf{i}\in[n]^{\mathcal{V}}:\pi(\mathbf{i})=\pi} \prod_{v\in\mathcal{V}} q_v(x_{1:k}[i_v]) \prod_{e=(u,v)\in\mathcal{E}} \Xi[i_u]\cdot\Xi[i_v]\cdot M[\sigma(i_u),\sigma(i_v)].$$

Then, defining $Q_R = \prod_{u\in R} q_u$ for the blocks $R \in \mathcal{K}_\pi \equiv \pi$, and identifying the sum over $\{\mathbf{i}\in[n]^{\mathcal{V}} : \pi(\mathbf{i}) = \pi\}$ as a sum over one distinct index for each block $R \in \mathcal{K}_\pi$, we have

$$\mathrm{val}_T(\mathbf{W};\mathbf{x}_{1:k}) = \sum_{\pi\in\mathcal{P}} \frac{1}{n} \sideset{}{^*}\sum_{\mathbf{i}\in[n]^{\mathcal{K}_\pi}} \prod_{R\in\mathcal{K}_\pi} Q_R(x_{1:k}[i_R]) \prod_{(R,S)\in\mathcal{F}_\pi} \Xi[i_R]\cdot\Xi[i_S]\cdot M[\sigma(i_R),\sigma(i_S)]$$

where we recall the notation that $\overset{*}{\sum}$ restricts the summation to index tuples $\mathbf{i}$ having all indices distinct. For every $R \in \mathcal{K}_\pi$, let $\deg_{\mathrm{ext}}(R)$ be its *external degree* in G_π, i.e. the total number of edges of $\mathcal{F}_\pi$ containing R (counting multiplicity) that are not self-loops. Then for every $R \in \mathcal{K}_\pi$, the number of times the factor $\Xi[i_R]$ appears in the above product is exactly $\deg_{\mathrm{ext}}(R)$ plus twice the number of self-loops on R. Since $\Xi[i_R]^2 = 1$, this implies

$$\mathrm{val}_T(\mathbf{W};\mathbf{x}_{1:k}) = \sum_{\pi\in\mathcal{P}} \frac{1}{n} \sideset{}{^*}\sum_{\mathbf{i}\in[n]^{\mathcal{K}_\pi}} \prod_{R\in\mathcal{K}_\pi} Q_R(x_{1:k}[i_R])\Xi[i_R]^{\deg_{\mathrm{ext}}(R)} \prod_{(R,S)\in\mathcal{F}_\pi} M[\sigma(i_R),\sigma(i_S)]. \tag{2.3.17}$$

Lemma 2.3.6. *Let $\mathbb{E}$ be the expectation over $\mathbf{\Pi}$ conditional on $\mathbf{M}$ and $\mathbf{x}_1,\ldots,\mathbf{x}_k$. Then Lemma 2.2.14 holds for $\mathbb{E}[\mathrm{val}_T(\mathbf{W};\mathbf{x}_1,\ldots,\mathbf{x}_k)]$ in place of $\mathrm{val}_T(\mathbf{W};\mathbf{x}_1,\ldots,\mathbf{x}_k)$.*

Proof. Taking expectations in (2.3.17) with respect to the independent signs $\mathbf{\Xi}$ and permutation σ, observe that

- If $R \in \mathcal{K}_\pi$ is such that $\deg_{\mathrm{ext}}(R)$ is odd, then $\mathbb{E}[\Xi[i_R]^{\deg_{\mathrm{ext}}(R)}] = 0$. Thus by independence of the diagonal entries of $\mathbf{\Xi}$ and distinctness of the indices of $\mathbf{i}$,

$$\mathbb{E}\Bigg[\prod_{R\in\mathcal{K}_\pi} \Xi[i_R]^{\deg_{\mathrm{ext}}(R)}\Bigg] = \mathbf{1}\{\deg_{\mathrm{ext}}(R) \text{ is even for all } R\in\mathcal{K}_\pi\}. \tag{2.3.18}$$

- Since σ is a uniformly random permutation on $[n]$, for any fixed tuple $\mathbf{i} \in [n]^{\mathcal{K}_\pi}$ with all entries distinct,

$$\mathbb{E}\Bigg[\prod_{(R,S)\in\mathcal{F}_\pi} M[\sigma(i_R),\sigma(i_S)]\Bigg] = \frac{(n-|\mathcal{K}_\pi|)!}{n!} \sum_{\mathbf{j}\in[n]^{\mathcal{K}_\pi}}^{*} \prod_{(R,S)\in\mathcal{F}_\pi} M[j_R,j_S] \quad (2.3.19)$$

  where $n!/(n-|\mathcal{K}_\pi|)!$ counts the total number of tuples $\mathbf{j} \in [n]^{\mathcal{K}_\pi}$ having distinct entries, and the right side represents a uniform average over such tuples $\mathbf{j}$.

Let us call a partition $\pi \in \mathcal{P}$ *even* if every vertex of $\mathcal{K}_\pi$ has even external degree. Then applying the above observations to take the expectation in (2.3.17), we obtain

$$\mathbb{E}[\mathrm{val}_T(\mathbf{W};\mathbf{x}_{1:k})] = \sum_{\text{even } \pi\in\mathcal{P}} B_n(\pi)\cdot Q_n(\pi)\cdot M_n(\pi) \quad (2.3.20)$$

where we set

$$B_n(\pi) = n^{|\mathcal{K}_\pi|}\cdot\frac{(n-|\mathcal{K}_\pi|)!}{n!}, \quad (2.3.21)$$

$$Q_n(\pi) = \frac{1}{n^{|\mathcal{K}_\pi|}} \sum_{\mathbf{i}\in[n]^{\mathcal{K}_\pi}}^{*} \prod_{R\in\mathcal{K}_\pi} Q_R(x_{1:k}[i_R]), \quad (2.3.22)$$

$$M_n(\pi) = \frac{1}{n} \sum_{\mathbf{j}\in[n]^{\mathcal{K}_\pi}}^{*} \prod_{(R,S)\in\mathcal{F}_\pi} M[j_R,j_S]. \quad (2.3.23)$$

It is clear that $\lim_{n\to\infty} B_n(\pi) = 1$ for every fixed π. For $Q_n(\pi)$, we may apply Lemma 2.3.3 with the identifications $\mathcal{S} \leftrightarrow \mathcal{K}_\pi$ and $\{q_s : s\in\mathcal{S}\} \leftrightarrow \{Q_R : R\in\mathcal{K}_\pi\}$. Then

$$\lim_{n\to\infty} Q_n(\pi) = \prod_{R\in\mathcal{K}_\pi} \mathbb{E}[Q_R(X_{1:k})].$$

For $M_n(\pi)$, since the original tensor network is connected, the graph $G_\pi = (\mathcal{K}_\pi, \mathcal{F}_\pi)$ corresponding to each partition π must also be connected. Consequently, applying Lemma 2.3.7 below (with $p_e(\mathbf{M}) = \mathbf{M}$ for every edge e), there exists a deterministic limit value $M(G_\pi, \mathcal{D}_{\mathrm{diag}})$ depending only on G_π and $\mathcal{D}_{\mathrm{diag}}$ such that, almost surely,

$\lim_{n\to\infty} M_n(\pi) = M(G_\pi, \mathcal{D}_{\mathrm{diag}})$. Applying these statements to every π in (2.3.20), we obtain

$$\lim_{n\to\infty} \mathbb{E}[\mathrm{val}_T(\mathbf{W};\mathbf{x}_{1:k})] = \sum_{\text{even } \pi\in\mathcal{P}} \left(\prod_{R\in\mathcal{K}_\pi} \mathbb{E}[Q_R(X_{1:k})]\right) M(G_\pi, \mathcal{D}_{\mathrm{diag}}) =: \text{lim-val}_T(X_{1:k}, \mathcal{D}_{\mathrm{diag}})$$

This limit value depends only on T, the joint law of $(X_1, \ldots, X_k)$, and the limit diagonal distribution $\mathcal{D}_{\mathrm{diag}}$, and does not depend on the specific matrix $\mathbf{M}$, concluding the proof. ■

Lemma 2.3.7. *Let $\mathbf{M} \in \mathbb{R}^{n\times n}$ be a deterministic symmetric matrix with limit diagonal distribution $\mathcal{D}_{\mathrm{diag}}$, satisfying the following condition: For any fixed $\varepsilon > 0$, any diagonal monomial $p(\mathbf{x}) \in \Delta\langle\mathbf{x}\rangle$, and all large n,*

$$\max_{i\neq j} |p(\mathbf{M})[i,j]| < n^{-1/2+\varepsilon}.$$

Let $G = (\mathcal{K}, \mathcal{F})$ be a connected multi-graph such that the external degree $\deg_{\mathrm{ext}}(R)$ is even for every vertex $R \in \mathcal{K}$. For every edge $e \in \mathcal{F}$, let $p_e(\mathbf{x})$ be a diagonal monomial labeling this edge. Then there exists a value $M(G, \mathcal{D}_{\mathrm{diag}})$ depending only on G and $\mathcal{D}_{\mathrm{diag}}$ such that

$$\lim_{n\to\infty} \frac{1}{n} \sum_{\mathbf{j}\in[n]^{\mathcal{K}}}^{*} \prod_{e=(R,R')\in\mathcal{F}} p_e(\mathbf{M})[j_R, j_{R'}] = M(G, \mathcal{D}_{\mathrm{diag}}).$$

Proof. For convenience, we denote

$$M_n(G) := \frac{1}{n} \sum_{\mathbf{j}\in[n]^{\mathcal{K}}}^{*} \prod_{e=(R,R')\in\mathcal{F}} p_e(\mathbf{M})[j_R, j_{R'}].$$

We proceed by induction over the number of vertices $|\mathcal{K}|$. For the base case $|\mathcal{K}| = 1$, all edges of $\mathcal{F}$ must be self-loops, and we have

$$M_n(G) = \frac{1}{n}\sum_{j=1}^{n}\prod_{e\in\mathcal{F}} p_e(\mathbf{M})[j,j] = \frac{1}{n}\operatorname{Tr}\prod_{e\in\mathcal{F}}\Delta(p_e(\mathbf{M}))$$

Here $\prod_{e\in\mathcal{F}}\Delta(p_e(\mathbf{x}))$ is a diagonal monomial. Then, since $\mathbf{M}$ has a limit diagonal distribution, the above quantity admits a limit value as $n\to\infty$.

Next, supposing that the result is true for every multi-graph $G=(\mathcal{K},\mathcal{F})$ with $|\mathcal{K}|\le K$, we prove the result for $|\mathcal{K}|=K+1$. Define $\mathcal{K}_* := \{R\in\mathcal{K} : \deg_{\text{ext}}(R)=2\}$.

First, consider the case where $|\mathcal{K}_*|=0$. Then

- Since every $\deg_{\text{ext}}(R)$ is even, we must have $\deg_{\text{ext}}(R)\ge 4$ for all $R\in\mathcal{K}$. Therefore, denoting by $\mathcal{F}_{\text{ext}}\subseteq\mathcal{F}$ those edges that are not self-loops, we have $4|\mathcal{K}|\le 2|\mathcal{F}_{\text{ext}}|$.

- We may assume without loss of generality that each vertex $R\in\mathcal{K}$ has exactly one self-loop: For R without a self-loop, we may add the self-loop $e=(R,R)$ with the identity label $p_e(\mathbf{M})=\mathrm{Id}$. For R with multiple self-loops $\{e\in\mathcal{F} : e=(R,R)\}$, we may replace these by a single self-loop $e'=(R,R)$ having label $p_{e'}(\mathbf{M})=\prod_{e\in\mathcal{F}:e=(R,R)}\Delta(p_e(\mathbf{M}))$. These operations do not change the value of $M_n(G)$.

We denote by e_R the unique self-loop on each vertex $R\in\mathcal{K}$. Then it follows that

$$\begin{aligned}
|M_n(G)| &\le \frac{1}{n}\sum_{\mathbf{j}\in[n]^{\mathcal{K}}}^{*}\prod_{e=(R,R')\in\mathcal{F}}|p_e(\mathbf{M})[j_R,j_{R'}]|\\
&\le \frac{1}{n}\sum_{\mathbf{j}\in[n]^{\mathcal{K}}}\prod_{R\in\mathcal{K}}|p_{e_R}(\mathbf{M})[j_R,j_R]|\cdot\prod_{e\in\mathcal{F}_{\text{ext}}}\max_{i\neq j}|p_e(\mathbf{M})[i,j]|\\
&\le \frac{1}{n}\cdot n^{(-1/2+\varepsilon)|\mathcal{F}_{\text{ext}}|}\cdot\sum_{\mathbf{j}\in[n]^{\mathcal{K}}}\prod_{R\in\mathcal{K}}|p_{e_R}(\mathbf{M})[j_R,j_R]|\\
&\le n^{-1+\varepsilon|\mathcal{F}_{\text{ext}}|}\prod_{R\in\mathcal{K}}\left(\frac{1}{n}\sum_{j=1}^{n}|p_{e_R}(\mathbf{M})[j,j]|\right).
\end{aligned}$$

Here, the second inequality uses the constraint that indices $j_R, j_{R'}$ are distinct if $R \neq R'$, the third inequality holds for any fixed $\varepsilon > 0$ and all large n by the given assumption on $\mathbf{M}$, and the last inequality applies $n^{-|\mathcal{F}_{\text{ext}}|/2} \leq n^{-|\mathcal{K}|}$ as follows from the above bound $4|\mathcal{K}| \leq 2|\mathcal{F}_{\text{ext}}|$. By Cauchy-Schwarz, we have

$$\left(\frac{1}{n}\sum_{j=1}^{n} |p_{e_R}(\mathbf{M})[j,j]|\right)^2 \leq \frac{1}{n}\sum_{j=1}^{n} p_{e_R}(\mathbf{M})[j,j]^2 = \frac{1}{n}\operatorname{Tr}\Delta(p_{e_R}(\mathbf{M}))\Delta(p_{e_R}(\mathbf{M}))$$

where $\Delta(p_{e_R}(\mathbf{x}))\Delta(p_{e_R}(\mathbf{x}))$ is a diagonal monomial. Then this quantity has a limit value as $n \to \infty$, for each $R \in \mathcal{K}$. Choosing $\varepsilon < 1/|\mathcal{F}_{\text{ext}}|$, we conclude that $M_n(G) \to 0$.

Next, consider the case where $|\mathcal{K}_*| > 0$. We pick an arbitrary vertex $R_* \in \mathcal{K}_*$, and let $R_1, R_2 \in \mathcal{K}$ be its two neighbors (where $R_1 \neq R_*$ and $R_2 \neq R_*$, but possibly $R_1 = R_2$). Denote $e_1 = (R_*, R_1)$, $e_2 = (R_*, R_2)$, and assume without loss of generality as above that R_* has a unique self-loop $e_* = (R_*, R_*)$. Then

$$\begin{aligned}
M_n(G) = \frac{1}{n} \sum^{*}_{\mathbf{j}\in[n]^{\mathcal{K}\setminus R_*}} &\prod_{e=(R,R')\in\mathcal{F}\setminus\{e_1,e_2\}} p_e(\mathbf{M})[j_R, j_{R'}] \\
&\times \sum_{\substack{j_{R_*}=1 \\ j_{R_*}\notin\{j_S : S\in\mathcal{K}\setminus R_*\}}}^{n} p_{e_1}(\mathbf{M})[j_{R_*}, j_{R_1}] p_{e_2}(\mathbf{M})[j_{R_*}, j_{R_2}] p_{e_*}(\mathbf{M})[j_{R_*}, j_{R_*}] \\
&:= \mathrm{I} - \sum_{S\in\mathcal{K}\setminus R_*} \mathrm{II}(S)
\end{aligned}$$

where we set

$$\begin{aligned}
\mathrm{I} &= \frac{1}{n}\sum^{*}_{\mathbf{j}\in[n]^{\mathcal{K}\setminus R_*}} \left(\prod_{e=(R,R')\in\mathcal{F}\setminus\{e_1,e_2\}} p_e(\mathbf{M})[j_R, j_{R'}]\right) \cdot \Big(p_{e_1}\Delta(p_{e_*})p_{e_2}\Big)(\mathbf{M})[j_{R_1}, j_{R_2}], \\
\mathrm{II}(S) &= \frac{1}{n}\sum^{*}_{\mathbf{j}\in[n]^{\mathcal{K}\setminus R_*}} \left(\prod_{e=(R,R')\in\mathcal{F}\setminus\{e_1,e_2\}} p_e(\mathbf{M})[j_R, j_{R'}]\right) \\
&\qquad\qquad \times p_{e_1}(\mathbf{M})[j_S, j_{R_1}] p_{e_2}(\mathbf{M})[j_S, j_{R_2}] p_{e_*}(\mathbf{M})[j_S, j_S].
\end{aligned}$$

Here I corresponds to the full summation over $j_{R_*} \in [n]$ without restriction, and each term $-\mathrm{II}(S)$ removes the contribution from the case $j_{R_*} = j_S$.

The term I is exactly equal to $M_n(G')$ for a multi-graph G' obtained from G by removing vertex R_* and the edges e_1, e_2, adding a new edge between R_1 and R_2 with label $p_{e_1}(\mathbf{x})\Delta(p_{e_*}(\mathbf{x}))p_{e_2}(\mathbf{x})$. This graph G' is connected and has one fewer vertex than G. Each remaining vertex in G' has the same external degree as in G if $R_1 \neq R_2$, and if $R_1 = R_2$ then the external degree of $R_1 = R_2$ is reduced by 2. In both cases, all external degrees in G' remain even. Then applying the inductive hypothesis to G', $\lim_{n\to\infty} \mathrm{I}$ exists and depends only on $(G', \mathcal{D}_{\mathrm{diag}})$.

Each term $\mathrm{II}(S)$ is exactly equal to $M_n(G')$ for a multi-graph G' that merges the vertices S and R_* of G into a single vertex S_* in G', and preserves all edges and their labels. The new vertex S_* in G' has external degree equal to $\deg_{\mathrm{ext}}(S) + \deg_{\mathrm{ext}}(R_*) - 2|\{e \in \mathcal{F} : e = (S, R_*)\}|$, which is even. It is clear that G' remains connected, and the external degrees of all other vertices of G' remain the same as in G. Then applying the inductive hypothesis to G', also $\lim_{n\to\infty} \mathrm{II}(S)$ exists and depends only on $(G', \mathcal{D}_{\mathrm{diag}})$, completing the induction. ■

Remark 2.3.8. In the language of [Mal20], our proof of Lemma 2.3.6 shows that if $\mathbf{W}$ is invariant in law under conjugation by permutations, then the expected tensor network value has a limit if $\mathbf{W}$ converges in traffic distribution, and this value is universal across matrices having the same limiting traffic distribution. Our arguments of Lemmas 2.3.6 and 2.3.7 further establish that if $\mathbf{W}$ is also invariant under conjugation by random signs and satisfies the additional delocalization conditions of Definition 2.2.6, then it has a limit traffic distribution that is uniquely determined by its limit diagonal law.

We provide in Section 2.5 an alternative computation of $\text{lim-val}_T(X_1, \ldots, X_k, \mathcal{D}_{\mathrm{diag}})$ for the special case where $\mathbf{W}$ is orthogonally invariant in law, using the orthogonal Weingarten calculus [CŚ06]. We establish the asymptotic freeness statement of Proposition 2.2.16(b) also in Section 2.5 via this computation.

Finally, we conclude the proof of Lemma 2.2.14 by showing concentration of the tensor network value.

Lemma 2.3.9. *Let $\mathbb{E}$ be the expectation over $\mathbf{\Pi}$ conditional on $\mathbf{M}$ and $\mathbf{x}_1, \ldots, \mathbf{x}_k$. Under the setting of Lemma 2.2.14, almost surely as $n \to \infty$,*

$$\mathrm{val}_T(\mathbf{W}; \mathbf{x}_1, \ldots, \mathbf{x}_k) - \mathbb{E}[\mathrm{val}_T(\mathbf{W}; \mathbf{x}_1, \ldots, \mathbf{x}_k)] \to 0.$$

Proof. Let us write as shorthand $\mathrm{val}(\mathbf{W}) = \mathrm{val}_T(\mathbf{W}; \mathbf{x}_{1:k})$. By Jensen's inequality,

$$\mathbb{E}[(\mathrm{val}(\mathbf{W}) - \mathbb{E}\,\mathrm{val}(\mathbf{W}))^4] \leq \mathbb{E}[(\mathrm{val}(\mathbf{W}) - \mathrm{val}(\bar{\mathbf{W}}))^4]$$

where $\bar{\mathbf{W}} = \bar{\mathbf{\Pi}}\mathbf{M}\bar{\mathbf{\Pi}}^\top$, $\bar{\mathbf{\Pi}}$ is an independent copy of $\mathbf{\Pi}$, and the expectation on the right side is over $(\mathbf{\Pi}, \bar{\mathbf{\Pi}})$. We proceed to bound this expectation.

Recall the tensor network $T = (\mathcal{V}, \mathcal{E}, \{q_v\}_{v \in \mathcal{V}})$. Let $(\mathcal{V}^{(1)}, \mathcal{E}^{(1)}), \ldots, (\mathcal{V}^{(4)}, \mathcal{E}^{(4)})$ denote four copies of the tree $(\mathcal{V}, \mathcal{E})$. For any subset $A \subseteq \{1, 2, 3, 4\}$, let

$$(\mathcal{V}_A, \mathcal{E}_A) \cong \bigsqcup_{a \in A} (\mathcal{V}^{(a)}, \mathcal{E}^{(a)})$$

denote the graph that is the disjoint union of those copies corresponding to $a \in A$, i.e. $(\mathcal{V}_A, \mathcal{E}_A)$ has $|A|$ connected components, each a copy of $(\mathcal{V}, \mathcal{E})$. We label each vertex $v \in \mathcal{V}^{(a)} \subseteq \mathcal{V}_A$ with the same label q_v as in the original tensor network T. We write $\bar{A} = \{1, 2, 3, 4\} \setminus A$ as the complement of A, and $\bar{\mathbf{\Xi}}$ and $\bar{\sigma}$ for the random sign matrix and random permutation corresponding to $\bar{\mathbf{\Pi}}$. Then we have, similarly to (2.3.16),

$$\begin{aligned}
&(\mathrm{val}(\mathbf{W}) - \mathrm{val}(\bar{\mathbf{W}}))^4 \\
&= \sum_{A \subseteq \{1,2,3,4\}} (-1)^{|A|} \prod_{a \in A} \mathrm{val}(\mathbf{W}) \prod_{a \in \bar{A}} \mathrm{val}(\bar{\mathbf{W}}) \\
&= \sum_{A \subseteq \{1,2,3,4\}} (-1)^{|A|} \frac{1}{n^4} \sum_{\mathbf{i} \in [n]^{\mathcal{V}_A}} \prod_{v \in \mathcal{V}_A} q_v(x_{1:k}[i_v]) \prod_{(u,v) \in \mathcal{E}_A} \Xi[i_u] \cdot \Xi[i_v] \cdot M[\sigma(i_u), \sigma(i_v)]
\end{aligned}$$

$$\times \sum_{\mathbf{j}\in[n]^{\mathcal{V}_{\bar{A}}}} \prod_{v\in\mathcal{V}_{\bar{A}}} q_v(x_{1:k}[j_v]) \prod_{(u,v)\in\mathcal{E}_{\bar{A}}} \bar{\Xi}[j_u]\cdot\bar{\Xi}[j_v]\cdot M[\bar{\sigma}(j_u),\bar{\sigma}(j_v)]. \quad (2.3.24)$$

Let $\mathcal{P}_A$ be the set of partitions of $\mathcal{V}_A$, and denote by $\pi(\mathbf{i}) \in \mathcal{P}_A$ the partition induced by $\mathbf{i} \in [n]^{\mathcal{V}_A}$. For each $\pi \in \mathcal{P}_A$, let $G_\pi = (\mathcal{K}_\pi, \mathcal{F}_\pi)$ be the image of $(\mathcal{V}_A, \mathcal{E}_A)$ under π, in the sense of Definition 2.3.1. Note that here, G_π is not necessarily connected but can consist of up to $|A| \le 4$ connected components. Define $B_n(\pi)$ and $Q_n(\pi)$ exactly as in (2.3.21–2.3.22), let $\mathcal{C}(\pi)$ denote the set of connected components of G_π, and define

$$M_n(\pi) = \frac{1}{n^{|\mathcal{C}(\pi)|}} \sum_{\mathbf{j}\in[n]^{\mathcal{K}_\pi}}^{*} \prod_{(R,S)\in\mathcal{F}_\pi} M[j_R, j_S]. \quad (2.3.25)$$

This coincides with our previous definition of (2.3.23) when $\mathcal{C}(\pi) = 1$. Define similarly $B_n(\bar{\pi}), Q_n(\bar{\pi}), M_n(\bar{\pi})$ via the graph $G_{\bar{\pi}} = (\mathcal{K}_{\bar{\pi}}, \mathcal{F}_{\bar{\pi}})$ that is the image of $(\mathcal{V}_{\bar{A}}, \mathcal{E}_{\bar{A}})$ under $\bar{\pi} \in \mathcal{P}_{\bar{A}}$. Then, stratifying the sums over $\mathbf{i}$ and $\mathbf{j}$ by $\pi(\mathbf{i}) \in \mathcal{P}_A$ and $\pi(\mathbf{j}) \in \mathcal{P}_{\bar{A}}$, and taking the expectation in (2.3.24) over $(\mathbf{\Pi}, \bar{\mathbf{\Pi}})$ using (2.3.18–2.3.19), we get analogously to (2.3.20)

$$\begin{aligned}
&\mathbb{E}[(\mathrm{val}(\mathbf{W}) - \mathrm{val}(\bar{\mathbf{W}}))^4] \\
&\quad = \sum_{A\subseteq\{1,2,3,4\}} (-1)^{|A|} \sum_{\substack{\text{even } \pi\in\mathcal{P}_A \\ \text{even } \bar{\pi}\in\mathcal{P}_{\bar{A}}}} \frac{n^{|\mathcal{C}(\pi)|+|\mathcal{C}(\bar{\pi})|}}{n^4} B_n(\pi)B_n(\bar{\pi})\cdot Q_n(\pi)Q_n(\bar{\pi})\cdot M_n(\pi)M_n(\bar{\pi}).
\end{aligned} \quad (2.3.26)$$

For $\pi \in \mathcal{P}_A$ and $\bar{\pi} \in \mathcal{P}_{\bar{A}}$, we define $\tau = \pi \oplus \bar{\pi} \in \mathcal{P}_{\{1,2,3,4\}}$ as the combined partition of all vertices in $\mathcal{V}_{\{1,2,3,4\}}$ given by taking the blocks of both π and $\bar{\pi}$. We write $G_\tau = (\mathcal{K}_\tau, \mathcal{F}_\tau)$ as the image of $(\mathcal{V}_{\{1,2,3,4\}}, \mathcal{E}_{\{1,2,3,4\}})$ under τ; this is the disjoint union of G_π and $G_{\bar{\pi}}$, so in particular

$$|\mathcal{K}_\tau| = |\mathcal{K}_\pi| + |\mathcal{K}_{\bar{\pi}}|, \qquad |\mathcal{C}(\tau)| = |\mathcal{C}(\pi)| + |\mathcal{C}(\bar{\pi})|.$$

We now proceed to approximate $B_n(\pi)B_n(\bar{\pi})$, $Q_n(\pi)Q_n(\bar{\pi})$, and $M_n(\pi)M_n(\bar{\pi})$ by quantities that depend only on τ, and not on the individual partitions $\pi, \bar{\pi}$. We write $O(n^{-\nu})$ for any error of magnitude at most C/n^ν for a constant $C := C(\pi, \bar{\pi}) > 0$ and all large n.

For B_n, observe from the definition (2.3.21) that

$$\begin{aligned} B_n(\tau) &= \frac{n}{n} \cdot \frac{n}{n-1} \cdot \frac{n}{n-2} \cdot \ldots \cdot \frac{n}{n - |\mathcal{K}_\tau| + 1} \\ &= 1 + \frac{\sum_{k=0}^{|\mathcal{K}_\tau|-1} k}{n} + O(n^{-2}) = 1 + n^{-1}\binom{|\mathcal{K}_\tau|}{2} + O(n^{-2}). \end{aligned}$$

Similarly,

$$B_n(\pi)B_n(\bar{\pi}) = 1 + n^{-1}\left(\binom{|\mathcal{K}_\pi|}{2} + \binom{|\mathcal{K}_{\bar{\pi}}|}{2}\right) + O(n^{-2}).$$

In particular,

$$B_n(\pi)B_n(\bar{\pi}) = B_n(\tau) + O(n^{-1}) = 1 + O(n^{-1}). \tag{2.3.27}$$

In the case where $G_\tau = (\mathcal{K}_\tau, \mathcal{F}_\tau)$ has 4 connected components, i.e. each block of both π and $\bar{\pi}$ is contained within a single copy $\mathcal{V}^{(a)}$ of $\mathcal{V}$, let us write $G_\tau(a) = (\mathcal{K}_\tau(a), \mathcal{F}_\tau(a))$ for the component corresponding to the partition of $\mathcal{V}^{(a)}$. Given any $\pi \in \mathcal{P}_A$, $\bar{\pi} \in \mathcal{P}_{\bar{A}}$, and $\tau = \pi \oplus \bar{\pi}$, we then have

$$\binom{|\mathcal{K}_\tau|}{2} = \binom{|\mathcal{K}_\pi|}{2} + \binom{|\mathcal{K}_{\bar{\pi}}|}{2} + \sum_{a \in A, b \notin A} |\mathcal{K}_\tau(a)| \cdot |\mathcal{K}_\tau(b)|$$

because to choose two elements of $\mathcal{K}_\tau$, we may choose them both from $\mathcal{K}_\pi$, both from $\mathcal{K}_{\bar{\pi}}$, or one from $\mathcal{K}_\tau(a) \subseteq \mathcal{K}_\pi$ and the other from $\mathcal{K}_\tau(b) \subseteq \mathcal{K}_{\bar{\pi}}$ for some $a \in A, b \in \bar{A}$. This gives a refinement of (2.3.27),

$$B_n(\pi)B_n(\bar{\pi}) = B_n(\tau) + \sum_{a \in A, b \notin A} B_n(\tau, a, b) + O(n^{-2}) \tag{2.3.28}$$

where we define $B_n(\tau, a, b) = -n^{-1}|\mathcal{K}_\tau(a)| \cdot |\mathcal{K}_\tau(b)|$. Here $B_n(\tau, a, b) = O(n^{-1})$.

For Q_n, observe from the definition (2.3.22) that the distinction between $Q_n(\pi)Q_n(\bar{\pi})$ and $Q_n(\tau)$ is that the former does not restrict indices of summation corresponding to π to be distinct from those corresponding to $\bar{\pi}$, i.e.

$$Q_n(\pi)Q_n(\bar{\pi}) = Q_n(\tau) + \frac{1}{n^{|\mathcal{K}_\tau|}} \sum^*_{\mathbf{i} \in [n]^{\mathcal{K}_\pi}} \sum^*_{\mathbf{j} \in [n]^{\mathcal{K}_{\bar{\pi}}}} \mathbf{1}\{\text{there is at least 1 pair of coinciding indices between } \mathbf{i} \text{ and } \mathbf{j}\} \times \prod_{R \in \mathcal{K}_\pi} Q_R(x_{1:k}[i_R]) \prod_{R' \in \mathcal{K}_{\bar{\pi}}} Q_{R'}(x_{1:k}[j_{R'}]).$$

By Lemma 2.3.3, the quantities $Q_n(\pi)$, $Q_n(\bar{\pi})$, and $Q_n(\tau)$ all have deterministic limit values as $n \to \infty$. Furthermore, by a simple inclusion-exclusion argument together with Lemma 2.3.3, in the above double summation the contribution from pairs $(\mathbf{i}, \mathbf{j})$ coinciding in exactly k pairs of indices is of size $O(n^{-k})$. Then in particular,

$$Q_n(\pi)Q_n(\bar{\pi}) = Q_n(\tau) + O(n^{-1}) = O(1). \tag{2.3.29}$$

In the case where G_τ has 4 connected components, let us write more explicitly

$$Q_n(\pi)Q_n(\bar{\pi}) = Q_n(\tau) + \frac{1}{n^{|\mathcal{K}_\tau|}} \sum^*_{\mathbf{i} \in [n]^{\mathcal{K}_\pi}} \sum^*_{\mathbf{j} \in [n]^{\mathcal{K}_{\bar{\pi}}}} \mathbf{1}\{\text{there is exactly 1 pair of coinciding indices between } \mathbf{i} \text{ and } \mathbf{j}\} \times \prod_{R \in \mathcal{K}_\pi} Q_R(x_{1:k}[i_R]) \prod_{R' \in \mathcal{K}_{\bar{\pi}}} Q_{R'}(x_{1:k}[j_{R'}]) + O(n^{-2}).$$

We may choose the coinciding index pair by choosing 1 vertex $R \in \mathcal{K}_\tau(a) \subseteq \mathcal{K}_\pi$ for some $a \in A$, and 1 vertex $R' \in \mathcal{K}_\tau(b) \subseteq \mathcal{K}_{\bar{\pi}}$ for some $b \in \bar{A}$. Now viewing $R \in \pi$ and $R' \in \bar{\pi}$ as disjoint blocks of vertices of $\mathcal{V}$, note that if $S = R \cup R' \subseteq \mathcal{V}$ is the block obtained upon merging R, R', then by definition $Q_S = Q_R \cdot Q_{R'}$. Thus, the above is equivalent to

$$Q_n(\pi)Q_n(\bar{\pi}) = Q_n(\tau) + \sum_{a \in A, b \notin A} Q_n(\tau, a, b) + O(n^{-2}) \tag{2.3.30}$$

where we define

$$Q_n(\tau,a,b)=\frac{1}{n^{|\mathcal{K}_\tau|}}\sum_{\mathbf{j}\in[n]^{\mathcal{K}_\tau}}\mathbf{1}\{\mathbf{j}\text{ has }|\mathcal{K}_\tau|-1\text{ distinct indices, and 1 index from }$$

$$\mathcal{K}_\tau(a)\text{ coincides with 1 index from }\mathcal{K}_\tau(b)\}\times\prod_{R\in\mathcal{K}_\tau}Q_R(x_{1:k}[j_R]).$$

By the preceding arguments, $Q_n(\tau,a,b)=O(n^{-1})$.

For M_n, consider any $A\subseteq\{1,2,3,4\}$ and $\pi\in\mathcal{P}_A$. Recall that $\mathcal{C}(\pi)$ is the set of connected components of G_π. Each component in $\mathcal{C}(\pi)$ takes the form $G_\sigma=(\mathcal{K}_\sigma,\mathcal{F}_\sigma)$ where σ is a partition that contains a subset of the blocks of π. Let us write $\sum^{**}_{\mathbf{j}\in[n]^{\mathcal{K}_\pi}}$ for the summation over tuples $\mathbf{j}$ such that indices corresponding to each component $\mathcal{K}_\sigma\subseteq\mathcal{K}_\pi$ are distinct, but they are not necessarily distinct across different components. Recalling (2.3.25), define

$$M_n^{**}(\pi):=\prod_{G_\sigma\in\mathcal{C}(\pi)}M_n(\sigma)=\frac{1}{n^{|\mathcal{C}(\pi)|}}\sum^{**}_{\mathbf{j}\in[n]^{\mathcal{K}_\pi}}\prod_{(R,S)\in\mathcal{F}_\pi}M[j_R,j_S]$$

where $M_n^{**}(\pi)$ is now a multiplicative function over connected components of π. Since each G_σ is connected, Lemma 2.3.7(a) implies the existence of the limit

$$M_n^{**}(\pi)\to\prod_{G_\sigma\in\mathcal{C}(\pi)}M(G_\sigma,\mathcal{D}_{\text{diag}}).\tag{2.3.31}$$

Comparing the definitions of $M_n^{**}(\pi)$ and $M_n(\pi)$, we have $M_n^{**}(\pi)=M_n(\pi)$ if G_π has a single connected component $|\mathcal{C}(\pi)|=1$, and more generally

$$M_n(\pi)=M_n^{**}(\pi)-\frac{1}{n^{|\mathcal{C}(\pi)|}}\sum^{**}_{\mathbf{j}\in[n]^{\mathcal{K}_\pi}}\mathbf{1}\{\text{some indices of }\mathbf{j}\text{ for different connected}$$

$$\text{components of }G_\pi\text{ coincide}\}\times\prod_{(R,S)\in\mathcal{F}_\pi}M[j_R,j_S].$$

For each $\mathbf{j}$ where this summand is non-zero, define $\pi'(\mathbf{j})\in\mathcal{P}_A$ as the partition that merges those blocks of π where the corresponding indices of $\mathbf{j}$ coincide. Let $\mathcal{P}(\pi)$ be the set of all possible such partitions $\pi'(\mathbf{j})$. (If $|\mathcal{C}(\pi)| = 1$, then $\mathcal{P}(\pi) = \emptyset$.) Then, stratifying the summation over $\mathbf{j}$ by $\pi'(\mathbf{j}) \in \mathcal{P}(\pi)$, letting $G_{\pi'} = (\mathcal{K}_{\pi'}, \mathcal{F}_{\pi'})$ be the image of $(\mathcal{V}_A, \mathcal{E}_A)$ under π', and identifying the sum over $\{\mathbf{j} : \pi'(\mathbf{j}) = \pi'\}$ as a sum over one distinct index for each $R \in \mathcal{K}_{\pi'}$,

$$\begin{aligned} M_n(\pi) &= M_n^{**}(\pi) - \frac{1}{n^{|\mathcal{C}(\pi)|}} \sum_{\pi' \in \mathcal{P}(\pi)} \sum_{\mathbf{j} \in [n]^{\mathcal{K}_{\pi'}}}^{*} \prod_{(R,S) \in \mathcal{F}_{\pi'}} M[j_R, j_S] \\ &= M_n^{**}(\pi) - \sum_{\pi' \in \mathcal{P}(\pi)} \frac{1}{n^{|\mathcal{C}(\pi)| - |\mathcal{C}(\pi')|}} M_n(\pi'). \end{aligned} \tag{2.3.32}$$

For any $\pi' \in \mathcal{P}(\pi)$, its number of connected components satisfies $|\mathcal{C}(\pi')| \leq |\mathcal{C}(\pi)| - 1$. In particular, if $|\mathcal{C}(\pi)| = 2$, then $|\mathcal{C}(\pi')| = 1$ for all $\pi' \in \mathcal{P}(\pi)$, so $M_n(\pi') = M_n^{**}(\pi')$ on the right side of (2.3.32). If $|\mathcal{C}(\pi)| \geq 3$, then we may apply this identity (2.3.32) recursively to further approximate $M_n(\pi')$ on the right side of (2.3.32) by $M_n^{**}(\pi')$, until only instances of M_n^{**} and no instances of M_n remain. Applying (2.3.31) to each instance of M_n^{**} in this final expression, this shows that

$$M_n(\pi) = M_n^{**}(\pi) + O(n^{-1}) = O(1).$$

Applying this for $\pi \in \mathcal{P}_A$, $\bar{\pi} \in \mathcal{P}_{\bar{A}}$, and $\tau = \pi \oplus \bar{\pi}$, and recalling that M_n^{**} is multiplicative across connected components so that $M_n^{**}(\tau) = M_n^{**}(\pi) M_n^{**}(\bar{\pi})$, this yields

$$M_n(\pi) M_n(\bar{\pi}) = M_n(\tau) + O(n^{-1}) = O(1). \tag{2.3.33}$$

When G_τ has 4 connected components, let us derive a more explicit expression for this $O(n^{-1})$ error. Applying (2.3.32) and the above arguments to τ, we have

$$M_n(\tau) = M_n^{**}(\tau) - \frac{1}{n} \sum_{\tau' \in \mathcal{P}(\tau) : |\mathcal{C}(\tau')| = |\mathcal{C}(\tau)| - 1} M_n^{**}(\tau') + O(n^{-2}).$$

If $\tau' \in \mathcal{P}(\tau)$ and $|\mathcal{C}(\tau')| = |\mathcal{C}(\tau)| - 1$, then τ' is obtained by picking exactly two connected components of G_τ, say $G_\tau(a) = (\mathcal{K}_\tau(a), \mathcal{F}_\tau(a))$ and $G_\tau(b) = (\mathcal{K}_\tau(b), \mathcal{F}_\tau(b))$, and merging one or more pairs of blocks $R \in \mathcal{K}_\tau(a)$ with $R' \in \mathcal{K}_\tau(b)$. We write the set of such partitions $\tau' \in \mathcal{P}(\tau)$ corresponding to the two fixed indices $a \neq b$ as $\mathcal{P}(\tau, a, b)$. Then

$$M_n(\tau) = M_n^{**}(\tau) - \frac{1}{n} \sum_{1 \leq a < b \leq 4} \sum_{\tau' \in \mathcal{P}(\tau,a,b)} M_n^{**}(\tau') + O(n^{-2}).$$

Similarly

$$M_n(\pi) = M_n^{**}(\pi) - \frac{1}{n} \sum_{\substack{a<b \\ a,b \in A}} \sum_{\pi' \in \mathcal{P}(\pi,a,b)} M_n^{**}(\pi') + O(n^{-2}),$$

$$M_n(\bar{\pi}) = M_n^{**}(\bar{\pi}) - \frac{1}{n} \sum_{\substack{a<b \\ a,b \in \bar{A}}} \sum_{\bar{\pi}' \in \mathcal{P}(\bar{\pi},a,b)} M_n^{**}(\bar{\pi}') + O(n^{-2}).$$

Taking the product of these two expressions and applying multiplicativity of M_n^{**}, we deduce

$$M_n(\pi)M_n(\bar{\pi}) = M_n(\tau) + \sum_{a \in A, b \in \bar{A}} M_n(\tau, a, b) + O(n^{-2}) \tag{2.3.34}$$

where we define $M_n(\tau, a, b) = \frac{1}{n} \sum_{\tau' \in \mathcal{P}(\tau,a,b)} M_n^{**}(\tau')$. Here again, $M_n(\tau, a, b) = O(n^{-1})$.

Equipped with these approximations, we now bound (2.3.26). Given $\tau \in \mathcal{P}_{\{1,2,3,4\}}$, let $\mathcal{A}(\tau)$ be the set of $A \subseteq \{1, 2, 3, 4\}$ for which $\tau = \pi \oplus \bar{\pi}$ for some $\pi \in \mathcal{P}_A$ and $\bar{\pi} \in \mathcal{P}_{\bar{A}}$, i.e. $A \in \mathcal{A}(\tau)$ if and only if each connected component of G_τ corresponds to vertices belonging entirely to $\mathcal{V}_A$ or entirely to $\mathcal{V}_{\bar{A}}$. Note that given $\tau = \pi \oplus \bar{\pi}$ and $A \in \mathcal{A}(\tau)$, this uniquely determines $\pi \in \mathcal{P}_A$ and $\bar{\pi} \in \mathcal{P}_{\bar{A}}$. Then, stratifying the summation in (2.3.26) by the number of connected components $|\mathcal{C}(\tau)| = |\mathcal{C}(\pi)| + |\mathcal{C}(\bar{\pi})|$, we have

$$\mathbb{E}[(\mathrm{val}(\mathbf{W}) - \mathrm{val}(\bar{\mathbf{W}}))^4] = E_1 + E_2 + E_3 + E_4$$

where

$$E_j = \frac{1}{n^{4-j}} \sum_{\substack{\text{even } \tau \in \mathcal{P}_{\{1,2,3,4\}} \\ |\mathcal{C}(\tau)|=j}} \sum_{A \in \mathcal{A}(\tau)} (-1)^{|A|} B_n(\pi)B_n(\bar{\pi}) \cdot Q_n(\pi)Q_n(\bar{\pi}) \cdot M_n(\pi)M_n(\bar{\pi}).$$

Here, π and $\bar{\pi}$ on the right side denote those partitions that are uniquely determined by $\tau = \pi \oplus \bar{\pi}$ and $A \in \mathcal{A}(\tau)$; we omit their dependence on (τ, A) for brevity.

Applying the simple the bound $B_n(\pi)B_n(\bar{\pi})Q_n(\pi)Q_n(\bar{\pi})M_n(\pi)M_n(\bar{\pi}) = O(1)$ from (2.3.27), (2.3.29), and (2.3.33), we get $E_1 = O(n^{-3})$ and $E_2 = O(n^{-2})$.

For E_3, applying the approximation $B_n(\pi)B_n(\bar{\pi}) = B_n(\tau)+O(n^{-1})$, $Q_n(\pi)Q_n(\bar{\pi}) = Q_n(\tau) + O(n^{-1})$, and $M_n(\pi)M_n(\bar{\pi}) = M_n(\tau) + O(n^{-1})$ from (2.3.27), (2.3.29), and (2.3.33), we have

$$E_3 = \frac{1}{n} \sum_{\text{even } \tau \in \mathcal{P}_{\{1,2,3,4\}} : |\mathcal{C}(\tau)|=3} B_n(\tau)Q_n(\tau)M_n(\tau) \sum_{A \in \mathcal{A}(\tau)} (-1)^{|A|} + O(n^{-2}).$$

Importantly, the leading term $B_n(\tau)Q_n(\tau)M_n(\tau)$ does not depend on A, so we have factored it outside of the sum over A, and the lower order terms all contribute to the $O(n^{-2})$ error. For any τ where $|\mathcal{C}(\tau)| = 3$, we have $\sum_{A \in \mathcal{A}(\tau)} (-1)^{|A|} = 0$: For example, if the 3 connected components of G_τ correspond to vertices in $\mathcal{V}_1$, $\mathcal{V}_2$, and $\mathcal{V}_{\{3,4\}}$, then $\mathcal{A}(\tau) = \{\emptyset, \{1\}, \{2\}, \{1,2\}, \{3,4\}, \{1,3,4\}, \{2,3,4\}, \{1,2,3,4\}\}$. Thus, we get $E_3 = O(n^{-2})$.

Finally, for E_4, we apply the finer approximations (2.3.28), (2.3.30), and (2.3.34) which hold when $|\mathcal{C}(\tau)| = 4$. In this case $\mathcal{A}(\tau)$ consists of all subsets of $\{1,2,3,4\}$, so

$$\begin{aligned} E_4 = \sum_{\text{even } \tau \in \mathcal{P}_{\{1,2,3,4\}} : |\mathcal{C}(\tau)|=4} \Bigg(& B_n(\tau)Q_n(\tau)M_n(\tau) \sum_{A \subseteq \{1,2,3,4\}} (-1)^{|A|} \\ & + \sum_{a \neq b \in \{1,2,3,4\}} \Big[B_n(\tau)Q_n(\tau)M_n(\tau,a,b) + B_n(\tau)Q_n(\tau,a,b)M_n(\tau) \end{aligned}$$

$$+ B_n(\tau, a, b) Q_n(\tau) M_n(\tau)\Big] \sum_{\substack{A \subseteq \{1,2,3,4\} \\ a \in A, b \in \bar{A}}} (-1)^{|A|} \Bigg) + O(n^{-2}).$$

Importantly, we have exchanged the order of summations over A and over $(a \in A, b \in \bar{A})$, and used that each term $B_n(\tau, a, b), Q_n(\tau, a, b), M_n(\tau, a, b)$ does not depend on the assignment of the remaining indices $\{1, 2, 3, 4\} \setminus \{a, b\}$ to A and $\bar{A}$. Then, applying $\sum_{A \subseteq \{1,2,3,4\}} (-1)^{|A|} = 0$ and also $\sum_{A \subseteq \{1,2,3,4\}: a \in A, b \in \bar{A}} (-1)^{|A|} = 0$ for each fixed pair a, b, we get $E_4 = O(n^{-2})$.

Combining the above, we have $\mathbb{E}[(\mathrm{val}(\mathbf{W}) - \mathbb{E}\,\mathrm{val}(\mathbf{W}))^4] \leq C/n^2$ for a constant $C > 0$ and all large n. Then Lemma 2.3.9 follows from Markov's inequality and the Borel-Cantelli lemma. ■

Combining Lemmas 2.3.6 and 2.3.9 completes the proof of Lemma 2.2.14.

2.3.3 Universality of AMP via polynomial approximation

We now prove Lemma 2.2.12, showing that the universality of AMP for Lipschitz non-linearities[5] can be obtained from universality of tensor network values by polynomial approximation.

For the given AMP algorithm with Lipschitz non-linearities $u_{t+1}(\cdot)$, we approximate it by an auxiliary AMP algorithm with polynomial non-linearities $\tilde{u}_{t+1}(\cdot)$, where each $\tilde{u}_{t+1}$ is an L_2-approximation for u_{t+1} with respect to the state evolution of its arguments. A similar method of approximation was recently used in [DLS22]. Combining this approximation, the validity of state evolution for polynomial AMP applied to $\mathbf{G}$, and the universality of tensor network values for $\mathbf{G}$ and $\mathbf{W}$, we show that iterates of the Lipschitz and polynomial AMP algorithms applied to $\mathbf{W}$ are close in (normalized) ℓ_2 distance. This will imply the desired W_2-convergence of the AMP iterates (2.2.1) to their state evolution.

5. By this we mean that each non-linearity $u_{t+1}(\cdot)$ is Lipschitz in its first t arguments $z_1, \ldots, z_t$.

We construct the auxiliary AMP algorithm as follows: Fix any $\varepsilon > 0$. For the same initialization $\tilde{\mathbf{u}}_1 = \mathbf{u}_1$ and vectors of side information $\mathbf{f}_1, \ldots, \mathbf{f}_k$ as in the given Lipschitz AMP algorithm (2.2.1), define the iterates for $t = 1, 2, 3, \ldots$

$$\begin{aligned} \tilde{\mathbf{z}}_t &= \mathbf{W}\tilde{\mathbf{u}}_t - \sum_{s=1}^{t} \tilde{b}_{ts}\tilde{\mathbf{u}}_s \\ \tilde{\mathbf{u}}_{t+1} &= \tilde{u}_{t+1}(\tilde{\mathbf{z}}_1, \ldots, \tilde{\mathbf{z}}_t, \mathbf{f}_1, \ldots, \mathbf{f}_k) \end{aligned} \tag{2.3.35}$$

such that

1. Each coefficient $\tilde{b}_{ts}$ is defined by $\tilde{u}_2, \tilde{u}_3, \ldots, \tilde{u}_t$ and the orthogonally invariant prescription (2.2.6).

2. Let $\widetilde{\mathbf{\Sigma}}_t$ be the orthogonally invariant prescription (2.2.5), and let $(U_1, F_{1:k}, \widetilde{Z}_{1:t})$ be the state evolution where $\widetilde{Z}_{1:t} \sim \mathcal{N}(0, \widetilde{\mathbf{\Sigma}}_t)$ is independent of $(U_1, F_{1:k})$. Then each polynomial $\tilde{u}_{t+1}(\cdot)$ is chosen to satisfy

$$\mathbb{E}\Big[\Big(\tilde{u}_{t+1}(\widetilde{Z}_{1:t}, F_{1:k}) - u_{t+1}(\widetilde{Z}_{1:t}, F_{1:k})\Big)^2\Big] < \varepsilon. \tag{2.3.36}$$

3. For any fixed arguments $z_{1:(t-1)}$ and $f_{1:k}$, the function $z_t \mapsto \tilde{u}_{t+1}(z_{1:t}, f_{1:k})$ has non-linear dependence in z_t.

We write the iterates as $\tilde{\mathbf{z}}_t(\mathbf{W}), \tilde{\mathbf{u}}_t(\mathbf{W})$ if we want to make explicit that the algorithm is evaluated on the matrix $\mathbf{W}$.

We also need the following lemma to ensure the existence of the desired polynomial approximation.

Lemma 2.3.10. *Let μ_X and μ_Y be probability laws on $\mathbb{R}^m$ and $\mathbb{R}^n$ having finite moments of all orders, such that multivariate polynomials are dense in the real L^2-spaces $L^2(\mu_X)$ and $L^2(\mu_Y)$. Then multivariate polynomials are also dense in $L^2(\mu_X \times \mu_Y)$.*

Proof. Consider any measurable $A \subseteq \mathbb{R}^m$ and $B \subseteq \mathbb{R}^n$, and let $\chi_A, \chi_B, \chi_{A\times B}$ be the indicator functions of A, B, and $A \times B$. For any $\varepsilon > 0$, by the density conditions for $L^2(\mu_X)$ and $L^2(\mu_Y)$, we may take polynomials p_A, p_B such that $\|\chi_A - p_A\|_{L^2(\mu_X)} < \varepsilon/2$ and $\|\chi_B - p_B\|_{L^2(\mu_Y)} < \varepsilon/(2\|p_A\|_{L^2(\mu_X)})$. Then

$$\begin{aligned}
&\|\chi_{A\times B} - p_A p_B\|_{L^2(\mu_X\times\mu_Y)} \\
&\leq \|\chi_A - p_A\|_{L^2(\mu_X)}\|\chi_B\|_{L^2(\mu_Y)} + \|p_A\|_{L^2(\mu_X)}\|\chi_B - p_B\|_{L^2(\mu_Y)} < \varepsilon.
\end{aligned}$$

Taking $\varepsilon \to 0$ shows that polynomials are dense in the linear span of indicator functions $\{\chi_{A\times B} : \text{measurable } A \subseteq \mathbb{R}^m, B \subseteq \mathbb{R}^n\}$. This linear span is in turn dense in $L^2(\mu_X \times \mu_Y)$, showing the lemma. ■

The choice of $\tilde{u}_{t+1}$ in condition (2) above is possible by the polynomial density condition in Assumption 2.2.1, and by Lemma 2.3.10 which ensures that the same density condition holds for $(U_1, F_{1:k}, \widetilde{Z}_{1:t})$. If condition (3) does not hold for this polynomial $\tilde{u}_{t+1}$, then it must hold upon adding to $\tilde{u}_{t+1}$ a small multiple of z_t^2. The conditions of [Fan22, Assumption 4.2] are verified by Assumption 2.2.1, the condition $\text{Var}[D] > 0$ given in Lemma 2.2.12, and the above condition (3). Then [Fan22, Theorem 4.3] ensures, almost surely as $n \to \infty$,

$$(\mathbf{u}_1, \mathbf{f}_1, \ldots, \mathbf{f}_k, \tilde{\mathbf{z}}_1(\mathbf{G}), \ldots, \tilde{\mathbf{z}}_t(\mathbf{G})) \xrightarrow{W} (U_1, F_1, \ldots, F_k, \widetilde{Z}_1, \ldots, \widetilde{Z}_t). \tag{2.3.37}$$

Lemma 2.3.11. *Fix any $t \geq 1$. Let $(\tilde{\mathbf{u}}_1, \tilde{\mathbf{z}}_1, \ldots, \tilde{\mathbf{z}}_t, \tilde{\mathbf{u}}_t)$ be the iterates of any algorithm of the form (2.3.35), where $\{\tilde{b}_{ts}\}$ are scalar constants and $\tilde{u}_{t+1} : \mathbb{R}^{t+k} \to \mathbb{R}$ are polynomial functions applied row-wise. Then for any polynomial $p : \mathbb{R}^{2t+k} \to \mathbb{R}$ and for some finite set $\mathcal{F}$ of diagonal tensor networks in $k+1$ variables,*

$$\langle p(\tilde{\mathbf{u}}_1, \ldots, \tilde{\mathbf{u}}_t, \tilde{\mathbf{z}}_1, \ldots, \tilde{\mathbf{z}}_t, \mathbf{f}_1, \ldots, \mathbf{f}_k)\rangle = \sum_{T\in\mathcal{F}} \text{val}_T(\mathbf{W}; \mathbf{u}_1, \mathbf{f}_1, \ldots, \mathbf{f}_k).$$

Proof. First note that

$$\langle p(\tilde{\mathbf{u}}_{1:t}, \tilde{\mathbf{z}}_{1:t}, \mathbf{f}_{1:k})\rangle = \mathrm{val}_T(\mathbf{W}; \tilde{\mathbf{u}}_{1:t}, \tilde{\mathbf{z}}_{1:t}, \mathbf{f}_{1:k})$$

where T is a tensor network with only one vertex v whose associated polynomial is $q_v = p$.

We claim that given any tensor network $T = (\mathcal{V}, \mathcal{E}, \{q_v\}_{v\in\mathcal{V}})$ in the variables $(\tilde{u}_{1:t}, \tilde{z}_{1:t}, f_{1:k})$, we can decompose

$$\mathrm{val}_T(\mathbf{W}; \tilde{\mathbf{u}}_{1:t}, \tilde{\mathbf{z}}_{1:t}, \mathbf{f}_{1:k}) = \sum_{T'\in\mathcal{F}} \mathrm{val}_{T'}(\mathbf{W}; \tilde{\mathbf{u}}_{1:t}, \tilde{\mathbf{z}}_{1:(t-1)}, \mathbf{f}_{1:k}) \tag{2.3.38}$$

where $\mathcal{F}$ is a finite set of tensor networks in the variables $(\tilde{u}_{1:t}, \tilde{z}_{1:(t-1)}, f_{1:k})$. To show this, recall that

$$\mathrm{val}_T(\mathbf{W}; \tilde{\mathbf{u}}_{1:t}, \tilde{\mathbf{z}}_{1:t}, \mathbf{f}_{1:k}) = \frac{1}{n} \sum_{\mathbf{i}\in[n]^{\mathcal{V}}} \prod_{v\in\mathcal{V}} q_v(\tilde{u}_{1:t}[i_v], \tilde{z}_{1:t}[i_v], f_{1:k}[i_v]) W_{\mathbf{i}|T}.$$

Applying $\tilde{\mathbf{z}}_t = \mathbf{W}\tilde{\mathbf{u}}_t - \sum_{s=1}^t \tilde{b}_{ts}\tilde{\mathbf{u}}_s$ and expanding each q_v in terms of $(\tilde{\mathbf{u}}_{1:t}, \tilde{\mathbf{z}}_{1:(t-1)}, \mathbf{f}_{1:k})$ and $\mathbf{W}\tilde{\mathbf{u}}_t$, we have

$$q_v(\tilde{u}_{1:t}[i_v], \tilde{z}_{1:t}[i_v], f_{1:k}[i_v]) = \sum_{\theta=0}^{\Theta_v} q_{v,\theta}(\tilde{u}_{1:t}[i_v], \tilde{z}_{1:(t-1)}[i_v], f_{1:k}[i_v]) \cdot \left(\sum_{j=1}^n W[i_v, j]\tilde{u}_t[j]\right)^\theta$$

where Θ_v is the maximum degree of q_v in $\tilde{z}_t$, and $q_{v,0}, q_{v,1}, \ldots, q_{v,\Theta_v}$ are polynomials that depend on q_v and $\{\tilde{b}_{ts}\}$. Therefore

$$\begin{aligned}\mathrm{val}_T(\mathbf{W}; \tilde{\mathbf{u}}_{1:t}, \tilde{\mathbf{z}}_{1:t}, \mathbf{f}_{1:k}) = \frac{1}{n} \sum_{\boldsymbol{\theta}\in\prod_{v\in\mathcal{V}}\{0,\ldots,\Theta_v\}} \sum_{\mathbf{i}\in[n]^{\mathcal{V}}} \prod_{v\in\mathcal{V}} q_{v,\theta_v}(\tilde{u}_{1:t}[i_v], \tilde{z}_{1:(t-1)}[i_v], f_{1:k}[i_v]) \\ \cdot \left(\sum_{j=1}^n W[i_v, j]\tilde{u}_t[j]\right)^{\theta_v} \prod_{(u,v)\in\mathcal{E}} W[i_u, i_v]\end{aligned}$$

For each $\boldsymbol{\theta} \in \prod_{v\in\mathcal{V}}\{0, \ldots, \Theta_v\}$, we define a new tensor network $T_{\boldsymbol{\theta}}$ from T as follows:

(1) for each $v \in \mathcal{V}$, replace the associated polynomial q_v by q_{v,θ_v}; (2) for each $v \in \mathcal{V}$, connect v with θ_v new vertices, where the associated polynomial for each new vertex is $q(\tilde{u}_{1:t}, \tilde{z}_{1:(t-1)}, f_{1:k}) = \tilde{u}_t$. Then the above is precisely

$$\mathrm{val}_T(\mathbf{W}; \tilde{\mathbf{u}}_{1:t}, \tilde{\mathbf{z}}_{1:t}, \mathbf{f}_{1:k}) = \sum_{\boldsymbol{\theta} \in \prod_{v \in \mathcal{V}} \{0, \ldots, \Theta_v\}} \mathrm{val}_{T_{\boldsymbol{\theta}}}(\mathbf{W}; \tilde{\mathbf{u}}_{1:t}, \tilde{\mathbf{z}}_{1:(t-1)}, \mathbf{f}_{1:k})$$

which shows the claim (2.3.38).

We next claim that for any tensor network T in the variables $(\tilde{u}_{1:t}, \tilde{z}_{1:(t-1)}, f_{1:k})$, we have

$$\mathrm{val}_T(\mathbf{W}; \tilde{\mathbf{u}}_{1:t}, \tilde{\mathbf{z}}_{1:(t-1)}, \mathbf{f}_{1:k}) = \mathrm{val}_{T'}(\mathbf{W}; \tilde{\mathbf{u}}_{1:(t-1)}, \tilde{\mathbf{z}}_{1:(t-1)}, \mathbf{f}_{1:k}) \tag{2.3.39}$$

for a tensor network T' in the variables $(\tilde{u}_{1:(t-1)}, \tilde{z}_{1:(t-1)}, f_{1:k})$. This holds because $\tilde{\mathbf{u}}_t = \tilde{u}_t(\tilde{\mathbf{z}}_{1:(t-1)}, \mathbf{f}_{1:k})$ is itself a polynomial of $(\tilde{\mathbf{z}}_{1:(t-1)}, \mathbf{f}_{1:k})$, so for each vertex v of T, we may write

$$q_v(\tilde{\mathbf{u}}_{1:(t-1)}, \tilde{u}_t(\tilde{\mathbf{z}}_{1:(t-1)}, \mathbf{f}_{1:k}), \tilde{\mathbf{z}}_{1:(t-1)}, \mathbf{f}_{1:k}) = \tilde{q}_v(\tilde{\mathbf{u}}_{1:(t-1)}, \tilde{\mathbf{z}}_{1:(t-1)}, \mathbf{f}_{1:k})$$

for some polynomial $\tilde{q}_v$. Then we can define T' by replacing each polynomial q_v with $\tilde{q}_v$ and preserving all other structures of T.

Having shown the reductions (2.3.38) and (2.3.39), the proof is completed by recursively applying these reductions for $t, t-1, t-2, \ldots, 1$. ■

Combining the above lemma, the state evolution (2.3.37) for the polynomial AMP algorithm applied to $\mathbf{G}$, and the given condition in Lemma 2.2.12 that tensor network values have the same limit for $\mathbf{G}$ and $\mathbf{W}$, we obtain the following state evolution guarantee for the polynomial AMP algorithm applied to $\mathbf{W}$.

Lemma 2.3.12. *In the setting of Lemma 2.2.12, for any fixed* $t \geq 1$, *almost surely as*

$n \to \infty$

$$(\mathbf{u}_1, \mathbf{f}_1, \ldots, \mathbf{f}_k, \tilde{\mathbf{z}}_1(\mathbf{W}), \ldots, \tilde{\mathbf{z}}_t(\mathbf{W})) \xrightarrow{W} (U_1, F_1, \ldots, F_k, \widetilde{Z}_1, \ldots, \widetilde{Z}_t).$$

Proof. By Lemma 2.3.11, for any polynomial $p : \mathbb{R}^{t+k+1} \to \mathbb{R}$, we have

$$\langle p(\mathbf{u}_1, \mathbf{f}_{1:k}, \tilde{\mathbf{z}}_{1:t}(\mathbf{W})) \rangle = \sum_{T \in \mathcal{F}} \mathrm{val}_T(\mathbf{W}; \mathbf{u}_1, \mathbf{f}_{1:k})$$

where $\mathcal{F}$ is a finite set of diagonal tensor networks, and the same decomposition holds for $\mathbf{G}$ in place of $\mathbf{W}$. Then by the condition given in Lemma 2.2.12 and the state evolution (2.3.37), almost surely

$$\lim_{n\to\infty} \langle p(\mathbf{u}_1, \mathbf{f}_{1:k}, \tilde{\mathbf{z}}_{1:t}(\mathbf{W})) \rangle = \lim_{n\to\infty} \langle p(\mathbf{u}_1, \mathbf{f}_{1:k}, \tilde{\mathbf{z}}_{1:t}(\mathbf{G})) \rangle = \mathbb{E}[p(U_1, F_{1:k}, \widetilde{Z}_{1:t})]. \quad (2.3.40)$$

In particular, this shows that on an event $\mathcal{E}$ having probability 1, all mixed moments of the empirical distribution of rows of $(\mathbf{u}_1, \mathbf{f}_{1:k}, \tilde{\mathbf{z}}_{1:t}(\mathbf{W}))$ converge to those of $(U_1, F_{1:k}, \widetilde{Z}_{1:t})$. Lemma 2.3.10 implies that the joint law of $(U_1, F_{1:k}, \widetilde{Z}_{1:t})$ is uniquely determined by its mixed moments, so on this event $\mathcal{E}$, the empirical distribution of rows converges weakly to $(U_1, F_{1:k}, \widetilde{Z}_{1:t})$ (cf. [Bil95, Theorem 30.2], which extends to the multivariate setting by the same proof). On this event $\mathcal{E}$, also

$$\lim_{n\to\infty} \frac{1}{n} \sum_{i=1}^{n} \|(u_1[i], f_{1:k}[i], \tilde{z}_{1:t}(\mathbf{W})[i])\|_2^{2d} = \mathbb{E}[\|(U_1, F_{1:k}, \widetilde{Z}_{1:t})\|_2^{2d}]$$

for each integer $d \geq 1$, which shows (cf. [Vil09, Definition 6.8 and Theorem 6.9]) that

$$(\mathbf{u}_1, \mathbf{f}_{1:k}, \tilde{\mathbf{z}}_{1:t}(\mathbf{W})) \xrightarrow{W} (U_1, F_{1:k}, \widetilde{Z}_{1:t}).$$

■

Remark 2.3.13. Lemmas 2.3.12, 2.2.13, and 2.2.14 already imply universality of the state evolution for polynomial AMP algorithms, without requiring the assumption $\|\mathbf{W}\|_{\mathrm{op}} < C$.

We now proceed with an inductive comparison of the given Lipschitz AMP algorithm (2.2.1) and the polynomial AMP algorithm (2.3.35), both applied to $\mathbf{W}$. For each $t \geq 1$, let $(U_1, F_{1:k}, Z_{1:t})$ describe the state evolution of the given Lipschitz AMP algorithm (2.2.1), where $Z_{1:t} \sim \mathcal{N}(0, \mathbf{\Sigma}_t)$ and $\mathbf{\Sigma}_t$ is non-singular by assumption in Lemma 2.2.12. Let $(U_1, F_{1:k}, \widetilde{Z}_{1:t})$ describe the state evolution of the auxiliary AMP algorithm (2.3.35) where $\widetilde{Z}_{1:t} \sim \mathcal{N}(0, \widetilde{\mathbf{\Sigma}}_t)$. We write as shorthand

$$\begin{aligned} U_{s+1} &= u_{s+1}(Z_{1:s}, F_{1:k}), \qquad & \nabla U_{s+1} &= (\partial_1 u_{s+1}, \ldots, \partial_s u_{s+1})(Z_{1:s}, F_{1:k}), \\ \widetilde{U}_{s+1} &= \tilde{u}_{s+1}(\widetilde{Z}_{1:s}, F_{1:k}), \qquad & \nabla \widetilde{U}_{s+1} &= (\partial_1 \tilde{u}_{s+1}, \ldots, \partial_s \tilde{u}_{s+1})(\widetilde{Z}_{1:s}, F_{1:k}) \end{aligned}$$

where the gradients are with respect to the first s arguments.

All subsequent constants may depend on the Lipschitz non-linearities $u_2, u_3, u_4, \ldots$, the corresponding Onsager coefficients $\{b_{ts}\}$ and state evolution covariances $\{\mathbf{\Sigma}_t\}$, and joint laws of $(U_1, F_{1:k}, Z_{1:t})$, which we treat as fixed throughout this argument.

Lemma 2.3.14. *Fix $t \geq 1$. Suppose (2.3.36) holds for $\varepsilon > 0$ and every polynomial $\tilde{u}_2, \ldots, \tilde{u}_{t+1}$. Suppose also $\|\mathbf{\Sigma}_t - \widetilde{\mathbf{\Sigma}}_t\|_{\mathrm{op}} < \delta$ for $\delta > 0$. Then for any sufficiently small δ, ε, we have*

$$\max_{s=1}^{t} \left\| \mathbb{E}[\nabla U_{s+1}] - \mathbb{E}[\nabla \widetilde{U}_{s+1}] \right\|_2 < \iota(\delta, \varepsilon), \qquad \max_{r,s=1}^{t+1} \left| \mathbb{E}[U_r U_s] - \mathbb{E}[\widetilde{U}_r \widetilde{U}_s] \right| < \iota(\delta, \varepsilon)$$

for a constant $\iota(\delta, \varepsilon) > 0$ satisfying $\iota(\delta, \varepsilon) \to 0$ as $(\delta, \varepsilon) \to (0, 0)$.

Proof. We write $\iota(\delta, \varepsilon)$ for any positive constant satisfying $\iota(\delta, \varepsilon) \to 0$ as $(\delta, \varepsilon) \to (0, 0)$ and changing from instance to instance. Since $\|\mathbf{\Sigma}_t - \widetilde{\mathbf{\Sigma}}_t\|_{\mathrm{op}} < \delta$, $\mathbf{\Sigma}_t$ is invertible, and $\mathbf{\Sigma}_s$ is the upper-left submatrix of $\mathbf{\Sigma}_t$ for $s \leq t$, for sufficiently small $\delta > 0$ and each

$s = 1, \ldots, t$ we have

$$\|\boldsymbol{\Sigma}_s^{-1} - \widetilde{\boldsymbol{\Sigma}}_s^{-1}\|_{\mathrm{op}} < \iota(\delta, \varepsilon), \qquad \mathbb{E}\Big[\Big\|Z_{1:s} - \widetilde{Z}_{1:s}\Big\|_2^2\Big] < \iota(\delta, \varepsilon) \tag{2.3.41}$$

for a coupling of $Z_{1:s}$ and $\widetilde{Z}_{1:s}$.

We introduce the additional abbreviations for the intermediate quantities

$$\overline{U}_{s+1} = u_{s+1}(\widetilde{Z}_{1:s}, F_{1:k}), \qquad \nabla\overline{U}_{s+1} = (\partial_1 u_{s+1}, \ldots, \partial_s u_{s+1})(\widetilde{Z}_{1:s}, F_{1:k}).$$

Then for any $s \in [t]$,

$$\Big\|\mathbb{E}[\nabla U_{s+1}] - \mathbb{E}[\nabla\widetilde{U}_{s+1}]\Big\|_2 \leq \Big\|\mathbb{E}[\nabla U_{s+1}] - \mathbb{E}[\nabla\overline{U}_{s+1}]\Big\|_2 + \Big\|\mathbb{E}[\nabla\overline{U}_{s+1}] - \mathbb{E}[\nabla\widetilde{U}_{s+1}]\Big\|_2 \tag{2.3.42}$$

Applying Stein's lemma, $(a+b+c)^2 \leq 3(a^2+b^2+c^2)$, and Cauchy-Schwarz, the first term of (2.3.42) is bounded as

$$\begin{aligned}
&\Big\|\mathbb{E}[\nabla U_{s+1}] - \mathbb{E}[\nabla\overline{U}_{s+1}]\Big\|_2^2 \\
&= \Big\|\mathbb{E}[U_{s+1} \cdot \boldsymbol{\Sigma}_s^{-1} Z_{1:s}^\top] - \mathbb{E}[\overline{U}_{s+1} \cdot \widetilde{\boldsymbol{\Sigma}}_s^{-1} \widetilde{Z}_{1:s}^\top]\Big\|_2^2 \\
&\leq 3\mathbb{E}\Big[\Big(U_{s+1} - \overline{U}_{s+1}\Big)^2\Big] \cdot \mathbb{E}\Big[\Big\|\boldsymbol{\Sigma}_s^{-1} Z_{1:s}^\top\Big\|_2^2\Big] + 3\mathbb{E}[\overline{U}_{s+1}^2] \cdot \mathbb{E}\Big[\Big\|(\boldsymbol{\Sigma}_s^{-1} - \widetilde{\boldsymbol{\Sigma}}_s^{-1}) Z_{1:s}^\top\Big\|_2^2\Big] \\
&\quad + 3\mathbb{E}[\overline{U}_{s+1}^2] \cdot \mathbb{E}\Big[\Big\|\widetilde{\boldsymbol{\Sigma}}_s^{-1}(Z_{1:s}^\top - \widetilde{Z}_{1:s}^\top)\Big\|_2^2\Big].
\end{aligned}$$

The latter two terms are at most $\iota(\delta, \varepsilon)$ by (2.3.41), and for the first term we have

$$\mathbb{E}\Big[\Big(U_{s+1} - \overline{U}_{s+1}\Big)^2\Big] \leq L_s^2 \cdot \mathbb{E}\Big[\Big\|Z_{1:s} - \widetilde{Z}_{1:s}\Big\|_2^2\Big] < \iota(\delta, \varepsilon) \tag{2.3.43}$$

where L_s is the Lipschitz constant of $u_{s+1}(\cdot)$. The second term of (2.3.42) is bounded similarly as

$$\begin{aligned}\left\|\mathbb{E}[\nabla\overline{U}_{s+1}]-\mathbb{E}[\nabla\widetilde{U}_{s+1}]\right\|_2^2&=\left\|\mathbb{E}[\overline{U}_{s+1}\cdot\widetilde{\boldsymbol{\Sigma}}_s^{-1}\widetilde{Z}_{1:s}^\top]-\mathbb{E}[\widetilde{U}_{s+1}\cdot\widetilde{\boldsymbol{\Sigma}}_s^{-1}\widetilde{Z}_{1:s}^\top]\right\|_2^2\\&\le\mathbb{E}\Big[\Big(\overline{U}_{s+1}-\widetilde{U}_{s+1}\Big)^2\Big]\cdot\mathbb{E}\Big[\Big\|\widetilde{\boldsymbol{\Sigma}}_s^{-1}\widetilde{Z}_{1:s}^\top\Big\|_2^2\Big],\end{aligned}$$

where by (2.3.36) we have

$$\mathbb{E}\Big[\Big(\overline{U}_{s+1}-\widetilde{U}_{s+1}\Big)^2\Big]=\mathbb{E}[(u_{s+1}(\widetilde{Z}_{1:s},F_{1:K})-\tilde{u}_{s+1}(\widetilde{Z}_{1:s},F_{1:K}))^2]<\varepsilon. \tag{2.3.44}$$

Combining these bounds and applying them to (2.3.42) yields the first claim of the lemma, $\|\mathbb{E}[\nabla U_{s+1}]-\mathbb{E}[\nabla\widetilde{U}_{s+1}]\|_2<\iota(\delta,\varepsilon)$. For the second claim, for any $r,s\in[t+1]$, we have

$$\Big|\mathbb{E}[U_{r+1}U_{s+1}]-\mathbb{E}[\widetilde{U}_{r+1}\widetilde{U}_{s+1}]\Big|\le\Big|\mathbb{E}[(U_{r+1}-\widetilde{U}_{r+1})U_{s+1}]\Big|+\Big|\mathbb{E}[\widetilde{U}_{r+1}(U_{s+1}-\widetilde{U}_{s+1})]\Big|.$$

Applying again Cauchy-Schwarz and the bounds (2.3.43) and (2.3.44) yields the second claim $|\mathbb{E}[U_{r+1}U_{s+1}]-\mathbb{E}[\widetilde{U}_{r+1}\widetilde{U}_{s+1}]|<\iota(\delta,\varepsilon)$. ∎

Lemma 2.3.15. *Fix $t\ge 1$. Suppose (2.3.36) holds for $\varepsilon>0$ and every polynomial $\tilde{u}_2,\ldots,\tilde{u}_{t+1}$. Then for any sufficiently small ε, almost surely for all large n, we have*

$$\max_{s=1}^{t}\frac{1}{\sqrt{n}}\|\mathbf{z}_s(\mathbf{W})-\tilde{\mathbf{z}}_s(\mathbf{W})\|_2<\iota(\varepsilon),\qquad \max_{s=1}^{t}\frac{1}{\sqrt{n}}\|\mathbf{z}_s(\mathbf{W})\|_2<C,\qquad \|\boldsymbol{\Sigma}_t-\widetilde{\boldsymbol{\Sigma}}_t\|_{\mathrm{op}}<\iota(\varepsilon)$$

for constants $C>0$ and $\iota(\varepsilon)>0$ satisfying $\iota(\varepsilon)\to 0$ as $\varepsilon\to 0$.

Proof. We write $\mathbf{z}_t,\tilde{\mathbf{z}}_t,\mathbf{u}_t,\tilde{\mathbf{u}}_t$ for the iterates of the Lipschitz and polynomial AMP algorithms applied to $\mathbf{W}$. We prove the extended claim that there are constants $\iota(\varepsilon)>0$ and $C>0$, satisfying $\iota(\varepsilon)\to 0$ as $\varepsilon\to 0$, for which almost surely for all large n,

(a) $\max_{s=1}^t|b_{ts}-\tilde{b}_{ts}|<\iota(\varepsilon)$;

(b) $\max_{s=1}^t \frac{1}{\sqrt{n}}\|\mathbf{z}_s - \tilde{\mathbf{z}}_s\|_2 < \iota(\varepsilon)$ and $\max_{s=1}^t \frac{1}{\sqrt{n}}\|\mathbf{z}_s\|_2 < C$;

(c) $\|\mathbf{\Sigma}_t - \widetilde{\mathbf{\Sigma}}_t\|_{\text{op}} < \iota(\varepsilon)$;

(d) $\max_{s=0}^t \frac{1}{\sqrt{n}}\|\mathbf{u}_{s+1} - \tilde{\mathbf{u}}_{s+1}\|_2 < \iota(\varepsilon)$ and $\max_{s=0}^t \frac{1}{\sqrt{n}}\|\mathbf{u}_{s+1}\|_2 < C$.

We induct on t. For the base case $t = 0$, statements (a–c) are vacuous, and (d) holds by the equality of initializations $\tilde{\mathbf{u}}_1 = \mathbf{u}_1$ and by the convergence of $\mathbf{u}_1$ in Assumption 2.2.1.

Consider any $t \geq 1$, and suppose inductively that claims (a–d) all hold for $t-1$. Denoting the constants in this inductive claim for $t-1$ as $\iota_{t-1}(\varepsilon)$ and C_{t-1}, let us write $\iota_t(\varepsilon)$ and C_t for any positive constants depending on $\iota_{t-1}(\varepsilon)$ and C_{t-1}, satisfying $\iota_t(\varepsilon) \to 0$ as $\varepsilon \to 0$ and $\iota_{t-1}(\varepsilon) \to 0$, and changing from instance to instance.

For (a), by the prescription (2.2.6), each b_{ts} is a continuous function of $\mathbb{E}[\nabla U_{s+1}]$ for $s \leq t-1$ and $\mathbb{E}[U_r U_s]$ for $1 \leq r \leq s \leq t$. Then $\max_{s=1}^t |b_{ts} - \tilde{b}_{ts}| < \iota_t(\varepsilon)$ by statement (c) of the inductive hypothesis and Lemma 2.3.14.

For (b), by the definition of $\mathbf{z}_t$ and $\tilde{\mathbf{z}}_t$, we have

$$\frac{\|\mathbf{z}_t - \tilde{\mathbf{z}}_t\|_2}{\sqrt{n}} \leq \frac{\|\mathbf{W}(\mathbf{u}_t - \tilde{\mathbf{u}}_t)\|_2}{\sqrt{n}} + \sum_{s=1}^t |b_{ts}| \cdot \frac{\|\mathbf{u}_s - \tilde{\mathbf{u}}_s\|_2}{\sqrt{n}} + \sum_{s=1}^t |b_{ts} - \tilde{b}_{ts}| \cdot \frac{\|\tilde{\mathbf{u}}_s\|_2}{\sqrt{n}}.$$

By the assumption that $\|\mathbf{W}\|_{\text{op}} \leq C$ almost surely for all large n, by claim (d) of the inductive hypothesis, and by claim (a) already shown, this is at most $\iota_t(\varepsilon)$. Also,

$$\frac{\|\mathbf{z}_t\|_2}{\sqrt{n}} \leq \frac{\|\mathbf{W}\mathbf{u}_t\|_2}{\sqrt{n}} + \sum_{s=1}^t |b_{ts}| \cdot \frac{\|\mathbf{u}_s\|_2}{\sqrt{n}}$$

which similarly is at most C_t.

For (c), by the prescription (2.2.5), the matrix $\mathbf{\Sigma}_t$ is (as in the proof of (a) above) a continuous function of $\mathbb{E}[\nabla U_{s+1}]$ for $s \leq t-1$ and $\mathbb{E}[U_r U_s]$ for $1 \leq r \leq s \leq t$. Then $\|\mathbf{\Sigma}_t - \widetilde{\mathbf{\Sigma}}_t\|_{\text{op}} < \iota_t(\varepsilon)$ again by statement (c) of the inductive hypothesis and Lemma 2.3.14.

For (d), it follows from the definitions of $\mathbf{u}_{t+1}$ and $\tilde{\mathbf{u}}_{t+1}$ that

$$\begin{aligned}\frac{\|\mathbf{u}_{t+1}-\tilde{\mathbf{u}}_{t+1}\|_2}{\sqrt{n}} &= \frac{\|u_{t+1}(\mathbf{z}_{1:t},\mathbf{f}_{1:k})-\tilde{u}_{t+1}(\tilde{\mathbf{z}}_{1:t},\mathbf{f}_{1:k})\|_2}{\sqrt{n}} \\ &\leq \frac{\|u_{t+1}(\mathbf{z}_{1:t},\mathbf{f}_{1:k})-u_{t+1}(\tilde{\mathbf{z}}_{1:t},\mathbf{f}_{1:k})\|_2}{\sqrt{n}} + \frac{\|u_{t+1}(\tilde{\mathbf{z}}_{1:t},\mathbf{f}_{1:k})-\tilde{u}_{t+1}(\tilde{\mathbf{z}}_{1:t},\mathbf{f}_{1:k})\|_2}{\sqrt{n}}.\end{aligned}$$

The first term is at most $\iota_t(\varepsilon)$ by statement (b) already proved and the fact that $u_{t+1}(\cdot)$ is Lipschitz. For the second term, Lemma 2.3.12 shows $(\tilde{\mathbf{z}}_{1:t},\mathbf{f}_{1:k}) \xrightarrow{W} (\widetilde{Z}_{1:t},F_{1:k})$ almost surely as $n\to\infty$, and the function $(u_{t+1}(\cdot)-\tilde{u}_{t+1}(\cdot))^2$ satisfies the polynomial growth condition (2.1.1) by the given conditions for $u_{t+1}(\cdot)$. Then

$$\lim_{n\to\infty}\frac{\|u_{t+1}(\tilde{\mathbf{z}}_{1:t},\mathbf{f}_{1:k})-\tilde{u}_{t+1}(\tilde{\mathbf{z}}_{1:t},\mathbf{f}_{1:k})\|_2^2}{n} = \mathbb{E}[(u_{t+1}(\widetilde{Z}_{1:t},F_{1:k})-\tilde{u}_{t+1}(\widetilde{Z}_{1:t},F_{1:k}))^2] < \varepsilon \tag{2.3.45}$$

where the inequality is due to (2.3.36) in the construction of $\tilde{u}_{t+1}$. Thus $\|\mathbf{u}_{t+1}-\tilde{\mathbf{u}}_{t+1}\|_2/\sqrt{n} < \iota_t(\varepsilon)$. Similarly,

$$\frac{\|\mathbf{u}_{t+1}\|_2}{\sqrt{n}} \leq \frac{\|u_{t+1}(\mathbf{z}_{1:t},\mathbf{f}_{1:k})-u_{t+1}(\mathbf{0},\mathbf{f}_{1:k})\|_2}{\sqrt{n}} + \frac{\|u_{t+1}(\mathbf{0},\mathbf{f}_{1:k})\|_2}{\sqrt{n}}.$$

The first term is at most C_t by statement (b) already proved and the fact that $u_{t+1}(\cdot)$ is Lipschitz. For the second term, we have $\lim_{n\to\infty}\frac{1}{n}\|u_{t+1}(\mathbf{0},\mathbf{f}_{1:k})\|_2^2 = \mathbb{E}[u_{t+1}(0,F_{1:k})^2]$ which is also at most a constant C_t. This concludes the proof of (d) and completes the induction. ∎

Finally, we apply Lemma 2.3.15 to prove Lemma 2.2.12.

Proof of Lemma 2.2.12. Let $\mathbf{u}_{1:t},\tilde{\mathbf{u}}_{1:t},\mathbf{z}_{1:t},\tilde{\mathbf{z}}_{1:t}$ be the iterates of the Lipschitz AMP algorithm and polynomial AMP algorithm applied to $\mathbf{W}$. We write $\iota(\varepsilon)$ for a positive constant such that $\iota(\varepsilon)\to 0$ as $\varepsilon\to 0$, and changing from instance to instance.

To show W_2-convergence of $(\mathbf{u}_1,\mathbf{f}_{1:k},\mathbf{z}_{1:t})$, consider any function $g:\mathbb{R}^{t+k+1}\to\mathbb{R}$

satisfying $|g(\mathbf{x}) - g(\mathbf{y})| \leq C(1 + \|\mathbf{x}\|_2 + \|\mathbf{y}\|_2)\|\mathbf{x} - \mathbf{y}\|_2$ for a constant $C > 0$. Then

$$\begin{aligned}
&\Big|\langle g(\mathbf{u}_1, \mathbf{f}_{1:k}, \mathbf{z}_{1:t}) - g(\mathbf{u}_1, \mathbf{f}_{1:k}, \tilde{\mathbf{z}}_{1:t})\rangle\Big| \\
&\leq \frac{C}{n}\sum_{i=1}^{n}(1 + \|(u_1[i], f_{1:k}[i], z_{1:t}[i])\|_2 + \|(u_1[i], f_{1:k}[i], \tilde{z}_{1:t}[i])\|_2) \cdot \|z_{1:t}[i] - \tilde{z}_{1:t}[i]\|_2 \\
&\leq \frac{C}{n}\sqrt{3\sum_{i=1}^{n}(1 + \|(u_1[i], f_{1:k}[i], z_{1:t}[i])\|_2^2 + \|(u_1[i], f_{1:k}[i], \tilde{z}_{1:t}[i])\|_2^2)} \cdot \sqrt{\sum_{i=1}^{n}\|z_{1:t}[i] - \tilde{z}_{1:t}[i]\|_2^2} \\
&= \frac{C}{n}\sqrt{3n + 6\|\mathbf{u}_1\|_2^2 + 6\sum_{j=1}^{k}\|\mathbf{f}_j\|_2^2 + 3\sum_{s=1}^{t}(\|\mathbf{z}_s\|_2^2 + \|\tilde{\mathbf{z}}_s\|_2^2)} \cdot \sqrt{\sum_{s=1}^{t}\|\mathbf{z}_s - \tilde{\mathbf{z}}_s\|_2^2}
\end{aligned}$$

This implies, by the statements for $\mathbf{z}_{1:t}$ in Lemma 2.3.15 and the convergence of $(\mathbf{u}_1, \mathbf{f}_1, \ldots, \mathbf{f}_k)$ in Assumption 2.2.1, that almost surely for all large n,

$$\Big|\langle g(\mathbf{u}_1, \mathbf{f}_{1:k}, \mathbf{z}_{1:t}) - g(\mathbf{u}_1, \mathbf{f}_{1:k}, \tilde{\mathbf{z}}_{1:t})\rangle\Big| < \iota(\varepsilon). \tag{2.3.46}$$

Since $(\mathbf{u}_1, \mathbf{f}_{1:k}, \tilde{\mathbf{z}}_{1:t}) \xrightarrow{W} (U_1, F_{1:k}, \widetilde{Z}_{1:t})$ by Lemma 2.3.12, we have

$$\lim_{n\to\infty}\langle g(\mathbf{u}_1, \mathbf{f}_{1:k}, \tilde{\mathbf{z}}_{1:t})\rangle = \mathbb{E}[g(U_1, F_{1:k}, \widetilde{Z}_{1:t})]. \tag{2.3.47}$$

By the statement for $\mathbf{\Sigma}_t$ in Lemma 2.3.15, there is a coupling of $Z_{1:t}$ and $\widetilde{Z}_{1:t}$ such that $\mathbb{E}[\|Z_{1:t} - \widetilde{Z}_{1:t}\|_2^2] < \iota(\varepsilon)$. Then similarly

$$\begin{aligned}
&\Big|\mathbb{E}[g(U_1, F_{1:k}, Z_{1:t})] - \mathbb{E}[g(U_1, F_{1:k}, \widetilde{Z}_{1:t})]\Big| \\
&\qquad \leq C \cdot \mathbb{E}\Big[(1 + \|(U_1, F_{1:k}, Z_{1:t})\|_2 + \|(U_1, F_{1:k}, \widetilde{Z}_{1:t})\|_2) \cdot \|Z_{1:t} - \widetilde{Z}_{1:t}\|_2\Big] \\
&\qquad \leq C\sqrt{\mathbb{E}[3 + 6U_1^2 + 6\|F_{1:k}\|_2^2 + 3\|Z_{1:t}\|_2^2 + 3\|\widetilde{Z}_{1:t}\|_2^2} \cdot \sqrt{\mathbb{E}[\|Z_{1:t} - \widetilde{Z}_{1:t}\|_2^2]} < \iota(\varepsilon).
\end{aligned} \tag{2.3.48}$$

Combining (2.3.46), (2.3.47), and (2.3.48), we obtain for a (different) constant $\iota(\varepsilon) > 0$, almost surely for all large n, $|\langle g(\mathbf{u}_1, \mathbf{f}_{1:k}, \mathbf{z}_{1:t})\rangle - \mathbb{E}[g(U_1, F_{1:k}, Z_{1:t})]| < \iota(\varepsilon)$. Since $\varepsilon > 0$ is arbitrary and $\iota(\varepsilon) \to 0$ as $\varepsilon \to 0$, we conclude that $\lim_{n\to\infty}\langle g(\mathbf{u}_1, \mathbf{f}_{1:k}, \mathbf{z}_{1:t})\rangle =$

$\mathbb{E}[g(U_1, F_{1:k}, Z_{1:t})]$. This holds for all bounded Lipschitz functions $g(\cdot)$ as well as for $g(U_1, F_{1:k}, Z_{1:t}) = \|(U_1, F_{1:k}, Z_{1:t})\|_2^2$, which implies $(\mathbf{u}_1, \mathbf{f}_{1:k}, \mathbf{z}_{1:t}) \xrightarrow{W_2} (U_1, F_{1:k}, Z_{1:t})$ (cf. [Vil09, Definition 6.8 and Theorem 6.9]). ■

Combining Lemmas 2.2.12 and 2.2.13 for $\mathbf{G} \sim \text{GOE}(n)$ concludes the proof of Theorem 2.2.4, and combining Lemmas 2.2.12 and 2.2.14 for an orthogonally invariant matrix $\mathbf{G}$ with limit spectral distribution D concludes the proof of Theorem 2.2.8.

2.4 Sufficient conditions for generalized invariance

In this section, we prove Proposition 2.2.7, providing examples of matrix models that satisfy the generalized invariance condition of Definition 2.2.6.

Lemma 2.4.1. *Let $\mathbf{M} \in \mathbb{R}^{n\times n}$ be a symmetric matrix having eigenvalues $\mathbf{d} \in \mathbb{R}^n$. Suppose $\mathbf{d} \xrightarrow{W} D$ almost surely as $n \to \infty$, where D has finite moments of all orders. Suppose $\mathbf{M}$ satisfies (2.2.9) almost surely for all large n. Then for any $p(\mathbf{x}) \in \Delta\langle\mathbf{x}\rangle$,*

(a) $\lim_{n\to\infty} \frac{1}{n} \operatorname{Tr} p(\mathbf{M})$ exists almost surely, is finite, and depends only on the law of D.

(b) For any $\varepsilon > 0$ and all large n, we have

$$\max_{i=1}^{n} |p(\mathbf{M})[i,i] - \tfrac{1}{n}\operatorname{Tr} p(\mathbf{M})| < n^{-1/2+\varepsilon}, \qquad \max_{i\neq j} |p(\mathbf{M})[i,j]| < n^{-1/2+\varepsilon}$$

Proof. By the definition of diagonal monomials, every $p(\mathbf{x}) \in \Delta\langle\mathbf{x}\rangle$ is a word of the form

$$p(\mathbf{x}) = \mathbf{x}^{r_1}\Delta(p_1(\mathbf{x}))\mathbf{x}^{r_2}\Delta(p_2(\mathbf{x}))\cdots\mathbf{x}^{r_L}\Delta(p_L(\mathbf{x}))\mathbf{x}^{r_{L+1}} \tag{2.4.1}$$

where each $p_\ell(\mathbf{x}) \in \Delta\langle\mathbf{x}\rangle$ and each $r_\ell \geq 0$. We define the *depth* of $p(\mathbf{x})$, denoted by $\delta(p)$, as $\delta(p) = 0$ if $L = 0$ (so that $p(\mathbf{x}) = \mathbf{x}^r$ for some $r \geq 0$), and $\delta(p) = 1 + \max_{\ell=1}^{L}\delta(p_\ell)$

if $L \geq 1$. Thus $\delta(p)$ is the maximum number of "nested" applications of $\Delta(\cdot)$. We induct on $\delta(p)$.

For the base case where $\delta(p) = 0$ and $p(\mathbf{x}) = \mathbf{x}^r$, we have

$$\tfrac{1}{n} \operatorname{Tr} p(\mathbf{M}) = \tfrac{1}{n} \operatorname{Tr}(\mathbf{M}^r) = \frac{1}{n} \sum_{i=1}^{n} d[i]^r \to \mathbb{E}[D^r]$$

almost surely, by the assumption $\mathbf{d} \stackrel{W}{\to} D$. Thus statement (a) holds, and statement (b) holds by the assumed condition (2.2.9).

Suppose inductively that the lemma is true for all $p(\mathbf{x})$ with $\delta(p) \leq K$, and consider $p(\mathbf{x})$ with $\delta(p) = K+1$. Fix any $\varepsilon > 0$. By the definition of depth, every $p_\ell(\mathbf{x})$ in (2.4.1) satisfies $\delta(p_\ell) \leq K$. Then for every $\ell = 1, 2, \ldots, L$, by claim (b) of the induction hypothesis, we can decompose $p_\ell(\mathbf{M}) = \frac{1}{n} \operatorname{Tr} p_\ell(\mathbf{M}) \cdot \mathrm{Id} + \mathbf{E}_\ell$ where $\mathbf{E}_\ell$ satisfies $\max_{i,j \in [n]} |E_\ell[i,j]| < n^{-1/2+\varepsilon}$ almost surely for all large n. Fix any $i, j \in [n]$ and write $i_0 \equiv i$ and $i_{L+1} \equiv j$. Then, applying this decomposition to every $p_\ell(\mathbf{M})$, we obtain

$$\begin{aligned}
p(\mathbf{M})[i,j] &= \sum_{\mathbf{i} \in [n]^L} M^{r_1}[i_0, i_1] p_1(\mathbf{M})[i_1, i_1] M^{r_2}[i_1, i_2] \cdots p_L(\mathbf{M})[i_L, i_L] M^{r_{L+1}}[i_L, i_{L+1}] \\
&= \sum_{\mathbf{i} \in [n]^L} \prod_{\ell=1}^{L+1} M^{r_\ell}[i_{\ell-1}, i_\ell] \prod_{\ell=1}^{L} \left(\frac{1}{n} \operatorname{Tr} p_\ell(\mathbf{M}) + E_\ell[i_\ell, i_\ell] \right) \\
&= \sum_{\mathcal{J} \subseteq [L]} \left(\prod_{\ell \in [L] \setminus \mathcal{J}} \frac{1}{n} \operatorname{Tr} p_\ell(\mathbf{M}) \right) \sum_{\mathbf{i} \in [n]^L} \prod_{\ell=1}^{L+1} M^{r_\ell}[i_{\ell-1}, i_\ell] \prod_{\ell \in \mathcal{J}} E_\ell[i_\ell, i_\ell].
\end{aligned}$$

By the induction hypothesis, the limit

$$M_{\mathcal{J}} := \lim_{n \to \infty} \prod_{\ell \in [L] \setminus \mathcal{J}} \frac{1}{n} \operatorname{Tr} p_\ell(\mathbf{M}) \tag{2.4.2}$$

exists, is finite, and depends only on the law of D. We set $M_{\mathcal{J}} = 1$ if $\mathcal{J} = [L]$. Note that this convergence is uniform over pairs $i, j \in [n]$. Therefore, for an error $\xi_{\mathcal{J}} = o(1)$

independent of i and j,

$$p(\mathbf{M})[i,j] = \sum_{\mathcal{J}\subseteq[L]} (M_{\mathcal{J}} + \xi_{\mathcal{J}}) \sum_{\mathbf{i}\in[n]^L} \prod_{\ell=1}^{L+1} M^{r_\ell}[i_{\ell-1}, i_\ell] \prod_{\ell\in\mathcal{J}} E_\ell[i_\ell, i_\ell].$$

We first sum over all indices $\{i_\ell : \ell \notin \mathcal{J}\}$: Write explicitly $\mathcal{J} = \{\ell_1, \ldots, \ell_{|\mathcal{J}|}\}$ where $1 \leq \ell_1 < \ldots < \ell_{|\mathcal{J}|} \leq L$. Let $\ell_0 = 0$ and $\ell_{|\mathcal{J}|+1} = L+1$, and denote $R_\rho = r_{\ell_{\rho-1}+1} + \ldots + r_{\ell_\rho}$. Then this gives

$$p(\mathbf{M})[i,j] = \sum_{\mathcal{J}\subseteq[L]} (M_{\mathcal{J}} + \xi_{\mathcal{J}}) \sum_{\mathbf{i}\in[n]^{\mathcal{J}}} \prod_{\rho=1}^{|\mathcal{J}|+1} M^{R_\rho}[i_{\ell_{\rho-1}}, i_{\ell_\rho}] \prod_{\ell\in\mathcal{J}} E_\ell[i_\ell, i_\ell]. \tag{2.4.3}$$

We denote by $C > 0$ a constant depending only on $p(\mathbf{x})$, $\mathcal{J}$, and the law of D, and changing from instance to instance. By (2.2.9), we have $\max_{i\in[n]} |M^{R_\rho}[i,i]| < \frac{1}{n}\operatorname{Tr}\mathbf{M}^{R_\rho} + n^{-1/2+\varepsilon} < C$ and $\max_{i\neq j} |M^{R_\rho}[i,j]| < n^{-1/2+\varepsilon}$ for each $\rho \in [|\mathcal{J}|+1]$, almost surely for all large n. For any $\mathbf{i} \in [n]^{\mathcal{J}}$, define $\Psi(\mathbf{i}) = \{\rho \in [|\mathcal{J}|+1] : i_{\ell_{\rho-1}} \neq i_{\ell_\rho}\}$. Then this implies

$$\prod_{\rho=1}^{|\mathcal{J}|+1} \left|M^{R_\rho}[i_{\ell_{\rho-1}}, i_{\ell_\rho}]\right| \prod_{\ell\in\mathcal{J}} |E_\ell[i_\ell, i_\ell]| \leq Cn^{(-1/2+\varepsilon)(|\Psi(\mathbf{i})|+|\mathcal{J}|)}. \tag{2.4.4}$$

Moreover, if $\psi \geq 1$, then note that $|\{\mathbf{i} \in [n]^{\mathcal{J}} : |\Psi(\mathbf{i})| = \psi\}| \leq Cn^{\psi-1}$, because $i_{\ell_0} = i_0 = i$ and $i_{\ell_{|\mathcal{J}|+1}} = i_{L+1} = j$ are fixed, so there is freedom to choose $\psi - 1$ remaining index values. Combining this with (2.4.4), for any $\psi \geq 1$,

$$\begin{aligned}\sum_{\mathbf{i}\in[n]^{\mathcal{J}}:|\Psi(\mathbf{i})|=\psi} \prod_{\rho=1}^{|\mathcal{J}|+1} \left|M^{R_\rho}[i_{\ell_{\rho-1}}, i_{\ell_\rho}]\right| \prod_{\ell\in\mathcal{J}} |E_\ell[i_\ell, i_\ell]| &\leq Cn^{\psi-1} \cdot n^{(-1/2+\varepsilon)(\psi+|\mathcal{J}|)} \\ &\leq Cn^{-1/2+\varepsilon(2|\mathcal{J}|+1)}\end{aligned} \tag{2.4.5}$$

where the last inequality follows from the observation that we always have $|\Psi(\mathbf{i})| \leq |\mathcal{J}| + 1$. For $\psi = 0$, we must have $i = j$ and $|\{\mathbf{i} \in [n]^{\mathcal{J}} : |\Psi(\mathbf{i})| = \psi\}| = 1$. Then by (2.4.4), this bound (2.4.5) still holds as long as $|\mathcal{J}| \geq 1$, i.e. $\mathcal{J} \neq \emptyset$.

Applying (2.4.5) for all non-empty $\mathcal{J} \subseteq [L]$, it follows from (2.4.3) that

$$\left| p(\mathbf{M})[i,j] - \mathbf{1}\{i=j\}(M_\emptyset + \xi_\emptyset) M^R[i,i] \right| \leq C n^{-1/2+\varepsilon(2L+1)} \tag{2.4.6}$$

where we set $R = r_1 + \ldots + r_{L+1}$. The above bounds all hold uniformly over $i,j \in [n]$, and hence (2.4.6) holds simultaneously for all pairs $i,j \in [n]$, almost surely for all large n. Thus, combining with the condition (2.2.9) for $M^R[i,i]$, we conclude that both $\max_{i \neq j} |p(\mathbf{M})[i,j]|$ and $\max_{i=1}^n |p(\mathbf{M})[i,i] - (M_\emptyset + \xi_\emptyset) \cdot \frac{1}{n} \operatorname{Tr} \mathbf{M}^R|$ are at most $n^{-1/2+\varepsilon(2L+2)}$ for all large n. Then $\{p(\mathbf{M})[i,i] : i \in [n]\}$ are uniformly close to a value independent of $i \in [n]$, which implies also $\max_{i=1}^n |p(\mathbf{M})[i,i] - \frac{1}{n} \operatorname{Tr} p(\mathbf{M})| < 2n^{-1/2+\varepsilon(2L+2)}$. These statements hold for any $\varepsilon > 0$, showing the inductive claim (b) for $p(\mathbf{x})$. Moreover, averaging (2.4.6) over $i = j$ gives

$$\lim_{n \to \infty} \frac{1}{n} \operatorname{Tr} p(\mathbf{M}) = \lim_{n \to \infty} (M_\emptyset + \xi_\emptyset) \cdot \frac{1}{n} \operatorname{Tr} M^R = M_\emptyset \cdot \mathbb{E}[D^R],$$

and we recall from (2.4.2) that $M_\emptyset$ depends only on the law of D. This shows the inductive claim (a) for $p(\mathbf{x})$, completing the induction. ■

Proof of Proposition 2.2.7. Lemma 2.4.1 implies that the matrix model in Proposition 2.2.7(b2) satisfies Definition 2.2.6, where the limit diagonal law $\mathcal{D}_{\text{diag}}$ is determined uniquely by the limit spectral distribution D. To complete the proof of Proposition 2.2.7, it suffices to verify that the orthogonally invariant matrix model of part (a) and the model of part (b1) are both special cases of the model in part (b2).

If $\mathbf{W} = \mathbf{O}\mathbf{D}\mathbf{O}^\top$ is orthogonally invariant, i.e. $\mathbf{O} \sim \operatorname{Haar}(\mathbb{O}(n))$ is independent of $\mathbf{D}$, then also $\mathbf{O} \overset{L}{=} \mathbf{\Pi}_V \mathbf{O} \mathbf{\Pi}_E$ where $\mathbf{\Pi}_V, \mathbf{\Pi}_E$ are uniformly random signed permutations independent of $\mathbf{O}$. The entries of $\mathbf{O}$ satisfy the delocalization condition (2.2.8) almost surely for all large n, as is implied by [Jia05, Theorem 1]. Thus $\mathbf{W}$ is a special case of the model in part (b1).

Now suppose $\mathbf{W}$ is any matrix satisfying the description of part (b1). Then $\mathbf{W}$

has the simpler form $\mathbf{W} = \mathbf{\Pi}\mathbf{M}\mathbf{\Pi}^\top$ where $\mathbf{\Pi} = \mathbf{\Pi}_V$,

$$\mathbf{M} = \mathbf{H}\mathbf{P}\mathbf{D}\mathbf{P}^\top\mathbf{H}^\top,$$

and $\mathbf{P} = \mathbf{P}_E$ is the random permutation corresponding to $\mathbf{\Pi}_E = \mathbf{P}_E\mathbf{\Xi}_E$. Here, we have eliminated the diagonal sign matrices $\mathbf{\Xi}_E$ from the expression using $\mathbf{\Xi}_E\mathbf{D}\mathbf{\Xi}_E^\top = \mathbf{D}$. To show that $\mathbf{W}$ is an example of the model in part (b2), it remains to show that this matrix $\mathbf{M}$ satisfies the condition (2.2.9) almost surely for all large n.

Consider $\mathbf{M}^\nu$ for any fixed integer $\nu \geq 1$. Let $\mathbf{h}_i \in \mathbb{R}^n$ denote the i^{th} row of $\mathbf{H}$, and let σ be the permutation of $[n]$ for which $P[i, \sigma(i)] = 1$ for all $i \in [n]$. Then

$$M^\nu[i,j] = (\mathbf{H}\mathbf{P}\mathbf{D}^\nu\mathbf{P}^\top\mathbf{H}^\top)[i,j] = \sum_{k=1}^{n} h_i[k]D^\nu[\sigma(k), \sigma(k)]h_j[k] \tag{2.4.7}$$

We condition on $(\mathbf{D}, \mathbf{H})$, and write $\mathbb{E}$ for the expectation over only the permutation σ. Then for each fixed $k \in [n]$, we have $\mathbb{E}[D^\nu[\sigma(k), \sigma(k)]] = n^{-1} \operatorname{Tr} \mathbf{D}^\nu$, so

$$\mathbb{E}[M^\nu[i,j]] = \tfrac{1}{n} \operatorname{Tr} \mathbf{D}^\nu \cdot \mathbf{h}_i^\top \mathbf{h}_j = \tfrac{1}{n} \operatorname{Tr} \mathbf{M}^\nu \cdot 1\{i = j\}.$$

We now show concentration of $M^\nu[i,j]$ around this expectation by computing its high moments: Consider first any fixed $i \neq j \in [n]$, and abbreviate $\tilde{h}[k] = h_i[k]h_j[k]$ and $\tilde{d}[k] = D^\nu[k,k]$. Then from (2.4.7), $M^\nu[i,j] = \sum_{k=1}^n \tilde{h}[k]\tilde{d}[\sigma(k)]$, so for any even integer $p \geq 2$,

$$\mathbb{E}[(M^\nu[i,j])^p] = \sum_{\mathbf{k} \in [n]^p} \tilde{h}[k_1] \ldots \tilde{h}[k_p] \mathbb{E}\Big[\tilde{d}[\sigma(k_1)] \ldots \tilde{d}[\sigma(k_p)]\Big].$$

Let $\mathcal{P}$ be the lattice of partitions of $[p]$, endowed with the usual partial ordering by refinement. For each $\mathbf{k} \in [n]^p$, let $\pi(\mathbf{k}) \in \mathcal{P}$ be the partition induced by $\mathbf{k}$, i.e. $i, j \in [p]$

belong to a common block of π if and only if $k_i = k_j$. Then

$$\begin{aligned}\mathbb{E}[(M^\nu[i,j])^p] &= \sum_{\pi \in \mathcal{P}} \sum_{\mathbf{k} \in [n]^p : \pi(\mathbf{k}) = \pi} \tilde{h}[k_1] \dots \tilde{h}[k_p] \mathbb{E}\Big[\tilde{d}[\sigma(k_1)] \dots \tilde{d}[\sigma(k_p)]\Big] \\ &= \sum_{\pi \in \mathcal{P}} \sum_{\substack{\mathbf{k} \in [n]^p \\ \pi(\mathbf{k}) = \pi}} \tilde{h}[k_1] \dots \tilde{h}[k_p] \cdot \frac{(n - |\pi|)!}{n!} \sum_{\substack{\mathbf{l} \in [n]^p \\ \pi(\mathbf{l}) = \pi}} \tilde{d}[l_1] \dots \tilde{d}[l_p], \qquad (2.4.8)\end{aligned}$$

the second equality using that the permutation σ is uniformly random, so the expectation over σ yields a uniform average over new choices for the $|\pi|$ distinct index values of $\mathbf{k}$.

Let $\mu(\pi, \pi')$ for $\pi \le \pi'$ be the Möbius function over $\mathcal{P}$, satisfying the inversion relation (see e.g. [NS06, Eq. (10.10)]) $\sum_{\tau \in \mathcal{P} : \pi \le \tau \le \pi'} \mu(\pi, \tau) = \mathbf{1}\{\pi = \pi'\}$. Then for any function f,

$$\sum_{\substack{\mathbf{k} \in [n]^p \\ \pi(\mathbf{k}) = \pi}} f(\mathbf{k}) = \sum_{\substack{\mathbf{k} \in [n]^p \\ \pi(\mathbf{k}) \ge \pi}} f(\mathbf{k}) \cdot \sum_{\substack{\tau \in \mathcal{P} \\ \pi \le \tau \le \pi(\mathbf{k})}} \mu(\pi, \tau) = \sum_{\substack{\tau \in \mathcal{P} \\ \tau \ge \pi}} \mu(\pi, \tau) \sum_{\substack{\mathbf{k} \in [n]^p \\ \pi(\mathbf{k}) \ge \tau}} f(\mathbf{k}). \qquad (2.4.9)$$

Applying this to the term involving $\tilde{h}$ in (2.4.8),

$$\sum_{\mathbf{k} \in [n]^p : \pi(\mathbf{k}) = \pi} \tilde{h}[k_1] \dots \tilde{h}[k_p] = \sum_{\tau \in \mathcal{P} : \tau \ge \pi} \mu(\pi, \tau) \prod_{R \in \tau} \sum_{k=1}^n \tilde{h}[k]^{|R|}.$$

Recalling $\tilde{h}[k] = h_i[k] h_j[k]$ where $i \ne j$, we have $\sum_{k=1}^n \tilde{h}[k] = \mathbf{h}_i^\top \mathbf{h}_j = 0$. Thus the summand for τ vanishes if τ has a singleton block. For all other partitions $\tau \in \mathcal{P}$, its number of blocks satisfies $|\tau| \le p/2$. Then applying $|\tilde{h}[k]| \le n^{2(-1/2+\varepsilon)}$ by the delocalization condition (2.2.8) for $\mathbf{H}$, for any fixed $\varepsilon > 0$ and all large n,

$$\left| \prod_{R \in \tau} \sum_{k=1}^n \tilde{h}[k]^{|R|} \right| \le n^{2p(-1/2+\varepsilon)} \cdot n^{|\tau|} \le n^{-p/2 + 2p\varepsilon}.$$

Thus $|\sum_{\mathbf{k} \in [n]^p : \pi(\mathbf{k}) = \pi} \tilde{h}[k_1] \dots \tilde{h}[k_p]| \le C n^{-p/2 + 2p\varepsilon}$ where, here and below, we denote by $C > 0$ a (π, D)-dependent constant that may change from instance to instance. By the assumption $\mathbf{d} \xrightarrow{W} D$ and Lemma 2.3.3 (applied with $\mathcal{S}$ being the blocks of π and $q_S(x) = \tilde{d}(x)^{|S|}$ for $S \in \pi$), also $n^{-|\pi|}|\sum_{\mathbf{l}\in[n]^p:\pi(\mathbf{l})=\pi} \tilde{d}[l_1]\dots\tilde{d}[l_p]| \leq C$. Applying these to (2.4.8), we obtain $\mathbb{E}[(M^\nu[i,j])^p] \leq Cn^{-p/2+2p\varepsilon}$, so $\mathbb{P}[|M^\nu[i,j]| > n^{-1/2+3\varepsilon}] \leq Cn^{-p\varepsilon}$ by Markov's inequality. Choosing even $p \geq 2$ sufficiently large and taking a union bound over all $i \neq j$, this shows that the second condition of (2.2.9) holds almost surely for all large n.

The case $i = j$ is similar: Fix $i \in [n]$ and now abbreviate $\tilde{h}[k] = h_i[k]^2$ and $\tilde{d}[k] = D^\nu[k,k] - n^{-1}\operatorname{Tr}\mathbf{D}^\nu$. Then from (2.4.7), $M^\nu[i,i] - n^{-1}\operatorname{Tr}\mathbf{M}^\nu = \sum_k \tilde{h}[k]\tilde{d}[k]$, so we obtain analogously to (2.4.8)

$$\mathbb{E}[(M^\nu[i,i] - n^{-1}\operatorname{Tr}\mathbf{M}^\nu)^p] = \sum_{\substack{\mathbf{k}\in[n]^p \\ \pi(\mathbf{k})=\pi}} \tilde{h}[k_1]\dots\tilde{h}[k_p] \cdot \frac{(n-|\pi|)!}{n!} \sum_{\substack{\mathbf{l}\in[n]^p \\ \pi(\mathbf{l})=\pi}} \tilde{d}[l_1]\dots\tilde{d}[l_p].$$

Applying the Möbius inversion relation (2.4.9) now to the second summation over $\mathbf{l}$,

$$\sum_{\mathbf{l}\in[n]^p:\pi(\mathbf{l})=\pi} \tilde{d}[l_1]\dots\tilde{d}[l_p] = \sum_{\tau\in\mathcal{P}:\tau\geq\pi} \mu(\pi,\tau)\prod_{R\in\tau}\sum_{l=1}^n \tilde{d}[l]^{|R|}.$$

Using that $\sum_{k=1}^n \tilde{d}[k] = 0$, the summand for τ vanishes if τ has a singleton block. For all other partitions $\tau \in \mathcal{P}$, applying $\mathbf{d} \xrightarrow{W} D$, we obtain

$$\left|\prod_{R\in\tau}\sum_{l=1}^n \tilde{d}[l]^{|R|}\right| \leq Cn^{|\tau|} \leq Cn^{p/2}.$$

Then $|\sum_{\mathbf{l}\in[n]^p:\pi(\mathbf{l})=\pi} \tilde{d}[l_1]\dots\tilde{d}[l_p]| \leq Cn^{p/2}$. From (2.2.8), we have also

$$n^{-|\pi|}\sum_{\mathbf{k}\in[n]^p:\pi(\mathbf{k})=\pi} |\tilde{h}[k_1]\dots\tilde{h}[k_p]| \leq n^{-2p(1/2+\varepsilon)}.$$

Then $\mathbb{E}[(M^\nu[i,i] - n^{-1}\operatorname{Tr}\mathbf{M}^\nu)^p] \leq Cn^{-p/2+2p\varepsilon}$, so the first condition of (2.2.9) follows also by Markov's inequality and a union bound. This verifies that $\mathbf{W}$ satisfying part (b1) also satisfies part (b2), as desired. ■

2.5 Tensor network value under orthogonal invariance

In this section, we derive a more explicit combinatorial form for the tensor network value of Lemma 2.2.14 when $\mathbf{W}$ is an orthogonally invariant matrix, using the orthogonal Weingarten calculus. We then prove the asymptotic freeness statement of Proposition 2.2.16(b).

Let T be a tensor network with $w+1$ vertices and w edges. Then there are $2w$ vertex-edge pairs (v,e) where edge e is incident to vertex v. We label these vertex-edge pairs arbitrarily as $1,2,\ldots,2w$. Let $\mathcal{P}$ be the lattice of partitions of $[2w]$, endowed with the usual partial ordering by refinement. We define two distinguished partitions $\pi_V, \pi_E \in \mathcal{P}$, such that vertex-edge pairs $\rho, \tau \in [2w]$ belong to the same block of π_V if and only if they have the same vertex v, and to the same block of π_E if and only if they have the same edge e. (Thus π_V has $w+1$ blocks, one for each vertex of T, and π_E is a pairing with w pairs, one for each edge of T.)

Define a metric over $\mathcal{P}$ by

$$d(\pi,\pi') = |\pi| + |\pi'| - 2|\pi \vee \pi'| \tag{2.5.1}$$

where $\pi \vee \pi'$ is the join (i.e. least upper bound) of π and π'. This is shown in [AB73, BO73] to be equivalent to the smallest number of merge and divide operations needed to transform π into π', where a merge operation combines any two blocks into one block, and a divide operation splits any one block into two blocks. From this characterization, it is immediate that $d(\cdot,\cdot)$ satisfies the triangle inequality $d(\pi,\pi') + d(\pi',\pi'') \geq d(\pi,\pi'')$. We call a path $\pi_0 \to \pi_1 \to \ldots \to \pi_k$ of partitions a *d-geodesic* if it is a shortest path from π_0 to π_k in the metric $d(\cdot,\cdot)$, i.e. if

$$d(\pi_0,\pi_k) = d(\pi_0,\pi_1) + d(\pi_1,\pi_2) + \ldots + d(\pi_{k-1},\pi_k).$$

The main result of this section is the following proposition.

Proposition 2.5.1. *In the setting of Lemma 2.2.14, suppose in addition that* $\mathbf{W} = \mathbf{O}\mathbf{D}\mathbf{O}^\top$ *is orthogonally invariant, where* $\mathbf{D} = \operatorname{diag}(\mathbf{d})$ *and* $\mathbf{d} \xrightarrow{W} D$ *almost surely as* $n \to \infty$. *For* $\pi \geq \pi_V$ *and* $\pi' \geq \pi_E$, *define*

$$q(\pi) = \prod_{S \in \pi} \mathbb{E}\left[\prod_{\textit{distinct vertices } v \textit{ in vertex-edge pairs of } S} q_v(X_1,\ldots,X_k)\right] \tag{2.5.2}$$

$$D(\pi') = \prod_{S \in \pi'} \mathbb{E}\left[D^{\textit{number of distinct edges in vertex-edge pairs of } S}\right]. \tag{2.5.3}$$

Then

$$\lim_{n\to\infty} \operatorname{val}_T(\mathbf{W};\mathbf{x}_1,\ldots,\mathbf{x}_k) = \sum_{j\geq 0} \sum_{\substack{\textit{distinct pairings } \pi_0,\ldots,\pi_j \textit{ of } [2w] \\ \pi_V \to \pi_0 \to \ldots \to \pi_j \to \pi_E \textit{ is a d-geodesic}}} (-1)^j q(\pi_V \vee \pi_0) D(\pi_E \vee \pi_j).$$

(Here π_j *is not required to be distinct from* π_E.*)*

To show this result, we apply the following statements derived from the orthogonal Weingarten calculus of [CŚ06] for mixed moments of entries of Haar-orthogonal random matrices.

Lemma 2.5.2. *Let* $\mathbf{O} \sim \operatorname{Haar}(\mathbb{O}(n))$. *Let* $\mathbf{i} = (i_1,\ldots,i_{2w})$ *and* $\mathbf{j} = (j_1,\ldots,j_{2w})$ *be any index tuples in* $[n]^{2w}$. *Then*

$$\mathbb{E}\left[\prod_{p=1}^{2w} O[i_p,j_p]\right] = \sum_{\substack{\textit{pairings } \pi,\pi' \textit{ of } [2w] \\ \pi \leq \pi(\mathbf{i}),\ \pi' \leq \pi(\mathbf{j})}} \operatorname{Wg}_n[\pi,\pi'] \tag{2.5.4}$$

where Wg_n *is the orthogonal Weingarten function. For fixed* w, *as* $n \to \infty$, *this* *satisfies*

$$\mathrm{Wg}_n[\pi, \pi'] = n^{-w-d(\pi,\pi')/2} \cdot \mu_{\mathrm{NC}}(\pi, \pi') + O(n^{-w-d(\pi,\pi')/2-1}) \tag{2.5.5}$$

where $d(\pi, \pi')$ is the metric (2.5.1), and $\mu_{\mathrm{NC}}(\pi, \pi')$ is the Möbius function on the non-crossing partition lattice, given by

$$\mu_{\mathrm{NC}}(\pi, \pi') = \sum_{k \geq 0} \sum_{\substack{\text{distinct pairings } \pi_0, \pi_1, \ldots, \pi_k \text{ of } [2w] \\ \pi_0 \to \pi_1 \to \ldots \to \pi_k \text{ is a } d\text{-geodesic from } \pi_0 = \pi \text{ to } \pi_k = \pi'}} (-1)^k. \tag{2.5.6}$$

Proof. We may identify pairings π, π' of $[2w]$ as permutations in the symmetric group S_{2w}, each a product of w disjoint transpositions corresponding to the w pairs. The cycle decomposition of their product $\pi\pi'$ in S_{2w} has exactly two cycles for each set of their join partition $\pi \vee \pi'$. Then, the metric $l(\pi, \pi') = |\pi\pi'|/2$ used in [CŚ06, Section 3] (where $|\cdot|$ is the Cayley distance to the identity permutation in S_{2w}, given by $2w$ minus the number of cycles) is equivalently

$$l(\pi, \pi') = \frac{2w - 2|\pi \vee \pi'|}{2} = \frac{d(\pi, \pi')}{2} \tag{2.5.7}$$

where the right side is our metric $d(\cdot, \cdot)$ restricted to pairings. The statements (2.5.4) and (2.5.5) then follow from [CŚ06, Corollary 3.4 and Theorem 3.13]. The form (2.5.6) for the Möbius function follows from comparing [CŚ06, Theorem 3.13] with [CŚ06, Lemma 3.12], noting that the leading-order terms of [CŚ06, Lemma 3.12] come from paths of pairings satisfying $\pi_i \neq \pi_{i+1}$ for each $i = 0, \ldots, k-1$ and also $l(\pi_0, \pi_1) + \ldots + l(\pi_{k-1}, \pi_k) = l(\pi_0, \pi_k)$. Any such path must be a geodesic of $k+1$ unique pairings in the metric $l(\cdot, \cdot)$, and hence also in the metric $d(\cdot, \cdot)$ by the equivalence (2.5.7), and this shows (2.5.6). ■

Proof of Proposition 2.5.1. Expanding the product $\mathbf{W} = \mathbf{O}\mathbf{D}\mathbf{O}^\top$, the tensor network value is given by

$$\mathrm{val}_T(\mathbf{W};\mathbf{x}_{1:k})=\frac{1}{n}\sum_{\mathbf{i}\in[n]^{\mathcal{V}}}\sum_{\mathbf{j}\in[n]^{\mathcal{E}}}\prod_{v\in\mathcal{V}}q_v(x_{1:k}[i_v])\prod_{e=(u,v)\in\mathcal{E}}O[i_u,j_e]D[j_e,j_e]O[i_v,j_e].$$

For each vertex v or edge e, let $\rho(v),\rho(e)\in[2w]$ be an arbitrary choice of vertex-edge pair containing this vertex or this edge. Then this is equivalently expressed as

$$\mathrm{val}_T(\mathbf{W};\mathbf{x}_{1:k})=\frac{1}{n}\sum_{\substack{\mathbf{i}\in[n]^{2w}\\ \pi(\mathbf{i})\geq\pi_V}}\sum_{\substack{\mathbf{j}\in[n]^{2w}\\ \pi(\mathbf{j})\geq\pi_E}}\prod_{v\in\mathcal{V}}q_v(x_{1:k}[i_{\rho(v)}])\prod_{e\in\mathcal{E}}D[j_{\rho(e)},j_{\rho(e)}]\prod_{\rho=1}^{2w}O[i_\rho,j_\rho]. \tag{2.5.8}$$

Note that by the constraints $\pi(\mathbf{i})\geq\pi_V$ and $\pi(\mathbf{j})\geq\pi_E$, this expression is the same for any choices of vertex-edge pairs $\rho(v),\rho(e)\in[2w]$.

Let $\mathbb{E}$ be the expectation over $\mathbf{O}$, conditional on $\mathbf{x}_1,\ldots,\mathbf{x}_k$ and $\mathbf{D}$. By Lemma 2.5.2, we have

$$\mathbb{E}\left[\prod_{p=1}^{2w}O[i_p,j_p]\right]=\sum_{\text{pairings }\pi,\pi'\text{ of }[2w]}\mathbf{1}_{\pi(\mathbf{i})\geq\pi}\mathbf{1}_{\pi(\mathbf{j})\geq\pi'}\cdot n^{-w-d(\pi,\pi')}(\mu_{\mathrm{NC}}(\pi,\pi')+o(1)).$$

Note that

$$\mathbf{1}_{\pi(\mathbf{i})\geq\pi_V}\mathbf{1}_{\pi(\mathbf{i})\geq\pi}=\mathbf{1}_{\pi(\mathbf{i})\geq\pi_V\vee\pi},\qquad\mathbf{1}_{\pi(\mathbf{j})\geq\pi_E}\mathbf{1}_{\pi(\mathbf{j})\geq\pi'}=\mathbf{1}_{\pi(\mathbf{j})\geq\pi_E\vee\pi'}.$$

Identifying summations over $\mathbf{i},\mathbf{j}\in[n]^{2w}$ with $\pi(\mathbf{i})\geq\pi_V\vee\pi$ and $\pi(\mathbf{j})\geq\pi_E\vee\pi'$ as a summation over one index in $[n]$ for each block of $\pi_V\vee\pi$ and $\pi_E\vee\pi'$, and applying the given conditions that $\mathbf{x}_{1:k}\overset{W}{\to}X_{1:k}$ and $\mathrm{diag}(\mathbf{D})\overset{W}{\to}D$ almost surely, observe that

$$\frac{1}{n^{|\pi_V\vee\pi|}}\sum_{\mathbf{i}\in[n]^{2w}}\mathbf{1}_{\pi(\mathbf{i})\geq\pi_V\vee\pi}\prod_{v\in\mathcal{V}}q_v(x_{1:k}[i_{\rho(v)}])\to q(\pi_V\vee\pi),$$
$$\frac{1}{n^{|\pi_E\vee\pi|}}\sum_{\mathbf{j}\in[n]^{2w}}\mathbf{1}_{\pi(\mathbf{j})\geq\pi_E\vee\pi'}\prod_{e\in\mathcal{E}}D[j_{\rho(e)},j_{\rho(e)}]\to D(\pi_E\vee\pi')$$

where $q(\cdot)$ and $D(\cdot)$ are as defined in (2.5.2) and (2.5.3). Then, taking the expectation over $\mathbf{O}$ in (2.5.8) and applying these observations,

$$\mathbb{E}[\mathrm{val}_T(\mathbf{W};\mathbf{x}_{1:k})] = \sum_{\text{pairings } \pi,\pi' \text{ of } [2w]} \frac{1}{n} \cdot n^{|\pi_V \vee \pi|} \cdot n^{|\pi_E \vee \pi'|} \cdot n^{-w-d(\pi,\pi')} \cdot \Big(\mu_{\mathrm{NC}}(\pi,\pi') \cdot q(\pi_V \vee \pi) \cdot D(\pi_E \vee \pi') + o(1)\Big). \tag{2.5.9}$$

Recall that $|\pi_V| = w+1$ and $|\pi| = |\pi'| = |\pi_E| = w$ as these are all pairings of $[2w]$. Then by definition of the metric $d(\cdot,\cdot)$,

$$|\pi_V \vee \pi| = \frac{2w+1-d(\pi_V,\pi)}{2}, \qquad |\pi_E \vee \pi| = \frac{2w-d(\pi_E,\pi)}{2}.$$

So the above value simplifies to

$$\sum_{\text{pairings } \pi,\pi' \text{ of } [2w]} n^{\frac{2w-1-d(\pi_V,\pi)-d(\pi,\pi')-d(\pi',\pi_E)}{2}} \Big(\mu_{\mathrm{NC}}(\pi,\pi')q(\pi_V \vee \pi)D(\pi_E \vee \pi') + o(1)\Big).$$

Applying the triangle inequality for $d(\cdot,\cdot)$ and the identity $|\pi_V \vee \pi_E| = 1$ since T is a connected tree, we have

$$d(\pi_V,\pi) + d(\pi,\pi') + d(\pi',\pi_E) \geq d(\pi_V,\pi_E) = (w+1) + w - 2 = 2w-1,$$

and equality holds if and only if $\pi_V \to \pi \to \pi' \to \pi_E$ is a d-geodesic. Thus, we obtain the limit value

$$\lim_{n\to\infty} \mathbb{E}[\mathrm{val}_T(\mathbf{W};\mathbf{x}_{1:k})] = \sum_{\substack{\text{pairings } \pi,\pi' \text{ of } [2w] \\ \pi_V\to\pi\to\pi'\to\pi_E \text{ is a } d\text{-geodesic}}} \mu_{\mathrm{NC}}(\pi,\pi')q(\pi_V \vee \pi)D(\pi_E \vee \pi'). \tag{2.5.10}$$

Here, π and π' are pairings of $[2w]$ that may coincide with each other and/or with π_E.

Finally, we apply (2.5.6) to express $\mu_{\mathrm{NC}}(\pi,\pi')$ also as a summation over geodesic paths of pairings from π to π', giving

$$\lim_{n\to\infty}\mathbb{E}[\mathrm{val}_T(\mathbf{W};\mathbf{x}_{1:k})]=\sum_{j\geq 0}\sum_{\substack{\text{distinct pairings }\pi_0,\ldots,\pi_j\text{ of }[2w]\\ \pi_V\to\pi_0\to\ldots\to\pi_j\to\pi_E\text{ is a }d\text{-geodesic}}}(-1)^j q(\pi_V\vee\pi_0)D(\pi_E\vee\pi_j).$$

We have set $\pi_0=\pi$ and $\pi_j=\pi'$, and the terms of the sum with $j=0$ correspond to $\pi=\pi'$. This shows that the stated form is the almost-sure limit of $\mathbb{E}[\mathrm{val}_T(\mathbf{W};\mathbf{x}_{1:k})]$ where $\mathbb{E}$ is the expectation over $\mathbf{O}$. Comparing with the result of Lemma 2.2.14, we conclude that this must be lim-val$_T(X_{1:k},\mathcal{D}_{\mathrm{diag}})$. ■

Proof of Proposition 2.2.16(b). By the universality established in Lemma 2.2.14, it suffices to check that the limit of $\mathbb{E}[\mathrm{val}_T(\mathbf{W};\mathbf{x}_{1:k})]$ for orthogonally invariant matrices $\mathbf{W}$, as computed in the preceding Proposition 2.5.1, equals 0 under the given conditions.

The given condition $\frac{1}{n}\operatorname{Tr}\mathbf{W}\to 0$ implies $\mathbb{E}[D]=0$. If $\pi_E\vee\pi'$ has any block containing only the two vertex-edge pairs for a single edge, then this implies $D(\pi_E\vee\pi')=0$ in (2.5.3). Otherwise, each block must correspond to at least two edges, so $|\pi_E\vee\pi'|\leq w/2$. Similarly, if $\pi_V\vee\pi$ is such that any block contains the vertex-edge pairs for only a single vertex, then the condition (2.2.13) implies $q(\pi)=0$ in (2.5.2). Otherwise,each block must correspond to at least two vertices, so $|\pi_V\vee\pi|\leq(w+1)/2$. Thus if $q(\pi_V\vee\pi)D(\pi_E\vee\pi')\neq 0$, then

$$\frac{1}{n}\cdot n^{|\pi_V\vee\pi|}\cdot n^{|\pi_E\vee\pi'|}\cdot n^{-w-d(\pi,\pi')}\leq n^{-1+(w+1)/2+w/2-w}\leq n^{-1/2}.$$

Applying this to (2.5.9), we get $\mathbb{E}[\mathrm{val}_T(\mathbf{W};\mathbf{x}_{1:k})]\to 0$ as desired. ■

2.6 Details for rectangular matrices

In Section 2.6.1, we provide further details on the Onsager corrections and state evolutions of AMP algorithms for rectangular matrices. In Section 2.6.2, we state and prove a result analogous to Proposition 2.2.7 in the symmetric setting, providing examples of matrices $\mathbf{W} \in \mathbb{R}^{m\times n}$ that satisfy the generalized invariance conditions of Definition 2.2.20.

The remaining subsections prove Theorems 2.2.21 and 2.2.22 on AMP universality: In Section 2.6.3, we introduce a definition of tensor networks with dimensions alternating between m and n, and show that AMP universality may be reduced to universality of the values of such tensor networks. Then, in Section 2.6.4 and 2.6.5, we establish universality of the tensor network values for the classes of generalized white noise and rectangular generalized invariant matrices.

2.6.1 Onsager corrections and state evolution

In an AMP algorithm, the coefficients $\{a_{ts}\}$ and $\{b_{ts}\}$ in (2.2.14) are chosen so that $\{\mathbf{y}_t\}$ and $\{\mathbf{z}_t\}$ are described by simple state evolutions in the asymptotic limit as $m,n\to\infty$ with $m/n\to\gamma\in(0,\infty)$. When $\mathbf{W}$ has i.i.d. $\mathcal{N}(0,1/n)$ entries, this may be done by setting $\mathbf{\Omega}_1=\gamma\cdot\mathbb{E}[U_1^2]\in\mathbb{R}^{1\times 1}$ and iteratively defining

$$\mathbf{\Sigma}_t=(\mathbb{E}[V_rV_s])_{r,s=1}^t\in\mathbb{R}^{t\times t},\qquad \mathbf{\Omega}_{t+1}=(\gamma\cdot\mathbb{E}[U_rU_s])_{r,s=1}^{t+1}\in\mathbb{R}^{(t+1)\times(t+1)},\tag{2.6.1}$$

where we set $Z_{1:t}\sim\mathcal{N}(0,\mathbf{\Omega}_t)$ independent of $G_{1:\ell}$; $V_s=v_s(Z_{1:s},G_{1:\ell})$ for each $s=1,\ldots,t$; $Y_{1:t}\sim\mathcal{N}(0,\mathbf{\Sigma}_t)$ independent of $(U_1,F_{1:k})$; and $U_{s+1}=u_{s+1}(Y_{1:s},F_{1:k})$ for each $s=1,\ldots,t$. We then define a_{ts} and b_{ts} as

$$a_{ts}=\mathbb{E}[\partial_s v_t(Z_{1:t},G_{1:\ell})],\qquad b_{ts}=\gamma\cdot\mathbb{E}[\partial_s u_t(Y_{1:(t-1)},F_{1:k})].\tag{2.6.2}$$

We call (2.6.1) and (2.6.2) the *white noise prescriptions* for $\mathbf{\Omega}_t,\mathbf{\Sigma}_t$ and a_{ts},b_{ts}. Results of [BM11, JM13] imply the state evolutions, for Lipschitz functions $v_t(\cdot)$ and $u_t(\cdot)$, almost surely as $n \to \infty$ for any fixed $t \geq 1$,

$$(\mathbf{u}_1, \mathbf{f}_1, \ldots, \mathbf{f}_k, \mathbf{y}_1, \ldots, \mathbf{y}_t) \xrightarrow{W} (U_1, F_1, \ldots, F_k, Y_1, \ldots, Y_t),$$
$$(\mathbf{g}_1, \ldots, \mathbf{g}_\ell, \mathbf{z}_1, \ldots, \mathbf{z}_t) \xrightarrow{W} (G_1, \ldots, G_\ell, Z_1, \ldots, Z_t).$$

For a bi-orthogonally invariant matrix $\mathbf{W} = \mathbf{O}\mathbf{D}\mathbf{Q}^\top \in \mathbb{R}^{m\times n}$, let $\mathbf{D} = \operatorname{diag}(\mathbf{d}) \in \mathbb{R}^{m\times n}$ be the matrix of singular values where $\mathbf{d} \in \mathbb{R}^{\min(m,n)}$, and define

$$\bar{\mathbf{d}} = \mathbf{d} \in \mathbb{R}^m \text{ if } m \leq n, \qquad \bar{\mathbf{d}} = (\mathbf{d}, 0, \ldots, 0) \in \mathbb{R}^m \text{ if } m > n \tag{2.6.3}$$

Suppose that $\bar{\mathbf{d}} \xrightarrow{W} D$ as $m, n \to \infty$. We refer to D as the limit singular value distribution of $\mathbf{W}$. The above may then be extended as follows: Set $\mathbf{\Omega}_1 = \gamma \cdot \mathbb{E}[D^2] \cdot \mathbb{E}[U_1^2]$. For two continuous functions $\mathbf{\Sigma}_t(\cdot)$ and $\mathbf{\Omega}_{t+1}(\cdot)$ whose forms depend only on γ and the law of D, define iteratively

$$\begin{aligned}\mathbf{\Sigma}_t = \mathbf{\Sigma}_t\Big(&\{\mathbb{E}[V_rV_s], \mathbb{E}[U_rU_s]\}_{r,s\leq t},\\ &\{\mathbb{E}[\partial_r v_s(Z_{1:s}, G_{1:\ell})]\}_{r\leq s\leq t}, \{\mathbb{E}[\partial_r u_{s+1}(Y_{1:s}, F_{1:k})]\}_{r\leq s<t}\Big),\end{aligned} \tag{2.6.4a}$$

$$\begin{aligned}\mathbf{\Omega}_{t+1} = \mathbf{\Omega}_{t+1}\Big(&\{\mathbb{E}[V_rV_s]\}_{r,s\leq t}, \{\mathbb{E}[U_rU_s]\}_{r,s\leq t+1},\\ &\{\mathbb{E}[\partial_r v_s(Z_{1:s}, G_{1:\ell})], \mathbb{E}[\partial_r u_{s+1}(Y_{1:s}, F_{1:k})]\}_{r\leq s\leq t}\Big).\end{aligned} \tag{2.6.4b}$$

Then for continuous functions $a_{ts}(\cdot)$ and $b_{ts}(\cdot)$ whose forms also depend only on γ and the law of D, define

$$\begin{aligned}a_{ts} = a_{ts}\Big(&\{\mathbb{E}[V_qV_r], \mathbb{E}[U_qU_r]\}_{q,r\leq t},\\ &\{\mathbb{E}[\partial_q v_r(Z_{1:r}, G_{1:\ell})]\}_{q\leq r\leq t}, \{\mathbb{E}[\partial_q u_{r+1}(Y_{1:r}, F_{1:k})]\}_{q\leq r<t}\Big),\end{aligned} \tag{2.6.5a}$$

$$\begin{aligned}b_{ts} = b_{ts}\Big(&\{\mathbb{E}[V_qV_r]\}_{q,r<t}, \{\mathbb{E}[U_qU_r]\}_{q,r\leq t},\\ &\{\mathbb{E}[\partial_q v_r(Z_{1:r}, G_{1:\ell})], \mathbb{E}[\partial_q u_{r+1}(Y_{1:r}, F_{1:k})]\}_{q\leq r<t}\Big).\end{aligned} \tag{2.6.5b}$$

We call (2.6.4) and (2.6.5) the *bi-orthogonally invariant prescriptions* for $\mathbf{\Omega}_t, \mathbf{\Sigma}_t$ and a_{ts}, b_{ts}, and we refer to [Fan22, Section 5] for their exact forms. For weakly differentiable functions $u_t(\cdot), v_t(\cdot)$ with derivatives having at most polynomial growth, it is shown in [Fan22] that the iterates of (2.2.14) satisfy the state evolution, almost surely as $m, n \to \infty$ for any fixed $t \geq 1$,

$$(\mathbf{u}_1, \mathbf{f}_1, \ldots, \mathbf{f}_k, \mathbf{y}_1, \ldots, \mathbf{y}_t) \xrightarrow{W} (U_1, F_1, \ldots, F_k, Y_1, \ldots, Y_t),$$
$$(\mathbf{g}_1, \ldots, \mathbf{g}_\ell, \mathbf{z}_1, \ldots, \mathbf{z}_t) \xrightarrow{W} (G_1, \ldots, G_\ell, Z_1, \ldots, Z_t).$$

2.6.2 Sufficient conditions for generalized invariance

Proposition 2.6.1. *Let $\mathbf{W} \in \mathbb{R}^{m \times n}$ have singular values $\mathbf{d} \in \mathbb{R}^{\min(m,n)}$, such that $\bar{\mathbf{d}}$ defined by (2.6.3) satisfies $\bar{\mathbf{d}} \xrightarrow{W} D$ almost surely as $m, n \to \infty$ with $m/n \to \gamma \in (0, \infty)$. Suppose D has finite moments of all orders.*

(a) If $\mathbf{W}$ is bi-orthogonally invariant in law, then $\mathbf{W}$ is a rectangular generalized invariant matrix in the sense of Definition 2.2.20, and its limit diagonal distribution $\mathcal{D}_{\mathrm{diag}}$ is determined uniquely by γ and the law of D.

(b) Suppose that either

1. *$\mathbf{W} = \mathbf{ODQ}^\top$ where $\mathbf{D} = \mathrm{diag}(\mathbf{d}) \in \mathbb{R}^{m \times n}$, $\mathbf{O} = \mathbf{\Pi}_U \mathbf{H} \mathbf{\Pi}_E$, and $\mathbf{Q} = \mathbf{\Pi}_V \mathbf{K} \mathbf{\Pi}_F$. Here, $\mathbf{\Pi}_U, \mathbf{\Pi}_E \in \mathbb{R}^{m \times m}$ and $\mathbf{\Pi}_V, \mathbf{\Pi}_F \in \mathbb{R}^{n \times n}$ are uniformly random signed permutations independent of each other and of $(\mathbf{D}, \mathbf{H}, \mathbf{K})$, and $\mathbf{H} \in \mathbb{R}^{m \times m}$ and $\mathbf{K} \in \mathbb{R}^{n \times n}$ are orthogonal matrices whose entries satisfy (2.2.8) for any fixed $\varepsilon > 0$, almost surely for all large m, n.*

2. *$\mathbf{W} = \mathbf{\Pi}_U \mathbf{M} \mathbf{\Pi}_V^\top$ such that $\mathbf{\Pi}_U \in \mathbb{R}^{m \times m}$ and $\mathbf{\Pi}_V \in \mathbb{R}^{n \times n}$ are uniformly random signed permutations independent of $\mathbf{M} \in \mathbb{R}^{m \times n}$. For any integer $\nu \geq 0$ and any fixed $\varepsilon > 0$, the matrices $(\mathbf{MM}^\top)^\nu$, $(\mathbf{M}^\top \mathbf{M})^\nu$, and $(\mathbf{MM}^\top)^\nu \mathbf{M}$*

satisfy

$$\max_{\alpha\in[m]}\left|(MM^\top)^\nu[\alpha,\alpha]-\tfrac{1}{m}\operatorname{Tr}(\mathbf{M}\mathbf{M}^\top)^\nu\right|<n^{-1/2+\varepsilon},$$
$$\max_{\alpha\neq\beta}\left|(MM^\top)^\nu[\alpha,\beta]\right|<n^{-1/2+\varepsilon},$$
$$\max_{i\in[n]}\left|(M^\top M)^\nu[i,i]-\tfrac{1}{n}\operatorname{Tr}(\mathbf{M}^\top\mathbf{M})^\nu\right|<n^{-1/2+\varepsilon}, \tag{2.6.6}$$
$$\max_{i\neq j}\left|(M^\top M)^\nu[i,j]\right|<n^{-1/2+\varepsilon},$$
$$\max_{\alpha\in[m],i\in[n]}\left|(MM^\top)^\nu M[\alpha,i]\right|<n^{-1/2+\varepsilon}$$

almost surely for all large m,n.

Then $\mathbf{W}$ is a rectangular generalized invariant matrix in the sense of Definition 2.2.20, and its limit diagonal distribution $\mathcal{D}_{\mathrm{diag}}$ coincides with that of the bi-orthogonally invariant matrix in part (a).

To prove Proposition 2.6.1, we first slightly relax the assumptions in Lemma 2.4.1 to obtain the following lemma for the rectangular setting.

Lemma 2.6.2. *Let $\mathbf{M}\in\mathbb{R}^{m\times n}$ be a rectangular matrix having singular values $\mathbf{d}\in\mathbb{R}^{\min(m,n)}$. Suppose $\bar{\mathbf{d}}$ defined by (2.6.3) satisfies $\bar{\mathbf{d}}\overset{W}{\to}D$ almost surely as $m,n\to\infty$ with $m/n\to\gamma\in(0,\infty)$, and D has finite moments of all orders. Suppose $\mathbf{M}$ satisfies* (2.6.6) *almost surely for all large m,n, and let $\widetilde{\mathbf{M}}$ be its symmetric embedding in (2.2.15). Then for any $p(\mathbf{x})\in\Delta\langle\mathbf{x},\mathbf{I}_m,\mathbf{I}_n\rangle$,*

(a) The limits

$$\lim_{m,n\to\infty}\frac{1}{m}\sum_{i=1}^{m}p(\widetilde{\mathbf{M}})[i,i],\qquad \lim_{m,n\to\infty}\frac{1}{n}\sum_{i=m+1}^{m+n}p(\widetilde{\mathbf{M}})[i,i]$$

both exist almost surely, are finite, and depend only on γ and the law of D.

(b) For any $\varepsilon>0$, almost surely for all large m,n,

$$\max_{1\le i\le m}\left|p(\widetilde{\mathbf{M}})[i,i]-\frac{1}{m}\sum_{j=1}^{m}p(\widetilde{\mathbf{M}})[j,j]\right|<n^{-1/2+\varepsilon},$$

$$\max_{m+1 \le i \le m+n} \left| p(\widetilde{\mathbf{M}})[i,i] - \frac{1}{n} \sum_{j=m+1}^{m+n} p(\widetilde{\mathbf{M}})[j,j] \right| < n^{-1/2+\varepsilon},$$

$$\max_{i \neq j \in [n+m]} |p(\widetilde{\mathbf{M}})[i,j]| < n^{-1/2+\varepsilon}.$$

Proof. The proof is similar to that of Lemma 2.4.1. Analogously to (2.4.1), we may represent

$$p(\mathbf{x}) = w_1(\mathbf{x})\Delta(p_1(\mathbf{x}))w_2(\mathbf{x})\Delta(p_2(\mathbf{x})) \cdots w_L(\mathbf{x})\Delta(p_L(\mathbf{x}))w_{L+1}(\mathbf{x}) \tag{2.6.7}$$

where each $w_\ell(\mathbf{x})$ is a word in $\{\mathbf{x}, \mathbf{I}_m, \mathbf{I}_n\}$. We again define the depth $\delta(p)$ as $\delta(p) = 0$ if $L = 0$ and $\delta(p) = 1 + \max_{\ell=1}^{L} \delta(p_\ell)$ if $L \ge 1$, and we induct on the value of $\delta(p)$. For the base case $\delta(p) = 0$, i.e. $p(\mathbf{x}) = w_1(\mathbf{x})$, we note that $p(\widetilde{\mathbf{M}})$ must be either a block diagonal matrix or a block off-diagonal matrix, and in the first case, it must have its upper-left block equal to $(\mathbf{M}\mathbf{M}^\top)^\nu$ or 0, lower-right block equal to $(\mathbf{M}^\top\mathbf{M})^\nu$ or 0, while in the second case, it must have upper-right block equal to $(\mathbf{M}\mathbf{M}^\top)^\nu\mathbf{M}$ or 0, and lower-left block equal to $\mathbf{M}^\top(\mathbf{M}\mathbf{M}^\top)^\nu$ or 0 for some (possibly different) integer values $\nu \ge 0$. Then claim (a) holds by the convergence $\bar{\mathbf{d}} \xrightarrow{W} D$, and claim (b) holds by the conditions in (2.6.6).

Next, suppose inductively the lemma is true for all $p(\mathbf{x}) \in \Delta\langle \mathbf{x}, \mathbf{I}_m, \mathbf{I}_n \rangle$ such that $\delta(p) \le K$, and consider any $p(\mathbf{x}) \in \Delta\langle \mathbf{x}, \mathbf{I}_m, \mathbf{I}_n \rangle$ with $\delta(p) = K+1$, of the form (2.6.7). Fix any $i, j \in [m+n]$ and write $i_0 \equiv i$ and $i_{L+1} \equiv j$. Then $p(\widetilde{\mathbf{M}})[i,j]$ may be expressed as

$$p(\widetilde{\mathbf{M}})[i,j] = \sum_{\mathbf{i} \in [m+n]^L} w_1(\widetilde{\mathbf{M}})[i_0, i_1] p_1(\widetilde{\mathbf{M}})[i_1, i_1] w_2(\widetilde{\mathbf{M}})[i_1, i_2] \cdots p_L(\widetilde{\mathbf{M}})[i_L, i_L] w_{L+1}(\widetilde{\mathbf{M}})[i_L, i_{L+1}].$$

Let us abbreviate $\mathrm{Tr}_m, \mathrm{Tr}_n$ for the traces of the upper-left $m \times m$ and lower-right $n \times n$ submatrices, respectively. Then by the induction hypothesis, $p_\ell(\widetilde{\mathbf{M}}) = \mathrm{diag}(\frac{1}{m} \mathrm{Tr}_m\, p_\ell(\widetilde{\mathbf{M}}) \cdot \mathrm{Id}_m, \frac{1}{n} \mathrm{Tr}_n\, p_\ell(\widetilde{\mathbf{M}}) \cdot \mathrm{Id}_n) + \mathbf{E}_\ell$, where Id_m and Id_n are $m \times m$ and $n \times n$ identity matrices and $\mathbf{E}_\ell$ satisfies $\max_{i,j\in[m+n]}|E_\ell[i,j]| < n^{-1/2+\varepsilon}$ almost surely for all large m,n.

For $\mathcal{J}\subseteq[L]$, we write $[L]\setminus\mathcal{J}=\bar{\mathcal{J}}_1\sqcup\bar{\mathcal{J}}_2$ where $\bar{\mathcal{J}}_1$ will contain indices of $\mathbf{i}$ taking values in $\{1,\dots,m\}$ and $\bar{\mathcal{J}}_2$ will contain indices of $\mathbf{i}$ taking values in $\{m+1,\dots,m+n\}$. Further define $\mathcal{I}(\bar{\mathcal{J}}_1,\bar{\mathcal{J}}_2)=\{\mathbf{i}\in[m+n]^L : i_\ell\le m \text{ for all } \ell\in\bar{\mathcal{J}}_1 \text{ and } i_\ell\ge m+1 \text{ for all } \ell\in\bar{\mathcal{J}}_2\}$. Then

$$\begin{aligned}
&p(\widetilde{\mathbf{M}})[i,j]\\
&=\sum_{\mathbf{i}\in[m+n]^L}\prod_{\ell=1}^{L+1}w_\ell(\widetilde{\mathbf{M}})[i_{\ell-1},i_\ell]\\
&\qquad\qquad\cdot\prod_{\ell=1}^{L}\left(\frac{1}{m}\operatorname{Tr}_m p_\ell(\widetilde{\mathbf{M}})\cdot\mathbf{1}_{1\le i_\ell\le m}+\frac{1}{n}\operatorname{Tr}_n p_\ell(\widetilde{\mathbf{M}})\cdot\mathbf{1}_{m+1\le i_\ell\le m+n}+E_\ell[i_\ell,i_\ell]\right)\\
&=\sum_{\mathcal{J}\sqcup\bar{\mathcal{J}}_1\sqcup\bar{\mathcal{J}}_2=[L]}\left(\prod_{\ell\in\bar{\mathcal{J}}_1}\frac{1}{m}\operatorname{Tr}_m p_\ell(\widetilde{\mathbf{M}})\right)\\
&\qquad\qquad\cdot\left(\prod_{\ell\in\bar{\mathcal{J}}_2}\frac{1}{n}\operatorname{Tr}_n p_\ell(\widetilde{\mathbf{M}})\right)\sum_{\mathbf{i}\in\mathcal{I}(\bar{\mathcal{J}}_1,\bar{\mathcal{J}}_2)}\prod_{\ell=1}^{L+1}w_\ell(\widetilde{\mathbf{M}})[i_{\ell-1},i_\ell]\prod_{\ell\in\mathcal{J}}E_\ell[i_\ell,i_\ell].
\end{aligned}$$

Similarly to (2.4.2), by part (a) of the induction hypothesis we have $\prod_{\ell\in\bar{\mathcal{J}}_1}\frac{1}{m}\operatorname{Tr}_m p_\ell(\widetilde{\mathbf{M}})=M_{\bar{\mathcal{J}}_1}+\xi_{\bar{\mathcal{J}}_1}$ and $\prod_{\ell\in\bar{\mathcal{J}}_2}\frac{1}{n}\operatorname{Tr}_n p_\ell(\widetilde{\mathbf{M}})=M_{\bar{\mathcal{J}}_2}+\xi_{\bar{\mathcal{J}}_2}$, where $M_{\bar{\mathcal{J}}_1},M_{\bar{\mathcal{J}}_2}$ are finite limit values depending only on γ and the law of D, and the convergence $\xi_{\bar{\mathcal{J}}_1},\xi_{\bar{\mathcal{J}}_2}\to 0$ holds uniformly over $i,j\in[m+n]$. This gives

$$p(\widetilde{\mathbf{M}})[i,j]=\sum_{\mathcal{J}\sqcup\bar{\mathcal{J}}_1\sqcup\bar{\mathcal{J}}_2=[L]}(M_{\bar{\mathcal{J}}_1}+\xi_{\bar{\mathcal{J}}_1})(M_{\bar{\mathcal{J}}_2}+\xi_{\bar{\mathcal{J}}_2})\sum_{\mathbf{i}\in\mathcal{I}(\bar{\mathcal{J}}_1,\bar{\mathcal{J}}_2)}\prod_{\ell=1}^{L+1}w_\ell(\widetilde{\mathbf{M}})[i_{\ell-1},i_\ell]\prod_{\ell\in\mathcal{J}}E_\ell[i_\ell,i_\ell]. \tag{2.6.8}$$

Next, let us write explicitly $\mathcal{J}=\{\ell_1,\dots,\ell_{|\mathcal{J}|}\}$ where $1\le\ell_1<\ldots<\ell_{|\mathcal{J}|}\le L$, and set $\ell_0=0$ and $\ell_{|\mathcal{J}|+1}=L+1$. We can contract the summation over indices $\{i_\ell:\ell\notin\mathcal{J}\}$, incorporating the constraints $i_\ell\le m$ for $\ell\in\bar{\mathcal{J}}_1$ and $i_\ell\ge m+1$ for $\ell\in\bar{\mathcal{J}}_2$ by introducing copies of $\mathbf{I}_m$ and $\mathbf{I}_n$. For example, if $\ell_{\rho-1},\ell_\rho\in\mathcal{J}$ where $\ell_\rho=\ell_{\rho-1}+3$, and $\ell_{\rho-1}+1 \in \bar{\mathcal{J}}_1$ and $\ell_{\rho-1}+2 \in \bar{\mathcal{J}}_2$, then

$$\sum_{i_{\ell_{\rho-1}+1}=1}^{m} \sum_{i_{\ell_{\rho-1}+2}=m+1}^{m+n} w_{\ell_{\rho-1}+1}(\widetilde{\mathbf{M}})[i_{\ell_{\rho-1}}, i_{\ell_{\rho-1}+1}] w_{\ell_{\rho-1}+2}(\widetilde{\mathbf{M}})[i_{\ell_{\rho-1}+1}, i_{\ell_{\rho-1}+2}] w_{\ell_\rho}(\widetilde{\mathbf{M}})[i_{\ell_{\rho-1}+2}, i_{\ell_\rho}]$$
$$= \Big(w_{\ell_{\rho-1}+1}(\widetilde{\mathbf{M}}) \cdot \mathbf{I}_m \cdot w_{\ell_{\rho-1}+2}(\widetilde{\mathbf{M}}) \cdot \mathbf{I}_n \cdot w_{\ell_\rho}(\widetilde{\mathbf{M}})\Big)[i_{\ell_{\rho-1}}, i_{\ell_\rho}].$$

Applying this argument gives, for some new words $w_1^{\bar{\mathcal{J}}_1, \bar{\mathcal{J}}_2}(\mathbf{x}), \ldots, w_{|\mathcal{J}|+1}^{\bar{\mathcal{J}}_1, \bar{\mathcal{J}}_2}(\mathbf{x})$ in $\{\mathbf{x}, \mathbf{I}_m, \mathbf{I}_n\}$ whose definitions depend on $\bar{\mathcal{J}}_1, \bar{\mathcal{J}}_2$,

$$p(\widetilde{\mathbf{M}})[i,j] = \sum_{\mathcal{J} \sqcup \bar{\mathcal{J}}_1 \sqcup \bar{\mathcal{J}}_2 = [L]} (M_{\bar{\mathcal{J}}_1} + \xi_{\bar{\mathcal{J}}_1})(M_{\bar{\mathcal{J}}_2} + \xi_{\bar{\mathcal{J}}_2}) \sum_{\mathbf{i} \in [m+n]^{\mathcal{J}}} \prod_{\rho=1}^{|\mathcal{J}|+1} w_{R_\rho}^{\bar{\mathcal{J}}_1, \bar{\mathcal{J}}_2}(\widetilde{\mathbf{M}})[i_{\ell_{\rho-1}}, i_{\ell_\rho}] \prod_{\ell \in \mathcal{J}} E_\ell[i_\ell, i_\ell].$$

The same argument as in Lemma 2.4.1 then yields that this is at most $Cn^{-1/2+\varepsilon(2|\mathcal{J}|+1)}$ unless $i = j$ and $\mathcal{J} = \emptyset$. Thus,

$$\left| p(\widetilde{\mathbf{M}})[i,j] - \mathbf{1}\{i=j\} \sum_{\bar{\mathcal{J}}_1 \sqcup \bar{\mathcal{J}}_2 = [L]} (M_{\bar{\mathcal{J}}_1} + \xi_{\bar{\mathcal{J}}_1})(M_{\bar{\mathcal{J}}_2} + \xi_{\bar{\mathcal{J}}_2}) w_{R_1}^{\bar{\mathcal{J}}_1, \bar{\mathcal{J}}_2}(\widetilde{\mathbf{M}})[i,i] \right| \leq Cn^{-1/2+\varepsilon(2L+1)}.$$

In the case $i = j \leq m$ or $i = j \geq m+1$, further approximating $w_{R_1}^{\bar{\mathcal{J}}_1, \bar{\mathcal{J}}_2}(\widetilde{\mathbf{M}})[i,i]$ by $\frac{1}{m} \operatorname{Tr}_m w_{R_1}^{\bar{\mathcal{J}}_1, \bar{\mathcal{J}}_2}$ or $\frac{1}{n} \operatorname{Tr}_n w_{R_1}^{\bar{\mathcal{J}}_1, \bar{\mathcal{J}}_2}$ respectively and applying the same argument as in Lemma 2.4.1, we obtain the inductive claim (b), and averaging over $i = j \leq m$ and $i = j \geq m+1$ then shows the inductive claim (a). ■

Proof of Proposition 2.6.1. Lemma 2.6.2 implies that Definition 2.2.20 holds for matrices satisfying the description of part (b2), where the limit diagonal law $\mathcal{D}_{\text{diag}}$ depends only on γ and the law of D.

In the setting of part (a) where $\mathbf{W} = \mathbf{O}\mathbf{D}\mathbf{Q}^\top$ is bi-orthogonally invariant, we have the equalities in law $\mathbf{O} \overset{L}{=} \mathbf{\Pi}_U \mathbf{O} \mathbf{\Pi}_E$ and $\mathbf{Q} \overset{L}{=} \mathbf{\Pi}_V \mathbf{Q} \mathbf{\Pi}_F$ for uniformly random signed permutations $\mathbf{\Pi}_U, \mathbf{\Pi}_E, \mathbf{\Pi}_V, \mathbf{\Pi}_F$ independent of each other and of $\mathbf{O}, \mathbf{Q}$. The entries of $\mathbf{O}, \mathbf{Q}$ satisfy (2.2.8) almost surely for all large m, n by [Jia05, Theorem 1], so $\mathbf{W}$ is an example of the matrix model in part (b1).

To conclude the proof, it remains to verify that if $\mathbf{W}$ is a matrix described by part (b1), then it also is an example of the matrix model in part (b2). For this, write $\mathbf{W} = \mathbf{\Pi}_U \mathbf{M} \mathbf{\Pi}_V^\top$ where

$$\mathbf{M} = \mathbf{H}\mathbf{\Pi}_E \mathbf{D} \mathbf{\Pi}_F^\top \mathbf{K}^\top.$$

Observe that $\mathbf{M}^\top\mathbf{M} = \mathbf{K}\mathbf{\Pi}_F\mathbf{D}^\top\mathbf{D}\mathbf{\Pi}_F^\top\mathbf{K}^\top$ and $\mathbf{M}\mathbf{M}^\top = \mathbf{H}\mathbf{\Pi}_E\mathbf{D}\mathbf{D}^\top\mathbf{\Pi}_E^\top\mathbf{H}^\top$. Then for any integer $\nu \geq 0$, $(\mathbf{M}^\top\mathbf{M})^\nu$ and $(\mathbf{M}\mathbf{M}^\top)^\nu$ satisfy the conditions of (2.6.6) by the proof of Proposition 2.2.7 in the symmetric setting.

Now fix any integer $\nu \geq 0$ and consider

$$(\mathbf{M}\mathbf{M}^\top)^\nu\mathbf{M} = \mathbf{H}\mathbf{\Pi}_E(\mathbf{D}\mathbf{D}^\top)^\nu\mathbf{D}\mathbf{\Pi}_F^\top\mathbf{K}^\top.$$

We suppose for notational simplicity that $m \leq n$; the case $m \geq n$ is analogous. Write $\mathbf{\Pi}_E = \mathbf{\Xi}_E\mathbf{P}_E$ and $\mathbf{\Pi}_F = \mathbf{\Xi}_F\mathbf{P}_F$ for permutation and sign matrices defining $\mathbf{\Pi}_E, \mathbf{\Pi}_F$, and let σ_E, σ_F be the permutations of $[m], [n]$ such that $P_E[\beta, \sigma_E(\beta)] = 1$ and $P_F[j, \sigma_F(j)] = 1$ for all $\beta \in [m]$ and $j \in [n]$. Fixing $\alpha \in [m]$ and $i \in [n]$, let $\mathbf{h}, \mathbf{k}$ be the α^{th} row of $\mathbf{H}$ and i^{th} row of $\mathbf{K}$. Let $\tilde{\mathbf{D}} = (\mathbf{D}\mathbf{D}^\top)^\nu\mathbf{D} \in \mathbb{R}^{m\times n}$. Then

$$\Big((\mathbf{M}\mathbf{M}^\top)^\nu\mathbf{M}\Big)[\alpha, i] = \sum_{\beta\in[m]}\sum_{j\in[n]} h[\beta]\Xi_E[\beta,\beta]\tilde{D}[\sigma_E(\beta), \sigma_F(j)]\Xi_F[j,j]k[j],$$

so for any even power $p \geq 2$,

$$\begin{aligned}\mathbb{E}\left[\Big((\mathbf{M}\mathbf{M}^\top)^\nu\mathbf{M}\Big)[\alpha, i]^p\right] = \sum_{\boldsymbol{\beta}\in[m]^p}\sum_{\mathbf{j}\in[n]^p} &h[\beta_1]\dots h[\beta_p]k[j_1]\dots k[j_p]\\ &\cdot\mathbb{E}\Big[\Xi_E[\beta_1,\beta_1]\dots\Xi_E[\beta_p,\beta_p]\Big]\cdot\mathbb{E}\Big[\Xi_F[j_1,j_1]\dots\Xi_F[j_p,j_p]\Big]\\ &\cdot\mathbb{E}\Big[\tilde{D}[\sigma_E(\beta_1),\sigma_F(j_1)]\dots\tilde{D}[\sigma_E(\beta_p),\sigma_F(j_p)]\Big].\end{aligned}$$

Let $\mathcal{P}$ be the lattice of partitions of $[p]$, and let $\pi(\boldsymbol{\beta}), \pi(\mathbf{j}) \in \mathcal{P}$ be those partitions induced by $\boldsymbol{\beta}, \mathbf{j}$. Observe that $\mathbb{E}[\Xi_E[\beta_1,\beta_1]\dots\Xi_E[\beta_p,\beta_p]] = 1$ if $\pi(\boldsymbol{\beta})$ is even (i.e. all blocks have even cardinality) and 0 otherwise, and similarly for $\mathbf{\Xi}_F$. Since $\tilde{\mathbf{D}}$ is diagonal, the product $\tilde{D}[\sigma_E(\beta_1), \sigma_F(j_1)] \dots \tilde{D}[\sigma_E(\beta_p), \sigma_F(j_p)]$ is zero unless $\sigma_E(\beta_a) = \sigma_F(j_a)$ for every $a = 1, \dots, p$, which can occur only when $\pi(\boldsymbol{\beta}) = \pi(\mathbf{j})$. Fixing $\boldsymbol{\beta}, \mathbf{j}$ for which $\pi(\boldsymbol{\beta}) = \pi(\mathbf{j}) = \pi$ and supposing $m \leq n$, observe that

$$\begin{aligned} &\mathbb{E}\Big[\tilde{D}[\sigma_E(\beta_1), \sigma_F(j_1)] \dots \tilde{D}[\sigma_E(\beta_p), \sigma_F(j_p)]\Big] \\ &\quad = \frac{(m - |\pi|)!}{m!} \cdot \frac{(n - |\pi|)!}{n!} \sum_{\boldsymbol{\gamma} \in [m]^p : \pi(\boldsymbol{\gamma}) = \pi} \tilde{d}[\gamma_1] \dots \tilde{d}[\gamma_p] \end{aligned}$$

where we set $\tilde{d}[\gamma] = \tilde{D}[\gamma, \gamma]$, because

- Given any σ_E, the probability over σ_F that $\sigma_F(j_a) = \sigma_E(\beta_a)$ for every $a = 1, \dots, p$ is $(n - |\pi|)!/n!$.
- The expectation of $\tilde{D}[\sigma_E(\beta_1), \sigma_E(\beta_1)] \dots \tilde{D}[\sigma_E(\beta_p), \sigma_E(\beta_p)]$ over σ_E is a uniform average over all $m!/(m - |\pi|)!$ relabelings of the $|\pi|$ distinct indices of $\boldsymbol{\beta}$.

Thus we have

$$\begin{aligned} \mathbb{E}\left[\Big((\mathbf{M}\mathbf{M}^\top)^\nu \mathbf{M}\Big)[\alpha, i]^p\right] = \sum_{\text{even } \pi \in \mathcal{P}} \frac{(m - |\pi|)!}{m!} \cdot \frac{(n - |\pi|)!}{n!} \sum_{\boldsymbol{\beta} \in [m]^p : \pi(\boldsymbol{\beta}) = \pi} h[\beta_1] \dots h[\beta_p] \\ \times \sum_{\mathbf{j} \in [n]^p : \pi(\mathbf{j}) = \pi} k[j_1] \dots k[j_p] \sum_{\boldsymbol{\gamma} \in [m]^p : \pi(\boldsymbol{\gamma}) = \pi} \tilde{d}[\gamma_1] \dots \tilde{d}[\gamma_p] \end{aligned}$$

Lemma 2.3.3 and the assumption $\bar{\mathbf{d}} \overset{W}{\to} D$ imply $|\sum_{\boldsymbol{\gamma} \in [m]^p : \pi(\boldsymbol{\gamma}) = \pi} \tilde{d}[\gamma_1] \dots \tilde{d}[\gamma_p]| \leq Cn^{|\pi|} \leq Cn^{p/2}$, where $|\pi| \leq p/2$ because π is even. Applying (2.2.8) for $\mathbf{H}$ and $\mathbf{K}$, we have

$$\begin{aligned} n^{-|\pi|} \sum_{\boldsymbol{\beta} \in [m]^p : \pi(\boldsymbol{\beta}) = \pi} |h[\beta_1] \dots h[\beta_p]| \leq Cn^{p(-1/2 + \varepsilon)}, \\ n^{-|\pi|} \sum_{\mathbf{j} \in [n]^p : \pi(\mathbf{j}) = \pi} |k[j_1] \dots k[j_p]| \leq Cn^{p(-1/2 + \varepsilon)}. \end{aligned}$$

Thus $\mathbb{E}[((\mathbf{M}\mathbf{M}^\top)^\nu \mathbf{M})[\alpha, i]^p] \leq Cn^{-p/2 + 2p\varepsilon}$. Choosing even $p \geq 2$ sufficiently large, this implies by Markov's inequality and a union bound over all $\alpha \in [m]$ and $i \in [n]$ that

$$\max_{\alpha \in [m], i \in [n]} |((\mathbf{M}\mathbf{M}^\top)^\nu \mathbf{M})[\alpha, i]| < n^{-1/2+3\varepsilon}$$

almost surely for all large m, n. Then $(\mathbf{M}\mathbf{M}^\top)^\nu \mathbf{M}$ also satisfies (2.6.6). So any $\mathbf{W}$ as described by part (b1) also satisfies the conditions of part (b2), concluding the proof. ■

2.6.3 Reduction to tensor networks

Definition 2.6.3. An *alternating diagonal tensor network* $T = (\mathcal{U}, \mathcal{V}, \mathcal{E}, \{p_u\}_{u \in \mathcal{U}}, \{q_v\}_{v \in \mathcal{V}})$ in (k, ℓ) variables is an undirected tree graph with vertices $\mathcal{U} \sqcup \mathcal{V}$ and edges $\mathcal{E} \subset \mathcal{U} \times \mathcal{V}$ where

- Each edge connects a vertex $u \in \mathcal{U}$ to a vertex $v \in \mathcal{V}$.
- Each $u \in \mathcal{U}$ is labeled by a polynomial function $p_u : \mathbb{R}^k \to \mathbb{R}$.
- Each $v \in \mathcal{V}$ is labeled by a polynomial function $q_v : \mathbb{R}^\ell \to \mathbb{R}$.

The *value* of T on a rectangular matrix $\mathbf{W} \in \mathbb{R}^{m \times n}$ and vectors $\mathbf{x}_1, \ldots, \mathbf{x}_k \in \mathbb{R}^m$ and $\mathbf{y}_1, \ldots, \mathbf{y}_\ell \in \mathbb{R}^n$ is

$$\mathrm{val}_T(\mathbf{W}; \mathbf{x}_1, \ldots, \mathbf{x}_k; \mathbf{y}_1, \ldots, \mathbf{y}_\ell) = \frac{1}{n} \sum_{\boldsymbol{\alpha} \in [m]^{\mathcal{U}}} \sum_{\mathbf{i} \in [n]^{\mathcal{V}}} p_{\boldsymbol{\alpha}|T} \cdot q_{\mathbf{i}|T} \cdot W_{\boldsymbol{\alpha}, \mathbf{i}|T}$$

where, for each index tuple $\boldsymbol{\alpha} = (\alpha_u : u \in \mathcal{U}) \in [m]^{\mathcal{U}}$ and $\mathbf{i} = (i_v : v \in \mathcal{V}) \in [n]^{\mathcal{V}}$,

$$p_{\boldsymbol{\alpha}|T} = \prod_{u \in \mathcal{U}} p_u(x_1[\alpha_u], \ldots, x_k[\alpha_u]),\ q_{\mathbf{i}|T} = \prod_{v \in \mathcal{V}} q_v(y_1[i_v], \ldots, y_\ell[i_v]),\ W_{\boldsymbol{\alpha}, \mathbf{i}|T} = \prod_{(u,v) \in \mathcal{E}} W[\alpha_u, i_v].$$

This value may be understood as:

1. Associating to vertices $u \in \mathcal{U}$ and $v \in \mathcal{V}$ the diagonal tensors $\mathbf{T}_u = \mathrm{diag}(p_u(\mathbf{x}_1, \ldots, \mathbf{x}_k)) \in$

$\mathbb{R}^{m\times\cdots\times m}$ and $\mathbf{T}_v = \mathrm{diag}(q_v(\mathbf{y}_1,\ldots,\mathbf{y}_\ell)) \in \mathbb{R}^{n\times\cdots\times n}$, whose orders equal the degrees of u and v in the tree.

2. Associating to each edge the matrix $\mathbf{W}$.

3. Iteratively contracting all tensor-matrix-tensor products represented by edges of the tree, where each product involves a tensor in dimension m, $\mathbf{W} \in \mathbb{R}^{m\times n}$, and a tensor in dimension n.

For example, if $\mathcal{U} = \{1, 3, \ldots, w-1, w+1\}$ and $\mathcal{V} = \{2, 4, \ldots, w\}$ for an even integer w, and T is the line graph $1 - 2 - \ldots - w - (w+1)$, then $\mathbf{T}_1, \mathbf{T}_{w+1} \in \mathbb{R}^m$ are vectors, $\mathbf{T}_3, \mathbf{T}_5, \ldots, \mathbf{T}_{w-1} \in \mathbb{R}^{m\times m}$ and $\mathbf{T}_2, \mathbf{T}_4, \ldots, \mathbf{T}_w \in \mathbb{R}^{n\times n}$ are matrices of alternating dimensions, and the value is

$$\mathrm{val}_T(\mathbf{W}; \mathbf{x}_1, \ldots, \mathbf{x}_k; \mathbf{y}_1, \ldots, \mathbf{y}_\ell) = \frac{1}{n}\mathbf{T}_1^\top \mathbf{W}\mathbf{T}_2\mathbf{W}^\top\mathbf{T}_3\mathbf{W}\cdots\mathbf{W}\mathbf{T}_w\mathbf{W}^\top\mathbf{T}_{w+1}.$$

The following lemma is analogous to Lemma 2.2.12, and reduces the universality of AMP to the universality of values of alternating diagonal tensor networks.

Lemma 2.6.4. *Let $\mathbf{u}_1, \mathbf{f}_1, \ldots, \mathbf{f}_k \in \mathbb{R}^m$ and $\mathbf{g}_1, \ldots, \mathbf{g}_\ell \in \mathbb{R}^n$ satisfy Assumption 2.2.17. Let $\mathbf{W}, \mathbf{G} \in \mathbb{R}^{m\times n}$ be random matrices independent of $\mathbf{u}_1, \mathbf{f}_1, \ldots, \mathbf{f}_k, \mathbf{g}_1, \ldots, \mathbf{g}_\ell$ such that*

1. *$\mathbf{G} = \mathbf{O}\mathbf{D}\mathbf{Q}^\top$ is a bi-orthogonally invariant matrix such that $\mathbf{D} = \mathrm{diag}(\mathbf{d})$ and $\bar{\mathbf{d}} \xrightarrow{W} D$, where $\bar{\mathbf{d}}$ is defined by (2.6.3) and D is a compactly supported limit law with $\mathbb{E}[D^2] > 0$.*

2. *$\|\mathbf{W}\|_{\mathrm{op}} < C$ for a constant $C > 0$, almost surely for all large m, n.*

3. *For every alternating diagonal tensor network T in $(k+1, \ell)$ variables, almost surely as $m, n \to \infty$,*

$$\mathrm{val}_T(\mathbf{W}; \mathbf{u}_1, \mathbf{f}_1, \ldots, \mathbf{f}_k; \mathbf{g}_1, \ldots, \mathbf{g}_\ell) - \mathrm{val}_T(\mathbf{G}; \mathbf{u}_1, \mathbf{f}_1, \ldots, \mathbf{f}_k; \mathbf{g}_1, \ldots, \mathbf{g}_\ell) \to 0.$$

Let $v_t : \mathbb{R}^{t+\ell} \to \mathbb{R}$ and $u_{t+1} : \mathbb{R}^{t+k} \to \mathbb{R}$ be continuous functions satisfying the polynomial growth condition (2.1.1) for some order $p \geq 1$, and are Lipschitz in their first t arguments. Let $\{a_{ts}\}, \{b_{ts}\}, \{\mathbf{\Omega}_t\}$ and $\{\mathbf{\Sigma}_t\}$ be defined by the bi-orthogonally invariant prescriptions in (2.6.4) and (2.6.5) for the limit law D, where each $\mathbf{\Omega}_t$ and $\mathbf{\Sigma}_t$ is non-singular. Then the iterates (2.2.14) applied to $\mathbf{W}$ satisfy, almost surely as $n \to \infty$ for any fixed $t \geq 1$,

$$(\mathbf{u}_1, \mathbf{f}_1, \ldots, \mathbf{f}_k, \mathbf{y}_1, \ldots, \mathbf{y}_t) \xrightarrow{W_2} (U_1, F_1, \ldots, F_k, Y_1, \ldots, Y_t)$$
$$(\mathbf{g}_1, \ldots, \mathbf{g}_\ell, \mathbf{z}_1, \ldots, \mathbf{z}_t) \xrightarrow{W_2} (G_1, \ldots, G_\ell, Z_1, \ldots, Z_t)$$

where these limits have the same joint laws as described by the AMP state evolution for $\mathbf{G}$.

Lemma 2.6.4 may be proven in the same way as Lemma 2.2.12. For the same initialization $\tilde{\mathbf{u}}_1 = \mathbf{u}_1$ and vectors of side information $\mathbf{f}_1, \ldots, \mathbf{f}_k, \mathbf{g}_1, \ldots, \mathbf{g}_\ell$ as in the given Lipschitz AMP algorithm, we consider an auxiliary AMP algorithm with polynomial non-linearities

$$\begin{aligned} \tilde{\mathbf{z}}_t &= \mathbf{W}^\top \tilde{\mathbf{u}}_t - \sum_{s=1}^{t-1} \tilde{b}_{ts} \tilde{\mathbf{v}}_s \\ \tilde{\mathbf{v}}_t &= \tilde{v}_t(\tilde{\mathbf{z}}_1, \ldots, \tilde{\mathbf{z}}_t, \mathbf{g}_1, \ldots, \mathbf{g}_\ell) \\ \tilde{\mathbf{y}}_t &= \mathbf{W} \tilde{\mathbf{v}}_t - \sum_{s=1}^{t} \tilde{a}_{ts} \tilde{\mathbf{u}}_s \\ \tilde{\mathbf{u}}_{t+1} &= \tilde{u}_{t+1}(\tilde{\mathbf{y}}_1, \ldots, \tilde{\mathbf{y}}_t, \mathbf{f}_1, \ldots, \mathbf{f}_k) \end{aligned} \tag{2.6.9}$$

Fixing $\varepsilon > 0$, this is defined such that

1. Each coefficient $\tilde{a}_{ts}$ and $\tilde{b}_{ts}$ is defined by the polynomials $\{\tilde{u}_{t+1}(\cdot)\}$, $\{\tilde{v}_t(\cdot)\}$ and the bi-orthogonally invariant prescriptions (2.6.5).

2. Let $\widetilde{\mathbf{\Omega}}_t$, $\widetilde{\mathbf{\Sigma}}_t$ be the bi-orthogonally invariant prescriptions (2.6.4), and let $(U_1, F_{1:k}, \widetilde{Y}_{1:t})$

and $(G_{1:\ell}, \widetilde{Z}_{1:t})$ be the corresponding state evolutions. Then each polynomial $\tilde{u}_{t+1}(\cdot)$ and $\tilde{v}_t(\cdot)$ is chosen to satisfy

$$\mathbb{E}\Big[\Big(\tilde{u}_{t+1}(\widetilde{Y}_{1:t}, F_{1:k}) - u_{t+1}(\widetilde{Y}_{1:t}, F_{1:k})\Big)^2\Big] < \varepsilon$$
$$\mathbb{E}\Big[\Big(\tilde{v}_t(\widetilde{Z}_{1:t}, G_{1:\ell}) - v_t(\widetilde{Z}_{1:t}, G_{1:\ell})\Big)^2\Big] < \varepsilon$$

3. For any fixed arguments $y_{1:(t-1)}$, $f_{1:k}$, $z_{1:(1-t)}$, and $g_{1:\ell}$, the functions $y_t \mapsto \tilde{u}_{t+1}(y_{1:t}, f_{1:k})$ and $z_t \mapsto v_t(z_{1:t}, g_{1:\ell})$ have non-linear dependences in y_t and z_t.

Again we write the iterates as $\tilde{\mathbf{u}}_t(\mathbf{W}), \tilde{\mathbf{z}}_t(\mathbf{W}), \tilde{\mathbf{v}}_t(\mathbf{W}), \tilde{\mathbf{y}}_t(\mathbf{W})$ if we want to specify that the algorithm is applied to $\mathbf{W}$. Assumption 2.2.17, the given condition $\mathbb{E}[D^2] > 0$, and condition (3) above verify the conditions of [Fan22, Assumption 5.2]. Then [Fan22, Theorem 5.3] applies to show that, for each fixed $t \geq 1$, almost surely as $n \to \infty$,

$$(\mathbf{u}_1, \mathbf{f}_{1:k}, \tilde{\mathbf{y}}_{1:t}(\mathbf{G})) \xrightarrow{W} (U_1, F_{1:k}, \widetilde{Y}_{1:t}), \qquad (\mathbf{g}_{1:\ell}, \tilde{\mathbf{z}}_{1:t}(\mathbf{G})) \xrightarrow{W} (G_{1:\ell}, \widetilde{Z}_{1:t}) \tag{2.6.10}$$

when this algorithm is applied to the bi-orthogonally invariant matrix $\mathbf{G}$.

The following lemma, analogous to Lemma 2.3.11, shows that empirical averages of polynomial test functions evaluated on the iterates of this polynomial AMP algorithm may be decomposed as a sum of values of alternating diagonal tensor networks.

Lemma 2.6.5. *Fix any $t \geq 1$ and let $\tilde{\mathbf{u}}_1, \tilde{\mathbf{z}}_1, \tilde{\mathbf{v}}_1, \tilde{\mathbf{y}}_1, \ldots, \tilde{\mathbf{u}}_t, \tilde{\mathbf{z}}_t, \tilde{\mathbf{v}}_t, \tilde{\mathbf{y}}_t$ be the iterates of any algorithm of the form (2.6.9), where $\{\tilde{a}_{ts}, \tilde{b}_{ts}\}$ are scalar constants and $\tilde{u}_{t+1} : \mathbb{R}^{t+k} \to \mathbb{R}$ and $\tilde{v}_{t+1} : \mathbb{R}^{t+\ell} \to \mathbb{R}$ are polynomial functions applied row-wise. For any polynomials $p : \mathbb{R}^{2t+k} \to \mathbb{R}$ and $q : \mathbb{R}^{2t+l} \to \mathbb{R}$, and for two finite sets $\mathcal{F}_1$ and $\mathcal{F}_2$ of alternating diagonal tensor networks in $(k+1, \ell)$ variables,*

$$\langle p(\tilde{\mathbf{u}}_1, \ldots, \tilde{\mathbf{u}}_t, \tilde{\mathbf{y}}_1, \ldots, \tilde{\mathbf{y}}_t, \mathbf{f}_1, \ldots, \mathbf{f}_k) \rangle = \sum_{T \in \mathcal{F}_1} \mathrm{val}_T(\mathbf{W}; \mathbf{u}_1, \mathbf{f}_1, \ldots, \mathbf{f}_k; \mathbf{g}_1, \ldots, \mathbf{g}_\ell), \tag{2.6.11}$$

$$\langle q(\tilde{\mathbf{v}}_1, \ldots, \tilde{\mathbf{v}}_t, \tilde{\mathbf{z}}_1, \ldots, \tilde{\mathbf{z}}_t, \mathbf{g}_1, \ldots, \mathbf{g}_\ell) \rangle = \sum_{T \in \mathcal{F}_2} \mathrm{val}_T(\mathbf{W}; \mathbf{u}_1, \mathbf{f}_1, \ldots, \mathbf{f}_k; \mathbf{g}_1, \ldots, \mathbf{g}_\ell). \tag{2.6.12}$$

Proof of Lemma 2.6.5. We have

$$\langle p(\tilde{\mathbf{u}}_{1:t}, \tilde{\mathbf{y}}_{1:t}, \mathbf{f}_{1:k}) \rangle = \mathrm{val}_T(\mathbf{W}; \tilde{\mathbf{u}}_{1:t}, \tilde{\mathbf{y}}_{1:t}, \mathbf{f}_{1:k})$$

where T is a tensor network with only one vertex $\{u\} = \mathcal{U}$ and polynomial label $p_u = p$.

We claim that for any tensor network T in variables $(\tilde{y}_{1:t}, \tilde{u}_{1:t}, f_{1:k}; \tilde{z}_{1:t}, \tilde{v}_{1:t}, g_{1:\ell})$, there exists a set of tensor networks $\mathcal{F}$ such that

$$\mathrm{val}_T(\mathbf{W}; \tilde{\mathbf{y}}_{1:t}, \tilde{\mathbf{u}}_{1:t}, \tilde{\mathbf{f}}_{1:k}; \tilde{\mathbf{z}}_{1:t}, \tilde{\mathbf{v}}_{1:t}, \tilde{\mathbf{g}}_{1:\ell}) = \sum_{T' \in \mathcal{F}} \mathrm{val}_{T'}(\mathbf{W}; \tilde{\mathbf{y}}_{1:(t-1)}, \tilde{\mathbf{u}}_{1:t}, \tilde{\mathbf{f}}_{1:k}; \tilde{\mathbf{z}}_{1:t}, \tilde{\mathbf{v}}_{1:t}, \tilde{\mathbf{g}}_{1:\ell}). \tag{2.6.13}$$

To show this claim, recall $\tilde{\mathbf{y}}_t = \mathbf{W}\tilde{\mathbf{v}}_t - \sum_{s=1}^t \tilde{a}_{ts}\tilde{\mathbf{u}}_s$. Expanding each polynomial p_u for $u \in \mathcal{U}$ in terms of $(\tilde{\mathbf{y}}_{1:(t-1)}, \tilde{\mathbf{u}}_{1:t}, \mathbf{f}_{1:k})$ and $\mathbf{W}\tilde{\mathbf{v}}_t$, we have

$$p_u(\tilde{y}_{1:t}[\alpha], \tilde{u}_{1:t}[\alpha], f_{1:k}[\alpha]) = \sum_{\theta=0}^{\Theta_u} p_{u,\theta}(\tilde{y}_{1:(t-1)}[\alpha], \tilde{u}_{1:t}[\alpha], f_{1:k}[\alpha]) \cdot \left(\sum_{i=1}^n W[\alpha, i]\tilde{v}_t[i] \right)^\theta$$

for some polynomials $p_{u,0}, \ldots, p_{u,\Theta_u}$, where Θ_u is the maximum degree of p_u in $\tilde{y}_t$. Then we may write

$$\begin{aligned}
&\mathrm{val}_T(\mathbf{W}; \tilde{\mathbf{y}}_{1:t}, \tilde{\mathbf{u}}_{1:t}, \mathbf{f}_{1:k}; \tilde{\mathbf{z}}_{1:t}, \tilde{\mathbf{v}}_{1:t}, \mathbf{g}_{1:\ell}) \\
&\quad = \frac{1}{n} \sum_{\boldsymbol{\alpha} \in [m]^{\mathcal{U}}} \sum_{\mathbf{i} \in [n]^{\mathcal{V}}} p_{\boldsymbol{\alpha}|T} \cdot q_{\mathbf{i}|T} \cdot W_{\boldsymbol{\alpha},\mathbf{i}|T} \\
&\quad = \frac{1}{n} \sum_{\boldsymbol{\theta} \in \prod_{u \in \mathcal{U}} \{0, \ldots, \Theta_u\}} \sum_{\boldsymbol{\alpha} \in [m]^{\mathcal{U}}} \sum_{\mathbf{i} \in [n]^{\mathcal{V}}} \left(\prod_{u \in \mathcal{U}} p_{u,\theta}(\tilde{y}_{1:(t-1)}[\alpha_u], \tilde{u}_{1:t}[\alpha_u], f_{1:k}[\alpha_u]) \right)
\end{aligned}$$

$$\cdot\, q_{\mathbf{i}|T} \cdot \left(\sum_{i=1}^{n} W[\alpha_u, i]\tilde{v}_t[i]\right)^{\theta_u} \cdot W_{\boldsymbol{\alpha},\mathbf{i}|T}.$$

For each $\boldsymbol{\theta} \in \prod_{u\in\mathcal{U}}\{0,\ldots,\Theta_u\}$, we define a new tensor network $T_{\boldsymbol{\theta}}$ from T as follows: For each $u \in \mathcal{U}$, replace the associated polynomial p_u by p_{u,θ_u}. Then, to each $u \in \mathcal{U}$, add θ_u new edges connecting to θ_u new vertices in $\mathcal{V}$, where each new vertex v has the label $q_v(\tilde{z}_{1:t}, \tilde{v}_{1:t}, g_{1:\ell}) = \tilde{v}_t$. Then the above is exactly

$$\begin{aligned}
&\mathrm{val}_T(\mathbf{W}; \tilde{\mathbf{y}}_{1:t}, \tilde{\mathbf{u}}_{1:t}, \mathbf{f}_{1:k}; \tilde{\mathbf{z}}_{1:t}, \tilde{\mathbf{v}}_{1:t}, \mathbf{g}_{1:\ell}) \\
&\quad= \sum_{\boldsymbol{\theta}\in\prod_{u\in\mathcal{U}}\{0,\ldots,\Theta_u\}} \mathrm{val}_{T_{\boldsymbol{\theta}}}(\mathbf{W}; \tilde{\mathbf{y}}_{1:(t-1)}, \tilde{\mathbf{u}}_{1:t}, \mathbf{f}_{1:k}; \tilde{\mathbf{z}}_{1:t}, \tilde{\mathbf{v}}_{1:t}, \mathbf{g}_{1:\ell})
\end{aligned}$$

which verifies (2.6.13).

Next, for any tensor network T in the variables $(\tilde{y}_{1:(t-1)}, \tilde{u}_{1:t}, f_{1:k}; \tilde{z}_{1:t}, \tilde{v}_{1:t}, g_{1:\ell})$, applying $\tilde{v}_t = \tilde{v}_t(\tilde{z}_{1:t}, g_{1:\ell})$ in the polynomial label p_v for each vertex $v \in \mathcal{V}$, there exists a tensor network T' such that

$$\mathrm{val}_T(\mathbf{W}; \tilde{\mathbf{y}}_{1:(t-1)}, \tilde{\mathbf{u}}_{1:t}, \mathbf{f}_{1:k}; \tilde{\mathbf{z}}_{1:t}, \tilde{\mathbf{v}}_{1:t}, \mathbf{g}_{1:\ell}) = \mathrm{val}_{T'}(\mathbf{W}; \tilde{\mathbf{y}}_{1:(t-1)}, \tilde{\mathbf{u}}_{1:t}, \mathbf{f}_{1:k}; \tilde{\mathbf{z}}_{1:t}, \tilde{\mathbf{v}}_{1:(t-1)}, \mathbf{g}_{1:\ell}).$$

Since $\tilde{\mathbf{z}}_t = \mathbf{W}^\top \tilde{\mathbf{u}}_t - \sum_{s=1}^{t-1} \tilde{b}_{ts}\tilde{\mathbf{v}}_s$, by the same argument as in the above for $\tilde{\mathbf{y}}_t$, for any tensor network T in variables $(\tilde{y}_{1:(t-1)}, \tilde{u}_{1:t}, f_{1:k}; \tilde{z}_{1:t}, \tilde{v}_{1:(t-1)}, g_{1:\ell})$, there exists a set of tensor networks $\mathcal{F}$ such that

$$\begin{aligned}
&\mathrm{val}_T(\mathbf{W}; \tilde{\mathbf{y}}_{1:(t-1)}, \tilde{\mathbf{u}}_{1:t}, \tilde{\mathbf{f}}_{1:k}; \tilde{\mathbf{z}}_{1:t}, \tilde{\mathbf{v}}_{1:(t-1)}, \tilde{\mathbf{g}}_{1:\ell}) \\
&\quad= \sum_{T'\in\mathcal{F}} \mathrm{val}_{T'}(\mathbf{W}; \tilde{\mathbf{y}}_{1:(t-1)}, \tilde{\mathbf{u}}_{1:t}, \tilde{\mathbf{f}}_{1:k}; \tilde{\mathbf{z}}_{1:(t-1)}, \tilde{\mathbf{v}}_{1:(t-1)}, \tilde{\mathbf{g}}_{1:\ell}),
\end{aligned}$$

and applying the fact that $\tilde{u}_t(\cdot)$ is a polynomial function, for any tensor network T in the variables $(\tilde{y}_{1:(t-1)}, \tilde{u}_{1:t}, f_{1:k}; \tilde{z}_{1:(t-1)}, \tilde{v}_{1:(t-1)}, g_{1:\ell})$, there exists a tensor network T'

such that

$$\begin{aligned}&\mathrm{val}_T(\mathbf{W};\tilde{\mathbf{y}}_{1:(t-1)},\tilde{\mathbf{u}}_{1:t},\mathbf{f}_{1:k};\tilde{\mathbf{z}}_{1:(t-1)},\tilde{\mathbf{v}}_{1:(t-1)},\mathbf{g}_{1:\ell})\\&\qquad=\mathrm{val}_{T'}(\mathbf{W};\tilde{\mathbf{y}}_{1:(t-1)},\tilde{\mathbf{u}}_{1:(t-1)},\mathbf{f}_{1:k};\tilde{\mathbf{z}}_{1:(t-1)},\tilde{\mathbf{v}}_{1:(t-1)},\mathbf{g}_{1:\ell}).\end{aligned}$$

Iteratively applying these four reductions shows (2.6.11), and the proof of (2.6.12) is analogous. ■

The remainder of the proof of Lemma 2.6.4 parallels that of Lemma 2.2.12: As in Lemma 2.3.12, the above result together with (2.6.10) and the given condition for universality of tensor network values between $\mathbf{W}$ and $\mathbf{G}$ implies the almost-sure Wasserstein convergence

$$(\mathbf{u}_1,\mathbf{f}_{1:k},\tilde{\mathbf{y}}_{1:t}(\mathbf{W}))\xrightarrow{W}(U_1,F_{1:k},\widetilde{Y}_{1:t}),\qquad(\mathbf{g}_{1:\ell},\tilde{\mathbf{z}}_{1:t}(\mathbf{W}))\xrightarrow{W}(G_{1:\ell},\widetilde{Z}_{1:t})$$

also for the polynomial AMP algorithm applied to $\mathbf{W}$. A similar inductive polynomial approximation argument as in Lemma 2.3.15, using the conditions that $\|\mathbf{W}\|_{\mathrm{op}}<C$ almost surely for all large m,n and $u_{t+1}(\cdot)$ and $v_t(\cdot)$ are Lipschitz, establishes

$$\max_{s=1}^{t}\frac{1}{\sqrt{n}}\|\mathbf{z}_t(\mathbf{W})-\tilde{\mathbf{z}}_t(\mathbf{W})\|_2<\iota(\varepsilon),\qquad\max_{s=1}^{t}\frac{1}{\sqrt{n}}\|\mathbf{y}_t(\mathbf{W})-\tilde{\mathbf{y}}_t(\mathbf{W})\|_2<\iota(\varepsilon),$$

$$\|\mathbf{\Omega}_t-\widetilde{\mathbf{\Omega}}_t\|_{\mathrm{op}}<\iota(\varepsilon),\qquad\|\mathbf{\Sigma}_t-\widetilde{\mathbf{\Sigma}}_t\|_{\mathrm{op}}<\iota(\varepsilon)$$

for a constant $\iota(\varepsilon)\to0$ as $\varepsilon\to0$. This then implies the desired W_2-convergence in Lemma 2.6.4. We omit the details of the argument for brevity.

2.6.4 Universality for generalized white noise matrices

In this section, we prove the following result showing that the value of an alternating diagonal tensor network is universal for the class of generalized white noise matrices in Definition 2.2.18.

Lemma 2.6.6. *Let* $\mathbf{x}_1, \ldots, \mathbf{x}_k \in \mathbb{R}^m$ *and* $\mathbf{y}_1, \ldots, \mathbf{y}_\ell \in \mathbb{R}^n$ *be (random or deterministic) vectors and let* $(X_1, \ldots, X_k)$ *and* $(Y_1, \ldots, Y_\ell)$ *have finite moments of all orders, such that almost surely as* $m, n \to \infty$ *with* $m/n \to \gamma \in (0, \infty)$,

$$(\mathbf{x}_1, \ldots, \mathbf{x}_k) \xrightarrow{W} (X_1, \ldots, X_k) \qquad \text{and} \qquad (\mathbf{y}_1, \ldots, \mathbf{y}_\ell) \xrightarrow{W} (Y_1, \ldots, Y_\ell).$$

Let $\mathbf{W} \in \mathbb{R}^{m \times n}$ *be a generalized white noise matrix, independent of* $\mathbf{x}_1, \ldots, \mathbf{x}_k$ *and* $\mathbf{y}_1, \ldots, \mathbf{y}_\ell$, *with variance profile* $\mathbf{S}$. *Let* $\mathbf{s}_\alpha$ *be the* α^{th} *row of* $\mathbf{S}$, *let* $\mathbf{s}^i$ *be the* i^{th} *column of* $\mathbf{S}$, *and suppose for any fixed polynomial functions* $p : \mathbb{R}^k \to \mathbb{R}$ *and* $q : \mathbb{R}^\ell \to \mathbb{R}$ *that*

$$\max_{i=1}^{n} \left| \langle p(\mathbf{x}_1, \ldots, \mathbf{x}_k) \odot \mathbf{s}^i \rangle - \langle p(\mathbf{x}_1, \ldots, \mathbf{x}_k) \rangle \cdot \langle \mathbf{s}^i \rangle \right| \to 0, \tag{2.6.14}$$

$$\max_{\alpha=1}^{m} \left| \langle q(\mathbf{y}_1, \ldots, \mathbf{y}_\ell) \odot \mathbf{s}_\alpha \rangle - \langle q(\mathbf{y}_1, \ldots, \mathbf{y}_\ell) \rangle \cdot \langle \mathbf{s}_\alpha \rangle \right| \to 0. \tag{2.6.15}$$

Then for any alternating diagonal tensor network T *in* (k, ℓ) *variables, there is a deterministic limit value*

$$\text{lim-val}_T(X_1, \ldots, X_k, Y_1, \ldots, Y_\ell)$$

depending only on T, γ, *and the joint laws of* $(X_1, \ldots, X_k)$ *and* $(Y_1, \ldots, Y_\ell)$ *such that almost surely,*

$$\lim_{m,n \to \infty} \text{val}_T(\mathbf{W}; \mathbf{x}_1, \ldots, \mathbf{x}_k; \mathbf{y}_1, \ldots, \mathbf{y}_\ell) = \text{lim-val}_T(\gamma, X_1, \ldots, X_k, Y_1, \ldots, Y_\ell).$$

In particular, this limit value is the same for $\mathbf{W}$ *as for a Gaussian white noise matrix* $\mathbf{G}$.

The proof of Lemma 2.6.6 is quite similar to that of Lemma 2.2.13, and we highlight only the differences: Fix the tensor network $T = (\mathcal{U}, \mathcal{V}, \mathcal{E}, \{p_u\}_{u \in \mathcal{U}}, \{q_v\}_{v \in \mathcal{V}})$.

Let $\mathcal{P} = \mathcal{P}_{\mathcal{U}} \times \mathcal{P}_{\mathcal{V}}$ be the set of pairs of partitions (π_U, π_V) of $\mathcal{U}$ and $\mathcal{V}$. For index tuples $\boldsymbol{\alpha} \in [m]^{\mathcal{U}}$ and $\mathbf{i} \in [n]^{\mathcal{V}}$, consider their induced partitions $\pi(\boldsymbol{\alpha}) \in \mathcal{P}_{\mathcal{U}}$ and $\pi(\mathbf{i}) \in \mathcal{P}_{\mathcal{V}}$. Then

$$\mathrm{val}_T(\mathbf{W}; \mathbf{x}_{1:k}; \mathbf{y}_{1:\ell}) = \frac{1}{n} \sum_{(\pi_U, \pi_V) \in \mathcal{P}} \sum_{\boldsymbol{\alpha} \in [m]^{\mathcal{U}} : \pi(\boldsymbol{\alpha}) = \pi_U} \sum_{\mathbf{i} \in [n]^{\mathcal{V}} : \pi(\mathbf{i}) = \pi_V} p_{\boldsymbol{\alpha}|T} \cdot q_{\mathbf{i}|T} \cdot W_{\boldsymbol{\alpha}, \mathbf{i}|T}.$$

Corresponding to $(\pi_U, \pi_V) \in \mathcal{P}$, let $G_{\pi_U, \pi_V} = (\mathcal{K}_{\pi_U} \sqcup \mathcal{K}_{\pi_V}, \mathcal{F}_{\pi_U, \pi_V})$ be the image of $(\mathcal{U}, \mathcal{V}, \mathcal{E})$ under (π_U, π_V): This is the bipartite multi-graph with vertices $\mathcal{K}_{\pi_U} \sqcup \mathcal{K}_{\pi_V}$ being the blocks of π_U and π_V, and having one edge $(U, V) \in \mathcal{F}_{\pi_U, \pi_V}$ for each edge $(u, v) \in \mathcal{E}$ where U, V are the blocks containing u, v. Denote $P_U = \prod_{u \in U} p_u$ and $Q_V = \prod_{v \in V} q_v$. Then this gives

$$\mathrm{val}_T(\mathbf{W}; \mathbf{x}_{1:k}; \mathbf{y}_{1:\ell}) = \frac{1}{n} \sum_{(\pi_U, \pi_V) \in \mathcal{P}} \sum_{\boldsymbol{\alpha} \in [m]^{\mathcal{K}_{\pi_U}}}^{*} \sum_{\mathbf{i} \in [n]^{\mathcal{K}_{\pi_V}}}^{*} P_{\boldsymbol{\alpha}|G_{\pi_U, \pi_V}} \cdot Q_{\mathbf{i}|G_{\pi_U, \pi_V}} \cdot W^e_{\boldsymbol{\alpha}, \mathbf{i}|G_{\pi_U, \pi_V}} \tag{2.6.16}$$

where

$$P_{\boldsymbol{\alpha}|G_{\pi_U, \pi_V}} = \prod_{U \in \mathcal{K}_{\pi_U}} P_U(x_{1:k}[\alpha_U]), \quad Q_{\mathbf{i}|G_{\pi_U, \pi_V}} = \prod_{V \in \mathcal{K}_{\pi_V}} Q_V(y_{1:\ell}[i_V]),$$

$$W^e_{\boldsymbol{\alpha}, \mathbf{i}|G_{\pi_U, \pi_V}} = \prod_{\text{unique edges } (U,V) \text{ of } \mathcal{G}_{\pi_U, \pi_V}} W[\alpha_U, i_V]^{e(U,V)}$$

and $e(U, V)$ is the number of times the unique edge (U, V) appears in $\mathcal{F}_{\pi_U, \pi_V}$.

Lemma 2.6.7. *Let $\mathbb{E}$ denote the expectation over $\mathbf{W}$ conditional on $\mathbf{x}_1, \ldots, \mathbf{x}_k, \mathbf{y}_1, \ldots, \mathbf{y}_\ell$. Then Lemma 2.6.6 holds for $\mathbb{E}[\mathrm{val}_T(\mathbf{W}; \mathbf{x}_1, \ldots, \mathbf{x}_k; \mathbf{y}_1, \ldots, \mathbf{y}_\ell)]$ in place of $\mathrm{val}_T(\mathbf{W}; \mathbf{x}_1, \ldots, \mathbf{x}_k; \mathbf{y}_1, \ldots, \mathbf{y}_\ell)$.*

Proof. Taking expectation on both sides of (2.6.16), observe that any summand has 0 expectation unless each unique edge of G_{π_U, π_V} appears at least twice. For the partitions (π_U, π_V) corresponding to summands with non-zero expectation, the numbers of vertices and unique edges of G_{π_U,π_V} must satisfy

$$|\mathcal{K}_{\pi_U}| + |\mathcal{K}_{\pi_V}| \leq |\mathcal{F}_{\pi_U,\pi_V}|_* + 1 \leq |\mathcal{E}|/2 + 1,$$

the first inequality holding because the graph G_{π_U,π_V} is connected. By the same argument as in Lemma 2.3.4, the summands where $|\mathcal{K}_{\pi_U}| + |\mathcal{K}_{\pi_V}| \leq |\mathcal{E}|/2$ have vanishing contributions in the limit as $m, n \to \infty$. Thus the only non-vanishing contributions come from partitions (π_U, π_V) where

$$|\mathcal{K}_{\pi_U}| + |\mathcal{K}_{\pi_V}| = |\mathcal{F}_{\pi_U,\pi_V}|_* + 1 = |\mathcal{E}|/2 + 1. \tag{2.6.17}$$

For such classes, each unique edge of G_{π_U,π_V} appears exactly twice by the second equality of (2.6.17), these edges form a tree by the first equality of (2.6.17), and we have

$$\begin{aligned}
&\frac{1}{n} \sideset{}{^*}\sum_{\boldsymbol{\alpha} \in [m]^{\mathcal{K}_{\pi_U}}} \sideset{}{^*}\sum_{\mathbf{i} \in [n]^{\mathcal{K}_{\pi_V}}} P_{\boldsymbol{\alpha}|G_{\pi_U,\pi_V}} \cdot Q_{\mathbf{i}|G_{\pi_U,\pi_V}} \cdot \mathbb{E}[W^e_{\boldsymbol{\alpha},\mathbf{i}|G_{\pi_U,\pi_V}}] \\
&\quad = \frac{1}{n} \sideset{}{^*}\sum_{\boldsymbol{\alpha} \in [m]^{\mathcal{K}_{\pi_U}}} \sideset{}{^*}\sum_{\mathbf{i} \in [n]^{\mathcal{K}_{\pi_V}}} P_{\boldsymbol{\alpha}|G_{\pi_U,\pi_V}} \cdot Q_{\mathbf{i}|G_{\pi_U,\pi_V}} \cdot \prod_{\text{unique edges } (U,V) \text{ of } G_{\pi_U,\pi_V}} \frac{S[\alpha_U, i_V]}{n}.
\end{aligned}$$

Applying Lemma 2.3.3 and the first equality of (2.6.17), this has the same asymptotic limit as

$$\frac{1}{n} \sum_{\boldsymbol{\alpha} \in [m]^{\mathcal{K}_{\pi_U}}} \sum_{\mathbf{i} \in [n]^{\mathcal{K}_{\pi_V}}} P_{\boldsymbol{\alpha}|G_{\pi_U,\pi_V}} \cdot Q_{\mathbf{i}|G_{\pi_U,\pi_V}} \cdot \prod_{\text{unique edges } (U,V) \text{ of } G_{\pi_U,\pi_V}} \frac{S[\alpha_U, i_V]}{n}$$

which removes the distinctness requirement for the indices of $\boldsymbol{\alpha}$ and $\mathbf{i}$. Applying the conditions (2.6.14–2.6.15) and the same argument as in Lemma 2.3.2, this quantity has the limit, as $m, n \to \infty$ with $m/n \to \gamma$,

$$\gamma^{|\mathcal{K}_{\pi_U}|} \prod_{U \in \mathcal{K}_{\pi_U}} \mathbb{E}[P_U(X_{1:k})] \prod_{V \in \mathcal{K}_{\pi_V}} \mathbb{E}[Q_V(Y_{1:\ell})]$$

(A factor of γ arises when summing over each vertex of $\mathcal{K}_{\pi_U}$ because $S[\alpha_U, i_V]$ is normalized by n but the summation is over $[m]$.) Thus

$$\begin{aligned}
&\lim_{m,n\to\infty} \mathbb{E}[\mathrm{val}_T(\mathbf{W}; \mathbf{x}_{1:k}; \mathbf{y}_{1,\ell})] \\
&= \sum_{\substack{(\pi_U,\pi_V)\in\mathcal{P} \\ |\mathcal{K}_{\pi_U}|+|\mathcal{K}_{\pi_V}|=|\mathcal{F}_{\pi_U,\pi_V}|_*+1=|\mathcal{E}|/2+1}} \gamma^{|\mathcal{K}_{\pi_U}|} \prod_{U \in \mathcal{K}_{\pi_U}} \mathbb{E}[P_U(X_{1:k})] \prod_{V \in \mathcal{K}_{\pi_V}} \mathbb{E}[Q_V(Y_{1:\ell})] \\
&=: \text{lim-val}_T(\gamma, X_1, \ldots, X_k, Y_1, \ldots, Y_\ell)
\end{aligned}$$

This limit depends only on T, γ, and the joint laws of $X_{1:k}$ and $Y_{1:\ell}$, concluding the proof. ■

Lemma 2.6.8. *Let $\mathbb{E}$ denote the expectation over $\mathbf{W}$, conditional on $\mathbf{x}_1, \ldots, \mathbf{x}_k, \mathbf{y}_1, \ldots, \mathbf{y}_\ell$. Under the setting of Lemma 2.6.6, almost surely as $m, n \to \infty$,*

$$\mathrm{val}_T(\mathbf{W}; \mathbf{x}_1, \ldots, \mathbf{x}_k; \mathbf{y}_1, \ldots, \mathbf{y}_\ell) - \mathbb{E}[\mathrm{val}_T(\mathbf{W}; x_1, \ldots, \mathbf{x}_k; \mathbf{y}_1, \ldots, \mathbf{y}_\ell)] \to 0.$$

Proof. The proof is similar to Lemma 2.3.5. Let $\mathbf{W}^{(1)}, \mathbf{W}^{(2)}, \mathbf{W}^{(3)}, \mathbf{W}^{(4)}$ be four independent copies of $\mathbf{W}$, define index tuples $\boldsymbol{\alpha}_1, \boldsymbol{\alpha}_2, \boldsymbol{\alpha}_3, \boldsymbol{\alpha}_4 \in [m]^{\mathcal{U}}$ and $\mathbf{i}_1, \mathbf{i}_2, \mathbf{i}_3, \mathbf{i}_4 \in [n]^{\mathcal{V}}$, and set

$$p_{\boldsymbol{\alpha}_{1:4}} = p_{\boldsymbol{\alpha}_1|T} p_{\boldsymbol{\alpha}_2|T} p_{\boldsymbol{\alpha}_3|T} p_{\boldsymbol{\alpha}_4|T}, \quad q_{\mathbf{i}_{1:4}} = q_{\mathbf{i}_1|T} q_{\mathbf{i}_2|T} q_{\mathbf{i}_3|T} q_{\mathbf{i}_4|T},$$

$$W^{(a_1,a_2,a_3,a_4)}_{\boldsymbol{\alpha}_{1:4},\mathbf{i}_{1:4}} = W^{(a_1)}_{\boldsymbol{\alpha}_1,\mathbf{i}_1|T} W^{(a_2)}_{\boldsymbol{\alpha}_2,\mathbf{i}_2|T} W^{(a_3)}_{\boldsymbol{\alpha}_3,\mathbf{i}_3|T} W^{(a_4)}_{\boldsymbol{\alpha}_4,\mathbf{i}_4|T}.$$

For index tuples $\boldsymbol{\alpha}_{1:4}$ and $\mathbf{i}_{1:4}$, consider a bi-partitite multigraph $G(\boldsymbol{\alpha}_{1:4}, \mathbf{i}_{1:4}) = (\mathcal{U}_G, \mathcal{V}_G, \mathcal{E}_G)$ whose vertices $\mathcal{U}_G$ are the unique index values in $\boldsymbol{\alpha}_{1:4}$ and vertices $\mathcal{V}_G$ are the unique index values in $\mathbf{i}_{1:4}$, and having one edge $(\alpha_{a,u}, i_{a,v})$ for every combination of $a = 1, 2, 3, 4$ and edge $(u, v) \in \mathcal{E}$. Define

$$\mathcal{I}_2 = \{(\boldsymbol{\alpha}_{1:4}, \mathbf{i}_{1:4}) : G(\boldsymbol{\alpha}_{1:4}, \mathbf{i}_{1:4}) \text{ has 1 or 2 connected components}\},$$
$$\mathcal{I}_3 = \{(\boldsymbol{\alpha}_{1:4}, \mathbf{i}_{1:4}) : G(\boldsymbol{\alpha}_{1:4}, \mathbf{i}_{1:4}) \text{ has 3 connected components}\},$$
$$\mathcal{I}_4 = \{(\boldsymbol{\alpha}_{1:4}, \mathbf{i}_{1:4}) : G(\boldsymbol{\alpha}_{1:4}, \mathbf{i}_{1:4}) \text{ has 4 connected components}\},$$

and define correspondingly for each $j = 2, 3, 4$

$$A_j = \frac{1}{n^4} \sum_{(\boldsymbol{\alpha}_{1:4}, \mathbf{i}_{1:4}) \in \mathcal{I}_j} p_{\boldsymbol{\alpha}_{1:4}} q_{\mathbf{i}_{1:4}} \Big(\mathbb{E}[W^{(1,1,1,1)}_{\boldsymbol{\alpha}_{1:4}, \mathbf{i}_{1:4}}] - 4 \cdot \mathbb{E}[W^{(1,1,1,2)}_{\boldsymbol{\alpha}_{1:4}, \mathbf{i}_{1:4}}] + 6 \cdot \mathbb{E}[W^{(1,1,2,3)}_{\boldsymbol{\alpha}_{1:4}, \mathbf{i}_{1:4}}] - 3 \cdot \mathbb{E}[W^{(1,2,3,4)}_{\boldsymbol{\alpha}_{1:4}, \mathbf{i}_{1:4}}] \Big).$$

Then $\mathbb{E}[(\text{val}(\mathbf{W}) - \text{val}(\mathbf{W}))^4] = A_2 + A_3 + A_4$. By the same argument as in Lemma 2.3.5, we have $A_3 = A_4 = 0$, while $|A_2| \le C/n^2$ for a constant $C > 0$ and all large n. Then the result follows from Markov's inequality and the Borel-Cantelli lemma. ■

Lemmas 2.6.7 and 2.6.8 show Lemma 2.6.6. Finally, combining Lemmas 2.6.4 and 2.6.6 concludes the proof of Theorem 2.2.21.

2.6.5 Universality for rectangular generalized invariant matrices

In this section, we prove the following result showing that the value of an alternating diagonal tensor network is universal across the class of rectangular generalized invariant matrices.

Lemma 2.6.9. *Let $\mathbf{x}_1, \ldots, \mathbf{x}_k \in \mathbb{R}^m$ and $\mathbf{y}_1, \ldots, \mathbf{y}_\ell \in \mathbb{R}^n$ be (random or deterministic) vectors and let $(X_1, \ldots, X_k)$ and $(Y_1, \ldots, Y_\ell)$ have finite moments of all orders, such that almost surely as $m, n \to \infty$ with $m/n \to \gamma \in (0, \infty)$,*

$$(\mathbf{x}_1, \ldots, \mathbf{x}_k) \xrightarrow{W} (X_1, \ldots, X_k) \qquad \textit{and} \qquad (\mathbf{y}_1, \ldots, \mathbf{y}_\ell) \xrightarrow{W} (Y_1, \ldots, Y_\ell).$$

Let $\mathbf{W} \in \mathbb{R}^{m \times n}$ be a rectangular generalized invariant matrix independent of $\mathbf{x}_1, \ldots, \mathbf{x}_k$ and $\mathbf{y}_1, \ldots, \mathbf{y}_\ell$, with limit diagonal distribution $\mathcal{D}_{\text{diag}}$. Then for any alternating diagonal tensor network T in (k, ℓ) variables, there is a deterministic limit value

$$\text{lim-val}_T(\gamma, X_1, \ldots, X_k, Y_1, \ldots, Y_\ell, \mathcal{D}_{\text{diag}})$$

depending only on T, γ, the joint laws of $(X_1, \ldots, X_k)$ and $(Y_1, \ldots, Y_\ell)$, and $\mathcal{D}_{\text{diag}}$, such that almost surely

$$\lim_{m,n \to \infty} \text{val}_T(\mathbf{W}; \mathbf{x}_1, \ldots, \mathbf{x}_k; \mathbf{y}_1, \ldots, \mathbf{y}_\ell) = \text{lim-val}_T(\gamma, X_1, \ldots, X_k, Y_1, \ldots, Y_\ell, \mathcal{D}_{\text{diag}}).$$

In particular, if there exists a bi-orthogonally invariant matrix $\mathbf{G}$ having the same limit diagonal distribution $\mathcal{D}_{\text{diag}}$, then this limit value is the same for $\mathbf{W}$ as for $\mathbf{G}$.

The proof of Lemma 2.6.9 is similar to Lemma 2.3.6. Fix the tensor network $T = (\mathcal{U}, \mathcal{V}, \mathcal{E}, \{p_u\}_{u \in \mathcal{U}}, \{q_v\}_{v \in \mathcal{V}})$. Expanding $\mathbf{W} = \mathbf{\Pi}_U \mathbf{M} \mathbf{\Pi}_V^\top$, the tensor network value is given by

$$\begin{aligned} &\text{val}_T(\mathbf{W}; \mathbf{x}_{1:k}; \mathbf{y}_{1:\ell}) \\ &\quad = \frac{1}{n} \sum_{\boldsymbol{\alpha} \in [m]^{\mathcal{U}}} \sum_{\mathbf{i} \in [n]^{\mathcal{V}}} \sum_{\boldsymbol{\beta} \in [m]^{\mathcal{E}}} \sum_{\mathbf{j} \in [n]^{\mathcal{E}}} \prod_{u \in \mathcal{U}} p_u(x_{1:k}[\alpha_u]) \prod_{v \in \mathcal{V}} q_v(y_{1:\ell}[i_v]) \\ &\qquad\qquad \cdot \prod_{e=(u,v) \in \mathcal{E}} \Pi_U[\alpha_u, \beta_e] M[\beta_e, j_e] \Pi_V[i_v, j_e]. \end{aligned}$$

Write $\mathbf{\Pi}_U = \mathbf{\Xi}_U \mathbf{P}_U$ and $\mathbf{\Pi}_V = \mathbf{\Xi}_V \mathbf{P}_V$ for the random sign and permutation matrices defining $\mathbf{\Pi}_U, \mathbf{\Pi}_V$, and let σ_U, σ_V be the permutations of $[m], [n]$ for which $P_U[\alpha, \sigma_U(\alpha)] = 1$ and $P_V[i, \sigma_V(i)] = 1$. Let $\mathcal{P} = \mathcal{P}_U \times \mathcal{P}_V$ be the set of partitions (π_U, π_V) of $\mathcal{U}$ and $\mathcal{V}$, and let $G_{\pi_U, \pi_V} = (\mathcal{K}_{\pi_U} \sqcup \mathcal{K}_{\pi_V}, \mathcal{F}_{\pi_U, \pi_V})$ be the image of $(\mathcal{U}, \mathcal{V}, \mathcal{E})$

under (π_U, π_V) as defined in Section 2.6.4. Then we may obtain analogously to (2.3.17)

$$\begin{aligned}\mathrm{val}_T(\mathbf{W}; \mathbf{x}_{1:k}; \mathbf{y}_{1:\ell}) = \sum_{(\pi_U,\pi_V)\in\mathcal{P}} \frac{1}{n} \sideset{}{^*}\sum_{\boldsymbol{\alpha}\in[m]^{\mathcal{K}_{\pi_U}}} \sideset{}{^*}\sum_{\mathbf{i}\in[n]^{\mathcal{K}_{\pi_V}}} \prod_{R\in\mathcal{K}_{\pi_U}} P_R(x_{1:k}[\alpha_R])\Xi_U[\alpha_R]^{\deg_{\mathrm{ext}}(R)} \\ \cdot \prod_{S\in\mathcal{K}_{\pi_V}} Q_S(y_{1:\ell}[i_S])\Xi_V[i_S]^{\deg_{\mathrm{ext}}(S)} \prod_{(R,S)\in\mathcal{F}_{\pi_U,\pi_V}} M[\sigma_U(\alpha_R), \sigma_V(i_S)].\end{aligned} \tag{2.6.18}$$

Here $P_R = \prod_{u\in R} p_u$, $Q_S = \prod_{v\in S} q_v$, and $\deg_{\mathrm{ext}}(R)$ is the number of non-self-loop edges in $\mathcal{F}_{\pi_U,\pi_V}$ (counting multiplicity) that contain R.

Lemma 2.6.10. *Let $\mathbb{E}$ be the expectation over $\mathbf{\Pi}_U, \mathbf{\Pi}_V$ conditional on $\mathbf{M}, \mathbf{x}_1, \ldots, \mathbf{x}_k, \mathbf{y}_1, \ldots, \mathbf{y}_\ell$. Then Lemma 2.6.9 holds with the $\mathbb{E}[\mathrm{val}_T(\mathbf{W}; \mathbf{x}_1, \ldots, \mathbf{x}_k; \mathbf{y}_1, \ldots, \mathbf{y}_\ell)]$ in place of the $\mathrm{val}_T(\mathbf{W}; \mathbf{x}_1, \ldots, \mathbf{x}_k; \mathbf{y}_1, \ldots, \mathbf{y}_\ell)$.*

Proof. Define analogously to (2.3.21–2.3.23)

$$B_n(\pi_U, \pi_V) = m^{|\mathcal{K}_{\pi_U}|} \cdot \frac{(m - |\mathcal{K}_{\pi_U}|)!}{m!} \cdot n^{|\mathcal{K}_{\pi_V}|} \cdot \frac{(n - |\mathcal{K}_{\pi_V}|)!}{n!} \tag{2.6.19}$$

$$Q_n(\pi_U, \pi_V) = \frac{1}{m^{|\mathcal{K}_{\pi_U}|}} \sideset{}{^*}\sum_{\boldsymbol{\alpha}\in[m]^{\mathcal{K}_{\pi_U}}} \prod_{R\in\mathcal{K}_{\pi_U}} P_R(x_{1:k}[\alpha_R]) \cdot \frac{1}{n^{|\mathcal{K}_{\pi_V}|}} \sideset{}{^*}\sum_{\mathbf{i}\in[n]^{\mathcal{K}_{\pi_V}}} \prod_{S\in\mathcal{K}_{\pi_V}} Q_S(y_{1:\ell}[i_S]) \tag{2.6.20}$$

$$M_n(\pi_U, \pi_V) = \frac{1}{n} \sideset{}{^*}\sum_{\boldsymbol{\alpha}\in[m]^{\mathcal{K}_{\pi_U}}} \sideset{}{^*}\sum_{\mathbf{i}\in[n]^{\mathcal{K}_{\pi_V}}} \prod_{(R,S)\in\mathcal{F}_{\pi_U,\pi_V}} M[\alpha_R, i_S] \tag{2.6.21}$$

Then, taking the expectation conditional on $\mathbf{M}$ using the identities (2.3.18) and (2.3.19) yields

$$\mathbb{E}[\mathrm{val}_T(\mathbf{W}; \mathbf{x}_{1:k}; \mathbf{y}_{1:\ell})] = \sum_{(\text{even } \pi_U,\ \text{even } \pi_V)\in\mathcal{P}} B_n(\pi_U, \pi_V) \cdot Q_n(\pi_U, \pi_V) \cdot M_n(\pi_U, \pi_V)$$

We note that in terms of the symmetric embedding $\widetilde{\mathbf{M}}$ from (2.2.15), the above quantity $M_n(\pi_U, \pi_V)$ has the equivalent form

$$M_n(\pi_U, \pi_V) = \frac{1}{n} \sum_{\boldsymbol{\alpha} \in [m]^{\mathcal{K}_{\pi_U}}}^{*} \sum_{\mathbf{i} \in [n]^{\mathcal{K}_{\pi_V}}}^{*} \prod_{(R,S) \in \mathcal{F}_{\pi_U, \pi_V}} \widetilde{M}[\alpha_R, m + i_S].$$

We further add edges to G_{π_U, π_V} by attaching to every vertex in $\mathcal{K}_{\pi_U}$ a self-loop labeled with $p_e(\mathbf{x}) = \mathbf{I}_m$, attaching to every vertex in $\mathcal{K}_{\pi_V}$ a self-loop labeled with $p_e(\mathbf{x}) = \mathbf{I}_n$, and labeling each original edge $e = (R, S) \in \mathcal{F}_{\pi_U, \pi_V}$ with $p_e(\mathbf{x}) = \mathbf{x}$. Denote the resulting graph as $\widetilde{G}_{\pi_U, \pi_V} = (\mathcal{K}_{\pi_U} \sqcup \mathcal{K}_{\pi_V}, \widetilde{\mathcal{F}}_{\pi_U, \pi_V})$. Then

$$M_n(\pi_U, \pi_V) = \frac{1}{n} \sum_{\mathbf{j} \in [m+n]^{\mathcal{K}_{\pi_U} \sqcup \mathcal{K}_{\pi_V}}}^{*} \prod_{(R,S) \in \widetilde{\mathcal{F}}_{\pi_U, \pi_V}} p_e(\widetilde{\mathbf{M}})[j_R, j_S].$$

By the conditions on $\mathbf{M}$ in Definition 2.2.20, $\widetilde{\mathbf{M}}$ satisfies the assumptions for Lemma 2.6.11 below. Note that $\widetilde{G}_{\pi_U, \pi_V}$ is connected, and all external degrees are even when π_U, π_V are both even. Then, applying Lemma 2.6.11, there exists a value $\widetilde{M}(G_{\pi_U, \pi_V}, \mathcal{D}_{\text{diag}})$ depending only on G_{π_U, π_V} and $\mathcal{D}_{\text{diag}}$ such that

$$\begin{aligned}
\lim_{m,n \to \infty} \frac{n}{n+m} M_n(\pi_U, \pi_V) &= \lim_{m,n \to \infty} \frac{1}{n+m} \sum_{\mathbf{j} \in [m+n]^{\mathcal{K}_{\pi_U} \sqcup \mathcal{K}_{\pi_V}}}^{*} \prod_{(R,S) \in \widetilde{\mathcal{F}}_{\pi_U, \pi_V}} p_e(\widetilde{\mathbf{M}})[j_R, j_S] \\
&= \widetilde{M}(G_{\pi_U, \pi_V}, \mathcal{D}_{\text{diag}})
\end{aligned}$$

Combining with the limit value for Q_n, which exists by Lemma 2.3.3 and the assumptions $\mathbf{x}_{1:k} \xrightarrow{W} X_{1:k}$ and $\mathbf{y}_{1:\ell} \xrightarrow{W} Y_{1:\ell}$, we conclude that $\lim_{m,n \to \infty} \mathbb{E}[\text{val}_T(\mathbf{W}; \mathbf{x}_{1:k}; \mathbf{y}_{1:\ell})]$ exists and is given by

$$\begin{aligned}
&\text{lim-val}_T(\gamma, X_{1:k}, Y_{1:\ell}, \mathcal{D}_{\text{diag}}) \\
&:= \sum_{(\text{even } \pi_U,\, \text{even } \pi_V) \in \mathcal{P}} \prod_{R \in \mathcal{K}_{\pi_U}} \mathbb{E}[P_R(X_{1:k})] \prod_{S \in \mathcal{K}_{\pi_V}} \mathbb{E}[Q_S(Y_{1:\ell})] \cdot (1 + \gamma)\, \widetilde{M}(G_{\pi_U, \pi_V}, \mathcal{D}_{\text{diag}}).
\end{aligned}$$

This depends only on γ, T, the joint laws of $X_{1:k}$ and $Y_{1:\ell}$, and $\mathcal{D}_{\text{diag}}$, concluding the proof. $\blacksquare$

Lemma 2.6.11. *Let* $\widetilde{\mathbf{M}} \in \mathbb{R}^{(m+n)\times(m+n)}$ *be a deterministic symmetric matrix, such that for any diagonal monomial* $p(\mathbf{x}) \in \Delta\langle \mathbf{x}, \mathbf{I}_m, \mathbf{I}_n \rangle$, $\lim_{m,n\to\infty} \frac{1}{m+n} \operatorname{Tr} p(\widetilde{\mathbf{M}})$ *exists (and is finite) almost surely, and for any fixed* $\varepsilon > 0$ *and all large* m, n,

$$\max_{i \neq j} |p(\widetilde{\mathbf{M}})[i,j]| < n^{-1/2+\varepsilon}.$$

Let $G = (\mathcal{K}, \mathcal{F})$ *be a connected multi-graph such that the external degree* $\deg_{\text{ext}}(R)$ *is even for every vertex* $R \in \mathcal{K}$. *For every edge* $e \in \mathcal{F}$, *let* $p_e(\mathbf{x}) \in \Delta\langle \mathbf{x}, \mathbf{I}_m, \mathbf{I}_n \rangle$ *be a diagonal monomial labeling this edge. Then there exists a value* $M(G, \mathcal{D}_{\text{diag}})$ *depending only on* G *and* $\mathcal{D}_{\text{diag}}$ *such that*

$$\lim_{m,n\to\infty} \frac{1}{m+n} \sum_{\mathbf{j}\in[m+n]^{\mathcal{K}}}^{*} \prod_{e=(R,R')\in\mathcal{F}} p_e(\widetilde{\mathbf{M}})[j_R, j_{R'}] = M(G, \mathcal{D}_{\text{diag}}).$$

Proof. The proof is identical to that of Lemma 2.3.7. $\blacksquare$

Lemma 2.6.12. *Let* $\mathbb{E}$ *be the expectation over* $\mathbf{\Pi}_U, \mathbf{\Pi}_V$ *conditional on* $\mathbf{M}, \mathbf{x}_1, \ldots, \mathbf{x}_k, \mathbf{y}_1, \ldots, \mathbf{y}_\ell$. *Under the setting of Lemma 2.6.9, almost surely as* $m, n \to \infty$,

$$\operatorname{val}_T(\mathbf{W}; \mathbf{x}_1, \ldots, \mathbf{x}_k; \mathbf{y}_1, \ldots, \mathbf{y}_\ell) - \mathbb{E}[\operatorname{val}_T(\mathbf{W}; \mathbf{x}_1, \ldots, \mathbf{x}_k; \mathbf{y}_1, \ldots, \mathbf{y}_\ell)] \to 0.$$

Proof. The proof is similar to that of Lemma 2.3.9. We write $\operatorname{val}(\mathbf{W})$ for $\operatorname{val}_T(\mathbf{W}; \mathbf{x}_{1:k}; \mathbf{y}_{1:\ell})$, and $\operatorname{val}(\bar{\mathbf{W}})$ for the value corresponding to $\bar{\mathbf{W}}$ defined by independent copies $\bar{\mathbf{\Pi}}_U, \bar{\mathbf{\Pi}}_V$ of $\mathbf{\Pi}_U, \mathbf{\Pi}_V$.

Let $(\mathcal{U}^{(1)}, \mathcal{V}^{(1)}, \mathcal{E}^{(1)}), \ldots, (\mathcal{U}^{(4)}, \mathcal{V}^{(4)}, \mathcal{E}^{(4)})$ denote four copies of $(\mathcal{U}, \mathcal{V}, \mathcal{E})$. For subsets $A \subseteq \{1, \ldots, 4\}$, let $(\mathcal{U}_A, \mathcal{V}_A, \mathcal{E}_A) = \bigsqcup_{a\in A}(\mathcal{U}^{(a)}, \mathcal{V}^{(a)}, \mathcal{E}^{(a)})$ be the graph obtained as the disjoint union of those copies in A. Let $\mathcal{P}_A = \mathcal{P}_{\mathcal{U}_A} \times \mathcal{P}_{\mathcal{V}_A}$ be the set of pairs of partitions of $\mathcal{U}_A$ and $\mathcal{V}_A$, and let $\bar{A} = \{1,2,3,4\} \setminus A$. Then, we have analogously to (2.3.26),

$$\mathbb{E}[(\mathrm{val}(\mathbf{W}) - \mathrm{val}(\bar{\mathbf{W}}))^4] = \sum_{A \subseteq \{1,2,3,4\}} (-1)^{|A|} \sum_{\substack{(\text{even } \pi_U,\text{ even } \pi_V) \in \mathcal{P}_A \\ (\text{even } \bar{\pi}_U,\text{ even } \bar{\pi}_V) \in \mathcal{P}_{\bar{A}}}} \frac{n^{|\mathcal{C}(\pi_U,\pi_V)| + |\mathcal{C}(\bar{\pi}_U,\bar{\pi}_V)|}}{n^4}$$
$$\cdot B_n(\pi_U,\pi_V) B_n(\bar{\pi}_U,\bar{\pi}_V) \cdot Q_n(\pi_U,\pi_V) Q_n(\bar{\pi}_U,\bar{\pi}_V)$$
$$\cdot M_n(\pi_U,\pi_V) M_n(\bar{\pi}_U,\bar{\pi}_V) \tag{2.6.22}$$

Here, B_n, Q_n are as defined in (2.6.19–2.6.20), $|\mathcal{C}(\pi_U,\pi_V)|$ denotes the number of connected components of the bipartite graph $G_{\pi_U,\pi_V} = (\mathcal{K}_{\pi_U} \sqcup \mathcal{K}_{\pi_V}, \mathcal{F}_{\pi_U,\pi_V})$, and we extend the definition (2.6.21) to

$$M_n(\pi_U,\pi_V) = \frac{1}{n^{|\mathcal{C}(\pi_U,\pi_V)|}} \sum^{*}_{\boldsymbol{\alpha} \in [m]^{\mathcal{K}_{\pi_U}}} \sum^{*}_{\mathbf{i} \in [n]^{\mathcal{K}_{\pi_V}}} \prod_{(R,S) \in \mathcal{F}_{\pi_U,\pi_V}} M[\alpha_R, i_S]$$

when G_{π_U,π_V} has more than 1 connected component.

We define $(\tau_U, \tau_V) = (\pi_U \oplus \bar{\pi}_U, \pi_V \oplus \bar{\pi}_V) \in \mathcal{P}_{\{1,2,3,4\}}$ as the partitions of $\mathcal{U}_{\{1,2,3,4\}}$ and $\mathcal{V}_{\{1,2,3,4\}}$ obtained by combining π_U with $\bar{\pi}_U$ and π_V with $\bar{\pi}_V$. Then the same arguments as in the proof of Lemma 2.3.9 show

$$B_n(\pi_U,\pi_V) B_n(\bar{\pi}_U,\bar{\pi}_V) = B_n(\tau_U,\tau_V) + O(n^{-1})$$
$$Q_n(\pi_U,\pi_V) Q_n(\bar{\pi}_U,\bar{\pi}_V) = Q_n(\tau_U,\tau_V) + O(n^{-1})$$
$$M_n(\pi_U,\pi_V) M_n(\bar{\pi}_U,\bar{\pi}_V) = M_n(\tau_U,\tau_V) + O(n^{-1})$$

When G_{τ_U,τ_V} has 4 connected components, which we denote by $G_{\tau_U,\tau_V}(a) = (\mathcal{K}_{\tau_U}(a) \sqcup \mathcal{K}_{\tau_V}(a), \mathcal{F}_{\tau_U,\tau_V}(a))$ for $a = 1,2,3,4$, the arguments of Lemma 2.3.9 show also

$$B_n(\pi_U,\pi_V) B_n(\bar{\pi}_U,\bar{\pi}_V) = B_n(\tau_U,\tau_V) + \sum_{a \in A, b \notin A} B_n(\tau_U,\tau_V,a,b) + O(n^{-2})$$
$$Q_n(\pi_U,\pi_V) Q_n(\bar{\pi}_U,\bar{\pi}_V) = Q_n(\tau_U,\tau_V) + \sum_{a \in A, b \notin A} Q_n(\tau_U,\tau_V,a,b) + O(n^{-2})$$

$$M_n(\pi_U,\pi_V)M_n(\bar{\pi}_U,\bar{\pi}_V) = M_n(\tau_U,\tau_V) + \sum_{a\in A, b\notin A} M_n(\tau_U,\tau_V,a,b) + O(n^{-2})$$

for the quantities

$$B_n(\tau_U,\tau_V,a,b) = -m^{-1}|\mathcal{K}_{\tau_U}(a)|\cdot|\mathcal{K}_{\tau_U}(b)| - n^{-1}|\mathcal{K}_{\tau_V}(a)|\cdot|\mathcal{K}_{\tau_V}(b)|$$

$$\begin{aligned}Q_n(\tau_U,\tau_V,a,b) = \frac{1}{m^{|\mathcal{K}_{\tau_U}|}}\frac{1}{n^{|\mathcal{K}_{\tau_V}|}}\sum_{\boldsymbol{\alpha}\in[m]^{\mathcal{K}_{\tau_U}}}\sum_{\mathbf{i}\in[n]^{\mathcal{K}_{\tau_V}}} &\mathbf{1}\{\text{indices of } \mathbf{i} \text{ are distinct}\\ &\text{and exactly 2 indices of } \boldsymbol{\alpha} \text{ from } \mathcal{K}_{\tau_U}(a),\mathcal{K}_{\tau_U}(b) \text{ coincide,}\\ &\text{or indices of } \boldsymbol{\alpha} \text{ are distinct and exactly 2 indices of } \mathbf{i}\\ &\text{from } \mathcal{K}_{\tau_V}(a),\mathcal{K}_{\tau_V}(b) \text{ coincide}\}\cdot\prod_{R\in\mathcal{K}_{\tau_U}} P_R(x_{1:k}[\alpha_R])\prod_{S\in\mathcal{K}_{\tau_V}} Q_S(y_{1:\ell}[i_S])\end{aligned}$$

$$M_n(\tau_U,\tau_V,a,b) = \frac{1}{n}\sum_{(\tau'_U,\tau'_V)\in\mathcal{P}(\tau_U,\tau_V,a,b)} M_n^{**}(\tau'_U,\tau'_V)$$

where $\mathcal{P}(\tau_U,\tau_V,a,b)$ is the set of partitions (τ'_U,τ'_V) obtained by merging one or more pairs of blocks $R\in\mathcal{K}_{\tau_U}(a)$ with $R'\in\mathcal{K}_{\tau_U}(b)$ or $S\in\mathcal{K}_{\tau_V}(a)$ with $S'\in\mathcal{K}_{\tau_V}(b)$, and $M_n^{**}(\tau_U,\tau_V)$ is the product across all connected components $G_{\sigma_U,\sigma_V}\subseteq G_{\tau_U,\tau_V}$ of $M_n(\sigma_U,\sigma_V)$.

Then the same argument as in Lemma 2.3.9 shows $\mathbb{E}[(\mathrm{val}(\mathbf{W})-\mathbb{E}\,\mathrm{val}(\mathbf{W}))^4]\leq\mathbb{E}[(\mathrm{val}(\mathbf{W})-\mathrm{val}(\bar{\mathbf{W}}))^4]\leq C/n^2$ for a constant $C>0$ and all large n, so that the result follows by Markov's inequality and the Borel-Cantelli lemma. ■

Lemmas 2.6.10 and 2.6.12 show Lemma 2.6.9, and combining Lemmas 2.6.4 and 2.6.9 concludes the proof of Theorem 2.2.22.

Chapter 3

Approximate Message Passing Algorithms for Orthogonally Invariant Ensembles with Spectral Initialization

In this chapter, we study a class of Approximate Message Passing (AMP) algorithms for symmetric and rectangular spiked random matrix models with orthogonally invariant noise. The AMP iterates have fixed dimension $K \geq 1$, a multivariate non-linearity is applied in each AMP iteration, and the algorithm is spectrally initialized with K super-critical sample eigenvectors. We derive the forms of the Onsager debiasing coefficients and corresponding AMP state evolution, which depend on the free cumulants of the noise spectral distribution. This extends previous results for such models with $K = 1$ and an independent initialization.

Applying this approach to Bayesian principal components analysis, we introduce a Bayes-OAMP algorithm that uses as its non-linearity the posterior mean conditional on all preceding AMP iterates. We describe a practical implementation of this algorithm, where all debiasing and state evolution parameters are estimated from the observed data, and we illustrate the accuracy and stability of this approach in simulations.

The rest of the chapter is organized as follows. In Section 3.1, we describe extensions of the AMP algorithms and state evolution characterizations from vector-valued to matrix-valued iterates, for orthogonally invariant matrices and an independent initialization. In Section 3.2, we develop a method of spectral initialization for the general AMP algorithms, using the sample eigenvectors or singular vectors of the data. We also discuss signal-plus-noise models and spectral initializations. In Section 3.3, we propose a Bayes-OAMP algorithm for PCA with a Bayesian prior for the PCs, and demonstrate its performance in simulations. The proofs of the main results are given in Section 3.4 and Section 3.5, with auxiliary lemmas deferred to Section 3.6.

We will present the results for only symmetric matrices, and the analyses can be naturally extended to rectangular matrices. See [ZWF21] for full details in the rectangular case.

Notational conventions. For random variables X and Y, $X \perp\!\!\!\perp Y$ denotes that they are independent. $\|\cdot\|$ denotes the ℓ_2-norm for vectors and the $\ell_2 \to \ell_2$ operator norm for matrices. $\|\cdot\|_{\mathrm{F}}$ is the Frobenius norm for matrices. $\mathrm{diag}(v) \in \mathbb{R}^{K\times K}$ is the diagonal matrix with $v \in \mathbb{R}^K$ on its diagonal, and we write $\mathrm{diag}(v) \in \mathbb{R}^{K\times K'}$ to indicate this matrix right-padded by $K' - K$ columns of 0. We adopt the convention $M^0 = \mathrm{Id}$ for the 0^{th} power of any square matrix M. For a block matrix $M \in \mathbb{R}^{tK\times tK}$ and $\kappa \in \mathbb{R}^{K\times K}$, $M \odot \kappa \in \mathbb{R}^{tK\times tK}$ and $\kappa \odot M \in \mathbb{R}^{tK\times tK}$ denote the block-wise right- and left- multiplication by κ.

Similar to the previous chapter, for a matrix $\mathbf{U} \in \mathbb{R}^{n\times K}$ and random vector $U \in \mathbb{R}^K$, we write $\mathbf{U} \overset{W_2}{\to} U$ for the Wasserstein-2 convergence of the empirical distribution of rows of $\mathbf{U}$, as $n \to \infty$. We write $\langle \mathbf{U} \rangle = n^{-1}\sum_{i=1}^n u_i \in \mathbb{R}^K$ for the empirical average of rows of $\mathbf{U}$.

3.1 Symmetric AMP with independent initialization

In this section, we first describe extensions of the AMP algorithms and state evolution characterizations of [Fan20] from vector-valued to matrix-valued iterates, for orthogonally invariant matrices and an independent initialization. We will then discuss signal-plus-noise models and spectral initializations in Section 3.2.

Let $\mathbf{W} \in \mathbb{R}^{n \times n}$ be a symmetric matrix, with eigen-decomposition $\mathbf{W} = \mathbf{O}^\top \mathbf{\Lambda} \mathbf{O}$ where $\mathbf{\Lambda} = \mathrm{diag}(\boldsymbol{\lambda})$. We will assume that $\mathbf{O}$ is a Haar-distributed orthogonal basis of eigenvectors, so $\mathbf{W}$ is orthogonally invariant in law.

For fixed dimensions $J \geq 0$ and $K \geq 1$, consider a possible additional matrix $\mathbf{E} \in \mathbb{R}^{n \times J}$ of "side information", and a sequence of Lipschitz functions $u_2, u_3, \ldots$ where each $u_{t+1} : \mathbb{R}^{tK+J} \to \mathbb{R}^K$. (We may set $J = 0$ if there is no such side information.) We consider an AMP algorithm with initialization $\mathbf{U}_1 \in \mathbb{R}^{n \times K}$ independent of $\mathbf{W}$, having the iterates

$$\mathbf{Z}_t = \mathbf{W}\mathbf{U}_t - \mathbf{U}_1 b_{t1}^\top - \mathbf{U}_2 b_{t2}^\top - \ldots - \mathbf{U}_t b_{tt}^\top \tag{3.1.1}$$

$$\mathbf{U}_{t+1} = u_{t+1}(\mathbf{Z}_1, \ldots, \mathbf{Z}_t, \mathbf{E}). \tag{3.1.2}$$

Here $b_{ts} \in \mathbb{R}^{K \times K}$ is a matrix-valued Onsager debiasing coefficient for each $s = 1, \ldots, t$, and $u_{t+1}(\cdot)$ is applied row-wise to $(\mathbf{Z}_1, \ldots, \mathbf{Z}_t, \mathbf{E}) \in \mathbb{R}^{n \times (tK+J)}$ to yield each next iterate $\mathbf{U}_{t+1} \in \mathbb{R}^{n \times K}$.

Debiasing coefficients. Let $\partial_s u_{t+1}(Z_1, \ldots, Z_t, E) \in \mathbb{R}^{K \times K}$ denote the Jacobian of u_{t+1} in its vector argument Z_s, which exists Lebesgue-a.e. since $u_{t+1}(\cdot)$ is Lipschitz [Zie12, Theorem 2.2.1]. Denote

$$\langle \partial_s \mathbf{U}_{t+1} \rangle = \frac{1}{n} \sum_{i=1}^{n} \partial_s u_{t+1}(z_{1,i}, \ldots, z_{t,i}, e_i)$$

where $z_{t,i} \in \mathbb{R}^K$ and $e_i \in \mathbb{R}^J$ are the i^{th} rows of $\mathbf{Z}_t$ and $\mathbf{E}$. For each $T \geq 1$, define the $TK \times TK$ block-lower-triangular matrix

$$\boldsymbol{\phi}_T = \begin{pmatrix} 0 & 0 & \cdots & 0 & 0 \\ \langle \partial_1 \mathbf{U}_2 \rangle & 0 & \cdots & 0 & 0 \\ \langle \partial_1 \mathbf{U}_3 \rangle & \langle \partial_2 \mathbf{U}_3 \rangle & \cdots & 0 & 0 \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ \langle \partial_1 \mathbf{U}_T \rangle & \langle \partial_2 \mathbf{U}_T \rangle & \cdots & \langle \partial_{T-1} \mathbf{U}_T \rangle & 0 \end{pmatrix}. \tag{3.1.3}$$

Let Λ be a random variable on $\mathbb{R}$ with compact support, which will be the limit eigenvalue distribution of $\mathbf{W}$ as $n \to \infty$. Let $\{\kappa_j\}_{j \geq 1}$ be the free cumulants of Λ—see e.g. [Fan20, Section 2.3] for definitions. Applying the convention $\boldsymbol{\phi}_t^0 = \mathrm{Id}$, we take the debiasing coefficient matrices $\{b_{ts}\}$ in (3.1.1) up to iteration T to be the blocks of

$$\mathbf{b}_T = \sum_{j=0}^{\infty} \kappa_{j+1} \boldsymbol{\phi}_T^j \stackrel{\text{def}}{=} \begin{pmatrix} b_{11} & & & \\ b_{21} & b_{22} & & \\ \vdots & \vdots & \ddots & \\ b_{T1} & b_{T2} & \cdots & b_{TT} \end{pmatrix} \in \mathbb{R}^{TK \times TK}.$$

This may be interpreted as the R-transform of Λ applied to $\boldsymbol{\phi}_T$ (cf. Section 3.2.1). Note that this is a finite sum, because $\boldsymbol{\phi}_T^j = 0$ for all $j \geq T$. We have $b_{tt} = \kappa_1 \mathrm{Id}$ for every $t \geq 1$, which vanishes if Λ has mean $\kappa_1 = 0$.

State evolution. This choice of debiasing coefficients leads to the empirical distribution of rows of $(\mathbf{Z}_1, \ldots, \mathbf{Z}_T)$ having an asymptotically mean-zero multivariate Gaussian limit, as $n \to \infty$ for any fixed iteration T. The limit Gaussian law is described by its covariance matrix $\mathbf{\Sigma}_T \in \mathbb{R}^{TK \times TK}$, which may be defined recursively via the following state evolution:

Let (U_1, E) be an initial random vector with $U_1 \in \mathbb{R}^K$ and $E \in \mathbb{R}^J$, representing the limit empirical distribution of rows of $(\mathbf{U}_1, \mathbf{E})$. Inductively for $t = 1, 2, 3, \ldots$, having defined the joint law of $(U_1, \ldots, U_t, Z_1, \ldots, Z_{t-1}, E)$, define the $tK \times tK$ matrices

$$\boldsymbol{\Delta}_t = \begin{pmatrix} \mathbb{E}[U_1 U_1^\top] & \mathbb{E}[U_1 U_2^\top] & \cdots & \mathbb{E}[U_1 U_t^\top] \\ \mathbb{E}[U_2 U_1^\top] & \mathbb{E}[U_2 U_2^\top] & \cdots & \mathbb{E}[U_2 U_t^\top] \\ \vdots & \vdots & \ddots & \vdots \\ \mathbb{E}[U_t U_1^\top] & \mathbb{E}[U_t U_2^\top] & \cdots & \mathbb{E}[U_t U_t^\top] \end{pmatrix}, \tag{3.1.4}$$

$$\boldsymbol{\Phi}_t = \begin{pmatrix} 0 & 0 & \cdots & 0 & 0 \\ \mathbb{E}[\partial_1 u_2(Z_1, E)] & 0 & \cdots & 0 & 0 \\ \mathbb{E}[\partial_1 u_3(Z_1, Z_2, E)] & \mathbb{E}[\partial_2 u_3(Z_1, Z_2, E)] & \cdots & 0 & 0 \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ \mathbb{E}[\partial_1 u_t(Z_1, \ldots, Z_{t-1}, E)] & \mathbb{E}[\partial_2 u_t(Z_1, \ldots, Z_{t-1}, E)] & \cdots & \mathbb{E}[\partial_{t-1} u_t(Z_1, \ldots, Z_{t-1}, E)] & 0 \end{pmatrix}. \tag{3.1.5}$$

Here, $\boldsymbol{\Phi}_t$ corresponds to the large-n limit of $\boldsymbol{\phi}_t$ defined in (3.1.3). Then define the covariance $\boldsymbol{\Sigma}_t$ by

$$\boldsymbol{\Sigma}_t = \sum_{j=0}^{\infty} \boldsymbol{\Theta}^{(j)}[\boldsymbol{\Phi}_t, \kappa_{j+2} \boldsymbol{\Delta}_t] \quad \text{where} \quad \boldsymbol{\Theta}^{(j)}[\boldsymbol{\Phi}, \kappa \boldsymbol{\Delta}] = \sum_{i=0}^{j} \boldsymbol{\Phi}^i (\kappa \boldsymbol{\Delta}) (\boldsymbol{\Phi}^\top)^{j-i}. \tag{3.1.6}$$

Define the next joint law of $(U_1, \ldots, U_{t+1}, Z_1, \ldots, Z_t, E)$ by

$$(Z_1, \ldots, Z_t) \sim \mathcal{N}(0, \boldsymbol{\Sigma}_t) \perp\!\!\!\perp (U_1, E), \quad U_{s+1} = u_{s+1}(Z_1, \ldots, Z_s, E) \text{ for each } s = 1, \ldots, t. \tag{3.1.7}$$

Under these inductive definitions, it may be checked that the upper-left $(t-1) \times (t-1)$ blocks of $\boldsymbol{\Sigma}_t$ coincide with $\boldsymbol{\Sigma}_{t-1}$.

This state evolution characterizes the iterates of the AMP algorithm (3.1.1–3.1.2), under the following assumptions.

Assumption 3.1.1. The matrix $\mathbf{W} = \mathbf{O}^\top \mathrm{diag}(\boldsymbol{\lambda}) \mathbf{O}$ and random variable Λ satisfy

(a) $\mathbf{O}$ is random and Haar-distributed over the orthogonal group.

(b) $\boldsymbol{\lambda}$ is independent of $\mathbf{O}$, and its empirical distribution converges weakly a.s. to Λ as $n \to \infty$.

(c) Λ has compact support $\mathrm{supp}(\Lambda)$. Denoting $(\lambda_-, \lambda_+) = (\min \mathrm{supp}(\Lambda), \max \mathrm{supp}(\Lambda))$, we have $\min(\boldsymbol{\lambda}) \to \lambda_-$ and $\max(\boldsymbol{\lambda}) \to \lambda_+$ a.s. as $n \to \infty$.

Assumption 3.1.2. The AMP initialization $\mathbf{U}_1$, functions $u_2, u_3, \ldots$, and random vectors (U_1, E) satisfy

(a) $(\mathbf{U}_1, \mathbf{E}) \in \mathbb{R}^{n \times (K+J)}$ is independent of $\mathbf{O}$, and $(\mathbf{U}_1, \mathbf{E}) \xrightarrow{W_2} (U_1, E)$ a.s. as $n \to \infty$.

(b) Each $u_{t+1}(\cdot)$ is Lipschitz in all arguments. For each $s = 1, \ldots, t$, $\partial_s u_{t+1}(Z_1, \ldots, Z_t, E)$ exists and is continuous on a set of probability 1 under the law of $(Z_1, \ldots, Z_t, E)$ defined by (3.1.7).

Theorem 3.1.3. *Suppose Assumptions 3.1.1 and 3.1.2 hold. For any $T \geq 1$, consider the AMP algorithm (3.1.1–3.1.2) up to iteration T, and define $(U_1, \ldots, U_{T+1}, Z_1, \ldots, Z_T, E)$ by the state evolution (3.1.7). Then almost surely as $n \to \infty$,*

$$(\mathbf{U}_1, \ldots, \mathbf{U}_{T+1}, \mathbf{Z}_1, \ldots, \mathbf{Z}_T, \mathbf{E}) \xrightarrow{W_2} (U_1, \ldots, U_{T+1}, Z_1, \ldots, Z_T, E).$$

The proof of Theorem 3.1.3 is an extension of that of [Fan20, Theorem 4.3 and Corollary 4.4]. Compared with [Fan20, Corollary 4.4], Theorem 3.1.3 considers matrix-valued iterates having dimension $n \times K$, relaxes the needed convergence $(\mathbf{U}_1, \mathbf{E}) \to (U_1, E)$ from Wasserstein-p for all orders $p \geq 1$ to only Wasserstein-2, and relaxes the continuous-differentiability requirement for each function $u_{t+1}(\cdot)$ to the weaker condition of Assumption 3.1.2(b). We describe the modifications of the proofs of [Fan20] needed to establish Theorem 3.1.3 in Section 3.5.

Remark 3.1.4. We have defined b_{ts} in (3.1.1) using the free cumulants of the limit spectral distributions. Theorem 3.1.3 then also holds for any AMP algorithm where

b_{ts} are replaced by b'_{ts} such that $\|b_{ts} - b'_{ts}\| \to 0$ a.s. as $n \to \infty$. In particular, they hold if b_{ts} are defined with $\{\kappa_j\}$ replaced by consistent estimates of these limit free cumulants.

3.2 Spectral initialization for the symmetric spiked model

We now develop versions of the preceding AMP algorithms for "spiked" signal-plus-noise models, with spectral initialization. Consider a rank-K' symmetric spiked model

$$\mathbf{X} = \sum_{k=1}^{K'} \frac{\theta_k}{n} \mathbf{u}_*^k \mathbf{u}_*^{k\top} + \mathbf{W} \in \mathbb{R}^{n \times n}, \tag{3.2.1}$$

where $\mathbf{u}_*^1, \ldots, \mathbf{u}_*^{K'}$ are K' orthogonal "signal" eigenvectors, $\theta_1, \ldots, \theta_{K'}$ are non-zero signal eigenvalues, and the noise matrix $\mathbf{W} = \mathbf{O}^\top \mathbf{\Lambda} \mathbf{O}$ is symmetric and rotationally-invariant in law. We distinguish this rank K' from the dimension K of the AMP iterates, in anticipation of applications where K' may be modeled as large, and the practitioner may wish to apply AMP to estimate small subsets of K signal eigenvectors at a time. We develop a version of AMP where any K super-critical sample eigenvectors of $\mathbf{X}$ may be chosen as the spectral initialization.

In the model (3.2.1), we fix the normalization

$$\|\mathbf{u}_*^k\|^2 = n, \qquad \mathbf{u}_*^{j\top} \mathbf{u}_*^k = 0 \qquad \text{for all } j \neq k \in \{1, \ldots, K'\}. \tag{3.2.2}$$

We order the signal components such that the first K will correspond to the spectral initialization, and the remaining $K' - K$ are ordered arbitrarily. (Thus $\theta_1, \ldots, \theta_{K'}$ are not sorted, and they may have arbitrary signs.) Supposing that K_+ values $\theta_1, \ldots, \theta_{K'}$

are positive and $K_- = K' - K_+$ values are negative, we denote by

$$\lambda_1(\mathbf{X}), \ldots, \lambda_{K'}(\mathbf{X}) \tag{3.2.3}$$

the largest K_+ and smallest K_- sample eigenvalues of $\mathbf{X}$, sorted in the same order as $\theta_1, \ldots, \theta_{K'}$. We denote the associated sample eigenvectors of $\mathbf{X}$ by $\mathbf{f}^1_{\text{pca}}, \ldots, \mathbf{f}^{K'}_{\text{pca}}$, with the normalization and sign convention

$$\|\mathbf{f}^k_{\text{pca}}\|^2 = n, \quad {\mathbf{f}^k_{\text{pca}}}^\top \mathbf{u}^k_* \geq 0, \quad {\mathbf{f}^j_{\text{pca}}}^\top \mathbf{f}^k_{\text{pca}} = 0 \quad \text{for all } j \neq k \in \{1, \ldots, K'\}. \tag{3.2.4}$$

3.2.1 Preliminaries on sample eigenvectors and the R-transform

Eigenvectors and spectral phase transition. For the signal-plus-noise model (3.2.1), the quantitative behavior of the leading (positive and negative) sample eigenvalues/eigenvectors of $\mathbf{X}$ and associated phenomena of spectral phase transitions were studied in [BGN11], extending the work of [BBAP05, BS06, Pau07] for models with i.i.d. noise. These depend on the Cauchy-transform of the limit spectral distribution of $\mathbf{W}$, and we briefly review these results here.

Let Λ be the limit spectral distribution of $\mathbf{W}$. Denote the Cauchy-transform of Λ by

$$G(z) = \mathbb{E}[(z - \Lambda)^{-1}] \qquad \text{for } z \in (\lambda_+, \infty) \cup (-\infty, \lambda_-)$$

where $\lambda_\pm$ are the endpoints of support of Λ as defined in Assumption 3.1.1(c). $G(z)$ is strictly decreasing and positive on (λ_+, ∞) and strictly decreasing and negative on $(-\infty, \lambda_-)$, and hence admits a functional inverse $G^{-1}(z)$ on $(0, G(\lambda_+)) \cup (G(\lambda_-), 0)$ where $G(\lambda_\pm) = \lim_{z \to \lambda_\pm} G(z)$. For $k \in \{1, \ldots, K'\}$ such that $1/\theta_k$ belongs to this domain of $G^{-1}(z)$, define

$$\lambda_{\text{pca},k} = G^{-1}(1/\theta_k), \qquad \mu^2_{\text{pca},k} = \frac{-1}{\theta_k^2 G'(\lambda_{k,\text{pca}})}. \tag{3.2.5}$$

The following theorem summarizes several results of [BGN11, Theorems 2.1 and 2.2], which establish the first-order behavior of the leading sample eigenvalues and eigenvectors.

Theorem 3.2.1 ([BGN11])**.** *Suppose* $\mathbf{W}$ *satisfies Assumption 3.1.1, and* $\theta_1, \ldots, \theta_{K'}$ *are distinct and fixed as* $n \to \infty$*. Then for each* $k \in \{1, \ldots, K'\}$ *where* $\theta_k > 1/G(\lambda_+) \geq 0$ *or* $\theta_k < 1/G(\lambda_-) \leq 0$*, almost surely*

$$\lim_{n\to\infty} \lambda_k(\mathbf{X}) = \lambda_{\mathrm{pca},k}, \quad \lim_{n\to\infty} \left(\frac{{\mathbf{f}_{\mathrm{pca}}^k}^\top \mathbf{u}_*^k}{n}\right)^2 = \mu_{\mathrm{pca},k}^2,$$
$$\lim_{n\to\infty} \left(\frac{{\mathbf{f}_{\mathrm{pca}}^k}^\top \mathbf{u}_*^j}{n}\right)^2 = 0 \quad \textit{for all } j \in \{1, \ldots, K'\} \setminus \{k\}.$$

For each other $k \in \{1, \ldots, K'\}$*,* $\lim_{n\to\infty} \lambda_k(\mathbf{X}) \in \{\lambda_-, \lambda_+\}$*.*

Thus, for a "super-critical" signal eigenvalue θ_k exceeding the positive and negative phase transition thresholds $1/G(\lambda_+)$ and $1/G(\lambda_-)$, the corresponding sample eigenvalue $\lambda_k(\mathbf{X})$ converges to a deterministic value $\lambda_{\mathrm{pca},k}$ outside the interval $[\lambda_-, \lambda_+]$, the sample eigenvector $\mathbf{f}_{\mathrm{pca}}^k$ achieves asymptotically non-vanishing alignment with its corresponding signal vector $\mathbf{u}_*^k$, and it has asymptotically 0 alignment with the other signal vectors $\mathbf{u}_*^j$. For a "sub-critical" signal eigenvalue θ_k below these phase transition thresholds, $\lambda_k(\mathbf{X})$ converges to the spectral edges $\lambda_\pm$ of the noise spectral distribution. Note that $G(\lambda_+)$ and $G(\lambda_-)$ may be infinite, in which case the phase transition thresholds are 0, and all signal eigenvalues are super-critical. Whether this occurs depends on the rates of decay of the distribution of Λ at its spectral edges $\lambda_\pm$, and this is discussed further in [BGN11, Proposition 2.4].

R**-transform.** The above first-order limits may be re-expressed via the R-transform and free cumulants of Λ, which linearize free addition of independent random matrices. Define

$$R(z) = G^{-1}(z) - \frac{1}{z}$$

on the same domain as G^{-1}. Differentiating on both sides yields

$$R'(z) = \frac{1}{G'(G^{-1}(z))} + \frac{1}{z^2}.$$

For z small enough, $R(z)$ and its derivative have the convergent series expansions defined through the free cumulants $\{\kappa_j\}_{j\geq 1}$ of Λ,

$$R(z) = \sum_{j=0}^{\infty} \kappa_{j+1} z^j, \qquad R'(z) = \sum_{j=0}^{\infty} (j+1)\kappa_{j+2} z^j, \tag{3.2.6}$$

see e.g. [MS17, Theorem 17]. Thus for sufficiently large $|\theta_k|$, the limits $\lambda_{\mathrm{pca},k}$ and $\mu^2_{\mathrm{pca},k}$ in (3.2.5) also have the convergent series forms

$$\lambda_{\mathrm{pca},k} = \theta_k + R(1/\theta_k) = \theta_k + \sum_{j=0}^{\infty} \frac{\kappa_{j+1}}{\theta_k^j}, \quad 1 - \mu^2_{\mathrm{pca},k} = \frac{1}{\theta_k^2} R'(1/\theta_k) = \sum_{j=0}^{\infty} (j+1) \frac{\kappa_{j+2}}{\theta_k^{j+2}}. \tag{3.2.7}$$

3.2.2 AMP algorithm

Isolating the first $K \leq K'$ of the (unsorted) signal components for the spectral initialization, denote

$$S = \mathrm{diag}(\theta_1, \ldots, \theta_K) \in \mathbb{R}^{K \times K}, \qquad S' = \mathrm{diag}(\theta_1, \ldots, \theta_{K'}) \in \mathbb{R}^{K' \times K'},$$

$$\mathbf{F}_{\mathrm{pca}} = (\mathbf{f}^1_{\mathrm{pca}}, \ldots, \mathbf{f}^K_{\mathrm{pca}}) \in \mathbb{R}^{n \times K}, \qquad \mathbf{U}'_* = (\mathbf{u}^1_*, \ldots, \mathbf{u}^{K'}_*) \in \mathbb{R}^{n \times K'}.$$

Here we assume that $\theta_1, \ldots, \theta_K$ are known for simplicity. The following procedure is also applicable when $\theta_1, \ldots, \theta_K$ are unknown, as one can consistently estimate these values using Theorem 3.2.1, and we discuss this estimation in Section 3.3.2. We consider an AMP algorithm with iterates of dimension $n \times K$, initialized spectrally at

$$\mathbf{U}_0 = \mathbf{F}_{\text{pca}}\, S^{-1}, \qquad \mathbf{F}_0 = \mathbf{F}_{\text{pca}}. \tag{3.2.8}$$

For a sequence of Lipschitz functions $u_1, u_2, \ldots$ where $u_t : \mathbb{R}^{tK} \to \mathbb{R}^K$, this algorithm then iteratively computes for $t \geq 1$

$$\mathbf{U}_t = u_t(\mathbf{F}_0, \mathbf{F}_1, \ldots, \mathbf{F}_{t-1}), \tag{3.2.9}$$

$$\mathbf{F}_t = \mathbf{X}\mathbf{U}_t - \mathbf{U}_0 b_{t0}^\top - \mathbf{U}_1 b_{t1}^\top - \ldots - \mathbf{U}_t b_{tt}^\top. \tag{3.2.10}$$

Thus each $u_t(\cdot)$ may depend on the preceding iterates $\mathbf{F}_0, \ldots, \mathbf{F}_{t-1}$, including the spectral initialization. An additional matrix of side information may be incorporated into each $u_t(\cdot)$ as in (3.1.2), but we omit this here for simplicity. To ease notation in the analysis, we have shifted the initialization index from 1 to 0.

Debiasing coefficients. For each fixed $s \geq 1$, define diagonal matrices $\tilde{\kappa}_s, \hat{\kappa}_s \in \mathbb{R}^{K \times K}$ by the matrix series

$$\tilde{\kappa}_s = \sum_{j=0}^{\infty} \kappa_{j+s} S^{-j}, \qquad \hat{\kappa}_s = \sum_{j=0}^{\infty} (j+1)\kappa_{j+s} S^{-j}. \tag{3.2.11}$$

For each $T \geq 1$, define the block-lower-triangular matrix

$$\boldsymbol{\phi}_T = \Big(\langle \partial_s \mathbf{U}_r \rangle\Big)_{r,s \in \{0,\ldots,T\}} \in \mathbb{R}^{(T+1)K \times (T+1)K}$$

with row blocks indexed by r and column blocks by s, where

$$\langle \partial_s \mathbf{U}_r \rangle = n^{-1} \sum_{i=1}^{n} \partial_s u_r(f_{0,i}, \ldots, f_{r-1,i}),$$

$\partial_s u_r \in \mathbb{R}^{K\times K}$ denotes the Jacobian of $u_r(f_0,\ldots,f_{r-1})$ in the argument f_s, and $\partial_s u_r = 0$ for $s \geq r$. Define the matrix series

$$\mathbf{b}_T = \sum_{j=0}^{\infty} \kappa_{j+1}\boldsymbol{\phi}_T^j, \qquad \widetilde{\mathbf{b}}_T = \sum_{j=0}^{\infty} \boldsymbol{\phi}_T^j \odot \tilde{\kappa}_{j+1},$$

where we recall our notation $M \odot \tilde{\kappa}_s$ and $\tilde{\kappa}_s \odot M$ for the right- and left- multiplication of each block of M. Indexing blocks by $\{0,\ldots,T\}$ and writing $[t,s]$ to denote the $K \times K$ submatrix corresponding to row block t and column block s, we set the debiasing coefficients of (3.2.10) up to iteration T as

$$b_{ts} = \begin{cases} \widetilde{\mathbf{b}}_T[t,s] & \text{if } s = 0, \\ \mathbf{b}_T[t,s] & \text{otherwise.} \end{cases} \tag{3.2.12}$$

State evolution. The state of this algorithm up to iteration T is characterized by

$$\boldsymbol{\mu}_T = \begin{pmatrix} \mu_0 \\ \vdots \\ \mu_T \end{pmatrix} \in \mathbb{R}^{(T+1)K\times K'}, \qquad \boldsymbol{\Sigma}_T = \begin{pmatrix} \sigma_{00} & \cdots & \sigma_{0T} \\ \vdots & \ddots & \vdots \\ \sigma_{T0} & \cdots & \sigma_{TT} \end{pmatrix} \in \mathbb{R}^{(T+1)K\times(T+1)K}$$

and a corresponding joint law for random vectors $(U'_*, U_0, \ldots, U_{T+1}, F_0, \ldots, F_T)$, where U'_*, U_t, F_t represent the limit distributions of $\mathbf{U}'_*, \mathbf{U}_t, \mathbf{F}_t$, the matrix $\boldsymbol{\Sigma}_T$ is the covariance of $(F_0,\ldots,F_T)$ conditional on U'_*, and $\boldsymbol{\mu}_T$ relates the conditional mean of $(F_0,\ldots,F_T)$ to U'_*. These are defined recursively as follows:

Let $U'_* \in \mathbb{R}^{K'}$ be a random vector satisfying $\mathbb{E}[U'_* U'^{\top}_*] = \mathrm{Id}$, representing the limit distribution of rows of $\mathbf{U}'_*$ under the normalization (3.2.2). Set

$$\mu_{\mathrm{pca}} = \mathrm{diag}(\mu_{\mathrm{pca},1},\ldots,\mu_{\mathrm{pca},K}) \in \mathbb{R}^{K\times K'},$$

where the last $K' - K$ columns are 0. Then $\mathrm{Id} - \mu_{\mathrm{pca}}\mu_{\mathrm{pca}}^\top = \mathrm{diag}(1 - \mu_{\mathrm{pca},1}^2, \ldots, 1 -$

$\mu_{\mathrm{pca},K}^2) \in \mathbb{R}^{K\times K}$. We initialize

$$\boldsymbol{\mu}_0 = \mu_0 = \mu_{\mathrm{pca}}, \quad \boldsymbol{\Sigma}_0 = \sigma_{00} = \mathrm{Id} - \mu_{\mathrm{pca}}\mu_{\mathrm{pca}}^\top.$$

Inductively, having defined $(\boldsymbol{\mu}_{t-1}, \boldsymbol{\Sigma}_{t-1})$, we define a joint law for $(U'_*, U_0, \ldots, U_t, F_0, \ldots, F_{t-1})$ by

$$\begin{gathered}(F_0, \ldots, F_{t-1}) \mid U'_* \sim \mathcal{N}\Big(\boldsymbol{\mu}_{t-1} \cdot U'_*,\ \boldsymbol{\Sigma}_{t-1}\Big), \\ U_0 = S^{-1}F_0, \quad U_s = u_s(F_0, \ldots, F_{s-1}) \text{ for } s = 1, \ldots, t.\end{gathered} \tag{3.2.13}$$

We then define the next mean transformation $\boldsymbol{\mu}_t$ to have the blocks

$$\mu_s = \mathbb{E}[U_s {U'_*}^\top] \cdot S' \text{ for each } s = 0, \ldots, t. \tag{3.2.14}$$

For $s = 0$, this may be checked to coincide with the above initialization μ_{pca}.

We decompose the second moment matrix of $(U_0, \ldots, U_t)$ into four parts,

$$\begin{aligned}\boldsymbol{\Delta}_t \stackrel{\text{def}}{=} &\underbrace{\begin{pmatrix} 0 & 0 & \cdots & 0 \\ 0 & \mathbb{E}[U_1U_1^\top] & \cdots & \mathbb{E}[U_1U_t^\top] \\ \vdots & \vdots & \ddots & \vdots \\ 0 & \mathbb{E}[U_tU_1^\top] & \cdots & \mathbb{E}[U_tU_t^\top] \end{pmatrix}}_{\overline{\boldsymbol{\Delta}}_t} + \underbrace{\begin{pmatrix} 0 & \mathbb{E}[U_0U_1^\top] & \cdots & \mathbb{E}[U_0U_t^\top] \\ 0 & 0 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & 0 \end{pmatrix}}_{\widetilde{\boldsymbol{\Delta}}_t} \\ &+ \underbrace{\begin{pmatrix} 0 & 0 & \cdots & 0 \\ \mathbb{E}[U_1U_0^\top] & 0 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ \mathbb{E}[U_tU_0^\top] & 0 & \cdots & 0 \end{pmatrix}}_{\widetilde{\boldsymbol{\Delta}}_t^\top} + \underbrace{\begin{pmatrix} \mathbb{E}[U_0U_0^\top] & 0 & \cdots & 0 \\ 0 & 0 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & 0 \end{pmatrix}}_{\widehat{\boldsymbol{\Delta}}_t},\end{aligned} \tag{3.2.15}$$

set

$$\boldsymbol{\Delta}_t^{(j)} = \kappa_{j+2}\overline{\boldsymbol{\Delta}}_t + \tilde{\kappa}_{j+2} \odot \widetilde{\boldsymbol{\Delta}}_t + \widetilde{\boldsymbol{\Delta}}_t^\top \odot \tilde{\kappa}_{j+2} + \hat{\kappa}_{j+2} \odot \widehat{\boldsymbol{\Delta}}_t,$$

and define analogously to (3.1.3)

$$\mathbf{\Phi}_t = \Big(\mathbb{E}[\partial_s u_r(F_0, \ldots, F_{r-1})]\Big)_{r,s \in \{0,\ldots,t\}} \in \mathbb{R}^{(t+1)K \times (t+1)K}.$$

Then, recalling the function $\mathbf{\Theta}^{(j)}[\cdot,\cdot]$ from (3.1.6), we define the next covariance matrix $\mathbf{\Sigma}_t$ by

$$\mathbf{\Sigma}_t = \sum_{j=0}^{\infty} \mathbf{\Theta}^{(j)}[\mathbf{\Phi}_t, \mathbf{\Delta}_t^{(j)}]. \tag{3.2.16}$$

It may be checked from these definitions that the first t blocks of $\boldsymbol{\mu}_t$ coincide with $\boldsymbol{\mu}_{t-1}$, and the upper-left $t \times t$ blocks of $\mathbf{\Sigma}_t$ coincide with $\mathbf{\Sigma}_{t-1}$.

Our main result in the context of model (3.2.1) shows that this state evolution provides a rigorous characterization of the AMP algorithm (3.2.9–3.2.10) with the spectral initialization (3.2.8), under the following assumptions.

Assumption 3.2.2. (a) $\mathbf{U}'_* = (\mathbf{u}_*^1, \ldots, \mathbf{u}_*^{K'})$ is independent of $\mathbf{O}$, satisfies (3.2.2), and $\mathbf{U}'_* \xrightarrow{W_2} U'_*$ a.s. as $n \to \infty$ where $\mathbb{E}[U'_* {U'_*}^\top] = \mathrm{Id}$.

(b) Each $u_{t+1}(\cdot)$ is Lipschitz in all arguments. For each $s = 0, \ldots, t$ and all $(\boldsymbol{\mu}, \mathbf{\Sigma})$ in a sufficiently small open neighborhood of $(\boldsymbol{\mu}_t, \mathbf{\Sigma}_t)$ defined by (3.2.14) and (3.2.16), $\partial_s u_{t+1}(F_0, \ldots, F_t)$ exists and is continuous on a set of probability 1 under the marginal law of $(F_0, \ldots, F_t)$ defined by $(F_0, \ldots, F_t) \mid U'_* \sim \mathcal{N}(\boldsymbol{\mu} \cdot U'_*, \mathbf{\Sigma})$.

(c) $\theta_1, \ldots, \theta_{K'}$ are distinct. For each $k \in \{1, \ldots, K\}$, either $\theta_k > G(1/\lambda_+) \geq 0$ or $\theta_k < G(1/\lambda_-) \leq 0$, and there exists some constant $\iota \in (0,1)$ such that

$$\frac{\max(|\lambda_+|, |\lambda_-|)}{|\theta_k|} + \sum_{j=1}^{\infty} \frac{|\kappa_j|}{|\theta_k|^j \cdot \iota^{j-1}} < 1. \tag{3.2.17}$$

Assumption 3.2.2(c) requires $\theta_1, \ldots, \theta_K$ for the first K selected signals to be "super-critical", as described by Theorem 3.2.1. Furthermore, (3.2.17) requires each $|\theta_k|$ to exceed some constant depending only on the law of Λ. (We believe that this additional requirement may be an artifact of the proof technique, and we do not optimize the value of this constant.) Note that this requirement (3.2.17) is sufficient to imply that the series (3.2.6) for $R(z)$ and $R'(z)$ are absolutely convergent at $z = 1/\theta_k$, and hence the series (3.2.11) defining $\tilde{\kappa}_s, \hat{\kappa}_s$ are also absolutely convergent.

We remark that we assume for simplicity the distinctness of $\theta_1, \ldots, \theta_{K'}$. If these signal values are not all distinct, then the AMP state evolution parameters may not concentrate around deterministic values, and this has been discussed and analyzed in [MV21b] when the noise matrix is Gaussian. The study of similar phenomena in our model with rotationally-invariant noise may also be of interest, but we will not pursue this direction in the current paper.

Theorem 3.2.3. *Consider the symmetric spiked model (3.2.1), where Assumptions 3.1.1 and 3.2.2 hold. For any $T \geq 1$, consider the spectrally initialized AMP algorithm (3.2.8–3.2.10) up to iteration T, and define $(U'_*, U_0, \ldots, U_{T+1}, F_0, \ldots, F_T)$ by (3.2.13). Then almost surely as $n \to \infty$,*

$$(\mathbf{U}'_*, \mathbf{U}_0, \ldots, \mathbf{U}_{T+1}, \mathbf{F}_0, \ldots, \mathbf{F}_T) \xrightarrow{W_2} (U'_*, U_0, \ldots, U_{T+1}, F_0, \ldots, F_T).$$

Remark 3.2.4. As in Remark 3.1.4, we have defined b_{ts} in (3.2.12) using the free cumulants $\{\kappa_j\}$ of the limit noise spectral distribution, as well as the true signal values $\theta_1, \ldots, \theta_K$. Theorem 3.2.3 then also holds when b_{ts} are replaced by b'_{ts} such that $\|b_{ts} - b'_{ts}\| \to 0$ a.s., and in particular if $\{\kappa_j\}$, $\{\tilde{\kappa}_j\}$, $\{\hat{\kappa}_j\}$, and $\theta_1, \ldots, \theta_K$ are replaced by consistent estimates of these quantities.

3.2.3 Proof idea

We provide the proof of Theorem 3.2.3 in Section 3.4, and describe here the main idea. We construct an auxiliary AMP algorithm consisting of two phases. For some $\tau \geq 1$, we index the iterates in the first phase from $-\tau$ to 0. We consider an initialization $\mathbf{U}_{-\tau}^{(\tau)}$ that is independent of $\mathbf{W}$, and apply linear AMP iterations with $\mathbf{U}_t^{(\tau)} = \mathbf{F}_{t-1}^{(\tau)} S^{-1}$ for $t = -\tau + 1, \ldots, 0$. This has the effect of implementing a version of the power method to compute the sample eigenvectors $\mathbf{F}_{\mathrm{pca}}$ of $\mathbf{X}$. A main step of the proof is to show that as $\tau \to \infty$, the final iterate $\mathbf{F}_0^{(\tau)}$ obtained from these linear updates approaches the spectral initialization $\mathbf{F}_0 = \mathbf{F}_{\mathrm{pca}}$. Here, the assumption in (3.2.17) guarantees the desired convergence of $\mathbf{F}_0^{(\tau)}$ and the corresponding state evolution. Our analysis directly characterizes the alignment between $\mathbf{F}_0^{(\tau)}$ and $\mathbf{F}_{\mathrm{pca}}$ (which then implies the convergence of the state evolution) under general spectral distributions. This is different from [MV21a], which studied directly the asymptotic state evolution, and instead established the optimal signal-to-noise condition for convergence of the linear AMP iterations under a more restrictive class of spectral distributions having positive free cumulants.

In the second phase, starting from $\mathbf{F}_0^{(\tau)}$, we then apply the same functions $u_t(\cdot)$ as in the spectrally-initialized AMP algorithm for $t = 1, 2, \ldots$. The combined auxiliary AMP algorithm can thus be summarized as follows:

$$\mathbf{F}_t^{(\tau)} = \mathbf{X}\mathbf{U}_t^{(\tau)} - \sum_{s=-\tau}^{t} \mathbf{U}_s^{(\tau)} b_{ts}^{(\tau)\top}, \qquad \mathbf{U}_{t+1}^{(\tau)} = u_{t+1}(\mathbf{F}_{-\tau}^{(\tau)}, \ldots, \mathbf{F}_t^{(\tau)})$$

where

$$u_{t+1}(\mathbf{F}_{-\tau}^{(\tau)}, \ldots, \mathbf{F}_t^{(\tau)}) = \begin{cases} \mathbf{F}_t^{(\tau)} S^{-1} & \text{for } -\tau + 1 \le t < 0, \\ u_{t+1}(\mathbf{F}_0^{(\tau)}, \ldots, \mathbf{F}_t^{(\tau)}) & \text{for } 0 \le t \le T. \end{cases} \tag{3.2.18}$$

This AMP algorithm may be analyzed using Theorem 3.1.3 with side information $\mathbf{E} = \mathbf{U}_*'$, to yield a state evolution for iterates $t \ge 0$ characterized by some $(\boldsymbol{\mu}_t^{(\tau)}, \boldsymbol{\Sigma}_t^{(\tau)})$, which we describe explicitly in Corollary 3.4.1. Then we prove that in the limit $\tau \to \infty$, the iterates of this auxiliary AMP algorithm for $t \ge 0$ will be close to those of the spectrally-initialized AMP algorithm, in the sense that

$$\lim_{\tau\to\infty}\limsup_{n\to\infty}\frac{\|\mathbf{F}_t^{(\tau)}-\mathbf{F}_t\|_{\mathrm{F}}}{\sqrt{n}}=0,\qquad \lim_{\tau\to\infty}\limsup_{n\to\infty}\frac{\|\mathbf{U}_t^{(\tau)}-\mathbf{U}_t\|_{\mathrm{F}}}{\sqrt{n}}=0.$$

We also prove that as $\tau\to\infty$, the state evolution characterizing (3.2.18) converges to the state evolution described by (3.2.14) and (3.2.16), i.e. for each fixed $t\geq 1$,

$$\lim_{\tau\to\infty}\boldsymbol{\mu}_t^{(\tau)}=\boldsymbol{\mu}_t\quad\text{and}\quad\lim_{\tau\to\infty}\boldsymbol{\Sigma}_t^{(\tau)}=\boldsymbol{\Sigma}_t.$$

Combining the above two facts, we then can characterize the behavior of the AMP algorithm with spectral initialization.

3.3 Orthogonal AMP for Bayesian PCA

We discuss in this section an application to estimating the signal vectors $\mathbf{u}_*^k$ in the preceding symmetric signal-plus-noise model in Eq. (3.2.1).

The distributions of $U'_* \in \mathbb{R}^{K'}$ for the row-wise limits of $\mathbf{U}'_*$ may be interpreted as Bayesian "priors" for these rows. Assuming these prior distributions are known, we describe a Bayes-OAMP method in Section 3.3.1 that uses Bayes posterior-mean denoisers as the non-linearities in the preceding AMP algorithms. We suggest ways of estimating the Onsager debiasing coefficients and state evolution parameters in Section 3.3.2, and illustrate the method in simulation in Section 3.3.3. In practice, the distributions of U'_* are typically also unknown, but may be estimated from the data $\mathbf{X}$ using empirical Bayes ideas. We will not consider this additional complexity here, but refer readers to [ZSF20] for an example of this approach.

Let us describe the Bayes-OAMP method in a setting where the AMP dimension K and signal rank K' satisfy $K\leq K'$, reflecting applications where K' may be large, and one may wish to estimate smaller subsets of dependent PCs at a time. In this setting, we consider the following additional assumption for the laws of U'_*.

Assumption 3.3.1. The last $K' - K$ coordinates of U'_* have mean 0, and are independent of the first K coordinates.

In applications, the data $\mathbf{X}$ is often centered to have row and column means approximately 0 before applying PCA, leading to PCs also having mean approximately 0. The components $\mathbf{u}_*^k$ should ideally be grouped into small subsets of dependent signals, with the signals within each subset estimated together to maximally leverage their joint structure. The above assumption of exact independence of one such subset $(1, \ldots, K)$ from the remaining signals $(K+1, \ldots, K')$ is a modeling approximation for this grouping.

Assumption 3.3.1 ensures that when AMP is initialized with the first K signal components, its state evolution will not depend on the remaining $K' - K$ components: In the symmetric setting of Section 3.2, let $U_* \in \mathbb{R}^K$ be the first K coordinates of U'_*. Then it is inductively verified using Assumption 3.3.1 that each mean transformation $\boldsymbol{\mu}_t \in \mathbb{R}^{(t+1)K \times K'}$ has last $K' - K$ columns equal to 0, and the law of each vector F_t and U_t depends on U'_* only via U_*. We may then write the conditional law of $(F_0, \ldots, F_t)$ more simply as

$$(F_0, \ldots, F_t) \mid U_* \sim \mathcal{N}(\boldsymbol{\mu}_t \cdot U_*, \, \boldsymbol{\Sigma}_t)$$

where, with slight abuse of notation, we write $\boldsymbol{\mu}_t \in \mathbb{R}^{(t+1)K \times K}$ for its first K columns.

3.3.1 Bayes-OAMP

In the symmetric spiked model (3.2.1), under Assumption 3.3.1, we consider an AMP algorithm which estimates the first K components $\mathbf{U}_* \in \mathbb{R}^{n \times K}$ of $\mathbf{U}'_*$, using as its non-linearities the posterior-mean denoising functions

$$u_{t+1}(f_0, \ldots, f_t) = \mathbb{E}[U_* \mid (F_0, \ldots, F_t) = (f_0, \ldots, f_t)]$$

computed under the above conditional law $(F_0, \ldots, F_t) \mid U_* \sim \mathcal{N}(\boldsymbol{\mu}_t \cdot U_*,\, \boldsymbol{\Sigma}_t)$. Explicitly, writing as shorthand $\mathbf{f}_t = (f_0, \ldots, f_t) \in \mathbb{R}^{(t+1)K}$,

$$u_{t+1}(\mathbf{f}_t) = \frac{\mathbb{E}[U_* \exp(-\frac{1}{2}(\mathbf{f}_t - \boldsymbol{\mu}_t \cdot U_*)^\top \boldsymbol{\Sigma}_t^{-1}(\mathbf{f}_t - \boldsymbol{\mu}_t \cdot U_*)]}{\mathbb{E}[\exp(-\frac{1}{2}(\mathbf{f}_t - \boldsymbol{\mu}_t \cdot U_*)^\top \boldsymbol{\Sigma}_t^{-1}(\mathbf{f}_t - \boldsymbol{\mu}_t \cdot U_*)]} \in \mathbb{R}^K. \tag{3.3.1}$$

We will refer to the AMP algorithm (3.2.8–3.2.10) using this choice of non-linearity as *Bayes-OAMP*.

If this posterior mean function $u_{t+1}(\cdot)$ is Lipschitz (which holds e.g. if U_* has bounded support or log-concave density) and the signal strengths $|\theta_1|, \ldots, |\theta_K|$ are sufficiently large, then Theorem 3.2.3 implies that the mean-squared-error risk of the estimate $\mathbf{U}_t$ has the asymptotic limit

$$\lim_{n\to\infty} \frac{1}{n}\|\mathbf{U}_t - \mathbf{U}_*\|_F^2 \overset{\text{def}}{=} \mathrm{MSE}(\mathbf{U}_t) = \mathbb{E}\Big[\|U_* - \mathbb{E}[U_* \mid F_0, \ldots, F_t]\|^2\Big] = \operatorname{Tr}\operatorname{Cov}[U_* \mid F_0, \ldots, F_t].$$

Thus these errors satisfy

1. $\mathrm{MSE}(\mathbf{U}_{t+1}) \le \mathrm{MSE}(\mathbf{U}_t)$, by the property $\operatorname{Cov}[U_* \mid F_0, \ldots, F_{t+1}] \preceq \operatorname{Cov}[U_* \mid F_0, \ldots, F_t]$. This implies also by the martingale property of $U_1, U_2, \ldots$ that the algorithm is convergent in the sense

$$\lim_{n\to\infty} \frac{1}{n}\|\mathbf{U}_{t+1} - \mathbf{U}_t\|_F^2 = \mathbb{E}\Big[\|U_{t+1} - U_t\|^2\Big] = \mathrm{MSE}(\mathbf{U}_t) - \mathrm{MSE}(\mathbf{U}_{t+1}) \overset{t\to\infty}{\to} 0.$$

2. $\mathrm{MSE}(\mathbf{U}_t) \le \mathrm{MSE}(\mathbf{F}_{\mathrm{pca}})$ for any iterate $t \ge 1$, because

$$\mathrm{MSE}(\mathbf{F}_{\mathrm{pca}}) = \mathrm{MSE}(\mathbf{F}_0) = \mathbb{E}[\|U_* - F_0\|_2^2] \ge \mathbb{E}[\|U_* - \mathbb{E}[U_* \mid F_0, \ldots, F_t]\|_2^2] = \mathrm{MSE}(\mathbf{U}_t).$$

Remark 3.3.2. This algorithm differs from the Bayes-AMP algorithms of [RF12, MV21b] for Gaussian noise and the single-iterate Bayes-OAMP algorithm that was studied in [Fan20, Section 3], which use instead $u_{t+1}(f_0, \ldots, f_t) = \mathbb{E}[U_* \mid F_t = f_t]$

based only on the single preceding iterate F_t.

When $\mathbf{W}$ is symmetric Gaussian noise and Λ is distributed as Wigner's semicircle law, these two approaches coincide: Indeed, in this case $\kappa_2 = 1$ and $\kappa_j = 0$ for all $j \geq 3$, so (3.2.16) reduces to $\mathbf{\Sigma}_t = \mathbf{\Delta}_t = \mathbb{E}[UU^\top]$ where $U = (U_0, \ldots, U_t)$, coinciding with the state evolution shown in [MV21b]. For any $t \geq 1$ and $0 \leq s \leq t$, from the martingale identity

$$\mathbb{E}[U_s U_t^\top] = \mathbb{E}[U_s \mathbb{E}[U_*^\top \mid F_0, \ldots, F_t]] = \mathbb{E}[\mathbb{E}[U_s U_*^\top \mid F_0, \ldots, F_t]] = \mathbb{E}[U_s U_*^\top]$$

and the definition of $\boldsymbol{\mu}_t$ in (3.2.14), we observe that $\mathbf{\Sigma}_t(\mathbf{e}_t \odot S) = \boldsymbol{\mu}_t$ and $\sigma_{tt} S = \mu_t$, where $S = \operatorname{diag}(\theta_1, \ldots, \theta_K) \in \mathbb{R}^{K \times K}$, $\mathbf{e}_t \odot S \in \mathbb{R}^{(t+1)K \times K}$ has first t row blocks equal to 0 and last row block equal to S, and $\sigma_{tt}, \mu_t \in \mathbb{R}^{K \times K}$ denote the last blocks of $\mathbf{\Sigma}_t$ and $\boldsymbol{\mu}_t$. Then $\mathbf{\Sigma}_t^{-1} \boldsymbol{\mu}_t = \mathbf{e}_t \odot S$ and $\sigma_{tt}^{-1} \mu_t = S$. So (3.3.1) is equivalent to the single-iterate posterior mean,

$$\begin{aligned} u_{t+1}(\mathbf{f}_t) &= \frac{\mathbb{E}[U_* \exp(f_t^\top S U_* - \frac{1}{2} U_*^\top \mu_t^\top S U_*)]}{\mathbb{E}[\exp(f_t^\top S U_* - \frac{1}{2} U_*^\top \mu_t^\top S U_*)]} \\ &= \frac{\mathbb{E}[U_* \exp(-\frac{1}{2}(f_t - \mu_t U_*)^\top \sigma_{tt}^{-1}(f_t - \mu_t U_*)]}{\mathbb{E}[\exp(-\frac{1}{2}(f_t - \mu_t U_*)^\top \sigma_{tt}^{-1}(f_t - \mu_t U_*)]} = \mathbb{E}[U_* \mid F_t = f_t]. \end{aligned}$$

Outside of this setting where Λ has Wigner semicircle law, these two approaches are different. The state evolution of Bayes-OAMP is more involved, and we will not pursue a theoretical characterization of its fixed point in this work, as was done for single-iterate Bayes-OAMP in [Fan20]. We observe in simulation that the above monotonicity property of $\mathrm{MSE}(\mathbf{U}_t)$ need not hold for single-iterate Bayes-OAMP in settings of small signal strength, as was observed also in [MV21a], and that Bayes-OAMP can substantially improve over the single-iterate approach in such settings.

The development and rigorous characterization of Bayes-optimal estimation procedures for this PCA problem is an interesting open question. Following the initial posting of [ZWF21], [BCMS23] has obtained the first results in this direction, deriving a conjectural form of the Bayes-optimal error in a symmetric rank-1 spiked model using the replica method, for certain examples of rotationally-invariant noise matrices defined by polynomial potentials. The authors of [BCMS23] suggested also two AMP methods that numerically attain the conjectured Bayes-optimal error in these examples, one of which is an extension of Bayes-OAMP (dubbed "AMP-AP") that alternates between posterior mean and identity nonlinearities. Our current results establish the rigorous state evolution of both Bayes-OAMP and AMP-AP under a spectral initialization and sufficiently large signal strength, constituting a first step towards an analytic characterization of the estimation errors attained by these procedures.

3.3.2 Estimating the debiasing corrections and state evolution

Numerical implementations of the Bayes-OAMP algorithms require estimating the debiasing coefficients and state evolution parameters that describe the conditional laws of $(F_0, \ldots, F_t)$. We describe here one approach for this estimation for the symmetric model.

We estimate the law of Λ by the observed empirical eigenvalue distribution of $\mathbf{X}$, with largest K_+ positive eigenvalues and smallest K_- negative eigenvalues removed. This estimate is weakly consistent as $n \to \infty$ by Weyl's eigenvalue interlacing inequality. We compute the empirical moments of this law, and estimate the free cumulants $\{\kappa_s\}$ through the non-crossing moment-cumulant relations, see e.g. [Fan20, Section 2.3]. (Alternative methods for estimating the free cumulants of Λ based on power iteration have also been discussed in [LHK21, Proposition 1] and [VKM22, Section 3].) For each signal value θ_k, based on (3.2.5) and (3.2.7), we then estimate

$$\theta_k \text{ by } \frac{1}{G(\lambda_{\text{pca},k})}, \quad \mu^2_{\text{pca},k} \text{ by } \frac{-1}{\theta_k^2 G'(\lambda_{\text{pca},k})},$$
$$R(1/\theta_k) \text{ by } \lambda_{\text{pca},k} - \theta_k, \quad R'(1/\theta_k) \text{ by } \theta_k^2(1 - \mu^2_{\text{pca},k})$$

where $\lambda_{\text{pca},k}$ is the observed eigenvalue of $\mathbf{X}$, and the integrals defining $G(\cdot)$ and $G'(\cdot)$ are computed using the above empirical estimate for the law of Λ. In particular, this provides the estimate of $S = \text{diag}(\theta_1, \ldots, \theta_K)$. Comparing (3.2.6) and (3.2.11), we then estimate

$$\tilde{\kappa}_1 \text{ by } \text{diag}(R(1/\theta_1), \ldots, R(1/\theta_K)), \qquad \hat{\kappa}_2 \text{ by } \text{diag}(R'(1/\theta_1), \ldots, R'(1/\theta_K)),$$

and estimate the remaining matrices $\tilde{\kappa}_s$ and $\hat{\kappa}_s$ using the following recursions derived from (3.2.11),

$$\begin{aligned}
\tilde{\kappa}_s &= \sum_{j=0}^{\infty} \kappa_{s+j} S^{-j} = \kappa_s I + \sum_{j=0}^{\infty} \kappa_{s+1+j} S^{-j} \cdot S^{-1} = \kappa_s I + \tilde{\kappa}_{s+1} S^{-1} \\
&\qquad \Rightarrow \tilde{\kappa}_{s+1} = (\tilde{\kappa}_s - \kappa_s I) S, \\
\hat{\kappa}_s &= \sum_{j=0}^{\infty} (j+1) \kappa_{s+j} S^{-j} = \tilde{\kappa}_s + \sum_{j=0}^{\infty} (j+1) \cdot \kappa_{s+1+j} S^{-j} \cdot S^{-1} = \tilde{\kappa}_s + \hat{\kappa}_{s+1} S^{-1} \\
&\qquad \Rightarrow \hat{\kappa}_{s+1} = (\hat{\kappa}_s - \tilde{\kappa}_s) S.
\end{aligned}$$

Next, explicitly differentiating (3.3.1) to compute the matrices $\langle \partial_s \mathbf{U}_t \rangle$ constituting $\boldsymbol{\phi}_t$, and combining with the above, we obtain consistent estimates of the debiasing coefficients b_{ts}. For the state evolution, we note that numerically evaluating the expectations defining $(\boldsymbol{\mu}_t, \boldsymbol{\Sigma}_t)$ may be prohibitive. We instead approximate the second-moment matrices $\mathbb{E}[U_s U_t^\top]$ by the empirical averages $n^{-1} \mathbf{U}_s^\top \mathbf{U}_t$. We approximate $\boldsymbol{\Phi}_t$ also by the empirical average $\boldsymbol{\phi}_t$, and combine with the above estimates of $\kappa_s, \tilde{\kappa}_s, \hat{\kappa}_s$ according to the definition (3.2.16) to obtain a consistent estimate of $\boldsymbol{\Sigma}_t$. We use the martingale identity $\mathbb{E}[U_t U_*^\top] = \mathbb{E}[U_t U_t^\top]$ for $t \geq 1$ to approximate $\mathbb{E}[U_t U_*^\top]$ by the empirical average $n^{-1} \mathbf{U}_t^\top \mathbf{U}_t$, and use this to consistently estimate $\boldsymbol{\mu}_t$.

We remark that when the AMP algorithm approaches convergence, the covariance matrix $\boldsymbol{\Sigma}_t$ becomes nearly singular, leading to potential instabilities in evaluating the posterior mean function (3.3.1) and its Jacobian. We use a simple early stopping rule of terminating the iterations when the smallest eigenvalue of our estimate for $\mathbf{\Sigma}_t$ falls below a small threshold. Alternatively, a small ridge regularization may be used when computing $\mathbf{\Sigma}_t^{-1}$.

3.3.3 Simulation studies

We compare the estimation accuracy of three AMP algorithms under various noise spectral distributions for $\mathbf{W}$:

1. **Bayes-OAMP**, as described and implemented in Sections 3.3.1 and 3.3.2 above.

2. **Single-iterate Bayes-OAMP**, using instead the single-iterate posterior mean non-linearities
$u_{t+1}(f_0, \ldots, f_t) = \mathbb{E}[U_* \mid F_t = f_t]$, with debiasing coefficients and state evolution parameters estimated as in Section 3.3.2.

3. **Gaussian Bayes-AMP**, using the debiasing coefficients and state evolution described in [MV21b] for i.i.d. noise. We also estimate the signal strengths θ_k and initial spectral alignments $\mu^2_{\text{pca},k}$ from (3.2.5) using functions $G(\cdot)$ that correspond to the semicircle law for symmetric i.i.d. noise.

Symmetric model. We consider the symmetric model (3.2.1) with $n = 4000$, in the settings

- (Semicircle) $\mathbf{W} \sim \text{GOE}(n)$ has i.i.d. $\mathcal{N}(0, 1/n)$ entries above the diagonal, and the limit spectral law Λ is the semicircle distribution supported on $[-2, 2]$.

- (Uniform) $\mathbf{W} = \mathbf{O}^\top \mathbf{\Lambda} \mathbf{O}$ where $\mathbf{\Lambda}$ has i.i.d. Uniform$[-\sqrt{3}, \sqrt{3}]$ diagonal entries, and $\mathbf{O}$ is uniformly random.

- (Centered Beta) $\mathbf{W} = \mathbf{O}^\top \mathbf{\Lambda} \mathbf{O}$ where $\mathbf{\Lambda}$ has i.i.d. $\sqrt{80/3} \cdot (\text{Beta}(3, 1) - 3/4)$ diagonal entries, and $\mathbf{O}$ is uniformly random.

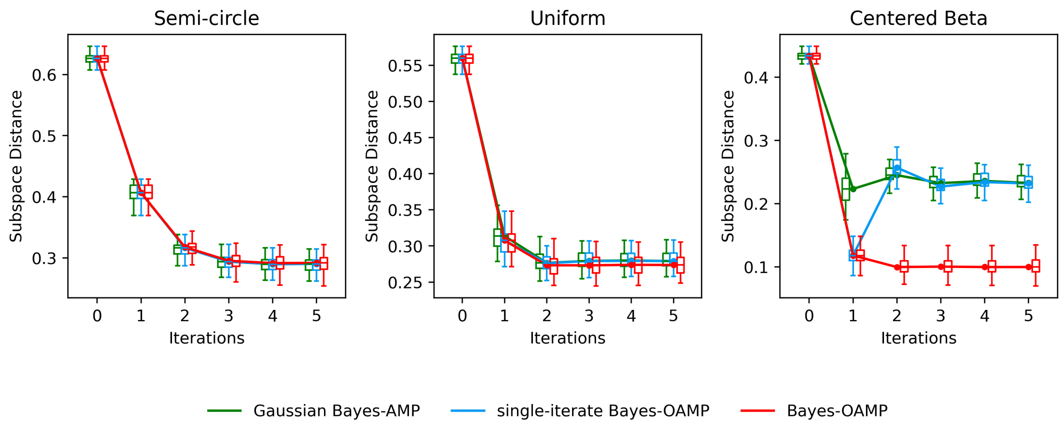


Figure 3.1: Estimation errors for AMP iterates $\mathbf{U}_t$ in the symmetric spiked model with $n = 4000$, rank-2 signal, and signal prior $U_* \sim \frac{1}{2}\delta_{(0,1)} + \frac{1}{4}\delta_{(\sqrt{2},-1)} + \frac{1}{4}\delta_{(-\sqrt{2},-1)}$. Boxes indicate the $\{25, 50, 75\}$-percentiles across 50 random trials, and whiskers indicate $1.5\times$inter-quartile range. Iteration 0 corresponds to the spectral initialization $\mathbf{U}_0$. The noise spectral distributions are (left) the semicircle law, (middle) Uniform$[-\sqrt{3}, \sqrt{3}]$, and (right) standarized Beta$(3, 1)$.

In all three settings, Λ is normalized to have mean $\kappa_1 = 0$ and variance $\kappa_2 = 1$. Figure 3.1 compares estimation error across AMP iterations, for an example with a rank-2 signal where $K' = K = 2$, the two signal strengths are $(\theta_1, \theta_2) = (2, 1.6)$, and the elements of the two signal vectors are drawn i.i.d. from the discrete three-point prior U_* defined by

$$\frac{1}{2}\delta_{(0,1)} + \frac{1}{4}\delta_{(\sqrt{2},-1)} + \frac{1}{4}\delta_{(-\sqrt{2},-1)}. \tag{3.3.2}$$

The error is defined by the subspace distance $\|\Pi_{\mathbf{U}_*} - \Pi_{\mathbf{U}_t}\|$ where $\Pi_{\mathbf{U}} \in \mathbb{R}^{n\times n}$ is the orthogonal projector onto the two-dimensional column span of $\mathbf{U}$. From the left panel, for GOE noise, we observe that the three algorithms yield comparable per-iteration error, and there is negligible degradation in accuracy from estimating the spectral free cumulants and the more complex state parameters in Bayes-OAMP. In the middle panel, for Uniform noise spectrum, Gaussian Bayes-AMP remains reasonably robust to the misspecification of the noise distribution. In the right panel, for Centered Beta noise spectrum, we observe a significant improvement of Bayes-OAMP over the other

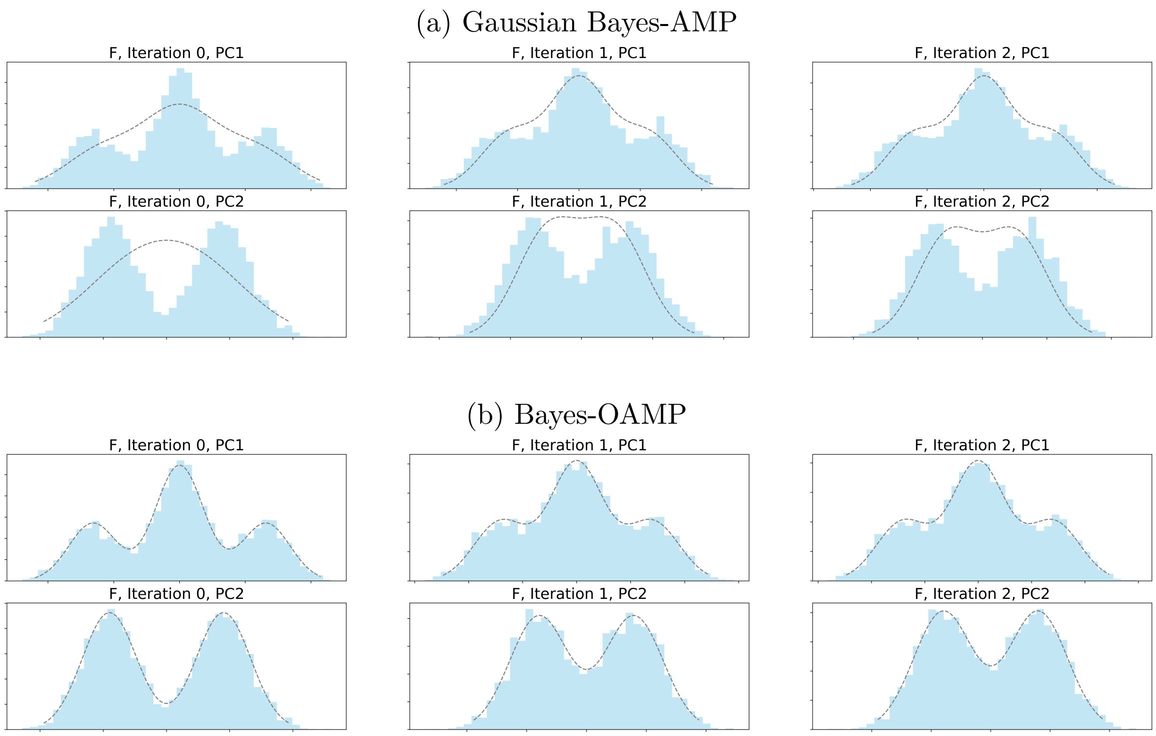


Figure 3.2: Distributions of iterates $\mathbf{F}_0, \mathbf{F}_1, \mathbf{F}_2$ in the Centered Beta noise example of Figure 3.1. Shown are histograms of the empirical distributions of the two columns of each iterate $\mathbf{F}_t$ (denoted PC1 and PC2), overlaid with the marginal density of the corresponding coordinate of the state evolution law $\mathcal{N}(\boldsymbol{\mu}_t \cdot U_*, \boldsymbol{\Sigma}_t)$. This density agrees closely for Bayes-OAMP, whereas a large discrepancy is observed for the state evolution prediction of Gaussian Bayes-AMP.

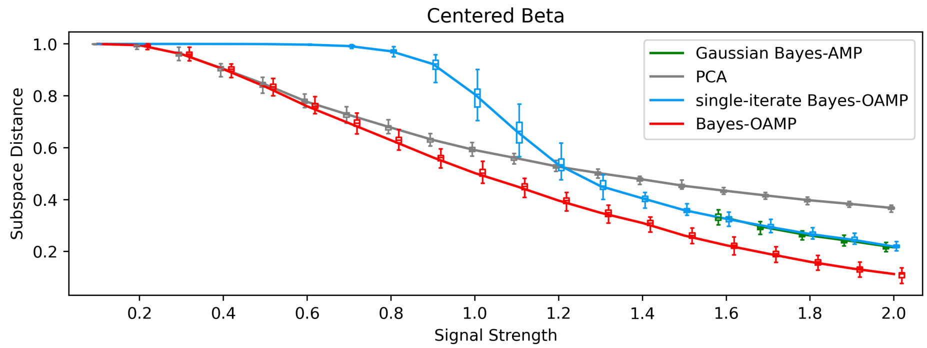


Figure 3.3: Estimation errors for the sample eigenvector and three AMP algorithms, in a symmetric model with Centered Beta noise spectrum, $n = 4000$, rank-1 signal with prior $U_* \sim \frac{1}{2}\delta_{-1} + \frac{1}{2}\delta_1$, and signal strength θ varying from 0.1 to 2.0. The spectral "phase transition" occurs at $\theta = 0$.

two approaches, which both exhibit non-monotonic error across iterations.

Figure 3.2 depicts the accuracy of the state evolution predictions for the distribution of AMP iterates $\mathbf{F}_t$ in the Centered Beta example. Marginal histograms of the two columns of $\mathbf{F}_t$ are overlaid with the corresponding densities of F_t, computed using state evolution parameters $(\boldsymbol{\mu}_t, \boldsymbol{\Sigma}_t)$ empirically estimated as in Section 3.3.2. Note that these densities are exactly the ones assumed for posterior-mean denoising in (3.3.1), to obtain the next iterate $\mathbf{U}_{t+1}$. We observe a close agreement for Bayes-OAMP, and a large discrepancy with the state evolution predictions of Gaussian Bayes-AMP.

Figure 3.3 illustrates estimation error across different signal strengths, in an example having rank-1 signal and the same Centered Beta noise. Note that since the density of Beta$(3, 1)$ does not decay to 0 at its right edge, the spectral phase transition point for this example is 0, and any positive signal strength is super-critical. We observe that Bayes-OAMP remains effective and improves over the spectral initialization for any positive signal strength. In contrast, single-iterate Bayes-OAMP exhibits the following behavior: below some critical signal strength, it diverges from the informative spectral initialization to an uninformative solution. This type of behavior was also reported in [MV21a, Section 4] for a different prior. Our implementation of Gaussian Bayes-AMP uses the Cauchy-transform $G(\cdot)$ based on the semicircle law to infer θ, and thus is only applicable when the largest sample eigenvalue exceeds 2, corresponding roughly to $\theta \geq 1.6$ in this example. For these values of θ, we observe its accuracy to be close to that of single-iterate Bayes-OAMP.

3.4 Proof for symmetric square matrices

3.4.1 State evolution for auxiliary AMP

As discussed in Section 3.2.3, in the setting of Theorem 3.2.3, we consider an auxiliary AMP algorithm starting at time index $-\tau$, with an initialization $\mathbf{U}_{-\tau}^{(\tau)} \in \mathbb{R}^{n \times K}$ independent of $\mathbf{W}$ and having the iterates, for $t = -\tau, -\tau+1, -\tau+2, \ldots$

$$\mathbf{F}_t^{(\tau)} = \mathbf{X}\mathbf{U}_t^{(\tau)} - \sum_{s=-\tau}^{t} \mathbf{U}_s^{(\tau)} b_{ts}^{(\tau)\top}, \qquad \mathbf{U}_{t+1}^{(\tau)} = u_{t+1}(\mathbf{F}_{-\tau}^{(\tau)}, \ldots, \mathbf{F}_t^{(\tau)}). \tag{3.4.1}$$

The coefficients $b_{ts}^{(\tau)}$ are defined as follows: For $T \geq 1$, we define

$$\boldsymbol{\phi}_{\text{all},T}^{(\tau)} = \Big(\langle \partial_s \mathbf{U}_r^{(\tau)} \rangle\Big)_{r,s \in \{-\tau,\ldots,T\}}, \qquad \mathbf{b}_{\text{all},T}^{(\tau)} = \sum_{j=0}^{\infty} \kappa_{j+1} \Big(\boldsymbol{\phi}_{\text{all},T}^{(\tau)}\Big)^j$$

where $\{\kappa_j\}_{j\geq 1}$ are the free cumulants of the limit eigenvalue distribution Λ for $\mathbf{W}$. We take the above debiasing coefficients up to iteration T to be the blocks, for $s, t \in \{-\tau, \ldots, T\}$,

$$b_{ts}^{(\tau)} = \mathbf{b}_{\text{all},T}^{(\tau)}[t, s] \tag{3.4.2}$$

Supposing that $(\mathbf{U}'_*, \mathbf{U}_{-\tau}^{(\tau)}) \xrightarrow{W_2} (U'_*, U_{-\tau}^{(\tau)})$, we define recursively the following state evolution: Having defined the joint law of $(U'_*, U_{-\tau}^{(\tau)}, \ldots, U_t^{(\tau)}, F_{-\tau}^{(\tau)}, \ldots, F_{t-1}^{(\tau)})$ for some $t \geq -\tau$, we define $\boldsymbol{\mu}_{\text{all},t}^{(\tau)} \in \mathbb{R}^{(t+\tau+1)K \times K'}$ having the blocks

$$\mu_s^{(\tau)} = \mathbb{E}[U_s U_*'^{\top}] \cdot S' \in \mathbb{R}^{K \times K'} \text{ for each } s = -\tau, \ldots, t.$$

Recalling the function $\mathbf{\Theta}^{(j)}[\cdot,\cdot]$ from (3.1.6), we define also the $(t+\tau+1)K\times(t+\tau+1)K$ matrices

$$\mathbf{\Delta}_{\text{all},t}^{(\tau)} = \Big(\mathbb{E}[U_r^{(\tau)}U_s^{(\tau)\top}]\Big)_{r,s\in\{-\tau,\ldots,t\}}, \qquad \mathbf{\Phi}_{\text{all},t}^{(\tau)} = \Big(\mathbb{E}[\partial_s u_r(F_{-\tau}^{(\tau)},\ldots,F_{r-1}^{(\tau)})]\Big)_{r,s\in\{-\tau,\ldots,t\}},$$

$$\mathbf{\Sigma}_{\text{all},t}^{(\tau)} = \sum_{j=0}^{\infty}\mathbf{\Theta}^{(j)}[\mathbf{\Phi}_{\text{all},t}^{(\tau)},\ \kappa_{j+2}\mathbf{\Delta}_{\text{all},t}^{(\tau)}].$$

Then we define the next joint law of $(U_*', U_{-\tau}^{(\tau)},\ldots,U_{t+1}^{(\tau)},F_{-\tau}^{(\tau)},\ldots,F_t^{(\tau)})$ by

$$(F_{-\tau}^{(\tau)},\ldots,F_t^{(\tau)}) \mid U_*' \sim \mathcal{N}\Big(\boldsymbol{\mu}_{\text{all},t}^{(\tau)}\cdot U_*',\ \mathbf{\Sigma}_{\text{all},t}^{(\tau)}\Big),$$
$$U_{s+1}^{(\tau)} = u_{s+1}(F_{-\tau}^{(\tau)},\ldots,F_s^{(\tau)}) \text{ for } s=-\tau,\ldots,t.$$

Corollary 3.4.1. *In the symmetric spiked model (3.2.1), suppose Assumption 3.1.1 holds for* $\mathbf{W}$*. Suppose the initialization* $\mathbf{U}_{-\tau}^{(\tau)}\in\mathbb{R}^{n\times K}$ *is independent of* $\mathbf{W}$*, and* $(\mathbf{U}_*',\mathbf{U}_{-\tau}^{(\tau)}) \xrightarrow{W_2} (U_*',U_{-\tau}^{(\tau)})$ *a.s. as* $n\to\infty$*. Suppose each function* $u_{t+1}(\cdot)$ *is Lipschitz, and each derivative* $\partial_s u_{t+1}(F_{-\tau}^{(\tau)},\ldots,F_t^{(\tau)})$ *exists and is continuous on a set of probability 1 under the above law of* $(F_{-\tau}^{(\tau)},\ldots,F_t^{(\tau)})$*. Then for any* $T\geq 1$*, a.s. as* $n\to\infty$*,*

$$(\mathbf{U}_*',\mathbf{U}_{-\tau}^{(\tau)},\ldots,\mathbf{U}_{T+1}^{(\tau)},\mathbf{F}_{-\tau}^{(\tau)},\ldots,\mathbf{F}_T^{(\tau)}) \xrightarrow{W_2} (U_*',U_{-\tau}^{(\tau)},\ldots,U_{T+1}^{(\tau)},F_{-\tau}^{(\tau)},\ldots,F_T^{(\tau)}).$$

Proof. The proof is similar to [Fan20, Theorem 3.1(a)]. Recalling $\mathbf{X}=n^{-1}\mathbf{U}_*'S'\mathbf{U}_*'^{\top}+\mathbf{W}$, we write the iterations (3.4.1) as

$$\mathbf{F}_t^{(\tau)} = \frac{1}{n}\mathbf{U}_*'\cdot S'\cdot\mathbf{U}_*'^{\top}\mathbf{U}_t^{(\tau)} + \mathbf{W}\mathbf{U}_t^{(\tau)} - \sum_{s=-\tau}^{t}\mathbf{U}_s^{(\tau)}b_{ts}^{(\tau)\top}.$$

Then, approximating $n^{-1}S'\cdot\mathbf{U}_*'^{\top}\mathbf{U}_t^{(\tau)}$ by $S'\cdot\mathbb{E}[U_*'U_t^{(\tau)\top}]=\mu_t^{(\tau)\top}$, we consider an alternative AMP sequence with the same initialization $\tilde{\mathbf{U}}_{-\tau}=\mathbf{U}_{-\tau}^{(\tau)}$ and "side information"

$\mathbf{U}'_*$, defined by

$$\tilde{\mathbf{Z}}_t = \mathbf{W}\tilde{\mathbf{U}}_t - \sum_{s=-\tau}^{t} \tilde{\mathbf{U}}_s \tilde{b}_{ts}^\top,$$
$$\tilde{\mathbf{F}}_t = \tilde{\mathbf{Z}}_t + \mathbf{U}'_* \mu_t^{(\tau)\top},$$
$$\tilde{\mathbf{U}}_{t+1} = \tilde{u}_{t+1}(\tilde{\mathbf{Z}}_{-\tau}, \ldots, \tilde{\mathbf{Z}}_t, \mathbf{U}'_*) \stackrel{\text{def}}{=} u_{t+1}(\tilde{\mathbf{F}}_{-\tau}, \ldots, \tilde{\mathbf{F}}_t).$$

Here, the debiasing coefficients are defined as

$$\tilde{b}_{ts} = \langle \partial_s \tilde{u}_{t+1}(\tilde{\mathbf{Z}}_{-\tau}, \ldots, \tilde{\mathbf{Z}}_t, \mathbf{U}'_*) \rangle = \langle \partial_s u_{t+1}(\tilde{\mathbf{F}}_{-\tau}, \ldots, \tilde{\mathbf{F}}_t) \rangle.$$

Using Theorem 3.1.3 to analyze this AMP algorithm, we have a.s. as $n \to \infty$ that

$$(\mathbf{U}'_*, \tilde{\mathbf{U}}_{-\tau}, \ldots, \tilde{\mathbf{U}}_{T+1}, \tilde{\mathbf{F}}_{-\tau}, \ldots, \tilde{\mathbf{F}}_T) \xrightarrow{W_2} (U'_*, U_{-\tau}^{(\tau)}, \ldots, U_{t+1}^{(\tau)}, F_{-\tau}^{(\tau)}, \ldots, F_t^{(\tau)})$$

for each fixed $t \geq -\tau$, as described by the above state evolution. Then, applying the same inductive argument as in [Fan20, Theorem 3.1(a)], we obtain a.s. as $n \to \infty$

$$n^{-1} \|\mathbf{U}_t^{(\tau)} - \tilde{\mathbf{U}}_t\|_F^2 \to 0, \quad n^{-1} \|\mathbf{F}_t^{(\tau)} - \tilde{\mathbf{F}}_t\|_F^2 \to 0$$

for each fixed $t \geq -\tau$, which implies this corollary. ■

As discussed in Section 3.2.3, we now specialize this auxiliary AMP algorithm (3.4.1) to the two-phase algorithm

$$u_{t+1}(\mathbf{F}_{-\tau}^{(\tau)}, \ldots, \mathbf{F}_t^{(\tau)}) = \begin{cases} \mathbf{F}_t^{(\tau)} S^{-1} & \text{for } -\tau + 1 \leq t < 0, \\ u_{t+1}(\mathbf{F}_0^{(\tau)}, \ldots, \mathbf{F}_t^{(\tau)}) & \text{for } t \geq 0. \end{cases}$$

The auxiliary algorithm is initialized at

$$\mathbf{U}_{-\tau}^{(\tau)} = (\mathbf{u}_{-\tau}^{1}, \dots, \mathbf{u}_{-\tau}^{K}) \text{ with } \mathbf{u}_{-\tau}^{k} = \left(\mu_{\text{pca},k}\mathbf{u}_{*}^{k} + \sqrt{1-\mu_{\text{pca},k}^{2}} \cdot \mathbf{z}_{k}\right) \Big/ \theta_{k} \text{ for each } k = 1, \dots, K,$$

where $\mu_{\text{pca},k}$ is as defined in Theorem 3.2.1, and $\mathbf{z}_1, \dots, \mathbf{z}_K$ are independent standard Gaussian random vectors also independent of $\mathbf{W}$.

For each $t \geq 1$, we adopt the following block decomposition of $\boldsymbol{\phi}_{\text{all},t}^{(\tau)}$, where the first block corresponds to indices $\{-\tau, \dots, -1\}$ and the second to indices $\{0, \dots, t\}$:

$$\boldsymbol{\phi}_{\text{all},t}^{(\tau)} = \begin{pmatrix} \boldsymbol{\phi}_{--}^{(\tau)} & \boldsymbol{\phi}_{-t}^{(\tau)} \\ \boldsymbol{\phi}_{t-}^{(\tau)} & \boldsymbol{\phi}_{t}^{(\tau)} \end{pmatrix}, \quad \text{where } \boldsymbol{\phi}_{--}^{(\tau)} \in \mathbb{R}^{\tau K \times \tau K} \text{ and } \boldsymbol{\phi}_{t}^{(\tau)} \in \mathbb{R}^{(t+1)K \times (t+1)K}.$$

Note that, due to the lower-triangular form of $\boldsymbol{\phi}_{\text{all},t}^{(\tau)}$ and the linear update rule for the first τ steps, we have $\boldsymbol{\phi}_{-t}^{(\tau)} = 0$ and

$$\boldsymbol{\phi}_{--}^{(\tau)} = \begin{pmatrix} 0 & 0 & \cdots & 0 & 0 \\ S^{-1} & 0 & \cdots & 0 & 0 \\ 0 & S^{-1} & \cdots & 0 & 0 \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & \cdots & S^{-1} & 0 \end{pmatrix}, \qquad \boldsymbol{\phi}_{t-}^{(\tau)} = \begin{pmatrix} 0 & \cdots & 0 & S^{-1} \\ 0 & \cdots & 0 & 0 \\ \vdots & \ddots & \vdots & \vdots \\ 0 & \cdots & 0 & 0 \end{pmatrix}.$$

Applying Lemma 3.6.2 with $\mathbf{A} = \boldsymbol{\phi}_{\text{all},t}^{(\tau)}$ and $\mathbf{B} = S^{-1}$, for all $r \in \{0, \dots, t\}$ and $c \in \{1, \dots, \tau\}$,

$$(\boldsymbol{\phi}_{\text{all},t}^{(\tau)})^{j}[r, -c] = \begin{cases} (\boldsymbol{\phi}_{t}^{(\tau)})^{j-c}[r, 0] S^{-c} & 1 \leq c \leq j, \\ 0 & j < c. \end{cases} \tag{3.4.3}$$

Similarly, for the state evolution, we decompose

$$\boldsymbol{\mu}_{\mathrm{all},t}^{(\tau)} = \begin{pmatrix} \boldsymbol{\mu}_{-}^{(\tau)} \\ \boldsymbol{\mu}_{t}^{(\tau)} \end{pmatrix}, \quad \boldsymbol{\Delta}_{\mathrm{all},t}^{(\tau)} = \begin{pmatrix} \boldsymbol{\Delta}_{--}^{(\tau)} & \boldsymbol{\Delta}_{-t}^{(\tau)} \\ \boldsymbol{\Delta}_{t-}^{(\tau)} & \boldsymbol{\Delta}_{t}^{(\tau)} \end{pmatrix},$$
$$\boldsymbol{\Phi}_{\mathrm{all},t}^{(\tau)} = \begin{pmatrix} \boldsymbol{\Phi}_{--}^{(\tau)} & \boldsymbol{\Phi}_{-t}^{(\tau)} \\ \boldsymbol{\Phi}_{t-}^{(\tau)} & \boldsymbol{\Phi}_{t}^{(\tau)} \end{pmatrix}, \quad \boldsymbol{\Sigma}_{\mathrm{all},t}^{(\tau)} = \begin{pmatrix} \boldsymbol{\Sigma}_{--}^{(\tau)} & \boldsymbol{\Sigma}_{-t}^{(\tau)} \\ \boldsymbol{\Sigma}_{t-}^{(\tau)} & \boldsymbol{\Sigma}_{t}^{(\tau)} \end{pmatrix}. \tag{3.4.4}$$

3.4.2 Phase I – linear AMP

We first establish the convergence of the iterates and the associated state evolution of the first τ steps of this auxiliary AMP algorithm, as $\tau \to \infty$. For notational convenience, in this section only, we re-index these iterates as $1, 2, \ldots, \tau$ and provide a standalone result for such a linear AMP algorithm.

Let $\mathbf{U}_1 = (\mathbf{u}_1^1, \ldots, \mathbf{u}_1^K)$ with each $\mathbf{u}_1^k = (\mu_{\mathrm{pca},k}\mathbf{u}_*^k + \sqrt{1-\mu_{\mathrm{pca},k}^2} \cdot \mathbf{z}_k)/\theta_k$. The first τ iterates of the above auxiliary AMP algorithm has the structure of the following linear AMP:

$$\mathbf{F}_t = \mathbf{X}\mathbf{U}_t - \sum_{i=1}^{t} \kappa_{t-i+1}\mathbf{U}_i S^{-(t-i)}, \qquad \mathbf{U}_{t+1} = \mathbf{F}_t S^{-1}. \tag{3.4.5}$$

We write $\mathbf{F}_0 = \mathbf{U}_1 S$. Up to iterate τ, let $\boldsymbol{\mu}_\tau = (\mu_t)_{1\le t\le\tau}$ and $\boldsymbol{\Sigma}_\tau = (\sigma_{st})_{1\le s,t\le\tau}$ be the parameters of the state evolution describing this linear AMP, where $\mu_t \in \mathbb{R}^{K\times K'}$ and $\sigma_{st} \in \mathbb{R}^{K\times K}$. Recall

$$\mu_{\mathrm{pca}} = \mathrm{diag}(\mu_{\mathrm{pca},1}, \ldots, \mu_{\mathrm{pca},K}) \in \mathbb{R}^{K\times K'},$$
$$\mathrm{Id} - \mu_{\mathrm{pca}}\mu_{\mathrm{pca}}^\top = \mathrm{diag}(1-\mu_{\mathrm{pca},1}^2, \ldots, 1-\mu_{\mathrm{pca},K}^2) \in \mathbb{R}^{K\times K}.$$

Then Corollary 3.4.1 ensures

$$(\mathbf{U}_*', \mathbf{U}_1, \ldots, \mathbf{U}_\tau, \mathbf{F}_1, \ldots, \mathbf{F}_\tau) \xrightarrow{W_2} (U_*', U_1, \ldots, U_\tau, F_1, \ldots, F_\tau)$$

where the limiting distribution is defined by

$$\begin{aligned}(F_1, \ldots, F_\tau) \mid U'_* &\sim \mathcal{N}(\boldsymbol{\mu}_\tau \cdot U'_*, \boldsymbol{\Sigma}_\tau),\\ U_1 \sim S^{-1} \cdot \mathcal{N}(\mu_{\mathrm{pca}} \cdot U'_*, \mathrm{Id} - \mu_{\mathrm{pca}}\mu_{\mathrm{pca}}^\top), \quad U_t &= S^{-1}F_{t-1} \text{ for } 2 \leq t \leq \tau.\end{aligned} \tag{3.4.6}$$

Lemma 3.4.2. *Under Assumptions 3.1.1 and 3.2.2(a) and (c), the following hold for the linear AMP algorithm (3.4.5):*

(a) $\lim_{t\to\infty} \limsup_{n\to\infty} \|\mathbf{F}_t - \mathbf{F}_{\mathrm{pca}}\|_{\mathrm{F}}/\sqrt{n} = 0$ *a.s.*

(b) The state evolution satisfies $\mu_t = \mu_{\mathrm{pca}}$ *for every* $t \geq 1$*, and* $\lim_{\min(s,t)\to\infty} \sigma_{st} = \mathrm{Id} - \mu_{\mathrm{pca}}\mu_{\mathrm{pca}}^\top$.

Proof. Recall the sample eigenvalues $\lambda_k(\mathbf{X})$ for $k = 1, \ldots, K'$ from (3.2.3), constituting the K_+ largest and K_- smallest eigenvalues of $\mathbf{X}$, with associated eigenvectors $\mathbf{f}_{\mathrm{pca}}^k$ satisfying (3.2.4). Denote the remaining eigenvalues and eigenvectors as $\lambda_i(\mathbf{X})$ and $\mathbf{f}_{\mathrm{pca}}^i$ for $i = K' + 1, \ldots, n$ in any order, with the same normalization $\|\mathbf{f}_{\mathrm{pca}}^i\| = \sqrt{n}$. Let

$$\mathcal{S} = \Big\{k \in \{1, \ldots, K'\} : \theta_k > 1/G(\lambda_+) \text{ or } \theta_k < 1/G(\lambda_-)\Big\}$$

be the indices corresponding to "super-critical" signal eigenvalues as characterized by Theorem 3.2.1. Denote $\|\Lambda\|_\infty = \max(|\lambda_+|, |\lambda_-|)$, fix a small constant $\delta > 0$, and define the event

$$\mathcal{E}_n = \Big\{|\lambda_i(\mathbf{X})| < \|\Lambda\|_\infty + \delta \text{ for all } i \notin \mathcal{S}\Big\}.$$

Then $\mathcal{E}_n$ occurs almost surely for all large n, where this bound for $i \in \{1, \ldots, K'\} \setminus \mathcal{S}$ follows from Theorem 3.2.1, and that for $i \in \{K' + 1, \ldots, n\}$ follows from Assumption 3.1.1(c) and Weyl's eigenvalue interlacing inequality.

Let $\mathbf{f}_t^k$ be the k^{th} column of the linear AMP iterate $\mathbf{F}_t$. For part (a), it suffices to show

$$\lim_{t\to\infty} \limsup_{n\to\infty} \|\mathbf{f}_t^k - \mathbf{f}_{\mathrm{pca}}^k\|/\sqrt{n} = 0$$

for each $k \in \{1, \dots, K\}$. Fixing any such k, by the definition of linear AMP in (3.4.5), we have

$$\mathbf{f}_t^k = \frac{1}{\theta_k} \cdot \mathbf{X}\mathbf{f}_{t-1}^k - \sum_{j=1}^{t} \frac{\kappa_{t-j+1}}{\theta_k^{t-j+1}} \cdot \mathbf{f}_{j-1}^k. \tag{3.4.7}$$

We first show that the component of $\mathbf{f}_t^k$ orthogonal to $\mathbf{f}_{\text{pca}}^k$ vanishes a.s. in the limits $t \to \infty$ and $n \to \infty$. Define $r_t^{k,i} = {\mathbf{f}_{\text{pca}}^i}^\top \mathbf{f}_t^k / n$ for each $i \in \{1, \dots, n\}$. Then applying (3.4.7),

$$\begin{aligned} r_t^{k,i} &= \frac{1}{\theta_k} \cdot \frac{(\mathbf{f}_{\text{pca}}^i)^\top \mathbf{X}\mathbf{f}_{t-1}^k}{n} - \sum_{j=0}^{t-1} \frac{\kappa_{t-j}}{\theta_k^{t-j}} \cdot \frac{(\mathbf{f}_{\text{pca}}^i)^\top \mathbf{f}_j^k}{n} \\ &= \frac{\lambda_i(\mathbf{X})}{\theta_k} \cdot r_{t-1}^{k,i} - \sum_{j=0}^{t-1} \frac{\kappa_{t-j}}{\theta_k^{t-j}} \cdot r_j^{k,i}. \end{aligned} \tag{3.4.8}$$

For any $i \in \mathcal{S} \setminus \{k\}$, by the initialization $\mathbf{f}_0^k = \theta_k \mathbf{u}_1^k = \mu_{\text{pca},k} \mathbf{u}_*^k + \sqrt{1 - \mu_{\text{pca},k}^2} \cdot \mathbf{z}_k$, we have

$$r_0^{k,i} = \frac{\mu_{\text{pca},k}}{n} (\mathbf{f}_{\text{pca}}^i)^\top \mathbf{u}_*^k + \frac{\sqrt{1 - \mu_{\text{pca},k}^2}}{n} (\mathbf{f}_{\text{pca}}^i)^\top \mathbf{z}_k \to 0$$

a.s. as $n \to \infty$, where the first term converges to 0 by Theorem 3.2.1, and the second term converges to 0 since ${\mathbf{f}_{\text{pca}}^i}^\top \mathbf{z}_k / n \sim \mathcal{N}(0, 1/n)$. Thus, it follows from the recursion (3.4.8) that

$$\lim_{n \to \infty} r_t^{k,i} = 0 \text{ a.s. for each fixed } t \geq 0 \text{ and } i \in \mathcal{S} \setminus \{k\}. \tag{3.4.9}$$

For any $i \notin \mathcal{S}$, consider a space $\mathcal{X}$ of bounded infinite-dimensional vectors with elements in $[0, \infty)$. For each $t \geq 0$, we define an element $\boldsymbol{\rho}_t^{k,i} \in \mathcal{X}$ by padding 0's after $(r_t^{k,i}, r_{t-1}^{k,i}, \dots, r_0^{k,i})$, i.e.,

$$\boldsymbol{\rho}_t^{k,i} = (r_t^{k,i}, r_{t-1}^{k,i}, \dots, r_0^{k,i}, 0, 0, \dots).$$

For some $\iota \in (0,1)$ chosen as in Assumption 3.2.2(c), let us consider a norm $\|\cdot\|$ on $\mathcal{X}$ defined by

$$\|(x_0, x_{-1}, x_{-2}, \ldots)\| = \sup_{j\geq 0} |x_{-j}| \cdot \iota^j.$$

Consider a map $g : \mathcal{X} \to \mathcal{X}$ defined as $g(\boldsymbol{\rho}_{t-1}^{k,i}) = \boldsymbol{\rho}_t^{k,i}$ for each $t \geq 1$. We verify that g is contractive with respect to the norm $\|\cdot\|$: Let $\{\boldsymbol{\rho}_t\}_{t\geq 1}$ and $\{\tilde{\boldsymbol{\rho}}_t\}_{t\geq 1}$ be two sequences of vectors in $\mathcal{X}$ given by

$$\begin{cases} \boldsymbol{\rho}_t = (r_t, r_{t-1}, \ldots, r_0, 0, \ldots) \\ \tilde{\boldsymbol{\rho}}_t = (\tilde{r}_t, \tilde{r}_{t-1}, \ldots, \tilde{r}_0, 0, \ldots) \end{cases}$$

where both $\{r_t\}_{t\geq 1}$ and $\{\tilde{r}_t\}_{t\geq 1}$ satisfy the same recursion as in (3.4.8). Then we have

$$\|g(\boldsymbol{\rho}_{t-1}) - g(\tilde{\boldsymbol{\rho}}_{t-1})\| = \|\boldsymbol{\rho}_t - \tilde{\boldsymbol{\rho}}_t\| = \sup_{0\leq j\leq t} |r_j - \tilde{r}_j| \cdot \iota^{t-j} = \max\left\{|r_t - \tilde{r}_t|, \iota \cdot \|\boldsymbol{\rho}_{t-1} - \tilde{\boldsymbol{\rho}}_{t-1}\|\right\}.$$

We then need to control $|r_t - \tilde{r}_t|$. It follows from (3.4.8) that

$$\begin{aligned} |r_t - \tilde{r}_t| &= \left| \frac{\lambda_i(\mathbf{X})}{\theta_k} \cdot (r_{t-1} - \tilde{r}_{t-1}) - \sum_{j=0}^{t-1} \frac{\kappa_{t-j}}{\theta_k^{t-j}} (r_j - \tilde{r}_j) \right| \\ &\leq \frac{|\lambda_i(\mathbf{X})|}{|\theta_k|} |r_{t-1} - \tilde{r}_{t-1}| + \sum_{j=0}^{t-1} \frac{|\kappa_{t-j}|}{|\theta_k|^{t-j} \iota^{t-1-j}} |r_j - \tilde{r}_j| \iota^{t-1-j} \\ &\leq \left(\frac{|\lambda_i(\mathbf{X})|}{|\theta_k|} + \sum_{j=1}^{t} \frac{|\kappa_j|}{|\theta_k|^j \iota^{j-1}} \right) \cdot \max_{0\leq j\leq t-1} |r_j - \tilde{r}_j| \iota^{t-1-j} \\ &\leq \underbrace{\left(\frac{\|\Lambda\|_\infty + \delta}{|\theta_k|} + \sum_{j=1}^{\infty} \frac{|\kappa_j|}{|\theta_k|^j \iota^{j-1}} \right)}_{\eta} \cdot \|\boldsymbol{\rho}_{t-1} - \tilde{\boldsymbol{\rho}}_{t-1}\| \end{aligned}$$

where the last inequality holds on the event $\mathcal{E}_n$ defined above. For sufficiently small $\delta > 0$, we have $\eta \in (0,1)$ by Assumption 3.2.2(c). Then denoting $\rho = \max(\eta, \iota) \in (0,1)$, we obtain

$$\|g(\boldsymbol{\rho}_{t-1}) - g(\tilde{\boldsymbol{\rho}}_{t-1})\| \le \rho \cdot \|\boldsymbol{\rho}_{t-1} - \tilde{\boldsymbol{\rho}}_{t-1}\|.$$

Therefore, g is a ρ-contraction. Applying this property to $\{\boldsymbol{\rho}_t^{k,i}\}_{t\ge1}$ yields

$$|r_t^{k,i}| \le \|(r_t^{k,i}, r_{t-1}^{k,i}, \ldots, r_0^{k,i}, 0, \ldots) - (0, 0, \ldots)\| \le \rho^t \cdot \|(r_0^{k,i}, 0, \ldots) - (0, 0, \ldots)\| = \rho^t \cdot |r_0^{k,i}|. \tag{3.4.10}$$

This holds simultaneously for all $i \notin \mathcal{S}$ on the event $\mathcal{E}_n$.

Now write $\mathbf{f}_t^k = \xi_t^k \mathbf{f}_{\text{pca}}^k + \mathbf{r}_t^k$ where $\mathbf{r}_t^k$ is orthogonal to $\mathbf{f}_{\text{pca}}^k$. Since $\{\mathbf{f}_{\text{pca}}^i/\sqrt{n}\}_{i=1,\ldots,n}$ is an orthonormal basis of $\mathbb{R}^n$, we can expand $\mathbf{r}_t^k$ as

$$\mathbf{r}_t^k = \sum_{i=1, i\ne k}^n \frac{{\mathbf{f}_{\text{pca}}^i}^\top \mathbf{f}_t^k}{n} \cdot \mathbf{f}_{\text{pca}}^i = \underbrace{\sum_{i\in\mathcal{S}\setminus\{k\}} r_t^{k,i}\mathbf{f}_{\text{pca}}^i}_{\mathcal{I}_1} + \underbrace{\sum_{i\notin\mathcal{S}} r_t^{k,i}\mathbf{f}_{\text{pca}}^i}_{\mathcal{I}_2}.$$

By (3.4.9), we have $\lim_{n\to\infty}\|\mathcal{I}_1\|/\sqrt{n} = 0$ for each fixed $t \ge 1$. For $\mathcal{I}_2$, we apply (3.4.10) on the event $\mathcal{E}_n$:

$$\frac{\|\mathcal{I}_2\|^2}{n} = \sum_{i\notin\mathcal{S}} (r_t^{k,i})^2 \le \rho^{2t} \cdot \sum_{i\notin\mathcal{S}} (r_0^{k,i})^2 \le \rho^{2t} \cdot \frac{\|\mathbf{r}_0^k\|^2}{n} \le \rho^{2t} \cdot \frac{\|\mathbf{f}_0^k\|^2}{n}.$$

By the initialization $\mathbf{f}_0^k = \mu_{\text{pca},k}\mathbf{u}_*^k + \sqrt{1-\mu_{\text{pca},k}^2} \cdot \mathbf{z}_k$, we have $\lim_{n\to\infty}\|\mathbf{f}_0^k\|^2/n = 1$. Thus, as desired,

$$\lim_{t\to\infty}\limsup_{n\to\infty} \frac{\|\mathbf{r}_t^k\|}{\sqrt{n}} = 0. \tag{3.4.11}$$

We now show that $\xi_t^k \overset{\text{def}}{=} {\mathbf{f}_t^k}^\top \mathbf{f}_{\text{pca}}^k / n \to 1$ by using the state evolution (3.4.6) of linear AMP. By this state evolution, we have

$$\mu_{t+1} = \mathbb{E}[U_{t+1}{U_*'}^\top] \cdot S' = S^{-1} \cdot \mathbb{E}[F_t{U_*'}^\top] \cdot S' = S^{-1} \cdot \mathbb{E}[(\mu_t U_*' + Z_t){U_*'}^\top] \cdot S' = S^{-1}\mu_t S'$$

where the second equality follows from $U_{t+1} = S^{-1}F_t$ and the last equality is due to $Z_t \perp\!\!\!\perp U_*'$ and $\mathbb{E}[U_*'{U_*'}^\top] = \mathrm{Id}$. From the definition of $\mathbf{U}_1$, it may be checked that $\mu_1 = \mu_{\text{pca}} \in \mathbb{R}^{K\times K'}$. Then it follows from the above that $\mu_t = \mu_{\text{pca}}$ for all $t \geq 1$. Thus for each $k \in \{1, \dots, K\}$, we have

$$\mu_{\text{pca},k} = \lim_{n\to\infty} \frac{{\mathbf{f}_{\text{pca}}^k}^\top \mathbf{u}_*^k}{n}, \qquad \mu_{\text{pca},k} = \lim_{n\to\infty} \frac{{\mathbf{f}_t^k}^\top \mathbf{u}_*^k}{n} = \lim_{n\to\infty} \xi_t^k \cdot \frac{{\mathbf{f}_{\text{pca}}^k}^\top \mathbf{u}_*^k}{n} + \frac{{\mathbf{r}_t^k}^\top \mathbf{u}_*^k}{n}$$

where the left equality follows from Theorem 3.2.1, and the right equality applies $\mu_t = \mu_{\text{pca}}$. This further implies that

$$\limsup_{n\to\infty} \left| (\xi_t^k - 1)\frac{{\mathbf{f}_{\text{pca}}^k}^\top \mathbf{u}_*^k}{n} \right| \leq \limsup_{n\to\infty} \frac{\|\mathbf{r}_t^k\|}{\sqrt{n}}.$$

Since $\lim_{n\to\infty} {\mathbf{f}_{\text{pca}}^k}^\top \mathbf{u}_*^k/n = \mu_{\text{pca},k} \neq 0$ a.s. by Theorem 3.2.1, taking the limit as $t \to \infty$ on both sides, it follows from (3.4.11) that

$$\lim_{t\to\infty} \limsup_{n\to\infty} |\xi_t^k - 1| = 0. \tag{3.4.12}$$

Then, recalling $\mathbf{f}_t^k = \xi_t^k \mathbf{f}_{\text{pca}}^k + \mathbf{r}_t^k$ and combining with (3.4.11),

$$\lim_{t\to\infty} \limsup_{n\to\infty} \frac{\|\mathbf{f}_t^k - \mathbf{f}_{\text{pca}}^k\|}{\sqrt{n}} = 0.$$

This shows both part (a) and the claim about μ_t in part (b).

It remains to show the convergence of σ_{st} in part (b). By (3.4.6) and the above identities $\mathbb{E}[F_t {U'_*}^\top] = \mu_t = \mu_{\rm pca}$, we have

$$\sigma_{st} = \mathbb{E}[(F_s - \mu_s U'_*)(F_t - \mu_t U'_*)^\top] = \mathbb{E}[F_s F_t^\top] - \mu_{\rm pca}\mu_{\rm pca}^\top = \lim_{n\to\infty} \frac{\mathbf{F}_s^\top \mathbf{F}_t}{n} - \mu_{\rm pca}\mu_{\rm pca}^\top.$$

Thus, using the notation $\langle \mathbf{uv}\rangle = \mathbf{u}^\top \mathbf{v}/n$ and recalling the decomposition $\mathbf{f}_t^k = \xi_t^k \mathbf{f}_{\rm pca}^k + \mathbf{r}_t^k$, we can bound the difference between $(1-\mu_{{\rm pca},k}^2)\,\mathbb{1}\{k=k'\}$ and the $(k,k')^{\rm th}$ entry of σ_{st} as

$$\begin{aligned}
&|\sigma_{st}[k,k'] - (1-\mu_{{\rm pca},k}^2)\,\mathbb{1}\{k=k'\}| \\
&\leq \limsup_{n\to\infty} \left|\xi_s^k \xi_t^{k'} \langle \mathbf{f}_{\rm pca}^k \mathbf{f}_{\rm pca}^{k'}\rangle - \mathbb{1}\{k=k'\} + \xi_s^k \langle \mathbf{f}_{\rm pca}^k \mathbf{r}_t^{k'}\rangle + \xi_t^{k'} \langle \mathbf{f}_{\rm pca}^{k'} \mathbf{r}_s^{k}\rangle + \langle \mathbf{r}_s^k \mathbf{r}_t^{k'}\rangle\right| \\
&\leq \limsup_{n\to\infty} \mathbb{1}\{k=k'\}|\xi_s^k \xi_t^{k'} - 1| + |\xi_s^k \langle \mathbf{f}_{\rm pca}^k \mathbf{r}_t^{k'}\rangle| + |\xi_t^{k'} \langle \mathbf{f}_{\rm pca}^{k'} \mathbf{r}_s^{k}\rangle| + |\langle \mathbf{r}_s^k \mathbf{r}_t^{k'}\rangle|
\end{aligned}$$

where the last inequality applies the triangle inequality and the orthogonality and normalization of $\mathbf{f}_{\rm pca}^k$ in (3.2.4). Finally, by the convergence of $\mathbf{r}_t$ in (3.4.11) and that of ξ_t in (3.4.12) as $t\to\infty$, we get

$$\lim_{\min(s,t)\to\infty} \sigma_{st}[k,k'] = (1-\mu_{{\rm pca},k}^2)\cdot \mathbb{1}\{k=k'\},$$

i.e. $\lim_{\min(s,t)\to\infty} \sigma_{st} = \mathrm{Id} - \mu_{\rm pca}\mu_{\rm pca}^\top$. ∎

The auxiliary AMP iterations $\mathbf{U}_{-\tau}^{(\tau)}, \mathbf{F}_{-\tau}^{(\tau)}, \mathbf{U}_{-\tau+1}^{(\tau)}, \ldots$ are defined by this linear AMP algorithm up to the iterates $\mathbf{U}_0^{(\tau)}$ and $\mathbf{F}_0^{(\tau)}$. Thus, translating Lemma 3.4.2 back to the indexing of this auxiliary AMP algorithm, and recalling the spectral initializations $\mathbf{F}_0 = \mathbf{F}_{\rm pca}$ and $\mathbf{U}_0 = \mathbf{F}_{\rm pca} S^{-1}$, the lemma implies the following:

$$\lim_{\tau\to\infty}\limsup_{n\to\infty} \frac{\|\mathbf{F}_i^{(\tau)} - \mathbf{F}_0\|_{\rm F}}{\sqrt{n}} = 0, \quad \lim_{\tau\to\infty}\limsup_{n\to\infty} \frac{\|\mathbf{U}_i^{(\tau)} - \mathbf{U}_0\|_{\rm F}}{\sqrt{n}} = 0 \text{ for all fixed } i \leq 0,$$
$$\mu_i^{(\tau)} = \mu_{\rm pca}, \quad \lim_{\tau\to\infty} \mathbf{\Sigma}_{{\rm all},t}^{(\tau)}[i,j] = \mathrm{Id} - \mu_{\rm pca}\mu_{\rm pca}^\top \text{ for all fixed } i,j\leq 0. \quad (3.4.13)$$

3.4.3 Phase II - auxiliary AMP

Now we proceed to prove Theorem 3.2.3 which provides a precise characterization of the state evolution of the AMP algorithm with spectral initialization for symmetric matrices.

Proof of Theorem 3.2.3. We show by induction that the following statements hold a.s. for each $t \geq 0$. In particular, part (b) is the main result that we want to prove, and all of the other parts serve as a road map for the proof. Parts (a)–(d) will apply standard comparison arguments, and the specific definitions (3.2.12) and (3.2.16) will be used to verify parts (e) and (f).

(a) $\lim_{\tau\to\infty} \limsup_{n\to\infty} \|\mathbf{U}_t^{(\tau)} - \mathbf{U}_t\|_{\mathrm{F}}/\sqrt{n} = 0$ and $\lim_{n\to\infty} \|\mathbf{U}_t\|_{\mathrm{F}}/\sqrt{n} < C_t$ for a constant $C_t > 0$.

(b) $(\mathbf{U}'_*, \mathbf{U}_0, \dots, \mathbf{U}_t, \mathbf{F}_0, \dots, \mathbf{F}_{s-1}) \xrightarrow{W_2} (U'_*, U_0, \dots, U_t, F_0, \dots, F_{t-1})$ where the limit distribution is defined by the recursion (3.2.13).

(c) $\lim_{\tau\to\infty} \mathbf{\Phi}_t^{(\tau)} = \mathbf{\Phi}_t$ and $\lim_{\tau\to\infty} \mathbf{\Delta}_t^{(\tau)} = \mathbf{\Delta}_t$.

(d) $\lim_{\tau\to\infty} \lim_{n\to\infty} \|\boldsymbol{\phi}_t^{(\tau)} - \boldsymbol{\phi}_t\| = 0$.

(e) $\lim_{\tau\to\infty} \limsup_{n\to\infty} \|\mathbf{F}_t^{(\tau)} - \mathbf{F}_t\|_{\mathrm{F}}/\sqrt{n} = 0$ and $\lim_{n\to\infty} \|\mathbf{F}_s\|_{\mathrm{F}}/\sqrt{n} < C_t$ for a constant $C_t > 0$.

(f) $\lim_{\tau\to\infty} \boldsymbol{\mu}_t^{(\tau)} = \boldsymbol{\mu}_t$ and $\lim_{\tau\to\infty} \mathbf{\Sigma}_t^{(\tau)} = \mathbf{\Sigma}_t$.

Denote by $t^{(a)}, t^{(b)}, \dots, t^{(f)}$ the claims of parts (a-f) at iteration t. We induct on t. For the base case $t = 0$, parts (a), (e), and (f) follow from (3.4.13) proved in the previous section, and the initializations $\mathbf{U}_0 = \mathbf{F}_{\mathrm{pca}} S^{-1}$ and $\mathbf{F}_0 = \mathbf{F}_{\mathrm{pca}}$. For (c) and (d), we have $\boldsymbol{\phi}_0^{(\tau)} = \boldsymbol{\phi}_0 = \mathbf{\Phi}_0^{(\tau)} = \mathbf{\Phi}_0 = 0$, while $\lim_{\tau\to\infty} \mathbf{\Delta}_0^{(\tau)} = \mathbf{\Delta}_0$ follows from part (a).

For (b), $(\mathbf{U}'_*, \mathbf{U}_0) \xrightarrow{W_2} (U'_*, U_0)$ follows from the convergence $(\mathbf{U}'_*, \mathbf{U}_0^{(\tau)}) \xrightarrow{W_2} (U'_*, U_0^{(\tau)})$ for the auxiliary AMP algorithm, together with (3.4.13). Thus all statements hold for $t = 0$.

For the induction step, let $t \geq 1$, and assume $s^{(a-f)}$ holds for $0 \leq s \leq t-1$. Let us then prove statements (a–f) for iteration t.

Part (a) By Assumption 3.2.2(b), u_t is Lipschitz, so there exists some $L > 0$ such that

$$\frac{\|\mathbf{U}_t^{(\tau)} - \mathbf{U}_t\|_\mathrm{F}}{\sqrt{n}} = \frac{\|u_t(\mathbf{F}_0^{(\tau)}, \dots, \mathbf{F}_{t-1}^{(\tau)}) - u_t(\mathbf{F}_0, \dots, \mathbf{F}_{t-1})\|_\mathrm{F}}{\sqrt{n}} \leq L \cdot \sum_{s=0}^{t-1} \frac{\|\mathbf{F}_s^{(\tau)} - \mathbf{F}_s\|_\mathrm{F}}{\sqrt{n}}.$$

Then by the induction hypothesis $t-1^{(e)}$, $\lim_{\tau\to\infty} \limsup_{n\to\infty} \|\mathbf{U}_t^{(\tau)} - \mathbf{U}_t\|_\mathrm{F}/\sqrt{n} = 0$. Moreover, for a constant $C_t > 0$,

$$\lim_{n\to\infty} \frac{\|\mathbf{U}_t\|_\mathrm{F}^2}{n} = \lim_{n\to\infty} \frac{\|u_t(\mathbf{F}_0, \dots, \mathbf{F}_{t-1})\|_\mathrm{F}^2}{n} = \mathbb{E}\left[\|u_t(F_0, \dots, F_{t-1})\|^2\right] < C_t.$$

Part (b) Let $u_{*,i} \in \mathbb{R}^{K'}$ denote the i^{th} row of $\mathbf{U}'_* \in \mathbb{R}^{n\times K'}$, and similarly for the other matrix variables. Let g be any pseudo-Lipschitz function, i.e. $|g(x) - g(y)| \leq C(1 + \|x\| + \|y\|)\|x - y\|$ for a constant $C > 0$. Then there exist some constant $C' > 0$ such that

$$\begin{aligned}
&\frac{1}{n}\sum_{i=1}^n \left| g(u_{*,i}, u_{0,i}^{(\tau)}, \dots, u_{t,i}^{(\tau)}, f_{0,i}^{(\tau)}, \dots, f_{t-1,i}^{(\tau)}) - g(u_{*,i}, u_{0,i}, \dots, u_{t,i}, f_{0,i}, \dots, f_{t-1,i}) \right| \\
&\leq \frac{C}{n}\sum_{i=1}^n \left(1 + \|u_{*,i}\| + \sum_{s=0}^{t}(\|u_{s,i}\| + \|u_{s,i}^{(\tau)}\|) + \sum_{s=0}^{t-1}(\|f_{s,i}\| + \|f_{s,i}^{(\tau)}\|)\right) \\
&\qquad\qquad \cdot \left(\sum_{s=0}^{t} \|u_{s,i} - u_{s,i}^{(\tau)}\| + \sum_{s=0}^{t-1} \|f_{s,i} - f_{s,i}^{(\tau)}\|\right) \\
&\leq \frac{C'}{n}\left(\sum_{i=1}^n \left(1 + \|u_{*,i}\|^2 + \sum_{s=0}^{t}(\|u_{s,i}\|^2 + \|u_{s,i}^{(\tau)}\|^2) + \sum_{s=0}^{t-1}(\|f_{s,i}\|^2 + \|f_{s,i}^{(\tau)}\|^2)\right)\right)^{1/2} \\
&\qquad\qquad \cdot \left(\sum_{i=1}^n \sum_{s=0}^{t} \|u_{s,i} - u_{s,i}^{(\tau)}\|^2 + \sum_{s=0}^{t-1} \|f_{s,i} - f_{s,i}^{(\tau)}\|^2\right)^{1/2}
\end{aligned}$$

$$\leq C' \underbrace{\left(1 + \frac{\|\mathbf{U}'_*\|_{\mathrm{F}} + \sum_{s=0}^{t} \|\mathbf{U}_s\|_{\mathrm{F}} + \|\mathbf{U}_s^{(\tau)}\|_{\mathrm{F}} + \sum_{s=0}^{t-1} \|\mathbf{F}_s\|_{\mathrm{F}} + \|\mathbf{F}_s^{(\tau)}\|_{\mathrm{F}}}{\sqrt{n}}\right)}_{\mathcal{I}_1}$$
$$\cdot \underbrace{\frac{\sum_{s=0}^{t} \|\mathbf{U}_s - \mathbf{U}_s^{(\tau)}\|_{\mathrm{F}} + \sum_{s=0}^{t-1} \|\mathbf{F}_s - \mathbf{F}_s^{(\tau)}\|_{\mathrm{F}}}{\sqrt{n}}}_{\mathcal{I}_2}$$

where the last two inequalities follow from Cauchy-Schwartz and the triangle inequality for the Frobenius norm, respectively. By $t^{(a)}$ and the induction hypothesis $t-1^{(e)}$, $\lim_{\tau\to\infty} \lim_{n\to\infty} \mathcal{I}_1 < C_t$ for a constant $C_t > 0$, and $\lim_{\tau\to\infty} \limsup_{n\to\infty} \mathcal{I}_2 = 0$. It then follows that

$$\lim_{\tau\to\infty} \limsup_{n\to\infty} \frac{1}{n} \sum_{i=1}^{n} \left| g(u_{*,i}, u_{0,i}^{(\tau)}, \ldots, u_{t,i}^{(\tau)}, f_{0,i}^{(\tau)}, \ldots, f_{t-1,i}^{(\tau)}) - g(u_{*,i}, u_{0,i}, \ldots, u_{t,i}, f_{0,i}, \ldots, f_{t-1,i}) \right| = 0.$$

By the induction hypothesis $t-1^{(f)}$ and Assumption 3.2.2(b), for all sufficiently large τ and each $1 \leq s < r \leq t$, $\partial_s u_r(F_0^{(\tau)}, \ldots, F_{r-1}^{(\tau)})$ exists and is continuous on a set of probability 1 under the law of $F_0^{(\tau)}, \ldots, F_{r-1}^{(\tau)}$ prescribed by Corollary 3.4.1. Thus, for all sufficiently large τ, Corollary 3.4.1 applies to this auxiliary AMP algorithm up to iteration t, to yield

$$\lim_{n\to\infty} \frac{1}{n} \sum_{i=1}^{n} g(u_{*,i}, u_{0,i}^{(\tau)}, \ldots, u_{t,i}^{(\tau)}, f_{0,i}^{(\tau)}, \ldots, f_{t-1,i}^{(\tau)}) = \mathbb{E}\left[g(U'_*, U_0^{(\tau)}, \ldots, U_t^{(\tau)}, F_0^{(\tau)}, \ldots, F_{t-1}^{(\tau)})\right].$$

Recalling $(F_0^{(\tau)}, \ldots, F_{t-1}^{(\tau)}) \mid U'_* \sim \mathcal{N}(\boldsymbol{\mu}_{t-1}^{(\tau)} U'_*, \boldsymbol{\Sigma}_{t-1}^{(\tau)})$ and $(F_0, \ldots, F_{t-1}) \mid U'_* \sim \mathcal{N}(\boldsymbol{\mu}_{t-1} U'_*, \boldsymbol{\Sigma}_{t-1})$, let us couple these laws by setting $(F_0^{(\tau)}, \ldots, F_{t-1}^{(\tau)}) = \boldsymbol{\mu}_{t-1}^{(\tau)} U'_* + (\boldsymbol{\Sigma}_{t-1}^{(\tau)})^{1/2} Z$ and $(F_0, \ldots, F_{t-1}) = \boldsymbol{\mu}_{t-1} U'_* + \boldsymbol{\Sigma}_{t-1}^{1/2} Z$ for a standard Gaussian vector Z independent of U'_*. Since $\lim_{\tau\to\infty} \boldsymbol{\mu}_{t-1}^{(\tau)} = \boldsymbol{\mu}_{t-1}$ and $\lim_{\tau\to\infty} \boldsymbol{\Sigma}_{t-1}^{(\tau)} = \boldsymbol{\Sigma}_{t-1}$ by $t-1^{(f)}$, and $g(\cdot)$ is pseudo-Lipschitz, by the dominated convergence theorem,

$$\lim_{\tau\to\infty} \mathbb{E}\left[g(U'_*, U_0^{(\tau)}, \ldots, U_t^{(\tau)}, F_0^{(\tau)}, \ldots, F_{t-1}^{(\tau)})\right] = \mathbb{E}\left[g(U'_*, U_0, \ldots, U_t, F_0, \ldots, F_{t-1})\right].$$

Combining the above three displays, this shows for any pseudo-Lipschitz function g that

$$\lim_{n\to\infty}\frac{1}{n}\sum_{i=1}^{n} g(u_{*,i},u_{0,i},\ldots,u_{t,i},f_{0,i},\ldots,f_{t-1,i}) = g(U'_*,U_0,\ldots,U_t,F_0,\ldots,F_{t-1}),$$

which implies the desired Wasserstein-2 convergence in part (b).

Part (c) Since the upper-left $tK\times tK$ submatrix of $\mathbf{\Phi}_t^{(\tau)}$ is exactly equal to $\mathbf{\Phi}_{t-1}^{(\tau)}$ and the last column block of $\mathbf{\Phi}_t^{(\tau)}$ is 0, by the induction hypothesis $t-1^{(c)}$, it suffices to show that the last row block of $\mathbf{\Phi}_t^{(\tau)}$ converges to that of $\mathbf{\Phi}_t$, and similarly for $\mathbf{\Delta}_t^{(\tau)}$ and $\mathbf{\Delta}_t$. For any $s\in\{0,\ldots,t-1\}$, we have

$$\mathbf{\Phi}_t^{(\tau)}[t,s]=\mathbb{E}\left[\partial_s u_t(F_0^{(\tau)},\ldots,F_{t-1}^{(\tau)})\right] \quad\text{and}\quad \mathbf{\Phi}_t[t,s]=\mathbb{E}\left[\partial_s u_t(F_0,\ldots,F_{t-1})\right].$$

By Assumption 3.2.2(b), $\partial_s u_t$ is bounded and continuous on a set of probability 1 under $(F_0,\ldots,F_{t-1})$, so by weak convergence of $(F_0^{(\tau)},\ldots,F_{t-1}^{(\tau)})$ to $(F_0,\ldots,F_{t-1})$ under the above coupling,

$$\lim_{\tau\to\infty}\mathbf{\Phi}_t^{(\tau)}[t,s]=\mathbf{\Phi}_t[t,s].$$

Similarly, for any $s\in\{0,\ldots,t\}$, $\lim_{\tau\to\infty}\mathbf{\Delta}_t^{(\tau)}[t,s]=\mathbf{\Delta}_t[t,s]$ by the induced coupling of $(U_0^{(\tau)},\ldots,U_t^{(\tau)})$ and $(U_0,\ldots,U_t)$ and the dominated convergence theorem.

Part (d) As above, we only need to show that the last row block of $\boldsymbol{\phi}_t^{(\tau)}$ converges to that of $\boldsymbol{\phi}_t$. For any $s\in\{0,\ldots,t-1\}$, by the auxiliary AMP state evolution of Corollary 3.4.1,

$$\lim_{n\to\infty}\boldsymbol{\phi}_t^{(\tau)}[t,s]=\lim_{n\to\infty}\langle\partial_s u_t(\mathbf{F}_0^{(\tau)},\ldots,\mathbf{F}_{t-1}^{(\tau)})\rangle=\mathbb{E}\left[\partial_s u_t(F_0^{(\tau)},\ldots,F_{t-1}^{(\tau)})\right]=\mathbf{\Phi}_t^{(\tau)}[t,s] \tag{3.4.14}$$

where the second equality again follows from $\partial_s u_t$ being bounded and continuous on a set of probability 1 under $(F_0^{(\tau)}, \ldots, F_{t-1}^{(\tau)})$, for sufficiently large τ. Similarly, by $t^{(b)}$,

$$\lim_{n\to\infty} \boldsymbol{\phi}_t[t,s] = \mathbb{E}\left[\partial_s u_t(F_0, \ldots, F_{t-1})\right] = \boldsymbol{\Phi}_t[t,s]. \tag{3.4.15}$$

Combining (3.4.14) and (3.4.15), it then follows from $t^{(c)}$ that

$$\lim_{\tau\to\infty}\lim_{n\to\infty}\left(\boldsymbol{\phi}_t^{(\tau)}[t,s] - \boldsymbol{\phi}[t,s]\right) = 0.$$

Part (e) We now control the difference between $\mathbf{F}_t^{(\tau)}$ and $\mathbf{F}_t$. By their definitions in (3.4.1) and (3.2.10), applying the triangle inequality yields

$$\frac{\|\mathbf{F}_t^{(\tau)} - \mathbf{F}_t\|_{\mathrm{F}}}{\sqrt{n}} \le \frac{\|\mathbf{X}\|\|\mathbf{U}_t^{(\tau)} - \mathbf{U}_t\|_{\mathrm{F}}}{\sqrt{n}} + \frac{1}{\sqrt{n}}\left\|\sum_{i=-\tau}^{t}\mathbf{U}_i^{(\tau)}b_{t,i}^{(\tau)\top} - \sum_{i=0}^{t}\mathbf{U}_i b_{t,i}^{\top}\right\|_{\mathrm{F}}. \tag{3.4.16}$$

Since $\lim_{n\to\infty}\|\mathbf{X}\| < C$ a.s. for a constant $C > 0$, the first term vanishes in the limit $n\to\infty$ by $t^{(a)}$. For the second term in (3.4.16), we can further decompose and bound it by

$$\begin{aligned}\frac{1}{\sqrt{n}}\cdot\left\|\sum_{i=-\tau}^{t}\mathbf{U}_i^{(\tau)}b_{t,i}^{(\tau)\top} - \sum_{i=0}^{t}\mathbf{U}_i b_{t,i}^{\top}\right\|_{\mathrm{F}} &\le \underbrace{\frac{1}{\sqrt{n}}\cdot\left\|\sum_{i=-\tau}^{0}\mathbf{U}_i^{(\tau)}b_{t,i}^{(\tau)\top} - \mathbf{U}_0 b_{t,0}^{\top}\right\|_{\mathrm{F}}}_{\mathcal{I}_1} \\ &\quad + \underbrace{\sum_{i=1}^{t}\frac{1}{\sqrt{n}}\cdot\left\|\mathbf{U}_i^{(\tau)}b_{t,i}^{(\tau)\top} - \mathbf{U}_i b_{t,i}^{\top}\right\|_{\mathrm{F}}}_{\mathcal{I}_2}.\end{aligned} \tag{3.4.17}$$

Term $\mathcal{I}_1$. By the triangle inequality,

$$\frac{1}{\sqrt{n}}\cdot\left\|\sum_{i=-\tau}^{0}\mathbf{U}_i^{(\tau)}b_{t,i}^{(\tau)\top} - \mathbf{U}_0 b_{t,0}^{\top}\right\|_{\mathrm{F}} \le \underbrace{\left\|\sum_{i=-\tau}^{0}b_{t,i}^{(\tau)} - b_{t,0}\right\|}_{\mathcal{I}_{1,1}}\cdot\frac{\|\mathbf{U}_0\|_{\mathrm{F}}}{\sqrt{n}} + \underbrace{\sum_{i=-\tau}^{0}\|b_{t,i}^{(\tau)}\|\cdot\frac{\|\mathbf{U}_i^{(\tau)} - \mathbf{U}_0\|_{\mathrm{F}}}{\sqrt{n}}}_{\mathcal{I}_{1,2}}. \tag{3.4.18}$$

Since $\|\mathbf{U}_0\|_{\mathrm{F}}/\sqrt{n} = \|\mathbf{F}_{\mathrm{pca}} S^{-1}\|_{\mathrm{F}}/\sqrt{n} < C$ for a constant $C > 0$, it suffices to bound $\mathcal{I}_{1,1}$ and $\mathcal{I}_{1,2}$. To bound $\mathcal{I}_{1,1}$, recall the definition of debiasing coefficients in (3.2.12):

$$b_{t,0} = \widetilde{\mathbf{B}}_t[t,0] = \sum_{j=0}^{\infty} \boldsymbol{\phi}_t^j[t,0]\tilde{\kappa}_{j+1} = \sum_{j=0}^{t} \boldsymbol{\phi}_t^j[t,0]\tilde{\kappa}_{j+1},$$

where the second equality applies $\boldsymbol{\phi}_t^j = 0$ when $j > t$. Recall also the debiasing coefficients in the auxiliary AMP sequence from (3.4.2): For every $i \in \{-\tau, \dots, 0\}$,

$$b_{t,i}^{(\tau)} = \mathbf{B}_t^{(\tau)}[t,i] = \sum_{j=0}^{\infty} \kappa_{j+1} \left(\boldsymbol{\phi}_{\mathrm{all},t}^{(\tau)}\right)^j [t,i] = \sum_{j=-i}^{-i+t} \kappa_{j+1} \left(\boldsymbol{\phi}_t^{(\tau)}\right)^{j+i} [t,0] S^i \tag{3.4.19}$$

where the last equality follows from (3.4.3) and the fact that $(\boldsymbol{\phi}_t^{(\tau)})^j = 0$ when $j > t$. Therefore,

$$\begin{aligned}
\sum_{i=-\tau}^{0} b_{t,i}^{(\tau)} &= \sum_{i=-\tau}^{0} \sum_{j=-i}^{-i+t} \kappa_{j+1} \left(\boldsymbol{\phi}_t^{(\tau)}\right)^{j+i} [t,0] S^i \\
&= \sum_{j=0}^{t} \left(\boldsymbol{\phi}_t^{(\tau)}\right)^j [t,0] \left(\sum_{i=-\tau}^{0} \kappa_{j-i+1} S^i\right) \\
&= \sum_{j=0}^{t} \left(\boldsymbol{\phi}_t^{(\tau)}\right)^j [t,0] \tilde{\kappa}_{j+1} + \sum_{j=0}^{t} \left(\boldsymbol{\phi}_t^{(\tau)}\right)^j [t,0] \sum_{i=\tau+1}^{\infty} \kappa_{j+i+1} S^{-i}
\end{aligned}$$

and thus

$$\sum_{i=-\tau}^{0} b_{t,i}^{(\tau)} - b_{t,0} = \sum_{j=0}^{t} \left((\boldsymbol{\phi}_t^{(\tau)})^j[t,0] - \boldsymbol{\phi}_t^j[t,0]\right) \tilde{\kappa}_{j+1} + \sum_{j=0}^{t} (\boldsymbol{\phi}_t^{(\tau)})^j[t,0] \left(\sum_{i=\tau+1}^{\infty} \kappa_{j+i+1} S^{-i}\right).$$

Taking the limits $\tau \to \infty$ and $n \to \infty$, the first term converges to 0 by $t^{(d)}$. Since each u_t is Lipschitz, we have $\|(\boldsymbol{\phi}_t^{(\tau)})^j[t,0]\| \le C_t$ for some constant $C_t > 0$ and all $j = 0, \dots, t$, and thus the second term will also converge to 0 as $\tau \to \infty$ by the absolute convergence of the series defining $\tilde{\kappa}_{j+1}$ in (3.2.11), as a consequence of Assumption

3.2.2(c). We therefore conclude

$$\lim_{\tau\to\infty}\lim_{n\to\infty}\sum_{i=-\tau}^{0} b_{t,i}^{(\tau)} - b_{t,0} = 0. \tag{3.4.20}$$

To bound $\mathcal{I}_{1,2}$, we apply Lemma 3.6.1 with $x_i^{(\tau)} = \|b_{t,-i}^{(\tau)}\|$ and $y_i^{(\tau)} = \limsup_{n\to\infty}\|\mathbf{U}_{-i}^{(\tau)} - \mathbf{U}_0\|_{\mathrm{F}}/\sqrt{n}$ for every $i = 0,\dots,\tau$. We need to verify that $\{x_i^{(\tau)}\}_{i=0,\dots,\tau}$ and $\{y_i^{(\tau)}\}_{i=0,\dots,\tau}$ satisfy the conditions of Lemma 3.6.1. By (3.4.19) and Assumption 3.2.2(c), for some constants $C_t, C_t' > 0$ independent of i and τ,

$$\begin{aligned}\sum_{i=0}^{\tau}|x_i^{(\tau)}| = \sum_{i=0}^{\tau}\|b_{t,-i}^{(\tau)}\| &\leq \sum_{i=0}^{\tau}\sum_{j=i}^{i+t}|\kappa_{j+1}|\cdot\left\|(\boldsymbol{\phi}_t^{(\tau)})^{j-i}[t,0]S^{-i}\right\| \\ &\leq C_t\sum_{i=0}^{\tau}\sum_{j=0}^{t}\frac{|\kappa_{i+j+1}|}{\min_{k\in\{1,\dots,K\}}\theta_k^i} \leq C_t\sum_{j=0}^{t}\sum_{i=0}^{\infty}\frac{|\kappa_{i+j+1}|}{\min_{k\in\{1,\dots,K\}}\theta_k^i} < C_t'.\end{aligned}$$

Therefore $\{x_i^{(\tau)}\}_{i=0,\dots,\tau}$ is uniformly bounded. Furthermore,

$$\sum_{i=\mathcal{T}}^{\tau}|x_i^{(\tau)}| \leq C_t\sum_{j=0}^{t}\sum_{i=\mathcal{T}}^{\infty}\frac{|\kappa_{i+j+1}|}{\min_{k\in\{1,\dots,K\}}\theta_k^i},$$

so for any $\epsilon > 0$, there exists some $\mathcal{T} > 0$ such that $\sum_{i=\mathcal{T}}^{\tau}|x_i^{(\tau)}| < \epsilon$ for all $\tau \geq \mathcal{T}$. For $y_i^{(\tau)}$, we have

$$|y_i^{(\tau)}| \leq \lim_{n\to\infty}\frac{\|\mathbf{U}_{-i}^{(\tau)}\|_{\mathrm{F}}}{\sqrt{n}} + \lim_{n\to\infty}\frac{\|\mathbf{U}_0\|_{\mathrm{F}}}{\sqrt{n}}.$$

Lemma 3.4.2 implies that the first limit exists, depends on (i,τ) only via the difference $\tau - i$, and $\lim_{\tau-i\to\infty}\lim_{n\to\infty}\|\mathbf{U}_{-i}^{(\tau)}\|_{\mathrm{F}}/\sqrt{n} < C$ for a constant $C > 0$. Then there is a constant $C' > 0$ independent of (i,τ) for which $|y_i^{(\tau)}| < C'$, so $\{y_i^{(\tau)}\}_{i=0,\dots,\tau}$ is uniformly bounded. By (3.4.13), $\lim_{\tau\to\infty} y_i^{(\tau)} = 0$ for each $0 \leq i \leq \mathcal{T}$. Hence, applying Lemma 3.6.1,

$$\lim_{\tau\to\infty}\limsup_{n\to\infty}\sum_{i=-\tau}^{0}\|b_{t,i}^{(\tau)}\|\cdot\frac{\|\mathbf{U}_i^{(\tau)} - \mathbf{U}_0\|_{\mathrm{F}}}{\sqrt{n}} = 0. \tag{3.4.21}$$

Combining (3.4.18), (3.4.20) and (3.4.21), we obtain that

$$\lim_{\tau\to\infty}\limsup_{n\to\infty}\frac{1}{\sqrt{n}}\cdot\left\|\sum_{i=-\tau}^{0}\mathbf{U}_i^{(\tau)}b_{t,i}^{(\tau)\top}-\mathbf{U}_0 b_{t,0}^{\top}\right\|_{\mathrm{F}}=0. \tag{3.4.22}$$

Term $\mathcal{I}_2$. The convergence of the term $\mathcal{I}_2$ in (3.4.17) is a more straightforward comparison: By the triangle inequality,

$$\sum_{i=1}^{t}\frac{1}{\sqrt{n}}\cdot\left\|\mathbf{U}_i^{(\tau)}b_{t,i}^{(\tau)\top}-\mathbf{U}_i b_{t,i}^{\top}\right\|_{\mathrm{F}}\leq\sum_{i=1}^{t}\|b_{t,i}^{(\tau)}\|\cdot\frac{\|\mathbf{U}_i^{(\tau)}-\mathbf{U}_i\|_{\mathrm{F}}}{\sqrt{n}}+\sum_{i=1}^{t}\|b_{t,i}^{(\tau)}-b_{t,i}\|\cdot\frac{\|\mathbf{U}_i\|_{\mathrm{F}}}{\sqrt{n}}.$$

For each $i\in\{0,\dots,t\}$, by the definition of the debiasing coefficients in (3.4.2), we have

$$b_{t,i}^{(\tau)}=\sum_{j=0}^{\infty}\kappa_{j+1}\left(\boldsymbol{\phi}_t^{(\tau)}\right)^j[t,i]=\sum_{j=0}^{t}\kappa_{j+1}\left(\boldsymbol{\phi}_t^{(\tau)}\right)^j[t,i].$$

Since each u_t is Lipschitz, this implies $\|b_{t,i}^{(\tau)}\|<C_t$ for a constant $C_t>0$. Recalling the definitions of $b_{t,i}^{(\tau)}$ in (3.4.2) and $b_{t,i}$ in (3.2.12), and applying $t^{(d)}$, also $\lim_{\tau\to\infty}\lim_{n\to\infty}b_{t,i}^{(\tau)}-b_{t,i}=0$. Then combining with $t^{(a)}$,

$$\lim_{\tau\to\infty}\limsup_{n\to\infty}\sum_{i=1}^{t}\frac{1}{\sqrt{n}}\cdot\left\|\mathbf{U}_i^{(\tau)}b_{t,i}^{(\tau)\top}-\mathbf{U}_i b_{t,i}^{\top}\right\|_{\mathrm{F}}=0. \tag{3.4.23}$$

Combining (3.4.16), (3.4.17), (3.4.22), and (3.4.23) shows $\lim_{\tau\to\infty}\limsup_{n\to\infty}\|\mathbf{F}_t^{(\tau)}-\mathbf{F}_t\|_{\mathrm{F}}/\sqrt{n}=0$ as desired for part (e).

Part (f) Recall that $\boldsymbol{\Sigma}_t^{(\tau)}$ is the lower-right $(t+1)K\times(t+1)K$ submatrix of $\boldsymbol{\Sigma}_{\mathrm{all},t}^{(\tau)}$ as in (3.4.4). By the block decompositions of $\boldsymbol{\Phi}_{\mathrm{all},t}^{(\tau)}$ and $\boldsymbol{\Delta}_{\mathrm{all},t}^{(\tau)}$ in (3.4.4), we obtain

$$\boldsymbol{\Sigma}_t^{(\tau)}=\underbrace{\sum_{j=0}^{\infty}\kappa_{j+2}\sum_{i=0}^{j}(\boldsymbol{\Phi}_{\mathrm{all},t}^{(\tau)})_{t-}^{i}\boldsymbol{\Delta}_{--}^{(\tau)}(\boldsymbol{\Phi}_{\mathrm{all},t}^{(\tau)\top})_{-t}^{j-i}}_{\widehat{\boldsymbol{\Sigma}}_t^{(\tau)}}+\underbrace{\sum_{j=0}^{\infty}\kappa_{j+2}\sum_{i=0}^{j}(\boldsymbol{\Phi}_t^{(\tau)})^{i}\boldsymbol{\Delta}_{t-}^{(\tau)}(\boldsymbol{\Phi}_{\mathrm{all},t}^{(\tau)\top})_{-t}^{j-i}}_{\widetilde{\boldsymbol{\Sigma}}_t^{(\tau)}}$$

$$+\underbrace{\sum_{j=0}^{\infty}\kappa_{j+2}\sum_{i=0}^{j}(\mathbf{\Phi}_{\text{all},t}^{(\tau)})^{i}_{t-}\mathbf{\Delta}_{-t}^{(\tau)}(\mathbf{\Phi}_t^{(\tau)\top})^{j-i}}_{(\widetilde{\mathbf{\Sigma}}_t^{(\tau)})^\top}+\underbrace{\sum_{j=0}^{\infty}\kappa_{j+2}\sum_{i=0}^{j}(\mathbf{\Phi}_t^{(\tau)})^{i}\mathbf{\Delta}_t^{(\tau)}(\mathbf{\Phi}_t^{(\tau)\top})^{j-i}}_{\overline{\mathbf{\Sigma}}_t^{(\tau)}} \tag{3.4.24}$$

where we observe that the third term is the transpose of the second term. To show $\lim_{\tau\to\infty}\mathbf{\Sigma}_t^{(\tau)}=\mathbf{\Sigma}_t$, we recall $\mathbf{\Delta}_t=\overline{\mathbf{\Delta}}_t+\widetilde{\mathbf{\Delta}}_t+\widetilde{\mathbf{\Delta}}_t^\top+\widehat{\mathbf{\Delta}}_t$, and correspondingly decompose $\mathbf{\Sigma}_t$ in (3.2.16) as follows:

$$\begin{aligned}
&\mathbf{\Sigma}_t\\
&=\sum_{j=0}^{\infty}\mathbf{\Theta}^{(j)}[\mathbf{\Phi}_t,\kappa_{j+2}\overline{\mathbf{\Delta}}_t+\tilde{\kappa}_{j+2}\odot\widetilde{\mathbf{\Delta}}_t+\widetilde{\mathbf{\Delta}}_t^\top\odot\tilde{\kappa}_{j+2}+\hat{\kappa}_{j+2}\odot\widehat{\mathbf{\Delta}}_t]\\
&=\underbrace{\sum_{j=0}^{\infty}\sum_{i=0}^{j}\mathbf{\Phi}_t^i\left[\hat{\kappa}_{j+2}\odot\widehat{\mathbf{\Delta}}_t-\tilde{\kappa}_{j+2}\odot\widehat{\mathbf{\Delta}}_t-\widehat{\mathbf{\Delta}}_t^\top\odot\tilde{\kappa}_{j+2}+\kappa_{j+2}\widehat{\mathbf{\Delta}}_t\right](\mathbf{\Phi}_t^\top)^{j-i}}_{\widehat{\mathbf{\Sigma}}_t}\\
&\quad+\underbrace{\sum_{j=0}^{\infty}\kappa_{j+2}\sum_{i=0}^{j}\mathbf{\Phi}_t^i\mathbf{\Delta}_t(\mathbf{\Phi}_t^\top)^{j-i}}_{\overline{\mathbf{\Sigma}}_t}\\
&\quad+\underbrace{\sum_{j=0}^{\infty}\sum_{i=0}^{j}\mathbf{\Phi}_t^i\left[(\tilde{\kappa}_{j+2}-\kappa_{j+2}\mathrm{Id})\odot(\widetilde{\mathbf{\Delta}}_t+\widehat{\mathbf{\Delta}}_t)+(\widetilde{\mathbf{\Delta}}_t^\top+\widehat{\mathbf{\Delta}}_t^\top)\odot(\tilde{\kappa}_{j+2}-\kappa_{j+2}\mathrm{Id})\right](\mathbf{\Phi}_t^\top)^{j-i}}_{\widetilde{\mathbf{\Sigma}}_t}.
\end{aligned} \tag{3.4.25}$$

We will show that, for any $r,c\in\{0,1,\dots,t\}$,

$$\lim_{\tau\to\infty}\widehat{\mathbf{\Sigma}}_t^{(\tau)}[r,c]=\widehat{\mathbf{\Sigma}}_t[r,c],\quad \lim_{\tau\to\infty}\left(\widetilde{\mathbf{\Sigma}}_t^{(\tau)}+\widetilde{\mathbf{\Sigma}}_t^{(\tau)\top}\right)[r,c]=\widetilde{\mathbf{\Sigma}}_t[r,c],\quad \lim_{\tau\to\infty}\overline{\mathbf{\Sigma}}_t^{(\tau)}[r,c]=\overline{\mathbf{\Sigma}}_t[r,c].$$

Convergence of $\widehat{\mathbf{\Sigma}}_t^{(\tau)}$ We have

$$\widehat{\mathbf{\Sigma}}_t^{(\tau)}[r,c]=\sum_{j=0}^{\infty}\kappa_{j+2}\sum_{i=0}^{j}\sum_{\alpha,\beta=1}^{\tau}(\mathbf{\Phi}_{\text{all},t}^{(\tau)})^i[r,-\alpha]\mathbf{\Delta}_{\text{all},t}^{(\tau)}[-\alpha,-\beta](\mathbf{\Phi}_{\text{all},t}^{(\tau)\top})^{j-i}[-\beta,c]$$

$$= \sum_{j=0}^{\infty}\sum_{i=0}^{j}\sum_{\alpha=1}^{i\wedge\tau}\sum_{\beta=1}^{(j-i)\wedge\tau} \kappa_{j+2}(\mathbf{\Phi}_t^{(\tau)})^{i-\alpha}[r,0]S^{-\alpha}\mathbf{\Delta}_{--}^{(\tau)}[-\alpha,-\beta]S^{-\beta}(\mathbf{\Phi}_t^{(\tau)\top})^{j-i-\beta}[0,c]$$

where the second equality follows from (3.4.3). Let us write $\mathbf{\Delta}_{--}^{(\tau)}[-\alpha,-\beta] = (\mathbf{\Delta}_{--}^{(\tau)}[-\alpha,-\beta] - \mathbf{\Delta}_t[0,0]) + \mathbf{\Delta}_t[0,0]$, and introduce $p = i-\alpha$ and $q = j-i-\beta$ to re-index this summation by (p,q,α,β). This yields

$$\begin{aligned}
&\widehat{\mathbf{\Sigma}}_t^{(\tau)}[r,c] \\
&= \underbrace{\sum_{p,q=0}^{\infty}(\mathbf{\Phi}_t^{(\tau)})^p[r,0]\left(\sum_{\alpha,\beta=1}^{\tau}\kappa_{p+q+\alpha+\beta+2}S^{-\alpha}(\mathbf{\Delta}_t^{(\tau)}[-\alpha,-\beta]-\mathbf{\Delta}_t[0,0])S^{-\beta}\right)(\mathbf{\Phi}_t^{(\tau)\top})^q[0,c]}_{\widehat{\mathcal{I}}_1^{(\tau)}} \\
&\quad + \underbrace{\sum_{p,q=0}^{\infty}(\mathbf{\Phi}_t^{(\tau)})^p[r,0]\left(\sum_{\alpha,\beta=1}^{\tau}\kappa_{p+q+\alpha+\beta+2}S^{-\alpha}\mathbf{\Delta}_t[0,0]S^{-\beta}\right)(\mathbf{\Phi}_t^{(\tau)\top})^q[0,c]}_{\widehat{\mathcal{I}}_2^{(\tau)}}.
\end{aligned}$$

Note that the summations over p,q are in fact finite and may be restricted to $p,q \in [0,t]$, because $(\mathbf{\Phi}_t^{(\tau)})^p = 0$ for all $p > t$.

We first show $\widehat{\mathcal{I}}_1^{(\tau)}$ vanishes as $\tau \to \infty$. Since each block $\|\mathbf{\Phi}_t^{(\tau)}[r,0]\|$ is bounded by a constant and the summation may be restricted to $p,q \in [0,t]$, there exists some constant $C_t > 0$ such that

$$\begin{aligned}
|\widehat{\mathcal{I}}_1^{(\tau)}| &\leq C_t \max_{\ell=0}^{2t}\sum_{\alpha,\beta=1}^{\tau}\frac{|\kappa_{\ell+\alpha+\beta+2}|}{\min_{k\in\{1,\ldots,K\}}|\theta_k|^{\alpha+\beta}}\cdot\left\|\mathbf{\Delta}_{--}^{(\tau)}[-\alpha,-\beta]-\mathbf{\Delta}_t[0,0]\right\| \\
&\leq C_t \max_{\ell=0}^{2t}\sum_{i=2}^{2\tau}\frac{i\cdot|\kappa_{\ell+i+2}|}{\min_{k\in\{1,\ldots,K\}}|\theta_k|^{i}}\cdot\sup_{\substack{\alpha,\beta\geq 1\\ \alpha+\beta=i}}\left\|\mathbf{\Delta}_{--}^{(\tau)}[-\alpha,-\beta]-\mathbf{\Delta}_t[0,0]\right\|.
\end{aligned}$$

Fixing any $\ell \in [0,2t]$, we apply Lemma 3.6.1 with $x_i^{(2\tau)} = i\cdot|\kappa_{\ell+i+2}|/\min_{k\in\{1,\ldots,K\}}|\theta_k|^i$ and

$$y_i^{(2\tau)} = \sup_{\substack{\alpha,\beta\geq 1\\ \alpha+\beta=i}}\left\|\mathbf{\Delta}_{--}^{(\tau)}[-\alpha,-\beta]-\mathbf{\Delta}_t[0,0]\right\|.$$

By Assumption 3.2.2(c), $\{x_i^{(2\tau)}\}_{i=1,\dots,2\tau}$ satisfies the condition of Lemma 3.6.1. By Lemma 3.4.2, for any fixed $\alpha, \beta \geq 0$,

$$\begin{aligned}\lim_{\tau\to\infty} \mathbf{\Delta}_{--}^{(\tau)}[-\alpha,-\beta] &= \lim_{\tau\to\infty} \mathbb{E}[U_{-\alpha}^{(\tau)} U_{-\beta}^{(\tau)\top}] \\ &= \lim_{\tau\to\infty} S^{-1}\mathbb{E}[F_{-\alpha-1}^{(\tau)} F_{-\beta-1}^{(\tau)\top}]S^{-1} \\ &= S^{-1}\cdot\left(\lim_{\tau\to\infty} \mathbf{\Sigma}_{\text{all},t}^{(\tau)}[-\alpha-1,-\beta-1] + \mu_{-\alpha-1}^{(\tau)}\cdot\mu_{-\beta-1}^{(\tau)\top}\right)\cdot S^{-1} = S^{-2},\end{aligned} \tag{3.4.26}$$

where the last equality applies (3.4.13). Identifying also

$$\mathbf{\Delta}_t[0,0] = \mathbb{E}[U_0 U_0^\top] = S^{-2}$$

because $\mathbf{U}_0 = \mathbf{F}_{\text{pca}} S^{-1} \xrightarrow{W_2} U_0$, this gives $\lim_{\tau\to\infty} y_i^{(2\tau)} = 0$ for every fixed i. Observe that $\mathbb{E}[\|F_{-\alpha}^{(\tau)}\|]$ depends on (α,τ) only via the difference $\tau-\alpha$, and (3.4.13) implies $\lim_{\tau-\alpha\to\infty} \mathbb{E}[\|F_{-\alpha}^{(\tau)}\|] < C$ for a constant $C > 0$. Then $\mathbb{E}[\|F_{-\alpha}^{(\tau)}\|] < C'$ for a constant $C' > 0$ and all (α,τ), implying that $\{y_i^{(2\tau)}\}$ is uniformly bounded. Then it follows from Lemma 3.6.1 that $\lim_{\tau\to\infty} |\widehat{\mathcal{I}}_1^{(\tau)}| = 0$.

Next, we show that $\lim_{\tau\to\infty} \widehat{\mathcal{I}}_2^{(\tau)} = \widehat{\mathbf{\Sigma}}_t[r,c]$. Taking $\tau\to\infty$, by the convergence of $\mathbf{\Phi}_t^{(\tau)}$ to $\mathbf{\Phi}_t$ in $t^{(c)}$,

$$\lim_{\tau\to\infty} \widehat{\mathcal{I}}_2^{(\tau)} = \sum_{p,q=0}^{\infty} \mathbf{\Phi}_t^p[r,0]\left(\sum_{\alpha,\beta=1}^{\infty} \kappa_{p+q+\alpha+\beta+2} S^{-\alpha}\mathbf{\Delta}_t[0,0]S^{-\beta}\right)(\mathbf{\Phi}_t^\top)^q[0,c].$$

Recall that $\mathbf{\Delta}_t[0,0] = S^{-2}$, so this commutes with $S^{-\beta}$. Then, applying

$$\begin{aligned}&\sum_{\alpha,\beta=1}^{\infty} \kappa_{p+q+\alpha+\beta+2} S^{-\alpha}\mathbf{\Delta}_t[0,0]S^{-\beta} \\ &= \sum_{\alpha,\beta=0}^{\infty} \kappa_{p+q+\alpha+\beta+2} S^{-(\alpha+\beta)}\mathbf{\Delta}_t[0,0] - 2\sum_{\alpha=0}^{\infty} \kappa_{p+q+\alpha+2} S^{-\alpha}\mathbf{\Delta}_t[0,0] + \kappa_{p+q+2}\mathbf{\Delta}_t[0,0] \\ &= (\hat{\kappa}_{p+q+2} - 2\tilde{\kappa}_{p+q+2} + \kappa_{p+q+2}\text{Id})\,\mathbf{\Delta}_t[0,0],\end{aligned}$$

we identify this limit as $\widehat{\mathbf{\Sigma}}_t[r,c]$. So

$$\lim_{\tau\to\infty}\widehat{\mathbf{\Sigma}}_t^{(\tau)}=\widehat{\mathbf{\Sigma}}_t. \tag{3.4.27}$$

Convergence of $\widetilde{\mathbf{\Sigma}}_t^{(\tau)}$ Next, we determine the limit of $\widetilde{\mathbf{\Sigma}}_t^{(\tau)}$. Applying (3.4.3) and re-indexing the summation similarly as above by setting $p=i$ and $q=j-i-\beta$,

$$\begin{aligned}
\widetilde{\mathbf{\Sigma}}_t^{(\tau)}[r,c] &= \sum_{j=0}^{\infty}\sum_{i=0}^{j}\sum_{\alpha=0}^{t}\sum_{\beta=1}^{\tau}\kappa_{j+2}(\mathbf{\Phi}_t^{(\tau)})^i[r,\alpha]\mathbf{\Delta}_{t-}^{(\tau)}[\alpha,-\beta](\mathbf{\Phi}_{\text{all},t}^{(\tau)\top})^{j-i}[-\beta,c]\\
&= \sum_{j=0}^{\infty}\sum_{i=0}^{j}\sum_{\alpha=0}^{t}\sum_{\beta=1}^{\tau}\kappa_{j+2}(\mathbf{\Phi}_t^{(\tau)})^i[r,\alpha]\mathbf{\Delta}_{t-}^{(\tau)}[\alpha,-\beta]S^{-\beta}(\mathbf{\Phi}_t^{(\tau)\top})^{j-i-\beta}[0,c]\\
&= \underbrace{\sum_{p,q=0}^{\infty}(\mathbf{\Phi}_t^{(\tau)})^p[r,\alpha]\left(\sum_{\alpha=0}^{t}\sum_{\beta=1}^{\tau}\kappa_{p+q+\beta+2}(\mathbf{\Delta}_{t-}^{(\tau)}[\alpha,-\beta]-\mathbf{\Delta}_t[\alpha,0])S^{-\beta}\right)(\mathbf{\Phi}_t^{(\tau)\top})^q[0,c]}_{\widetilde{\mathcal{I}}_1^{(\tau)}}\\
&\qquad+\underbrace{\sum_{p,q=0}^{\infty}(\mathbf{\Phi}_t^{(\tau)})^p[r,\alpha]\left(\sum_{\alpha=0}^{t}\sum_{\beta=1}^{\tau}\kappa_{p+q+\beta+2}\mathbf{\Delta}_t[\alpha,0]S^{-\beta}\right)(\mathbf{\Phi}_t^{(\tau)\top})^q[0,c]}_{\widetilde{\mathcal{I}}_2^{(\tau)}}.
\end{aligned}$$

For each fixed $\alpha\in[0,t]$ and $\beta>0$, note that by $t^{(a)}$,

$$\lim_{\tau\to\infty}\mathbb{E}[\|U_\alpha^{(\tau)}-U_\alpha\|^2]=0$$

and by $t^{(a)}$ and (3.4.26),

$$\begin{aligned}
\lim_{\tau\to\infty}\mathbb{E}[\|U_{-\beta}^{(\tau)}-U_0\|^2] &= \lim_{\tau\to\infty}\mathbb{E}[\|U_{-\beta}^{(\tau)}-U_0^{(\tau)}\|^2]\\
&= \lim_{\tau\to\infty}\operatorname{Tr}\left(\mathbf{\Delta}_{\text{all},t}^{(\tau)}[-\beta,-\beta]-2\mathbf{\Delta}_{\text{all},t}^{(\tau)}[-\beta,0]+\mathbf{\Delta}_{\text{all},t}^{(\tau)}[0,0]\right)=0.
\end{aligned}$$

So

$$\lim_{\tau\to\infty}\mathbf{\Delta}_{t-}^{(\tau)}[\alpha,-\beta]-\mathbf{\Delta}_t[\alpha,0]=\lim_{\tau\to\infty}\mathbb{E}[U_\alpha^{(\tau)}U_{-\beta}^{(\tau)\top}]-\mathbb{E}[U_\alpha U_0^\top]=0.$$

Then by Lemma 3.6.1 and a similar argument as used previously for $\widehat{\mathcal{I}}_1^{(\tau)}$, this implies $\lim_{\tau\to\infty} \widetilde{\mathcal{I}}_1^{(\tau)} = 0$. For $\widetilde{\mathcal{I}}_2^{(\tau)}$, we take $\tau \to \infty$ and write $\sum_{\beta=1}^{\infty} \kappa_{p+q+\beta+2} S^{-\beta} = \tilde{\kappa}_{p+q+2} - \kappa_{p+q+2}\mathrm{Id}$. Then

$$\begin{aligned}\lim_{\tau\to\infty} \widetilde{\boldsymbol{\Sigma}}_t^{(\tau)}[r,c] &= \sum_{p,q=0}^{\infty} \sum_{\alpha=0}^{t} \boldsymbol{\Phi}_t^p[r,\alpha]\Big(\boldsymbol{\Delta}_t[\alpha,0](\tilde{\kappa}_{p+q+2} - \kappa_{p+q+2}\mathrm{Id})\Big)(\boldsymbol{\Phi}_t^\top)^q[0,c] \\ &= \sum_{p,q=0}^{\infty} \Big[\boldsymbol{\Phi}_t^p\Big((\widetilde{\boldsymbol{\Delta}}_t^\top + \widehat{\boldsymbol{\Delta}}_t^\top) \odot (\tilde{\kappa}_{p+q+2} - \kappa_{p+q+2}\mathrm{Id})\Big)(\boldsymbol{\Phi}_t^\top)^q\Big][r,c].\end{aligned}$$

Summing this limit with its transpose, we obtain

$$\lim_{\tau\to\infty} (\widetilde{\boldsymbol{\Sigma}}_t^{(\tau)} + \widetilde{\boldsymbol{\Sigma}}_t^{(\tau)\top}) = \widetilde{\boldsymbol{\Sigma}}_t. \tag{3.4.28}$$

Convergence of $\overline{\boldsymbol{\Sigma}}_t^{(\tau)}$ By the convergence of $\boldsymbol{\Phi}_t^{(\tau)}, \boldsymbol{\Delta}_t^{(\tau)}$ to $\boldsymbol{\Phi}_t, \boldsymbol{\Delta}_t$ in $t^{(c)}$ and the fact that the summation in $\overline{\boldsymbol{\Sigma}}_t^{(\tau)}$ is finite and may be restricted to $j \le 2t$, we immediately have

$$\begin{aligned}\lim_{\tau\to\infty} \overline{\boldsymbol{\Sigma}}_t^{(\tau)} &= \lim_{\tau\to\infty} \sum_{j=0}^{\infty} \kappa_{j+2} \sum_{i=0}^{j} (\boldsymbol{\Phi}_t^{(\tau)})^i \boldsymbol{\Delta}_t^{(\tau)} (\boldsymbol{\Phi}_t^{(\tau)\top})^{j-i} \\ &= \sum_{j=0}^{\infty} \kappa_{j+2} \sum_{i=0}^{j} \boldsymbol{\Phi}_t^i \boldsymbol{\Delta}_t (\boldsymbol{\Phi}_t^\top)^{j-i} = \overline{\boldsymbol{\Sigma}}_t.\end{aligned} \tag{3.4.29}$$

Collecting the decomposition of $\boldsymbol{\Sigma}_t^{(\tau)}$ in (3.4.24), that of $\boldsymbol{\Sigma}_t$ in (3.4.25), and the convergence of each component in (3.4.27), (3.4.28) and (3.4.29), we obtain $\lim_{\tau\to\infty} \boldsymbol{\Sigma}_t^{(\tau)} = \boldsymbol{\Sigma}_t$ as desired for part (f). ■

3.5 Proof for independent initialization

The proof of Theorem 3.1.3 follows closely the arguments of [Fan20], with minor modifications needed to extend the results of [Fan20, Theorem 4.3 and Corollary 4.4] from vector-valued iterates in $\mathbb{R}^n$ to matrix-valued iterates in $\mathbb{R}^{n\times K}$. As in

[Fan20], the theorem is first shown under an additional nondegeneracy condition of Assumption 3.5.1 below, which ensures that the state evolution covariance matrices $\mathbf{\Sigma}_t$ are invertible for all $t \geq 1$. This enables an inductive argument of partial conditioning on the randomness of the Haar-orthogonal matrix $\mathbf{O}$, to establish state evolution characterizations for the original AMP iterates $(\mathbf{U}_1, \ldots, \mathbf{U}_{t+1}, \mathbf{Z}_1, \ldots, \mathbf{Z}_t, \mathbf{E})$ and also for the above auxiliary iterates $(\mathbf{R}_1, \ldots, \mathbf{R}_t)$ and

$$\boldsymbol{\lambda} = \operatorname{diag}(\mathbf{\Lambda}) \in \mathbb{R}^n.$$

The combinatorial arguments needed to close this induction are the same as in [Fan20]. The theorem without Assumption 3.5.1 is then obtained by applying this result to a slightly perturbed AMP sequence, and taking the limit of the perturbation to 0. We describe these arguments in further detail below.

Write the AMP iterations (3.1.1–3.1.2) as

$$\mathbf{R}_t = \mathbf{O}\mathbf{U}_t, \quad \mathbf{S}_t = \mathbf{O}^\top \mathbf{\Lambda} \mathbf{R}_t, \quad \mathbf{Z}_t = \mathbf{S}_t - \mathbf{U}_1 b_{t1}^\top - \ldots - \mathbf{U}_t b_{tt}^\top, \quad \mathbf{U}_{t+1} = u_{t+1}(\mathbf{Z}_1, \ldots, \mathbf{Z}_t, \mathbf{E}).$$

For each $t \geq 1$ and $k \geq 0$, define the $tK \times tK$ matrix

$$\mathbf{L}_t^{(k)} = \sum_{j=0}^{\infty} c_{k,j} \mathbf{\Theta}_t^{(j)}, \qquad \mathbf{\Theta}_t^{(j)} = \mathbf{\Theta}^{(j)}[\mathbf{\Phi}_t, \kappa_{j+2} \mathbf{\Delta}_t]$$

where $\{c_{k,j}\}_{k,j \geq 0}$ are the partial moment coefficients defined in [Fan20, Section A.1], and $\mathbf{\Theta}^{(j)}[\cdot, \cdot]$ is as defined in (3.1.6). We write

$$\mathbf{B}_t = \sum_{j=0}^{\infty} \kappa_{j+1} \mathbf{\Phi}_t^j$$

as the large-n limit of $\mathbf{b}_t$, replacing $\boldsymbol{\phi}_t$ by its large-n limit $\mathbf{\Phi}_t$. (To simplify notation, we are using $\boldsymbol{\phi}_t, \mathbf{b}_t^\top$ and $\mathbf{\Phi}_t, \mathbf{B}_t^\top$ in place of $\mathbf{\Phi}_t, \mathbf{B}_t$ and $\mathbf{\Phi}_t^\infty, \mathbf{B}_t^\infty$ respectively in [Fan20].

We have also defined $\mathbf{\Sigma}_t, \mathbf{\Theta}_t^{(j)}, \mathbf{L}_t^{(k)}$ directly in the large-n limit, instead of using the notation $\mathbf{\Sigma}_t^\infty, \mathbf{\Theta}_t^{(j,\infty)}, \mathbf{L}_t^{(k,\infty)}$.) The identities and proofs of [Fan20, Lemmas A.2 and A.3] on the relations between $\mathbf{B}_t^\top, \mathbf{\Sigma}_t, \mathbf{\Theta}_t^{(j)}, \mathbf{L}_t^{(k)}$ then hold without any modification, as they rely only on the definitions of these matrices in terms of $\mathbf{\Delta}_t, \mathbf{\Phi}_t$ and on the combinatorial relations between $\{c_{k,j}\}_{k,j\geq 0}$.

We first show an extended form of Theorem 3.1.3 under the following additional assumption.

Assumption 3.5.1. The random variables Λ and U_1 satisfy $\mathrm{Var}[\Lambda] > 0$ and $\mathbb{E}[U_1 U_1^\top] \succ 0$ strictly. Defining $(U_1, \ldots, U_{t+1}, Z_1, \ldots, Z_t)$ by the state evolution (3.1.7), for each $t \geq 1$ and any deterministic vector $v \in \mathbb{R}^K$, there do not exist constants $\alpha_1, \ldots, \alpha_t, \beta_1, \ldots, \beta_t$ for which $v^\top U_{t+1}$ is almost surely equal to $\alpha_1 \cdot v^\top U_1 + \ldots + \alpha_t \cdot v^\top U_t + \beta_1 \cdot v^\top Z_1 + \ldots + \beta_t \cdot v^\top Z_t$.

Lemma 3.5.2. *Suppose Assumptions 3.1.1, 3.1.2, and 3.5.1 hold. Then almost surely for each fixed $t \geq 1$,*

(a) $\lim_{n\to\infty} (\langle \mathbf{U}_r^\top \mathbf{U}_s \rangle)_{r,s=1}^t = \mathbf{\Delta}_t$ *and* $\lim_{n\to\infty} \boldsymbol{\phi}_t = \mathbf{\Phi}_t$.

(b) For some random vectors $R_1, \ldots, R_t \in \mathbb{R}^K$ having finite second moment, it holds that $(\mathbf{R}_1, \ldots, \mathbf{R}_t, \boldsymbol{\lambda}) \xrightarrow{W_2} (R_1, \ldots, R_t, \Lambda)$. Furthermore, for each $k \geq 0$, $\mathbb{E}[(R_1, \ldots, R_t)^\top \Lambda^k (R_1, \ldots, R_t)] = \mathbf{L}_t^{(k)}$.

(c) $(\mathbf{U}_1, \ldots, \mathbf{U}_{t+1}, \mathbf{Z}_1, \ldots, \mathbf{Z}_t, \mathbf{E}) \xrightarrow{W_2} (U_1, \ldots, U_{t+1}, Z_1, \ldots, Z_t, E)$ *as described in Theorem 3.1.3.*

(d) The matrix

$$\begin{pmatrix} \mathbf{\Delta}_t & \mathbf{\Phi}_t \mathbf{\Sigma}_t \\ \mathbf{\Sigma}_t \mathbf{\Phi}_t^\top & \mathbf{\Sigma}_t \end{pmatrix}$$

is non-singular.

Proof. The proof is an extension of that of [Fan20, Lemma A.4]. We denote $t^{(a),(b),(c),(d)}$ for the claims of (a–d) up to iteration t.

Applying Lemma 3.6.5(a) to $\mathbf{R}_1 = \mathbf{O}\mathbf{U}_1$, we have $(\boldsymbol{\lambda}, \mathbf{R}_1) \xrightarrow{W_2} (\Lambda, R_1)$ where $R_1 \sim \mathcal{N}(0, \mathbb{E}[U_1 U_1^\top])$ is independent of Λ. Then applying Lemma 3.6.3, $n^{-1}\mathbf{R}_1^\top \boldsymbol{\Lambda}^k \mathbf{R}_1 \to m_k \cdot \mathbb{E}[U_1 U_1^\top] = m_k \cdot \boldsymbol{\Delta}_1$ where $m_k = \mathbb{E}[\Lambda^k]$. Then $1^{(a)}$ and $1^{(b)}$ follow as in [Fan20, Lemma A.4]. Conditional on $\mathbf{R}_1 = \mathbf{O}\mathbf{U}_1$, we may represent the law of $\mathbf{S}_1$ as

$$\mathbf{S}_1 = \mathbf{S}_\parallel + \mathbf{S}_\perp, \qquad \mathbf{S}_\parallel = \mathbf{U}_1(\mathbf{R}_1^\top \mathbf{R}_1)^{-1}\mathbf{R}_1^\top \boldsymbol{\Lambda}\mathbf{R}_1, \qquad \mathbf{S}_\perp = \Pi_{\mathbf{U}_1^\perp} \tilde{\mathbf{O}} \Pi_{\mathbf{R}_1^\perp}^\top \boldsymbol{\Lambda}\mathbf{R}_1$$

where $\Pi_{\mathbf{U}_1^\perp}, \Pi_{\mathbf{R}_1^\perp} \in \mathbb{R}^{n\times(n-K)}$ are projections orthogonal to the column spans of $\mathbf{U}_1, \mathbf{R}_1$, and $\tilde{\mathbf{O}}$ is a Haar-orthogonal matrix of dimension $n - K$. We have $n^{-1}\mathbf{R}_1^\top \mathbf{R}_1 \to \mathbb{E}[U_1 U_1^\top] = \boldsymbol{\Delta}_1$, which is invertible by Assumption 3.5.1, and

$$n^{-1}(\Pi_{\mathbf{R}_1^\perp}^\top \boldsymbol{\Lambda}\mathbf{R}_1)^\top(\Pi_{\mathbf{R}_1^\perp}^\top \boldsymbol{\Lambda}\mathbf{R}_1) = n^{-1}\mathbf{R}_1^\top \boldsymbol{\Lambda}^2 \mathbf{R}_1 - n^{-1}\mathbf{R}_1^\top \boldsymbol{\Lambda}\mathbf{R}_1 \cdot (n^{-1}\mathbf{R}_1^\top \mathbf{R}_1)^{-1} \cdot n^{-1}\mathbf{R}_1^\top \boldsymbol{\Lambda}\mathbf{R}_1$$

Then applying the same arguments as [Fan20, Lemma A.4], using Lemma 3.6.5(a) to analyze $\mathbf{S}_\perp$, we obtain

$$(\mathbf{Z}_1, \mathbf{U}_1, \mathbf{E}) \xrightarrow{W_2} (Z_1, U_1, E), \qquad (U_1, E) \perp\!\!\!\perp Z_1 \sim \mathcal{N}(0, \kappa_2 \boldsymbol{\Delta}_1).$$

Here, $\kappa_2 \boldsymbol{\Delta}_1 = \boldsymbol{\Sigma}_1$. Since $u_2(Z_1, E)$ is Lipschitz, Lemma 3.6.4 then implies $1^{(c)}$. Assumption 3.5.1 and the identities $\boldsymbol{\Sigma}_1 = \kappa_2 \boldsymbol{\Delta}_1$ and $\boldsymbol{\Phi}_1 = 0$ then imply $1^{(d)}$.

Now suppose $t^{(a),(b),(c),(d)}$ hold. For $t+1^{(a)}$, convergence to $\boldsymbol{\Delta}_{t+1}$ in $t+1^{(a)}$ is immediate from W_2-convergence of $(\mathbf{U}_1, \ldots, \mathbf{U}_{t+1})$ in $t^{(c)}$ and Lemma 3.6.3 (applied with $k = 0$). Since each $u_s(\cdot)$ is Lipschitz, its derivatives are bounded and also continuous with probability 1 with respect to the limit law of $(Z_1, \ldots, Z_t, E)$ in $t^{(c)}$, by Assumption 3.1.2. Then the convergence $\boldsymbol{\phi}_t \to \boldsymbol{\Phi}_t$ follows also from weak convergence of $(\mathbf{Z}_1, \ldots, \mathbf{Z}_t, \mathbf{E})$ in $t^{(c)}$, which shows $t+1^{(a)}$.

For $t+1^{(b)}$, observe that by the W_2-convergence in $t^{(c)}$, weak-differentiability of the Lipschitz function $u_s(\cdot)$, and [Fan20, Proposition E.5],

$$n^{-1}(\mathbf{Z}_1,\ldots,\mathbf{Z}_t)^\top(\mathbf{U}_1,\ldots,\mathbf{U}_t) \to \boldsymbol{\Sigma}_t \cdot \boldsymbol{\Phi}_t^\top.$$

Then conditioning $\mathbf{O}$ on

$$(\mathbf{R}_1,\ldots,\mathbf{R}_t) = \mathbf{O}(\mathbf{U}_1,\ldots,\mathbf{U}_t) \text{ and } \mathbf{O}(\mathbf{Z}_1,\ldots,\mathbf{Z}_t) = \boldsymbol{\Lambda}(\mathbf{R}_1,\ldots,\mathbf{R}_t) - (\mathbf{R}_1,\ldots,\mathbf{R}_t)\mathbf{B}_t^\top,$$

the same argument that shows [Fan20, Eq. (A.23)] yields $(\mathbf{R}_1,\ldots,\mathbf{R}_{t+1},\boldsymbol{\lambda}) \xrightarrow{W_2} (R_1,\ldots,R_{t+1},\Lambda)$ where

$$R_{t+1} = \begin{pmatrix} R_1 & \cdots & R_t & \Lambda R_1 & \cdots & \Lambda R_t \end{pmatrix} \boldsymbol{\Upsilon}_t^{-1} \begin{pmatrix} \boldsymbol{\Delta}_{t+1}[1:t,\ t+1] \\ \boldsymbol{\Phi}_{t+1}^\top[1:t,\ t+1] \end{pmatrix} + R_\perp.$$

Here

$$\boldsymbol{\Upsilon}_t = \begin{pmatrix} \boldsymbol{\Delta}_t & \boldsymbol{\Delta}_t\mathbf{B}_t^\top + \boldsymbol{\Phi}_t\boldsymbol{\Sigma}_t \\ \boldsymbol{\Phi}_t^\top & \boldsymbol{\Phi}_t^\top\mathbf{B}_t^\top + \mathrm{Id} \end{pmatrix} \in \mathbb{R}^{2tK\times 2tK},$$

and in the decomposition into blocks of size $K\times K$, $\boldsymbol{\Delta}_{t+1}[1:t,\ t+1]$ and $\boldsymbol{\Phi}_{t+1}^\top[1:t,\ t+1]$ denote the first t row blocks of the last column block $t+1$ of $\boldsymbol{\Delta}_{t+1}$ and $\boldsymbol{\Phi}_{t+1}^\top$. The random vector $R_\perp \in \mathbb{R}^K$ has the Gaussian law

$$\mathcal{N}\left(0,\ \mathbb{E}[U_{t+1}U_{t+1}^\top] - \begin{pmatrix} \boldsymbol{\Delta}_{t+1}[1:t,\ t+1] \\ \boldsymbol{\Sigma}_t\boldsymbol{\Phi}_{t+1}^\top[1:t,\ t+1] \end{pmatrix}^\top \begin{pmatrix} \boldsymbol{\Delta}_t & \boldsymbol{\Phi}_t\boldsymbol{\Sigma}_t \\ \boldsymbol{\Sigma}_t\boldsymbol{\Phi}_t^\top & \boldsymbol{\Sigma}_t \end{pmatrix}^{-1} \begin{pmatrix} \boldsymbol{\Delta}_{t+1}[1:t,\ t+1] \\ \boldsymbol{\Sigma}_t\boldsymbol{\Phi}_{t+1}^\top[1:t,\ t+1] \end{pmatrix}\right).$$

For any fixed $v \in \mathbb{R}^K$, we may check that $v^\top \operatorname{Cov}[R_\perp]v$ is the residual variance of the L_2-projection of $v^\top U_{t+1}$ onto the linear span of $(v^\top Z_1,\ldots,v^\top Z_t, v^\top U_1,\ldots,v^\top U_t)$.

Thus, by Assumption 3.5.1, we have analogously to [Fan20, Eq. (A.24)] that

$$\mathrm{Cov}[R_\perp] \succ 0 \tag{3.5.1}$$

Note that Lemma 3.6.3 implies

$$n^{-1}(\mathbf{R}_1, \ldots, \mathbf{R}_{t+1})^\top \mathbf{\Lambda}^k (\mathbf{R}_1, \ldots, \mathbf{R}_{t+1}) \to \mathbb{E}[(R_1, \ldots, R_{t+1})^\top \Lambda^k (R_1, \ldots, R_{t+1})].$$

Using [Fan20, Lemmas A.1 and A.2] and the same arguments as in [Fan20, Lemma A.4] to analyze $\mathbb{E}[(R_1, \ldots, R_t)^\top \Lambda^k R_{t+1}]$ and $\mathbb{E}[R_{t+1}^\top \Lambda^k R_{t+1}]$, we obtain

$$\mathbb{E}[(R_1, \ldots, R_{t+1})^\top \Lambda^k (R_1, \ldots, R_{t+1})] = \mathbf{L}_{t+1}^{(k)}$$

and hence conclude $t+1^{(b)}$.

For $t+1^{(c)}$, we define $\tilde{\mathbf{\Phi}}_t = \mathbf{\Phi}_{t+1}[1:t+1,\ 1:t]$ and $\tilde{\mathbf{B}}_t^\top = \mathbf{B}_{t+1}^\top[1:t+1,\ 1:t]$ as the first t column blocks of $\mathbf{\Phi}_{t+1}$ and $\mathbf{B}_{t+1}^\top$, and condition $\mathbf{O}$ on

$$\begin{aligned}
(\mathbf{R}_1, \ldots, \mathbf{R}_{t+1}) &= \mathbf{O}(\mathbf{U}_1, \ldots, \mathbf{U}_{t+1}), \\
\mathbf{O}(\mathbf{Z}_1, \ldots, \mathbf{Z}_t) &= \mathbf{\Lambda}(\mathbf{R}_1, \ldots, \mathbf{R}_{t+1}) - (\mathbf{R}_1, \ldots, \mathbf{R}_{t+1})\tilde{\mathbf{B}}_t^\top.
\end{aligned}$$

Applying again the W_2-convergence in $t^{(c)}$ and [Fan20, Proposition E.5], we have

$$n^{-1}(\mathbf{Z}_1, \ldots, \mathbf{Z}_t)^\top (\mathbf{U}_1, \ldots, \mathbf{U}_{t+1}) \to \mathbf{\Sigma}_t \cdot \tilde{\mathbf{\Phi}}_t^\top.$$

Observe that

$$\begin{pmatrix} \mathbf{\Delta}_{t+1} & \tilde{\mathbf{\Phi}}_t \mathbf{\Sigma}_t \\ \mathbf{\Sigma}_t \tilde{\mathbf{\Phi}}_t^\top & \mathbf{\Sigma}_t \end{pmatrix}$$

is invertible, as its submatrix removing row block $t+1$ and column block $t+1$ is invertible by $t^{(d)}$, and the Schur-complement of the $(t+1, t+1)$ block is invertible by

(3.5.1). Then applying the same arguments as leading to [Fan20, Eq. (A.33)], we get $(\mathbf{U}_1, \ldots, \mathbf{U}_{t+1}, \mathbf{Z}_1, \ldots, \mathbf{Z}_t, \mathbf{E}, \mathbf{S}_{t+1}) \xrightarrow{W_2} (U_1, \ldots, U_{t+1}, Z_1, \ldots, Z_t, E, S_{t+1})$ where

$$S_{t+1} = \begin{pmatrix} U_1 & \cdots & U_{t+1} \end{pmatrix} \mathbf{B}_{t+1}^\top[1:t,\ t+1] + \begin{pmatrix} Z_1 & \cdots & Z_t \end{pmatrix} \mathbf{\Sigma}_t^{-1} \cdot \mathbf{\Sigma}_{t+1}[1:t,\ t+1] + S_\perp$$

and

$$S_\perp \sim \mathcal{N}\left(0,\ \left(\mathbf{L}_{t+1}^{(2)}[t+1, t+1] - \begin{pmatrix} \mathbf{L}_{t+1}^{(1)}[1:t+1,\ t+1] \\ \mathbf{L}_{t+1}^{(2)}[1:t,\ t+1] \end{pmatrix}^\top \right.\right.$$
$$\left.\left. \times \begin{pmatrix} \mathbf{L}_{t+1}^{(0)} & \mathbf{L}_{t+1}^{(1)}[1:t,\ 1:t+1] \\ \mathbf{L}_{t+1}^{(1)}[1:t,\ 1:t+1]^\top & \mathbf{L}_t^{(2)} \end{pmatrix}^{-1} \begin{pmatrix} \mathbf{L}_{t+1}^{(1)}[1:t+1,\ t+1] \\ \mathbf{L}_{t+1}^{(2)}[1:t,\ t+1] \end{pmatrix} \right)\right).$$

Here again, we use $[1:t,\ t+1]$ to denote the first t row blocks of the last column block $t+1$, and similarly for $[t+1,\ t+1]$ and $[1:t+1,\ t+1]$. Since $\mathbf{Z}_{t+1} = \mathbf{S}_{t+1} - \begin{pmatrix} \mathbf{U}_1 & \cdots & \mathbf{U}_{t+1} \end{pmatrix} \mathbf{B}_{t+1}^\top[1:t,\ t+1]$, the same computations as in [Fan20, Lemma A.4] show convergence of $(\mathbf{U}_1, \ldots, \mathbf{U}_{t+1}, \mathbf{Z}_1, \ldots, \mathbf{Z}_{t+1}, \mathbf{E})$ as in $t+1^{(c)}$, where the covariance of $(Z_1, \ldots, Z_{t+1})$ is exactly $\mathbf{\Sigma}_{t+1}$. Since $\mathbf{U}_{t+2} = u_{t+2}(\mathbf{Z}_1, \ldots, \mathbf{Z}_{t+1}, \mathbf{E})$ and $u_{t+2}(\cdot)$ is Lipschitz, $t+1^{(c)}$ follows from Lemma 3.6.4.

Finally, for $t+1^{(d)}$, observe from the definition of $\mathbf{L}_t^{(k)}$ that for any fixed vector $v \in \mathbb{R}^K$, $v^\top \operatorname{Cov}[S_\perp]v$ is the residual variance of the L_2-projection of $v^\top \Lambda R_{t+1}$ onto the linear span of $(v^\top R_1, \ldots, v^\top R_{t+1}, v^\top \Lambda R_1, \ldots, v^\top \Lambda R_t)$. If $v^\top \operatorname{Cov}[S_\perp]v = 0$, then this would imply

$$v^\top \Lambda R_{t+1} = \alpha_1 \cdot v^\top R_1 + \ldots + \alpha_{t+1} \cdot v^\top R_{t+1} + \beta_1 \cdot v^\top \Lambda R_1 + \ldots + \beta_t \cdot v^\top \Lambda R_t,$$

and hence $(\Lambda - \alpha_{t+1}) v^\top R_\perp = f(v^\top R_1, \ldots, v^\top R_t, \Lambda)$ for some function f. This implies that $R_\perp$ is constant conditional on $(R_1, \ldots, R_t, \Lambda)$ and the positive-probability event $\Lambda \neq \alpha_{t+1}$. This contradicts that $R_\perp$ is independent of $(R_1, \ldots, R_t, \Lambda)$ with a Gaussian law of positive variance. So $v^\top \operatorname{Cov}[S_\perp]v > 0$ implying $\operatorname{Cov}[S_\perp] \succ 0$ strictly. Then $t+1^{(d)}$ follows as in [Fan20, Lemma A.4], concluding the proof. ■

Proof of Theorem 3.1.3. We may remove Assumption 3.5.1 by studying a perturbed AMP sequence and applying a continuity argument, as in [Fan20, Appendix D] and [BMN20]. Consider the perturbed AMP iterations

$$\mathbf{Z}_t^\varepsilon = \mathbf{W}^\varepsilon \mathbf{U}_t^\varepsilon - \mathbf{U}_1^\varepsilon b_{t1}^{\varepsilon\top} - \ldots - \mathbf{U}_t^\varepsilon b_{tt}^{\varepsilon\top}, \quad \mathbf{U}_{t+1}^\varepsilon = u_{t+1}(\mathbf{Z}_1^\varepsilon, \ldots, \mathbf{Z}_t^\varepsilon, \mathbf{E}) + \varepsilon \mathbf{G}_{t+1}$$

where $\mathbf{W}^\varepsilon = \mathbf{O}^\top \operatorname{diag}(\boldsymbol{\lambda} + \varepsilon\boldsymbol{\gamma})\mathbf{O}$, the vector $\boldsymbol{\gamma} \in \mathbb{R}^n$ has i.i.d. Uniform$(-1,1)$ entries, each b_{ts}^ε is the version of b_{ts} defined with the limit free cumulants of $\mathbf{W}^\varepsilon$ in place of $\mathbf{W}$, and $\mathbf{G}_{t+1} \in \mathbb{R}^{n\times K}$ is an independent matrix with i.i.d. $\mathcal{N}(0,1)$ entries in each iteration. Then the same argument as in [Fan20, Appendix D] shows

$$\lim_{\varepsilon\to 0} \limsup_{n\to\infty} n^{-1}\|\mathbf{Z}_t^\varepsilon - \mathbf{Z}_t\|_F^2 = 0, \qquad \lim_{\varepsilon\to 0} \limsup_{n\to\infty} n^{-1}\|\mathbf{U}_t^\varepsilon - \mathbf{U}_t\|_F^2 = 0,$$

$$\lim_{\varepsilon\to 0}(\boldsymbol{\Delta}_t^\varepsilon, \boldsymbol{\Phi}_t^\varepsilon, \mathbf{B}_t^{\varepsilon\top}, \boldsymbol{\Sigma}_t^\varepsilon) = (\boldsymbol{\Delta}_t, \boldsymbol{\Phi}_t, \mathbf{B}_t^\top, \boldsymbol{\Sigma}_t)$$

where $\boldsymbol{\Delta}_t^\varepsilon, \boldsymbol{\Phi}_t^\varepsilon, \mathbf{B}_t^\varepsilon, \boldsymbol{\Sigma}_t^\varepsilon$ are the matrices corresponding to this perturbed AMP sequence. Note that in [Fan20, Appendix D], the continuous-differentiability of each function $u_s(\cdot)$ is used only to show the convergence $\boldsymbol{\Phi}_t \to \boldsymbol{\Phi}_t^\infty$ at the end of the proof (i.e. $\boldsymbol{\phi}_t \to \boldsymbol{\Phi}_t$ in our notation), and the relaxed condition of Assumption 3.1.2(b) is sufficient for this statement. On the other hand, Assumption 3.5.1 holds for this perturbed AMP sequence, so this perturbed sequence is characterized by Lemma 3.5.2. Combining these shows Theorem 3.1.3. ■

3.6 Auxiliary lemmas

Lemma 3.6.1. *For each $\tau > 0$, let $\{x_i^{(\tau)}\}_{0 \le i \le \tau}$ and $\{y_i^{(\tau)}\}_{0 \le i \le \tau}$ be two sequences where $|x_i^{(\tau)}|$ and $|y_i^{(\tau)}|$ uniformly bounded by a constant $C > 0$. Suppose, for any $\epsilon > 0$, there exists some $\mathcal{T} > 0$ such that for all $\tau > \mathcal{T}$,*

$$\sum_{i=\mathcal{T}}^{\tau} |x_i^{(\tau)}| \le \epsilon \quad \text{and} \quad \lim_{\tau \to \infty} |y_i^{(\tau)}| = 0 \text{ for all } 0 \le i \le \mathcal{T}.$$

Then $\lim_{\tau \to \infty} \sum_{i=0}^{\tau} x_i^{(\tau)} y_i^{(\tau)} = 0$.

Proof. For any $\epsilon > 0$, let $\mathcal{T}$ be as given. Then for all $\tau > \mathcal{T}$, we have

$$\begin{aligned} \left| \sum_{i=0}^{\tau} x_i^{(\tau)} y_i^{(\tau)} \right| &\le \max_{0 \le i \le \mathcal{T}} |x_i^{(\tau)}| \cdot \sum_{i=0}^{\mathcal{T}} |y_i^{(\tau)}| + \max_{\mathcal{T} < i \le \tau} |y_i^{(\tau)}| \cdot \sum_{i=\mathcal{T}}^{\tau} |x_i^{(\tau)}| \\ &\le C \cdot \sum_{i=0}^{\mathcal{T}} |y_i^{(\tau)}| + C \cdot \sum_{i=\mathcal{T}}^{\tau} |x_i^{(\tau)}| \end{aligned}$$

where C is the given uniform bound. Taking the limit $\tau \to \infty$ on both sides, we obtain

$$\limsup_{\tau \to \infty} \left| \sum_{i=0}^{\tau} x_i^{(\tau)} y_i^{(\tau)} \right| \le C\epsilon.$$

Then since ϵ is arbitrary, the desired result follows. ■

Lemma 3.6.2. *For any integers $\tau, t, K \ge 1$, let $\mathbf{A} \in \mathbb{R}^{(\tau+t+1)K \times (\tau+t+1)K}$ have the block decomposition*

$$\mathbf{A} = \begin{pmatrix} \mathbf{A}_{--} & 0 \\ \mathbf{A}_{+-} & \mathbf{A}_{++} \end{pmatrix}$$

where $\mathbf{A}_{--} \in \mathbb{R}^{\tau K \times \tau K}$ and $\mathbf{A}_{++} \in \mathbb{R}^{(t+1)K \times (t+1)K}$. Suppose $\mathbf{A}_{--}$ and $\mathbf{A}_{+-}$ have the *further block decompositions*

$$\mathbf{A}_{--} = \begin{pmatrix} 0 & 0 & \cdots & 0 & 0 \\ \mathbf{B} & 0 & \cdots & 0 & 0 \\ 0 & \mathbf{B} & \cdots & 0 & 0 \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & \cdots & \mathbf{B} & 0 \end{pmatrix}, \quad \mathbf{A}_{+-} = \begin{pmatrix} 0 & \cdots & 0 & \mathbf{B} \\ 0 & \cdots & 0 & 0 \\ \vdots & \ddots & \vdots & \vdots \\ 0 & \cdots & 0 & 0 \end{pmatrix}$$

for some $\mathbf{B} \in \mathbb{R}^{K\times K}$*. Then, indexing the row and column blocks by* $\{-\tau, -\tau+1, \ldots, t\}$*, for all* $r = 0, \ldots, t$ *and* $c = 1, \ldots, \tau$*,*

$$\mathbf{A}^j[r, -c] = \begin{cases} \mathbf{A}_{++}^{j-c}[r, 0] \cdot \mathbf{B}^c & 1 \leq c \leq j, \\ 0 & j < c \end{cases}$$

where $[r, -c]$ *denotes the* $K \times K$ *submatrix corresponding to row block* r *and column block* $-c$*.*

Proof. Due to the special structure of $\mathbf{A}$, it is straightforward to verify (by induction) that for any $j \geq 1$, we have

$$\mathbf{A}^j = \begin{pmatrix} \mathbf{A}_{--}^j & 0 \\ \sum_{i=0}^{j-1} \mathbf{A}_{++}^i \mathbf{A}_{+-} \mathbf{A}_{--}^{j-i-1} & \mathbf{A}_{++}^j \end{pmatrix}.$$

Moreover, observe that for each $j - i - 1 \geq 0$,

$$\mathbf{A}_{+-}\mathbf{A}_{--}^{j-i-1} = \begin{pmatrix} 0 & \cdots & 0 & \mathbf{B}^{j-i} & 0 & \cdots & 0 \\ 0 & \cdots & 0 & 0 & 0 & \cdots & 0 \\ \vdots & \ddots & \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & \cdots & 0 & 0 & 0 & \cdots & 0 \end{pmatrix}.$$

Following the convention that $\mathbf{A}_{++}^0 = \mathrm{Id}$, we further have

$$\sum_{i=0}^{j-1} \mathbf{A}_{++}^i \mathbf{A}_{+-} \mathbf{A}_{--}^{j-i-1} = \begin{pmatrix} 0 & \cdots & 0 & \mathbf{A}_{++}^0[:,0]\mathbf{B}^j & \cdots & \mathbf{A}_{++}^{j-1}[:,0]\mathbf{B} \end{pmatrix},$$

or equivalently, for all $r = 0, \ldots, t$ and $c = 1, \ldots, \tau$,

$$\mathbf{A}^j[r, -c] = \begin{cases} \mathbf{A}_{++}^{j-c}[r, 0] \cdot \mathbf{B}^c & 1 \leq c \leq j, \\ 0 & j < c. \end{cases}$$

■

Lemma 3.6.3. *If $\mathbf{\Lambda} = \mathrm{diag}(\boldsymbol{\lambda}) \in \mathbb{R}^{n \times n}$ and the random variable Λ satisfy Assumption 3.1.1, and $\mathbf{R} \in \mathbb{R}^{n \times j}$ is such that $(\mathbf{R}, \boldsymbol{\lambda}) \xrightarrow{W_2} (R, \Lambda)$ as $n \to \infty$ for a random vector $R \in \mathbb{R}^j$, then for any $k \geq 0$,*

$$\lim_{n \to \infty} \frac{1}{n} \mathbf{R}^\top \mathbf{\Lambda}^k \mathbf{R} = \mathbb{E}[\Lambda^k \cdot RR^\top] \in \mathbb{R}^{j \times j}.$$

Proof. Fix any $a, b \in \{1, \ldots, j\}$, let $C > \max(|\lambda_-|, |\lambda_+|)$, and define the function $f_C(\lambda, r_a, r_b) = \min(\max(\lambda, -C), C)^k \cdot r_a r_b$. Then $|f_C(\lambda, r_a, r_b)| \leq C^k(r_a^2 + r_b^2)$, so by the given Wasserstein-2 convergence, a.s.

$$\frac{1}{n} \sum_{i=1}^n f_C(\lambda_i, r_{a,i}, r_{b,i}) \to \mathbb{E}[f_C(\Lambda, R_a, R_b)].$$

The left side coincides with the (a, b) entry of $n^{-1}\mathbf{R}^\top \mathbf{\Lambda}^k \mathbf{R}$ a.s. for all large n, while the right side coincides with that of $\mathbb{E}[\Lambda^k \cdot RR^\top]$. ■

Lemma 3.6.4. *Suppose $\mathbf{Z} \in \mathbb{R}^{n \times j}$ satisfies $\mathbf{Z} \xrightarrow{W_2} Z$, and $u : \mathbb{R}^j \to \mathbb{R}^k$ is Lipschitz. Then $(\mathbf{Z}, u(\mathbf{Z})) \xrightarrow{W_2} (Z, u(Z))$.*

Proof. Let L be the Lipschitz constant of u. For any continuous function $f : \mathbb{R}^{j+k} \to \mathbb{R}$

satisfying $|f(z,u)| \leq C(1+\|z\|^2+\|u\|^2)$, we have

$$|f(z,u(z))| \leq C(1+\|z\|^2+(L\|z\|+u(0))^2) \leq C'(1+\|z\|^2)$$

for a different constant $C' > 0$. Then $n^{-1}\sum_i f(z_i,u(z_i)) \to \mathbb{E}[f(Z,u(Z))]$, so the result follows. ∎

Lemma 3.6.5. *Fix $J, L \geq 0$ and $K \geq 1$. Let $\mathbf{O} \in \mathbb{R}^{(n-L)\times(n-L)}$ be a random Haar-uniform orthogonal matrix. Let $\mathbf{E} \in \mathbb{R}^{n\times J}$ and $\mathbf{V} \in \mathbb{R}^{(n-L)\times K}$ satisfy $\mathbf{E} \xrightarrow{W_2} E$ and $n^{-1}\mathbf{V}^\top\mathbf{V} \to \Sigma \in \mathbb{R}^{K\times K}$. Let $\Pi \in \mathbb{R}^{n\times(n-L)}$ be any deterministic matrix with orthonormal columns. Then almost surely as $n \to \infty$,*

$$(\Pi\mathbf{O}\mathbf{V}, \mathbf{E}) \xrightarrow{W_2} (Z, E)$$

where $Z \sim \mathcal{N}(0,\Sigma) \in \mathbb{R}^K$ is independent of $E \in \mathbb{R}^J$.

Proof. Let $\tilde{\mathbf{O}}$ be the first K columns of $\mathbf{O}$. Writing the singular value decomposition $\mathbf{V} = \mathbf{Q}\mathbf{D}\mathbf{U}^\top$ and applying the equality in law $\mathbf{O}\mathbf{Q} \stackrel{L}{=} \tilde{\mathbf{O}}\mathbf{U}$, we have the equality in law $\mathbf{O}\mathbf{V} \stackrel{L}{=} \tilde{\mathbf{O}}(\mathbf{V}^\top\mathbf{V})^{1/2}$, where $(\mathbf{V}^\top\mathbf{V})^{1/2} = \mathbf{U}\mathbf{D}\mathbf{U}^\top$ is the positive-semidefinite matrix square-root. Then introducing $\mathbf{Z} \in \mathbb{R}^{n\times K}$ with i.i.d. $\mathcal{N}(0,1)$ entries, and applying $\tilde{\mathbf{O}} \stackrel{L}{=} \Pi^\top\mathbf{Z}(\mathbf{Z}^\top\Pi\Pi^\top\mathbf{Z})^{-1/2}$, also

$$\mathbf{O}\mathbf{V} \stackrel{L}{=} \Pi^\top\mathbf{Z}(n^{-1}\mathbf{Z}^\top\Pi\Pi^\top\mathbf{Z})^{-1/2}(n^{-1}\mathbf{V}^\top\mathbf{V})^{1/2}.$$

So

$$\Pi\mathbf{O}\mathbf{V} \stackrel{L}{=} (\mathrm{Id}-\Pi^\perp)\mathbf{Z}(n^{-1}\mathbf{Z}^\top\Pi\Pi^\top\mathbf{Z})^{-1/2}(n^{-1}\mathbf{V}^\top\mathbf{V})^{1/2}, \tag{3.6.1}$$

where $\Pi^\perp = \mathrm{Id} - \Pi\Pi^\top \in \mathbb{R}^{n\times n}$ is a projection onto a subspace of fixed dimension L. Since $n^{-1}\mathbf{Z}^\top\Pi\Pi^\top\mathbf{Z} \to \mathrm{Id}_{K\times K}$ and $n^{-1}\mathbf{V}^\top\mathbf{V} \to \Sigma$, and the matrix square-root is continuous, we obtain

$$\Big(\mathbf{Z}(n^{-1}\mathbf{Z}^\top \Pi\Pi^\top \mathbf{Z})^{-1/2}(n^{-1}\mathbf{V}^\top\mathbf{V})^{1/2},\ \mathbf{E}\Big) \xrightarrow{W_2} (Z, E) \tag{3.6.2}$$

by [Fan20, Propositions E.1 and E.4]. Now write $\Pi^\perp = \mathbf{U}\mathbf{U}^\top$ where $\mathbf{U} \in \mathbb{R}^{n\times L}$ has orthonormal columns. Then $\Pi^\perp \mathbf{Z} \overset{L}{=} \mathbf{U}G$ where $G \in \mathbb{R}^{L\times K}$ has i.i.d. $\mathcal{N}(0,1)$ entries. Let u_i be the i^{th} row of $\mathbf{U}$, and g_j be the j^{th} column of G. Then a.s.

$$\frac{1}{n}\sum_{i=1}^n (u_i^\top g_j)^2 \le \frac{1}{n}\sum_{i=1}^n \|u_i\|^2 \cdot \|g_j\|^2 = \frac{L}{n}\cdot \|g_j\|^2 \to 0.$$

This holds for each column j of $\Pi^\perp \mathbf{Z} \overset{L}{=} \mathbf{U}G$, so $\Pi^\perp \mathbf{Z} \xrightarrow{W_2} 0$. Then combining with (3.6.2) and applying this to (3.6.1), we obtain $(\Pi\mathbf{O}\mathbf{V}, \mathbf{E}) \xrightarrow{W_2} (Z, E)$ by [Fan20, Proposition E.4]. ■

Chapter 4

Implicit Regularization of Stochastic Gradient Descent

Understanding the implicit bias of Stochastic Gradient Descent (SGD) is one of the key challenges in deep learning, especially for overparametrized models, where the local minimizers of the loss function L can form a manifold. Intuitively, with a sufficiently small learning rate η, SGD tracks Gradient Descent (GD) until it gets close to such manifold, where the gradient noise prevents further convergence. In such regime, [BGVV20] proved that SGD with label noise locally decreases a regularizer-like term, the sharpness of loss, $\mathrm{tr}[\nabla^2 L]$.

In this chapter, we give a general framework for such analysis by adapting ideas from [Kat91]. It allows in principle a complete characterization for the regularization effect of SGD around such manifold—i.e., the "implicit bias"—using a stochastic differential equation (SDE) describing the limiting dynamics of the parameters, which is determined jointly by the loss function and the noise covariance. This yields some new results: (1) a *global* analysis of the implicit bias valid for η^{-2} steps, in contrast to the local analysis of [BGVV20] that is only valid for $\eta^{-1.6}$ steps and (2) allowing *arbitrary* noise covariance. As an application, we show with arbitrary large initialization, label noise SGD can always escape the kernel regime and only requires $O(\kappa \ln d)$ samples for learning a κ-sparse overparametrized linear model in $\mathbb{R}^d$ [WGL+20], while GD initialized in the kernel regime requires $\Omega(d)$ samples. This upper bound is minimax optimal and improves the previous $\widetilde{O}(\kappa^2)$ upper bound [HWLM20].

The framework developed here is general and can be applied to other variants of SGD, such as SGD with weight decay for models with normalization layers [LWY22] and SGD with momentum [WMW+24].

4.1 Introduction on SGD and implicit regularization

The implicit bias underlies the generalization ability of machine learning models trained by stochastic gradient descent (SGD). But it still remains a mystery to mathematically characterize such bias. We study SGD in the following formulation

$$x_\eta(k+1) = x_\eta(k) - \eta(\nabla L(x_\eta(k)) + \sqrt{\Xi} \cdot \sigma_{\xi_k}(x_\eta(k))) \tag{4.1.1}$$

where η is the learning rate (LR), $L : \mathbb{R}^D \to \mathbb{R}$ is the training loss and $\sigma(x) = [\sigma_1(x), \sigma_2(x), \ldots, \sigma_\Xi(x)] \in \mathbb{R}^{D\times\Xi}$ is a continuously differentiable deterministic noise function. Here ξ_k is sampled uniformly from $\{1, 2, \ldots, \Xi\}$ and it satisfies $\mathbb{E}_{\xi_k}[\sigma_{\xi_k}(x)] = 0$, $\forall x \in \mathbb{R}^d$ and k.

It is widely believed that large LR (or equivalently, small batch size) helps SGD find better minima. For instance, some previous works argued that large noise enables SGD to select a flatter attraction basin of the loss landscape which potentially benefits generalization [LWM19, JKA+17]. However, there is also experimental evidence [LLA20] that small LR also has equally good implicit bias (albeit with higher training time), and that is the case studied here. Presumably low LR precludes SGD jumping between different basins since under general conditions this should require $\Omega(\exp(1/\eta))$ steps [SSJ20]. In other words, there should be a mechanism to reach better generalization while staying within a single basin. For deterministic GD similar mechanisms have been demonstrated in simple cases [SHN$^+$18, LL19] and referred to as *implicit bias of gradient descent*. The results in this chapter can be seen as study of implicit bias of Stochastic GD, which turns out to be quite different, mathematically.

Recent work [BGVV20] shed light on this direction by analyzing effects of stochasticity in the gradient. For sufficiently small LR, SGD will reach and be trapped around some manifold of local minimizers, denoted by Γ (see Figure 4.2). The effect is shown to be an implicit deterministic drift in a direction corresponding to lowering a regularizer-like term along the manifold. They showed SGD with label noise locally decreases the sharpness of loss, $\mathrm{tr}[\nabla^2 L]$, by $\Theta(\eta^{0.4})$ in $\eta^{-1.6}$ steps. However, such an analysis is actually *local*, since the natural time scale of analysis should be η^{-2}, not $\eta^{-1.6}$.

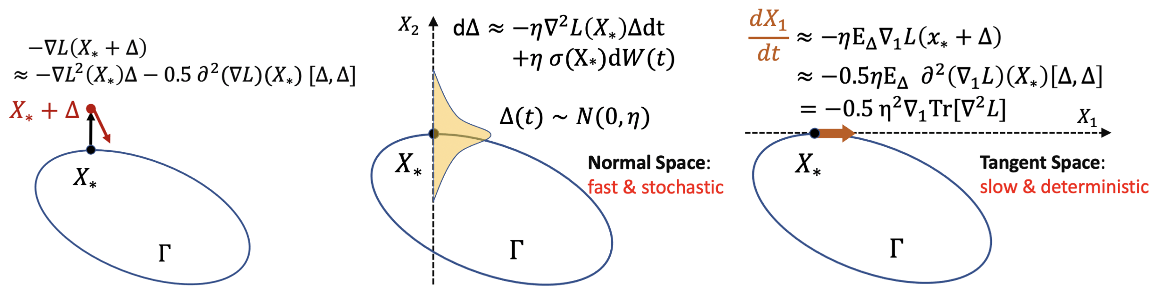
Our contribution is a more general and global analysis of this type. We introduce a more powerful framework inspired by the classic paper [Kat91].

4.1.1 Intuitive explanation of regularization effect due to SGD

We start with an intuitive description of the implicit regularization effect described in [BGVV20]. For simplification, we show it for the canonical SDE approximation (4.1.2) (See Section 4.7.1 for more details) of SGD (4.1.1) [LTW17, CYBJ20].

$$\mathrm{d}\tilde{X}_\eta(t) = -\eta \nabla L(\tilde{X}_\eta(t))\mathrm{d}t + \eta \cdot \sigma(\tilde{X}_\eta(t))\mathrm{d}W(t). \tag{4.1.2}$$

Here $W(t)$ is the standard Ξ-dimensional Brownian motion. The only property about label noise SGD we will use is that the noise covariance $\sigma\sigma^\top(x) = \nabla^2 L(x)$ for every x



(a) Taylor Expansion of ∇L (b) Normal Space Dynamics (c) Tangent Space Dynamics

Figure 4.1: Illustration for limiting flow in $\mathbb{R}^2$. Γ is an 1D manifold of minimizers of loss L.

in the manifold Γ (See derivation in Section 4.4).

Suppose $\tilde{X}_\eta(0)$ is already close to some local minimizer point $X_* \in \Gamma$. The goal is to show $\tilde{X}_\eta(t)$ will move in the tangent space and steadily decrease $\mathrm{tr}[\nabla^2 L]$. At first glance, this seems impossible as the gradient ∇L vanishes around Γ, and the noise has zero mean, implying SGD should be like random walk instead of a deterministic drift. The key observation of [BGVV20] is that the local dynamics of $\tilde{X}_\eta(t)$ is completely different in tangent space and normal space — the fast random walk in normal space causes $\tilde{X}_\eta(t)$ to move slowly (with velocity $\Theta(\eta^2)$) but deterministically in certain direction. To explain this, letting $\Delta(t) = \tilde{X}_\eta(t) - X_*$, Taylor expansion of (4.1.2) gives $\mathrm{d}\Delta(t) \approx -\eta\nabla^2 L(X_*)\Delta(t)\mathrm{d}t + \eta\sigma(X_*)\mathrm{d}W(t)$, meaning Δ is behaving like an Ornstein-Uhlenbeck (OU) process locally in the normal space. Its mixing time is $\Theta(\eta^{-1})$ and the stationary distribution is the standard multivariate gaussian in the normal space scaled by $\sqrt{\eta}$ (see Figure 4.1b), because noise covariance $\sigma\sigma^\top = \nabla^2 L$. Though this OU process itself doesn't form any regularization, it activates the second order Taylor expansion of $\nabla L(X_* + \Delta(t))$, i.e., $-\frac{1}{2}\partial^2(\nabla L)(X_*)[\Delta(t), \Delta(t)]$, creating a $\Theta(\eta^2)$ velocity in the tangent space. Since there is no push back force in the tangent space, the small velocity accumulates over time, and in a longer time scale of $\Omega(\eta^{-1})$, the time average of the stochastic velocity is roughly the same as the expected velocity when Δ is sampled from its stationary distribution. This simplifies the expression

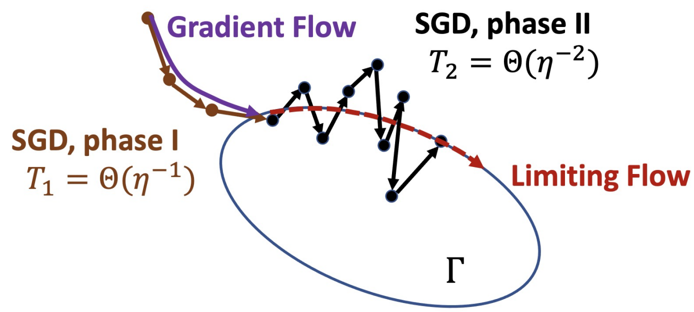


Figure 4.2: Illustration for two-phase dynamics of SGD with the same example as in Figure 4.1 . Γ is an 1D manifold of minimizers of loss L.

of the velocity in tangent space to $\frac{\eta^2}{2}\nabla_\Gamma \operatorname{tr}[\nabla^2 L]$ (see Figure 4.1c), where ∇_Γ means the Riemmanian gradient on Γ (with respect to the standard euclidean metric), or equivalently, the ordinary gradient projected into the tangent space of Γ.

However, the above approach only gives a local analysis for $O(\eta^{-1.6})$ time, where the total movement due to implicit regularization is $O(\eta^{2-1.6}) = O(\eta^{0.4})$ and thus is negligible when $\eta \to 0$. In order to get a non-trivial limiting dynamics when $\eta \to 0$, a global analysis for $\Omega(\eta^{-2})$ steps is necessary and it cannot be done by Taylor expansion with a single reference point. Recent work by [DML21] glues analyses of multiple local phases into a global guarantee that SGD finds a (ε, γ)-stationary point for the regularized loss, but still doesn't show convergence for trajectory when $\eta \to 0$ and cannot deal with general noise types, *e.g.*, noise lying in the tangent space of the manifold. The main technical difficulty here is that it's not clear how to separate the slow and fast dynamics in different spaces and how to only take limit for the slow dynamics, especially when shifting to a new reference point in the Taylor series calculation.

4.1.2 Our Approach: Separating the slow from the fast

We tackle this problem via a different angle. First, since the anticipated limiting dynamics is of speed $\Theta(\eta^2)$, we change the time scaling to accelerate (4.1.2) by η^{-2}

times, which yields

$$\mathrm{d}X_\eta(t) = -\eta^{-1}\nabla L(X_\eta(t))\mathrm{d}t + \sigma(X_\eta(t))\mathrm{d}W(t). \tag{4.1.3}$$

The key idea here is that we only need to track the slow dynamic, or equivalently, some *projection* of X onto the manifold Γ, $\Phi(X)$. Here $\Phi : \mathbb{R}^D \to \Gamma$ is some function to be specified and hopefully we can simplify the dynamics (4.1.3) via choosing suitable Φ. To track the dynamics of $\Phi(X_\eta)$, we apply Ito's lemma (a.k.a. stochastic chain rule, see Lemma 4.6.11) to Equation (4.1.3), which yields

$$\begin{aligned}\mathrm{d}\Phi(X_\eta(t)) = &-\eta^{-1}\partial\Phi(X_\eta(t))\nabla L(X_\eta(t))\mathrm{d}t + \partial\Phi(X_\eta(t))\sigma(X_\eta(t))\mathrm{d}W(t)\\ &+ \frac{1}{2}\sum_{i,j=1}^{D}\partial_{ij}\Phi(X_\eta(t))(\sigma(X_\eta(t))\sigma(X_\eta(t))^\top)_{ij}\mathrm{d}t.\end{aligned}$$

Note the first term $-\eta^{-1}\partial\Phi(X_\eta)\nabla L(X_\eta)$ is going to diverge to ∞ when $\eta \to 0$, so a natural choice for Φ is to kill the first term. Further note $-\partial\Phi(X)\nabla L(X)$ is indeed the directional derivative of Φ at X towards $-\nabla L$, killing the first term becomes equivalent to making Φ invariant under Gradient Flow (GF) of $-\nabla L(X)$! Thus it suffices to take $\Phi(X)$ to be the limit of GF starting at X. (Formally defined in Section 4.2; see Lemma 4.8.2 for a proof of $\partial\Phi(X)\nabla L(X) \equiv 0$.) Indeed, such choice is also unique under two nature requirements for Φ: Φ is continuous and $\Phi(X) = X$ for every $X \in \Gamma$.

Also intuitively X_η will be infinitely close to Γ, i.e., $d(X_\eta(t), \Gamma) := \inf_{x\in\Gamma}\|X_\eta(t) - x\|_2 \to 0$ for any $t > 0$ as $\eta \to 0$, so we have $\Phi(X_\eta) \approx X_\eta$. Thus we can rewrite the above equation as

$$\mathrm{d}X_\eta(t) \approx \partial\Phi(X_\eta(t))\sigma(X_\eta(t))\mathrm{d}W(t) + \frac{1}{2}\sum_{i,j=1}^{D}\partial_{ij}\Phi(X_\eta(t))(\sigma(X_\eta(t))\sigma(X_\eta(t))^\top)_{ij}\mathrm{d}t, \tag{4.1.4}$$

and the solution of (4.1.4) shall converge to that of the following (in an intuitive sense):

$$\mathrm{d}Y(t) = \partial\Phi(Y(t))\sigma(Y(t))\mathrm{d}W(t) + \frac{1}{2}\sum_{i,j=1}^{D}\partial_{ij}\Phi(Y(t))(\sigma(Y(t))\sigma(Y(t))^\top)_{ij}\mathrm{d}t, \quad (4.1.5)$$

The above argument for SDE was first formalized and rigorously proved by [Kat91]. It included an extension of the analysis to the case of asymptotic continuous dynamics (Theorem 4.3.1) including SGD with infinitesimal LR, but the result is weaker in this case and no convergence is shown. Another obstacle for applying this analysis is that 2nd order partial derivatives of Φ are unknown. We solve these issues in Section 4.3 and our main result Theorem 4.3.6 gives a clean and complete characterization for the implicit bias of SGD with infinitesimal LR in $\Theta(\eta^{-2})$ steps. Finally, our Corollary 4.4.2 shows (4.1.5) gives exactly the same regularization as $\mathrm{tr}[\nabla^2 L]$ for label noise SGD.

4.2 Notation and preliminaries

For any natural number k, we denote $\mathcal{C}^k$ as the set of the k times continuously differentiable functions and $\overline{\mathcal{C}}^k$ as the set of functions whose kth order derivatives are locally Lipschitz. It holds that $\mathcal{C}^{k+1} \subseteq \overline{\mathcal{C}}^k \subseteq \mathcal{C}^k$. We denote $a \wedge b = \min\{a, b\}$. For any vector u, v and $\alpha \in \mathbb{R}$, we define $[u \odot v]_i = u_i v_i$ and $[v^{\odot\alpha}]_i = v_i^\alpha$. We denote $\mathbb{1}_\xi \in \mathbb{R}^\Xi$ as the one-hot vector where ξ-th coordinate is 1, and $\mathbb{1}$ the all 1 vector. For any matrix A, we denote its pseudo-inverse by $A^\dagger$. For mapping $F : \mathbb{R}^D \to \mathbb{R}^D$, we denote the *Jacobian* of F at x by $\partial F(x) \in \mathbb{R}^{D\times D}$ where the (i, j)-th entry is $\partial_j F_i(x)$. We also use $\partial F(x)[u]$ and $\partial^2 F(x)[u, v]$ to denote the first and second order directional derivative of F at x along the derivation of u (and v). We abuse the notation of $\partial^2 F$ by viewing it as a linear mapping defined on $\mathbb{R}^D \otimes \mathbb{R}^D \cong \mathbb{R}^{D^2}$, in the sense that $\partial^2 F(x)[\Sigma] = \sum_{i,j=1}^{D} \partial^2 F(x)[e_i, e_j]\Sigma_{ij}$, for any $\Sigma \in \mathbb{R}^{D\times D}$. For any submanifold $\Gamma \subset \mathbb{R}^D$ and $x \in \Gamma$, we denote by $T_x(\Gamma)$ the tangent space of Γ at x and $T_x^\perp(\Gamma)$ the normal space of Γ at x.

Given loss L, GF governed by L can be described through a mapping $\phi : \mathbb{R}^D \times [0, \infty) \to \mathbb{R}^D$ satisfying $\phi(x, t) = x - \int_0^t \nabla L(\phi(x, s))\mathrm{d}s$. We further denote the limiting mapping $\Phi(x) = \lim_{t\to\infty} \phi(x, t)$ whenever the limit exists.

4.2.1 Manifold of local minimizers

Assumption 4.2.1. Assume that the loss $L : \mathbb{R}^D \to \mathbb{R}$ is a $\mathcal{C}^4$ function, and that Γ is a $(D - M)$-dimensional $\mathcal{C}^2$-submanifold of $\mathbb{R}^D$ for some integer $0 \le M \le D$, where for all $x \in \Gamma$, x is a local minimizer of L and $\mathrm{rank}(\nabla^2 L(x)) = M$.

Assumption 4.2.2. Assume that U is an open neighborhood of Γ satisfying that gradient flow starting in U converges to some point in Γ, i.e., $\forall x \in U$, $\Phi(x) \in \Gamma$. (Then Φ is $\overline{\mathcal{C}}^2$ on U by [Fal83]. See Lemma 4.7.1 for details.)

When does such a manifold exist? The vast overparametrization in modern deep learning is a major reason for the set of global minimizers to appear as a Riemannian manifold (possibly with multiple connected components), instead of isolated ones. Suppose all global minimizers interpolate the training dataset, i.e., $\forall x \in \mathbb{R}^D$, $L(x) = \min_{x' \in \mathbb{R}^D} L(x')$ implies $f_i(x) = y_i$ for all $i \in [n]$, then by preimage theorem [BH13], the manifold $\Gamma := \{x \in \mathbb{R}^D \mid f_i(x) = y_i, \forall i \in [n]\}$ is of dimension $D - n$ if the Jacobian matrix $[\nabla f_1(x), \ldots, \nabla f_n(x)]$ has rank n for all $x \in \Gamma$. Note this condition is equivalent to that NTK at x has full rank, which is very common in literature.

4.3 Limiting diffusion of SGD

In Section 4.3.1 we first recap the main result of [Kat91]. In Section 4.3.2 we derive the closed-form expressions of $\partial\Phi$ and $\partial^2\Phi$. We present our main result in Section 4.3.3. We remark that sometimes we omit the dependency on t to make things clearer.

4.3.1 Recap of Katzenberger's theorem

Let $\{A_n\}_{n\geq 1}$ be a sequence of integrators, where each $A_n : \mathbb{R} \to \mathbb{R}$ is a non-decreasing function with $A_n(0) = 0$. Let $\{Z_n\}_{n\geq 1}$ be a sequence of $\mathbb{R}^{\Xi}$-valued stochastic processes. Given a loss function L, a $\mathbb{R}^{D\times\Xi}$-valued function σ and an initialization $x_{\text{init}} \in \mathbb{R}^D$, we consider stochastic processes $\{X_n(\cdot)\}_{n\geq 1}$ which satisfies the following equation Equation (4.3.1):

$$X_n(t) = x_{\text{init}} + \int_{s=0}^{t} \sigma(X_n(s))\mathrm{d}Z_n(s) + \int_{s=0}^{t} -\nabla L(X_n(s))\mathrm{d}A_n(s) \tag{4.3.1}$$

In particular, when Z_n converges in distribution to some stochastic process and the integrator sequence $\{A_n\}_{n\geq 1}$ increases infinitely fast, meaning that $\forall \epsilon > 0, \inf_{t\geq 0}(A_n(t+\epsilon) - A_n(t)) \to \infty$ as $n \to \infty$, we call (4.3.1) a *Katzenberger process*. In this paper, we are interested in the case where Z_n converges to a Ξ-dimensional Brownian motion and σ is related to the stochastic gradient noise as defined in Equation (4.1.1).

One difficulty for directly studying the limiting dynamics of $X_n(t)$ is that the point-wise limit as $n \to \infty$ become discontinuous at $t = 0$ if $x_{\text{init}} \notin \Gamma$. The reason is that clearly $\lim_{n\to\infty} X_n(0) = x_{\text{init}}$, but for any $t > 0$, since $\{A_n\}_{n\geq 1}$ increases infinitely fast, one can prove $\lim_{n\to\infty} X_n(t) \in \Gamma$! To circumvent this issue, we consider $Y_n(t) = X_n(t) - \phi(x_{\text{init}}, A_n(t)) + \Phi(x_{\text{init}})$. Then for each $n \geq 1$, we have $Y_n(0) = \Phi(x_{\text{init}})$ and $\lim_{n\to\infty} Y_n(t) = \lim_{n\to\infty} X_n(t)$. Thus $Y_n(t)$ has the same limit on $(0, \infty)$ as $X_n(t)$, but the limit of the former is further continuous at $t = 0$.

Theorem 4.3.1 (Informal version of Theorem 4.7.7, [Kat91])**.** *Suppose the loss L, manifold Γ and neighborhood U satisfies Assumptions 4.2.1 and 4.2.2. Let $\{X_n\}_{n\geq 1}$ be a sequence of Katzenberger process with $\{A_n\}_{n\geq 1}, \{Z_n\}_{n\geq 1}$. Let $Y_n(t) = X_n(t) - \phi(x_{init}, A_n(t)) + \Phi(X_0)$. Under technical assumptions, it holds that if (Y_n, Z_n) converges to some (Y, W) in distribution, where $\{W(t)\}_{t\geq 0}$ is the standard Brownian motion,* *then Y stays on Γ and admits*

$$Y(t) = Y(0) + \int_0^t \partial\Phi(Y)\sigma(Y)\mathrm{d}W(s) + \frac{1}{2}\sum_{i,j=1}^{D}\int_0^t \partial_{ij}\Phi(Y)(\sigma(Y)\sigma(Y)^\top)_{ij}\mathrm{d}s. \quad (4.3.2)$$

Indeed, SGD (4.1.1) can be rewritten into a Katzenberger process as in the following lemma.

Lemma 4.3.2. *Let $\{\eta_n\}_{n=1}^{\infty}$ be any positive sequence with $\lim_{n\to\infty}\eta_n = 0$, $A_n(t) = \eta_n\lfloor t/\eta_n^2\rfloor$, and $Z_n(t) = \eta_n\sum_{k=1}^{\lfloor t/\eta_n^2\rfloor}\sqrt{\Xi}(\mathbb{1}_{\xi_k} - \frac{1}{\Xi}\mathbb{1})$, where $\xi_1, \xi_2, \ldots \overset{i.i.d.}{\sim}$ Unif($[\Xi]$). Then with the same initialization $X_n(0) = x_{\eta_n}(0) \equiv x_{init}$, $X_n(k\eta_n^2)$ defined by (4.3.1) is a Katzenberger process and is equal to $x_{\eta_n}(k)$ defined in (4.1.1) with LR equal to η_n for all $k \geq 1$. Moreover, the counterpart of (4.3.2) is*

$$Y(t) = \Phi(x_{init}) + \int_0^t \partial\Phi(Y)\sigma(Y)\mathrm{d}W(s) + \frac{1}{2}\int_0^t \partial^2\Phi(Y)[\Sigma(Y)]\mathrm{d}s, \quad (4.3.3)$$

where $\Sigma \equiv \sigma\sigma^\top$ and $\{W(t)\}_{t\geq 0}$ is a Ξ-dimensional standard Brownian motion.

However, there are two obstacles preventing us from directly applying Theorem 4.3.1 to SGD. First, the stochastic integral in (4.3.3) depends on the derivatives of Φ, $\partial\Phi$ and $\partial_{ij}\Phi$, but [Kat91] did not give their dependency on loss L. To resolve this, we explicitly calculate the derivatives of Φ on Γ in terms of the derivatives of L in Section 4.3.2.

The second difficulty comes from the convergence of (Y_n, Z_n) which we assume as granted for brevity in Theorem 4.3.1. In fact, the full version of Theorem 4.3.1 (see Theorem 4.7.7) concerns the stopped version of Y_n with respect to some compact $K \subset U$, i.e., $Y_n^{\mu_n(K)}(t) = Y_n(t \wedge \mu_n(K))$ where $\mu_n(K)$ is the stopping time of Y_n leaving K. As noted in [Kat91], we need the convergence of $\mu_n(K)$ for $Y_n^{\mu_n(K)}$ to converge, which is a strong condition and difficult to prove in our cases. We circumvent this issue by proving Theorem 4.7.9, a user-friendly interface for the original theorem in

[Kat91], and it only requires the information about the limiting diffusion. Building upon these, we present our final result as Theorem 4.3.6.

4.3.2 Closed-form expression of the limiting diffusion

We can calculate the derivatives of Φ by relating to those of L. Here the key observation is the invariance of Φ along the trajectory of GF. The proofs of this section are deferred into Section 4.8.

Lemma 4.3.3. *For any $x \in \Gamma$, $\partial\Phi(x)$ is the orthogonal projection matrix onto tangent space $T_x(\Gamma)$.*

To express the second-order derivatives compactly, we introduce the notion of *Lyapunov operator*.

Definition 4.3.4 (Lyapunov Operator)**.** For a symmetric matrix H, we define $W_H = \{\Sigma \in \mathbb{R}^{D\times D} \mid \Sigma = \Sigma^\top, HH^\dagger\Sigma = \Sigma = \Sigma HH^\dagger\}$ and *Lyapunov Operator* $\mathcal{L}_H : W_H \to W_H$ as $\mathcal{L}_H(\Sigma) = H^\top\Sigma + \Sigma H$. It's easy to verify $\mathcal{L}_H^{-1}$ is well-defined on W_H.

Lemma 4.3.5. *Let x be any point in Γ and $\Sigma = \Sigma(x) = \sigma\sigma^\top(x) \in \mathbb{R}^{D\times D}$ be the noise covariance at x[1]. Then Σ can be decomposed as $\Sigma = \Sigma_\parallel + \Sigma_\perp + \Sigma_{\parallel,\perp} + \Sigma_{\perp,\parallel}$, where $\Sigma_\parallel := \partial\Phi\Sigma\partial\Phi$, $\Sigma_\perp := (I_D - \partial\Phi)\Sigma(I_D - \partial\Phi)$ and $\Sigma_{\parallel,\perp} = \Sigma_{\perp,\parallel}^\top = \partial\Phi\Sigma(I_D - \partial\Phi)$ are the noise covariance in tangent space, normal space and across both spaces, respectively. Then it holds that*

$$\partial^2\Phi[\Sigma] = -(\nabla^2 L)^\dagger\partial^2(\nabla L)\left[\Sigma_\parallel\right] - \partial\Phi\partial^2(\nabla L)\left[\mathcal{L}_{\nabla^2 L}^{-1}(\Sigma_\perp)\right] - 2\partial\Phi\partial^2(\nabla L)\left[(\nabla^2 L)^\dagger\Sigma_{\perp,\parallel}\right]. \tag{4.3.4}$$

1. For notational convenience, we drop dependency on x.

4.3.3 Main result

Now we are ready to present our main result. It's a direct combination of Theorem 4.7.9 and Lemma 4.3.5.

Theorem 4.3.6. *Suppose the loss function L, the manifold of local minimizer Γ and the open neighborhood U satisfy Assumptions 4.2.1 and 4.2.2, and $x_\eta(0) = x_{init} \in U$ for all $\eta > 0$. If SDE (4.3.5) has a global solution Y in U with $Y(0) = \Phi(x_{init})$ and Y never leaves U, i.e., $Y(t) \in U, \forall t \geq 0$, then for any $T > 0$, $x_\eta(\lfloor T/\eta^2 \rfloor)$ converges in distribution to $Y(T)$ as $\eta \to 0$.*

$$\begin{aligned} \mathrm{d}Y(t) = {} & \underbrace{\Sigma_{\parallel}^{1/2}(Y)\mathrm{d}W(t)}_{\text{Tangent Noise}} - \underbrace{\frac{1}{2}\nabla^2 L(Y)^\dagger \partial^2(\nabla L)(Y)\left[\Sigma_{\parallel}(Y)\right]\mathrm{d}t}_{\text{Tangent Noise Compensation}} \\ & - \frac{1}{2}\partial\Phi(Y)\Big(2\underbrace{\partial^2(\nabla L)(Y)\left[\nabla^2 L(Y)^\dagger \Sigma_{\perp,\parallel}(Y)\right]}_{\text{Mixed Regularization}} + \underbrace{\partial^2(\nabla L)(Y)\left[\mathcal{L}^{-1}_{\nabla^2 L}(\Sigma_\perp(Y))\right]}_{\text{Normal Regularization}}\Big)\mathrm{d}t, \end{aligned} \tag{4.3.5}$$

where $\Sigma \equiv \sigma\sigma^\top$ and $\Sigma_\parallel, \Sigma_\perp, \Sigma_{\perp,\parallel}$ are defined in Lemma 4.3.5.

Based on the above theorem, the limiting dynamics of SGD can be understood as follows: (a) the **Tangent Noise**, $\Sigma_{\parallel}^{1/2}(Y)\mathrm{d}W(t)$, is preserved, and the second term of (4.3.5) can be viewed as the necessary **Tangent Noise Compensation** for the limiting dynamics to stay on Γ. Indeed, Lemma 4.8.7 shows that the value of the second term only depends on Γ itself, i.e., it's same for all loss L which locally defines the same Γ. (b) The noise in the normal space is killed since the limiting dynamics always stay on Γ. However, its second order effect (Itô correction term) takes place as a vector field on Γ, which induces the **Noise Regularization** and **Mixed Regularization** term, corresponding to the mixed and normal noise covariance respectively.

Remark 4.3.7. In Section 4.7.4 we indeed prove a stronger version of Theorem 4.3.6 that the sample paths of SGD converge in distribution, i.e., let $X_\eta(t) = x_\eta(\lfloor t/\eta^2 \rfloor)$, then X_η weakly converges to Y on $[0, T]$. Moreover, we only assume the existence of a global solution for ease of presentation. As long as there exists a compact $K \subseteq \Gamma$ such that Y stays in K on $[0, T]$ with high probability, Theorem 4.7.9 still provides the convergence of SGD iterates (stopped at the boundary of K) before time T with high probability.

4.4 Implications and examples

In this section, we derive the limiting dynamics for two notable noise types, where we fix the expected loss L and the noise distribution, and only drive η to 0. The proofs are deferred into Section 4.8.3.

Type I: Isotropic Noise. Isotropic noise means $\Sigma(x) \equiv I_D$ for any $x \in \Gamma$ [SSJ20]. The following theorem shows that the limiting diffusion with isotropic noise can be viewed as a Brownian Motion plus Riemannian Gradient Flow with respect to the pseudo-determinant of $\nabla^2 L$.

Corollary 4.4.1 (Limiting Diffusion for Isotropic Noise)**.** *If $\Sigma \equiv I_D$ on Γ, SDE (4.3.5) is then*

$$\mathrm{d}Y(t) = \underbrace{\partial\Phi(Y)\mathrm{d}W - \frac{1}{2}\nabla^2 L(Y)^{\dagger}\partial^2(\nabla L)(Y)\left[\partial\Phi(Y)\right]\mathrm{d}t}_{\text{Brownian Motion on Manifold}} - \underbrace{\frac{1}{2}\partial\Phi(Y)\nabla(\ln|\nabla^2 L(Y)|_+)\mathrm{d}t}_{\text{Normal Regularization}} \tag{4.4.1}$$

where $|\nabla^2 L(Y)|_+ = \lim_{\alpha\to 0}\frac{|\nabla^2 L(Y)+\alpha I_D|}{\alpha^{D-\mathrm{rank}(\nabla^2 L(Y))}}$ *is the* pseudo-determinant *of* $\nabla^2 L(Y)$. $|\nabla^2 L(Y)|_+$ *is also equal to the sum of log of non-zero eigenvalue values of* $\nabla^2 L(Y)$.

Type II: Label Noise. When doing SGD for ℓ_2-regression on dataset $\{(z_i, y_i)\}_{i=1}^n$, adding label noise [BGVV20, DML21] means replacing the true label at iteration k, y_{i_k}, by a fresh noisy label $\tilde{y}_{i_k} := y_{i_k} + \delta_k$, where $\delta_k \overset{\text{i.i.d.}}{\sim} \mathrm{Unif}\{-\delta, \delta\}$ for some constant

$\delta > 0$. Then the corresponding loss becomes $\frac{1}{2}(f_{i_k}(x) - \tilde{y}_{i_k})^2$, where $f_{i_k}(x)$ is the output of the model with parameter x on data z_{i_k}. So the label noise SGD update is

$$x_{k+1} = x_k - \frac{\eta}{2} \cdot \nabla_x \left(f_{i_k}(x_k) - y_{i_k} + \delta_k\right)^2 = x_k - \eta(f_{i_k}(x_k) - y_{i_k} + \delta_k)\nabla_x f_{i_k}(x_k). \tag{4.4.2}$$

Suppose the model can achieve the global minimum of the loss $L(x) := \frac{1}{2}\mathbb{E}[(f_i(x) - \tilde{y}_i)^2]$ at x_*, then the model must interpolate the whole dataset, i.e., $f_i(x_*) = y_i$ for all $i \in [n]$, and thus here the manifold Γ is a subset of $\{x \in \mathbb{R}^D \mid f_i(x) = y_i, \forall i \in [n]\}$. Here the key property of the label noise used in previous works is $\Sigma(x) = \frac{\delta^2}{n}\sum_{i=1}^n \nabla_x f_i(x) \nabla_x f_i(x)^\top = \delta^2 \nabla^2 L(x)$. Lately, [DML21] further generalizes the analysis to other losses, e.g., logistic loss and exponential loss, as long as they satisfy $\Sigma(x) = c\nabla^2 L(x)$ for some constant $c > 0$.

In sharp contrast to the delicate discrete-time analysis in [BGVV20] and [DML21], the following corollary recovers the same result but with much simpler analysis – taking derivatives is all you need. Under our framework, we no longer need to do Taylor expansion manually nor carefully control the infinitesimal variables of different orders together. It is also worth mentioning that our framework immediately gives a global analysis of $\Theta(\eta^{-2})$ steps for SGD, far beyond the local coupling analysis in previous works. In Section 4.5, we will see how such global analysis allows us to prove a concrete generalization upper bound in a non-convex problem, the overparametrized linear model [WGL+20, HWLM20].

Corollary 4.4.2 (Limiting Flow for Label Noise)**.** *If $\Sigma \equiv c\nabla^2 L$ on Γ for some constant $c > 0$, SDE (4.3.5) can be simplified into (4.4.3) where the regularization is from the noise in the normal space.*

$$\mathrm{d}Y(t) = -\frac{1}{4} \cdot \partial\Phi(Y(t))\nabla \operatorname{tr}[c\nabla^2 L(Y(t))]\mathrm{d}t. \tag{4.4.3}$$

Example: k-Phase Motor. We also give an example with rigorous proof where the implicit bias induced by noise in the normal space cannot be characterized by a fixed regularizer, which was first discovered by [DML21] but was only verified via experiments.

Note the normal regularization in both cases of label noise and isotropic noise induces Riemmanian gradient flow against some regularizer, and it's natural to wonder if the limiting flow induced by the normal noise can always be characterized by certain regularizer. Interestingly, [DML21] answers this question negatively via experiments in their Section E.2. We adapt their example into the following one, and rigorously prove the limiting flow moves around a cycle at a constant speed and never stops using our framework.

Suppose dimension $D \geq 5$ and we let $k = D - 2$. For each $x \in \mathbb{R}^D$, we decompose $x = \binom{x_{1:2}}{x_{3:D}}$ where $x_{1:2} \in \mathbb{R}^2$ and $x_{3:D} \in \mathbb{R}^{D-2}$. Let $Q_\theta \in \mathbb{R}^{2\times 2}$ be the rotation matrix of angle θ, i.e., $Q_\theta = \begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix}$ and the loss $L(x) := \frac{1}{8}(\|x_{1:2}\|_2^2 - 1)^2 + \frac{1}{2}\sum_{j=3}^{D}(2 + \langle Q_\alpha^{j-3} v, x_{1:2}\rangle x_j^2$, where $\alpha = \frac{2\pi}{D-2}$ and v is any vector in $\mathbb{R}^2$ with unit norm. Here the manifold is given by $\Gamma := \{x \mid L(x) = 0\} = \{x \in \mathbb{R}^D \mid x_1^2 + x_2^2 = 1, x_j = 0, \forall j = 3, \ldots, D\}$.

The basic idea is that we can add noise in the 'auxiliary dimensions' for $j = 3, \ldots, D$ to get the regularization force on the circle $\{x_1^2 + x_2^2 = 1\}$, and the goal is to make the vector field induced by the normal regularization always point to the same direction, say anti-clockwise. However, this cannot be done with a single auxiliary dimension because from the analysis for label noise, we know when $\mathcal{L}^{-1}_{\nabla^2 L}(\Sigma_\perp)$ is identity, the *normal regularization* term in Equation (4.3.5) has 0 path integral along the unit circle and thus it must have both directions. The key observation here is that we can align the magnitude of noise with the strength of the regularization to make the path integral positive. By using $k \geq 3$ auxiliary dimensions, we can further ensure the normal regularization force is anti-clockwise and of constant magnitude, which is reminiscent of how a three-phase induction motor works. The result is summarized in the following Lemma 4.4.3. We provide its proof sketch here, and the complete proof is deferred to Section 4.8.4.

Lemma 4.4.3. *Let $\Sigma \in \mathbb{R}^{D\times D}$ be given by $\Sigma_{ij}(x) = (1 + \langle Q_\alpha^{j-3} v, Q_{-\pi/2} x_{1:2}\rangle)(2 + \langle Q_\alpha^{j-3} v, x_{1:2}\rangle)$, if $i = j \geq 3$ or 0 otherwise, then the solution of SDE (4.3.5) is the following (4.4.4) , which implies that $Y(t)$ moves anti-clockwise with a constant angular speed of $(D-2)/2$.*

$$Y_{1:2}(t) = Q_{t(D-2)/2} Y_{1:2}(0) \quad \textit{and} \quad Y_{3:D}(t) \equiv 0. \tag{4.4.4}$$

Proof sketch of Lemma 4.4.3. Note that for any $x \in \Gamma$, it holds that

$$\left(\nabla^2 L(x)\right)_{ij} = \begin{cases} 2 + \langle Q_\alpha^{j-3} v, x_{1:2}\rangle & \text{if } i = j \geq 3, \\ x_i x_j & \text{if } i, j \in \{1, 2\}, \\ 0 & \text{otherwise.} \end{cases}$$

Then clearly Σ only brings about noise in the normal space, and specifically, it holds that $\mathcal{L}^{-1}_{\nabla^2 L(x)}(\Sigma(x)) = \mathrm{diag}(0, 0, 1 + \langle Q_\alpha^0 v, Q_{-\pi/2} x_{1:2}\rangle, \ldots, 1 + \langle Q_\alpha^{D-3} v, Q_{-\pi/2} x_{1:2}\rangle)$. Further note that, by the special structure of the hessian in (4.8.8) and Lemma 4.8.3, for any $x \in \Gamma$, we have $\partial\Phi(x) = (x_2, -x_1, 0, \ldots, 0)^\top (x_2, -x_1, 0, \ldots, 0) = \binom{Q_{-\pi/2} x_{1:2}}{0}\binom{Q_{-\pi/2} x_{1:2}}{0}^\top$. Combining these facts, the dynamics of the first two coordinates in SDE (4.3.5) can be simplified into

$$\frac{\mathrm{d}x_{1:2}(t)}{\mathrm{d}t} = -\left(\frac{1}{2}\partial\Phi(x(t))\partial^2(\nabla L)(x(t))[\mathcal{L}^{-1}_{\nabla^2 L}(\Sigma(x(t))]\right)_{1:2} = \frac{D-2}{2} Q_{\pi/2} x_{1:2}.$$

On the other hand, we have $\frac{\mathrm{d}x_{3:D}(t)}{\mathrm{d}t} = 0$ as $\partial\Phi$ kills the movement on that component.

The proof is completed by noting that the solution of $x_{1:2}$ is

$$x_{1:2}(t) = \exp\left(t \cdot \frac{D-2}{2} Q_{\pi/2}\right) x_{1:2}(0) = Q_{\frac{t(D-2)}{2}} x_{1:2}(0).$$

■

4.5 Provable generalization benefit with label noise

In this section, we show provable generalization benefit of label noise using our framework (Theorem 4.7.7) in a concrete setting, the *overparametrized linear models* (OLM) [WGL+20]. While the existing implicit regularization results for Gradient Flow often relates the generalization quality to initialization, e.g., [WGL+20] shows that for OLM, small initialization corresponds to the *rich* regime and prefers solutions with small ℓ_1 norm while large initialization corresponds to the *kernel* regime and prefers solutions with small ℓ_2 norm, our result Theorem 4.5.1 surprisingly proves that even if an OLM is initialized in the kernel regime, label noise SGD can still help it escape and then enter the rich regime by minimizing its weighted ℓ_1 norm. When the groundtruth is κ-sparse, this provides a $\widetilde{O}(\kappa \ln d)$ vs $\Omega(d)$ sample complexity separation between SGD with label noise and GD when both initialized in the kernel regime. Here d is the dimension of the groundtruth. The lower bound for GD in the kernel regime is folklore, but for completeness, we state the result as Theorem 4.5.9 in Section 4.5.3 and append its proof in Section 4.9.6.

Setting of OLM: Let $\{(z_i, y_i)\}_{i\in[n]}$ be the training dataset where $z_1, \ldots, z_n \overset{\text{i.i.d.}}{\sim}$ $\mathrm{Unif}(\{\pm 1\}^d)$ or $\mathcal{N}(0, I_d)$ and each $y_i = \langle z_i, w^*\rangle$ for some unknown $w^* \in \mathbb{R}^d$. We assume that w^* is κ-sparse for some $\kappa < d$. Denote $x = \binom{u}{v} \in \mathbb{R}^D = \mathbb{R}^{2d}$, and we will use x and (u, v) exchangeably as the parameter of functions defined on $\mathbb{R}^D$ in the sequel. For each $i \in [n]$, define $f_i(x) = f_i(u, v) = z_i^\top (u^{\odot 2} - v^{\odot 2})$. Then we fit

$\{(z_i, y_i)\}_{i\in[n]}$ with an overparametrized model through the following loss function:

$$L(x) = L(u,v) = \frac{1}{n}\sum_{i=1}^{n}\ell_i(u,v), \quad \text{where } \ell_i(u,v) = \frac{1}{2}(f_i(u,v) - y_i)^2. \tag{4.5.1}$$

For simplicity, we define $Z = (z_1, \ldots, z_n)^\top \in \mathbb{R}^{n\times d}$, then $Zw^* = (y_1, \ldots, y_n)^\top$. Consider the following manifold Γ and its open neighborhood $U \supset \Gamma$:

$$U = (\mathbb{R}\setminus\{0\})^D \quad \text{and} \quad \Gamma = \left\{x = (u^\top, v^\top)^\top \in U \mid Z(u^{\odot 2} - v^{\odot 2}) = Zw^*\right\}. \tag{4.5.2}$$

Theorem 4.5.1. *In the setting of OLM, suppose the groundtruth is κ-sparse and $n \geq \Omega(\kappa \ln d)$ training data are sampled from either i.i.d. Gaussian or Boolean distribution. Then for almost all initialization x_{init} (*i.e.*, for all initialization in U) and any $\varepsilon > 0$, there exist $\eta_0, T > 0$ such that for any $\eta < \eta_0$, OLM trained with label noise SGD (4.4.2) with LR equal to η for $\lfloor T/\eta^2 \rfloor$ steps returns an ε-optimal solution, with probability of $1 - e^{-\Omega(n)}$ over the randomness of the training dataset.*

Our setting is more general than that of [HWLM20], which assumes $w^* \in \{0,1\}^d$ and their reparametrization can only express positive linear functions, i.e., $w = u^{\odot 2}$. Their $\widetilde{O}(\kappa^2)$ rate is achieved with a delicate three-phase LR schedule, while our $O(\kappa \ln d)$ rate only uses a constant LR.

Next, we provide the proof sketch of Theorem 4.5.1. The full proofs of Theorem 4.5.1 and other lemmas in this section are deferred into Section 4.9.

Proof sketch of Theorem 4.5.1. First, we verify Assumption 4.2.1 and Assumption 4.2.2. This requires showing that the set of local minimizers, Γ, is indeed a manifold, the hessian $\nabla^2 L(x)$ is non-degenerate on Γ, and that $\Phi(U) \subset \Gamma$. These are summarized in Lemma 4.5.2 and Lemma 4.5.3.

Lemma 4.5.2. *Consider the loss L defined in (4.5.1) and manifold Γ defined in (4.5.2). If the data Z is full rank, i.e., $\operatorname{rank}(Z) = n$, then it holds that*

- Γ *is a smooth manifold of dimension* $D-n$*;*

- $\operatorname{rank}(\nabla^2 L(x)) = n$ *for all* $x \in \Gamma$*.*

In particular, $\operatorname{rank}(Z) = n$ *holds with probability 1 for the Gaussian case where* $z_1, \dots, z_n \overset{\text{i.i.d.}}{\sim} \mathcal{N}(0, I_d)$*, and with probability* $1-c^d$ *for the boolean case where* $z_1, \dots, z_n \overset{\text{i.i.d.}}{\sim} \operatorname{Unif}(\{\pm 1\}^d)$ *for some constant* $c \in (0,1)$*.*

Lemma 4.5.3. *Consider the loss function* L *defined in (4.5.1), manifold* Γ *and its open neighborhood defined in (4.5.2). For any initialization* $x_0 \in U$*,* $\Phi(x_0) \in \Gamma$*.*

Then, by the result from the previous section, the implicit regularizer on the manifold is $R(x) = \operatorname{tr}(\Sigma(x)) = \operatorname{tr}(\delta^2 \nabla^2 L(x))$. It is straightforward to verify that

$$\nabla^2 L(x) = \frac{4}{n} \sum_{i=1}^{n} \begin{pmatrix} z_i \odot u \\ -z_i \odot v \end{pmatrix} \begin{pmatrix} z_i \odot u \\ -z_i \odot v \end{pmatrix}^\top, \ \forall x = \begin{pmatrix} u \\ v \end{pmatrix} \in \Gamma.$$

Therefore,

$$R(x) = \operatorname{tr}(\nabla^2 L(x)) = \frac{4\delta^2}{n} \sum_{j=1}^{D} \left(\sum_{i=1}^{n} z_{i,j}^2 \right) (u_j^2 + v_j^2), \ \forall x = \begin{pmatrix} u \\ v \end{pmatrix} \in \Gamma. \tag{4.5.3}$$

Recalling Corollary 4.4.2, the limiting behavior of label noise SGD is described by the Riemannian gradient flow on Γ with respect to the regularizer $R(x)$:

$$\mathrm{d}Y(t) = -\frac{1}{4} \partial \Phi(Y(t)) \nabla R(Y(t)) \mathrm{d}t, \ \text{with } Y(0) = \Phi(x_{\text{init}}) \in \Gamma. \tag{4.5.4}$$

More specifically, for every fixed $T > 0$, we have $x(\lfloor T/\eta^2 \rfloor)$ converges in distribution to $Y(T)$ for sufficiently small η by Theorem 4.3.6.

Furthermore, we show that the limiting flow (4.5.4) converges to some global minimizer of the regularizer $R(x)$ under the constraint that $Z(u^{\odot 2} - v^{\odot 2}) = Zw^*$ by Lemma 4.5.5. Moreover, we prove that any global minimizer $x^* = \binom{u^*}{v^*}$ of $R(x)$

recovers the groundtruth w^*, in the sense that $w^* = (u^*)^{\odot 2} - (v^*)^{\odot 2}$. This result is presented in Lemma 4.5.8.

Hence, combining the above, let T be large enough such that $Y(T) \approx x^*$, and it follows that $x(\lfloor T/\eta^2 \rfloor) \approx x^*$ for sufficiently small η. ■

The above proof sketch provides a general picture of our argument, and in the rest of this section, we discuss in detail the main ingredients of the proof.

Remark 4.5.4. In previous works [WGL+20, AMN+21], the convergence of gradient flow is only assumed but not proved. Recently [PPVF21] rigorously proved it for a specific balanced initialization $\binom{u}{v}$ satisfying that $u_j = v_j = \alpha, \forall j \in [n]$, for some $\alpha > 0$. Our Lemma 4.5.3 provides a complete convergence guarantee for this setting.

4.5.1 Limiting flow converges to minimizers of regularizer

In this subsection, we show that the limiting flow (4.5.4) starting from anywhere on Γ converges to the minimizer of regularizer R. The result is given in the following lemma.

Lemma 4.5.5. *Let $Y(t) \in \mathbb{R}^D$ be the unique solution of Equation (4.5.4) with any given initialization $Y(0) \in \Gamma$. Then $Y(\infty) = \lim_{t\to\infty} Y(t)$ exists. Moreover, $Y(\infty) = x^* = \binom{u^*}{v^*}$ is the optimal solution to the following optimization problem:*

$$\begin{aligned} \text{minimize} \quad & R(u,v) = \frac{4}{n}\sum_{j=1}^{d}\left(\sum_{i=1}^{n} z_{i,j}^2\right)(u_j^2 + v_j^2), \\ \text{subject to} \quad & Z(u^{\odot 2} - v^{\odot 2}) = Zw^*. \end{aligned} \tag{4.5.5}$$

The proof of Lemma 4.5.5 contains two parts: (a) showing that the limiting flow converges; (b) showing that the convergence point of the flow cannot be a sub-optimal stationary point. These are indeed the most technical and difficult parts of proving the $O(\kappa \ln d)$ upper bound, where the difficulty comes from the fact that the manifold

Γ is not compact, and the stationary points of the limiting flow are in fact all located on the boundary of Γ. However, the limiting flow itself is not even defined on the boundary of the manifold Γ. Even if we can extend $\partial\Phi(\cdot)\nabla R(\cdot)$ continuously to entire $\mathbb{R}^D$, the continuous extension is not everywhere differentiable.

Thus the non-compactness of Γ brings challenges for both (a) and (b). For (a), the convergence for standard gradient flow is often for free, as long as the trajectory is bounded and the objective is semialgebraic or analytic. The latter ensures the so-called Kurdyka-Łojasiewicz (KL) inequality [Loj63], which implies finite trajectory length and thus the convergence. However, since our flow does not satisfy those nice properties, we have to show that the limiting flow satisfies Polyak-Łojasiewicz condition (a special case of KL condition) [Pol64] via careful calculation (by Lemma 4.9.14).

For (b), the standard analysis based on center stable manifold theorem shows that gradient descent/flow converges to strict saddle (stationary point where the hessian has least one negative eigenvalue) only for a zero-measure set of initialization [LSJR16, LPP+17]. However, such analyses cannot deal with the case where the flow is not differentiable at the sub-optimal stationary point. To circumvent this issue, we prove the non-convergence to sub-optimal stationary points with a novel approach: we show that for any stationary point x, whenever there exists a descent direction of the regularizer R at x, we can construct a potential function which increases monotonically along the flow around x, while the potential function is equal to $-\infty$ at x, leading to a contradiction.

4.5.2 Minimizer of the regularizer recovers the sparse groundtruth

It then remains to show that the minimizer of the regularizer recovers the sparse groundtruth. Note that $\frac{1}{n}\sum_{i=1}^n z_{i,j}^2 = 1$ when $z_{i,j} \overset{\text{i.i.d.}}{\sim} \text{Unif}\{-1, 1\}$, and in the Gaussian case we also have $\frac{1}{n}\sum_{i=1}^n z_{i,j}^2 \in (\frac{1}{2}, \frac{3}{2})$ with high probability. Based on this, we can show that minimizing $R(x)$ on Γ, (4.5.5), is equivalent to finding the minimum ℓ_1

norm solution of the underlying d-dimensional regression problem. More specifically, we consider the following:

$$\begin{aligned} \text{minimize} \quad & R(w) = \frac{4}{n} \sum_{j=1}^{d} \left(\sum_{i=1}^{n} z_{i,j}^2 \right) |w_j|, \\ \text{subject to} \quad & Zw = Zw^*. \end{aligned} \tag{4.5.6}$$

Here we slightly abuse the notation of R and the parameter dimension will be clear from the context. We can relate the optimal solution of (4.5.5) to that of (4.5.6) via a canonical parametrization defined as follows.

Definition 4.5.6 (Canonical Parametrization)**.** For any $w \in \mathbb{R}^d$, we define the *canonical parametrization* of w as

$$\psi(w) = \begin{pmatrix} [w]_+^{\odot 1/2} \\ [-w]_+^{\odot 1/2} \end{pmatrix}.$$

Clearly, if we let $\binom{u}{v} = \psi(w)$, then it holds that $u^{\odot 2} - v^{\odot 2} = w$.

Indeed, we can show that if (4.5.6) has a unique optimal solution, then the optimal solution to (4.5.5) is also unique up to sign flips of each coordinate, as summarized in the following lemma.

Lemma 4.5.7. *Suppose the optimal solution to* (4.5.6) *is unique and equal to* w^**. Then the optimal solution to* (4.5.5) *is also unique up to sign flips of each coordinate. In particular, one of them is given by* $\psi(w^*)$*, that is, the canonical parametrization of* w^**.*

Therefore, it suffices to show that the optimization problem (4.5.6) has a unique solution w^*. For the boolean case, standard results in sparse recovery imply that the minimum ℓ_1 norm solution recovers the sparse groundtruth. The gaussian case is more complicated but still can be proved with techniques from [Tro15]. Combining the above gives rise to the following recovery guarantee.

Lemma 4.5.8. *Let $z_1, \dots, z_n \overset{i.i.d.}{\sim} \mathrm{Unif}(\{\pm 1\}^d)$ or $\mathcal{N}(0, I_d)$. There exist some constants $C, c > 0$ such that if $n \geq C\kappa \ln d$, then with probability at least $1 - e^{-cn}$, the optimal solution to (4.5.5), denoted by $\binom{\hat{u}}{\hat{v}}$, is unique up to sign flips of each coordinate and recovers the groundtruth, i.e., $\hat{u}^{\odot 2} - \hat{v}^{\odot 2} = w^*$.*

4.5.3 Lower bound for gradient descent in the kernel regime

Finally, to exhibit the generalization separation between label noise SGD and GD, we show that GD needs at least $\Omega(d)$ samples to learn OLM, when initialized in the kernel regime. This lower bound holds for all learning rate schedules and numbers of steps. This is in sharp contrast to the $\widetilde{O}(\kappa \ln d)$ sample complexity upper bound of SGD with label noise. Following the setting of kernel regime in [WGL+20], we consider the limit of $u_0 = v_0 = \alpha \mathbb{1}$, with $\alpha \to \infty$. It holds that $f_i(u_0, v_0) = 0$ and $\nabla f_i(u_0, v_0) = [\alpha z_i, -\alpha z_i]$ for each $i \in [n]$. Standard convergence analysis for NTK (Neural Tangent Kernel, [JGH18]) shows that upon convergence, the distance traveled by parameter converges to 0, and thus the learned model shall converge in function space, so is the generalization performance. For ease of illustration, we directly consider the lower bound for test loss when the NTK is fixed throughout the training.

Theorem 4.5.9. *Assume $z_1, \dots, z_n \overset{i.i.d.}{\sim} \mathcal{N}(0, I_d)$ and $y_i = z_i^\top w^*$, for all $i \in [n]$. Define the loss with linearized model as $L(x) = \sum_{i=1}^n (f_i(x_0) + \langle \nabla f_i(x_0), x - x_0 \rangle - y_i)^2$, where $x = \binom{u}{v}$ and $x_0 = \binom{u_0}{v_0} = \alpha \binom{\mathbb{1}}{\mathbb{1}}$. Then for any groundtruth w^*, any learning rate schedule $\{\eta_t\}_{t \geq 1}$, and any fixed number of steps T, the expected ℓ_2 loss of $x(T)$ is at least $(1 - \frac{n}{d}) \|w^*\|_2^2$, where $x(T)$ is the T-th iterate of GD on L, i.e., $x(t+1) = x(t) - \eta_t \nabla L(x(t))$, for all $t \geq 0$.*

4.6 Preliminaries on stochastic processes

Before we present the proofs in Section 4.7, 4.8 and 4.9, we first review a few basics of stochastic processes that will be useful for proving our results, so that our paper will be self-contained. We refer the reader to classics like [KS14, Bil13, Pol84] for more systematic derivations.

Throughout the rest of this section, let $\mathcal{E}$ be a Banach space equipped with norm $\|\cdot\|$, e.g., $(\mathbb{R}, |\cdot|)$ and $(\mathbb{R}^D, \|\cdot\|_2)$.

4.6.1 Càdlàg function and metric

Definition 4.6.1 (Càdlàg function)**.** Let $T \in [0, \infty]$. A function $g : [0, T) \to E$ is *càdlàg* if for all $t \in [0, T)$ it is right-continuous at t and its left limit $g(t-)$ exists. Let $\mathcal{D}_{\mathcal{E}}[0, T)$ be the set of all càdlàg function mapping $[0, T)$ into $\mathcal{E}$. We also use $\mathcal{C}_{\mathcal{E}}[0, T)$ to denote the set of all continuous function mapping $[0, T)$ into $\mathcal{E}$. By definition, $\mathcal{C}_{\mathcal{E}}[0, T) \subset \mathcal{D}_{\mathcal{E}}[0, T)$.

Definition 4.6.2 (Continuity modulus)**.** For any function $f : [0, \infty) \to \mathcal{E}$ and any interval $I \subseteq [0, \infty)$, we define

$$\omega(f; I) = \sup_{s,t \in I} \|f(s) - f(t)\|.$$

For any $N \in \mathbb{N}$ and $\theta > 0$, we further define the continuity modulus of continuous f as

$$\omega_N(f, \theta) = \sup_{0 \le t \le t+\theta \le N} \{\omega(f; [t, t+\theta])\}.$$

Moreover, the continuity modulus of càdlàg $f \in \mathcal{D}_{\mathcal{E}}[0, \infty)$ is defined as

$$\omega'_N(f, \theta) = \inf \left\{ \max_{i \le r} \omega(f; [t_{i-1}, t_i) : 0 \le t_0 < \cdots < t_r = N, \inf_{i<r} (t_i - t_{i-1}) \ge \theta \right\}.$$

Definition 4.6.3 (Jump)**.** For any $g \in \mathcal{D}_{\mathcal{E}}[0,T)$, we define the jump of g at t to be

$$\Delta g(t) = g(t) - g(t-).$$

For any $\delta > 0$, we define $h_\delta : [0,\infty) \to [0,\infty)$ by

$$h_\delta(r) = \begin{cases} 0 & \text{if } r \leq \delta \\ 1 - \delta/r & \text{if } r \geq \delta \end{cases}.$$

We then further define $J_\delta : \mathcal{D}_{\mathbb{R}^D}[0,\infty) \to \mathcal{D}_{\mathbb{R}^D}[0,\infty)$ [Kat91] as

$$J_\delta(g)(t) = \sum_{0<s\leq t} h_\delta(\|\Delta g(s)\|)\Delta g(s). \tag{4.6.1}$$

Definition 4.6.4 (Skorokhod metric on $\mathcal{D}_{\mathcal{E}}[0,\infty)$)**.** For each finite $T > 0$ and each pair of functions $f, g \in \mathcal{D}_{\mathcal{E}}[0,\infty)$, define $d_S(f,g;T)$ as the infimum of all those values of δ for which there exist grids $0 \leq t_0 < t_1 < \cdots < t_k$ and $0 < s_0 < s_1 < \cdots < \cdots < s_k$, with $t_k, s_k \geq T$, such that $|t_i - s_i| \leq \delta$ for $i = 0, \ldots, k$, and

$$\|f(t) - g(s)\| \leq \delta \qquad \text{if } (t,s) \in [t_i, t_{i+1}) \times [s_i, s_{i+1})$$

for $i = 0, \ldots, k-1$. The *Skorokhod metric* on $\mathcal{D}_{\mathcal{E}}[0,\infty)$ is defined to be

$$d_S(f,g) = \sum_{T=1}^{\infty} 2^{-T} \min\{1, d_S(f,g;T)\}.$$

4.6.2 Stochastic processes and stochastic integral

Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space and $\{\mathcal{F}_t\}_{t\geq 0}$ be a filtration of $\mathcal{F}$.

Definition 4.6.5 (Stochastic Process)**.** $Y := \{Y(t)\}_{t\in[0,\infty]}$ is said to be a *stochastic process* if for each $t \geq 0$, $Y(t)$ is a random variable, that is, $Y(t,\cdot) : \Omega \to \mathbb{R}$ is a measurable function. The definition extends to vector-valued stochastic process as well.

Definition 4.6.6 (Cross variation)**.** Let X and Y be two $\{\mathcal{F}_t\}_{t\geq 0}$-adapted stochastic processes such that X has sample paths in $\mathcal{D}_{\mathbb{R}^{D\times e}}[0,\infty)$ and Y has samples paths in $\mathcal{D}_{\mathbb{R}^e}[0,\infty)$, then the cross variation of X and Y on $(0,t]$, denoted by $[X,Y](t)$, is defined to be the limit of

$$\sum_{i=0}^{m-1}(X(t_{i+1})-X(t_i))(Y(t_{i+1})-Y(t_i))$$

in probability as the mesh size of $0=t_0<t_1<\cdots<t_m=t$ goes to 0, if it exists. Moreover, for Y itself, we write

$$[Y]=\sum_{i=1}^{e}[Y_i,Y_i]$$

Definition 4.6.7 (Martingale)**.** Let $\{X(t)\}_{t\geq 0}$ be a $\{\mathcal{F}_t\}_{t\geq 0}$-adapted stochastic process. If for all $0\leq s\leq t$, it holds that

$$\mathbb{E}[X(t)\mid\mathcal{F}_s]=X(s),$$

then X is called a martingale.

Definition 4.6.8 (Local martingale)**.** Let $\{X(t)\}_{t\geq 0}$ be a $\{\mathcal{F}_t\}_{t\geq 0}$-adapted stochastic process. If there exists a sequence of $\{\mathcal{F}_t\}_{t\geq 0}$-stopping time, $\{\tau_k\}_{k\geq 0}$, such that

- $\mathbb{P}[\tau_k<\tau_{k+1}]=1$, $\mathbb{P}[\lim_{k\to\infty}\tau_k=\infty]=1$,
- and $\{X^{\tau_k}(t)\}_{t\geq 0}$ is a $\{\mathcal{F}_t\}_{t\geq 0}$-adapted martingale,

then X is called a local martingale.

Definition 4.6.9 (Semimartingale)**.** Let $\{X(t)\}_{t\geq 0}$ be a $\{\mathcal{F}_t\}_{t\geq 0}$-adapted stochastic process. If there exists a local martingale $\{M(t)\}_{t\geq 0}$ and a càdlàg $\{\mathcal{F}_t\}_{t\geq 0}$-adapted process $\{A(t)\}_{t\geq 0}$ with bounded total variation that $X(t) = M(t) + A(t)$, then X is called a semimartingale.

Definition 4.6.10 (Itô's Stochastic Integral)**.** If $\{X(t)\}_{t\geq 0}$ and $\{Y(t)\}_{t\geq 0}$ are adapted stochastic processes, X has sample paths in $D_{\mathbb{R}^{d\times e}}[0,\infty)$, sample paths in $D_{\mathbb{R}^e}[0,\infty)$ and Y is a semimartingale, then the integral $\int_s^t XdY$ is defined, as the limit of $\sum_{i=0}^{n-1} X(r_i)(Y(r_{i+1}) - Y(r_i))$ where $s = r_0 < r_1 < \ldots < r_n = t$, the limit being in probability as the mesh size goes to 0. Standard results in stochastic calculus imply that this limit exists. We call X the *integrand* and Y the *integrator*.

Since all deterministic process are adapted, the above definition of integral also makes sense for deterministic functions and is a generalization of standard Riemman-Stieltjes Integral. The difference is that in the above Itô's Stochastic Integral we use the left-end value of the integrand but the existence of Riemman-Stieltjes Integral requires the limit exists for any point within the interval. When X and Y don't jump together, Riemman-Stieltjes Integral exists and coincides with the Itô's Integral.

Lemma 4.6.11 (Itô's Lemma)**.** *Let $\{X(t)\}_{t\geq 0}$ be defined through the following Itô drift-diffusion process:*

$$\mathrm{d}X(t) = \mu(t)\mathrm{d}t + \sigma(t)\mathrm{d}W(t).$$

where $\{W(t)\}_{t\geq 0}$ is the standard Brownian motion. Then for any twice differentiable function f, it holds that

$$\mathrm{d}f(t, X(t)) = \left(\frac{\partial f}{\partial t} + (\nabla_x f)^\top \mu_t + \frac{1}{2}\operatorname{tr}[\sigma^\top \nabla_x^2 f \sigma]\right)\mathrm{d}t + (\nabla_x f)^\top \sigma(t)\mathrm{d}W(t).$$

4.6.3 Weak convergence for stochastic processes

Let $(\mathcal{D}_{\mathcal{E}}[0,\infty), d)$ be a metric space with metric d and $d_{\mathbb{R}}$ be the standard euclidean metric over $\mathbb{R}$. We say a function $f : \mathcal{D}_{\mathcal{E}}[0,\infty) \to \mathbb{R}$ is *d-continuous* if and only if for all $X \in \mathcal{D}_{\mathcal{E}}[0,\infty)$ and $\varepsilon > 0$, there exists $\delta > 0$, such that for any $Y \in \mathcal{D}_{\mathcal{E}}[0,\infty)$, it holds that $d(X,Y) < \delta \implies |f(X) - f(Y)| < \epsilon$.

Definition 4.6.12 (Weak convergence of stochastic processes with càdlàg sample paths)**.** Let $\{X_n\}_{n\geq 0}$ be a sequence of stochastic processes on a sequence of probability spaces $\{(\Omega_n, \mathcal{F}_n, \mathbb{P}_n)\}_{n\geq 0}$ such that each X_n has sample paths in $\mathcal{D}_{\mathcal{E}}[0,\infty)$. Also, let X be a stochastic process on $(\Omega, \mathcal{F}, \mathbb{P})$ with sample paths on $\mathcal{D}_{\mathcal{E}}[0,\infty)$. Then $\{X_n\}_{n\geq 0}$ is said to *converge in distribution* or *weakly converge* to X under metric d(written as $X_n \Rightarrow X$) if and only if for all bounded and d-continuous function $f : \mathcal{D}_{\mathcal{E}}[0,\infty) \to \mathbb{R}$, it holds that

$$\lim_{n\to\infty} \mathbb{E}\left[f(X_n)\right] = \mathbb{E}\left[f(X)\right]. \tag{4.6.2}$$

Though we define weak convergence for a countable sequence of stochastic processes, but it is still valid if we index the stochastic processes by real numbers, e.g., $\{X_\eta\}_{\eta\geq 0}$, and consider the weak convergence of X_η as $\eta \to 0$. This is because the convergence in (4.6.2) is for a sequence of real numbers, which is also well-defined if we replace $\lim_{n\to\infty}$ by $\lim_{\eta\to 0}$.

By viewing each X_n as a random variable on $(\Omega_n, \mathcal{F}_n, \mathbb{P}_n)$ taking value in $D_{\mathcal{E}}[0,\infty)$, the above definition corresponds to the usual definition of weak convergence. However, to ensure the $f(X_n)$ is Borel measurable on $\mathbb{R}$, we typically need the measurability of each X_n with respect to Borel σ-algebra generated by open sets induced by the Skorokhod metric, which is nontrivial here. Fortunately, it turns out that Borel σ-algebra generated by open sets induced by the Skorokhod metric actually coincides with the projection σ-algebra, which refers to the σ-algebra generated by all projection maps $\pi_S : x \mapsto (x(t) : t \in S)$ where S is a finite subset of $[0,\infty)$. See Theorem 6 in Section 6 in [Pol84]. Hence, from now on we consider stochastic processes with càdlàg sample paths as random variables taking values in $\mathcal{D}_{\mathcal{E}}[0,\infty)$ equipped with the Skorokhod metric, and we can use tools regarding weak convergence for random variables taking values in general metric space.

The weak convergence has several equivalent formulations, and below we introduce a characterization based on the Prokhorov metric that will be useful in later proofs. For any random variable X, we denote by $\mathcal{L}(X)$ its law. For any two probability measures P and Q on a metric space, we denote by $\mathcal{C}(P,Q)$ the set of all couplings of random variables (X,Y) such that the marginal law of X is $\mathcal{L}(X) = P$ and the marginal law of Y is $\mathcal{L}(Y) = Q$.

Definition 4.6.13 (δ-Prokhorov distance)**.** Let $\delta > 0$. For any two probability measures P and Q on a metric space equipped with metric d, the δ-Prokhorov distance between P and Q is defined as

$$\rho_d^\delta(P,Q) := \inf\Big\{\epsilon > 0 \mid \exists (X,Y) \in \mathcal{C}(P,Q), \mathbb{P}[d(X,Y) \geq \epsilon] \leq \delta\Big\}.$$

Note this distance is not a metric because it does not satisfy triangle inequality.

Definition 4.6.14 (Prokhorov metric)**.** For any two probability measure P and Q on a metric space equipped with metric d, the Prokhorov metric between P and Q is defined as

$$\rho_d(P,Q) := \inf\Big\{\epsilon > 0 \mid \exists (X,Y) \in \mathcal{C}(P,Q), \mathbb{P}[d(X,Y) \geq \epsilon] \leq \varepsilon\Big\}.$$

With a slight abuse of notation, for any two random variables X and Y taking value in the same metric space, we write $\rho(X,Y) := \rho(\mathcal{L}(X), \mathcal{L}(Y))$. It can be shown that $X_n \Rightarrow X$ is equivalent to $\lim_{n\to\infty} \rho(X_n, X) = 0$.

Note that the weak convergence concerns only the distributions of $\{X_n\}_{n\geq 1}$, regardless of the probability space that each X_n lies in, and these probability spaces can be completely different. Fortunately, whenever $X_n \Rightarrow X$, we can always align the distributions of these random variables with some other random variables in a common probability space such that the latter admits a stronger almost sure convergence. This is known as the Skorokhod's representation theorem.

Theorem 4.6.15 (Skorokhod's Representation Theorem)**.** *Let $\{P_n\}_{n\geq 1}$ be a sequence of probability measures on a metric space S such that P_n weakly converges to some probability measure P as $n \to \infty$. Suppose that the support of P is separable. Then there exist S-valued random variables X_n defined a common probability space $(\Omega, \mathcal{F}, \mathbb{P})$ such that $\mathcal{L}(X) = P$ and $\mathcal{L}(X_n) = P_n$ for all $n \geq 1$ and such that $\lim_{n\to\infty} X_n(\omega) = X(\omega)$ for every $\omega \in \Omega$.*

The main convergence result in [Kat91] (Theorem 4.7.7) are in the sense of Skorokhod metric in Definition 4.6.4, which is harder to understand and use compared to the more common *uniform metric* (Definition 4.6.16). However, convergence in Skorokhod metric and uniform metric indeed coincide with each other when the limit is in $\mathcal{C}_{\mathbb{R}^D}[0, \infty)$, i.e., the continuous functions.

Definition 4.6.16 (Uniform metric on $\mathcal{D}_{\mathcal{E}}[0, \infty)$)**.** For each finite $T > 0$ and each pair of functions $f, g \in \mathcal{D}_{\mathcal{E}}[0, T)$, the *uniform metric* is defined to be

$$d_U(f, g; T) = \sup_{t\in[0,T)} \|f(t) - g(t)\|.$$

The *uniform metric* on $\mathcal{D}_{\mathcal{E}}[0, \infty)$ is defined to be

$$d_U(f, g) = \sum_{T=1}^{\infty} 2^{-T} \min\{1, d_U(f, g; T)\}.$$

Lemma 4.6.17 (Problem 7, Section 6, [Pol84])**.** *If $X_n \Rightarrow X$ under the Skorokhod metric d_S, and X has sample paths in $\mathcal{C}_{\mathbb{R}^D}[0, \infty)$, then $X_n \Rightarrow X$ under the uniform metric d_U.*

Remark 4.6.18. We shall note the uniform metric defined above is weaker than the metric $\sup_{t\in[0,\infty)} \|f(t) - g(t)\|$. Convergence in the uniform metric on $[0, \infty]$ defined in Definition 4.6.16 is equivalent to convergence in the uniform metric on each compact set $[0, T]$ for $T \in \mathbb{N}^+$. The same holds for the Skorokhod topology.

4.7 Proof for the limiting diffusion of SGD

In this section, we give a complete derivation of the limiting diffusion of SGD. Here we use $\Rightarrow$ to denote the convergence in distribution. For any $U \subseteq \mathbb{R}^D$, we denote by $\mathring{U}$ its interior. For a subspace S, we use $S^\perp$ to denote its orthogonal complement.

First, as mentioned in Assumption 4.2.2, we verify that the mapping Φ is $\overline{\mathcal{C}}^2$ in Lemma 4.7.1. In Section 4.7.1 we discuss how different time scalings could affect the coefficients in SDE (4.1.2) and (4.1.3). Then we check the necessary conditions for applying the results in [Kat91] in Section 4.7.2 and recap the corresponding theorem for the asymptotically continuous case in Section 4.7.3. Finally, we provide a user-friendly interface for Katzenberger's theorem in Section 4.7.4.

Theorem (Adapted version of Theorem 5.1 of [Fal83]). Let $f : V \to \mathbb{R}^D$ be an $\overline{\mathcal{C}}^k$ mapping where $V \subseteq \mathbb{R}^D$ is open and $k > 0$. Suppose that $\Gamma = \{x \in V \mid f(x) = x\}$ is a $\mathcal{C}^1$ submanifold of $\mathbb{R}^D$ of dimension $D - M$ and that for every $y \in \Gamma$, the matrix $\partial f(y)$ has $D - M$ eigenvalues in $\{z \in \mid |z| < 1\}$. Write f^n for the nth iterate of f. Then

$$V_f = \{x \in V \mid \lim_{n\to\infty} f^n(x) \text{ exists and is in } \Gamma\}.$$

is a neighborhood of Γ and $f^\infty = \lim_{n\to\infty} f^n$ is $\overline{\mathcal{C}}^k$ on V_f.

Lemma 4.7.1. *Under Assumption 4.2.2, Φ is $\overline{\mathcal{C}}^2$ on U.*

Proof of Lemma 4.7.1. Assumption 4.2.1 assumes that L is $\mathcal{C}^4$. So ∇L is $\mathcal{C}^3$ and thus $\overline{\mathcal{C}}^2$. It is well known the flow of $-\nabla L(\cdot)$, $\phi(\cdot, t)$, has the same degree of smoothness as $\nabla L(\cdot)$ with respect to its initialization for any $t \geq 0$ (see Remark 1 in Section 2.3 of [Per01] for a reference). Thus $\phi(\cdot, 1)$ is also $\overline{C}^2$. Let $f(\cdot) = \phi(\cdot, 1)$. By Assumption 4.2.2, we know that $U \subset V_f = \{x \in V \mid \lim_{n\to\infty} f^n(x) \text{ exists and is in } \Gamma\}$ and $\Phi(x) = \lim_{n\to\infty} \phi(x, n)$ on U. Applying the above adapted version of Theorem 5.1 of [Fal83], we get $\Phi(x) = \lim_{n\to\infty} \phi(x, n)$ is $\overline{C}^2$ on U. ∎

4.7.1 Approximating SGD by SDE

Let's first clarify how we derive the SDEs, (4.1.2) and (4.1.3), that approximate SGD (4.1.1) under different time scalings. Recall $W(t)$ is Ξ-dimensional Brownian motion and that $\sigma(X) : \mathbb{R}^D \to \mathbb{R}^{D\times\Xi}$ is a deterministic continuously differentiable noise function. As proposed by [LTW17], one approach to approximate SGD (4.1.1) by SDE is to consider the following SDE:

$$\mathrm{d}\bar{X}(t) = -\nabla L(\bar{X}(t))\mathrm{d}t + \sqrt{\eta}\sigma(\bar{X}(t))\mathrm{d}W(t),$$

where the time correspondence is $t = k\eta$, i.e., $\bar{X}(k\eta) \approx x_\eta(k)$.

Now rescale the above SDE by considering $\tilde{X}(t) = \bar{X}(t\eta)$, which then yields

$$\begin{aligned}\mathrm{d}\tilde{X}(t) = \mathrm{d}\bar{X}(t\eta) &= -\nabla L(\bar{X}(t\eta))\mathrm{d}(t\eta) + \sqrt{\eta}\sigma(\bar{X}(t\eta))\mathrm{d}W(t\eta)\\ &= -\eta\nabla L(\bar{X}(t\eta))\mathrm{d}t + \sqrt{\eta}\sigma(\bar{X}(t\eta))\mathrm{d}W(t\eta).\end{aligned}$$

Note that $\frac{1}{\sqrt{\eta}}W(t\eta)$ is also a Ξ-dimensional Brownian motion that has the same distribution of sample paths in $\mathcal{C}_{\mathbb{R}^d}[0,\infty)$ as $W(t)$ does. Thus with a slight abuse of notation we write $W(t)$ in place of $\frac{1}{\sqrt{\eta}}W(t\eta)$ and obtain

$$\begin{aligned}\mathrm{d}\tilde{X}(t) &= -\eta\nabla L(\bar{X}(t\eta))\mathrm{d}t + \eta\sigma(\bar{X}(t\eta))\mathrm{d}W(t)\\ &= -\eta\nabla L(\tilde{X}(t))\mathrm{d}t + \eta\sigma(\tilde{X}(t))\mathrm{d}W(t),\end{aligned}$$

where the time correspondence is $t = k$, i.e., $\tilde{X}(k) \approx x_\eta(k)$. The above SDE is exactly the same as (4.1.2).

Then, to accelerate the above SDE by η^{-2} times, let's define $X(t) = \tilde{X}(t/\eta^2)$. Then it follows that

$$\begin{aligned} \mathrm{d}X(t) = \mathrm{d}\tilde{X}(t/\eta^2) &= -\eta \nabla L(\tilde{X}(t/\eta^2))\mathrm{d}t/\eta^2 + \eta\sigma(\tilde{X}(t/\eta^2))\mathrm{d}W(t/\eta^2) \\ &= -\frac{1}{\eta}\nabla L(X(t))\mathrm{d}t + \sigma(X(t))\mathrm{d}\left(\eta W(t/\eta^2)\right) \end{aligned}$$

Again we note that $\eta W(t/\eta^2)$ is also a Ξ-dimensional Brownian motion. Here the time correspondence is $t = k\eta^2$, i.e., evolving for constant time with the above SDE approximates $\Omega(1/\eta^2)$ steps of SGD. In this way, we derive SDE (4.1.3) in the main context.

4.7.2 Necessary conditions

Below we collect the necessary conditions imposed on $\{Z_n\}_{n\geq 1}$ and $\{A_n\}_{n\geq 1}$ in [Kat91]. Recall that we consider the following stochastic process

$$X_n(t) = x_{\text{init}} + \int_0^t \sigma(X_n(s))\mathrm{d}Z_n(s) - \int_0^t \nabla L(X_n(s))\mathrm{d}A_n(s).$$

For any stopping time τ, the stopped process is defined as $X_n^\tau(t) = X_n(t \wedge \tau)$. For any compact $K \subset U$, we define the stopping time of X_n leaving K as $\lambda_n(K) = \inf\{t \geq 0 \mid X_n(t-) \notin \mathring{K} \text{ or } X_n(t) \notin \mathring{K}\}$.

Condition 4.7.2. The integrator sequence $\{A_n\}_{n\geq 1}$ is *asymptotically continuous*: $\sup_{t>0} |A_n(t) - A_n(t-)| \Rightarrow 0$ where $A_n(t-) = \lim_{s\to t-} A_n(s)$ is the left limit of A_n at t.

Condition 4.7.3. The integrator sequence $\{A_n\}_{n\geq 1}$ *increases infinitely fast*: $\forall \epsilon > 0$, $\inf_{t\geq 0}(A_n(t+\epsilon) - A_n(t)) \Rightarrow \infty$.

Condition 4.7.4 (Eq.(5.1), [Kat91])**.** For every $T > 0$, as $n \to \infty$, it holds that

$$\sup_{0<t\le T\wedge\lambda_n(K)} \|\Delta Z_n(t)\|_2 \Rightarrow 0.$$

Condition 4.7.5 (Condition 4.2, [Kat91])**.** For each $n \geq 1$, let Y_n be a $\{\mathcal{F}_t^n\}$-semimartingale with sample paths in $\mathcal{D}_{\mathbb{R}^D}[0,\infty)$. Assume that for some $\delta > 0$ (allowing $\delta = \infty$) and every $n \geq 1$ there exist stopping times $\{\tau_n^m \mid m \geq 1\}$ and a decomposition of $Y_n - J_\delta(Y_n)$ into a local martingale M_n plus a finite variation process F_n such that $\mathbb{P}[\tau_n^m \leq m] \leq 1/m$, $\{[M_n](t\wedge\tau_n^m)+T_{t\wedge\tau_n^m}(F_n)\}_{n\geq 1}$ is uniformly integrable for every $t \geq 0$ and $m \geq 1$, and

$$\lim_{\gamma\to 0}\limsup_{n\to\infty}\mathbb{P}\left[\sup_{0\le t\le T}(T_{t+\gamma}(F_n) - T_t(F_n)) > \epsilon\right] = 0,$$

for every $\epsilon > 0$ and $T > 0$, where $T_t(\cdot)$ denotes total variation on the interval $[0, t]$.

Lemma 4.7.6. *For SGD iterates defined using the notation in Lemma 4.3.2, the sequences $\{A_n\}_{n\geq1}$ and $\{Z_n\}_{n\geq1}$ satisfy Condition 4.7.2, 4.7.3, 4.7.4 and 4.7.5.*

Proof of Lemma 4.7.6. Condition 4.7.2 is obvious from the definition of $\{A_n\}_{n\geq1}$.

Next, for any $\epsilon > 0$ and $t \in [0, T]$, we have

$$A_n(t+\epsilon) - A_n(t) = \eta_n \cdot \left\lfloor \frac{t+\epsilon}{\eta_n^2}\right\rfloor - \eta_n \cdot \left\lfloor \frac{t}{\eta_n^2}\right\rfloor \geq \frac{t+\epsilon-\eta_n^2}{\eta_n} - \frac{t}{\eta_n} = \frac{\epsilon-\eta_n^2}{\eta_n},$$

which implies that $\inf_{0\le t\le T}(A_n(t+\epsilon) - A_n(t)) > \epsilon/(2\eta_n)$ for small enough η_n. Then taking $n \to \infty$ yields the Condition 4.7.3.

For Condition 4.7.4, note that

$$\Delta Z_n(t) = \begin{cases} \eta_n\sqrt{\Xi}(\mathbb{1}_{\xi_k} - \frac{1}{\Xi}\mathbb{1}) & \text{if } t = k\cdot\eta_n^2, \\ 0 & \text{otherwise.}\end{cases}$$

Therefore, we have $\|\Delta Z_n(t)\|_2 \leq 2\eta_n\sqrt{\Xi}$ for all $t > 0$. This implies that $\|\Delta Z_n(t)\|_2 \to 0$ uniformly over $t > 0$ as $n \to \infty$, which verifies Condition 4.7.4.

We proceed to verify Condition 4.7.5. By the definition of Z_n, we know that $\{Z_n(t)\}_{t\geq 0}$ is a jump process with independent increments and thus is a martingale. Therefore, by decomposing $Z_n = M_n + F_n$ with M_n being a local martingale and F_n a finite variation process, we must have $F_n = 0$ and M_n is Z_n itself. It then suffices to show that $[M_n](t \wedge \tau_n^m)$ is uniformly integrable for every $t \geq 0$ and $m \geq 1$. Since M_n is a pure jump process, we have

$$
\begin{aligned}
[M_n](t \wedge \tau_n^m) &= \sum_{0<s\leq t\wedge\tau_n^m} \|\Delta M_n(s)\|_2^2 \leq \sum_{0<s\leq t} \|\Delta M_n(s)\|_2^2 \\
&= \sum_{k=1}^{\lfloor t/\eta_n^2 \rfloor} \left\| \eta_n\sqrt{\Xi}\left(\mathbb{1}_{\xi_k} - \frac{1}{\Xi}\mathbb{1}\right)\right\|_2^2 \leq 4\Xi \sum_{k=1}^{\lfloor t/\eta_n^2 \rfloor} \eta_n^2 \leq 4\Xi t.
\end{aligned}
$$

This implies that $[M_\eta](t \wedge \tau_\eta^m)$ is universally bounded by $4t$, and thus $[M_\eta](t \wedge \tau_\eta^m)$ is uniformly integrable. This completes the proof. ■

Proof of Lemma 4.3.2. For any $n \geq 1$, it suffices to show that given $X_n(k\eta_n^2) = x_{\eta_n}(k)$, we further have $X_n((k+1)\eta_n^2) = x_{\eta_n}(k+1)$. By the definition of $X_n(t)$ and note that $A_n(t), Z_n(t)$ are constants on $[k\eta_n^2, (k+1)\eta_n^2)$, we have that $X_n(t) = X_n(k\eta_n^2)$ for all $t \in [k\eta_n^2, (k+1)\eta_n^2)$, and therefore

$$
\begin{aligned}
&X_n((k+1)\eta_n^2) - X_n(k\eta_n^2) \\
=& -\int_{k\eta_n^2}^{(k+1)\eta_n^2} \nabla L(X_n(t))\mathrm{d}A_n(t) + \int_{k\eta_n^2}^{(k+1)\eta_n^2} \sigma(X_n(t))\mathrm{d}Z_n(t) \\
=& -\nabla L(X_n(k\eta_n^2))(A_n((k+1)\eta_n^2) - A_n(k\eta_n^2)) + \sigma(X_n(k\eta_n^2))(Z_n((k+1)\eta_n^2) - Z_n(k\eta_n^2)) \\
=& -\eta_n\nabla L(X_n(k\eta_n^2)) + \eta_n\sqrt{\Xi}\sigma_{\xi_k}(X_n(k\eta_n^2)) \\
=& -\eta_n\nabla L(x_{\eta_n}(k)) + \eta_n\sqrt{\Xi}\sigma_{\xi_k}(x_{\eta_n}(k)) = x_{\eta_n}(k+1) - x_{\eta_n}(k)
\end{aligned}
$$

where the second equality is because $A_n(t)$ and $Z_n(t)$ are constant on interval $[k\eta_n^2, (k+$

$1)\eta_n^2)$. This confirms the alignment between $\{X_n(k\eta_n^2)\}_{k\geq 1}$ and $\{x_{\eta_n}(k)\}_{k\geq 1}$.

For the second claim, note that $\sigma(x)\mathbb{E}Z_n(t) \equiv 0$ for all $x \in \mathbb{R}^D, t \geq 0$ (since the noise has zero-expectation) and that $\{Z_n(t) - \mathbb{E}Z_n(t)\}_{t\geq 0}$ will converge in distribution to a Brownian motion by the classic functional central limit theorem (see, for example, Theorem 4.3.5 in [Whi02]). Thus, the limiting diffusion of X_n as $n \to \infty$ can be obtained by substituting Z with the standard Brownian motion W in (4.7.2). This completes the proof. ∎

4.7.3 Katzenberger's theorem in the asymptotically continuous case

The full Katzenberger's theorem deals with a more general case, which only requires the sequence of intergrators to be *asymptotically continuous*, thus including SDE (4.1.3) and SGD (4.1.1) with η goes to 0.

To describe the results in [Kat91], we first introduce some definitions. For each $n \geq 1$, let $(\Omega^n, \mathcal{F}^n, \{\mathcal{F}_t^n\}_{t\geq 0}, \mathbb{P})$ be a filtered probability space, Z_n an $\mathbb{R}^\Xi$-valued cadlag $\{\mathcal{F}_t^n\}$-semimartingale with $Z_n(0) = 0$ and A_n a real-valued cadlag $\{\mathcal{F}_t^n\}$-adapted nondecreasing process with $A_n(0) = 0$. Let $\sigma_n : U \to \mathbb{R}^{D\times\Xi}$ be continuous with $\sigma_n \to \sigma$ uniformly on compact subsets of U. Let X_n be an $\mathbb{R}^D$-valued càdlàg $\{\mathcal{F}_t^n\}$-semimartingale satisfying, for all compact $K \subset U$,

$$X_n(t) = x_{\text{init}} + \int_0^t \sigma(X_n)\mathrm{d}Z_n + \int_0^t -\nabla L(X_n)\mathrm{d}A_n \tag{4.7.1}$$

for all $t \leq \lambda_n(K)$ where $\lambda_n(K) = \inf\{t \geq 0 \mid X_n(t-) \notin \mathring{K} \text{ or } X_n(t) \notin \mathring{K}\}$ is the stopping time of X_n leaving K.

Theorem 4.7.7 (Theorem 6.3, [Kat91]). *Suppose $x_{init} \in U$, Assumptions 4.2.1 and 4.2.2, Condition 4.7.2, 4.7.3, 4.7.4 and 4.7.5 hold. For any compact $K \subset U$, define $\mu_n(K) = \inf\{t \geq 0 \mid Y_n(t-) \notin \mathring{K}$ or $Y_n(t) \notin \mathring{K}\}$, then the sequence*

$\{(Y_n^{\mu_n(K)}, Z_n^{\mu_n(K)}, \mu_n(K)\}$ *is relatively compact in* $\mathcal{D}_{\mathbb{R}^{D\times e}}[0,\infty)\times[0,\infty)$. *If* (Y,Z,μ) *is a limit point of this sequence under the Skorokhod metric (Definition 4.6.4), then* (Y,Z) *is a continuous semimartingale,* $Y(t)\in\Gamma$ *for every* $t\geq 0$ *a.s.,* $\mu\geq\inf\{t\geq 0\mid Y(t)\notin\mathring{K}\}$ *a.s. and* $Y(t)$ *admits*

$$Y(t) = Y(0) + \int_0^{t\wedge\mu} \partial\Phi(Y(s))\sigma(Y(s))\mathrm{d}Z(s) + \frac{1}{2}\sum_{i,j=1}^{D}\sum_{k,l=1}^{e}\int_0^{t\wedge\mu} \partial_{ij}\Phi(Y(s))\sigma(Y(s))_{ik}\sigma(Y(s))_{jl}\mathrm{d}[Z_k,Z_l](s). \tag{4.7.2}$$

We note that by Lemma 4.6.17, convergence in distribution under Skorokhod metric is equivalent to convergence in distribution under uniform metric Definition 4.6.16, therefore in the rest of the paper we will only use the uniform metric in the rest of the paper, e.g., whenever we mention Prokhorov metric and δ-Prokhorov distance, the underlying metric is the uniform metric.

4.7.4 A user-friendly interface for Katzenberger's theorem

Based on the Lemma 4.7.6, we can immediately apply Theorem 4.7.7 to obtain the following limiting diffusion of SGD.

Theorem 4.7.8. *Let the manifold* Γ *and its open neighborhood* U *satisfy Assumptions 4.2.1 and 4.2.2. Let* $K\subset U$ *be any compact set and fix some* $x_{init}\in K$. *Consider the SGD formulated in Lemma 4.3.2 where* $X_{\eta_n}(0)\equiv x_{init}$. *Define*

$$Y_{\eta_n}(t) = X_{\eta_n}(t) - \phi(X_{\eta_n}(0), A_{\eta_n}(t)) + \Phi(X_{\eta_n}(0))$$

and $\mu_{\eta_n}(K) = \inf\{t\geq 0\mid Y_{\eta_n}(t)\notin\mathring{K}\}$. *Then the sequence* $\{(Y_{\eta_n}^{\mu_{\eta_n}(K)}, Z_{\eta_n}, \mu_{\eta_n}(K))\}_{n\geq 1}$ *is relatively compact in* $\mathcal{D}_{\mathbb{R}^D\times\mathbb{R}^n}[0,\infty)\times[0,\infty]$. *Moreover, if* (Y,Z,μ) *is a limit point of this sequence, it holds that* $Y(t)\in\Gamma$ *a.s for all* $t\geq 0$, $\mu\geq\inf\{t\geq 0\mid Y(t)\notin\mathring{K}\}$

and $Y(t)$ admits

$$Y(t) = Y(0) + \int_{s=0}^{t\wedge\mu} \partial\Phi(Y(s))\sigma(Y(s))\mathrm{d}W(s) \\ + \int_{s=0}^{t\wedge\mu} \frac{1}{2}\sum_{i,j=1}^{D} \partial_{ij}\Phi(Y(s))(\sigma(Y(s))\sigma(Y(s))^\top)_{ij}\mathrm{d}s \tag{4.7.3}$$

where $\{W(s)\}_{s\geq 0}$ is the standard Brownian motion and $\sigma(\cdot)$ is as defined in Lemma 4.3.2.

However, the above theorem is hard to parse and cannot be directly applied if we want to further study the implicit bias of SGD through this limiting diffusion. Therefore, we develop a user-friendly interface to it in below. In particular, Theorem 4.3.6 is the a special case of Theorem 4.7.9. In Theorem 4.3.6, we replace $\partial\Phi(Y(t))\sigma(Y(t))$ with $\Sigma_{\parallel}^{\frac{1}{2}}(Y(t))$ to simplify the equation, since $\partial\Phi(Y(t))\sigma(Y(t))\left(\partial\Phi(Y(t))\sigma(Y(t))\right)^\top = \Sigma_{\parallel}(Y(t))$ and thus this change doesn't affect the distribution of the sample paths of the solution.

Theorem 4.7.9. *Under the same setting as Theorem 4.7.8, we change the integer index back to $\eta > 0$ with a slight abuse of notation. Let $\{Y(t)\}_{t\geq 0}$ be a stochastic process with continuous sample paths, K be a compact set and time $T > 0$. Define $\mu(K) = \inf\{t \geq 0 \mid Y(t) \notin \mathring{K}\}$ and $\delta = \mathbb{P}(\mu(K) \leq T)$. Suppose there is a stochastic process $\{Y(t)\}_{t\geq 0}$ with continuous sample paths such that $Y(0) = \Phi(x_{init})$ and that $(Y, \mu(K))$ satisfy*

$$Y(t) = Y(0) + \int_{s=0}^{t\wedge\mu(K)} \partial\Phi(Y(s))\sigma(Y(s))\mathrm{d}W(s) \\ + \int_{s=0}^{t\wedge\mu(K)} \frac{1}{2}\sum_{i,j=1}^{D} \partial_{ij}\Phi(Y(s))(\sigma(Y(s))\sigma(Y(s))^\top)_{ij}\mathrm{d}s.$$

Then for any $\epsilon > 0$, it holds for all sufficiently small LR η that:

$$\rho_{d_U}^{2\delta}(Y_\eta^{\mu_\eta(K)\wedge T}, Y^{\mu(K)\wedge T}) \leq \epsilon, \tag{4.7.4}$$

which means there is a coupling between the distribution of the stopped processes $Y_\eta^{\mu_\eta(K)\wedge T}$ and $Y^{\mu(K)\wedge T}$, such that the uniform metric between them is smaller than ε with probability at least $1-2\delta$. In other words, $\lim_{\eta\to 0}\rho_{d_U}^{2\delta}(Y_\eta^{\mu_\eta(K)\wedge T}, Y^{\mu(K)\wedge T}) = 0$.

Moreover, when $\{Y(t)\}_{t\geq 0}$ is a global solution to the following limiting diffusion

$$Y(t) = \int_{s=0}^{t} \partial\Phi(Y(s))\sigma(Y(s))\mathrm{d}W(s) + \int_{s=0}^{t} \frac{1}{2}\sum_{i,j=1}^{D} \partial_{ij}\Phi(Y(s))(\sigma(Y(s))\sigma(Y(s))^\top)_{ij}\mathrm{d}s$$

and Y never leaves U, i.e. $\mathbb{P}[\forall t\geq 0, Y(t)\in U] = 1$, it holds that Y_η^T converges in distribution to Y^T as $\eta\to 0$ for any fixed $T>0$.

For clarity, we break the proof of Theorem 4.7.9 into two parts, devoted to the two claims respectively.

Proof of the first claim of Theorem 4.7.9. First, for any sequence of learning rates $\{\eta_m\}_{m\geq 1}$ such that $\eta_m\to 0$, Theorem 4.7.8 guarantees there exists a stopping time $\tilde{\mu}$ and a stochastic process $\{\widetilde{Y}(t)\}_{t\geq 0}$ such that

1. $(\widetilde{Y}, \tilde{\mu})$ satisfies Equation (4.7.3);

2. $\widetilde{Y}\in\Gamma$ *a.s.*;

3. $\tilde{\mu}\geq\tilde{\mu}(K) := \inf\{t\geq 0 \mid \widetilde{Y}(t)\notin \mathring{K}\}$.

The above conditions imply that $\widetilde{Y}^{\tilde{\mu}(K)}\in\Gamma$ almost surely.

Now, let $\{\eta'_m\}_{m\geq 1}$ be a subsequence of $\{\eta_m\}_{m\geq 1}$ such that $Y_{\eta'_m}^{\mu_{\eta'_m}(K)} \Rightarrow \widetilde{Y}^{\tilde{\mu}}$ as $m\to\infty$. Note that weak convergence for probability measures on $(\mathcal{D}_{\mathbb{R}^D}[0,\infty), d)$ is equivalent to convergence in Prokhorov metric. Thus for any $\epsilon>0$, there exists some $N>0$ such that for all $m>N$,

$$\mathbb{P}\left[d_U(Y_{\eta'_m}^{\mu_{\eta'_m}(K)\wedge T}, \widetilde{Y}^{\tilde{\mu}\wedge T})\geq\epsilon\right]\leq\epsilon.$$

Since $d_U(Y_{\eta'_m}^{\mu_{\eta'_m}(K)\wedge T}, \widetilde{Y}^{\tilde{\mu}\wedge T}) \leq d_U(Y_{\eta'_m}^{\mu_{\eta'_m}(K)}, \widetilde{Y}^{\tilde{\mu}})$, we have that for any $\epsilon < \delta$,

$$\mathbb{P}\left[d_U(Y_{\eta'_m}^{\mu_{\eta'_m}(K)\wedge T}, \widetilde{Y}^{\tilde{\mu}\wedge T}) \geq \epsilon\right] \leq \mathbb{P}\left[d_U(Y_{\eta'_m}^{\mu_{\eta'_m}(K)}, \widetilde{Y}^{\tilde{\mu}}) \geq \epsilon\right] \leq \epsilon \leq \delta.$$

Let $\mathcal{E}_T$ be the event such that $\tilde{\mu}(K) > T$ on $\mathcal{E}_T$. Then restricted on $\mathcal{E}_T$, we have $\widetilde{Y}(T \wedge \tilde{\mu}) = \widetilde{Y}(T \wedge \tilde{\mu}(K))$ almost surely as $\tilde{\mu} \geq \tilde{\mu}(K)$ almost surely. Restricted on $\mathcal{E}_T$, we have $d_U(Y_{\eta'_m}^{\mu_{\eta'_m}(K)\wedge T}, \widetilde{Y}^{\tilde{\mu}\wedge T}) = d_U(Y_{\eta'_m}^{\mu_{\eta'_m}(K)\wedge T}, \widetilde{Y}^{\tilde{\mu}(K)\wedge T})$, and it follows that for all $m > N$,

$$\begin{aligned}
\mathbb{P}\left[d_U(Y_{\eta'_m}^{\mu_{\eta'_m}(K)\wedge T}, \widetilde{Y}^{\tilde{\mu}(K)\wedge T}) \geq \epsilon\right] &\leq \mathbb{P}\left[\{d_U(Y_{\eta'_m}^{\mu_{\eta'_m}(K)\wedge T}, \widetilde{Y}^{\tilde{\mu}(K)\wedge T}) \geq \epsilon\} \cap \mathcal{E}_T\right] + \mathbb{P}\left[\mathcal{E}_T^c\right] \\
&= \mathbb{P}\left[\{d_U(Y_{\eta'_m}^{\mu_{\eta'_m}(K)\wedge T}, \widetilde{Y}^{\tilde{\mu}\wedge T}) \geq \epsilon\} \cap \mathcal{E}_T\right] + \mathbb{P}[\mathcal{E}_T^c] \\
&\leq \mathbb{P}\left[d_U(Y_{\eta'_m}^{\mu_{\eta'_m}(K)\wedge T}, \widetilde{Y}^{\tilde{\mu}\wedge T}) \geq \epsilon\right] + \mathbb{P}[\mathcal{E}_T^c] \\
&\leq 2\delta,
\end{aligned}$$

where we denote the complement of $\mathcal{E}_T$ by $\mathcal{E}_T^c$.

By the definition of the δ-Prokhorov distance in Definition 4.6.13, we then get $\rho_{d_U}^{2\delta}(Y_{\eta'_m}^{\mu_{\eta'_m}(K)\wedge T}, \widetilde{Y}^{\tilde{\mu}(K)\wedge T}) \leq \epsilon$ for all $m > N$. Therefore, we have

$$\lim_{m\to\infty} \rho_{d_U}^{2\delta}(Y_{\eta'_m}^{\mu_{\eta'_m}(K)\wedge T}, \widetilde{Y}^{\tilde{\mu}(K)\wedge T}) = 0.$$

Since the coefficients in Equation (4.7.3) are locally Lipschitz, we claim that $(\widetilde{Y}^{\tilde{\mu}(K)}, \tilde{\mu}(K)) \stackrel{d}{=} (Y^{\mu(K)}, \mu(K))$. To see this, note that for any compact $K \subseteq U$, the noise function σ, $\partial\Phi$, and $\partial^2\Phi$ are all Lipschitz on K, thus we can extend their definitions to $\mathbb{R}^D$ such that the resulting functions are still locally Lipschitz. Based on this extension, applying the classic theorem on weak uniqueness (e.g., Theorem 1.1.10, [Hsu02]) to the extended version of Equation (4.7.3) yields the equivalence in law. Thus we conclude that for every sequence of learning rates $\{\eta_m\}_m$ with $\lim_{m\to\infty} \eta_m = 0$, there is a subsequence,

$\{\eta'_m\}$ such that

$$\lim_{m\to\infty} \rho^{2\delta}_{d_U}(Y^{\mu_{\eta'_m}(K)\wedge T}_{\eta'_m}, Y^{\mu(K)\wedge T}) = 0. \tag{4.7.5}$$

Now we claim that it indeed holds that $\lim_{\eta\to 0} \rho^{2\delta}_{d_U}(Y^{\mu_\eta(K)\wedge T}_\eta, Y^{\mu(K)\wedge T}) = 0$. We prove this by contradiction. Suppose otherwise, then there exists some $\epsilon > 0$ such that for all $\eta_0 > 0$, there exists some $\eta < \eta_0$ with $\rho^{2\delta}_{d_U}(Y^{\mu_\eta(K)\wedge T}_\eta, Y^{\mu(K)\wedge T}) > \epsilon$. Consequently, there is a sequence $\{\eta_m\}_{m\geq 1}$ satisfying $\lim_{m\to\infty} \eta_m = 0$ and $\rho^{2\delta}_{d_U}(Y^{\mu_{\eta_m}(K)}_{\eta_m}, Y^{\mu(K)\wedge T}) > \epsilon$ for all m. However, this leads to a contradiction with Equation (4.7.5). This completes the proof. ∎

Proof of the second claim of Theorem 4.7.9. We will first show there exists a sequence of compact set $\{K_m\}_{m\geq 1}$ such that $\cup_{m=1}^\infty K_m = U$ and $K_m \subseteq K_{m+1}$. For $m \in \mathbb{N}^+$, we define $H_m = U \setminus (B_{1/m}(0) + \mathbb{R}^D \setminus U)$ and $K_m = \overline{H_m} \cap B_m(0)$. By definition it holds that $\forall m < m'$, $H_m \subseteq H_{m'}$ and $K_m \subseteq K_{m'}$. Moreover, since K_m is bounded and closed, K_m is compact for every m. Now we claim $\cup_{m=1}^\infty K_m = U$. Note that $\cup_{m=1}^\infty K_m = \cup_{m=1}^\infty \overline{H_m} \cap B_m(0) = \cup_{m=1}^\infty \overline{H_m}$. $\forall x \in U$, since U is open, we know that there exists $m_0 \in \mathbb{N}^+$, such that $\forall m \geq m_0$, $x \notin (B_{1/m}(0) + \mathbb{R}^D \setminus U)$ and thus $x \in H_m$, which implies $x \in \cup_{m=1}^\infty \overline{H_m}$. On the other hand, $\forall x \in \mathbb{R}^D \setminus U$, it holds that $x \in (B_{1/m}(0) + \mathbb{R}^D \setminus U)$ for all $m \in \mathbb{N}^+$, thus $x \notin H_m \subset K_m$.

Therefore, since $Y \in U$ and is continuous almost surely, random variables $\lim_{m\to\infty} \mu(K_m) = \infty$ a.s., which implies $\mu(K_m)$ converges to ∞ in distribution, i,e,, $\forall \delta > 0, T > 0$, $\exists m \in \mathbb{N}^+$, such that $\forall K \supseteq K_m$, it holds $\mathbb{P}[\mu(K) \leq T] \leq \delta$.

Now we will show for any $T > 0$ and $\varepsilon > 0$, there exists η_0 such that $\rho^{\varepsilon}_{d_U}(Y^T, Y^T_\eta) \leq \varepsilon$ for all $\eta \leq \eta_0$. Fixing any $T > 0$, for any $\varepsilon > 0$, let $\delta = \frac{\varepsilon}{4}$, then from above we know exists compact set K, such that $\mathbb{P}(\mu(K) \leq T) \leq \delta$. We further pick $K' = K + B_{2\epsilon'}(0)$, where ε' can be any real number satisfying $0 < \varepsilon' < \varepsilon$ and $K' \subseteq U$. Such ε' exists since U is open. Note $K \subseteq K'$, we have $\mathbb{P}(\mu(K') \leq T) \leq \mathbb{P}(\mu(K) \leq T) \leq \delta$. Thus by the first claim of Theorem 4.7.9, there exists $\eta_0 > 0$, such that for all $\eta \leq \eta_0$, we have $\rho^{2\delta}_{d_U}(Y_\eta^{\mu_\eta(K')\wedge T}, Y^{\mu(K')\wedge T}) \leq 2^{-\lceil T \rceil}\epsilon'$.

Note that $\rho^{\delta}_{d_U}(Y^{\mu(K)\wedge T}, Y^{\mu(K')\wedge T}) = 0$, so we have for all $\eta \leq \eta_0$,

$$\rho^{3\delta}_{d_U}(Y^{\mu(K)\wedge T}, Y_\eta^{\mu_\eta(K')\wedge T}) \leq 2^{-\lceil T \rceil}\varepsilon'.$$

By the definition of δ-Prokhorov distance in Definition 4.6.13, without loss of generality we can assume $(Y^{\mu(K)\wedge T}, Y_\eta^{\mu_\eta(K')\wedge T})$ is already the coupling such that

$$\mathbb{P}\left[d_U(Y^{\mu(K)\wedge T}, Y_\eta^{\mu_\eta(K')\wedge T}) \geq 2^{-\lceil T \rceil}\epsilon'\right] \leq 3\delta.$$

Below we want to show $\rho^{3\delta}_{d_U}(Y^{\mu(K)\wedge T}, Y^T_\eta) \leq 2^{-\lceil T \rceil}\varepsilon'$. Note that for all $t \geq 0$, $Y^{\mu(K)\wedge T}(t) \in K$, thus we know if $\mu_\eta(K') \leq T$, then

$$d_U(Y^{\mu(K)\wedge T}, Y_\eta^{\mu_\eta(K')\wedge T}) \geq 2^{-\lceil T \rceil} \inf_{x\in K, y \in \mathbb{R}^D\setminus K'} \|x-y\|_2 \geq 2^{-\lceil T \rceil}\varepsilon'.$$

On the other hand, if $\mu_\eta(K') > T$, then $Y^T_\eta = Y_\eta^{\mu_\eta(K')\wedge T}$. Thus we can conclude that $d_U(Y^{\mu(K)\wedge T}, Y^T_\eta) \geq 2^{-\lceil T \rceil}\epsilon'$ implies $d_U(Y^{\mu(K)\wedge T}, Y_\eta^{\mu_\eta(K')\wedge T}) \geq 2^{-\lceil T \rceil}\epsilon'$. Therefore, we further have

$$\mathbb{P}\left[d_U(Y^{\mu(K)\wedge T}, Y^T_\eta) \geq 2^{-\lceil T \rceil}\epsilon'\right] \leq \mathbb{P}\left[d_U(Y^{\mu(K)\wedge T}, Y_\eta^{\mu_\eta(K')\wedge T}) \geq 2^{-\lceil T \rceil}\epsilon'\right] \leq 3\delta,$$

that is,

$$\rho^{3\delta}_{d_U}(Y^{\mu(K)\wedge T}, Y^T_\eta) \leq 2^{-\lceil T \rceil}\epsilon'.$$

Finally, since $\rho^{\delta}_{d_U}(Y^T, Y^{\mu(K)\wedge T}) = 0$, we have for all $\eta \leq \eta_0$,

$$\rho^{\varepsilon}_{d_U}(Y^T, Y^T_\eta) = \rho^{4\delta}_{d_U}(Y^T, Y^T_\eta) \leq \rho^{3\delta}_{d_U}(Y^{\mu(K)\wedge T}, Y^T_\eta) + \rho^{\delta}_{d_U}(Y^T, Y^{\mu(K)\wedge T}) \leq 2^{-\lceil T \rceil}\varepsilon' + 0 \leq \varepsilon,$$

The proof is completed by the arbitrariness of ϵ and t. ∎

Now, we provide the proof of Theorem 4.3.6 as a direct application of Theorem 4.7.9.

Proof of Theorem 4.3.6. We first prove that Y never leaves Γ, i.e., $\mathbb{P}[Y(t) \in \Gamma, \forall t \geq 0] = 1$. By the result of Theorem 4.7.8, we know that for each compact set $K \subset \Gamma$, $Y^{\mu(K)}$ stays on Γ almost surely, where $\mu(K) := \inf\{t \geq 0 \mid \widetilde{Y}(t) \notin \mathring{K}\}$ is the earliest time that Y leaves K. In other words, for all compact set $K \subset \Gamma$, $\mathbb{P}[\exists t \geq 0, Y(t) \notin \Gamma, Y(t) \in K] = 0$. Let $\{K_m\}_{m\geq 1}$ be any sequence of compact sets such that $\cup_{m\geq 1} K_m = U$ and $K_m \subset U$, e.g., the ones constructed in the proof of the second claim of Theorem 4.7.9. Therefore, we have

$$\begin{aligned}\mathbb{P}[\exists t \geq 0, Y(t) \notin \Gamma] &= \mathbb{P}[\exists t \geq 0, Y(t) \notin \Gamma, Y(t) \in U] \\ &\leq \sum_{m=1}^{\infty} \mathbb{P}[\exists t \geq 0, Y(t) \notin \Gamma, Y(t) \in K_m] = 0,\end{aligned}$$

which means Y always stays on Γ.

Then recall the decomposition of $\Sigma = \Sigma_{\parallel} + \Sigma_{\perp} + \Sigma_{\parallel,\perp} + \Sigma_{\perp,\parallel}$ as defined in Lemma 4.3.5. Since Y never leaves Γ, by Lemma 4.3.5, we can rewrite Equation (4.3.5) as

$$\begin{aligned}\mathrm{d}Y(t) &= \Sigma_{\parallel}^{1/2}\mathrm{d}W(t) + \partial^2\Phi(Y(t))[\Sigma(Y(t))]\mathrm{d}t \\ &= \partial\Phi(Y(t))\sigma(Y(t))\mathrm{d}W(t) + \frac{1}{2}\sum_{i,j=1}^{D}\partial_{ij}\Phi(Y(t))(\sigma(Y(t))\sigma(Y(t))^{\top})_{ij}\mathrm{d}t\end{aligned}$$

where the second equality follows from the definition that $\Sigma_{\parallel} = \partial\Phi\Sigma\partial\Phi = \partial\Phi\sigma\sigma^{\top}\partial\Phi$. This coincides with the formulation of the limiting diffusion in Theorem 4.7.9. Therefore, further combining Lemma 4.3.2 and the second part of Theorem 4.7.9, we obtain the desired result. ∎

Remark 4.7.10. Our result suggests that for tiny LR η, SGD dynamics have two phases.

In Phase I of $\Theta(1/\eta)$ steps, the SGD iterates move towards the manifold Γ of local minimizers along GF. Then in Phase II which is of $\Theta(1/\eta^2)$ steps, the SGD iterates stay close to Γ and diffuse approximately according to (4.3.5). See Figure 4.2 for an illustration of this two-phase dynamics. However, since the length of Phase I gets negligible compared to that of Phase II when $\eta \to 0$, Theorem 4.3.6 only reflects the time scaling of Phase II.

4.8 Explicit formula of the limiting diffusion

In this section, we demonstrate how to compute the derivatives of Φ by relating to those of the loss function L, and then present the explicit formula of the limiting diffusion.

4.8.1 Explicit Expression of the Derivatives

For any $x \in \Gamma$, we choose an orthonormal basis of $T_x(\Gamma)$ as $\{v_1, \dots, v_{D-M}\}$. Let $\{v_{D-M+1}, \dots, v_D\}$ be an orthonormal basis of $T_x^{\perp}(\Gamma)$ so that $\{v_i\}_{i\in[D]}$ is an orthonormal basis of $\mathbb{R}^D$.

Lemma 4.8.1. *For any $x \in \Gamma$ and any $v \in T_x(\Gamma)$, it holds that $\nabla^2 L(x)v = 0$.*

Proof. For any $x \in T_x(\Gamma)$, let $\{x(t)\}_{t\geq 0}$ be a parametrized smooth curve on Γ such that $x(0) = x$ and $\frac{\mathrm{d}x(t)}{\mathrm{d}t}\Big|_{t=0} = v$. Then $\nabla L(x_t) = 0$ for all t. Thus $0 = \frac{\mathrm{d}\nabla L(x_t)}{\mathrm{d}t}\Big|_{t=0} = \nabla^2 L(x)v$. ∎

Lemma 4.8.2. *For any $x \in \mathbb{R}^D$, it holds that $\partial\Phi(x)\nabla L(x) = 0$ and*

$$\partial^2\Phi(x)[\nabla L(x), \nabla L(x)] = -\partial\Phi(x)\nabla^2 L(x)\nabla L(x).$$

Proof. Fixing any $x \in \mathbb{R}^D$, let $\frac{\mathrm{d}x(t)}{\mathrm{d}t} = -\nabla L(x(t))$ be initialized at $x(0) = x$. Since

$\Phi(x(t)) = \Phi(x)$ for all $t \geq 0$, we have

$$\frac{\mathrm{d}}{\mathrm{d}t}\Phi(x(t)) = -\partial\Phi(x(t))\nabla L(x(t)) = 0.$$

Evaluating the above equation at $t = 0$ yields $\partial\Phi(x)\nabla L(x) = 0$. Moreover, take the second order derivative and we have

$$\frac{\mathrm{d}^2}{\mathrm{d}t^2}\Phi(x_t) = -\partial^2\Phi(x(t))\left[\frac{\mathrm{d}x(t)}{\mathrm{d}t}, \nabla L(x(t))\right] - \partial\Phi(x(t))\nabla^2 L(x(t))\frac{\mathrm{d}x(t)}{\mathrm{d}t} = 0.$$

Evaluating at $t = 0$ completes the proof. ■

Now we can prove Lemma 4.3.3.

Proof of Lemma 4.3.3. For any $v \in T_x(\Gamma)$, let $\{v(t), t \geq 0\}$ be a parametrized smooth curve on Γ such that $v(0) = x$ and $\frac{\mathrm{d}v(t)}{\mathrm{d}t}\Big|_{t=0} = v$. Since $v(t) \in \Gamma$ for all $t \geq 0$, we have $\Phi(v(t)) = v(t)$, and thus

$$\frac{\mathrm{d}v(t)}{\mathrm{d}t}\bigg|_{t=0} = \frac{\mathrm{d}}{\mathrm{d}t}\Phi(v(t))\bigg|_{t=0} = \partial\Phi(x)\frac{\mathrm{d}v(t)}{\mathrm{d}t}\bigg|_{t=0}.$$

This implies that $\partial\Phi(x)v = v$ for all $v \in T_x(\Gamma)$.

Next, for any $u \in T_x^\perp(\Gamma)$ and $t \geq 0$, consider expanding $\nabla L(x + t\nabla^2 L(x)^\dagger u)$ at $t = 0$:

$$\begin{aligned}\nabla L\left(x + t\nabla^2 L(x)^\dagger u\right) &= \nabla^2 L(x) \cdot t\nabla^2 L(x)^\dagger u + o(t) \\ &= tu + o(t)\end{aligned}$$

where the second equality follows from the assumption that $\nabla^2 L(x)$ is full-rank when restricted on $T_x^\perp(\Gamma)$. Then since $\partial\Phi$ is continuous, it follows that

$$\lim_{t\to 0}\frac{\partial\Phi(x + t\nabla^2 L(x)^\dagger u)\nabla L(x + t\nabla^2 L(x)^\dagger u)}{t} = \lim_{t\to 0}\partial\Phi(x + t\nabla^2 L(x)^\dagger)(u + o(1))$$

$$= \partial\Phi(x)u.$$

By Lemma 4.8.2, we have $\partial\Phi(x+t(\nabla^2 L(x))^\dagger u))\nabla L(x+t(\nabla^2 L(x))^\dagger u) = 0$ for all $t > 0$, which then implies that $\partial\Phi(x)u = 0$ for all $u \in T_x^\perp(\Gamma)$.

Therefore, under the basis $\{v_1, \ldots, v_D\}$, $\partial\Phi(x)$ is given by

$$\partial\Phi(x) = \begin{pmatrix} I_{D-M} & 0 \\ 0 & 0 \end{pmatrix} \in \mathbb{R}^{D\times D},$$

that is, the projection matrix onto $T_x(\Gamma)$. ■

Lemma 4.8.3. *For any $x \in \Gamma$, it holds that $\partial\Phi(x)\nabla^2 L(x) = 0$.*

Proof. It directly follows from Lemma 4.8.1 and Lemma 4.3.3. ■

Next, we proceed to compute the second-order derivatives.

Lemma 4.8.4. *For any $x \in \Gamma$, $u \in \mathbb{R}^D$ and $v \in T_x(\Gamma)$, it holds that*

$$\partial^2\Phi(x)[v,u] = -\partial\Phi(x)\partial^2(\nabla L)(x)[v, \nabla^2 L(x)^\dagger u] - \nabla^2 L(x)^\dagger \partial^2(\nabla L)(x)[v, \partial\Phi(x)u].$$

Proof of Lemma 4.8.4. Consider a parametrized smooth curve $\{v(t)\}_{t\geq 0}$ on Γ such that $v(0) = x$ and $\frac{\mathrm{d}v(t)}{\mathrm{d}t}\Big|_{t=0} = v$. We define $P(t) = \partial\Phi(v(t))$, $P^\perp(t) = I_D - P(t)$ and $H(t) = \nabla^2 L(v(t))$ for all $t \geq 0$. By Lemma 4.8.1 and 4.3.3, we have

$$P^\perp(t)H(t) = H(t)P^\perp(t) = H(t), \tag{4.8.1}$$

Denote the derivative of $P(t), P^\perp(t)$ and $H(t)$ with respect to t as $P'(t), (P^\perp)'(t)$ and $H'(t)$. Then differentiating with respect to t, we have

$$(P^\perp)'(t)H(t) = H'(t) - P^\perp(t)H'(t) = P(t)H'(t). \tag{4.8.2}$$

Then combining (4.8.1) and (4.8.2) and evaluating at $t = 0$, we have

$$P'(0)H(0) = -(P^{\perp})'(0)H(0) = -P(0)H'(0)$$

We can decompose $P'(0)$ and $H(0)$ as follows

$$P'(0) = \begin{pmatrix} P'_{11}(0) & P'_{12}(0) \\ P'_{21}(0) & P'_{22}(0) \end{pmatrix}, \quad H(0) = \begin{pmatrix} 0 & 0 \\ 0 & H_{22}(0) \end{pmatrix}, \tag{4.8.3}$$

where $P'_{11}(0) \in \mathbb{R}^{(D-M)\times(D-M)}$ and H_{22} is the hessian of L restricted on $T_x^{\perp}(\Gamma)$. Also note that

$$P(0)H'(0)P^{\perp}(0) = \begin{pmatrix} I_{D-M} & 0 \\ 0 & 0 \end{pmatrix} \begin{pmatrix} H'_{11}(0) & H'_{12}(0) \\ H'_{21}(0) & H'_{22}(0) \end{pmatrix} \begin{pmatrix} 0 & 0 \\ 0 & I_M \end{pmatrix} = \begin{pmatrix} 0 & H'_{12}(0) \\ 0 & 0 \end{pmatrix},$$

and thus by (4.8.3) we have

$$P'(0)H(0) = \begin{pmatrix} 0 & P'_{12}(0)H_{22}(0) \\ 0 & P'_{22}(0)H_{22}(0) \end{pmatrix} = \begin{pmatrix} 0 & -H'_{12}(0) \\ 0 & 0 \end{pmatrix}.$$

This implies that we must have $P'_{22}(0) = 0$ and $P'_{12}(0)H_{22}(0) = H'_{12}(0)$. Similarly, by taking transpose in (4.8.3), we also have $H_{22}(0)P'_{21}(0) = -H'_{21}(0)$.

It then remains to determine the value of $P'_{11}(0)$. Note that since $P(t)P(t) = P(t)$, we have $P'(t)P(t) + P(t)P'(t) = P'(t)$, evaluating which at $t = 0$ yields

$$2P'_{11}(0) = P'_{11}(0).$$

Therefore, we must have $P'_{11}(0) = 0$. Combining the above results, we obtain

$$P'(0) = -P(0)H'(0)H(0)^{\dagger} - H(0)^{\dagger}H'(0)P(0).$$

Finally, recall that $P(t) = \partial\Phi(v(t))$, and thus

$$P'(0) = \frac{\mathrm{d}}{\mathrm{d}t}\partial\Phi(v(t))\bigg|_{t=0} = \partial^2\Phi(x)[v].$$

Similarly, we have $H'(0) = \partial^2(\nabla L)(x)[v]$, and it follows that

$$\partial^2\Phi(x)[v] = -\partial\Phi(x)\partial^2(\nabla L)(x)[v]\nabla^2 L(x)^\dagger - \nabla^2 L(x)^\dagger\partial^2(\nabla L)(x)[v]\partial\Phi(x).$$

■

Lemma 4.8.5. *For any $x \in \Gamma$ and $u \in T_x^\perp(\Gamma)$, it holds that*

$$\partial^2\Phi(x)[uu^\top + \nabla^2 L(x)^\dagger uu^\top\nabla^2 L(x)] = -\partial\Phi(x)\partial^2(\nabla L)(x)[\nabla^2 L(x)^\dagger uu^\top].$$

Proof of Lemma 4.8.5. For any $u \in T_x^\perp(\Gamma)$, we define $u(t) = x + t\nabla^2 L(x)^\dagger u$ for $t \geq 0$. By Taylor approximation, we have

$$\nabla L(u(t)) = t\nabla^2 L(x)\nabla^2 L(x)^\dagger u + o(t) = tu + o(t) \tag{4.8.4}$$

and

$$\nabla^2 L(u(t)) = \nabla^2 L(x) + t\partial^2(\nabla L)(x)[\nabla^2 L(x)^\dagger u] + o(t). \tag{4.8.5}$$

Combine (4.8.4) and (4.8.5) and apply Lemma 4.8.2, and it follows that

$$\begin{aligned}
0 &= \partial^2\Phi(u(t))[\nabla L(u(t)), \nabla L(u(t))] + \partial\Phi(u(t))\nabla^2 L(u(t))\nabla L(u(t)) \\
&= t^2\partial^2\Phi(u(t))[u + o(1)](u + o(1)) + t^2\partial\Phi(u(t))\partial^2(\nabla L)(x)[\nabla^2 L(x)^\dagger u](u + o(1)) \\
&\quad + t^2\frac{\partial\Phi(u(t))}{t}\nabla^2 L(x)(u + o(1)) \\
&= t^2\partial^2\Phi(u(t))[u + o(1)](u + o(1)) + t^2\partial\Phi(u(t))\partial^2(\nabla L)(x)[\nabla^2 L(x)^\dagger u](u + o(1))
\end{aligned}$$

$$+ t^2 \frac{\partial\Phi(u(t)) - \partial\Phi(x)}{t} \nabla^2 L(x)(u + o(1))$$

where the last equality follows from Lemma 4.8.3. Dividing both sides by t^2 and letting $t \to 0$, we get

$$\partial^2\Phi(x)[u]u + \partial\Phi(x)\partial^2(\nabla L)(x)[\nabla^2 L(x)^\dagger u]u + \partial^2\Phi(x)[\nabla^2 L(x)^\dagger u]\nabla^2 L(x)u = 0.$$

Rearranging the above equation completes the proof. ■

With the notion of Lyapunov Operator in Definition 4.3.4, Lemma 4.8.5 can be further simplified into Lemma 4.8.6.

Lemma 4.8.6. *For any $x \in \Gamma$ and $\Sigma \in \operatorname{span}\{uu^\top \mid u \in T_x^\perp(\Gamma)\}$,*

$$\langle \partial^2\Phi(x), \Sigma\rangle = -\partial\Phi(x)\partial^2(\nabla L)(x)[\mathcal{L}^{-1}_{\nabla^2 L(x)}(\Sigma)]. \tag{4.8.6}$$

Proof of Lemma 4.8.6. Let $A = uu^\top + \nabla^2 L(x)^\dagger uu^\top \nabla^2 L(x)$ and $B = \nabla^2 L(x)^\dagger uu^\top$. The key observation is that $A + A^\top = \mathcal{L}_{\nabla^2 L(x)}(B + B^\top)$. Therefore, by Lemma 4.8.5, it holds that

$$\begin{aligned}
\partial^2\Phi(x)[\mathcal{L}_{\nabla^2 L(x)}(B + B^\top)] &= \partial^2\Phi(x)[A + A^\top] \\
&= 2\partial\Phi(x)\partial^2(\nabla L)(x)[B] \\
&= \partial\Phi(x)\partial^2(\nabla L)(x)[B + B^\top].
\end{aligned}$$

Since $\nabla^2 L(x)^\dagger$ is full-rank when restricted to $T_x^\perp(\Gamma)$, we have $\operatorname{span}\{\nabla^2 L(x)^\dagger uu^\top + uu^\top \nabla^2 L(x)^\dagger \mid u \in T_x^\perp(\Gamma)\} = \operatorname{span}\{uu^\top \mid u \in T_x^\perp(\Gamma)\}$. Thus by the linearity of above equation, we can replace $B + B^\top$ by any $\Sigma \in \operatorname{span}\{uu^\top \mid u \in T_x^\perp(\Gamma)\}$, resulting in the desired equation. ■

Then Lemma 4.3.5 directly follows from Lemma 4.8.4 and 4.8.6.

4.8.2 Tangent noise compensation only depend on the manifold itself

Here we show that the second term of (4.3.5), i.e., the **tangent noise compensation** for the limiting dynamics to stay on Γ, only depends on Γ itself.

Lemma 4.8.7. *For any $x \in \Gamma$, suppose there exist a neighborhood U_x of x and two loss functions L and L' that define the same manifold Γ locally in U_x, i.e., $\Gamma \cap U_x = \{x \mid \nabla L(x) = 0\} = \{x \mid \nabla L'(x) = 0\}$. Then for any $v \in T_x(\Gamma)$, it holds that $(\nabla^2 L(x))^\dagger \partial^2(\nabla L)(x)\,[v, v] = (\nabla^2 L'(x))^\dagger \partial^2(\nabla L')(x)\,[v, v]$.*

Proof of Lemma 4.8.7. Let $\{v(t)\}_{t\geq 0}$ be a smooth curve on Γ with $v(0) = x$ and $\frac{\mathrm{d}v(t)}{\mathrm{d}t}\Big|_{t=0} = v$. Since $v(t)$ stays on Γ, we have $\nabla L(v(t)) = 0$ for all $t \geq 0$. Taking derivative for two times yields $\partial^2(\nabla L)(v(t))[\frac{\mathrm{d}v(t)}{\mathrm{d}t}, \frac{\mathrm{d}v(t)}{\mathrm{d}t}] + \nabla^2 L(v(t))\frac{\mathrm{d}^2 v(t)}{\mathrm{d}t^2} = 0$. Evaluating it at $t = 0$ and multiplying both sides by $\nabla^2 L(x)^\dagger$, we get

$$\nabla^2 L(x)^\dagger \partial^2(\nabla L)(x)\,[v, v] = -\nabla^2 L(x)^\dagger \nabla^2 L(x) \frac{\mathrm{d}^2 v(t)}{\mathrm{d}t^2}\bigg|_{t=0} = -\partial\Phi(x)\frac{\mathrm{d}^2 v(t)}{\mathrm{d}t^2}\bigg|_{t=0}.$$

Since $\partial\Phi(x)$ is the projection matrix onto $T_x(\Gamma)$ by Lemma 4.3.3, it does not depend on L, so analogously we also have $\nabla^2 L'(x)^\dagger \partial^2(\nabla L')(x)\,[v, v] = -\partial\Phi(x)\frac{\mathrm{d}^2 v(t)}{\mathrm{d}t^2}\Big|_{t=0}$ as well. The proof is thus completed. Note that $\partial\Phi(x)\frac{\mathrm{d}^2 v(t)}{\mathrm{d}t^2}\Big|_{t=0}$ is indeed the second fundamental form for v at x, and the value won't change if we choose another parametric smooth curve with a different second-order time derivative. (See Chapter 6 in [DC13] for a reference.) ■

4.8.3 Proof of results in Section 4.4

Now we are ready to give the missing proofs in Section 4.4 which yield explicit formula of the limiting diffusion for label noise and isotropic noise.

Proof of Corollary 4.4.1. Set $\Sigma_{\parallel} = \partial\Phi$, $\Sigma_{\perp} = I_D - \partial\Phi$ and $\Sigma_{\perp,\parallel} = \Sigma_{\parallel,\perp} = 0$ in the decomposition of Σ by Lemma 4.3.5, and we need to show $\nabla(\ln|\Sigma|_+) = \partial^2(\nabla L)[(\nabla^2 L)^\dagger]$.

[Hol18] shows that the gradient of pseudo-inverse determinant satisfies $\nabla|A|_+ = |A|_+ A^\dagger$. Thus we have for any vector $v \in \mathbb{R}^D$, $\langle v, \nabla \ln|\nabla^2 L|_+\rangle = \langle \frac{|\nabla^2 L|_+ \nabla^2 L}{|\nabla^2 L|_+}, \partial^2(\nabla L)[v]\rangle = \langle \nabla^2 L, \partial^2(\nabla L)[v]\rangle = \partial^2(\nabla L)[v, \nabla^2 L] = \langle v, \partial^2(\nabla L)[(\nabla^2 L)^\dagger]\rangle$, which completes the proof. ■

Proof of Corollary 4.4.2. Since $\Sigma = c\nabla^2 L$, here we have $\Sigma_\perp = \Sigma$ and $\Sigma_\parallel, \Sigma_{\perp,\parallel}, \Sigma_{\parallel,\perp} = 0$. Thus it suffices to show that $2\partial^2(\nabla L)\left[\mathcal{L}^{-1}_{\nabla^2 L}(\Sigma_\perp)\right] = \nabla \operatorname{tr}[\nabla^2 L]$. Note that for any $v \in \mathbb{R}^D$,

$$v^\top \nabla \operatorname{tr}[\nabla^2 L] = \langle I_D, \partial(\nabla^2 L)[v]\rangle = \langle I_D - \partial\Phi, \partial(\nabla^2 L)[v]\rangle, \tag{4.8.7}$$

where the second equality is because the the tangent space of symmetric rank-$(D-M)$ matrices at $\nabla^2 L$ is $\{A\nabla^2 L + \nabla^2 L A^\top \mid A \in \mathbb{R}^{D\times D}\}$, and every element in this tangent space has zero inner-product with $\partial\Phi$ by Lemma 4.3.3. Also note that $\mathcal{L}^{-1}_{\nabla^2 L}(\nabla^2 L) = \frac{1}{2}(I_D - \partial\Phi)$, thus $\langle I_D - \partial\Phi, \partial^2(\nabla L)[v]\rangle = 2\langle \mathcal{L}^{-1}_{\nabla^2 L}(\nabla^2 L), \partial^2(\nabla L)[v]\rangle = 2v^\top \partial^2(\nabla L)[\mathcal{L}^{-1}_{\nabla^2 L}(\nabla^2 L)]$. ■

4.8.4 Proof for the example of k-phase motor

Proof of Lemma 4.4.3. Note that for any $x \in \Gamma$, it holds that

$$\left(\nabla^2 L(x)\right)_{ij} = \begin{cases} 2 + \langle Q_\alpha^{j-3} v, x_{1:2}\rangle & \text{if } i = j \geq 3, \\ x_i x_j & \text{if } i, j \in \{1, 2\}, \\ 0 & \text{otherwise.} \end{cases} \tag{4.8.8}$$

Then clearly Σ only brings about noise in the normal space, and specifically, it holds that $\mathcal{L}^{-1}_{\nabla^2 L(x)}(\Sigma(x)) = \operatorname{diag}(0, 0, 1 + \langle Q_\alpha^0 v, Q_{-\pi/2} x_{1:2}\rangle, \ldots, 1 + \langle Q_\alpha^{D-3} v, Q_{-\pi/2} x_{1:2}\rangle)$.

Further note that, by the special structure of the hessian in (4.8.8) and Lemma 4.8.3, for any $x \in \Gamma$, we have $\partial\Phi(x) = (x_2, -x_1, 0, \dots, 0)^\top (x_2, -x_1, 0, \dots, 0) = \begin{pmatrix} Q_{-\pi/2}x_{1:2} \\ 0 \end{pmatrix}\begin{pmatrix} Q_{-\pi/2}x_{1:2} \\ 0 \end{pmatrix}^\top$. Combining these facts, the dynamics of the first two coordinates in SDE (4.3.5) can be simplified into

$$
\begin{aligned}
\frac{\mathrm{d}x_{1:2}(t)}{\mathrm{d}t} &= -\left(\frac{1}{2}\partial\Phi(x(t))\partial^2(\nabla L)(x(t))[\mathcal{L}^{-1}_{\nabla^2 L}(\Sigma(x(t))]\right)_{1:2} \\
&= -\frac{1}{2}Q_{-\pi/2}x_{1:2}x_{1:2}^\top Q_{-\pi/2}^\top \sum_{j=3}^{D}\left(1 + \langle Q_\alpha^{j-3}v, Q_{-\pi/2}x_{1:2}\rangle\right)\nabla_{1:2}(\partial_{jj}L)(x) \\
&= -\frac{1}{2}Q_{-\pi/2}x_{1:2}\left\langle Q_{-\pi/2}x_{1:2}, \sum_{j=3}^{D}\left(1 + \langle Q_\alpha^{j-3}v, Q_{-\pi/2}x_{1:2}\rangle\right)Q_\alpha^{j-3}v\right\rangle \\
&= -\frac{1}{2}Q_{-\pi/2}x_{1:2}\left(\left\langle Q_{-\pi/2}x_{1:2}, \sum_{j=3}^{D}Q_\alpha^{j-3}v\right\rangle + \sum_{j=3}^{D}\langle Q_\alpha^{j-3}v, Q_{-\pi/2}x_{1:2}\rangle^2\right) \\
&= -\frac{1}{2}Q_{-\pi/2}x_{1:2}\left(0 + \frac{D-2}{2}\left\|Q_{-\pi/2}x_{1:2}\right\|_2^2\right) = \frac{D-2}{2}Q_{\pi/2}x_{1:2},
\end{aligned}
$$

where the second to the last equality follows from the property of Q_α and the last equality follows from the fact that $\|x_{1:2}\|_2^2 = 1$ for all $x \in \Gamma$. Note we require $k \geq 3$ (or $D \geq 5$) to allow $\sum_{j=3}^{D}\langle Q_\alpha^{j-3}v, Q_{-\pi/2}x_{1:2}\rangle^2 = \frac{D-2}{2}\left\|Q_{-\pi/2}x_{1:2}\right\|_2^2$. On the other hand, we have $\frac{\mathrm{d}x_{3:D}(t)}{\mathrm{d}t} = 0$ as $\partial\Phi$ kills the movement on that component.

The proof is completed by noting that the solution of $x_{1:2}$ is

$$
x_{1:2}(t) = \exp\left(t \cdot \frac{D-2}{2}Q_{\pi/2}\right)x_{1:2}(0),
$$

and by Lemma 4.8.8,

$$
\exp\left(t \cdot \frac{D-2}{2}Q_{\pi/2}\right) = (\exp(Q_{\pi/2}))^{\frac{t(D-2)}{2}} = Q_1^{\frac{t(D-2)}{2}} = Q_{\frac{t(D-2)}{2}}.
$$

■

Lemma 4.8.8. $\exp(\begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}) = \begin{pmatrix} \cos 1 & -\sin 1 \\ \sin 1 & \cos 1 \end{pmatrix}$.

Proof. By definition, for matrix $A = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$, $\exp(A) = \sum_{t=0}^{\infty}\frac{A^t}{t!}$. Note that $A_2 = -I$,

$A^3 = -A$ and $A^4 = I$. Using this pattern, we can easily check that

$$\sum_{t=0}^{\infty} \frac{A^t}{t!} = \begin{pmatrix} \sum_{i=0}^{\infty}(-1)^i \frac{1}{(2i)!} & -\sum_{i=0}^{\infty}(-1)^i \frac{1}{(2i+1)!} \\ \sum_{i=0}^{\infty}(-1)^i \frac{1}{(2i+1)!} & \sum_{i=0}^{\infty}(-1)^i \frac{1}{(2i)!} \end{pmatrix} = \begin{pmatrix} \cos 1 & -\sin 1 \\ \sin 1 & \cos 1 \end{pmatrix}.$$

■

4.9 Proof of results in Section 4.5

In this section, we present the missing proofs in Section 4.5 regarding the over-parametrized linear model.

For convenience, for any $p, r \geq 0$ and $u \in \mathbb{R}^D$, we denote by $B_r^p(u)$ the ℓ_p norm ball of radius r centered at u. We also denote $v_{i:j} = (v_i, v_{i+1}, \ldots, v_j)^\top$ for $i, j \in [D]$.

4.9.1 Proof of Theorem 4.5.1

We first provide the proof of Theorem 4.5.1.

Proof of Theorem 4.5.1. First, by Lemma 4.5.8, it holds with probability at least $1 - e^{-\Omega(n)}$ that the solution to (4.5.5), x_*, is unique up to and satisfies $|x_*| = \psi(w_*)$. Then on this event, for any $\epsilon > 0$, by Lemma 4.5.5, there exists some $T > 0$ such that $Y(T)$ given by the Riemannian gradient flow (4.5.4) satisfies that $Y(T)$ is an $\epsilon/2$-optimal solution of the OLM. For this T, by Theorem 4.3.6, we know that the $\lfloor T/\eta^2 \rfloor$-th SGD iterate, $x_\eta(\lfloor T/\eta^2 \rfloor)$, satisfies $\|x_\eta(\lfloor T/\eta^2 \rfloor) - Y(T)\|_2 \leq \epsilon/2$ with probability at least $1 - e^{-\Omega(n)}$ for all sufficiently small $\eta > 0$, and thus $x_\eta(\lfloor T/\eta^2 \rfloor)$ is an ϵ-optimal solution of the OLM. Finally, the validity of applying Theorem 4.3.6 is guaranteed by Lemma 4.5.2 and 4.5.3. This completes the proof. ■

In the following subsections, we provide the proofs of all the components used in the above proof.

4.9.2 Proof of Lemma 4.5.2

Recall that for each $i \in [n]$ $f_i(x) = f(u,v) = z_i^\top(u^{\odot 2} - v^{\odot 2})$, $\nabla f_i(x) = 2\binom{z_i\odot u}{z_i \odot v}$, and $K(x) = (K_{ij}(x))_{i,j\in[n]}$ where each $K_{ij}(x) = \langle \nabla f_i(x), \nabla f_j(x)\rangle$. Then

$$\nabla^2 \ell_i(x) = 2\begin{pmatrix} z_i \odot u \\ -z_i \odot v\end{pmatrix}\begin{pmatrix}(z_i\odot u)^\top & -(z_i \odot v)^\top\end{pmatrix} + (f_i(u,v) - y_i)\cdot \mathrm{diag}(z_i, z_i).$$

So for any $x \in \Gamma$, it holds that

$$\nabla^2 L(x) = \frac{2}{n}\sum_{i=1}^n \begin{pmatrix} z_i \odot u \\ -z_i \odot v\end{pmatrix}\begin{pmatrix}(z_i\odot u)^\top & -(z_i \odot v)^\top\end{pmatrix}. \tag{4.9.1}$$

Lemma 4.9.1. *For any fixed $x \in \mathbb{R}^D$, suppose $\{\nabla f_i(x)\}_{i\in[n]}$ are linearly independent, then $K(x)$ is full-rank.*

Proof of Lemma 4.9.1. Suppose otherwise, then there exists some $\lambda \in \mathbb{R}^n$ such that $\lambda \neq 0$ and $\lambda^\top K(x)\lambda = 0$. However, note that

$$\begin{aligned}\lambda^\top K(x)\lambda &= \sum_{i,j=1}^n \lambda_i\lambda_j K_{ij}(x)\\ &= \sum_{i,j=1}^n \lambda_i\lambda_j\langle \nabla f_i(x), \nabla f_j(x)\rangle\\ &= \left\|\sum_{i=1}^n \lambda_i \nabla f_i(x)\right\|_2^2,\end{aligned}$$

which implies that $\sum_{i=1}^n \lambda_i \nabla f_i(x) = 0$. This is a contradiction since by assumption $\{\nabla f_i(x)\}_{i\in[n]}$ is linearly independent. ■

Proof of Lemma 4.5.2. (1) By preimage theorem [BH13], it suffices to check the jacobian $[\nabla f_1(x), \ldots, \nabla f_n(x)] = 2[\binom{z_1\odot u}{-z_1\odot v}, \ldots, \binom{z_n\odot u}{-z_n\odot v}]$ is full rank. Similarly, for the second claim, due to (4.9.1). it is also equivalent to show that $\{\binom{z_i\odot u}{-z_i\odot v}\}_{i\in[n]}$ is of rank n.

Since $\binom{u}{v} \in \Gamma \subset U$, each coordinate is non-zero, thus we only need to show that $\{z_i\}_{i\in[n]}$ is of rank n. This happens with probability 1 in the Gaussian case, and probability at least $1 - c^d$ for some constant $c \in (0, 1)$ by [KKS95]. This completes the proof. ■

4.9.3 Proof of Lemma 4.5.3

We first establish some auxiliary results. The following lemma shows the PL condition along the trajectory of gradient flow.

Lemma 4.9.2. *Let $x(t)$ be the gradient flow with respect to L starting from $x(0)$, that is, $\frac{\mathrm{d}x(t)}{\mathrm{d}t} = -\nabla L(x(t))$. Then it holds that*

$$\|\nabla L(x(t))\|^2 \geq \frac{16}{n} \cdot \lambda_{\min}(ZZ^\top) \cdot \min_{i\in[d]} |u_i(0)v_i(0)| L(x(t)), \quad \forall t \geq 0.$$

To prove Lemma 4.9.2, we need the following invariance along the gradient flow.

Lemma 4.9.3. *Let $x(t)$ be the gradient flow with respect to L starting from $x(0)$, that is, $\frac{\mathrm{d}x(t)}{\mathrm{d}t} = -\nabla L(x(t))$. Then it holds that $u_j(t)v_j(t) \equiv u_j(0)v_j(0)$ for all $j \in [d]$. Thus,* $\mathrm{sign}(u_j(t)) = \mathrm{sign}(u_j(0))$ *and* $\mathrm{sign}(v_j(t)) = \mathrm{sign}(v_j(0))$ *for any $j \in [d]$.*

Proof of Lemma 4.9.3.

$$\begin{aligned}
\frac{\partial}{\partial t}(u_j(t)v_j(t)) &= \frac{\partial u_j(t)}{\partial t} \cdot v_j(t) + u_j(t) \cdot \frac{\partial v_j(t)}{\partial t} \\
&= \nabla_u L(u(t), v(t))_j \cdot v_j(t) + u_j(t) \cdot \nabla_v L(u(t), v(t))_j \\
&= \frac{2}{n}\sum_{i=1}^{n}(f_i(u(t), v(t)) - y_i)z_{i,j}u_j(t)v_j(t) \\
&\qquad - \frac{2u_j(t)}{n}\sum_{i=1}^{n}(f_i(u(t), v(t)) - y_i)z_{i,j}v_j(t) \\
&= 0.
\end{aligned}$$

Therefore, any sign change of $u_j(t), v_j(t)$ would enforce $u_j(t) = 0$ or $v_j(t) = 0$ for some $t > 0$ since $u_j(t), v_j(t)$ are continuous in time t. This immediately leads to a contradiction to the invariance of $u_j(t)v_j(t)$. ■

We then can prove Lemma 4.9.2.

Proof of Lemma 4.9.2. Note that

$$\begin{aligned}
\|\nabla L(x)\|_2^2 &= \frac{1}{n^2}\sum_{i,j=1}^{n}(f_i(x) - y_i)(f_j(x) - y_j)\langle \nabla f_i(x), \nabla f_j(x)\rangle \\
&\geq \frac{1}{n^2}\sum_{i=1}^{n}(f_i(x) - y_i)^2 \lambda_{\min}(K(x)) \\
&= \frac{2}{n}L(x)\lambda_{\min}(K(x)),
\end{aligned}$$

where $K(x)$ is a $n \times n$ p.s.d. matrix with $K_{ij}(x) = \langle \nabla f_i(x), \nabla f_j(x)\rangle$. Below we lower bound $\lambda_{\min}(K(x))$, the smallest eigenvalue of $K(x)$. Note that $K_{ij}(x(t)) = 4\sum_{h=1}^{d} z_{i,h}z_{j,h}((u_h(t))^2 + (v_h(t))^2)$, and we have

$$\begin{aligned}
K(x(t)) &= 4Z\text{diag}((u(t))^{\odot 2} + (v(t))^{\odot 2})Z^\top \succeq 8Z\text{diag}(|u(t) \odot v(t)|)Z^\top \\
&\overset{(*)}{=} 8Z\text{diag}(|u(0) \odot v(0)|)Z^\top \succeq 8\min_{i\in[d]}|u_i(0)v_i(0)|ZZ^T
\end{aligned}$$

where $(*)$ is by Lemma 4.9.3. Thus $\lambda_{\min}(K(x(t)) \geq 8\min_{i\in[d]}|u_i(0)v_i(0)|\lambda_{\min}(ZZ^T)$ for all $t \geq 0$, which completes the proof. ■

We also need the following characterization of the manifold Γ.

Lemma 4.9.4. *All the stationary points in U are global minimizers, i.e., $\Gamma = \{x \in U \mid \nabla L(x) = 0\}$.*

Proof of Lemma 4.9.4. For any stationary point $x \in U$, applying Lemma 4.9.2 with $t = 0$ and $x(0) = x$, we have

$$\frac{16}{n}\cdot\lambda_{\min}(ZZ^\top)\cdot\min_{i\in[d]}|u_i v_i|L(x) \leq \|\nabla L(x)\|^2 = 0, \quad \forall t \geq 0.$$

By definition of U, we know that for each $x \in U$, we have that $\min_{i\in[d]} |u_i v_i| > 0$. Further note $\text{rank}(\{z_i\}_{i\in[n]}) = n$, it holds that $\lambda_{\min}(ZZ^\top) > 0$. Thus we conclude $L(x) = 0$.

The other direction is proved by noting that since Γ is the set of local minimizers, each x in Γ must satisfy $\nabla L(x) = 0$. ■

Now, we are ready to prove Lemma 4.5.3.

Proof of Lemma 4.5.3. It suffices to prove gradient flow $\frac{\mathrm{d}x(t)}{\mathrm{d}t} = -\nabla L(x(t))$ converges when $t \to \infty$, as long as $x(0) \in U$. Whenever it converges, it must converge to a stationary point in U. The proof will be completed by noting that all stationary point of L in U belongs to Γ (Lemma 4.9.4).

Below we prove $\lim_{t\to\infty} x(t)$ exists. Denote $C = \frac{16}{n} \cdot \lambda_{\min}(ZZ^\top) \cdot \min_{i\in[d]} |u_i(0)v_i(0)|$, then it follows from Lemma 4.9.2 that

$$\left\|\frac{\mathrm{d}x(t)}{\mathrm{d}t}\right\|_2 = \|\nabla L(x(t))\|_2 \leq \frac{\|\nabla L(x(t))\|_2^2}{\sqrt{CL(x(t))}} = \frac{-\frac{\mathrm{d}L(x(t))}{\mathrm{d}t}}{\sqrt{L(x(t))}} = -\frac{1}{2\sqrt{C}} \frac{\mathrm{d}\sqrt{L(x(t))}}{\mathrm{d}t}.$$

Thus the total length of the gradient flow trajectory is bounded by

$$\int_{t=0}^{\infty} \left\|\frac{\mathrm{d}x(t)}{\mathrm{d}t}\right\|_2 \mathrm{d}t \leq \int_{t=0}^{\infty} -\frac{1}{2\sqrt{C}} \frac{\mathrm{d}\sqrt{L}(x(t))}{\mathrm{d}dt} \mathrm{d}t \leq \frac{L(x(0))}{2\sqrt{C}},$$

where the last inequality uses that L is non-negative over $\mathbb{R}^D$. Therefore, the GF must converge. ■

4.9.4 Proof of Results in Section 4.5.2

Proof of Lemma 4.5.7. Let $\binom{\hat{u}}{\hat{v}}$ be any optimal solution of (4.5.5), and we define $\hat{w} = \hat{u}^{\odot 2} - \hat{v}^{\odot 2}$, which is also feasible to (4.5.6). By the optimality of w^*, we have

$$\sum_{j=1}^{d} \left(\sum_{i=1}^{n} z_{i,j}^2\right) |w_j^*| \leq \sum_{j=1}^{d} \left(\sum_{i=1}^{n} z_{i,j}^2\right) |\hat{w}_j| \leq \sum_{j=1}^{d} \left(\sum_{i=1}^{n} z_{i,j}^2\right) (\hat{u}_j^2 + \hat{v}_j^2). \tag{4.9.2}$$

Let us write $\psi(w^*) = \binom{\tilde{u}^*}{\tilde{v}^*}$, then $\binom{\tilde{u}^*}{\tilde{v}^*}$ is feasible to (4.5.5). Thus, it follows from the optimality of $(\hat{u}, \hat{v})$ that

$$\sum_{j=1}^{d}\left(\sum_{i=1}^{n} z_{i,j}^2\right)(\hat{u}_j^2 + \hat{v}_j^2) \le \sum_{j=1}^{d}\left(\sum_{i=1}^{n} z_{i,j}^2\right)((\tilde{u}_j^*)^2 + (\tilde{v}_j^*)^2) = \sum_{j=1}^{d}\left(\sum_{i=1}^{n} z_{i,j}^2\right)|w_j^*|. \quad (4.9.3)$$

Combining (4.9.2) and (4.9.3) yields

$$\sum_{j=1}^{d}\left(\sum_{i=1}^{n} z_{i,j}^2\right)(\hat{u}_j^2 + \hat{v}_j^2) = \sum_{j=1}^{d}\left(\sum_{i=1}^{n} z_{i,j}^2\right)|w_j^*| = \sum_{j=1}^{d}\left(\sum_{i=1}^{n} z_{i,j}^2\right)|\hat{u}_j^2 - \hat{v}_j^2| \quad (4.9.4)$$

which implies that $\hat{u}^{\odot 2} - \hat{v}^{\odot 2}$ is also an optimal solution of (4.5.6). Since w^* is the unique optimal solution to (4.5.6), we have $\hat{u}^{\odot 2} - \hat{v}^{\odot 2} = w^*$. Moreover, by (4.9.4), we must have $\hat{u}^{\odot 2} = [w^*]_+$ and $\hat{u}^{\odot 2} = [w^*]_+$, otherwise the equality would not hold. This completes the proof. ∎

Therefore, the unique optimality of (4.5.5) can be reduced to that of (4.5.6). In the sequel, we show that the latter holds for both Boolean and Gaussian random vectors. We divide Lemma 4.5.8 into to Lemma 4.9.6 and 4.9.5 for clarity.

Lemma 4.9.5 (Boolean Case). *Let $z_1, \ldots, z_n \overset{i.i.d.}{\sim} \mathrm{Unif}(\{\pm 1\}^d)$. There exist some constants $C, c > 0$ such that if the sample size n satisfies*

$$n \ge C[\kappa \ln(d/\kappa) + \kappa]$$

then with probability at least $1 - e^{-cn^2}$, the optimal solution of (4.5.5), denoted by $\binom{\hat{u}}{\hat{v}}$, is unique up to sign flips of each coordinate and recovers the groundtruth, i.e., $\hat{u}^{\odot 2} - \hat{v}^{\odot 2} = w^*$.

Proof of Lemma 4.9.5. By the assumption that $z_1, \ldots, z_n \overset{\text{i.i.d.}}{\sim} \mathrm{Unif}(\{\pm 1\}^d)$, we have $\sum_{i=1}^{n} z_{i,j}^2 = n$ for all $j \in [d]$. Then (4.5.6) is equivalent to the following optimization problem:

$$\begin{aligned} \text{minimize} \quad & g(w) = \|w\|_1, \\ \text{subject to} \quad & Zw = Zw^*. \end{aligned} \tag{4.9.5}$$

This model exactly fits the Example 6.2 in [Tro15] with $\sigma = 1$ and $\alpha = 1/\sqrt{2}$. Then applying Equation (4.2) and Theorem 6.3 in [Tro15], (4.9.5) has a unique optimal solution equal to $(u^*)^{\odot 2} - (v^*)^{\odot 2}$ with probability at least $1 - e^{-ch^2}$ for some constant $c > 0$, given that the sample size satisfies

$$n \geq C(\kappa \ln(d/\kappa) + \kappa + h)$$

for some absolute constant $C > 0$. Choosing $h = \frac{n}{2C}$ and then adjusting the choices of C, c appropriately yield the desired result. Finally, applying Lemma 4.5.7 finishes the proof. ■

The Gaussian case requires more careful treatment.

Lemma 4.9.6 (Gaussian Case)**.** *Let $z_1, \ldots, z_n \overset{i.i.d.}{\sim} \mathcal{N}(0, I_d)$. There exist some constants $C, c > 0$ such that if the sample size satisfies*

$$n \geq C\kappa \ln d,$$

then with probability at least $1 - (2d+1)e^{-cn}$, the optimal solution of (4.5.5), denoted by $\binom{\hat{u}}{\hat{v}}$, is unique up to sign flips of each coordinate and recovers the groundtruth, i.e., $\hat{u}^{\odot 2} - \hat{v}^{\odot 2} = w^*$.

Proof of Lemma 4.9.6. Since $z_1, \ldots, z_n \overset{\text{i.i.d.}}{\sim} \mathcal{N}(0, I_d)$, we have

$$\mathbb{P}\left[\sum_{i=1}^{n} z_{i,j}^2 \in [n/2, 3n/2], \forall j \in [d]\right] \geq 1 - 2de^{-cn}$$

for some constant $c > 0$, and we denote this event by $\mathcal{E}_n$. Therefore, on $\mathcal{E}_n$, we have

$$2\sum_{j=1}^{D}(u_j^2 + v_j^2) \leq R(x) \leq 6\sum_{j=1}^{D}(u_j^2 + v_j^2)$$

or equivalently,

$$2(\|u^{\odot 2}\|_1 + \|v^{\odot 2}\|_1) \leq R(x) \leq 6(\|u^{\odot 2}\|_1 + v^{\odot 2}\|_1).$$

Define $w^* = (u^*)^{\odot 2} - (v^*)^{\odot 2}$, and (4.5.6) is equivalent to the following convex optimization problem

$$\begin{aligned} \text{minimize} \quad & g(w) = \frac{4}{n}\sum_{j=1}^{d}\left(\sum_{i=1}^{n} z_{i,j}^2\right)|w_j + w_j^*|, \\ \text{subject to} \quad & Zw = 0. \end{aligned} \tag{4.9.6}$$

The point $w = 0$ is feasible for (4.9.6), and we claim that this is the unique optimal solution when n is large enough. In detail, assume that there exists a non-zero feasible point w for (4.9.6) in the descent cone [Tro15] $\mathcal{D}(g, w^*)$ of g, then

$$\lambda_{\min}(Z; \mathcal{D}(g, w^*)) \leq \frac{\|Zw\|_2}{\|w\|_2} = 0$$

where the equality follows from that w is feasible. Therefore, we only need to show that $\lambda_{\min}(Z; \mathcal{D}(g, x^*))$ is bounded from below for sufficiently large n.

On $\mathcal{E}_n$, it holds that g belongs to the following function class

$$\mathcal{G} = \left\{h : \mathbb{R}^d \to \mathbb{R} \mid h(w) = \sum_{j=1}^{d} \upsilon_j |w_j|, \upsilon \in \Upsilon\right\} \text{ with } \Upsilon = \{\upsilon \in \mathbb{R}^d : \upsilon_j \in [2,6], \forall j \in [d]\}.$$

We identify $g_\upsilon \in \mathcal{G}$ with $\upsilon \in \Upsilon$, then $\mathcal{D}(g, w^*) \subseteq \cup_{\upsilon \in \Upsilon}\mathcal{D}(g_\upsilon, w^*)) := \mathcal{D}_\Upsilon$, which further implies that

$$\lambda_{\min}(Z; \mathcal{D}(g, w^*)) \geq \lambda_{\min}(Z; \mathcal{D}_\Upsilon).$$

Recall the definition of minimum conic singular value [Tro15]:

$$\lambda_{\min}(Z; \mathcal{D}_\Upsilon) = \inf_{p \in \mathcal{D}_\Upsilon \cap \mathcal{S}^{d-1}} \sup_{q \in \mathcal{S}^{n-1}} \langle q, Zp \rangle.$$

where $\mathcal{S}^{n-1}$ denotes the unit sphere in $\mathbb{R}^n$. Applying the same argument as in [Tro15] yields

$$\mathbb{P}\left[\lambda_{\min}(Z; \mathcal{D}_\Upsilon) \geq \sqrt{n-1} - w(\mathcal{D}_\Upsilon) - h\right] \geq 1 - e^{-h^2/2}.$$

Take the intersection of this event with $\mathcal{E}_n$, and we obtain from a union bound that

$$\lambda_{\min}(Z; \mathcal{D}(g, w^*)) \geq \sqrt{n-1} - w(\mathcal{D}_\Upsilon) - h \tag{4.9.7}$$

with probability at least $1 - e^{-h^2/2} - 2de^{-cn}$. It remains to determine $w(\mathcal{D}_\Upsilon)$, which is defined as

$$w(\mathcal{D}_\Upsilon) = \mathbb{E}_{z\sim\mathcal{N}(0,I_d)}\left[\sup_{p \in \mathcal{D}_\Upsilon \cap \mathcal{S}^{d-1}} \langle z, p \rangle\right] = \mathbb{E}_{z\sim\mathcal{N}(0,I_d)}\left[\sup_{\upsilon \in \Upsilon} \sup_{p \in \mathcal{D}(g_\upsilon, x^*) \cap \mathcal{S}^{d-1}} \langle z, p \rangle\right]. \tag{4.9.8}$$

Without loss of generality, we assume that $w^* = (w_1^*, \ldots, w_\kappa^*, 0, \ldots, 0)^\top$ with $w_1^*, \ldots, w_\kappa^* > 0$, otherwise one only needs to specify the signs and the nonzero set of w^* in the sequel. For any $\upsilon \in \Upsilon$ and any $p \in \mathcal{D}(g_\upsilon, w^*) \cap \mathcal{S}^{d-1}$, there exists some $\tau > 0$ such that $g_\upsilon(w^* + \tau \cdot p) \leq g_\upsilon(w^*)$, i.e.,

$$\sum_{j=1}^d \upsilon_j |w_j^* + \tau p_j| \leq \sum_{j=1}^d \upsilon_j |w_j^*|$$

which further implies that

$$\tau \sum_{j=\kappa+1}^{d} \upsilon_j |p_j| \leq \sum_{j=1}^{\kappa} \upsilon_j (|w_j^*| - |w_j^* - \tau p_j|) \leq \tau \sum_{j=1}^{\kappa} \upsilon_j |p_j|$$

where the second inequality follows from the triangle inequality. Then since each $\upsilon_j \in [2, 6]$, it follows that

$$\sum_{j=\kappa+1}^{d} |p_j| \leq 3 \sum_{j=1}^{\kappa} |p_j|.$$

Note that this holds for all $\xi \in \Xi$ simultaneously. Now let us denote $p_{1:\kappa} = (p_1, \ldots, p_\kappa) \in \mathbb{R}^\kappa$ and $p_{(\kappa+1):d} = (p_{\kappa+1}, \ldots, p_d) \in \mathbb{R}^{d-\kappa}$, and similarly for other d-dimensional vectors. Then for all $p \in \mathcal{D}_\Upsilon \cap \mathcal{S}^{d-1}$, by Cauchy-Schwartz inequality, we have

$$\|p_{(\kappa+1):d}\|_1 \leq 3\|p_{1:\kappa}\|_1 \leq 3\sqrt{\kappa}\|p_{1:\kappa}\|_2.$$

Thus, for any $z \in \mathbb{R}^d$ and any $p \in \mathcal{D}_\Upsilon \cap \mathcal{S}^{d-1}$, it follows that

$$\begin{aligned}
\langle z, p \rangle &= \langle z_{1:\kappa}, p_{1:\kappa} \rangle + \langle z_{(\kappa+1):d}, p_{(\kappa+1):d} \rangle \\
&\leq \|z_{1:\kappa}\|_2 \|p_{1:\kappa}\|_2 + \|p_{(\kappa+1):d}\|_1 \cdot \max_{j \in \{\kappa+1, \ldots, d\}} |z_j| \\
&\leq \|z_{1:\kappa}\|_2 \|p_{1:\kappa}\|_2 + 3\sqrt{\kappa}\|p_{1:\kappa}\|_2 \cdot \max_{j \in \{\kappa+1, \ldots, d\}} |z_j| \\
&\leq \|z_{1:\kappa}\|_2 + 3\sqrt{\kappa} \cdot \max_{j \in \{\kappa+1, \ldots, d\}} |z_j|
\end{aligned}$$

where the last inequality follows from the fact that $p \in \mathcal{S}^{d-1}$. Therefore, combine the above inequality with (4.9.8), and we obtain that

$$w(\mathcal{D}_\Upsilon) \leq \mathbb{E}\left[\|z_{1:\kappa}\|_2 + 3\sqrt{\kappa} \cdot \max_{j \in \{\kappa+1, \ldots, d\}} |z_j|\right] \leq \sqrt{\kappa} + 3\sqrt{\kappa} \cdot \mathbb{E}\left[\max_{j \in \{\kappa+1, \ldots, d\}} |z_j|\right]. \tag{4.9.9}$$

where the second inequality follows from the fact that $\mathbb{E}[\|z_{1:\kappa}\|_2] \le \sqrt{\mathbb{E}[\|z_{1:\kappa}\|_2^2]} = \sqrt{\kappa}$. To bound the second term in (4.9.9), applying Lemma 4.9.7, it follows from (4.9.9) that

$$w(\mathcal{D}_\Upsilon) \le \sqrt{\kappa} + 3\sqrt{2\kappa \ln(2(d-\kappa))}. \tag{4.9.10}$$

Therefore, combining (4.9.10) and (4.9.7), we obtain

$$\lambda_{\min}(Z; \mathcal{D}(g, w^*)) \ge \sqrt{n-1} - \sqrt{\kappa} - 3\sqrt{2\kappa \ln(2(d-\kappa))} - h.$$

Therefore, choosing $h = \sqrt{n-1}/2$, as long as n satisfies that $n \ge C(\kappa \ln d)$ for some constant $C > 0$, we have $\lambda_{\min}(Z; \mathcal{D}(g, w^*)) > 0$ with probability at least $1-(2d+1)e^{-cn}$. Finally, the uniqueness of the optimal solution to (4.5.5) in this case follows from Lemma 4.5.7. ■

Lemma 4.9.7. *Let $z \sim \mathcal{N}(0, I_d)$, then it holds that $\mathbb{E}\left[\max_{i\in[d]} |z_i|\right] \le \sqrt{2\ln(2d)}$.*

Proof of Lemma 4.9.7. Denote $M = \max_{i\in[d]} |z_i|$. For any $\lambda > 0$, by Jensen's inequality, we have

$$e^{\lambda\cdot\mathbb{E}[M]} \le \mathbb{E}[e^{\lambda M}] = \mathbb{E}\left[\max_{i\in[d]} e^{\lambda|z_i|}\right] \le \sum_{i=1}^d \mathbb{E}\left[e^{\lambda|z_i|}\right].$$

Note that $\mathbb{E}[e^{\lambda|z_i|}] \le 2\cdot\mathbb{E}[e^{\lambda z_i}]$. Thus, by the expression of the Gaussian moment generating function, we further have

$$e^{\lambda\cdot\mathbb{E}[M]} \le 2\sum_{i=1}^d \mathbb{E}\left[e^{\lambda z_i}\right] = 2de^{\lambda^2/2},$$

from which it follows that

$$\mathbb{E}[M] \le \frac{\ln(2d)}{\lambda} + \frac{\lambda}{2}.$$

Choosing $\lambda = \sqrt{2\ln(2d)}$ yields the desired result. ■

4.9.5 Proof of Lemma 4.5.5

Instead of studying the convergence of the Riemannian gradient flow directly, it is more convenient to consider it in the ambient space $\mathbb{R}^D$. To do so, we define a Lagrangian function $\mathcal{L}(x;\lambda) = R(x) + \sum_{i=1}^n \lambda_i(f_i(x) - y_i)$ for $\lambda \in \mathbb{R}^n$. Based on this Lagrangian, we can continuously extend $\partial\Phi(x)\nabla R(x)$ to the whole space $\mathbb{R}^D$. In specific, we can find a continuous function $F : \mathbb{R}^D \to \mathbb{R}^D$ such that $F(\cdot)|_\Gamma = \partial\Phi(\cdot)\nabla R(\cdot)$. Such an F can be implicitly constructed via the following lemma.

Lemma 4.9.8. *The ℓ_2 norm has a unique minimizer among $\{\nabla_x\mathcal{L}(x;\lambda) \mid \lambda \in \mathbb{R}^n\}$ for any fixed $x \in \mathbb{R}^D$. Thus we can define $F : \mathbb{R}^D \to \mathbb{R}^D$ by $F(x) = \operatorname{argmin}_{g\in\{\nabla_x\mathcal{L}(x;\lambda)\mid\lambda\in\mathbb{R}^n\}} \|g\|_2$. Moreover, it holds that $\langle F(x), \nabla f_i(x)\rangle = 0$ for all $i \in [n]$.*

Proof of Lemma 4.9.8. Fix any $x \in \mathbb{R}^D$. Note that $\{\nabla_x\mathcal{L}(x;\lambda) \mid \lambda \in \mathbb{R}^n\}$ is the subspace spanned by $\{\nabla f_i(x)\}_{i\in[n]}$ shifted by $\nabla R(x)$, thus there is unique minimizer of the ℓ_2 norm in this set. This implies that $F(x) = \operatorname{argmin}_{g\in\{\nabla_x\mathcal{L}(x;\lambda)\mid\lambda\in\mathbb{R}^n\}} \|g\|_2$ is well-defined.

To show the second claim, denote $h(\lambda) = \|\nabla_x\mathcal{L}(x;\lambda)\|_2^2/2$, which is a quadratic function of $\lambda \in \mathbb{R}^n$. Then we have

$$\begin{aligned}
\nabla h(\lambda) &= \begin{pmatrix} \langle \nabla R(x), \nabla f_1(x)\rangle \\ \vdots \\ \langle \nabla R(x), \nabla f_n(x)\rangle \end{pmatrix} + \begin{pmatrix} \sum_{i=1}^n \lambda_i \langle \nabla f_1(x), \nabla f_i(x)\rangle \\ \vdots \\ \sum_{i=1}^n \lambda_i \langle \nabla f_n(x), \nabla f_i(x)\rangle \end{pmatrix} \\
&= \begin{pmatrix} \langle \nabla R(x), \nabla f_1(x)\rangle \\ \vdots \\ \langle \nabla R(x), \nabla f_n(x)\rangle \end{pmatrix} + K(x)\lambda.
\end{aligned}$$

For any λ such that $\nabla_x\mathcal{L}(x;\lambda) = F(x)$, we must have $\nabla h(\lambda) = 0$ by the definition of

$F(x)$, which by the above implies

$$(K(x)\lambda)_i = -\langle \nabla R(x), \nabla f_i(x) \rangle \qquad \text{for all } i \in [n].$$

Therefore, we further have

$$\begin{aligned}\langle F(x), \nabla f_i(x) \rangle &= \langle \nabla R(x), \nabla f_i(x) \rangle + \sum_{j=1}^{n} \lambda_j \langle \nabla f_i(x), \nabla f_j(x) \rangle \\ &= \langle \nabla R(x), \nabla f_i(x) \rangle + (K(x)\lambda)_i = 0\end{aligned}$$

for all $i \in [n]$. This finishes the proof. ∎

Hence, with any initialization $x(0) \in \Gamma$, the limiting flow (4.5.4) is equivalent to the following dynamics

$$\frac{\mathrm{d}x(t)}{\mathrm{d}t} = -\frac{1}{4} F(x(t)). \tag{4.9.11}$$

Thus Lemma 4.5.5 can be proved by showing that the above $x(t)$ converges to x^* as $t \to \infty$. We first present a series of auxiliary results in below.

Lemma 4.9.9 (Implications for $F(x) = 0$)**.** *Let $F : \mathbb{R}^D \to \mathbb{R}^D$ be as defined in Lemma 4.9.8. For any $x = \binom{u}{v} \in \mathbb{R}^D$ such that $F(x) = 0$, it holds that for each $j \in [d]$, either $u_j = 0$ or $v_j = 0$.*

Proof. Since $F(x) = 0$, it holds for all $j \in [d]$ that,

$$\begin{aligned}0 &= \frac{\partial R}{\partial u_j}(x) + \sum_{i=1}^{n} \lambda_i(x) \frac{\partial f_i}{\partial u_j}(x) = 2u_j \left[\frac{4}{n} \sum_{i=1}^{n} z_{i,j}^2 + \sum_{i=1}^{n} \lambda_i(x) z_{i,j} \right], \\ 0 &= \frac{\partial R}{\partial v_j}(x) + \sum_{i=1}^{n} \lambda_i(x) \frac{\partial f_i}{\partial v_j}(x) = 2v_j \left[\frac{4}{n} \sum_{i=1}^{n} z_{i,j}^2 - \sum_{i=1}^{n} \lambda_i(x) z_{i,j} \right].\end{aligned}$$

If there exists some $j \in [d]$ such that $u_j \neq 0$ and $v_j \neq 0$, then it follows from the above two identities that

$$\sum_{i=1}^{n} z_{i,j}^2 = 0$$

which happens with probability 0 in both the Boolean and Gaussian case. Therefore, we must have $u_j = 0$ or $v_j = 0$ for all $j \in [d]$. ■

Lemma 4.9.10. *Let $F : \mathbb{R}^D \to \mathbb{R}^D$ be as defined in Lemma 4.9.8. Then F is continuous on $\mathbb{R}^D$.*

Proof. **Case I.** We first consider the simpler case of any fixed $x^* \in U = (\mathbb{R} \setminus \{0\})^D$, assuming that $K(x^*)$ is full-rank. Lemma 4.9.8 implies that for any $\lambda \in \mathbb{R}^n$ such that $\nabla_x \mathcal{L}(x^*; \lambda) = F(x^*)$, we have

$$K(x^*)\lambda = -[\nabla f_1(x) \dots \nabla f_n(x)]^\top \nabla R(x).$$

Thus such λ is unique and given by

$$\lambda(x^*) = -K(x^*)^{-1}[\nabla f_1(x) \dots \nabla f_n(x)]^\top \nabla R(x).$$

Since $K(x)$ is continuous around x^*, there exists a sufficiently small $\delta > 0$ such that for any $x \in B_\delta(x^*)$, $K(x)$ is full-rank, which further implies that $K(x)^{-1}$ is also continuous in $B_\delta(x)$. Therefore, by the above characterization of λ, we see that $\lambda(x)$ is continuous for $x \in B_\delta(x^*)$, and so is $F(x) = \nabla R(x) + \sum_{i=1}^{n} \lambda_i(x) \nabla f_i(x)$.

Case II. Next, we consider all general $x^* \in \mathbb{R}^D$. Here for simplicity, we reorder the coordinates as $x = (u_1, v_1, u_2, v_2, \dots, u_d, v_d)$ with a slight abuse of notation. Without loss of generality, fix any x^* such that for some $q \in [d]$, $(u_i^*)^2 + (v_i^*)^2 > 0$ for all $i = 1, \dots, q$ and $u_i^* = v_i^* = 0$ for all $i = q+1, \dots, d$. Then $\nabla R(x^*)$ and $\{\nabla f_i(x^*)\}_{i \in [n]}$

only depend on $\{z_{i,j}\}_{i\in[n],j\in[q]}$, and for all $i \in [n]$, it holds that

$$(\nabla R(x^*))_{(2q+1):D} = (\nabla f_i(x^*))_{(2q+1):D} = 0.$$

Note that if we replace $\{\nabla f_i(x)\}_{i\in[n]}$ by any fixed and invertible linear transform of itself, it would not affect the definition of $F(x)$. In specific, we can choose an invertible matrix $Q \in \mathbb{R}^{n\times n}$ such that, for some $q' \in [q]$, $(\tilde{z}_1, \dots, \tilde{z}_n) = (z_1, \dots, z_n)Q$ satisfies that $\{\tilde{z}_{i,1:q}\}_{i\in[q']}$ is linearly independent and $\tilde{z}_{i,1:q} = 0$ for all $i = q'+1, \dots, n$. We then consider $[\nabla \tilde{f}_1(x), \dots, \nabla \tilde{f}_n(x)] = [\nabla f_1(x), \dots, \nabla f_n(x)]Q$ and the corresponding $F(x)$. For notational simplicity, we assume that Q can be chosen as the identity matrix, so that $(z_1, \dots, z_n)$ itself satisfies the above property, and we repeat it here for clarity

$$\{z_{i,1:q}\}_{i\in[q']} \text{ is linearly independent and } \tilde{z}_{i,1:q} = 0 \text{ for all } i = q'+1, \dots, n. \tag{4.9.12}$$

This further implies that

$$(\nabla f_i(x))_{1:(2q)} = 0, \quad \text{for all } i \in \{q'+1, \dots, n\} \text{ and } x \in \mathbb{R}^D. \tag{4.9.13}$$

In the sequel, we use λ for n-dimensional vectors and $\bar{\lambda}$ for q'-dimensional vectors. Denote[2]

$$\lambda(x) \in \operatorname*{argmin}_{\lambda\in\mathbb{R}^n} \left\| \nabla R(x) + \sum_{i=1}^{n} \lambda_i \nabla f_i(x) \right\|_2,$$
$$\bar{\lambda}(x) \in \operatorname*{argmin}_{\bar{\lambda}\in\mathbb{R}^{q'}} \left\| \left(\nabla R(x) + \sum_{i=1}^{q'} \bar{\lambda}_i \nabla (f_i(x)) \right)_{1:(2q)} \right\|_2.$$

2. We do not care about the specific choice of $\lambda(x)$ or $\bar{\lambda}(x)$ when there are multiple candidates, and we only need their properties according to Lemma 4.9.8, so they can be arbitrary. Also, the minimum of ℓ_2-norm of an affine space can always be attained so argmin exists.

Then due to (4.9.12) and (4.9.13), we have

$$\left\|\left(\nabla R(x^*)+\sum_{i=1}^{q'}\bar{\lambda}_i(x^*)\nabla f_i(x^*)\right)_{1:(2q)}\right\|_2=\left\|\nabla R(x^*)+\sum_{i=1}^{n}\lambda_i(x)\nabla f_i(x^*)\right\|_2=\|F(x^*)\|_2. \tag{4.9.14}$$

On the other hand, for any $x\in\mathbb{R}^D$, by (4.9.13), we have

$$\begin{aligned}\left\|\left(\nabla R(x)+\sum_{i=1}^{q'}\bar{\lambda}_i(x)\nabla f_i(x)\right)_{1:(2q)}\right\|_2&=\min_{\lambda\in\mathbb{R}^n}\left\|\left(\nabla R(x)+\sum_{i=1}^{n}\lambda_i(x)\nabla f_i(x)\right)_{1:(2q)}\right\|_2\\&\leq\left\|\left(\nabla R(x)+\sum_{i=1}^{n}\lambda_i(x)\nabla f_i(x)\right)_{1:(2q)}\right\|_2=\|F_{1:(2q)}(x)\|_2\\&\leq\|F(x)\|_2\leq\left\|\nabla R(x)+\sum_{i=1}^{n}\lambda_i(x^*)\nabla f_i(x)\right\|_2\end{aligned} \tag{4.9.15}$$

where the first and third inequalities follow from the definition of $F(x)$. Let $x\to x^*$, by the continuity of $\nabla R(x)$ and $\{\nabla f_i(x)\}_{i\in[n]}$, we have

$$\lim_{x\to x^*}\left\|\nabla R(x)+\sum_{i=1}^{n}\lambda_i(x^*)\nabla f_i(x)\right\|_2=\left\|\nabla R(x^*)+\sum_{i=1}^{n}\lambda_i(x^*)\nabla f_i(x^*)\right\|_2 \tag{4.9.16}$$

Denote $\tilde{K}(x)=(\tilde{K}_{ij}(x))_{(i,j)\in[q']^2}=(\langle\nabla f_i(x)_{1:(2q)},\nabla f_i(x)_{1:(2q)}\rangle)_{(i,j)\in[q']^2}$. By applying the same argument as in **Case I**, since $\tilde{K}(x^*)$ is full-rank, it also holds that $\lim_{x\to x^*}\bar{\lambda}(x)=\bar{\lambda}(x^*)$, and thus

$$\lim_{x\to x^*}\left\|\left(\nabla R(x)+\sum_{i=1}^{q'}\bar{\lambda}_i(x)\nabla f_i(x)\right)_{1:(2q)}\right\|_2=\left\|\left(\nabla R(x)+\sum_{i=1}^{q'}\bar{\lambda}_i(x^*)\nabla f_i(x^*)\right)_{1:(2q)}\right\|_2. \tag{4.9.17}$$

Combing (4.9.14), (4.9.15), (4.9.16) and (4.9.17) yields

$$\lim_{x\to x^*}\|F_{1:(2q)}(x)\|_2=\lim_{x\to x^*}\min_{\lambda\in\mathbb{R}^n}\left\|\left(\nabla R(x)+\sum_{i=1}^{n}\lambda_i\nabla f_i(x)\right)_{1:(2q)}\right\|_2=\|F(x^*)\|_2. \tag{4.9.18}$$

Moreover, since $\|F_{(2q+1):D}(x)\|_2 = \sqrt{\|F(x)\|_2^2 - \|F_{1:(2q)}(x)\|_2^2}$, we also have

$$\lim_{x \to x^*} \|F_{(2q+1):D}(x)\|_2 = 0. \tag{4.9.19}$$

It then remains to show that $\lim_{x\to x^*} F_{1:(2q)}(x) = F_{1:(2q)}(x^*)$, which directly follows from $\lim_{x\to x^*} \lambda_{1:q'}(x) = \lambda_{1:q'}(x^*) = \bar{\lambda}(x^*)$.

Now, for any $\epsilon > 0$, due to the convergence of $\bar{\lambda}(x)$ and that $\tilde{K}(x^*) \succ 0$, we can pick a sufficiently small δ_1 such that for some constant $\alpha > 0$ and all $x \in B_{\delta_1}(x^*)$, it holds that $\|\bar{\lambda}(x) - \bar{\lambda}(x^*)\|_2 \le \epsilon/2$ and

$$\left\| \left(\nabla R(x) + \sum_{i=1}^{q'} \bar{\lambda}_i \nabla f_i(x) \right)_{1:(2q)} \right\|_2^2 \ge \left\| \left(\nabla R(x) + \sum_{i=1}^{q'} \bar{\lambda}_i(x) \nabla f_i(x) \right)_{1:(2q)} \right\|_2^2 + \alpha \|\bar{\lambda} - \bar{\lambda}(x)\|_2^2. \tag{4.9.20}$$

for all $\bar{\lambda} \in \mathbb{R}^p$, where the inequality follows from the strong convexity. Meanwhile, due to (4.9.13), we have

$$\begin{aligned}
\lim_{x\to x^*} \left\| \left(\nabla R(x) + \sum_{i=1}^{q'} \lambda_i(x) \nabla f_i(x) \right)_{1:(2q)} \right\|_2 &= \lim_{x\to x^*} \left\| \left(\nabla R(x) + \sum_{i=1}^{n} \lambda_i(x) \nabla f_i(x) \right)_{1:(2q)} \right\|_2 \\
&= \left\| \left(\nabla R(x) + \sum_{i=1}^{q'} \bar{\lambda}_i(x^*) \nabla f_i(x^*) \right)_{1:(2q)} \right\|_2 \\
&= \lim_{x\to x^*} \left\| \left(\nabla R(x) + \sum_{i=1}^{q'} \bar{\lambda}_i(x) \nabla f_i(x) \right)_{1:(2q)} \right\|_2 .
\end{aligned}$$

where the second equality follows from (4.9.18) and the second equality is due to (4.9.17). Therefore, we can pick a sufficiently small δ_2 such that

$$\left\| \left(\nabla R(x) + \sum_{i=1}^{q'} \lambda_i(x) \nabla f_i(x) \right)_{1:(2q)} \right\|_2 \le \left\| \left(\nabla R(x) + \sum_{i=1}^{q'} \bar{\lambda}_i(x) \nabla f_i(x) \right)_{1:(2q)} \right\|_2 + \frac{\alpha \epsilon^2}{4} \tag{4.9.21}$$

for all $x \in B_{\delta_2}(x^*)$. Setting $\delta = \min(\delta_1, \delta_2)$, it follows from (4.9.20) and (4.9.21) that

$$\|\lambda_{1:q'}(x) - \bar{\lambda}(x)\|_2 \le \frac{\epsilon}{2}, \quad \text{for all } x \in B_\delta(x^*).$$

Recall that we already have $\|\bar{\lambda}(x) - \bar{\lambda}(x^*)\| \le \epsilon/2$, and thus

$$\|\lambda_{1:q'}(x) - \lambda(x^*)_{1:q'}\|_2 = \|\lambda_{1:q'}(x) - \bar{\lambda}(x^*)\|_2 \le \|\lambda_{1:q'}(x) - \bar{\lambda}(x)\|_2 + \|\bar{\lambda}(x) - \bar{\lambda}(x^*)\|_2 \le \epsilon$$

for all $x \in B_\delta(x^*)$. Therefore, we see that $\lim_{x \to x^*} \lambda_{1:q'}(x) = \lambda(x^*)_{1:q'}$.

Finally, it follows from the triangle inequality that

$$\begin{aligned}
&\|F(x) - F(x^*)\|_2 \\
\le& \left\| \left(F(x) - F(x^*) \right)_{1:(2q)} \right\|_2 + \|F_{(2q+1):D}(x)\|_2 + \underbrace{\|F_{(2q+1):D}(x^*)\|_2}_{=0} \\
=& \left\| \left(\nabla R(x) + \sum_{i=1}^{q'} \lambda_i(x) \nabla f_i(x) - \nabla R(x^*) - \sum_{i=1}^{q'} \lambda_i(x^*) \nabla f_i(x^*) \right)_{1:(2q)} \right\|_2 + \|F_{(2q+1):D}(x)\|_2 \\
\le& \left\| \sum_{i=1}^{q'} \lambda_i(x) \nabla f_i(x) - \lambda_i(x^*) \nabla f_i(x^*) \right\|_2 + \|\nabla R(x) - \nabla R(x^*)\|_2 + \|F_{(2q+1):D}(x)\|_2
\end{aligned}$$

where, as $x \to x^*$, the first term vanishes by the convergence of $\lambda_{1:q'}(x)$ and the continuity of each $\nabla f_i(x)$, the second term converges to 0 by the continuity of $\nabla R(x)$ and the third term vanishes by (4.9.19). Therefore, we conclude that

$$\lim_{x \to x^*} F(x) = F(x^*),$$

that is, F is continuous. ∎

Lemma 4.9.11. *For any initialization $Y(0) \in \Gamma$, the Riemmanian Gradient Flow* (4.5.4) *(or equivalently,* (4.9.11)*) has a unique solution on* $[0, \infty)$.

Proof of Lemma 4.9.11. We can view (4.5.4) as an ODE defined on open set $U \subseteq \mathbb{R}^D$.

Since $\partial\Phi(\cdot)\nabla R(\cdot)$ is continuously differentiable, we know that (4.5.4) has a unique solution in each interval in which it has a solution. Let $[0, T)$ be the right maximal interval of existence of the unique solution of (4.5.4), $Y(t) = \binom{u(t)}{v(t)}$ and suppose $T \neq \infty$. Since $R(Y(t))$ is monotone decreasing, thus $R(Y(t))$ is upper bounded by $R(Y(0))$ and therefore $\|\nabla R(Y(t))\|$ is also upper bounded. Since $\left\|\frac{dY(t)}{dt}\right\|_2 \leq \|\nabla R(Y(t))\|_2$ for any $t < T$, the left limit $Y(T-) := \lim_{\tau \to T^-} Y(\tau)$ must exist. By Corollary 1, [Per01], $x(T-)$ belongs to boundary of U, i.e., $u_j(T-) = 0$ or $v_j(T-) = 0$ for some $j \in [d]$ by Lemma 4.9.9. By the definition of the Riemannian gradient flow in (4.5.4), we have

$$\begin{aligned}\frac{\mathrm{d}}{\mathrm{d}t}(u_j(t)v_j(t)) &= \begin{pmatrix} v_j(t)e_j^\top & u_j(t)e_j^\top \end{pmatrix} \frac{\mathrm{d}x(t)}{\mathrm{d}t} \\ &= -\frac{1}{4}\begin{pmatrix} v_j(t)e_j^\top & u_j(t)e_j^\top \end{pmatrix} F(x(t)).\end{aligned}$$

By the expression of $F(x(t)) = \nabla R(x(t)) + \sum_{i=1}^n \lambda_i(x(t))\nabla f_i(x(t))$, we then have

$$\begin{aligned}\frac{\mathrm{d}}{\mathrm{d}t}(u_j(t)v_j(t)) = &-\left[\frac{2}{n}\sum_{i=1}^n z_{i,j}^2 + \frac{1}{2}\sum_{i=1}^n \lambda_i(x(t))z_{i,j}\right]u_j(t)v_j(t) \\ &- \left[\frac{2}{n}\sum_{i=1}^n z_{i,j}^2 - \frac{1}{2}\sum_{i=1}^n \lambda_i(x(t))z_{i,j}\right]u_j(t)v_j(t) \\ = &-\left(\frac{4}{n}\sum_{i=1}^n z_{i,j}^2\right)u_j(t)v_j(t).\end{aligned}$$

Denote $s_j = \frac{4}{n}\sum_{i=1}^n z_{i,j}^2$. It follows that $|u_j(t)v_j(t)| = |u_j(0)v_j(0)|e^{-s_j t}$ for all $t \in [0, T)$. Taking the limit we have $|u_j(T-)v_j(T-)| \geq |u_j(0)v_j(0)|e^{-s_j T} > 0$. Contradiction with $T \neq \infty$! ∎

Before showing that F satisfies the PL condition, we need the following two intermediate results. Given two points u and v in $\mathbb{R}^d$, we say u weakly dominate v (written as $u \leq v$) if and only if $u_i \leq v_i$, for all $i \in [d]$. Given two subsets A and B of $\mathbb{R}^D$, we say A weakly dominates B if and only if for any point v in B, there exists a point $u \in A$ such that $u \leq v$.

Lemma 4.9.12. *For some $q \in [D]$, let S be any q-dimensional subspace of $\mathbb{R}^D$ and $P = \{u \in \mathbb{R}^D \mid u_i \geq 0, \forall i \in [D]\}$. Let $u_\star$ be an arbitrary point in P and $Q = P \cap (u_\star + S)$. Then there exists a radius $r > 0$, such that $B_r^1(0) \cap Q$ is non-empty and weakly dominates Q, where $B_r^1(0)$ is the ℓ_1-norm ball of radius r centered at 0.*

As a direct implication, for any continuous function $f : P \to \mathbb{R}$, which is coordinate-wise non-decreasing, $\min_{x \in U} f(x)$ can always be achieved.

Proof of Lemma 4.9.12. We will prove by induction on the environment dimension D. For the base case of $D = 1$, either $S = \{0\}$ or $S = \mathbb{R}$, and it is straight-forward to verify the desired for both scenarios.

Suppose the proposition holds for $D - 1$, below we show it holds for D. For each $i \in [D]$, we apply the proposition with $D - 1$ to $Q \cap \{u \in P \mid u_i = 0\}$ (which can be seen as a subset of $\mathbb{R}^{D-1}$), and let r_i be the corresponding ℓ_1 radius. Set $r = \max_{i \in [D]} r_i$, and we show that choosing the radius to be r suffices.

For any $v \in Q$, we take a random direction in S, denoted by ω. If $\omega \geq 0$ or $\omega \leq 0$, we denote by y the first intersection (i.e., choosing the smallest λ) between the line $\{v - \lambda|\omega|\}_{\lambda \geq 0}$ and the boundary of U, i.e., $\cup_{i=1}^D \{z \in \mathbb{R}^D \mid z_i = 0\}$. Clearly $y \leq v$. By the induction hypothesis, there exists a $u \in B_r^1(0) \cap Q$ such that $u \leq y$. Thus $u \leq v$ and meets our requirement.

If ω has different signs across its coordinates, we take y_1, y_2 to be the first intersections of the line $\{v - \lambda|\omega|\}_{\lambda \in \mathbb{R}}$ and the boundary of U in directions of $\lambda > 0$ and $\lambda < 0$, respectively. Again by the induction hypothesis, there exist $u_1, u_2 \in B_r^1(0) \cap Q$ such that $u_1 \leq y_1$ and $u_2 \leq y_2$. Since v lies in the line connecting u_1 and u_2, there exists some $h \in [0, 1]$ such that $v = (1 - h)u_1 + hu_2$. It then follows that $(1 - h)u_1 + hu_2 \leq (1 - h)y_1 + hy_2 = v$. Now since Q is convex, we have $(1 - h)u_1 + hu_2 \in Q$, and by the triangle inequality it also holds that $\|(1 - h)u_1 + hu_2\|_1 \leq r$, so $(1 - h)u_1 + hu_2 \in B_r^1(0) \cap Q$. Therefore, we conclude that $B_r^1(0) \cap Q$ weakly dominates Q, and thus the proposition holds for D. This completes the proof by induction. ■

Lemma 4.9.13. *For some $q \in [D]$, let S be any q-dimensional subspace of $\mathbb{R}^D$ and $P = \{u \in \mathbb{R}^D \mid u_i \geq 0, \forall i \in [D]\}$. Let $u_\star$ be an arbitrary point in P and $Q = P \cap (u_\star + S)$. Then there exists a constant $c \in (0, 1]$ such that for any sufficiently small radius $r > 0$, $c \cdot Q$ weakly dominates $P \cap (u_\star + S + B_r^2(0))$, where $B_r^2(0)$ is the ℓ_2-norm ball of radius r centered at 0.*

Proof of Lemma 4.9.13. We will prove by induction on the environment dimension D. For the base case of $D = 1$, either $S = \{0\}$ or $S = \mathbb{R}$. $S = \mathbb{R}$ is straight-forward; for the case $S = \{0\}$, we just need to ensure $c|u_\star| \leq |u_\star| - r$, and it suffices to pick $r = |u_\star|$ and $c = 0.5$.

Suppose the proposition holds for $D - 1$, below we show it holds for D. For each $i \in [D]$, we first consider the intersection between $P \cap (u_\star + S + B_r^2(0))$ and $H_i := \{u \in \mathbb{R}^D \mid u_i = 0\}$. Let u_i be an arbitrary point in $P \cap (u_\star + S) \cap H_i$, then $P \cap (u_\star + S) \cap H_i = P \cap (u_i + S) \cap H_i = P \cap (u_i + S \cap H_i)$. Furthermore, there exists $\{\alpha_i\}_{i \in [D]}$ which only depends on S and satisfies $P \cap (u^* + S + B_r^2(0)) \cap H_i \subset P \cap (u_i + S \cap H_i + B_{\alpha_i r}^2(0) \cap H_i)$. Applying the induction hypothesis to $P \cap (u_i + S \cap H_i + B_{\alpha_i r}^2(0) \cap H_i)$, we know there exists a $c > 0$ such that for sufficiently small r, $c(P \cap (u_\star + S) \cap H_i) = c(P \cap (u_i + S \cap H_i))$ weakly dominates $P \cap (u_i + S \cap H_i + B_{\alpha_i r}^2(0) \cap H_i)$.

For any point v in Q and any $z \in B_r^2(0)$, we take a random direction in S, denoted by ω. If $\omega \geq 0$ or $\omega \leq 0$, we denote by y the first intersection between $\{v + z - \lambda|\omega|\}_{\lambda \geq 0}$ and the boundary of U. Clearly $y \leq v$. Since $y \in P \cap (u_\star + S + B_r^2(0)) \cap H_i \subset P \cap (u_i + S \cap H_i + B_{\alpha_i r}^2(0) \cap H_i)$, by the induction hypothesis, there exists a $u \in c(P \cap (u_\star + S) \cap H_i)$ such that $u \leq y$. Thus $z \leq v + z$ and $z \in c(P \cap (u_\star + S)) = c \cdot Q$.

If ω has different signs across its coordinates, we take y_1, y_2 to be the first intersections of the line $\{v + z - \lambda|\omega|\}_{\lambda \in \mathbb{R}}$ and the boundary of U in directions of $\lambda > 0$ and $\lambda < 0$, respectively. By the induction hypothesis, there exist $u_1, u_2 \in c \cdot Q$ such that $u_1 \leq y_1$ and $u_2 \leq y_2$. Since $v + z$ lies in the line connecting u_1 and u_2, there exists some $h \in [0,1]$ such that $v + z = (1-h)y_1 + hy_2$. It then follows that $(1-h)u_1 + hu_2 \le (1-h)y_1 + hy_2 = v + z$. Since Q is convex, we have $(1-h)u_1 + hu_2 \in cQ$. Therefore, we conclude that $cQ \cap Q$ weakly dominates $P \cap (u_\star + S + B_r^2(0))$ for all sufficiently small r, and thus the proposition holds for D. This completes the proof by induction. ■

Lemma 4.9.14. *(Polyak-Łojasiewicz condition for F.) For any x^* such that $L(x^*) = 0$, i.e., $x^* \in \overline{\Gamma}$, there exist a neighbourhood U' of x^* and a constant $c > 0$, such that $\|F(x)\|_2^2 \ge c \cdot \max(R(x) - R(x^*), 0)$ for all $x \in U' \cap \overline{\Gamma}$. Note this requirement is only non-trivial when $\|F(x^*)\|_2 = 0$ since F is continuous.*

Proof of Lemma 4.9.14. It suffices to show the PL condition for $\{x \mid F(x) = 0\}$. We need to show for any x^* satisfying $F(x^*) = 0$, there exist some $\varepsilon > 0$ and $C > 0$, such that for all $x \in \overline{\Gamma} \cap B_\epsilon^2(x^*)$ with $R(x) > R(x^*)$, it holds that $\|F(x)\|_2^2 \ge C(R(x) - R(x^*))$.

Canonical Case. We first prove the case where $x = \binom{u}{v}$ itself is a canonical parametrization of $w = u^{\odot 2} - v^{\odot 2}$, i.e., $u_j v_j = 0$ for all $j \in [d]$. Since x^* satisfies $\nabla F(x^*) = 0$, by Lemma 4.9.9, we have $x^* = \psi(w^*)$ where $w^* = (u^*)^{\odot 2} - (v^*)^{\odot 2}$. In this case, we can rewrite both R and F as functions of $w \in \mathbb{R}^d$. In detail, we define $R'(w) = R(\psi(w))$ and $F'(w) = F(\psi(w))$ for all $w \in \mathbb{R}^d$. For any w in a sufficiently small neighbourhood of w^*, it holds that $\operatorname{sign}(w_j) = \operatorname{sign}(w_j^*)$ for all $j \in [q]$. Below we show that for each possible sign pattern of $w_{(q+1):d}$, there exists some constant C which admits the PL condition in the corresponding orthant. Then we take the minimum of all C from different orthant and the proof is completed. W.L.O.G., we assume that $w_j \ge 0$, for all $j = q+1 \ldots, d$.

We temporarily reorder the coordinates as $x = (u_1, v_1, u_2, v_2, \ldots, u_d, v_d)^\top$. Recall that $Z = [z_1, \dots, z_n]^\top$ is a n-by-d matrix, and we have

$$\|F'(w)\|_2^2 = \min_{\lambda\in\mathbb{R}^n} \langle (a - \mathrm{sign}(w) \odot Z^\top \lambda)^{\odot 2}, |w|\rangle,$$

where $a = \frac{8}{n}\sum_{i=1}^n z_i^{\odot 2} \in \mathbb{R}^d$. Since $F(x^*) = 0$, there must exist $\lambda^* \in \mathbb{R}^n$, such that the first $2q$ coordinates of $\nabla R(x^*) + \sum_{i=1}^n \lambda_i^* \nabla f_i(x^*)$ are equal to 0. As argued in the proof of Lemma 4.9.10, we can assume the first q' rows of Z are linear independent on the first q coordinates for some $q' \in [q]$. In other words, Z can be written as $\begin{bmatrix} Z_A & Z_B \\ 0 & Z_D \end{bmatrix}$ where $Z_A \in \mathbb{R}^{q'\times q}$. We further denote $\lambda_a := \lambda_{1:q'}$, $\lambda_b := \lambda_{(q'+1):n}$, $a_a := a_{1:q}$ and $a_b := a_{(q+1):d}$, $w_a := w_{1:q}$ and $w_b := w_{(q+1):d}$ for convenience, then we have

$$\|F'(w)\|_2^2 = \min_{\lambda\in\mathbb{R}^n} \langle (a_a + \mathrm{sign}(w_a) \odot Z_A^\top \lambda_a)^{\odot 2}, |w_a|\rangle + \langle (a_b + Z_B^\top \lambda_a + Z_D^\top \lambda_b)^{\odot 2}, w_b\rangle. \tag{4.9.22}$$

Since every w in $\overline{\Gamma}$ is a global minimizer, $R'(w) = R'(w) + \sum_{i=1}^n \lambda_i^*(z_i^\top w - y_i) := g^\top w + R'(w_*)$, where $g = \mathrm{sign}(w) \odot a + Z^\top \lambda^*$. Similarly we define $g_a := g_{1:q}$ and $g_b := g_{(q+1):d}$. It holds that $g_a = 0$ and we assume $Z_D g_b = 0$ without loss of generality, because this can always be done by picking suitable λ_i^* for $i = q'+1, \dots, n$. (We have such freedom on $\lambda^*_{q'+1:n}$ because they doesn't affect the first $2q$ coordinates.)

We denote $\lambda_a - \lambda_a^*$ by $\Delta\lambda_a$, then since $0 = g_a = \mathrm{sign}(w_a) \odot Z_A^\top \lambda_a^* + a_a$, we further have

$$\begin{aligned}\langle (a_a + \mathrm{sign}(w_a) \odot Z_A^\top \lambda_a)^{\odot 2}, |w_a|\rangle &= \langle (a_a + \mathrm{sign}(w_a) \odot Z_A^\top \lambda_a^* + \mathrm{sign}(w_a) \odot Z_A^\top \Delta\lambda_a)^{\odot 2}, |w_a|\rangle \\ &= \langle (\mathrm{sign}(w_a) \odot Z_A^\top \Delta\lambda_a)^{\odot 2}, |w_a|\rangle.\end{aligned}$$

On the other hand, we have $g_b = \mathrm{sign}(w_b) \odot a_b + Z_B^\top \lambda_a^* + Z_D^\top \lambda_b^* = a_b + Z_B^\top \lambda_a^* + Z_D^\top \lambda_b^*$ by the assumption that each coordinate of w_b is non-negative. Combining this with the above identity, we can rewrite Equation (4.9.22) as:

$$\|F'(w)\|_2^2 = \min_{\lambda \in \mathbb{R}^D} \langle (Z_A^\top \Delta\lambda_a)^{\odot 2}, |w_a|\rangle + \langle (g_b + Z_B^\top \Delta\lambda_a + Z_D^\top \lambda_b)^{\odot 2}, w_b\rangle. \tag{4.9.23}$$

Now suppose $R'(w) - R'(w^*) = g_b^\top w_b = \delta$ for some sufficiently small δ (which can be controlled by ε). We will proceed in the following two cases separately.

- **Case I.1**: $\|\Delta\lambda_a\|_2 = \Omega(\sqrt{\delta})$. Since Z_A has full row rank, $\left\|(Z_A^\top \Delta\lambda_a)^{\odot 2}\right\|_1 = \left\|(Z_A^\top \Delta\lambda_a)\right\|_2^2 \geq \|\Delta\lambda_a\|_2^2 \lambda_{\min}^2(Z_A)$ is lower-bounded. On the other hand, we can choose ε small enough such that $\forall i \in [q] |(w_a)_i^2| \geq \frac{1}{2}(w_a^*)_i^2$. Thus the first term of Equation (4.9.23) is lower bounded by $\|\Delta\lambda_a\|_2^2 \lambda_{\min}^2(Z_A) \cdot \min_{i\in[q]} \frac{1}{2}(w_a^*)_i^2 = \Omega(\delta) = \Omega(R'(w) - R'(w^*))$.

- **Case I.2**: $\|\Delta\lambda_a\|_2 = O(\sqrt{\delta})$. Let $u = g_b + Z_B^\top \Delta\lambda_a + Z_D^\top \lambda_b$, then we have $u \in S + B^2_{c\sqrt{\delta}}(0)$ for some constant $c > 0$, where $S = \{g_b + Z_D^\top \lambda_b \mid \lambda_b \in \mathbb{R}^{n-q'}\}$. By Lemma 4.9.12, there exists some constant $c_0 \geq 1$, such that $\frac{1}{c_0} \cdot S$ weakly dominates $S + B^2_{c\sqrt{\delta}}(0)$. Thus we have $\|F'(w)\|_2^2 \geq \inf_{u \in S + B_{c\sqrt{\delta}}(0)} \langle u^{\odot 2}, w_b\rangle \geq \inf_{u \in \frac{1}{c_0}\cdot S} \langle s^{\odot 2}, w_b\rangle$, where the last step is because each coordinate of w_b is non-negative.

  Let A be the orthogonal complement of $\text{span}(Z_D, g_b)$, i.e., the spanned space of columns of Z_D and g_b, we know $w_b \in \frac{\delta}{\|g_b\|_2^2} g_b + A$, since $Z_D w_b = Z_D w_*^2 = 0$ and $g_b^\top w_b = \delta$. Therefore,

$$\begin{aligned}\inf_{w: R'(w) - R'(w^*) = \delta > 0} \frac{\|F'(w)\|_2^2}{R'(w) - R'(w^*)} &\geq \inf_{w_b: R'(w) - R'(w^*) = \delta > 0} \inf_{u \in \frac{1}{c_0}\cdot S} \langle u^{\odot 2}, \frac{w_b}{\delta}\rangle \\ &\geq \frac{1}{c_0^2} \inf_{w_b \in \frac{\delta}{\|g_b\|_2^2} g_b + A, w_b \geq 0, u \in S} \langle u^{\odot 2}, w_b\rangle. \end{aligned} \tag{4.9.24}$$

  Note $\langle u^{\odot 2}, w_b\rangle$ is a monotone non-decreasing function in the first joint orthant, i.e., $\{(u, w_b) \in \mathbb{R}^d \times \mathbb{R}^{d-q'} \mid u \geq 0, w_b \geq 0\}$, thus by Lemma 4.9.13 the infinimum can be achieved by some finite (u, w_b) in the joint first orthant. Applying the same argument to each other orthant of $u \in \mathbb{R}^d$, we conclude that the right-hand-side of (4.9.24) can be achieved.

On the other hand, we have $u^\top w_b = \delta > 0$ for all $w_b \in \frac{\delta}{\|g_b\|_2^2} g_b + A$ and $u \in S$, by $Z_D g_b = 0$ and the definition of A. This implies there exists at least one $i \in [d - q']$ such that $w_{2,i} u_i > 0$, which further implies $\langle u^{\odot 2}, w_b \rangle > 0$. Therefore, we conclude that $\|F'(w)\|_2^2 = \Omega(R'(w) - R'(w_0))$.

General Case. Next, for any general $x = \binom{u}{v}$, we define $w = u^{\odot 2} - v^{\odot 2}$ and $m = \min\{u^{\odot 2}, v^{\odot 2}\}$, where min is taken coordinate-wise. Then we can rewrite $\|F(x)\|_2^2$ as

$$
\begin{aligned}
\|F(x)\|_2^2 &= \min_{\lambda \in \mathbb{R}^n} \left\| \left(\begin{bmatrix} a \\ a \end{bmatrix} + \begin{bmatrix} Z \\ -Z \end{bmatrix} \lambda \right) \odot \begin{bmatrix} u \\ v \end{bmatrix} \right\|_2^2 \\
&= \min_{\lambda \in \mathbb{R}^n} \left\| \left(\begin{bmatrix} a \\ a \end{bmatrix} + \begin{bmatrix} Z \\ -Z \end{bmatrix} \lambda \right)^{\odot 2} \odot \begin{bmatrix} u^{\odot 2} \\ v^{\odot 2} \end{bmatrix} \right\|_1 \\
&= \min_{\lambda \in \mathbb{R}^n} \left\| \left(\begin{bmatrix} a \\ a \end{bmatrix} + \begin{bmatrix} Z \\ -Z \end{bmatrix} \lambda \right)^{\odot 2} \odot \left(\psi(w)^{\odot 2} + \begin{bmatrix} m \\ m \end{bmatrix} \right) \right\|_1 \\
&\geq \min_{\lambda \in \mathbb{R}^n} \left\| \left(\begin{bmatrix} a \\ a \end{bmatrix} + \begin{bmatrix} Z \\ -Z \end{bmatrix} \lambda \right)^{\odot 2} \odot \psi(w)^{\odot 2} \right\|_1 + \min_{\lambda \in \mathbb{R}^n} \left\| \left(\begin{bmatrix} a \\ a \end{bmatrix} + \begin{bmatrix} Z \\ -Z \end{bmatrix} \lambda \right)^{\odot 2} \odot \begin{bmatrix} m \\ m \end{bmatrix} \right\|_1 \\
&= \min_{\lambda \in \mathbb{R}^n} \left\| \left(\begin{bmatrix} a \\ a \end{bmatrix} + \begin{bmatrix} Z \\ -Z \end{bmatrix} \lambda \right) \odot \psi(w) \right\|_2^2 + \min_{\lambda \in \mathbb{R}^n} \left\| \left(\begin{bmatrix} a \\ a \end{bmatrix} + \begin{bmatrix} Z \\ -Z \end{bmatrix} \lambda \right) \odot \begin{bmatrix} \sqrt{m} \\ \sqrt{m} \end{bmatrix} \right\|_2^2 .
\end{aligned}
$$

Then applying the result for the previous case yields the following for some constant $C \in (0,1)$:

$$\begin{aligned}
\|F(x)\|_2^2 &\geq C(R(\psi(w)) - R(\psi(w^*)) + \min_{\lambda\in\mathbb{R}^n} \left\| \left(\begin{bmatrix} a \\ a \end{bmatrix} + \begin{bmatrix} Z \\ -Z \end{bmatrix} \lambda \right) \odot \begin{bmatrix} \sqrt{m} \\ \sqrt{m} \end{bmatrix} \right\|_2^2 \\
&= C(R(\psi(w)) - R(x^*) + 2\langle a^{\odot 2}, m\rangle \\
&\geq C(R(\psi(w)) - R(x^*) + 2 \min_{i\in[d]} a_i \langle a, m\rangle \\
&= C(R(\psi(w)) - R(x^*) + \min_{i\in[d]} a_i (R(x) - R(\psi(w))) \\
&\geq \min\left\{C, \min_{i\in[d]} a_i\right\} (R(x) - R(x^*)),
\end{aligned}$$

where the first equality follows from the fact that $x^* = \psi(w^*)$ and the last inequality is due to the fact that both $R(\psi(w) - R(\psi(w^*))$ and $R(x) - R(\psi(w))$ are non-negative. This completes the proof. ■

Now, based on the PL condition, we can show that (4.5.4) indeed converges.

Lemma 4.9.15. *The trajectory of the flow defined in* (4.5.4) *has finite length, i.e.,* $\int_{t=0}^{\infty} \|\frac{\mathrm{d}x}{\mathrm{d}t}\|_2 \mathrm{d}t < \infty$ *for any* $x^* \in \Gamma$*. Moreover,* $x(t)$ *converges to some* $x(\infty)$ *when* $t \to \infty$ *with* $F(x(\infty)) = 0$*.*

Proof of Lemma 4.9.15. Note that along the Riemannian gradient flow, $R(x(t))$ is non-increasing, thus $\|x(t)\|_2$ is bounded over time and $\{x(t)\}_{t\geq 0}$ has at least one limit point, which we will call x^*. Therefore, $R(x^*)$ is a limit point of $R(x(t))$, and again since $R(x(t))$ is non-increasing, it follows that $R(x(t)) \geq R(x^*)$ and $\lim_{t\to\infty} R(x(t)) = R(x^*)$. Below we will show $\lim_{t\to\infty} x(t) = x^*$.

Note that $\frac{\mathrm{d}R(x(t))}{\mathrm{d}t} = \langle \nabla R(x(t)), \frac{\mathrm{d}x(t)}{\mathrm{d}t}\rangle = -\langle \nabla R(x(t)), \frac{1}{4}F(x(t))\rangle = -\frac{1}{4}\|F(x(t))\|_2^2$ where the last equality applies Lemma 4.9.8. By Lemma 4.9.14, there exists a neighbourhood of x^*, U', in which PL condition holds of F. Since x^* is a limit point, there exists a time T_0, such that $x_{T_0} \in U$. Let $T_1 = \inf_{t\geq T_0}\{x(t) \notin U'\}$ (which is equal to ∞ if $x(t) \in U'$ for all $t \geq T_0$). Since $x(t)$ is continuous in t and U is open, we know $T_1 > T_0$ and for all $t \in [T_0, T_1)$, we have $\|F(x(t))\|_2 \geq \sqrt{c}(R(x(t)) - R(x^*))^{1/2}$.

Thus it holds that for $t \in [T_0, T_1)$,

$$\frac{\mathrm{d}(R(x(t)) - R(x^*))}{\mathrm{d}t} \leq -\frac{\sqrt{c}}{4}(R(x(t)) - R(x^*))^{1/2} \left\|F(x(t))\right\|_2,$$

that is,

$$\frac{\mathrm{d}(R(x(t)) - R(x^*))^{1/2}}{\mathrm{d}t} \leq -\frac{\sqrt{c}}{8} \left\|F(x(t))\right\|_2.$$

Therefore, we have

$$\int_{t=T_0}^{T_1} \left\|F(x(t))\right\|_2 \mathrm{d}t \leq \frac{8}{\sqrt{c}}(R(x(T_0)) - R(x^*))^{1/2}. \tag{4.9.25}$$

Thus if we pick T_0 such that $R(x(T_0)) - R(x^*)$ is sufficiently small, $R(T_1)$ will remain in U, which implies that T_1 cannot be finite and has to be ∞. Therefore, Equation (4.9.25) shows that the trajectory of $x(t)$ is of finite length, so $x(\infty) := \lim_{t\to\infty} x(t)$ exists and is equal to x^*. As a by-product, $F(x^*)$ must be 0. ■

Finally, collecting all the above lemmas, we are ready to prove Lemma 4.5.5. In Lemma 4.9.15 we already show the convergence of $x(t)$ as $t \to \infty$, the main part of the proof of Lemma 4.5.5 is to show the $x(\infty)$ cannot be sub-optimal stationary points of R on $\overline{\Gamma}$, the closure of Γ. The key idea here is that we can construct a different potential φ for each such sub-optimal stationary point x^*, such that (1) $\varphi(x_t)$ is increasing over time t in a sufficiently neighborhood of x^* and (2) $\lim_{x\to x^*} \varphi(x) = -\infty$.

Proof of Lemma 4.5.5. We will prove by contradiction. Suppose $x(\infty) = \binom{u(\infty)}{v(\infty)} = \lim_{t\to\infty} x(t)$ is not the optimal solution to (4.5.5). Denote $w(t) = (u(t))^{\odot 2} - (v(t))^{\odot 2}$, then $w(\infty) = \lim_{t\to\infty} w(t)$ is not the optimal solution to (4.5.6). Thus we have $R(w(t)) > R(w^*)$. Without loss of generality, suppose there is some $q \in [d]$ such that $(u_i(\infty))^2 + (v_i(\infty))^2 > 0$ for all $i = 1, \ldots, q$ and $u_i(\infty) = v_i(\infty) = 0$ for all $i = q+1, \ldots, d$. Again, as argued in the proof of Lemma 4.9.10, we can assume that, for some $q' \in [q]$,

$$\{z_{i,1:q}\}_{i\in[q']} \text{ is linearly independent and } z_{i,1:q} = 0 \text{ for all } i = q'+1, \ldots, n. \tag{4.9.26}$$

Since both $w(\infty)$ and w^* satisfy the constraint that $Zw(\infty) = Zw^* = Y$, we further have

$$0 = \langle z_i, w(\infty)\rangle = \langle z_i, w^*\rangle = \langle z_{i,(q+1):d}, w^*_{(q+1):d}\rangle, \qquad \text{for all } i = q'+1, \ldots, n. \tag{4.9.27}$$

Consider a potential function $\varphi : U \to \mathbb{R}$ defined as

$$\varphi(x) = \varphi(u, v) = \sum_{j=q+1}^{d} w_j^* \left[\ln(u_j)^2 \, \mathbb{1}\{w_j^* > 0\} - \ln(v_j)^2 \, \mathbb{1}\{w_j^* < 0\}\right].$$

Clearly $\lim_{t\to\infty} \varphi(x(t)) = -\infty$ if $\lim_{t\to\infty} x(t) = x(\infty)$. Below we will show contradiction if $x(\infty)$ is suboptimal. Consider the dynamics of $\varphi(x)$ along the Riemannian gradient flow:

$$\frac{\mathrm{d}\varphi}{\mathrm{d}t}(x(t)) = \left\langle \nabla\varphi(x(t)), \frac{\mathrm{d}x(t)}{\mathrm{d}t}\right\rangle = -\left\langle \nabla\varphi(x(t)), \frac{1}{4}F(x(t))\right\rangle \tag{4.9.28}$$

where F is defined previously in Lemma 4.9.8. Recall the definition of F, and we have

$$\begin{aligned}\langle \nabla\varphi(x(t)), F(x(t))\rangle = &\underbrace{\left\langle \nabla\varphi(x(t)), \frac{1}{4}\nabla R(x(t)) + \frac{1}{4}\sum_{i=1}^{q'} \lambda_i(x(t))\nabla f_i(x(t))\right\rangle}_{\mathcal{I}_1} \\ &+ \underbrace{\left\langle \nabla\varphi(x(t)), \frac{1}{4}\sum_{i=q'+1}^{n} \lambda_i(x(t))\nabla f_i(x(t))\right\rangle}_{\mathcal{I}_2}.\end{aligned} \tag{4.9.29}$$

To show $\langle \nabla\varphi(x(t)), F(x(t))\rangle < 0$, we analyze $\mathcal{I}_1$ and $\mathcal{I}_2$ separately. By the definition of $\varphi(x)$, we have

$$\nabla\varphi(x) = \sum_{j=q+1}^{d} 2w_j^* \left[\frac{\mathbb{1}\{w_j^*>0\}}{u_j}\cdot e_j - \frac{\mathbb{1}\{w_j^*<0\}}{v_j}\cdot e_{D+j}\right]$$

where e_j is the j-th canonical base of $\mathbb{R}^d$. Recall that $\nabla f_i(x) = 2\binom{z_i\odot u}{-z_i\odot v}$, and we further have

$$\begin{aligned}
\mathcal{I}_2 &= \sum_{i=q'+1}^{n}\lambda_i(x(t))\sum_{j=q+1}^{d} w_j^*\left[\frac{\mathbb{1}\{w_j^*>0\}}{u_j}\langle e_j, z_i\odot u\rangle + \frac{\mathbb{1}\{w_j^*<0\}}{v_j}\langle e_j, z_i\odot v\rangle\right]\\
&= \sum_{i=q'+1}^{n}\lambda_i(x(t))\sum_{j=q+1}^{d} w_j^*\left[\frac{\mathbb{1}\{w_j^*>0\}}{u_j}z_{i,j}u_j + \frac{\mathbb{1}\{w_j^*<0\}}{v_j}z_{i,j}v_j\right]\\
&= \sum_{i=q'+1}^{n}\lambda_i(x(t))\sum_{j=q+1}^{d} w_j^* z_{i,j} = \sum_{i=q'+1}^{n}\lambda_i(x(t))\langle z_{i,(q+1):d}, w^*_{(q+1):d}\rangle = 0 \qquad (4.9.30)
\end{aligned}$$

where the last equality follows from (4.9.27).

Next, we show that $\mathcal{I}_1 < 0$ by utilizing the fact that $w^* - w(\infty)$ is a descent direction of $R'(w)$. For $w \in \mathbb{R}^d$, define $\tilde{f}_i(w) = z_i^\top w$ and

$$\tilde{R}(w) = R(w) + \sum_{i=1}^{q'}\lambda_i(x(\infty))(\tilde{f}_i(w) - y_i).$$

Clearly, for any $w \in \mathbb{R}^D$ satisfying $Zw = Y$, it holds that $\tilde{f}_i(w) - y_i = 0$ for each $i \in [n]$, and thus $R(w) = \tilde{R}(w)$. In particular, we have $\tilde{R}(w(\infty)) = R(w(\infty)) > R(w^*) = \tilde{R}(w^*)$. Since $\tilde{R}(w)$ is a convex function, it follows that $\tilde{R}(w(\infty) + s(w^* - w(\infty))) \le s\tilde{R}(w^*) + (1-s)\tilde{R}(\infty) < \tilde{R}(w(\infty))$ for all $0 < s \le 1$, which implies $\frac{\mathrm{d}\tilde{R}}{\mathrm{d}t}(w(\infty) + s(w^* - w(\infty)))|_{s=0} < -2c < 0^+$ for some constant $c > 0$. Note that, for small enough $s > 0$, we have

$$\begin{aligned}
R(w(\infty) + s(w^* - w(\infty))) &= \frac{4}{n}\sum_{j=1}^{d}\left(\sum_{i=1}^{n} z_{i,j}^2\right)|w_j(\infty) + s(w_j^* - w_j(\infty))|\\
&= \frac{4}{n}\sum_{j=1}^{q}\left(\sum_{i=1}^{n} z_{i,j}^2\right)\mathrm{sign}(w_j(\infty))(w_j(\infty) + s(w_j^* - w_j(\infty)))
\end{aligned}$$

$$+ \frac{4}{n} \sum_{j=q+1}^{d} \left(\sum_{i=1}^{n} z_{i,j}^2 \right) s|w_j^*|.$$

Therefore, we can compute the derivative with respect to s at $s = 0$ as

$$\begin{aligned}
-2c &> \frac{\mathrm{d}\tilde{R}}{\mathrm{d}t}(w(\infty) + s(w^* - w(\infty)))\Big|_{s=0} \\
&= \frac{4}{n} \sum_{j=1}^{q} \left(\sum_{i=1}^{n} z_{i,j}^2 \right) \operatorname{sign}(w_j(\infty))(w_j^* - w_j(\infty)) + \frac{4}{n} \sum_{j=q+1}^{d} \left(\sum_{i=1}^{n} z_{i,j}^2 \right) |w_j^*| \\
&\quad + \sum_{i=1}^{q'} \lambda_i(x(\infty)) z_i^\top (w^* - w_j(\infty)) \\
&= \frac{4}{n} \sum_{j=1}^{q} \left(\sum_{i=1}^{n} z_{i,j}^2 \right) \operatorname{sign}(w_j(\infty))(w_j^* - w(\infty)) + \frac{4}{n} \sum_{j=q+1}^{d} \left(\sum_{i=1}^{n} z_{i,j}^2 \right) |w_j^*| \\
&\quad + \sum_{j=1}^{q} (w_j^* - w_j(\infty)) \sum_{i=1}^{q'} \lambda_i(x(\infty)) z_{i,j} + \sum_{j=q+1}^{d} w_j^* \sum_{i=1}^{q'} \lambda_i(x(\infty)) z_{i,j} \qquad (4.9.31)
\end{aligned}$$

where the second equality follows from the fact that $w_{(q+1):d}(\infty) = 0$. Since $x(t)$ converges to $x(\infty)$, we must have $F(x(\infty)) = 0$, which implies that for each $j \in \{1, \dots, q\}$,

$$\begin{aligned}
0 &= \frac{\partial R}{\partial u_j}(x(\infty)) + \sum_{i=1}^{q'} \lambda_i(x(\infty)) \frac{\partial f_i}{\partial u_j}(x(\infty)) = 2u_j(\infty) \left[\frac{4}{n} \sum_{i=1}^{n} z_{i,j}^2 + \sum_{i=1}^{q'} \lambda_i(x(\infty)) z_{i,j} \right], \\
0 &= \frac{\partial R}{\partial v_j}(x(\infty)) + \sum_{i=1}^{q'} \lambda_i(x(\infty)) \frac{\partial f_i}{\partial v_j}(x(\infty)) = 2v_j(\infty) \left[\frac{4}{n} \sum_{i=1}^{n} z_{i,j}^2 - \sum_{i=1}^{q'} \lambda_i(x(\infty)) z_{i,j} \right].
\end{aligned}$$

Combining the above two equalities yields

$$\frac{4}{n} \sum_{i=1}^{n} z_{i,j}^2 = -\operatorname{sign}(w_j(\infty)) \sum_{i=1}^{q'} \lambda_i(x(\infty)) z_{i,j}, \quad \text{for all } j \in [q].$$

Apply the above identity together with (4.9.31), and we obtain

$$-2c > \sum_{j=1}^{q} -\operatorname{sign}(w_j(\infty))^2 (w_j^* - w(\infty)) \sum_{i=1}^{q'} \lambda_i(x(\infty)) z_{i,j} + \frac{4}{n} \sum_{j=q+1}^{d} \left(\sum_{i=1}^{n} z_{i,j}^2 \right) |w_j^*|$$

$$
\begin{aligned}
&+\sum_{j=1}^{q}(w_j^*-w_j(\infty))\sum_{i=1}^{q'}\lambda_i(x(\infty))z_{i,j}+\sum_{j=q+1}^{d}w_j^*\sum_{i=1}^{q'}\lambda_i(x(\infty))z_{i,j}\\
&=\frac{4}{n}\sum_{j=q+1}^{d}\left(\sum_{i=1}^{n}z_{i,j}^2\right)|w_j^*|+\sum_{j=q+1}^{d}w_j^*\sum_{i=1}^{q'}\lambda_i(x(\infty))z_{i,j} \qquad (4.9.32)
\end{aligned}
$$

On the other hand, by directly evaluating $\nabla R(x(t))$ and each $\nabla f_i(x(t))$, we can compute $\mathcal{I}_1$ as

$$
\begin{aligned}
\mathcal{I}_1 &= \sum_{j=q+1}^{d}\frac{w_j^*\,\mathbb{1}\{w_j^*>0\}}{u_j(t)}\left[\frac{2}{n}\sum_{i=1}^{n}z_{i,j}^2u_j(t)+\frac{1}{2}\sum_{i=1}^{q'}\lambda_i(x(t))z_{i,j}u_j(t)\right]\\
&\quad-\sum_{j=q+1}^{d}\frac{w_j^*\,\mathbb{1}\{w_j^*<0\}}{v_j(t)}\left[\frac{2}{n}\sum_{i=1}^{n}z_{i,j}^2v_j(t)-\frac{1}{2}\sum_{i=1}^{q'}\lambda_i(x(t))z_{i,j}v_j(t)\right]\\
&=\frac{2}{n}\sum_{j=q+1}^{d}\left(\sum_{i=1}^{n}z_{i,j}^2\right)|w_j^*|+\frac{1}{2}\sum_{j=q+1}^{d}w_j^*\sum_{i=1}^{q'}\lambda_i(x(t))z_{i,j}\\
&=\frac{2}{n}\sum_{j=q+1}^{d}\left(\sum_{i=1}^{n}z_{i,j}^2\right)|w_j^*|+\frac{1}{2}\sum_{j=q+1}^{d}w_j^*\sum_{i=1}^{q'}\lambda_i(x(\infty))z_{i,j}\\
&\quad+\frac{1}{2}\sum_{j=q+1}^{d}w_j^*\sum_{i=1}^{q'}\left(\lambda_i(x(t))-\lambda_i(x(\infty))\right)z_{i,j}.
\end{aligned}
$$

We already know that $\lambda_{1:q'}(x)$ is continuous at $x(\infty)$ by the proof of Lemma 4.9.10, so the third term converges to 0 as $x(t)$ tends to $x(\infty)$. Now, applying (4.9.32), we immediately see that there exists some $\delta>0$ such that $\mathcal{I}_1<-c$ for $x(t)\in B_\delta(x(\infty))$. As we have shown in the above that $\mathcal{I}_2=0$, it then follows from (4.9.28) and (4.9.29) that

$$
\frac{\mathrm{d}\varphi}{\mathrm{d}t}(x(t))>c, \quad \text{for all } x(t)\in B_\delta(x(\infty)). \qquad (4.9.33)
$$

Since $\lim_{t\to\infty}x(t)=x(\infty)$, there exists some $T>0$ such that $x(t)\in B_\delta(x(\infty))$ for all $t>T$. By the proof ofLemma 4.9.11, we know that $\varphi(x(T))>-\infty$, then it follows from (4.9.33) that

$$\lim_{t\to\infty}\varphi(x(t)) = \varphi(x(T)) + \int_T^\infty \frac{\mathrm{d}\varphi(x(t))}{\mathrm{d}t}\mathrm{d}t > \varphi(x(T)) + \int_T^\infty c\mathrm{d}t = \infty$$

which is a contradiction. This finishes the proof. ■

4.9.6 Proof of Theorem 4.5.9

Here we present the lower bound on the sample complexity of GD in the kernel regime.

Proof of Theorem 4.5.9. We first simplify the loss function by substituting $x' = x - x(0)$, so correspondingly $x'_0 = 0$ and we consider $L'(x') := L(x) = (\langle \nabla f_i(x(0)), x'\rangle - y_i)^2$. We can think as if GD is performed on $L'(x')$. For simplicity, we still use the x and $L(x)$ notation in below.

In order to show test loss lower bound against a single fixed target function, we must take the properties of the algorithm into account. The proof is based on the observation that GD is rotationally equivariant [Ng04, LZA20] as an iterative algorithm, i.e., if one rotates the entire data distribution (including both the training and test data), the expected loss of the learned function remains the same. Since the data distribution and initialization are invariant under any rotation, it means the expected loss of $x(T)$ with ground truth being w^* is the same as the case where the ground truth is uniformly randomly sampled from all vectors of ℓ_2-norm $\|w^*\|_2$.

Thus the test loss of $x(T)$ is

$$\begin{aligned}\mathbb{E}_z\left[(\langle \nabla f_z(x(0)), x(T)\rangle - \langle z, w^*\rangle)^2\right] &= \mathbb{E}_z\left[(\langle z, w^* - (u(T) - v(T))\rangle)^2\right] \\ &= \|w^* - (u(T) - v(T))\|_2^2.\end{aligned}$$

Note $x(T) \in \operatorname{span}\{\nabla f_x(x(0))\}$, which is at most an n-dimensional space spanned by the gradients of model output at $x(0)$, so is $u(T) - v(T)$. We denote the corresponding space for $u(T)-v(T)$ by S, so $\dim(S) \leq n$ and it holds that $\|w^* - (u(T)-v(T))\|_2^2 \geq \|(I_D - P_S)w^*\|_2^2$, where P_S is projection matrix onto space S.

The expected test loss is then lower bounded by

$$\begin{aligned}
\mathbb{E}_{w^*}\left[\mathbb{E}_{z_i}\left[\|w^* - (u(T)-v(T))\|_2^2\right]\right] &= \mathbb{E}_{z_i}\left[\mathbb{E}_{w^*}\left[\|w^* - (u(T)-v(T))\|_2^2\right]\right] \\
&\geq \min_{\{z_i\}_{i\in[n]}} \mathbb{E}_{w^*}\left[\|(I_D - P_S)w^*\|_2^2\right] \\
&\geq \left(1 - \frac{n}{d}\right)\|w^*\|_2^2 .
\end{aligned}$$

This completes the proof. ■

Chapter 5

Implicit Regularization of Gradient Descent on Reparametrized Models

As part of the effort to understand implicit bias of gradient descent in overparametrized models, several results have shown how the training trajectory on the overparametrized model can be understood as mirror descent on a different objective. The main result in this chapter is a characterization of this phenomenon under a notion termed *commuting parametrization*, which encompasses all the previous results in this setting. It is shown that gradient flow with any commuting parametrization is equivalent to continuous mirror descent with a related Legendre function. Conversely, continuous mirror descent with any Legendre function can be viewed as gradient flow with a related commuting parametrization. The latter result relies upon Nash's embedding theorem.

5.1 Introduction on gradient descent and reparametrization

Implicit bias refers to the phenomenon in machine learning that the solution obtained from loss minimization has special properties that were not implied by value of the loss function and instead arise from the trajectory taken in parameter space by the optimization. Quantifying implicit bias necessarily has to go beyond the traditional black-box convergence analyses of optimization algorithms. Implicit bias can explain how choice of optimization algorithm can affect generalization [WGL+20, LWA22, LLL20].

Many existing results about implicit bias treat training (in the limit of infinitesimal step size) as a differential equation or process $\{x(t)\}_{t\geq 0} \subset \mathbb{R}^D$. To show the implicit bias of $x(t)$, the idea is to show for another (more intuitive or better understood) process $\{w(t)\}_{t\geq 0} \subset \mathbb{R}^d$ that $x(t)$ is *simulating* $w(t)$, in the sense that there exists a mapping $G : \mathbb{R}^D \to \mathbb{R}^d$ such that $w(t) = G(x(t))$. Then the implicit bias of $x(t)$ can be characterized by translating the special properties of $w(t)$ back to $x(t)$ through G. A related term, *implicit regularization*, refers to a handful of such results where particular update rules are shown to lead to regularized solutions; specifically, $x(t)$ is simulating $w(t)$ where $w(t)$ is solution to a regularized version of the original loss.

Here we develop a general framework involving optimization in the continuous-time regime of a loss $L : \mathbb{R}^d \to \mathbb{R}$ that has been re-parametrized before optimization[1] as $w = G(x)$ for some $G : \mathbb{R}^D \to \mathbb{R}^d$. Then the original loss $L(w)$ in the w-space induces the implied loss $(L \circ G)(x) \equiv L(G(x))$ in the x-space, and the gradient flow in the x-space is given by

$$\mathrm{d}x(t) = -\nabla(L \circ G)(x(t))\mathrm{d}t. \tag{5.1.1}$$

Using $w(t) = G(x(t))$ and the fact that $\nabla(L \circ G)(x) = \partial G(x)^\top \nabla L(G(x))$ where $\partial G(x) \in \mathbb{R}^{d\times D}$ denotes the Jacobian of G at x, the corresponding dynamics of (5.1.1)

1. Two examples from recent years, where G does not change expressiveness of the model, involve (a) overparametrized linear regression where the parameter vector w is reparametrized (for example as $w = u^{\odot 2} - v^{\odot 2}$ [WGL+20]) and (b) deep linear nets [ACHL19] where a matrix W is factorized as $W = W_1 W_2 \cdots W_L$ where each W_ℓ is the weight matrix for the ℓ-th layer.

in the w-space is

$$\mathrm{d}w(t) = \partial G(x(t))\mathrm{d}x(t) = -\partial G(x(t))\partial G(x(t))^\top \nabla L(w(t))\mathrm{d}t. \qquad (5.1.2)$$

Our framework is developed to fully understand phenomena in recent papers [GWB+18, VKR19, YKM20, AW20a, WGL+20, AW20b, AMN+21], which give examples suggesting that gradient flow in the x-space could end up simulating a more classical algorithm, mirror descent (specifically, the continuous analog, mirror flow) in the w-space. Recall that mirror flow is continuous-time limit of the classical mirror descent, written as $\mathrm{d}\nabla R(w(t)) = -\nabla L(w(t))\mathrm{d}t$ where $R : \mathbb{R}^d \to \mathbb{R} \cup \{\infty\}$ is a strictly convex function [NY83, BT03], which is called *mirror map* or *Lengendre function* in literature. Equivalently it is *Riemannian gradient flow* with metric tensor $\nabla^2 R$, an old notion in geometry:

$$\mathrm{d}w(t) = -\nabla^2 R(w(t))^{-1}\nabla L(w(t))\mathrm{d}t. \qquad (5.1.3)$$

If there exists a Legendre function R such that $\partial G(x(t))\partial G(x(t))^\top = \nabla^2 R(w(t))^{-1}$ for all t, then (5.1.2) becomes a simple mirror flow in the w-space. Many existing results about implicit bias indeed concern reparametrizations G that satisfy $\partial G(x)\partial G(x)^\top = \nabla^2 R(w)^{-1}$ for a strictly convex function R, and the implicit bias/regularization is demonstrated by showing that the convergence point satisfies the KKT conditions needed for minimizing R among all minimizers of the loss L. A concrete example is that $w_i(t) = G_i(x(t)) = (x_i(t))^2$ for all $i \in [d]$, so here $D = d$. In this case, the Legendre function R must satisfy $(\nabla^2 R(w(t)))^{-1} = \partial G(x(t))\partial G(x(t))^\top = 4\mathrm{diag}((x_1(t))^2, \ldots, (x_d(t))^2) = 4\mathrm{diag}(w_1(t), \ldots, w_d(t))$ which suggests R is the classical negative entropy function, i.e., $R(w) = \sum_{i=1}^d w_i(\ln w_i - 1)$.

However, in general, it is hard to decide *whether gradient flow for a given parametrization G can be written as mirror flow for some Legendre function R,* especially when $D > d$ and G is not an injective map. In such cases, there could be multiple x's mapping to the same $G(x)$ yet having different $\partial G(x)\partial G(x)^\top$. If more than one of such x can be reached by gradient flow, then the desired Legendre function cannot exist. [2] If only one of such x can be reached by gradient flow, we must decide which x it is in order to decide the value of $\nabla^2 R$ using $\partial G \partial G^\top$. Conversely, [AW20b] raises the following question: *for what Legendre function R can the corresponding mirror flow be the result of gradient flow after some reparametrization G?* Answering the questions in both directions requires a deeper understanding of the impact of parametrizations.

5.2 Notations and preliminaries

Notational convections in this chapter. We denote $\mathbb{N}$ as the set of natural numbers. For any positive integer n, we denote $\{1, 2, \ldots, n\}$ by $[n]$. For any vector $u \in \mathbb{R}^D$, we denote its i-th coordinate by u_i. For any vector $u, v \in \mathbb{R}^D$ and $\alpha \in \mathbb{R}$, we define $u \odot v = (u_1 v_1, \ldots, u_D v_D)^\top$ and $u^{\odot \alpha} = ((u_1)^\alpha, \ldots, (u_D)^\alpha)^\top$. For any $k \in \mathbb{N} \cup \{\infty\}$, we say a function f is $\mathcal{C}^k$ if it is k times continuously differentiable, and use $\mathcal{C}^k(M)$ to denote the set of all C^k functions from M to $\mathbb{R}$. We use $\circ$ to denote the composition of functions, *e.g.*, $f \circ g(x) = f(g(x))$. For any convex function $R : \mathbb{R}^D \to \mathbb{R} \cup \{\infty\}$, we denote its domain by $\operatorname{dom} R = \{w \in \mathbb{R}^D \mid R(w) < \infty\}$. For any set S, we denote its interior by $\operatorname{int}(S)$ and its closure by $\overline{S}$.

We assume that the model has parameter vector $w \in \mathbb{R}^d$ and $\mathcal{C}^1$ loss function $L : \mathbb{R}^d \to \mathbb{R}$. Training involves a reparametrized vector $x \in \mathbb{R}^D$, which is a reparametrization of w such that $w = G(x)$ for some differentiable parametrization function G, and the objective is $L(G(x))$. From now on, we follow the convention

2. To avoid such an issue, [AW20b] has to assume all the preimages of G at w have the same $\partial G(\partial G)^\top$ and a recent paper [GLH22] assumes that G is injective.

that d is the dimension of the original parameter w and D is the dimension of the reparametrized x. We also refer to $\mathbb{R}^d$ as the w-space and $\mathbb{R}^D$ as the x-space.

In particular, we are interested in understanding the dynamics of gradient flow under the objective $L \circ G$ on some submanifold $M \subseteq \mathbb{R}^D$. Most of our results also generalize to the following notion of *time-dependent* loss.

Definition 5.2.1 (Time-dependent loss)**.** A time-dependent loss $L_t(w)$ is a function piecewise constant in time t and continuously differentiable in $w \in \mathbb{R}^d$, that is, there exists $k \in \mathbb{N}$, $0 = t_1 < t_2 < \cdots < t_{k+1} = \infty$ and $\mathcal{C}^1$ loss functions $L^{(1)}, L^{(2)}, \ldots, L^{(k)}$ such that for each $i \in [k]$ and all $t \in [t_i, t_{i+1})$,

$$L_t(w) = L^{(i)}(w), \qquad \forall w \in \mathbb{R}^d.$$

We denote the set of such time-dependent loss functions by $\mathcal{L}$.

5.2.1 Manifold and vector field

Vector fields are a natural way to formalize the continuous-time gradient descent (a good reference is [Lee13]). Let M be any smooth submanifold of $\mathbb{R}^D$. A *vector field* X on M is a continuous map from M to $\mathbb{R}^D$ such that for any $x \in M$, $X(x)$ is in the tangent space of M at x, which is denoted by $T_x(M)$. Formally, $T_x(M) := \{\frac{\mathrm{d}\gamma}{\mathrm{d}t}\big|_{t=0} \mid \forall \text{ smooth curves } \gamma : \mathbb{R} \to M, \gamma(0) = x\}$.

Definition 5.2.2 (Complete vector field; p.215, [Lee13])**.** Let M be a smooth submanifold of $\mathbb{R}^D$ and X be a vector field on M. We say X is a *complete vector field* on M if and only if for any initialization $x_{\text{init}} \in M$, the differential equation $\mathrm{d}x(t) = X(x(t))\mathrm{d}t$ has a solution on $(-\infty, \infty)$ with $x(0) = x_{\text{init}}$.

When the smooth submanifold $M \subseteq \mathbb{R}^D$ is equipped with a metric tensor g, we then have a Riemannian manifold (M, g), where for each $x \in M$, $g_x : T_xM \times T_xM \to \mathbb{R}$

is a positive definite bilinear form. In particular, the standard Euclidean metric $\overline{g}$ corresponds to $\overline{g}_x(u, v) = u^\top v$ for each $x \in M$ and $u, v \in T_x M$, under which the length of any arc on M is given by its length as a curve in $\mathbb{R}^D$.

For any differentiable function $f : M \to \mathbb{R}$, we denote by $\nabla_g f$ its gradient vector field with respect to metric tensor g. More specifically, $\nabla_g f(x)$ is defined as the unique vector in $\mathbb{R}^D$ such that $\nabla_g f(x) \in T_x(M)$ and $\frac{\mathrm{d}f(\gamma(t))}{\mathrm{d}t}\Big|_{t=0} = g_x\Big(\nabla f(x), \frac{\mathrm{d}\gamma(t)}{\mathrm{d}t}\Big|_{t=0}\Big)$. Throughout the paper, we assume by default that the metric on the submanifold $M \subseteq \mathbb{R}^D$ is inherited from $(\mathbb{R}^D, \overline{g})$, and we will use ∇f as a shorthand for $\nabla_{\overline{g}} f$. If M is an open set of $\mathbb{R}^D$, ∇f is then simply the ordinary gradient of f.

For any $x \in M$ and $\mathcal{C}^1$ function $f : M \to \mathbb{R}$, we denote by $\phi_f^t(x)$ the point on M reached after time t by following the vector field $-\nabla f$ starting at x, i.e., the solution at time t (when it exists) of

$$\mathrm{d}\phi_f^t = -\nabla f(\phi_f^t)\mathrm{d}t, \qquad \phi_f^0(x) = x.$$

We say $\phi_f^t(x)$ is *well-defined* at time t when the above differential equation has a solution at time t. Moreover, for any differentiable function $X : M \to \mathbb{R}^d$, we denote its Jacobian by

$$\partial X(x) = (\nabla X_1(x), \nabla X_2(x), \ldots, \nabla X_d(x))^\top.$$

Definition 5.2.3 (Lie bracket)**.** Let M be a smooth submanifold of $\mathbb{R}^D$. Given two C^1 vector fields X, Y on M, we define the *Lie Bracket* of X and Y as $[X, Y](x) := \partial Y(x)X(x) - \partial X(x)Y(x)$.

5.2.2 Parametrizations

We use the term *parametrization* to refer to differentiable maps from a smooth submanifold of $\mathbb{R}^D$ (x-space) to $\mathbb{R}^d$ (w-space). We reserve G to denote parametrizations, and omit the dependence on G for notations of objects related to G when it is clear from the context.

The following notion of regular parametrization plays an important role in our analysis, and it is necessary for our main equivalence result between mirror flow and gradient flow with reparametrization. This is because if the null space of $\partial G(x)$ is non-trivial, i.e., it contains some vector $u \neq 0$, then the gradient flow with parametrization G obviously cannot simulate any mirror flow with nonzero velocity in the direction of u.

Definition 5.2.4 (Regular parametrization)**.** Let M be a smooth submanifold of $\mathbb{R}^D$. A *regular parametrization* $G : M \to \mathbb{R}^d$ is a $\mathcal{C}^1$ parametrization such that $\partial G(x)$ is of rank d for all $x \in M$.

Note that a regular parametrization G can become irregular when its domain is changed. For example, $G(x) = x^2$ is regular on $\mathbb{R}^+$, but it is not regular on $\mathbb{R}$ as $\partial G(0) = 0$.

Given a $\mathcal{C}^2$ parametrization $G : M \to \mathbb{R}^d$, for any $x \in M$ and $\mu \in \mathbb{R}^d$, we define

$$\psi(x; \mu) := \phi_{G_1}^{\mu_1} \circ \phi_{G_2}^{\mu_2} \circ \cdots \circ \phi_{G_d}^{\mu_d}(x) \tag{5.2.1}$$

when it is well-defined, i.e., the corresponding integral equation has a solution. For any $x \in M$, we define the domain of $\psi(x; \cdot)$ as

$$\mathcal{U}(x) = \Big\{\mu \in \mathbb{R}^d \mid \psi(x; \mu) \text{ is well-defined}\Big\}. \tag{5.2.2}$$

When every ∇G_i is a complete vector field on M as in Definition 5.2.2, we have $\mathcal{U}(x) = \mathbb{R}^d$. However, such completeness assumption is relatively strong, and most polynomials would violate it. For example, consider $G(x) = x^{\odot 3}$ for $x \in \mathbb{R}^d$, then the solution to $\mathrm{d}x_i(t) = 3x_i(t)^2 \mathrm{d}t$ explodes in finite time for each $i \in [d]$. To relax this, we consider parametrizations such that the domain of the flows induced by its gradient vector fields is pairwise symmetric. More specifically, for any $x \in M$ and $i, j \in [d]$, we define

$$\mathcal{U}_{ij}(x) = \Big\{(s,t) \in \mathbb{R}^2 \mid \phi^s_{G_i} \circ \phi^t_{G_j}(x) \text{ is well-defined}\Big\},$$

and we make the following assumption.

Assumption 5.2.5. Let M be a smooth submanifold of $\mathbb{R}^D$ and $G : M \to \mathbb{R}^d$ be a parametrization. We assume that for any $x \in M$ and $i \in [d]$, $\phi^t_{G_i}(x)$ is well-defined for $t \in (T_-, T_+)$ such that either $\lim_{t \to T_+} \|\phi^t_{G_i}(x)\|_2 = \infty$ or $T_+ = \infty$ and similarly for T_-. Also, we assume that for any $x \in M$ and $i, j \in [d]$, it holds that $\mathcal{U}_{ji}(x) = \{(t,s) \in \mathbb{R}^2 \mid (s,t) \in \mathcal{U}_{ij}(x)\}$, i.e., $\phi^s_{G_i} \circ \phi^t_{G_j}(x)$ is well-defined if and only if $\phi^t_{G_j} \circ \phi^s_{G_i}(x)$ does.

Indeed, under Assumption 5.2.5, we can show that for any $x \in M$, $\mathcal{U}(x)$ is a hyperrectangle, as summarized in the following lemma. See Section 5.7 for a proof.

Lemma 5.2.6. *Let M be a smooth submanifold of $\mathbb{R}^D$ and $G : M \to \mathbb{R}^d$ be a $\mathcal{C}^2$ parametrization satisfying Assumption 5.2.5. Then for any $x \in M$, $\mathcal{U}(x)$ is a hyperrectangle, i.e., $\mathcal{U}(x)$ can be decomposed as*

$$\mathcal{U}(x) = \mathcal{I}_1(x) \times \mathcal{I}_2(x) \times \cdots \times \mathcal{I}_d(x)$$

where $\mathcal{I}_j(x) := \{x'_j \mid x' \in \mathcal{U}(x)\}$ is an open interval.

For any initialization $x_{\text{init}} \in M$, the set of points that are reachable via gradient flow under G with respect to some time-dependent loss (see Definition 5.2.1) is a subset of M that depends on G and x_{init}.

Definition 5.2.7 (Reachable set)**.** Let M be a smooth submanifold of $\mathbb{R}^D$. For any $\mathcal{C}^2$ parametrization $G : M \to \mathbb{R}^d$ and any initialization $x_{\text{init}} \in M$, the reachable set

$\Omega_x(x_{\text{init}}; G)$ is defined as

$$\Omega_x(x_{\text{init}}; G) = \left\{ \phi^{\mu_1}_{L_1 \circ G} \circ \phi^{\mu_2}_{L_2 \circ G} \circ \cdots \circ \phi^{\mu_k}_{L_k \circ G}(x_{\text{init}}) \;\middle|\; \forall k \in \mathbb{N}, \forall i \in [k], L_i \in \mathcal{C}^1(\mathbb{R}^d), \mu_i \geq 0 \right\}.$$

It is clear that the above definition induces a transitive "reachable" relationship between points on M, and it is also reflexive since for all $L \in \mathcal{C}^1(\mathbb{R}^d)$ and $t > 0$, $\phi^t_{L\circ G} \circ \phi^t_{(-L)\circ G}$ is the identity map on the domain of $\phi^t_{-L\circ G}$. In this sense, the reachable sets are orbits of the family of gradient vector fields $\{\nabla(L \circ G) \mid L \in \mathcal{C}^1(\mathbb{R}^d)\}$, i.e., the reachable sets divide the domain M into equivalent classes. The above reachable set in the x-space further induces the corresponding reachable set in the w-space given by $\Omega_w(x_{\text{init}}; G) = G(\Omega_x(x_{\text{init}}; G))$.

In most natural examples, the parametrization G is smooth (though this is not necessary for our results), and by Sussman's Orbit Theorem [Sus73], each reachable set $\Omega_x(x_{\text{init}}; G)$ is an immersed submanifold of M. Moreover, it follows that $\Omega_x(x_{\text{init}}; G)$ can be generated by $\{\nabla G_i\}_{i=1}^d$, i.e., $\Omega_x(x_{\text{init}}; G) = \{\phi^{\mu_1}_{G_{j_1}} \circ \phi^{\mu_2}_{G_{j_2}} \circ \cdots \circ \phi^{\mu_k}_{G_{j_k}}(x_{\text{init}}) \mid \forall k \in \mathbb{N}, \forall i \in [k], j_i \in [d], \mu_i \geq 0\}$.

5.2.3 Mirror descent and mirror flow

Next, we introduce some basic notions for mirror descent [NY83, BT03]. We refer the readers to Section 5.6 for more preliminaries on convex analysis.

Definition 5.2.8 (Legendre function and mirror map)**.** Let $R : \mathbb{R}^d \to \mathbb{R} \cup \{\infty\}$ be a differentiable convex function. We say R is a *Legendre function* when the following holds:

(a) R is strictly convex on $\operatorname{int}(\operatorname{dom} R)$.

(b) For any sequence $\{w_i\}_{i=1}^\infty$ going to the boundary of $\operatorname{dom} R$, $\lim_{i\to\infty} \|\nabla R(w_i)\|_2 = \infty$.

In particular, we call R a *mirror map* if R further satisfies the following condition (see p.298 in [B+15]):

(c) The gradient map $\nabla R : \mathrm{int}(\mathrm{dom}\, R) \to \mathbb{R}^d$ is surjective.

Given a Legendre function $R : \mathbb{R}^d \to \mathbb{R} \cup \{\infty\}$, for any initialization $w_0 = w_{\mathrm{init}} \in \mathrm{int}(\mathrm{dom}\, R)$, mirror descent with step size η updates as follows:

$$\nabla R(w_{k+1}) = \nabla R(w_k) - \eta \nabla L(w_k). \tag{5.2.3}$$

Usually ∇R is required to be surjective so that after a discrete descent step in the dual space, it can be projected back to the primal space via $(\nabla R)^{-1}$. Nonetheless, as long as $\nabla R(w_k) - \eta \nabla L(w_k)$ is in the range of ∇R, the above discrete update is well-defined. In the limit of $\eta \to 0$, (5.2.3) becomes the continuous mirror flow:

$$\mathrm{d}\nabla R(w(t)) = -\nabla L(w(t))\mathrm{d}t. \tag{5.2.4}$$

Given a differentiable function R, the corresponding Bregman divergence D_R is defined as

$$D_R(w, w') = R(w) - R(w') - \langle \nabla R(w'), w - w' \rangle.$$

We recall a well-known implicit bias result for mirror flow (which holds for mirror descent as well) [GLSS18], which shows that for a specific type of loss, if mirror flow converges to some optimal solution, then the convergence point minimizes some convex regularizer among all optimal solutions.

Theorem 5.2.9. *Given any data $Z \in \mathbb{R}^{n \times d}$ and corresponding label $Y \in \mathbb{R}^n$, suppose the loss $L(w)$ is in the form of $L(w) = \widetilde{L}(Zw)$ for some differentiable $\widetilde{L} : \mathbb{R}^n \to \mathbb{R}$. Assume that initialized at $w(0) = w_{\mathrm{init}}$, the mirror flow (5.2.4) converges and the* *convergence point* $w_\infty = \lim_{t\to\infty} w(t)$ *satisfies* $Zw_\infty = Y$*, then*

$$D_R(w_\infty, w_0) = \min_{w:Zw=Y} D_R(w, w_0).$$

See Section 5.7 for a proof. The above theorem is the building block for proving the implicit bias induced by any commuting parametrization in overparametrized linear models (see Theorem 5.3.17).

5.3 Every gradient flow with commuting parametrization is a mirror flow

5.3.1 Commuting parametrization

We now formalize the notion of commuting parametrization. We remark that M is a smooth submanifold of $\mathbb{R}^D$, and it is the domain of the parametrization G.

Definition 5.3.1 (Commuting parametrization)**.** Let M be a smooth submanifold of $\mathbb{R}^D$. A $\mathcal{C}^2$ parametrization $G : M \to \mathbb{R}^d$ is *commuting* in a subset $S \subseteq M$ if and only if for any $i, j \in [d]$, the Lie bracket $[\nabla G_i, \nabla G_j](x) = 0$ for all $x \in S$. Moreover, we say G is a *commuting parametrization* if it is commuting in the entire M.

In particular, when M is an open subset of $\mathbb{R}^d$, $\{\nabla G_i\}_{i=1}^d$ are ordinary gradients in $\mathbb{R}^D$, and the Lie bracket between any pair of ∇G_i and ∇G_j is given by

$$[\nabla G_i, \nabla G_j](x) = \nabla^2 G_j(x)\nabla G_i(x) - \nabla^2 G_i(x)\nabla G_j(x).$$

This provides an easy way to check whether G is commuting or not.

The above definition of commuting parametrizations builds upon the differential properties of the gradient vector fields $\{\nabla G_i\}_{i=1}^d$, where each Lie bracket $[\nabla G_i, \nabla G_j]$

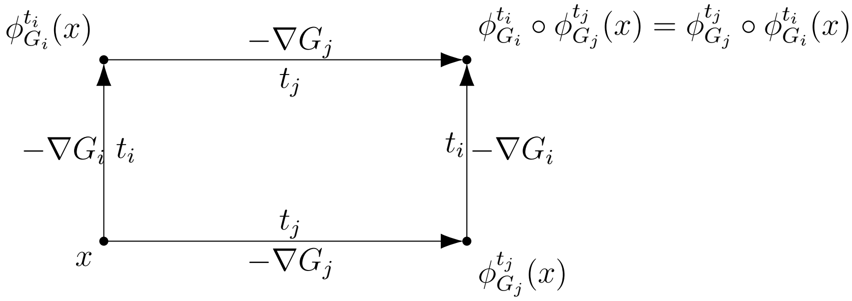


Figure 5.1: Illustration of commuting parametrizations. Suppose $G : M \to \mathbb{R}^d$ is a commuting parametrization satisfying Assumption 5.2.5, then starting from any $x \in M$, first moving along $-\nabla G_i$ for time t_i then moving along $-\nabla G_j$ for time t_j yields the same result as first moving along $-\nabla G_j$ for time t_j then moving along $-\nabla G_i$ for time t_i does, i.e., $\phi^{t_i}_{G_i} \circ \phi^{t_j}_{G_j}(x) = \phi^{t_j}_{G_j} \circ \phi^{t_i}_{G_i}(x)$.

characterizes the change of ∇G_j along the flow generated by ∇G_i. In particular, when G is a commuting parametrization satisfying Assumption 5.2.5, it is further equivalent to a characterization of 'commuting' in the integral form, as summarized in Theorem 5.3.2. Also see Figure 5.1 for an illustration.

Theorem 5.3.2 (Adapted from Theorem 9.44 in [Lee13])**.** *Let M be a smooth submanifold of $\mathbb{R}^D$ and $G : M \to \mathbb{R}^d$ be a $\mathcal{C}^2$ parametrization. For any $i, j \in [d]$, $[\nabla G_i, \nabla G_j](x) = 0$ for all $x \in M$ if and only if for any $x \in M$, whenever both $\phi^s_{G_i} \circ \phi^t_{G_j}(x)$ and $\phi^t_{G_j} \circ \phi^s_{G_i}(x)$ are well-defined for all (s, t) in some rectangle $\mathcal{I}_1 \times \mathcal{I}_2$ where $\mathcal{I}_1, \mathcal{I}_2 \subseteq \mathbb{R}$ are open intervals, it holds that $\phi^s_{G_i} \circ \phi^t_{G_j}(x) = \phi^t_{G_j} \circ \phi^s_{G_i}(x)$ for all $(s, t) \in \mathcal{I}_1 \times \mathcal{I}_2$.*

Under Assumption 5.2.5, Lemma 5.2.6 implies that the domain of $\phi^s_{G_i} \circ \phi^t_{G_j}(x)$ is exactly $\mathcal{I}_i(x) \times \mathcal{I}_j(x)$, and thus the above theorem simplifies into the following.

Theorem 5.3.3. *Let M be a smooth submanifold of $\mathbb{R}^D$ and $G : M \to \mathbb{R}^d$ be a $\mathcal{C}^2$ parametrization satisfying Assumption 5.2.5. For any $i, j \in [d]$, $[\nabla G_i, \nabla G_j](x) = 0$ for all $x \in M$ if and only if for any $x \in M$, it holds that $\phi^s_{G_i} \circ \phi^t_{G_j}(x) = \phi^t_{G_j} \circ \phi^s_{G_i}(x)$ for all $(s, t) \in \mathcal{I}_1(x) \times \mathcal{I}_2(x)$.*

The commuting condition clearly holds when each G_i only depends on a different subset of coordinates of x, because we then have $\nabla^2 G_i(\cdot) \nabla G_j(\cdot) \equiv 0$ for any distinct

$i,j \in [d]$ as $\nabla^2 G_i$ and ∇G_j live in different subspaces of $\mathbb{R}^D$. We call such G *separable parametrizations*, and this case covers all the previous examples [GWB+18, VKR19, AW20a, WGL+20, AW20b]. Another interesting example is the *quadratic parametrization*: We parametrize $w \in \mathbb{R}^d$ by $G : \mathbb{R}^D \to \mathbb{R}^d$ where for each $i \in [d]$, there is a symmetric matrix $A_i \in \mathbb{R}^{D\times D}$ such that $G_i(x) = \frac{1}{2}x^\top A_i x$. Then each Lie bracket $[G_i, G_j](x) = (A_j A_i - A_i A_j)x$, and thus G is a commuting parametrization if and only if matrices $\{A_i\}_{i=1}^d$ commute.

For concreteness, we analyze two examples below. The first one is both a separable parametrization and a commuting quadratic parametrization. The second one is a quadratic parametrization but not commuting.

Example 5.3.4 ($u^{\odot 2} - v^{\odot 2}$ parametrization, [WGL+20])**.** *Parametrize $w \in \mathbb{R}^d$ by $w = u^{\odot 2} - v^{\odot 2}$. Here $D = 2d$, and the parametrization G is given by $G(x) = u^{\odot 2} - v^{\odot 2}$ for $x = \binom{u}{v} \in \mathbb{R}^D$. Since each $G_i(x)$ involves only u_i and v_i, G is a separable parametrization and hence a commuting parametrization. Meanwhile, each $G_i(x)$ is a quadratic form in x, and it can be directly verified that the matrices underlying these quadratic forms commute with each other.*

Example 5.3.5 (Matrix factorization)**.** *As a counter-example, consider two parametrizations for matrix factorization: $G(U) = UU^\top$ and $G(U,V) = UV^\top$, where $U, V \in \mathbb{R}^{d\times r}$ and $d \geq 2, r \geq 1$. These are both* non-commuting *quadratic parametrizations. Here we only demonstrate for the parametrization $G(U) = UU^\top$, and $G(U,V) = UV^\top$ follows a similar argument. For each $i,j \in [d]$, we define $E_{ij} \in \mathbb{R}^d$ as the one-hot matrix with the (i,j)-th entry being 1 and the rest being 0, and denote $\overline{E}_{ij} = \frac{1}{2}(E_{ij} + E_{ji})$. For $r = 1$, we have $G_{ij}(U) = U_i U_j = U^\top \overline{E}_{ij} U$ for any $i,j \in [d]$, so G is a quadratic parametrization. Note that $\overline{E}_{ii}\overline{E}_{ij} = \frac{1}{2}E_{ij} \neq \frac{1}{2}E_{ji} = \overline{E}_{ij}\overline{E}_{ii}$ for all distinct $i,j \in [d]$, which implies that $[\nabla G_{ij}, \nabla G_{ii}] \neq 0$, so G is non-commuting. More generally, we can reshape U as a vector $\overrightarrow{U} := [U_{:1}^\top, \ldots, U_{:r}^\top]^\top \in \mathbb{R}^{rd}$ where each $U_{:j}$ is the j-th column of U, and the resulting quadratic form for the (i,j)-entry of $G(U)$ corresponds to a* *block-diagonal matrix:*

$$G_{ij}(U) = (\overrightarrow{U})^\top \begin{pmatrix} \overline{E}_{ij} & & \\ & \ddots & \\ & & \overline{E}_{ij} \end{pmatrix} \overrightarrow{U}.$$

Therefore, $\nabla^2 G_{ij}$ does not commute with $\nabla^2 G_{ii}$ due to the same reason as in the rank-1 case.

Remark 5.3.6. This non-commuting issue for general matrix factorization does not conflict with the theoretical analysis in [GWB+18] where the measurements are commuting, or equivalently, only involves diagonal elements, as $\{G_{ii}\}_{i=1}^d$ are indeed commuting parametrizations. [GWB+18] is the first to identify the above non-commuting issue and conjectured that the implicit bias result for diagonal measurements can be extended to the general case.

5.3.2 Main Equivalence Result

Next, we proceed to present our analysis for gradient flow with commuting parametrization. The following two lemmas highlight the special properties of commuting parametrizations. Lemma 5.3.7 shows that the point reached by gradient flow with any commuting parametrization is determined by the integral of the negative gradient of the loss along the trajectory.

Lemma 5.3.7. *Let M be a smooth submanifold of $\mathbb{R}^D$ and $G : M \to \mathbb{R}^d$ be a commuting parametrization. For any initialization $x_{\text{init}} \in M$, consider the gradient flow for any time-dependent loss $L_\cdot \in \mathcal{L}$ as in Definition 5.2.1:*

$$\mathrm{d}x(t) = -\nabla(L_t \circ G)(x(t))\mathrm{d}t, \qquad x(0) = x_{\text{init}}.$$

Further define $\mu(t) = \int_0^t -\nabla L_t(G(x(s)))\mathrm{d}s$. Suppose $\mu(t) \in \mathcal{U}(x_{\text{init}})$ for all $t \in [0, T)$

where $T \in \mathbb{R} \cup \{\infty\}$, then it holds that $x(t) = \psi(x_{\mathrm{init}}; \mu(t))$ for all $t \in [0, T)$.

Based on Lemma 5.3.7, the next key lemma reveals the essential approach to find the Legendre function.

Lemma 5.3.8. *Let M be a smooth submanifold of $\mathbb{R}^D$ and $G : M \to \mathbb{R}^d$ be a commuting and regular parametrization satisfying Assumption 5.2.5. Then for any $x_{\mathrm{init}} \in M$, there exists a Legendre function $Q : \mathbb{R}^d \to \mathbb{R} \cup \{\infty\}$ such that $\nabla Q(\mu) = G(\psi(x_{\mathrm{init}}; \mu))$ for all $\mu \in \mathcal{U}(x_{\mathrm{init}})$. Moreover, let R be the convex conjugate of Q, then R is also a Legendre function and satisfies that $\mathrm{int}(\mathrm{dom}\, R) = \Omega_w(x_{\mathrm{init}}; G)$ and*

$$\nabla^2 R(G(\psi(x_{\mathrm{init}}; \mu))) = \left(\partial G(\psi(x_{\mathrm{init}}; \mu)) \partial G(\psi(x_{\mathrm{init}}; \mu))^\top\right)^{-1}$$

for all $\mu \in \mathcal{U}(x_{\mathrm{init}})$.

Next, we present our main result on characterizing any gradient flow with commuting parametrization by a mirror flow.

Theorem 5.3.9. *Let M be a smooth submanifold of $\mathbb{R}^D$ and $G : M \to \mathbb{R}^d$ be a commuting and regular parametrization satisfying Assumption 5.2.5. For any initialization $x_{\mathrm{init}} \in M$, consider the gradient flow for any time-dependent loss function $L_t : \mathbb{R}^d \to \mathbb{R}$:*

$$\mathrm{d}x(t) = -\nabla(L_t \circ G)(x(t))\mathrm{d}t, \qquad x(0) = x_{\mathrm{init}}.$$

Define $w(t) = G(x(t))$ for all $t \geq 0$, then the dynamics of $w(t)$ is a mirror flow with respect to the Legendre function R given by Lemma 5.3.8, i.e.,

$$\mathrm{d}\nabla R(w(t)) = -\nabla L_t(w(t))\mathrm{d}t, \qquad w(0) = G(x_{\mathrm{init}}).$$

Moreover, this R only depends on the initialization x_{init} and the parametrization G, *and is independent of the loss function L_t.*

Proof of Theorem 5.3.9. Recall that the gradient flow in the x-space governed by $-\nabla(L_t \circ G)(x)$ is

$$\mathrm{d}x(t) = -\nabla(L_t \circ G)(x(t))\mathrm{d}t = -\partial G(x(t))^\top \nabla L_t(G(x(t)))\mathrm{d}t.$$

Using $w(t) = G(x(t))$, the corresponding dynamics in the w-space is

$$\mathrm{d}w(t) = \partial G(x(t))\mathrm{d}x(t) = -\partial G(x(t))\partial G(x(t))^\top \nabla L_t(w(t))\mathrm{d}t. \tag{5.3.1}$$

By Lemma 5.3.7, we know that the solution to the gradient flow satisfies $x(t) = \psi(x_{\text{init}}; \mu(t))$ where $\mu(t) = \int_0^t -\nabla L_t(G(x(s)))\mathrm{d}s$. Therefore, applying Lemma 5.3.8, we get a Legendre function $R : \mathbb{R}^d \to \mathbb{R} \cup \{\infty\}$ with domain $\Omega_w(x_{\text{init}}; G)$ such that

$$\nabla^2 R(w(t)) = \nabla^2 R(G(\psi(x_{\text{init}}; \mu(t)))) = \Big(\partial G(\psi(x_{\text{init}}; \mu(t)))\partial G(\psi(x_{\text{init}}; \mu(t)))\Big)^{-1}$$

for all $t \geq 0$. Then the dynamics of $w(t)$ in (5.3.1) can be rewritten as

$$\mathrm{d}w(t) = -\nabla^2 R(w(t))^{-1}\nabla L_t(w(t))\mathrm{d}t,$$

or equivalently,

$$\mathrm{d}\nabla R(w(t)) = -\nabla L_t(w(t))\mathrm{d}t,$$

which is exactly the mirror flow with respect to R initialized at $w(0) = G(x_{\text{init}})$. Further note that the result of Lemma 5.3.8 is completely independent of the loss function L_t, and thus R only depends on the initialization x_{init} and the parametrization G. This finishes the proof. ■

Theorem 5.3.9 provides a sufficient condition for when a gradient flow with certain parametrization G is simulating a mirror flow. The next question is then: What are the necessary conditions on the parametrization G so that it enables the gradient flow to simulate a mirror flow? We provide a (partial) characterization of such G in the following theorem.

Theorem 5.3.10 (Necessary condition on smooth parametrization to be commuting)**.** *Let M be a smooth submanifold of $\mathbb{R}^D$ and $G : M \to \mathbb{R}^d$ be a smooth parametrization. If for any $x_{\text{init}} \in M$, there is a Legendre function R such that for all time-dependent loss $L_t \in \mathcal{L}$, the gradient flow under $L_t \circ G$ initialized at x_{init} can be written as the mirror flow under L_t with respect to R, then G must be a regular parametrization, and it also holds that for each $x \in M$,*

$$\mathrm{Lie}^{\geq 2}(\partial G)\Big|_x \subseteq \ker(\partial G(x)), \tag{5.3.2}$$

where $\mathrm{Lie}^{\geq K}(\partial G) := \mathrm{span}\Big\{[[[[\nabla G_{j_1}, \nabla G_{j_2}], \ldots], \nabla G_{j_{k-1}}], \nabla G_{j_k}] \mid k \geq K, \forall i \in [k], j_i \in [d]\Big\}$ *is the subset of the Lie algebra generated by the gradients of coordinate functions of G only containing elements of order higher than K, and* $\ker(\partial G(x))$ *is the orthogonal complement of* $\mathrm{span}(\{\nabla G_i(x)\}_{i=1}^d)$ *in $\mathbb{R}^D$.*

Note the necessary condition in (5.3.2) is weaker than assuming that G is a commuting parametrization, and we conjecture that it is indeed sufficient.

Conjecture 5.3.11. The claim in Theorem 5.3.9 still holds, if we relax the commuting assumption to that $\mathrm{Lie}^{\geq 2}(\partial G)\Big|_x \subseteq \ker(\partial G(x))$ for all $x \in M$.

With the above necessary condition (5.3.2), we can formally refute the possibility that one can use mirror flow to characterize the implicit bias of gradient flow for matrix factorization in general settings, as summarized in Corollary 5.3.12. It is also worth mentioning that [LWM19] constructed a concrete counter example showing that the implicit bias for commuting measurements, that gradient flow finds the solution with minimal nuclear norm, does not hold for the general case, where gradient flow could prefer the solution with minimal rank instead.

Corollary 5.3.12 (Gradient flow for matrix factorization cannot be written as mirror flow)**.** *For any $d, r \in \mathbb{N}$, let M be an open set in $\mathbb{R}^{d\times r}$ and $G : M \to \mathbb{R}^{d\times d}$ be a smooth parametrization given by $G(U) = UU^\top$. Then there exists a initial point $x_{\text{init}} \in M$ and a time-dependent loss L_t such that the gradient flow under $L_t \circ G$ starting from U_{init} cannot be written as a mirror flow with respect to any Legendre function R under the loss L_t.*

Proof of Corollary 5.3.12. It turns out that the necessary condition in Theorem 5.3.10 is already violated by only considering the Lie algebra spanned by $\{\nabla G_{11}, \nabla G_{12}\}$. We follow the notation in Example 5.3.5 to define each $E_{ij} \in \mathbb{R}^d$ as the one-hot matrix with the (i, j)-th entry being 1, and denote $\overline{E}_{ij} = \frac{1}{2}(E_{ij} + E_{ji})$ and $\Delta_{ij} = E_{ij} - E_{ji}$. Then $[\nabla G_{11}, \nabla G_{12}](U) = 4(\overline{E}_{11}\overline{E}_{12} - \overline{E}_{12}\overline{E}_{11})U = \Delta_{12}U$ and $[\nabla G_{11}, [\nabla G_{11}, \nabla G_{12}]](U) = (\overline{E}_{11}\Delta_{12} - \Delta_{12}\overline{E}_{11})U = \overline{E}_{12}U$. Further noting that $\langle[\nabla G_{11}, [\nabla G_{11}, \nabla G_{12}]], \nabla G_{12}\rangle = 2\left\|\overline{E}_{12}U\right\|_F^2 = \frac{1}{2}\sum_{i=1}^r(U_{1i}^2 + U_{2i}^2)$ must be positive at some U in every open set M, by Theorem 5.3.10, we know such U_{init} and L_t exist. Moreover, L_t will only depend on $G_{11}(U)$ and $G_{12}(U)$. ∎

The following corollary shows that gradient flow with non-commuting parametrization cannot be mirror flow, when the dimension of the reachable set matches with that of the w-space.

Corollary 5.3.13. *Let M be a smooth submanifold of $\mathbb{R}^D$ and $G : M \to \mathbb{R}^d$ be a regular parametrization. Then G must be a commuting parametrization if for any $x_{\text{init}} \in M$, the following holds:*

(a) $\Omega_x(x_{\text{init}}; G)$ is a submanifold of dimension d.

(b) *There is a Legendre function R such that for any time-dependent loss $L_t \in \mathcal{L}$, the gradient flow governed by $-\nabla(L_t \circ G)$ with initialization x_{init} can be written as a mirror flow with respect to R.*

Proof of Corollary 5.3.13. By the condition (b) and Theorem 5.3.10, we know that each Lie bracket $[\nabla G_i, \nabla G_j] \in \ker(\partial G)$. By the condition (a), we know that each Lie bracket $[\nabla G_i, \nabla G_j] \in \operatorname{span}\{\nabla G_i\}_{i=1}^d$. Combining these two facts, we conclude that each $[\nabla G_i, \nabla G_j] \equiv 0$, so G is a commuting parametrization. ■

Next, we establish the convergence of $w(t) = G(x(t))$ when $x(t)$ is given by some gradient flow with the commuting parametrization G. Here we require that the convex function R given by Lemma 5.3.8 is a Bregman function (see definition in Section 5.6). The proofs of Theorem 5.3.14, Corollary 5.3.15 and Theorem 5.3.16 are in Section 5.8.

Theorem 5.3.14. *Under the setting of Theorem 5.3.9, further assume that the loss L is quasi-convex, ∇L is locally Lipschitz and $\operatorname{argmin}\{L(w) \mid w \in \operatorname{dom} R\}$ is non-empty where $R : \mathbb{R}^d \to \mathbb{R} \cup \{\infty\}$ is the convex function given by Lemma 5.3.8. Suppose R is a Bregman function, then as $t \to \infty$, $w(t)$ converges to some w^* such that $\nabla L(w^*)^\top (w - w^*) \geq 0$ for all $w \in \operatorname{dom} R$. Moreover, if the loss function L is convex, then $w(t)$ converges to a minimizer in $\overline{\operatorname{dom} R}$.*

Corollary 5.3.15. *Under the setting of Theorem 5.3.14, if the reachable set in the w-space satisfies $\Omega_w(x_{\text{init}}; G) = \mathbb{R}^d$, then R is a Bregman function and all the statements in Theorem 5.3.14 hold.*

Theorem 5.3.16. *Under the setting of Theorem 5.3.14, consider the commuting quadratic parametrization $G : \mathbb{R}^D \to \mathbb{R}^d$ where each $G_i(x) = \frac{1}{2} x^\top A_i x$, for symmetric matrices $A_1, A_2, \ldots, A_d \in \mathbb{R}^{D \times D}$ that commute with each other, i.e., $A_i A_j - A_j A_i = 0$ for all $i, j \in [d]$. For any $x_{\text{init}} \in \mathbb{R}^D$, if $\{\nabla G_i(x_{\text{init}})\}_{i=1}^d = \{A_i x_{\text{init}}\}_{i=1}^d$ are linearly independent, then the following holds:*

(a) *For all* $\mu \in \mathbb{R}^d$*,* $\psi(x_{\text{init}}; \mu) = \exp(\sum_{i=1}^d \mu_i A_i) x_{\text{init}}$ *where* $\exp(\cdot)$ *is the matrix exponential defined as* $\exp(A) := \sum_{k=0}^{\infty} \frac{A^k}{k!}$.

(b) *For each* $j \in [d]$ *and all* $\mu \in \mathbb{R}^d$*,* $G_j(\psi(x_{\text{init}}; \mu)) = \frac{1}{2} {x_{\text{init}}}^\top \exp(\sum_{i=1}^d 2\mu_i A_i) A_j x_{\text{init}}$.

(c) $Q(\mu) = \frac{1}{4} \|\psi(x_{\text{init}}; \mu)\|_2^2 = \frac{1}{4} \left\| \exp(\sum_{i=1}^d \mu_i A_i) x_{\text{init}} \right\|_2^2$ *is a Legendre function with domain* $\mathbb{R}^d$.

(d) R *is a Bregman function with* $\operatorname{dom} R = \overline{\operatorname{range} \nabla Q}$ *where* $\operatorname{range} \nabla Q$ *is the range of* ∇Q*, and thus all the statements in Theorem 5.3.14 hold.*

5.3.3 Solving underdetermined linear regression with commuting parametrization

Next, we specialize to underdetermined linear regression problems to showcase our framework.

Setting: underdetermined linear regression. Let $\{(z_i, y_i)\}_{i=1}^n \subset \mathbb{R}^d \times \mathbb{R}$ be a dataset of size n. Given any parametrization G, the output of the linear model on the i-th data is $z_i^\top G(x)$. The goal is to solve the regression for the label vector $Y = (y_1, y_2, \ldots, y_n)^\top$. For notational convenience, we define $Z = (z_1, z_2, \ldots, z_n) \in \mathbb{R}^{d \times n}$.

We can apply Theorem 5.2.9 to obtain the implicit bias of gradient flow with any commuting parametrization.

Theorem 5.3.17. *Let* M *be a smooth submanifold of* $\mathbb{R}^d$ *and* $G : M \to \mathbb{R}^d$ *be a commuting and regular parametrization satisfying Assumption 5.2.5. Suppose the loss function* L *satisfies* $L(w) = \widetilde{L}(Zw)$ *for some differentiable* $\widetilde{L} : \mathbb{R}^n \to \mathbb{R}$*. For any* $x_{\text{init}} \in M$*, consider the gradient flow*

$$\mathrm{d}x(t) = -\nabla(L \circ G)(x(t))\mathrm{d}t, \qquad x(0) = x_{\text{init}}.$$

There exists a convex function R (given by Lemma 5.3.8, depending only on the initialization x_{init} and the parametrization G), such that for any dataset $\{(z_i, y_i)\}_{i=1}^n \subset \mathbb{R}^d \times \mathbb{R}$, if $w(t) = G(x(t))$ converges as $t \to \infty$ and the convergence point $w_\infty = \lim_{t\to\infty} w(t)$ satisfies $Zw_\infty = Y$, then

$$R(w_\infty) = \min_{w:Zw=Y} R(w),$$

that is, gradient flow implicitly minimizes the convex regularizer R among all interpolating solutions.

Proof of Theorem 5.3.17. By Theorem 5.3.9, $w(t)$ obeys the following mirror flow:

$$\mathrm{d}\nabla R(w(t)) = -\nabla L(w(t))\mathrm{d}t, \qquad w(0) = G(x_{\mathrm{init}}).$$

Applying Theorem 5.2.9 yields

$$D_R(w_\infty, G(x_{\mathrm{init}})) = \min_{w:Zw=Y} D_R(w, G(x_{\mathrm{init}})).$$

Therefore, for any $w \in \mathrm{int}(\mathrm{dom}\, R)$ such that $Zw = Y$, we have

$$\begin{aligned} &R(w_\infty) - R(G(x_{\mathrm{init}})) - \langle \nabla R(G(x_{\mathrm{init}}), w_\infty - G(x_{\mathrm{init}}) \rangle \\ &\quad \le R(w) - R(G(x_{\mathrm{init}})) - \langle \nabla R(G(x_{\mathrm{init}}), w - G(x_{\mathrm{init}}) \rangle \end{aligned}$$

which can be reorganized as

$$R(w_\infty) \le R(w) - \langle \nabla R(G(x_{\mathrm{init}})), w - w_\infty \rangle. \tag{5.3.3}$$

Note that by Lemma 5.3.8, we also have

$$\nabla R(G(x_{\mathrm{init}})) = \nabla R(G(\psi(x_{\mathrm{init}}; 0))) = \nabla R(\nabla Q(0)) = 0 \tag{5.3.4}$$

where the last equality follows from the property of convex conjugate. Combining (5.3.3) and (5.3.4), we get $R(w_\infty) \leq R(w)$ for all $w \in \operatorname{int}(\operatorname{dom} R)$ such that $Zw = Y$. By the continuity of R, this property can be further extended to the entire $\operatorname{dom} R$, and for any $w \notin \operatorname{dom} R$, we have $R(w) = \infty$ by definition, so $R(w_\infty) \leq R(w)$ holds trivially. This finishes the proof. ∎

Note that the identity parametrization $w = G(x) = x$ is a commuting parametrization. Therefore, if we run the ordinary gradient flow on w itself and it converges to some interpolating solution, then the convergence point is closest to the initialization in Euclidean distance among all interpolating solutions. This recovers the well-known implicit bias of gradient flow for underdetermined regression.

Furthermore, we can recover the results on the quadratically overparametrized linear model studied in a series of papers [GWB+18, WGL+20, AMN+21], as summarized in the following Corollary 5.3.18. Note that their results assumed convergence in order to characterize the implicit bias, whereas our framework enables us to directly prove the convergence as in Theorem 5.3.16, where the convergence guarantee is also more general than existing convergence results for Example 5.3.4 in [PPVF21, LWA22].

Corollary 5.3.18. *Consider the underdetermined linear regression problem with data $Z \in \mathbb{R}^{d\times n}$ and $Y \in \mathbb{R}^n$. Let $\widetilde{L} : \mathbb{R}^n \to \mathbb{R}$ be a differentiable loss function such that $\widetilde{L}$ is quasi-convex, $\nabla \widetilde{L}$ is locally Lipschitz, and $Y \in \mathbb{R}^n$ is its unique global minimizer. Consider solving $\min_w \widetilde{L}(Zw)$ by running gradient flow on $L(w) = \widetilde{L}(Zw)$ with the quadratic parametrization $w = G(x) = u^{\odot 2} - v^{\odot 2}$ where $x = \binom{u}{v} \in \mathbb{R}^{2d}_+$, for any initialization $x_{\text{init}} \in \mathbb{R}^{2d}_+$:*

$$\mathrm{d}x(t) = -\nabla(L \circ G)(x(t))\mathrm{d}t, \qquad x(0) = x_{\text{init}}.$$

Then as $t \to \infty$, $w(t) = G(x(t))$ converges to some w_∞ such that $Zw_\infty = Y$ and

$$R(w_\infty) = \min_{w:Zw=Y} R(w)$$

where R is given by

$$R(w) = \frac{1}{4}\sum_{i=1}^{d}\left(w_i \operatorname{arcsinh}\left(\frac{w_i}{2u_{0,i}v_{0,i}}\right) - \sqrt{w_i^2 + 4u_{0,i}^2 v_{0,i}^2} - w_i \ln \frac{u_{0,i}}{v_{0,i}}\right).$$

5.4 Every mirror flow is a gradient flow with commuting parametrization

Consider any smooth Legendre function $R : \mathbb{R}^d \to \mathbb{R} \cup \{\infty\}$, and recall the corresponding mirror flow:

$$\mathrm{d}\nabla R(w(t)) = -\nabla L(w(t))\mathrm{d}t.$$

Note that $\operatorname{int}(\operatorname{dom} R)$ is a convex open set of $\mathbb{R}^d$, hence a smooth manifold (see Example 1.26 in [Lee13]). Then $\nabla^2 R$ is a continuous positive-definite metric on $\operatorname{int}(\operatorname{dom} R)$. As discussed previously, the above mirror flow can be further rewritten as the Riemannian gradient flow on the Riemannian manifold $(\operatorname{int}(\operatorname{dom} R), \nabla^2 R)$, i.e.,

$$\mathrm{d}w(t) = -\nabla^2 R(w(t))^{-1}\nabla L(w(t))\mathrm{d}t.$$

The goal is to find a parametrization $G : U \to \mathbb{R}^d$, where U is an open set of $\mathbb{R}^D$ and initialization $x_{\text{init}} \in U$, such that the dynamics of $w(t) = G(x(t))$ can be induced by the gradient flow on $x(t)$ governed by $-\nabla(L \circ G)(x)$. Formally, we have the following result:

Theorem 5.4.1. *Let $R : \mathbb{R}^d \to \mathbb{R} \cup \{\infty\}$ be a smooth Legendre function. There exist* *a smooth submanifold of* $\mathbb{R}^D$ *denoted by* M*, an open neighborhood* U *of* M *and a smooth and regular parametrization* $G : U \to \mathbb{R}^d$ *such that for mirror flow on any time-dependent loss function* L_t *with any initialization* $w_{\text{init}} \in \operatorname{int}(\operatorname{dom} R)$

$$\mathrm{d}\nabla R(w(t)) = -\nabla L_t(w(t))\mathrm{d}t, \quad w(0) = w_{\text{init}}, \tag{5.4.1}$$

it holds that $w(t) = G(x(t))$ *for all* $t \geq 0$ *where* $x(t)$ *is given by the gradient flow under the objective* $L_t \circ G$ *initialized at* x_{init}*, i.e.,*

$$\mathrm{d}x(t) = -\nabla(L_t \circ G)(x(t))\mathrm{d}t, \quad x(0) = x_{\text{init}}. \tag{5.4.2}$$

Moreover, G *restricted on* M*, denoted by* $G|_M$ *is a commuting and regular parametrization and* $\partial G = \partial G|_M$ *on* M*, which implies* $x(t) \in M$ *for all* $t \geq 0$*. If* R *is further a mirror map, then* $\{\nabla G_i|_M\}_{i=1}^d$ *are complete vector fields on* M*.*

To illustrate the idea, let us first suppose such a smooth and regular parametrization G exists and is a bijection between the reachable set $\Omega_x(x_{\text{init}}; G) \subset \mathbb{R}^D$ and $\operatorname{int}(\operatorname{dom} R)$, whose inverse is denoted by F. It turns out that we can show

$$\partial F(w)^\top \partial F(w) = (\partial G(F(w))\partial G(F(w))^\top)^{-1} = \nabla^2 R(w)$$

where the second equality follows from the relationship between R and G as discussed in the introduction on Equation (5.1.2). Note that this corresponds to expressing the metric tensor $\nabla^2 R$ using an explicit map F, which is further equivalent to embedding the Riemannian manifold $(\operatorname{int}(\operatorname{dom} R), \nabla^2 R)$ into a Euclidean space $(\mathbb{R}^D, \overline{g})$ in a way that preserves its metric. This refers to a notion called isometric embedding in differential geometry.

Definition 5.4.2 (Isometric embedding)**.** Let (M, g) be a Riemannian submanifold of $\mathbb{R}^d$. An *isometric embedding* from (M, g) to $(\mathbb{R}^D, \overline{g})$ is an differentiable injective map

$F : M \to \mathbb{R}^D$ that preserves the metric in the sense that for any two tangent vectors $v, w \in T_x(M)$ we have $g_x(v, w) = \overline{g}_x(\partial F(x)v, \partial F(x)w)$ where the standard euclidean metric tensor $\overline{g}$ is defined as $\overline{g}_x(u, v) = \langle u, v\rangle$ for all $u, v \in \mathbb{R}^d$.

Nash's embedding theorem is a classic result in differential geometry that guarantees the existence of isometric embedding of any Riemannian manifold into a Euclidean space with a plain geometry.

Theorem 5.4.3 (Nash's embedding theorem, [Nas56, Gun91])**.** *Any d-dimensional Riemannian manifold has an isometric embedding to $(\mathbb{R}^D, \overline{g})$ for some $D \geq d$.*

The other way to understand Theorem 5.3.9 is that we can view $\nabla^2 R(w)^{-1}\nabla L(w)$ as the gradient of L with respect to metric tensor g_R, where g^R is the Hessian metric induced by strictly convex function R in the sense $g^R_x(u, v) := u^\top \nabla^2 R(x) v$ for any $u, v \in \mathbb{R}^d$. It is well-known that gradient flow is invariant under isometric embedding and thus we can use Nash's embedding theorem to write the gradient flow on riemmanian manifold $(\mathrm{int}(\mathrm{dom}\, R), g^R)$ as that on $(\mathbb{R}^D, \overline{g})$.

5.4.1 Existence of non-separable commuting parametrization

Despite the recent line of works on the connection between mirror descent and gradient descent [GLSS18, AW20a, AW20b, AMN+21, GLH22], so far we have not seen any concrete example of non-separable parametrizaiton (in the sense of Definition 5.4.4) such that the reparametrized gradient flow can be written as a mirror flow. In this subsection, we discuss how we can use Theorem 5.4.1 to construct non-separable, yet commuting parametrizations.

Definition 5.4.4 (Separable parametrization in the general sense)**.** Let M be an open subset of $\mathbb{R}^D$. We say a function $G : M \to \mathbb{R}^d$ is a *generalized separable parametrization* if and only if there exist d projection matrices $\{P_i\}_{i=1}^d$ satisfying

$\sum_{i=1}^{d} P_i = I_d$, $P_iP_j = \mathbb{1}\{i = j\} \cdot P_i$, a function $\widehat{G} : M \to \mathbb{R}^d$ satisfying $\widehat{G}_i(x) = \widehat{G}_i(P_ix)$, a matrix $A \in \mathbb{R}^{d\times d}$ and a vector $b \in \mathbb{R}^d$, such that

$$G(x) = A\widehat{G}(x) + b, \qquad \forall x \in M.$$

Given the above definition, it is easy to check that $\widehat{G}$ is a commuting parametrization as $\nabla^2\widehat{G}_i\nabla\widehat{G}_j = P_i\nabla^2\widehat{G}_iP_i \cdot P_j\nabla\widehat{G}_j \equiv 0$ for all $i \neq j$, so each Lie bracket $[\nabla G_i, \nabla G_j]$ is also 0 by the linearity.

As a concrete example, for matrix sensing with commutable measurement $A_1, \ldots, A_m \in \mathbb{R}^{d\times d}$, let $V = (v_1, \ldots, v_d) \in \mathbb{R}^{d\times d}$ be a common eigenvector matrix for $\{A_i\}_{i=1}^m$ such that we can write $A_i = V\Sigma_iV^\top = \sum_{j=1}^{d} \sigma_{i,j}v_iv_i^\top$ for each $i \in [m]$. With parametrization $G : \mathbb{R}^{d\times r} \to d$ where each $G_i(U) = v_i^\top UU^\top v_i$, we can write $\langle A_i, UU^\top\rangle = \sum_{j=1}^{d} \sigma_{i,j}G_j(U)$.

However, the bad news is that separable commuting parametrizations can only express a restricted class of Legendre functions. It is easy to see $\partial\hat{G}(x)\partial\hat{G}(x)^\top$ must be diagonal for every x. Thus $\partial G(x)\partial G(x)^\top$ are simultaneously diagonalizable for all x, and so are the Hessian of the corresponding Legendre function (given by Lemma 5.3.8). There are interesting Legendre functions that does not always have their Hessians simultaneously diagonalizable, such as

$$R(w) = \sum_{i=1}^{d} w_i(\ln w_i - 1) + \left(1 - \sum_{i=1}^{d} w_i\right)\left(\ln\left(1 - \sum_{i=1}^{d} w_i\right) - 1\right),$$

where each $w_i > 0$ and $\sum_{i=1}^{d} w_i < 1$. We can check that $\nabla R(w) = \sum_{i=1}^{d} \ln \frac{w_i}{1-\sum_{i=1}^{d} w_i}$ and $\nabla^2R(w) = \mathrm{diag}(w^{\odot(-1)}) + \mathbb{1}_d\,\mathbb{1}_d^\top$. It is proposed as an open problem by [AW20b] that whether we can find a parametrization G such that the reparametrized gradient flow in the x-space simulates the mirror flow in the w-space with respect to the aforementioned Legendre function R.

Our Theorem 5.4.1 answers the open problem by [AW20b] affirmatively since it shows every mirror flow can be written as some reparametrized gradient flow. According to the previous discussion, every mirror flow for Lengendre function whose Hessian cannot be simultaneously diagonalized always induces a non-separable commuting parametrization. But this type of construction has two caveats: First, the construction of the Legendre function uses Nash's Embedding theorem, which is implicit and hard to implement; second, the parametrization given by Theorem 5.4.1, though defined on an open set in $\mathbb{R}^D$, is only commuting on the reachable set, which is a d-dimensional submanifold of $\mathbb{R}^D$. This is different from all the natural examples of commuting parametrizations which are commuting on an open set, leading to the following open question.

Open question: Is there any smooth, regular, commuting, yet non-separable (in the sense of Definition 5.4.4) parametrization from an open subset of $\mathbb{R}^D$ to $\mathbb{R}^d$, for some integers D and d?

Theorem 5.4.5. *All smooth, regular and commuting parametrizations are non-separable when $D = 1$.*

Proof of Theorem 5.4.5. Note that $[\nabla G_i, \nabla G_j] \equiv 0$ implies that all G_i share the same set of stationary points, i.e., $\{x \in \mathbb{R} \mid \nabla G_i(x) = 0\}$ is the same for all $i \in [d]$. Since $D = 1$, without loss of generality, we can assume $G_i'(x) = \nabla G_i(x) > 0$ for all $x \in M$ and $i \in [d]$ since G is regular. Then it holds that $\mathrm{sign}(G_i')(\ln |G_i'|)' = \mathrm{sign}(G_j')(\ln |G_j'|)'$, which implies that $|G_i'|/|G_j'|$ is equal to some constant independent of x. This completes the proof. ■

Remark 5.4.6. We note that the assumption that the parametrization is regular is necessary for the open question to be non-trivial. Otherwise, consider the following example with $D = 1$ and $d = 2$: Let $f_1, f_2 : \mathbb{R} \to \mathbb{R}$ be any smooth function supported on $(0, 1)$ and $(1, 2)$ respectively. Define $G_i(x) = \int_0^x f_i(t)\mathrm{d}t$ for all $x \in \mathbb{R}$. Then parametrization G is non-separable.

5.5 Beyond commuting parametrizations: Time warping

Our main equivalence result in Theorem 5.3.9 relies on G being a commuting parametrization. Indeed, it can also be generalized to non-commuting G in some special cases. In particular, [AMN+21] showed in their Section 5 that a time warping technique can be applied to get a time-rescaled mirror descent. This technique can be readily incorporated into our framework, following a streamlined argument same as that in the proof of Lemma 5.3.8. Moreover, our approach provides guidance on how to choose the correct time warping function.

Below, we discuss the extension of our results to this case, and then present explicit calculation for an example of two-layer neural network with one hidden neuron. The analyses in this section can be made fully rigorous, but we only provide key heuristics for illustration.

5.5.1 Time warped commuting gradient flow: general result

For a loss function $L : \mathbb{R}^d \to \mathbb{R}$, recall that the gradient flow governed by $-\nabla L \circ G$ is

$$\mathrm{d}x(t) = -\partial G(x(t))^\top \nabla L(G(x(t)))\mathrm{d}t,$$

and correspondingly,

$$\mathrm{d}w(t) = -\partial G(x(t))\partial G(x(t))^\top \nabla L(w(t))\mathrm{d}t.$$

Now suppose that $G : M \to \mathbb{R}^d$ is not necessarily a commuting parametrization, but there exists some scalar function $\tau : M \to (0, \infty)$ such that $\{\nabla G_i(x)/\tau(x)\}_{i=1}^d$ are commuting vector fields on M, i.e., $[\nabla G_i/\tau, \nabla G_j/\tau] \equiv 0$ on M for all $i \neq j$. For notational simplicity, define $g_i := \nabla G_i/\tau$ for each $i \in [d]$. With a slight abuse of notation, further write $\phi^t_{g_i}(x)$ as the point on M reached after time t by following the vector field $-g_i$ starting at x, and $\psi(x;\mu) = \phi^{\mu_1}_{g_1} \circ \cdots \circ \phi^{\mu_d}_{g_d}(x)$.

Next, we follow the argument in Lemma 5.3.8 to show that in this case, the dynamics of $w(t)$ follows a MF with time warped. More specifically, consider $\Psi(\mu) := G(\psi(x_{\text{init}};\mu))$, and it is easy to see that

$$\partial\Psi(\mu) = \frac{\partial G(\psi(x_{\text{init}};\mu))\partial G(\psi(x_{\text{init}};\mu))^\top}{\tau(\psi(x_{\text{init}};\mu))}$$

which is symmetric and positive definite. Thus, there exists a strictly convex function $Q : \mathbb{R}^d \to \mathbb{R} \cup \{\infty\}$ such that $\nabla Q(\mu) = \Psi(\mu) = G(\psi(x_{\text{init}};\mu))$ for all $\mu \in \mathcal{U}(x_{\text{init}})$. Further let R be the convex conjugate of Q, then we can rewrite the dynamics of $w(t)$ as

$$\mathrm{d}w(t) = -\nabla^2 R(w(t))^{-1}\tau(x(t))\nabla L(w(t))\mathrm{d}t,$$

or equivalently,

$$\mathrm{d}\nabla R(w(t)) = -\tau(x(t))\nabla L(w(t))\mathrm{d}t.$$

Finally, under the setting of Theorem 5.2.9, applying its proof yields that if the above mirror flow converges and $w_\infty = \lim_{t\to\infty} w(t)$ satisfies that $Zw_\infty = Y$, then

$$D_R(w_\infty, w_{\text{init}}) = \min_{w:Zw=Y} D_R(w, w_{\text{init}}).$$

5.5.2 Example: two-layer neural network with one hidden neuron

Consider a two-layer neural network with one hidden neuron as in Theorem 2 in [AMN+21]. Specifically, we parametrize $w \in \mathbb{R}^d$ as $w = a \cdot z$ where $a \in \mathbb{R}$ and $z \in \mathbb{R}^d$. In other words, let $x = \binom{a}{z} \in \mathbb{R}^{d+1}$, and set $w = G(x) := a \cdot z$, where $G : \mathbb{R}^{d+1} \to \mathbb{R}^d$ is the parametrization. Note that here G is not a commuting parametrization. Indeed, for each $i \in [d]$, we have $\nabla G_i(x) = (z_i, 0, \dots, 0, a, 0, \dots 0)^\top = z_i e_1 + a e_{i+1}$, and $\nabla^2 G_i(x) = e_1 e_{i+1}^\top + e_{i+1} e_1^\top$. Then for any distinct $i, j \in [d]$, $[\nabla G_i, \nabla G_j](x) = \nabla^2 G_j(x) \nabla G_i(x) - \nabla^2 G_i(x) \nabla G_j(x) = y_i e_{j+1} - y_j e_{i+1}$, which is not equal to 0 as long as $y_i \neq 0$ or $y_j \neq 0$. Therefore, G is not a commuting parametrization, and the result in Theorem 5.3.9 cannot be applied here. Instead, we use the procedure developed in the previous subsection.

Choose $\tau(x) = a$, and consider $g_i(x) = \nabla G_i(x)/a$ for each $i \in [d]$. We claim that $\{g_i\}_{i=1}^d$ are now commuting vector fields. To see this, we can easily verify that

$$\partial g_i(x) = \begin{pmatrix} -\frac{y_i}{a^2} & 0 & \cdots & 0 & \frac{1}{a} & 0 & \cdots & 0 \\ 0 & 0 & \cdots & 0 & 0 & 0 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & 0 & 0 & 0 & \cdots & 0 \end{pmatrix},$$

and thus

$$\begin{aligned} [g_i, g_j](x) &= \partial g_j(x) g_i(x) - \partial g_i(x) g_j(x) \\ &= (-y_j y_i/a^3 + 1, 0, \dots, 0)^\top - (-y_i y_j/a^3 + 1, 0, \dots, 0)^\top = 0. \end{aligned}$$

Hence, we see that $\{g_i\}_{i=1}^d$ are commuting, thus the argument in the previous subsection implies that the dynamics of $w(t)$ is given by

$$\mathrm{d}\nabla R(w(t)) = -a(t)\nabla L(w(t))\mathrm{d}t.$$

Moreover, we can provide an closed-form expression for the regularizer $D_R(w_\infty, w_{\text{init}})$, similar to the one derived in [AMN+21].

To do so, we first calculate the closed-form expression of $\psi(x_{\text{init}}; \mu)$. Since $g_{i,i+1}(x) = 1$ and $g_{i,j+1}(x) = 0$ for all $j \in [n]$ such that $j \neq i$, we immediately have $\psi(x_0; \mu)_{2:(d+1)} = y_0 + \mu$, and it remains to determine the first coordinate of $\psi(x_0; \mu)$. For any $i \in [d]$, since $g_{i,1}(x) = y_i/a$, we have

$$\frac{\partial \phi_{g_i}^{\mu_i}(x)_1}{\partial \mu_i} = \frac{\phi_{g_i}^{\mu_i}(x)_{i+1}}{\phi_{g_i}^{\mu_i}(x)_1} = \frac{y_i + \mu_i}{\phi_{g_i}^{\mu_i}(x)_1}.$$

Rearranging and integrating the above, we get

$$\phi_{g_i}^{\mu_i}(x)_1^2 = a^2 + 2\mu_i y_i + \mu_i^2.$$

Further applying the above to all $i \in [d]$, we finally obtain

$$\psi(x_0; \mu)_1 = \sqrt{a_0^2 + 2\langle y_0, \mu\rangle + \|\mu\|_2^2} = \sqrt{\|y_0 + \mu\|_2^2 + a_0^2 - \|y_0\|_2^2}$$

Therefore, the closed-from expression of $\psi(x_0; \mu)$ is given by

$$\psi(x_0; \mu) = \left(\sqrt{\|y_0 + \mu\|_2^2 + a_0^2 - \|y_0\|_2^2}, y_{0,1} + \mu_1, \ldots, y_{0,d} + \mu_d\right)^\top. \tag{5.5.1}$$

Next, for $\Psi(\mu) := G(\psi(x_0; \mu))$ and the strictly convex function $Q : \mathbb{R}^d \to \mathbb{R}^d$ such that $\nabla Q(\mu) = \Psi(\mu) = G(\psi(x_0; \mu))$, it follows from (5.5.1) that

$$\nabla Q(\mu) = (y_0 + \mu)\sqrt{\|y_0 + \mu\|_2^2 + a_0^2 - \|y_0\|_2^2}. \tag{5.5.2}$$

Integrating the above, we get, for some constant $C \in \mathbb{R}$,

$$Q(\mu) = \frac{1}{3}(\|y_0 + \mu\|_2^2 + a_0^2 - \|y_0\|_2^2)^{3/2} + C. \tag{5.5.3}$$

Note that here the domain of Q is given by $\operatorname{dom} Q = \{\mu \in \mathbb{R}^d \mid \|y_0+\mu\|_2^2+a_0^2-\|y_0\|_2^2 \geq 0\}$. Further let R be the convex conjugate of Q, then we can rewrite the dynamics of $w(t)$ as

$$\mathrm{d}\nabla R(w(t)) = -\tau(x(t))\nabla L(w(t))\mathrm{d}t.$$

Then under the setting of Theorem 5.2.9, if the above mirror flow converges and $w_\infty = \lim_{t\to\infty} w(t)$ satisfies that $Zw_\infty = Y$, we have

$$D_R(w_\infty, w_0) = \min_{w:Zw=Y} D_R(w, w_0).$$

We only need to determine the closed-form expression of $D_R(w, w_0)$.

Note that by the property of Bregman divergence,

$$\begin{aligned}
D_R(w, w_0) &= D_Q(\nabla R(w_0), \nabla R(w)) \\
&= Q(\nabla R(w_0)) - Q(\nabla R(w)) - \langle \nabla Q(\nabla R(w)), \nabla R(w_0) - \nabla R(w)\rangle \\
&= Q(\nabla R(w_0)) - Q(\nabla R(w)) - \langle w, \nabla R(w_0) - \nabla R(w)\rangle
\end{aligned}$$

where the last equality is due to the fact that $\nabla Q = (\nabla R)^{-1}$ by Lemma 5.6.4. Moreover, since $\nabla R(w_0) = (\nabla Q)^{-1}(G(\psi(x_0; 0)) = 0$, we have

$$D_R(w, w_0) = -Q(\nabla R(w)) + \langle w, \nabla R(w)\rangle + \frac{a_0^3}{3} + C. \tag{5.5.4}$$

Thus it remains to calculate $\nabla R(w)$. Again, since $\nabla R = (\nabla Q)^{-1}$, we can solve from

(5.5.2) that

$$\nabla R(w) = \frac{w}{\sqrt{\frac{a_0^2 - \|y_0\|_2^2}{2} + \sqrt{\|w\|_2^2 + \frac{(a_0^2 - \|y_0\|_2^2)^2}{4}}}} - y_0$$
$$= \frac{w}{\|w\|_2}\sqrt{\sqrt{\|w\|_2^2 + \frac{(a_0^2 - \|y_0\|_2^2)^2}{4}} - \frac{a_0^2 - \|y_0\|_2^2}{2}} - y_0. \tag{5.5.5}$$

Finally, combining (5.5.3), (5.5.4) and (5.5.5), we conclude that

$$D_R(w, w_0) = \frac{a_0^3}{3} - \frac{1}{3}\left(\sqrt{\|w\|_2^2 + \frac{(a_0^2 - \|y_0\|_2^2)^2}{4}} + \frac{a_0^2 - \|y_0\|_2^2}{2}\right)^{3/2}$$
$$+ \|w\|_2\sqrt{\sqrt{\|w\|_2^2 + \frac{(a_0^2 - \|y_0\|_2^2)^2}{4}} - \frac{a_0^2 - \|y_0\|_2^2}{2}} - \langle w, y_0 \rangle.$$

5.6 Related basics for convex analysis

Before we present the proofs of our main results in Sections 5.7, 5.8 and 5.9, it would be helpful to review some related basics for convex analysis.

We first introduce some additional notations. For any function f, we denote its range (or image) by $\operatorname{range} f$. For any set S, we use $\overline{S}$ to denote its closure. For any matrix $\Lambda \in \mathbb{R}^{d \times D}$ and set $S \subseteq \mathbb{R}^D$, we define $\Lambda S = \{\Lambda x \mid x \in S\} \subseteq \mathbb{R}^d$.

Below we collect some related basic definitions and results in convex analysis. We refer the reader to [Roc15] and [BB+97] as main reference sources. In particular, Sections 2, 3 and 4 in [BB+97] provide a clear summary of the related concepts.

Here we consider a convex function $f : \mathbb{R}^d \to \mathbb{R} \cup \{\infty\}$ whose domain is $\operatorname{dom} f = \{w \in \mathbb{R}^d \mid f(w) < \infty\}$. **From now on, we assume by default** that f is continuous on $\operatorname{dom} f$, the interior of its domain $\operatorname{int}(\operatorname{dom} f)$ is non-empty, and f is differentiable on $\operatorname{int}(\operatorname{dom} f)$.

The notions of essential smoothness and essential strict convexity defined below describe certain nice properties of a convex function (see Section 26 in [Roc15]).

Definition 5.6.1 (Essential smoothness and essential strict convexity)**.** If for any sequence $\{w_n\}_{n=1}^{\infty} \subset \mathrm{int}(\mathrm{dom}\, f)$ going to the boundary of $\mathrm{dom}\, f$ as $n \to \infty$, it holds that $\|\nabla f(w_n)\| \to \infty$, then we say f is *essentially smooth*. If f is strictly convex on every convex subset of $\mathrm{int}(\mathrm{dom}\, f)$, then we say f is *essentially strictly convex*.

The concept of *convex conjugate* is critical in our derivation. Specifically, given a convex function $f : \mathbb{R}^d \to \mathbb{R} \cup \{\infty\}$, its convex conjugate f^* is defined as

$$f^*(w) = \sup_{y \in \mathbb{R}^d} \langle w, y \rangle - f(y).$$

The following results characterize the relationship between a convex function and its conjugate.

Theorem 5.6.2 (Theorem 26.3, [Roc15])**.** *A convex function f is essentially strictly convex if and only if its convex conjugate f^* is essentially smooth.*

Proposition 5.6.3 (Proposition 2.5, [BB+97])**.** *If f is essentially strictly convex, then* $\mathrm{range}\, \partial f = \mathrm{int}(\mathrm{dom}\, f^*) = \mathrm{dom}\, \nabla f^*$*, where ∂f is the subgradient of f.*

Lemma 5.6.4 (Corollary 2.6, [BB+97])**.** *If f is essentially strictly convex, then it holds for all $w \in \mathrm{int}(\mathrm{dom}\, f)$ that $\nabla f(w) \in \mathrm{int}(\mathrm{dom}\, f^*)$ and $\nabla f^*(\nabla f(w)) = w$.*

The class of Legendre functions defined in Definition 5.2.8 contains convex functions that are both essentially smooth and essentially strictly convex.

Theorem 5.6.5 (Theorem 26.5, [Roc15])**.** *A convex function f is a Legendre function if and only if its conjugate f^* is. In this case, the gradient mapping $\nabla f : \mathrm{int}(\mathrm{dom}\, f) \to \mathrm{int}(\mathrm{dom}\, f^*)$ satisfies $(\nabla f)^{-1} = \nabla f^*$.*

Next, we introduce the notion of Bregman function [Bre67, CL81]. It has been shown in [BB+97] that the properties of Bregman functions are crucial to prove the trajectory convergence of Riemannian gradient flow where the metric tensor is given by the Hessian of some Bregman function f.

Definition 5.6.6 (Bregman functions; Definition 4.1, [ABB04])**.** A function f is called a *Bregman function* if it satisfies the following properties:

(a) $\operatorname{dom} f$ is closed. f is strictly convex and continuous on $\operatorname{dom} f$. f is $\mathcal{C}^1$ on $\operatorname{int}(\operatorname{dom} f)$.

(b) For any $w \in \operatorname{dom} f$ and $\alpha \in \mathbb{R}$, $\{y \in \operatorname{dom} f \mid D_R(w, y) \leq \alpha\}$ is bounded.

(c) For any $w \in \operatorname{dom} f$ and sequence $\{w_i\}_{i=1}^{\infty} \subset \operatorname{int}(\operatorname{dom} f)$ such that $\lim_{i\to\infty} w_i = w$, it holds that $\lim_{i\to\infty} D_R(w, w_i) \to 0$.

The following theorem provides a special sufficient condition for f to be a Bregman function.

Theorem 5.6.7 (Theorem 4.7, [ABB04])**.** *If f is a Legendre function with* $\operatorname{dom} f = \mathbb{R}^d$, *then* $\operatorname{dom} f^* = \mathbb{R}^d$ *implies that f is a Bregman function.*

The following theorem from [ABB04] provides a convenient tool for proving the convergence of a Riemannian gradient flow.

Theorem 5.6.8 (Theorem 4.2, [ABB04])**.** *Suppose $f : \mathbb{R}^d \to \mathbb{R} \cup \{\infty\}$ is a Bregman function and also a Legendre function, and satisfies that f is twice continuously differentiable on* $\operatorname{int}(\operatorname{dom} f)$ *and $\nabla^2 f$ is locally Lipschitz. Consider the following Riemannian gradient flow:*

$$\mathrm{d}w(t) = -\nabla^2 f(w(t))^{-1} \nabla L(w(t)) \mathrm{d}t, \qquad w(0) = w_{\text{init}} \in \operatorname{int}(\operatorname{dom} f)$$

where the loss $L : \mathbb{R}^d \to \mathbb{R}$ satisfies that L is quasi-convex, ∇L is locally Lipschitz, and $\operatorname{argmin}\{L(w) \mid w \in \operatorname{dom} f\}$ *is non-empty. Then as $t \to \infty$, $w(t)$ converges to some $w^* \in \operatorname{dom} f$ such that $\langle \nabla L(w^*), w - w^* \rangle \geq 0$ for all $w \in \operatorname{dom} f$. If the loss L is further convex, then w^* is a minimizer of L on* $\operatorname{dom} f$.

5.7 Proof of results in Section 5.2

Here we first present the proof for the result on the domain of the flow induced by G.

Proof of Lemma 5.2.6. Fix any $x \in M$. For each $i \in [d]$, let $\mathcal{I}_i(x)$ be the domain of $\phi^t_{G_j}(x)$ in terms of t. If ∇G_i is a complete vector field on M as in Definition 5.2.2, then $\mathcal{I}_i(x) = \mathbb{R}^d$, otherwise $\phi^t_{G_j}(x)$ is defined for t in an open interval containing 0 (see, e.g., Theorem 2.1 in [Lan06]). Then we claim that for any distinct $j_1, j_2, \ldots, j_k \in [d]$ where $k \in [d]$, the set of all $(\mu_{j_1}, \ldots, \mu_{j_k}) \in \mathbb{R}^k$ such that $\phi^{\mu_{j_1}}_{G_{j_1}} \circ \cdots \circ \phi^{\mu_{j_k}}_{G_{j_k}}(x)$ is well-defined is a hyperrectangle given by $\mathcal{I}_{j_1}(x) \times \mathcal{I}_{j_2}(x) \times \cdots \times \mathcal{I}_{j_k}(x)$. Then the desired result can be obtained by letting $(j_1, j_2, \ldots, j_d) = (1, 2, \ldots, d)$. We prove the claim by induction over $k \in [d]$.

The base case for $k = 1$ has already been established above. Next, assume the claim holds for $1, 2, \ldots, k-1$ where $k \geq 3$, and we proceed to show it for k. By the claim for $k-2$, $\phi^{\mu_{j_3}}_{G_{j_3}} \circ \cdots \circ \phi^{\mu_{j_k}}_{G_{j_k}}(x)$ is well-defined for $(\mu_{j_3}, \ldots, \mu_{j_k}) \in \mathcal{I}_{j_3}(x) \times \cdots \times \mathcal{I}_{j_k}(x)$. For any such $(\mu_{j_3}, \ldots, \mu_{j_k})$, $\phi^t_{G_{j_1}} \circ \phi^{\mu_3}_{G_{j_3}} \circ \cdots \circ \phi^{\mu_{j_k}}_{G_{j_k}}(x)$ is well-defined for t in and only in the open interval $\mathcal{I}_{j_1}(x)$ by applying the claim for $k-1$, and similarly $\phi^t_{G_{j_2}} \circ \phi^{\mu_3}_{G_{j_3}} \circ \cdots \circ \phi^{\mu_{j_k}}_{G_{j_k}}(x)$ is also well-defined for t in and only in the open interval $\mathcal{I}_{j_2}(x)$. Note that for any $(s,t) \in \mathcal{I}_{j_1}(x) \times \mathcal{I}_{j_2}(x)$,

$$\phi^s_{G_{j_1}} \circ \phi^{-t}_{G_{j_2}} \circ \phi^t_{G_{j_2}} \circ \phi^{\mu_{j_3}}_{G_{j_3}} \circ \cdots \circ \phi^{\mu_{j_k}}_{G_{j_k}}(x)$$

is well-defined, so by Assumption 5.2.5, we see that

$$\phi^{-t}_{G_{j_2}} \circ \phi^s_{G_{j_1}} \circ \phi^t_{G_{j_2}} \circ \phi^{\mu_{j_3}}_{G_{j_3}} \circ \cdots \circ \phi^{\mu_{j_k}}_{G_{j_k}}(x)$$

is also well-defined, which further implies that $\phi^s_{G_{j_1}} \circ \phi^t_{G_{j_2}} \circ \phi^{\mu_{j_3}}_{G_{j_3}} \circ \cdots \circ \phi^{\mu_{j_k}}_{G_{j_k}}(x)$ is well-defined. Therefore, we conclude that $\phi^{\mu_{j_1}}_{G_{j_1}} \circ \cdots \circ \phi^{\mu_{j_k}}_{G_{j_k}}(x)$ is well-defined for and only for $(\mu_{j_1}, \ldots, \mu_{j_k}) \in \mathcal{I}_{j_1}(x) \times \cdots \times \mathcal{I}_{j_k}(x)$. This completes the induction and hence finishes the proof. ∎

Next, we provide the proof for the implicit bias of mirror flow summarized in Theorem 5.2.9. We need the following lemma that characterizes the KKT conditions for minimizing a convex function R in a linear subspace.

Lemma 5.7.1. *For any convex function $R : \mathbb{R}^d \to \mathbb{R} \cup \{\infty\}$ and $Z \in \mathbb{R}^{n\times d}$, suppose $\nabla R(w^*) = Z^\top \lambda$ for some $\lambda \in \mathbb{R}^n$, then*

$$R(w^*) = \min_{w: Z(w-w^*)=0} R(w).$$

Proof of Lemma 5.7.1. Consider another convex function defined as $\widetilde{R}(w) = R(w) - w^\top Z^\top \lambda$, then $\nabla \widetilde{R}(w^*) = \nabla R(w^*) - Z^\top \lambda = 0$, which implies that

$$\begin{aligned} \tilde{R}(w^*) &= \min_{w\in\mathbb{R}^d} R(w) - w^\top Z^\top \lambda \\ &\le \min_{w: Z(w-w^*)=0} R(w) - w^\top Z^\top \lambda \\ &= \min_{w: Z(w-w^*)=0} R(w) - w^{*\top} Z^\top \lambda. \end{aligned}$$

Since $\widetilde{R}(w^*) = R(w^*) - w^{*\top} Z^\top \lambda$, it follows that

$$R(w^*) \le \min_{w: Z(w-w^*)=0} R(w),$$

and the equality is achieved at $w = w^*$. This finishes the proof. ∎

We then can prove Theorem 5.2.9 by using Lemma 5.7.1.

Proof of Theorem 5.2.9. Since $L(w) = \widetilde{L}(Zw - Y)$, the mirror flow (5.2.4) can be further written as

$$\mathrm{d}\nabla R(w(t)) = -Z^\top \nabla \widetilde{L}(Zw(t) - Y)\mathrm{d}t.$$

Integrating the above yields that for any $t \geq 0$,

$$\nabla R(w(t)) - \nabla R(w_0) = -Z^\top \int_0^t \nabla \widetilde{L}(Zw(s) - Y)\mathrm{d}s \in \mathrm{span}(X^\top),$$

which further implies that $\nabla R(w_\infty) - \nabla R(w_0) \in \mathrm{span}(Z^\top)$. Therefore,

$$\nabla D_R(w, w_0)|_{w=w_\infty} = \nabla R(w_\infty) - \nabla R(w_0) \in \mathrm{span}(Z^\top).$$

Then applying Lemma 5.7.1 yields

$$D_R(w_\infty, w_0) = \min_{w: Z(w-w_\infty)=0} D_R(w, w_0).$$

This finishes the proof. ■

5.8 Proof of results in Section 5.3

Here we provide the omitted proofs in Section 5.3, including four main parts:

(1) Properties of commuting parametrizations (Section 5.8.1);

(2) Necessary condition for a smooth parametrization to be commuting (Section 5.8.2);

(3) Convergence for gradient flow with commuting parametrization (Section 5.8.3);

(4) Results for the underdetermined linear regression (Section 5.8.4).

5.8.1 Properties of commuting parametrizations

We first show the representation formula for gradient flow with commuting parametrization given in Lemma 5.3.7.

Proof of Lemma 5.3.7. Let $\mu(t)$ be given by the following differential equation:

$$\mathrm{d}\mu(t) = -\nabla L_t(G(\psi(x_{\text{init}}; \mu(t))))\mathrm{d}t, \qquad \mu(0) = 0.$$

For any $\mu \in \mathcal{U}(x)$ and $j \in [d]$, $\mu + \delta e_j \in \mathcal{U}(x)$ for all sufficiently small δ, thus

$$\begin{aligned}
\frac{\partial}{\partial \mu_j}\psi(x_{\text{init}}; \mu) &= \lim_{\delta \to 0} \frac{\psi(x_{\text{init}}; \mu + \delta e_j) - \psi(x_{\text{init}}; \mu)}{\delta} \\
&= \lim_{\delta \to 0} \frac{\phi^{\delta}_{G_j}(\psi(x_{\text{init}}; \mu)) - \psi(x_{\text{init}}; \mu)}{\delta} \\
&= \nabla G_j(\psi(x_{\text{init}}; \mu))
\end{aligned}$$

where the second equality follows from the assumption that G is a commuting parametrization and Theorem 5.3.2. Then we have $\frac{\partial \psi(x_{\text{init}};\mu)}{\partial \mu} = \partial G(\psi(x_{\text{init}}; \mu))^\top$ for all $\mu \in \mathcal{U}(x_{\text{init}})$, and thus when $\mu(t) \in \mathcal{U}(x_{\text{init}})$,

$$\begin{aligned}
\mathrm{d}\psi(x_{\text{init}}; \mu(t)) &= \frac{\partial \psi(x_{\text{init}}; \mu(t))}{\partial \mu(t)} \mathrm{d}\mu(t) \\
&= -\partial G(x_{\text{init}}; \mu(t)) \nabla L_t(G(\psi(x_{\text{init}}; \mu(t))))\mathrm{d}t \\
&= -\nabla(L_t \circ G)(\psi(x_{\text{init}}; \mu(t)))\mathrm{d}t.
\end{aligned}$$

Then since $\psi(x_{\text{init}}; \mu(0)) = x_{\text{init}}$ and $\psi(x_{\text{init}}; \mu(t))$ follows the same differential equation and has the same initialization as $x(t)$, we have $x(t) \equiv \psi(x_{\text{init}}; \mu(t))$ for all $t \in [0, T)$. Therefore,

$$\mu(t) = \mu(0) + \int_0^t -\nabla L_t(G(\psi(x_{\text{init}}; \mu(s))))\mathrm{d}s = \int_0^t -\nabla L_t(G(x(s)))\mathrm{d}s$$

for all $t \in [0, T)$, which completes the proof. ■

Next, to prove Lemma 5.3.8, we need the following lemma which provides a sufficient condition for a vector function to be gradient of some other function.

Lemma 5.8.1. *Let $\Psi : C \to \mathbb{R}^d$ be a differentiable function where C is a simply connected open subset of $\mathbb{R}^d$. If for all $w \in C$ and any $i, j \in [d]$, $\frac{\partial}{\partial w_j}\Psi_i(w) = \frac{\partial}{\partial w_i}\Psi_j(w)$, then there exists some function $Q : C \to \mathbb{R}$ such that $\Psi = \nabla Q$.*

Proof of Lemma 5.8.1. This follows from a direct application of Corollary 16.27 in [Lee13]. ■

Based on the above results, we proceed to prove Lemma 5.3.8.

Proof of Lemma 5.3.8. By Lemma 5.2.6, $\mathcal{U}(x_{\text{init}})$ is hyperrectangle, and hence is convex. Next, recall that by the proof of Lemma 5.3.7, we have $\frac{\partial \psi(x_{\text{init}};\mu)}{\partial \mu} = \partial G(\psi(x_{\text{init}};\mu))^\top$ for all $\mu \in \mathcal{U}(x_{\text{init}})$. Denoting $\Psi(\mu) = G(\psi(x_{\text{init}};\mu))$, we further have

$$\partial \Psi(\mu) = \frac{\partial G(\psi(x_{\text{init}};\mu))}{\partial \psi(x_{\text{init}};\mu)} \frac{\partial \psi(x_{\text{init}};\mu)}{\partial \mu} = \partial G(\psi(x_{\text{init}};\mu)) \partial G(\psi(x_{\text{init}};\mu))^\top, \quad \forall \mu \in \mathcal{U}(x).$$

Since G is regular, $\partial G(\psi(x_{\text{init}};\mu))$ is of full-rank for all $\mu \in \mathcal{U}(x_{\text{init}})$, so $\partial \Psi$ is symmetric and positive definite for all $\mu \in \mathcal{U}(x_{\text{init}})$, which implies that Ψ is the gradient of some strictly convex function $Q : \mathbb{R}^d \to \mathbb{R} \cup \{\infty\}$ by Lemma 5.8.1. This Q satisfies that $\nabla Q(\mu) = \Psi(\mu) = G(\psi(x_{\text{init}};\mu))$ for all $\mu \in \mathcal{U}(x_{\text{init}})$. Therefore, Q is a strictly convex function with $\operatorname{dom} \nabla Q = \mathcal{U}(x_{\text{init}})$ and range $\nabla Q = \Omega_w(x_{\text{init}}; G)$.

Next, we show that Q is essentially smooth. If $\mathcal{U}(x_{\text{init}}) = \mathbb{R}^d$, then $\operatorname{dom} Q = \mathbb{R}^d$ and the boundary of $\operatorname{dom} Q$ is empty, so it is trivial that Q is essentially smooth. Otherwise, it suffices to show that for any μ on the boundary of $\operatorname{dom} Q$ and any sequence $\{\mu_k\}_{k=1}^\infty \subset \mathcal{U}(x_{\text{init}})$ such that $\lim_{k\to\infty} \mu_k = \mu_\infty$, we have $\lim_{k\to\infty} \|\nabla Q(\mu_k)\|_2 = \infty$. Since each $\nabla Q(\mu_k) = G(\psi(x_{\text{init}};\mu_k))$, we only need to show that $\lim_{k\to\infty} \|G(\psi(x_{\text{init}};\mu_k))\|_2 = \infty$. Suppose otherwise, then $\{G(\psi(x_{\text{init}};\mu_k)\}_{k=1}^\infty$ is bounded. Note that by Lemma 5.3.7, let $H_k(x) = \langle \mu_k, G(x) \rangle$, and we have

$$\psi(x_{\text{init}};\mu_k) = \phi^1_{-H_k}(x_{\text{init}}) = x_{\text{init}} + \int_0^1 \nabla H_k(\phi^s_{-H_k}(x_{\text{init}}))\mathrm{d}s.$$

Therefore,

$$\|\psi(x_{\text{init}};\mu_k) - x_{\text{init}}\|_2 \le \int_0^1 \left\|\nabla H_k(\phi^s_{-H_k}(x_{\text{init}}))\right\|_2 \mathrm{d}s \le \sqrt{\int_0^1 \left\|\nabla H_k(\phi^s_{-H_k}(x_{\text{init}}))\right\|_2^2 \mathrm{d}s}. \tag{5.8.1}$$

where the second inequality follows from Cauchy-Schwarz inequality. Further note that

$$\begin{aligned} H_k(\psi(x_{\text{init}};\mu_k)) - H_k(x_{\text{init}}) &= \int_0^1 \frac{\mathrm{d}}{\mathrm{d}s} H_k(\phi^s_{-H_k}(x_{\text{init}}))\mathrm{d}s \\ &= \int_0^1 \left\langle \nabla H_k(\phi^s_{-H_k}(x_{\text{init}})), \frac{\mathrm{d}\phi^s_{-H_k}(x_{\text{init}})}{\mathrm{d}s} \right\rangle \mathrm{d}s \\ &= \int_0^1 \|\nabla H_k(\phi^s_{-H_k}(x_{\text{init}}))\|_2^2 \mathrm{d}s. \end{aligned} \tag{5.8.2}$$

Then combining (5.8.1) and (5.8.2), we get

$$\begin{aligned} \|\psi(x_{\text{init}};\mu_k) - x_{\text{init}}\|_2 &\le \sqrt{\langle \mu_k, G(\psi(x_{\text{init}};\mu_k)) - G(x_{\text{init}})\rangle} \\ &\le \sqrt{\|\mu_k\|_2 \cdot \|G(\psi(x_{\text{init}};\mu_k)) - G(x_{\text{init}})\|_2}, \end{aligned}$$

which implies that $\{\psi(x_{\text{init}};\mu_k)\}_{k=1}^{\infty}$ is bounded. Then there exists a convergent subsequence of $\{\psi(x_{\text{init}};\mu_k)\}_{k=1}^{\infty}$, and without loss of generality we assume that $\psi(x_{\text{init}};\mu_k)$ itself converges to some $x_\infty \in M$ as $k \to \infty$. Note that $\psi(x_\infty;\mu)$ is well-defined for μ in a small open neighborhood of 0, and since $\lim_{k\to\infty} \psi(x_{\text{init}};\mu_k) = x_\infty$, for sufficiently large k, $\psi(\psi(x_{\text{init}};\mu_k);\mu)$ is well-defined for μ in a small neighborhood of 0 that does not depend on k. Thus there exists some $\mu \in \mathbb{R}^d$ such that $\mu_k + \mu \notin \mathcal{U}(x_{\text{init}})$ but $\psi(\psi(x_{\text{init}};\mu_k);\mu)$ is well-defined for sufficiently large k. But by Lemma 5.2.6 and Theorem 5.3.2, $\psi(\psi(x_{\text{init}};\mu_k);\mu) = \psi(x_{\text{init}};\mu_k+\mu)$ and thus $\mu_k + \mu \in \mathcal{U}(x_{\text{init}})$, which leads to a contradiction. Hence, we conclude that Q is essentially smooth.

Combining the above, it follows that Q is a Legendre function. Let $R : \mathbb{R}^d \to \mathbb{R} \cup \{\infty\}$ be the convex conjugate of Q. Then by Theorem 5.6.5, R is also a Legendre function. Note that for any $\mu \in \mathcal{U}(x_{\text{init}})$, by the result in [Cro77], we have

$$\nabla^2 R(G(\psi(x_{\text{init}};\mu))) = \nabla^2 R(\nabla Q(\mu)) = \nabla^2 Q(\mu)^{-1} = (\partial G(\psi(x_{\text{init}};\mu))\partial G(\psi(x_{\text{init}};\mu))^\top)^{-1}.$$

Therefore, R and Q are both Legendre functions, and by Proposition 5.6.3, we further have $\operatorname{range} \nabla R = \operatorname{int}(\operatorname{dom} Q) = \operatorname{dom} \nabla Q = \mathcal{U}(x)$ and conversely $\operatorname{dom} \nabla R = \operatorname{range} \nabla Q = \Omega_w(x_{\text{init}}; G)$. This finishes the proof. ■

5.8.2 Necessary condition for a smooth parametrization to be commuting

Proof of Theorem 5.3.10. Fix any initialization $x_{\text{init}} \in M$, and let the Legendre function R be given such that for all time-dependent loss L_t, the gradient flow under $L_t \circ G$ initialized at x can be written as the mirror flow under L_t with respect to the Legendre function R. We first introduce a few notations that will be useful for the proof. For any $s \in \mathbb{R}$, we define a time-shifting operator $\mathcal{T}_s$ such that for any time-dependent loss $L_t(\cdot)$, $(\mathcal{T}_s L)_t(\cdot) = L_{t-s}(\cdot)$. We say a time-dependent loss L_t is supported on finite time if $L_t = \sum_{i=1}^k \mathbb{1}_{t\in[t_i,t_{i+1})} L^{(i)}$ for some $k \geq 1$ where $t_1 = 0$, $t_{k+1} = \infty$ and $L^{(k)} \equiv 0$, and we denote $\operatorname{len}(L) = t_k$. We further define the concatenation of two time-dependent loss L_t, L'_t supported on finite time as $L \parallel L' = L + \mathcal{T}_{\operatorname{len}(L)} L'$. We also use $\overline{L}$ to denote the time-reverse of the time-dependent loss L which is supported on finite time, that is, $\overline{L}_t = L_{\operatorname{len}(L)-t}$ for all $t \geq 0$. For any $j \in [d]$ and $\delta > 0$, we define the following loss function

$$\ell_t^{j,\delta}(w) = \mathbb{1}\{0 \leq t \leq \delta\} \cdot \langle e_j, w\rangle \tag{5.8.3}$$

where e_j is the j-th canonical base of $\mathbb{R}^d$.

Now for any $k \geq 2$, let $\{j_i\}_{i=1}^k$ be any sequence where each $j_i \in [d]$. Then we recursively define a sequence of time-dependent losses as follows: First define $L^{1,\delta} = -\ell^{j_1,\delta}$, then sequentially for each $i = 2, 3, \ldots, k$, we define

$$L^{i,\delta} = L^{i-1,\sqrt{\delta}} \parallel \left(-\ell^{j_i,\sqrt{\delta}}\right) \parallel \left(-\overline{L}^{i-1,\sqrt{\delta}}\right) \parallel \ell^{j_i,\sqrt{\delta}} \tag{5.8.4}$$

where we write $\overline{L}^{i-1,\sqrt{\delta}} = \overline{L^{i-1,\sqrt{\delta}}}$ for convenience. Denote $\iota_i(\delta) = \mathrm{len}(L^{i,\delta})$ for each $i \in [k]$. Then $\iota_1(\delta) = \delta$ and $\iota_i(\delta) = 2\sqrt{\delta} + 2\iota_{i-1}(\sqrt{\delta})$ for $i = 2, 3, \ldots, k$, which further implies

$$\iota_i(\delta) = \sum_{m=1}^{i-1} 2^m \delta^{1/2^m} + 2^{i-1}\delta^{1/2^{i-1}} \text{ for all } i \in [k].$$

Moreover, for each $i = 2, 3, \ldots, k$, the gradient of $L^{i,\delta}$ with respect to w is given by

$$\nabla L_t^{i,\delta}(w) = \begin{cases} \nabla L_t^{i-1,\sqrt{\delta}}(w) & 0 \le t \le \iota_{i-1}(\sqrt{\delta}), \\ -e_{j_i} & \iota_{i-1}(\sqrt{\delta}) < t \le \iota_{i-1}(\sqrt{\delta}) + \sqrt{\delta}, \\ -\nabla \overline{L}_t^{i-1,\sqrt{\delta}}(w) & \iota_{i-1}(\sqrt{\delta}) + \sqrt{\delta} < t \le 2\iota_{i-1}(\sqrt{\delta}) + \sqrt{\delta}, \\ e_{j_i} & 2\iota_{i-1}(\sqrt{\delta}) + \sqrt{\delta} < t \le 2\iota_{i-1}(\sqrt{\delta}) + 2\sqrt{\delta}, \\ 0 & t > 2\iota_{i-1}(\sqrt{\delta}) + 2\sqrt{\delta}. \end{cases} \tag{5.8.5}$$

This inductively implies that for any $t \in [0, \iota_k(\delta)]$, $\nabla L_t^{k,\delta}(w) \in \{e_j\}_{j=1}^d$ does not depend on w and is only determined by t. Therefore, for any initialization $x \in M$, for all sufficiently small $\delta > 0$, the gradient flow under $L^{k,\delta}$ for $\iota_k(\delta)$ time, i.e., $\phi_{L^{k,\delta}}^{\iota_k(\delta)}(x)$, is well-defined. Moreover, it follows from (5.8.5) that

$$\begin{aligned} \int_0^{\iota_{k-1}(\delta)} \nabla L_t^{k,\delta}(w(t))\mathrm{d}t &= \int_0^{\iota_{k-1}(\sqrt{\delta})} \nabla L^{k-1,\sqrt{\delta}}(w(t))\mathrm{d}t + \int_{\iota_{k-1}(\sqrt{\delta})}^{\iota_{k-1}(\sqrt{\delta})+\sqrt{\delta}} -e_{j_k}\mathrm{d}t \\ &\quad + \int_{\iota_{k-1}(\sqrt{\delta})+\sqrt{\delta}}^{2\iota_{k-1}(\sqrt{\delta})+\sqrt{\delta}} -\nabla \overline{L}^{k-1,\sqrt{\delta}}(w(t))\mathrm{d}t + \int_{2\iota_{k-1}(\sqrt{\delta})+\sqrt{\delta}}^{2\iota_{k-1}(\sqrt{\delta})2\sqrt{\delta}} e_{j_k}\mathrm{d}t \\ &= \int_0^{\iota_{k-1}(\sqrt{\delta})} \left(\nabla L_t^{k-1,\sqrt{\delta}}(w(t)) - \nabla \overline{L}_t^{k-1,\sqrt{\delta}}(w(t))\right)\mathrm{d}t = 0 \end{aligned}$$

where the last two equalities follow from the fact that $\nabla L_t^{k-1,\sqrt{\delta}}(w)$ does not depend on w and is only determined by t by our construction.

Hence, the mirror flow with respect to the Legendre function R for the time-dependent loss $L^{k,\delta}$ will return to the initialization after $\iota_k(\delta)$ time since

$$\nabla R(w(\iota_k(\delta))) - \nabla R(w(0)) = \int_0^{\iota_k(\delta)} -\nabla L^{k,\delta}(w(t))\mathrm{d}t = 0.$$

This further implies that

$$G(x_{\text{init}}) = G\Big(\phi_{L^{k,\delta}\circ G}^{\iota_k(\delta)}(x_{\text{init}})\Big)$$

for all sufficiently small δ. Then differentiating with δ on both sides yields

$$\partial G(x) \cdot \left.\frac{\mathrm{d}\phi_{L^{k,\delta}\circ G}^{\iota_k(\delta)}(x_{\text{init}})}{\mathrm{d}\delta}\right|_{\delta=0} = 0. \tag{5.8.6}$$

Note that if the following holds:

$$\left.\frac{\mathrm{d}\phi_{L^{k,\delta}\circ G}^{\iota_k(\delta)}(x_{\text{init}})}{\mathrm{d}\delta}\right|_{\delta=0} = [[[[\nabla G_{j_1}, \nabla G_{j_2}], \ldots], \nabla G_{j_{k-1}}], \nabla G_{j_k}](x_{\text{init}}), \tag{5.8.7}$$

then combining (5.8.6) and (5.8.7) completes the proof, so it remains to verify (5.8.7).

We will prove by induction over k, and now let $\{j_i\}_{i=1}^{\infty}$ be an arbitrary sequence where each $j_i \in [d]$. For notational convenience, we denote for each $k \geq 1$,

$$\pi_{k,\delta}(\cdot) := \phi_{-\ell^{j_k},\delta}^{\delta}(\cdot) \quad \text{and} \quad \Pi_{k,\delta}(\cdot) := \phi_{L^{k,\delta}}^{\iota_k(\delta)}(\cdot).$$

Then their inverse maps are given by $\pi_{k,\delta}^{-1}(\cdot) = \phi_{\ell^{j_k},\delta}^{\delta}(\cdot)$ and $\Pi_{k,\delta}^{-1}(\cdot) = \phi_{-\overline{L}^{k,\delta}}^{\iota_k(\delta)}(\cdot)$ respectively. Since G is smooth, each $\Pi_{k,\sqrt{\delta}}$ is a $\mathcal{C}^\infty$ function of $\delta^{1/2^k}$, and we can expand it in $\delta^{1/2^k}$ as

$$\Pi_{k,\sqrt{\delta}}(x) = x + \sum_{i=1}^{2^k} \frac{\delta^{i/2^k}}{i!} \Delta_{k,i}(x) + r_{k,\delta}(x) \tag{5.8.8}$$

where the remainder term $r_{k,\delta}(x)$ is continuous in x and for each $x \in M$, $r_{k,\delta}(x) = o(\delta)$ (i.e., $\lim_{\delta\to 0} \frac{r_{k,\delta}(x)}{\delta} = 0$), and each $\Delta_{k,i}$ is defined as

$$\Delta_{k,i}(x) = \left.\frac{\mathrm{d}^i \Pi_{k,\sqrt{\delta}}(x)}{\mathrm{d}(\delta^{1/2^k})^i}\right|_{\delta=0}.$$

In particular, for $k = 1$, we have

$$\Pi_{1,\sqrt{\delta}}(x) = \pi_{1,\sqrt{\delta}}(x) = x + \sqrt{\delta}\nabla G_{j_1}(x) + \frac{\delta}{2}\partial(\nabla G_{j_1})(x)\nabla G_{j_1}(x) + r_{1,\delta}(x) \tag{5.8.9}$$

where the second equality holds as well for any other G_j in place of G_{j_1}, with a different but similar remainder term. For any fixed $K \geq 2$, there is a small open neighborhood of x_{init} on M, denoted by $\mathcal{N}_{x_{\text{init}}} \subseteq M$, such that for all $k \in [K]$, we have $r_{k,\delta}(x) = o(\delta)$ uniformly over all $x \in \mathcal{N}_{x_{\text{init}}}$, so we can replace all $r_{k,\delta}(x)$ by $o(\delta)$ when $x \in \mathcal{N}_{x_{\text{init}}}$. Then we claim that for each $k = 2, 3, \ldots, K$,

$$\lim_{\delta\to\infty} \frac{1}{\sqrt{\delta}} \sum_{i=1}^{2^{k-1}} \frac{\delta^{i/2^k}}{i!} \Delta_{k,i}(x) = [[[\nabla G_{j_1}, \nabla G_{j_2}], \ldots], \nabla G_{j_k}](x), \quad \forall x \in \mathcal{N}_{x_{\text{init}}}, \tag{5.8.10}$$

which directly implies (5.8.7). With a slight abuse of notation, the claim is also true for $k = 1$ since $\Delta_{1,1}(x) = \nabla G_{j_1}(x)$ by (5.8.9), so we use this as the base case of the induction. Then, assuming (5.8.10) holds for $k - 1 < K$, we proceed to prove it for k. For convenience, further define $\mathrm{Lie}_G(j_{1:k}) = [[[\nabla G_{j_1}, \nabla G_{j_2}], \ldots], \nabla G_{j_k}]$.

Combining the Taylor expansion in (5.8.8) and (5.8.10) for $k - 1$, we obtain for all

$x \in \mathcal{N}_{x_{\text{init}}}$ that

$$\Pi_{k-1,\sqrt{\delta}}(x) = x + \sqrt{\delta} \cdot \text{Lie}_G(j_{1:(k-1)})(x) + \sum_{i=2^{k-2}+1}^{2^{k-1}} \frac{\delta^{i/2^{k-1}}}{i!} \Delta_{k-1,i}(x) + o(\delta)$$

for sufficiently small δ. Further apply (5.8.9) with G_{j_k} in place of G_{j_1} for sufficiently small δ, and then

$$\begin{aligned}
&\Pi_{k-1,\sqrt{\delta}}\Big(\pi_{k,\sqrt{\delta}}(x)\Big) \\
&\quad = \Pi_{k-1,\sqrt{\delta}}\left(x + \sqrt{\delta}\nabla G_{j_k}(x) + \frac{\delta}{2}\partial(\nabla G_{j_k})(x)\nabla G_{j_k}(x) + o(\delta)\right) \\
&\quad = x + \sqrt{\delta}\nabla G_{j_k}(x) + \frac{\delta}{2}\partial(\nabla G_{j_k})(x)\nabla G_{j_k}(x) + o(\delta) \\
&\qquad + \sqrt{\delta} \cdot \text{Lie}_G(j_{1:(k-1)})\left(x + \sqrt{\delta}\nabla G_{j_k}(x) + \frac{\delta}{2}\partial(\nabla G_{j_k})(x)\nabla G_{j_k}(x) + o(\delta)\right) \\
&\qquad + \sum_{i=2^{k-2}+1}^{2^{k-1}} \frac{\delta^{i/2^{k-1}}}{i!} \Delta_{k-1,i}\left(x + \sqrt{\delta}\nabla G_{j_k}(x) + \frac{\delta}{2}\partial(\nabla G_{j_k})(x)\nabla G_{j_k}(x) + o(\delta)\right) \\
&\qquad + r_{k-1,\delta}\left(x + \sqrt{\delta}\nabla G_{j_k}(x) + \frac{\delta}{2}\partial(\nabla G_{j_k})(x)\nabla G_{j_k}(x) + o(\delta)\right)
\end{aligned}$$

where the second equality follows from the Taylor expansion of $\Pi_{k-1,\sqrt{\delta}}$ and that $\pi_{k,\sqrt{\delta}}(x) \in \mathcal{N}_{x_{\text{init}}}$ for sufficiently small δ. Then by the Taylor expansion of $\text{Lie}_G(j_{1:(k-1)})$ and each $\Delta_{k-1,i}$, we have for all $x \in \mathcal{N}_{x_{\text{init}}}$,

$$\begin{aligned}
\Pi_{k-1,\sqrt{\delta}}\Big(\pi_{k,\sqrt{\delta}}(x)\Big) &= x + \sqrt{\delta}\nabla G_{j_k}(x) + \sqrt{\delta} \cdot \text{Lie}_G(j_{1:(k-1)})(x) + \frac{\delta}{2}\partial(\nabla G_{j_k})(x)\nabla G_{j_k}(x) \\
&\quad + \delta \cdot \partial\text{Lie}_G(j_{1:(k-1)})(x)\nabla G_{j_k}(x) + \sum_{i=2^{k-2}+1}^{2^{k-1}} \frac{\delta^{i/2^{k-1}}}{i!} \Delta_{k-1,i}(x) + o(\delta)
\end{aligned} \tag{5.8.11}$$

for sufficiently small δ. For the other way around, we similarly have

$$\pi_{k,\sqrt{\delta}}\Big(\Pi_{k-1,\sqrt{\delta}}(x)\Big) = \pi_{k,\sqrt{\delta}}\left(x + \sqrt{\delta} \cdot \text{Lie}_G(j_{1:(k-1)})(x) + \sum_{i=2^{k-2}+1}^{2^{k-1}} \frac{\delta^{i/2^{k-1}}}{i!} \Delta_{k-1,i}(x) + o(\delta)\right)$$

$$
\begin{aligned}
&= x + \sqrt{\delta}\nabla G_{j_k}(x) + \sqrt{\delta}\cdot \mathrm{Lie}_G(j_{1:(k-1)}) + \frac{\delta}{2}\partial(\nabla G_{j_k})(x)\nabla G_{j_k}(x) \\
&\quad + \delta\partial(\nabla G_{j_k})(x)\mathrm{Lie}_G(j_{1:(k-1)})(x) + \sum_{i=2^{k-2}+1}^{2^{k-1}} \frac{\delta^{i/2^k}}{i!}\Delta_{k-1,i}(x) + o(\delta)
\end{aligned}
\tag{5.8.12}
$$

for all $x \in \mathcal{N}_{x_{\text{init}}}$, when δ is sufficiently small. Note that $x = \pi^{-1}_{k,\sqrt{\delta}} \circ \Pi^{-1}_{k-1,\sqrt{\delta}} \circ \Pi_{k-1,\sqrt{\delta}} \circ \pi_{k,\sqrt{\delta}}(x)$, thus

$$
\begin{aligned}
\Pi_{k,\delta}(x) - x &= \pi^{-1}_{k,\sqrt{\delta}} \circ \Pi^{-1}_{k-1,\sqrt{\delta}} \circ \pi_{k,\sqrt{\delta}} \circ \Pi_{k-1,\sqrt{\delta}}(x) - x \\
&= \pi^{-1}_{k,\sqrt{\delta}} \circ \Pi^{-1}_{k-1,\sqrt{\delta}} \circ \pi_{k,\sqrt{\delta}} \circ \Pi_{k-1,\sqrt{\delta}}(x) - \pi^{-1}_{k,\sqrt{\delta}} \circ \Pi^{-1}_{k,\sqrt{\delta}} \circ \Pi_{k,\sqrt{\delta}} \circ \pi_{k,\sqrt{\delta}}(x) \\
&= \pi^{-1}_{k,\sqrt{\delta}} \circ \Pi^{-1}_{k-1,\sqrt{\delta}} \circ \pi_{k,\sqrt{\delta}} \circ \Pi_{k-1,\sqrt{\delta}}(x) - \pi_{k,\sqrt{\delta}} \circ \Pi_{k-1,\sqrt{\delta}}(x) \\
&\qquad + \pi_{k,\sqrt{\delta}} \circ \Pi_{k-1,\sqrt{\delta}}(x) - \Pi_{k,\sqrt{\delta}} \circ \pi_{k,\sqrt{\delta}}(x) \\
&\qquad + \Pi_{k-1,\sqrt{\delta}}(x) \circ \pi_{k,\sqrt{\delta}} - \pi^{-1}_{k,\sqrt{\delta}} \circ \Pi^{-1}_{k,\sqrt{\delta}} \circ \Pi_{k,\sqrt{\delta}} \circ \pi_{k,\sqrt{\delta}}(x) \\
&= \Pi_{k-1,\sqrt{\delta}} \circ \pi_{k,\sqrt{\delta}}(x) - \pi_{k,\sqrt{\delta}} \circ \Pi_{k-1,\sqrt{\delta}}(x) + o(\delta)
\end{aligned}
\tag{5.8.13}
$$

where the last equality follows from the Taylor expansion of $\pi^{-1}_{k,\sqrt{\delta}} \circ \Pi^{-1}_{k-1,\sqrt{\delta}}(\cdot)$ in terms of $\sqrt{\delta}$. Now, combining (5.8.11), (5.8.12) and (5.8.13), we obtain

$$
\begin{aligned}
\Pi_{k,\delta}(x) - x &= \delta\left(\partial(\nabla G_{j_k})(x)\mathrm{Lie}_G(j_{1:(k-1)})(x) - \partial\mathrm{Lie}_G(j_{1:(k-1)})(x)\nabla G_{j_k}(x)\right) + o(\delta) \\
&= \delta\cdot[\mathrm{Lie}_G(j_{1:(k-1)}), \nabla G_{j_k}](x) + o(\delta)
\end{aligned}
\tag{5.8.14}
$$

where the second equality follows from the definition of Lie bracket. Comparing (5.8.14) with (5.8.8) yields (5.8.10). This completes the induction for $k \in [K]$ and hence finishes the proof as K is arbitrary. ■

5.8.3 Convergence for gradient flow with commuting parametrization

Proof of Theorem 5.3.14. Recall that the dynamics of $w(t)$ is given by

$$\mathrm{d}w(t) = -\nabla^2 R(w(t))^{-1}\nabla L(w(t))\mathrm{d}t, \qquad w(0) = G(x_{\text{init}}).$$

By Lemma 5.3.8, we know that R is a Legendre function. Therefore, when R is further a Bregman function, we can apply Theorem 5.6.8 to obtain the convergence of $w(t)$. This finishes the proof. ■

Based on Theorem 5.6.7, we can prove the trajectory convergence of $w(t)$ for the special case where $\Omega_w(x_{\text{init}}; G) = \mathbb{R}^d$ as summarized in Corollary 5.3.15.

Proof of Corollary 5.3.15. It suffices to verify that R is a Bregman function in this case. By Lemma 5.3.8, we know that R is a Legendre function and satisfies that $\mathbb{R}^d = \Omega_w(x_{\text{init}}; G) = \operatorname{dom}\nabla R \subseteq \operatorname{dom} R \subseteq \mathbb{R}^d$, which implies $\operatorname{dom} R = \mathbb{R}^d$. Moreover, the domain of its convex conjugate Q is also $\mathbb{R}^d$. Then by Theorem 5.6.7, we see that R is a Bregman function. This finishes the proof. ■

Next, we prove that for a class of commuting quadratic parametrizations, the corresponding Legendre function is also a Bregman function, thus guaranteeing the trajectory convergence.

Proof of Theorem 5.3.16. Since $A_1, A_2, \ldots, A_d$ commute with each other, these matrices can be simultaneously diagonalized. Thus we can assume without loss of generality that each $A_i = \operatorname{diag}(\lambda_i)$ where $\lambda_i \in \mathbb{R}^D$, then $G_i(x) = \lambda_i^\top x^{\odot 2}$. For convenience, we denote $\Lambda = (\lambda_1, \lambda_2, \ldots, \lambda_d)^\top \in \mathbb{R}^{d\times D}$, so the parametrization is given by $G(x) = \Lambda x^{\odot 2}$. Note that for each $i \in [d]$, $\nabla G_i(x) = 2\lambda_i \odot x$ and $\nabla^2 G_i(x) = 2\operatorname{diag}(\lambda_i)$, so for any

$i, j \in [d]$, we have

$$[\nabla G_i, \nabla G_j](x) = 4\mathrm{diag}(\lambda_i)\lambda_j \odot x - 4\mathrm{diag}(\lambda_j)\lambda_i \odot x = 0.$$

Therefore, we see that $G : \mathbb{R}_+^D \to \mathbb{R}^d$ is a commuting parametrization. Also, for any $t \in \mathbb{R}$, $x(t) = x_{\text{init}} - \int_0^t \nabla G_i(x(s))\mathrm{d}s = x_{\text{init}} \odot e^{-2\lambda_i t}$, which proves the first and the second claims. Moreover, if the sign of each coordinate of x will not change from that of initialization, (sign means $+$,$-$ or 0). Without loss of generality, below we will assume every coordinate is non-zero at initialization (otherwise we just ignore it). We can also assume the coordinates at initialization are all positive, as the negatives will induce the same trajectory in terms of $G(x)$. By Theorem 5.3.9, the dynamics of $w(t) = G(x(t))$ is given by

$$\mathrm{d}w(t) = -\nabla^2 R(w(t))^{-1}\nabla L(w(t))\mathrm{d}t, \qquad w(0) = G(x_{\text{init}})$$

for some Legendre function R whose conjugate is denoted by Q. To apply the results in Theorem 5.3.14, it suffices to show that this R is a Bregman function.

To do so, we further denote $\widetilde{w} = x^{\odot 2}$ and $\widetilde{G}(x) = x^{\odot 2}$, then $w = \Lambda\widetilde{w}$ and in this case $\widetilde{G}$ is a commuting parametrization for $\widetilde{w}$ defined on $M = \mathbb{R}_+^D$. Also, we have $\partial G(x) = \Lambda\partial\widetilde{G}(x)$. Let $\widetilde{L} : \mathbb{R}^d \to \mathbb{R}$ be defined by $\widetilde{L}(\widetilde{w}) = L(\Lambda\widetilde{w})$, which satisfies that $\nabla\widetilde{L}(\widetilde{w}) = \Lambda^\top\nabla L(\Lambda\widetilde{w})$. Then the gradient flow with parametrization $\widetilde{G}$ governed by $-\nabla(\widetilde{L} \circ \widetilde{G})(x)$ is given by

$$\begin{aligned}
\mathrm{d}x(t) &= -\nabla(\widetilde{L} \circ \widetilde{G})(x)\mathrm{d}t = -\partial\widetilde{G}(x(t))^\top\nabla\widetilde{L}(\widetilde{G}(x(t))\mathrm{d}t \\
&= -\partial\widetilde{G}(x(t))^\top\Lambda^\top\nabla L(\Lambda\widetilde{G}(x(t))\mathrm{d}t \\
&= -\partial G(x(t))^\top\nabla L(G(x(t))\mathrm{d}t,
\end{aligned}$$

which yields the same dynamics of the gradient flow with parametrization G governed by $-\nabla(L \circ G)(x)$. Therefore, we have $w(t) = G(x(t)) = \Lambda\widetilde{G}(x(t)) = \Lambda\widetilde{w}(t)$, where again by Theorem 5.3.9, the dynamics of $\widetilde{w}(t)$ is

$$\mathrm{d}\widetilde{w}(t) = -\nabla^2 \widetilde{R}(\widetilde{w}(t))^{-1}\nabla\widetilde{L}(\widetilde{w}(t))\mathrm{d}t, \qquad \widetilde{w}(0) = \widetilde{G}(x_{\text{init}})$$

for some Legendre function $\widetilde{R}$ whose conjugate is denoted by $\tilde{Q}$. For any $x \in M$ and $\tilde{\mu} \in \mathbb{R}^D$, we define $\tilde{\psi}(x;\tilde{\mu}) = \phi_{\tilde{G}_1}^{\tilde{\mu}_1} \circ \phi_{\tilde{G}_2}^{\tilde{\mu}_2} \circ \cdots \circ \phi_{\tilde{G}_D}^{\tilde{\mu}_D}(x)$. We need the following lemma.

Lemma 5.8.2. *In the setting of the proof of Theorem 5.3.16, for any $\mu \in \mathbb{R}^d$ and $x \in M$, we have $\psi(x;\mu) = \tilde{\psi}(x;\Lambda^\top\mu)$.*

Recall from Lemma 5.3.8 that $\nabla Q(\mu) = G(\psi(x_{\text{init}};\mu))$ for any $\mu \in \mathbb{R}^d$ and $\nabla\widetilde{Q}(\tilde{\mu}) = \widetilde{G}(\tilde{\psi}(x_{\text{init}};\tilde{\mu}))$ for any $\tilde{\mu} \in \mathbb{R}^D$. Note that

$$\nabla Q(\mu) = \Lambda\psi(x_{\text{init}};\mu)^{\odot 2} = \Lambda\tilde{\psi}(x_{\text{init}};\Lambda^\top\mu)^{\odot 2} = \Lambda\widetilde{G}(\tilde{\psi}(x_{\text{init}};\Lambda^\top\mu)) = \Lambda\nabla\widetilde{Q}(\Lambda^\top\mu) \tag{5.8.15}$$

where the second equality follows from Lemma 5.8.2. This implies that $Q(\mu) = \widetilde{Q}(\Lambda^\top\mu) + C$ for some constant C. Recall the definition of convex conjugate, and we have

$$\widetilde{R}(\widetilde{w}) = \sup_{\widetilde{\mu}\in\mathbb{R}^D} \langle\tilde{\mu},\tilde{w}\rangle - \tilde{Q}(\tilde{\mu}), \qquad R(w) = \sup_{\mu\in\mathbb{R}^d}\langle\mu, w\rangle - Q(\mu).$$

Then for any $\tilde{w} \in \mathbb{R}^D$, we have

$$\begin{aligned} R(\Lambda\tilde{w}) &= \sup_{\mu\in\mathbb{R}^d}\langle\mu,\Lambda\tilde{w}\rangle - Q(\mu) = \sup_{\mu\in\mathbb{R}^d}\langle\Lambda^\top\mu,\tilde{w}\rangle - \tilde{Q}(\Lambda^\top\mu) - C \\ &= \sup_{\tilde{\mu}\in\Lambda^\top\mathbb{R}^d}\langle\tilde{\mu},\tilde{w}\rangle - \tilde{Q}(\tilde{\mu}) - C \le \sup_{\tilde{\mu}\in\mathbb{R}^D}\langle\tilde{\mu},\tilde{w}\rangle - \tilde{Q}(\tilde{\mu}) - C = \widetilde{R}(\tilde{w}) - C \end{aligned} \tag{5.8.16}$$

Therefore, for any $\tilde{w} \in \operatorname{dom}\widetilde{R}$, it holds that $R(\Lambda\tilde{w}) \le \widetilde{R}(\tilde{w}) - C < \infty$, so $\Lambda \operatorname{dom}\widetilde{R} \subseteq$

$\mathrm{dom}\, R$, where $\Lambda\, \mathrm{dom}\, \widetilde{R}$ On the other hand, by (5.8.15) and Proposition 5.6.3, we have

$$\mathrm{dom}\, \nabla R = \mathrm{range}\, \nabla Q \subseteq \Lambda\, \mathrm{range}\, \nabla \tilde{Q} = \Lambda\, \mathrm{dom}\, \nabla \widetilde{R}$$

and it follows that

$$\mathrm{int}(\mathrm{dom}\, R) = \mathrm{dom}\, \nabla R \subseteq \Lambda\, \mathrm{dom}\, \nabla \widetilde{R} = \Lambda\ \mathrm{int}(\mathrm{dom}\, \widetilde{R}).$$

Combining the above, we see that $\mathrm{dom}\, R = \Lambda\, \mathrm{dom}\, \widetilde{R}$. As discussed in Section 5.1, here it is straightforward to verify that $\widetilde{R}(\tilde{w}) = \sum_{i=1}^{D} \tilde{w}_i (\ln \frac{\tilde{w}_i}{x_{\mathrm{init},i}^2} - 1)$, which is indeed a Bregman function with domain $\mathrm{dom}\, \widetilde{R} = \overline{\mathbb{R}_+^D}$. Thus $\mathrm{dom}\, R = \Lambda \overline{\mathbb{R}_+^D}$ is also a closed set. This yields the first condition in Definition 5.6.6.

Next, we verify the second condition in Definition 5.6.6. For any $\mu \in \mathbb{R}^d$, we have

$$\nabla R(G(\psi(x_{\mathrm{init}}; \mu))) = \nabla R(\nabla Q(\mu)) = \mu$$

and

$$\nabla \widetilde{R}(\tilde{G}(\psi(x_{\mathrm{init}}; \mu))) = \nabla \widetilde{R}(\tilde{G}(\tilde{\psi}(x_{\mathrm{init}}; \Lambda^\top \mu))) = \nabla \widetilde{R}(\nabla \tilde{Q}(\Lambda^\top \mu)) = \Lambda^\top \mu.$$

Comparing the above two equalities, we get

$$\nabla \widetilde{R}(\tilde{w}) = \Lambda^\top \nabla R(\Lambda \tilde{w}) \tag{5.8.17}$$

for all $\tilde{w} \in \mathbb{R}_+^D$. Then for any $\tilde{w} \in \overline{\mathbb{R}_+^D}$ and $y = \Lambda \tilde{y} \in \mathrm{int}(\mathrm{dom}\, R)$, we have

$$\begin{aligned} D_R(\Lambda \tilde{w}, y) &= R(\Lambda \tilde{w}) - R(y) - \langle \nabla R(y), \Lambda \tilde{w} - y \rangle \\ &= R(\Lambda \tilde{w}) - R(\Lambda \tilde{y}) - \langle \Lambda^\top \nabla R(\Lambda \tilde{y}), \tilde{w} - \tilde{y} \rangle \\ &= R(\Lambda \tilde{w}) - R(\Lambda \tilde{y}) - \langle \nabla \widetilde{R}(\tilde{y}), \tilde{w} - \tilde{y} \rangle \end{aligned}$$

$$= R(\Lambda\tilde{w}) - R(\Lambda\tilde{y}) - \widetilde{R}(\tilde{w}) + \widetilde{R}(\tilde{y}) + D_{\widetilde{R}}(\tilde{w}, \tilde{y}) \tag{5.8.18}$$
$$\geq R(\Lambda\tilde{w}) - \widetilde{R}(\tilde{w}) + C + D_{\widetilde{R}}(\tilde{w}, \tilde{y})$$

where the inequality follows from (5.8.16). Therefore, we further have for any $\alpha \in \mathbb{R}$

$$\{y \in \operatorname{int}(\operatorname{dom} R) \mid D_R(\Lambda\tilde{w}, y) \leq \alpha\} \subseteq \Lambda\{\tilde{y} \in \mathbb{R}_+^D \mid D_{\widetilde{R}}(\tilde{w}, \tilde{y}) \leq \alpha - R(\Lambda\tilde{w}) + \widetilde{R}(\tilde{w}) - C\}$$

where the right-hand side is bounded since $\widetilde{R}$ is a Bregman function, and so is the left-hand side.

Finally, we verify the third condition in Definition 5.6.6. Consider any $w \in \operatorname{dom} R$ and sequence $\{w_i\}_{i=1}^{\infty} \subset \operatorname{int}(\operatorname{dom} R)$ such that $\lim_{i\to\infty} w_i = w$. Since $\operatorname{dom} R = \Lambda \operatorname{dom} \widetilde{R}$, there is some $\tilde{w} \in \overline{\mathbb{R}_+^D}$ such that $w = \Lambda\tilde{w}$ and some $\tilde{w}_i \in \mathbb{R}_+^D$ for each $i \in \mathbb{N}^+$ such that $w_i = \Lambda\tilde{w}_i$. We have that

$$\begin{aligned}
R(w) - R(w_i) &= \int_0^1 \langle \nabla R((1-t)w_i + tw), w - w_i\rangle \mathrm{d}t \\
&= \int_0^1 \langle \Lambda^\top \nabla R(\Lambda((1-t)\tilde{w}_i + t\tilde{w})), \tilde{w} - \tilde{w}_i\rangle \mathrm{d}t \\
&= \int_0^1 \langle \nabla \widetilde{R}((1-t)\tilde{w}_i + t\tilde{w}), \tilde{w} - \tilde{w}_i\rangle \mathrm{d}t \\
&= \widetilde{R}(\tilde{w}) - \widetilde{R}(\tilde{w}_i).
\end{aligned}$$

Combining this with (5.8.18), we get $D_R(w, w_i) = D_{\widetilde{R}}(\tilde{w}, \tilde{w}_i)$. Note that we can always choose each $\tilde{w}_i$ properly such that $\lim_{i\to\infty} \tilde{w}_i = \tilde{w}$. Then since $\widetilde{R}$ is a Bregman function, we have

$$\lim_{i\to\infty} D_R(w, w_i) = \lim_{i\to\infty} D_{\widetilde{R}}(\tilde{w}, \tilde{w}_i) = 0.$$

Therefore, we conclude that R is also a Bregman function. This finishes the proof. ∎

Proof of Lemma 5.8.2. For each $i \in [D]$ and any $t > 0$, we have

$$\phi^t_{G_i}(x) = x + \int_{s=0}^{t} -\nabla G_i(\phi^s_{f_i}(x))\mathrm{d}s = x + \int_{s=0}^{t} -\sum_{j=1}^{D} \lambda_{i,j} \nabla \tilde{G}_j(\phi^s_{f_i}(x))\mathrm{d}s = \tilde{\psi}(x; t\lambda_i)$$

where the last equality follows from Lemma 5.3.7. Therefore, for any $\mu \in \mathbb{R}^d$, we further have

$$\begin{aligned}
\psi(x;\mu) &= \phi^{\mu_1}_{G_1} \circ \phi^{\mu_2}_{G_2} \circ \cdots \circ \phi^{\mu_d}_{G_d}(x) \\
&= \phi^{\mu_1\lambda_{1,1}}_{\tilde{G}_1} \circ \cdots \circ \phi^{\mu_1\lambda_{1,D}}_{\tilde{G}_D} \circ \cdots \circ \phi^{\mu_d\lambda_{d,1}}_{\tilde{G}_1} \circ \cdots \circ \phi^{\mu_d\lambda_{d,D}}_{\tilde{G}_D}(x) \\
&= \phi^{\sum_{i=1}^d \mu_i\lambda_{i,1}}_{\tilde{G}_1} \circ \cdots \circ \phi^{\sum_{i=1}^d \mu_i\lambda_{i,D}}_{\tilde{G}_D}(x) \\
&= \phi^{(\Lambda^\top\mu)_1}_{\tilde{G}_1} \circ \cdots \phi^{(\Lambda^\top\mu)_D}_{\tilde{G}_D}(x) = \tilde{\psi}(x; \Lambda^\top\mu).
\end{aligned}$$

where the third equality follows from the assumption that $\tilde{G}$ is a commuting parametrization. This finishes the proof. ∎

5.8.4 Results for underdetermined linear regression

Here we provide the proof for the implicit bias result for the quadratically over-parametrized linear model.

Proof of Corollary 5.3.18. By symmetry, we assume without loss of generality that all coordinates of x_{init} are positive. Note that for $M = \mathbb{R}^D_+$ with $D = 2d$, $G : M \to \mathbb{R}^d$ can be written as $G_i(x) = x^\top A_i x$ where each $A_i = e_i e_i^\top - e_{d+i}e_{d+i}^\top$. Therefore, this parametrization G satisfies the conditions in Theorem 5.3.16, which then implies the convergence of $w(t)$.

Next, we identify the function R given by Theorem 5.3.9. we have $\psi(x_{\text{init}};\mu) = \binom{u_0 \odot e^{-2\mu}}{v_0 \odot e^{2\mu}}$ and thus

$$G(\psi(x_{\text{init}};\mu)) = u_0^{\odot 2} \odot e^{-4\mu} - v_0^{\odot 2} \odot e^{4\mu}$$

$$= (u_0^{\odot 2} + v_0^{\odot 2}) \odot \sinh(4\mu) + (u_0^{\odot 2} - v_0^{\odot 2}) \odot \cosh(4\mu).$$

So $G(\psi(x_{\text{init}}; \mu))$ is the gradient of $Q(\mu) = \frac{1}{4}(u_0^{\odot 2} + v_0^{\odot 2}) \odot \cosh(4\mu) + \frac{1}{4}(u_0^{\odot 2} - v_0^{\odot 2}) \odot \sinh(4\mu) + C$ where C is an arbitrary constant. Also note that $(\nabla Q(\mu))_i$ only depends on μ_i, then we have

$$\begin{aligned}(\nabla R(w))_i = (\nabla Q(\mu))_i^{-1}(w) &= \frac{1}{4}\ln\left(\sqrt{1 + \left(\frac{w_i}{2u_{0,i}v_{0,i}}\right)^2} + \frac{w_i}{2u_{0,i}v_{0,i}}\right) + \frac{1}{4}\ln\frac{v_{0,i}}{u_{0,i}} \\ &= \frac{1}{4}\operatorname{arcsinh}\left(\frac{w_i}{2u_{0,i}v_{0,i}}\right) + \frac{1}{4}\ln\frac{v_{0,i}}{u_{0,i}}\end{aligned}$$

which further implies that

$$R(w) = \frac{1}{4}\sum_{i=1}^{d}\left(w_i \operatorname{arcsinh}\left(\frac{w_i}{2u_{0,i}v_{0,i}}\right) - \sqrt{w_i^2 + 4u_{0,i}^2 v_{0,i}^2} - w_i \ln\frac{u_{0,i}}{v_{0,i}}\right) + C.$$

This finishes the proof. ■

5.9 Proof of results in Section 5.4

We first prove the following intermediate result that will be useful in the proof of Theorem 5.4.1.

Lemma 5.9.1. *Under the setting of Theorem 5.4.1, let F be the smooth map that isometrically embeds $(\operatorname{int}(\operatorname{dom} R), g^R)$ into $(\mathbb{R}^D, \overline{g})$. Let $M = \operatorname{range}(F)$, and denote the inverse of F by $\tilde{G} : M \to \mathbb{R}^d$. Then for any $w \in \operatorname{int}(\operatorname{dom} R)$, it holds that*

$$\partial F(w)(\partial F(w)^\top \partial F(w))^{-1} = \partial \tilde{G}(F(w))^\top \quad \textit{and} \quad \partial \tilde{G}(F(w))\partial \tilde{G}(F(w))^\top = \nabla^2 R(w)^{-1}.$$

Proof of Lemma 5.9.1. For any $x \in M$ and $v \in T_x(M)$, consider a parametrized curve $\{x(t)\}_{t \geq 0} \subset M$ such that $x(0) = x$ and $\frac{\mathrm{d}x(t)}{\mathrm{d}t}\big|_{t=0} = v$. Since $x(t) = F(\tilde{G}(x(t)))$ for any

$t \geq 0$, differentiating with respect to t on both sides and evaluating at $t = 0$ yield

$$v = \partial F(\tilde{G}(x))\partial\tilde{G}(x)v. \tag{5.9.1}$$

Now, for any $w \in \operatorname{int}(\operatorname{dom} R)$, let $x = F(w)$, then for any $v \in T_x(M)$, it follows from (5.9.1) that

$$v^\top \partial F(w) = v^\top (\partial F(w)\partial\tilde{G}(F(w)))^\top \partial F(w) = v^\top \partial\tilde{G}(F(w))^\top \partial F(w)^\top \partial F(w).$$

Note that the span of the column space of $\partial F(w)$ is exactly $T_x(M)$, so for any v in the orthogonal complement of $T_x(M)$, it holds that

$$v^\top \partial F(w) = 0 = v^\top \partial\tilde{G}(F(w))^\top \partial F(w)^\top \partial F(w)$$

where the second equality follows from the fact that for any $i \in [d]$, $\nabla\tilde{G}_i(x) \in T_x(M)$. Therefore, combining the above two cases, we conclude that

$$\partial F(w) = \partial\tilde{G}(F(w))^\top \partial F(w)^\top \partial F(w).$$

Since $\partial F(w)^\top \partial F(w) = \nabla^2 R(w)$ is invertible, we then get

$$\partial\tilde{G}(F(w))^\top = \partial F(w)(\partial F(w)^\top \partial F(w))^{-1}.$$

Next, for any $w \in \operatorname{int}(\operatorname{dom} R)$, since $\tilde{G}(F(w)) = w$, differentiating on both sides yields

$$\partial\tilde{G}(F(w))\partial F(w) = I_d.$$

Therefore, using the identity proved above, we have

$$\begin{aligned}\partial\tilde{G}(F(w))\partial\tilde{G}(F(w))^\top &= \partial\tilde{G}(F(w))\partial F(w)(\partial F(w)^\top\partial F(w))^{-1}\\ &= (\partial F(w)^\top\partial F(w))^{-1} = \nabla^2 R(w)^{-1}.\end{aligned}$$

This finishes the proof. ■

Next, we prove Theorem 5.4.1.

Proof of Theorem 5.4.1. By Nash's embedding theorem, there is a smooth map $F : \mathrm{int}(\mathrm{dom}\, R) \to \mathbb{R}^D$ that isometrically embeds $(\mathrm{int}(\mathrm{dom}\, R), g^R)$ into $(\mathbb{R}^D, \overline{g})$. Denote $M = \mathrm{range}(F)$, i.e., the embedding of $\mathrm{int}(\mathrm{dom}\, R)$ in $\mathbb{R}^D$. We further denote the inverse of F on M by $\tilde{G} : M \to \mathbb{R}^d$. Note $(M, \tilde{G})$ is a global atlas for M, we have that $T_x(M) = \mathrm{span}(\{\nabla\tilde{G}_i(x)\}_{i=1}^d)$ for all $x \in M$. This $\tilde{G}$ is almost the commuting parametrization that we seek for, except now it is only defined on M but not on an open neighborhood of M. Yet we can extend $\tilde{G}$ to an open neighbourhood of M in the following way: First by [Foo84], for each $x \in M$, there is an open neighbourhood U_x of x such that projection function P defined by

$$P(y) = \operatorname*{argmin}_{y' \in M} \|y - y'\|_2$$

is smooth in U_x. Then we define $U = \cup_{x\in M} U_x$, and extend $\tilde{G}$ to U by defining $G(x) := \tilde{G}(P(x))$ for all $x \in U$. We have $G(x) = \tilde{G}(x)$ for all $x \in M$, and we can verify that $\partial G \equiv \partial\tilde{G}$ on M as well. For any $v \in T_x(M)$, let $\{\gamma(t)\}_{t\geq 0}$ be a parametrized curve on M such that $\gamma(0) = x$ and $\frac{\mathrm{d}\gamma(t)}{\mathrm{d}t}\Big|_{t=0} = v$, then for sufficiently small t, by Taylor expansion we have

$$\begin{aligned}\gamma(t) = P(\gamma(t)) &= P(x) + \partial P(x)(\gamma(t) - x) + o(\|\gamma(t) - x\|_2)\\ &= x + \partial P(x)(\gamma(t) - x) + o(\|\gamma(t) - x\|_2)\end{aligned}$$

which implies that $v = \partial P(x)v$ by letting $t \to 0$. While for any v in the orthogonal complement of $T_x(M)$, for sufficiently small $\delta > 0$, we have $P(x + \delta v)$ is smooth in δ. Then since $P(x + \delta v) \in M$ for all sufficiently small δ by its definition, we have

$$\partial P(x)v = \left.\frac{\mathrm{d}P(x + \delta v)}{\mathrm{d}\delta}\right|_{\delta=0} = \lim_{\delta \to 0} \frac{P(x + \delta v) - P(x)}{\delta} =: u \in T_x(M). \tag{5.9.2}$$

Note that $\|x+\delta v - P(x+\delta v)\|_2 \le \|x+\delta v - P(x)\|_2 = \delta\|v\|_2$, and by Taylor expansion, we have

$$\|x + \delta v - P(x + \delta v)\|_2 = \|x + \delta v - \delta \partial P(x)v + O(\delta^2)\|_2 = \|x + \delta v - \delta u + O(\delta^2)\|_2$$

where $O(\delta^2)$ denotes a term whose norm is bounded by $C\delta^2$ for a constant $C > 0$ for all sufficiently small δ, and the second equality follows from (5.9.2). Then dividing both sides by δ and letting $\delta \to 0$, we have $\|v\|_2 \ge \|v - u\|_2$. Since u is orthogonal to v, we must have $u = 0$. As v is arbitrary, we conclude that $\partial P(x)$ is the orthogonal projection matrix onto $T_x(M)$. Then differentiating both sides of $G(x) = \tilde{G}(P(x))$ with x yields

$$\partial G(x) = \partial \tilde{G}(P(x))\partial P(x) = \partial \tilde{G}(x) \tag{5.9.3}$$

where the second equality follows from the fact that $T_x(M) = \mathrm{span}(\{\nabla \tilde{G}_i(x)\}_{i=1}^d)$. This further implies that the solution of Equation (5.4.2) satisfies $\mathrm{d}x/\mathrm{d}t = -\nabla(L \circ \tilde{G})(x) \in T_x(M)$, and thus $x(t) \in M$ for all $t \ge 0$.

Now we consider the mirror flow

$$\mathrm{d}w(t) = -\nabla^2 R(w(t))^{-1} \nabla L_t(w(t))\mathrm{d}t, \qquad w(0) = w_{\text{init}}.$$

Since $\nabla^2 R(w) = \partial F(w)^\top \partial F(w)$ by the fact that F is an isometric embedding, we further have

$$\mathrm{d}w(t) = -\Big(\partial F(w(t))^\top \partial F(w(t))\Big)^{-1} \nabla L_t(w(t))\mathrm{d}t.$$

Now define $x(t) = F(w(t))$, and it follows that

$$\begin{aligned}\mathrm{d}x(t) = \partial F(w(t))\mathrm{d}w(t) &= -\partial F(w(t))(\partial F(w(t))^\top \partial F(w(t)))^{-1}\nabla L_t(w(t))\mathrm{d}t \\ &= -\partial G(F(w(t)))^\top \nabla L_t(w(t))\mathrm{d}t = -\nabla(L_t \circ G)(x(t))\mathrm{d}t\end{aligned}$$

where the third equality follows from Lemma 5.9.1 and (5.9.3).

Next, we verify that G restricted on M, $\tilde{G}$, is a commuting and regular parametrization. First, for any $x \in M$, we have $\partial \tilde{G}(x)^\top = \partial F(\tilde{G}(x))(\partial F(\tilde{G}(x))^\top \partial F(\tilde{G}(x)))^{-1}$ by Lemma 5.9.1 and (5.9.3). Since $\nabla^2 R(w) = \partial F(w)^\top \partial F(w)$ is of rank d for all $w \in \mathrm{int}(\mathrm{dom}\, R)$, it follows that $\partial F(w)$ is also of rank d for all $w \in \mathrm{int}(\mathrm{dom}\, R)$, thus $\partial \tilde{G}(x)$ is of rank d for all $x \in M$. The commutability of $\{\nabla \tilde{G}_i\}_{i=1}^d$ follows directly from Corollary 5.3.13. Here we just need to show $\mathrm{rank}(\Omega_x(x;\tilde{G})) = \mathrm{rank}(M)$. This is because on one hand $\mathrm{rank}(\Omega_x(x;\tilde{G})) \geq \mathrm{rank}(\mathrm{span}(\{\nabla \tilde{G}_i(x)\}_{i=1}^d)) = \mathrm{rank}(M)$, and on the other hand, $\mathrm{rank}(\Omega_x(x;\tilde{G})) \leq \mathrm{rank}(M)$ since $\Omega_x(x;\tilde{G}) \subset M$, for any $x \in M$.

Finally, we show that when R is a mirror map, each $\nabla \tilde{G}_j$ is a complete vector field on M. For any $x_{\mathrm{init}} \in M$, consider loss $L_t(w) = \langle e_j, w\rangle$, and the corresponding gradient flow is

$$\mathrm{d}x(t) = -\nabla(L_t \circ \tilde{G})(x(t))\mathrm{d}t = -\partial \tilde{G}(x(t))^\top \nabla L_t(\tilde{G}(x(t)))\mathrm{d}t = -\nabla \tilde{G}_j(x(t)),$$

so $x(t) = \phi^t_{\tilde{G}_j}(x_{\mathrm{init}})$ for all $t \geq 0$. On the other hand, $w(t) = \tilde{G}(x(t))$ satisfies that

$$\begin{aligned}\mathrm{d}w(t) = \partial \tilde{G}(x(t))\mathrm{d}x(t) &= -\partial \tilde{G}(x(t))\partial \tilde{G}(x(t))^\top \nabla L_t(w(t))\mathrm{d}t \\ &= -\nabla^2 R(w(t))^{-1}\nabla L_t(w(t))\mathrm{d}t = -\nabla^2 R(w(t))^{-1} e_j \mathrm{d}t\end{aligned}$$

where the third equality follows from Lemma 5.9.1 and Equation (5.9.3). Therefore, rewriting the above as a mirror flow yields

$$\mathrm{d}\nabla R(w(t)) = -e_j \mathrm{d}t,$$

the solution to which exists for all $t \in \mathbb{R}$ and is given by $\nabla R(w(t)) = e_j t$, so $w(t) = (\nabla R)^{-1}(e_j t)$ is defined for all $t \in \mathbb{R}$ as ∇R is surjective. This further implies that $x(t) = F(w(t))$ is well-defined for all $t \in \mathbb{R}$, hence $\nabla \tilde{G}_j$ is a complete vector field. ■

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